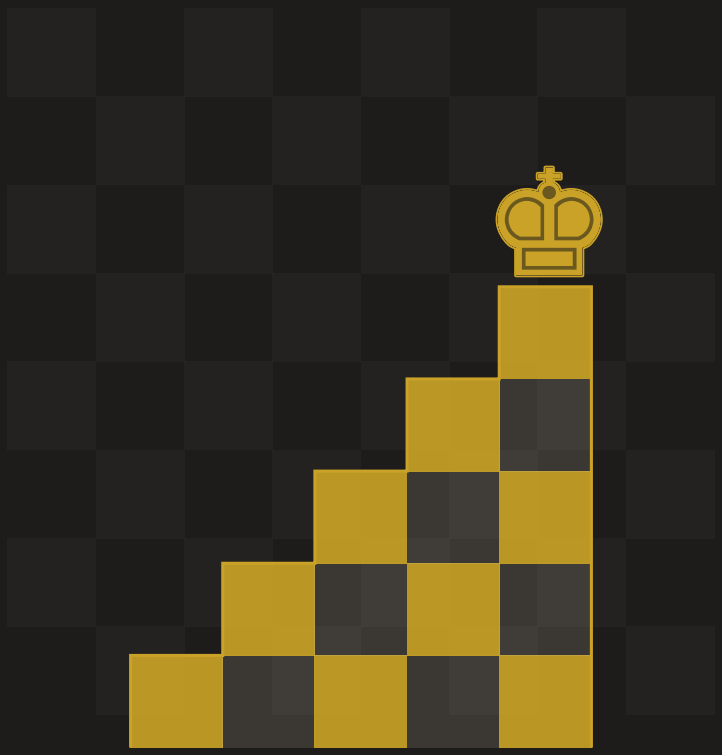
A COMPLETE CHESS IMPROVEMENT SYSTEM

THE ELO BLUEPRINT

Learn · Solve · Play · Analyze · Fix · Repeat

400 → 1500



ELO BLUEPRINT EDITIONS · FIRST EDITION

The Elo Blueprint

The Complete 400 → 1500 Chess Improvement System

ELO BLUEPRINT EDITIONS

The Elo Blueprint — The Complete 400 → 1500 Chess Improvement System
First Edition, version 1.0.0 (2026-11-11).

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How this book was made. The text was written with the assistance of an AI system (Claude, by Anthropic) under a detailed editorial specification. Every diagram, move sequence and all 383 exercises were checked programmatically for legality with python-chess and analysed with the Stockfish 18 chess engine; exercise solutions, listed wrong answers and recommended opening lines are engine-verified. Instructional judgments that go beyond what an engine can verify are labelled as coaching judgment.

Typeset in Source Serif 4, Cormorant Garamond and Inter (SIL Open Font License). Chess pieces drawn from the DejaVu Sans glyph outlines (DejaVu/Bitstream Vera licence).

Companion products: *The Elo Cheat Sheets*, *The 90-Day Chess Accelerator*, and *The Elo Blueprint Interactive* trainer (offline web app).

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How to Use This Book

You probably picked up this book after a run of games that went the same way. You had a decent position, and then a piece disappeared or you walked into checkmate. You started another game straight away, and the same thing happened again.

This book is a complete training system for going from about 400 to about 1500 Elo. It teaches you what to look at, what to decide and what to do at the board, and it gives you a routine that turns each game into improvement instead of just another result.

Who this book is for

This book is written for an adult or teenage player who:

- knows how the pieces move, or mostly knows (Chapter 1 fills every gap);
- plays online or over the board and is rated somewhere around 400 to 1000, or has no rating yet;
- wants a structured plan rather than random videos and puzzles;
- is willing to do some real work: solving positions with a pencil, playing slower games, and looking honestly at their own mistakes.

You need no coach, club or subscription: a chess set or any free app is enough. If you are already above 1000, take the stage tests and start where they show gaps.

The core loop: LEARN, SOLVE, PLAY, ANALYZE, FIX, REPEAT

Every chapter plugs into the same six-step loop. The loop matters more than any single chapter, because reading alone does not change how you play. Here is what each step means in practice.

LEARN. Read the chapter with a board in front of you. Set up every diagram and play through every move sequence. When the text says “First I check…”, stop and

do the check yourself before reading on. Plan on 45 to 90 minutes per chapter, in two or three sittings if you like.

SOLVE. Each chapter ends with 11 exercises. Solve them as described below: honestly, without moving pieces, writing your answer down before you look. The exercises are where the chapter's idea moves from "I understood it" to "I can find it".

PLAY. Play at least two or three slow games that week. Each chapter ends with a concrete assignment for your next game, such as "before every move, check all of your opponent's checks and captures". Try to carry out that one instruction in every game, even if you forget everything else.

ANALYZE. After each game, before you switch on an engine, go through the game on your own for ten minutes. Find the move where the game turned. Write down what you were thinking at that moment, if you can remember. Only then check the game with an engine, and compare what it says with what you thought.

FIX. Turn each serious mistake into a one-line entry in a mistake journal: the date, the position (a FEN or a photo is fine), what you played, what was better, and a category such as *hanging piece*, *missed tactic*, *king safety* or *endgame*. After ten games, count the categories. The biggest category tells you which chapter to go back to. Chapter 28 shows the full method.

REPEAT. Go round the loop again with the next chapter, and keep revisiting the mistakes you have logged. Most improvement at this level comes from fixing the same two or three habits many times over.

> KEY IDEA **The loop in one sentence**
>
> Learn an idea, drill it until you can find it, use it in real games, look honestly at what happened, and fix the pattern that cost you points.

The five stages

The book is divided into five stages. The rating ranges are milestones that describe typical needs. They are not entry requirements and they are not promises. People improve at very different speeds, and ratings on different sites are not directly comparable.

Stage	Rating range	Main goal	Chapters
1	400–600	Eliminate elementary blunders	1–6
2	600–800	Tactical awareness	7–13
3	800–1000	Calculation and conversion	14–18
4	1000–1200	Positional judgment	19–23
5	1200–1500	Calculation, planning and endgame technique	24–28

Stage 1 (400–600): eliminate elementary blunders. Most games at this level are decided by a piece left hanging or a one-move checkmate. You will learn the complete rules, read and write notation, count attackers and defenders, run a safety scan before every move, deliver the basic checkmates, and play sound opening moves.

Stage 2 (600–800): tactical awareness. Now the games are decided by two-move ideas: forks, pins, skewers, discovered attacks and simple mating patterns. You will also build a small, reliable opening repertoire for White and Black.

Stage 3 (800–1000): calculation and conversion. You will learn a move-selection routine, calculate short forcing lines accurately, turn winning positions into wins, defend worse positions stubbornly, and play basic king and pawn endgames.

Stage 4 (1000–1200): positional judgment. Here you learn what to do when there is no tactic: improving your worst piece, understanding pawn structures, attacking a weakened king, making a plan, and handling the most common rook endgames.

Stage 5 (1200–1500): calculation, planning and endgame technique. Harder tactics, middlegame strategy, technical endgames, practical play, and a lifelong method for analyzing your games.

Each stage ends with a test of 15 positions. Use it as a checkpoint: it shows whether you can move on or which chapters to revisit.

> COACHING JUDGMENT **Do not skip Stage 1**
>
> Coaching judgment: many players who feel "beyond the basics" still lose a large share of their games to pieces left undefended. Even if you are rated above 600, take the Stage 1 test before you skip anything. If you score well, you lose an hour. If you do not, you have found your fastest route to more wins.

How the exercises work

The exercises are the heart of the book. Each one gives a position and a task, such as "White to move. Win material" or "Black to move. Is the pawn on e4 safe to take? Play the right move." Every exercise has one move-based answer, though sometimes a second move is listed as equally good.

Work through them like this:

1. **Set up the position** on a board or in an app, from the side of the player to move.
2. **Do not move any pieces.** Look, think, and calculate in your head, as you would in a game.
3. **Write down your answer**, including the main reply you expect from your opponent. Writing forces you to commit. "I was going to play that" does not count.
4. **Check the solution** in the Solutions section at the back of the book. Read the explanation even when you were right; it often contains the reason you were right, which is the part you need next time.
5. **Mark the result**: correct first time, correct with the hint, or wrong.

Some exercises have no tactic at all: the answer is a sound, quiet move, and the trap is a capture that does not work. In real games most moves are not tactics, and you need to know when to stop looking for one.

When to move on

A good guideline is about **80% first-try accuracy** on a chapter's exercise set, which means 9 of 11. If you score lower, re-read the sections that cover the exercises you missed, wait a day, and try those exercises again. Do not grind the same

set ten times in one evening: once you remember the answers, the set no longer tests anything.

Revisiting your misses

Every exercise you get wrong goes onto a review list. Revisit each one after about **1 day, 3 days, 7 days and 21 days**. On each visit, set up the position fresh and solve it again without looking at your old notes. If you solve it cleanly twice in a row, take it off the list. A simple notebook works fine: write the exercise number and the four dates when you first miss it.

> IN YOUR NEXT GAME **Start your review list today**
>
> Open a notebook or a notes file and give it two sections: “Exercises to review” and “Mistakes from my games”. You will use both in every chapter.

Using a board or a free app

A physical set is the best way to read the chapters, because moving real pieces helps you keep track of the position. A screen works equally well for the exercises and for playing out positions.

Every diagram and exercise in the book comes with a **FEN**, a single line of text that describes a position exactly. Chapter 2 explains how to read one. To load a FEN into a free app:

1. Open the analysis board of any free chess website or app.
2. Look for an option called “Set up position”, “Load FEN”, “Board editor” or similar.
3. Paste or type the FEN and confirm. Check that the correct side is to move.

Once a position is on the screen, you can:

- **Solve first, then check.** Switch the engine off, solve the position, then switch the engine on to confirm your answer.
- **Play it out against a friend.** Set up the position on a real board, and take the side that should be winning. Your friend defends. If you cannot win a “winning” position against a human, you have found something to practise.

- **Play it out against an engine.** Many apps let you play from any position against a computer at a chosen strength, which is especially useful for endgames.

> COMMON MISTAKE **Letting the engine think for you**
>
> If you open the engine before deciding on your own move, you will feel that you understand the position and remember very little. Decide first, then check.

Play slow games

You do not need to play blitz or bullet at all to use this book. The book's routines (the safety scan, counting, looking for candidate moves) take time to run, especially at first, and you need games where you can afford that.

- **Recommended:** 15 minutes plus 10 seconds per move, or slower. Thirty-minute games are excellent.
- **Also excellent:** correspondence or "daily" chess, where you have a day or more per move. It is ideal for practising the full thinking routine without time pressure.
- **Over the board:** if you play at a club, longer time limits are usually standard, which suits this book perfectly.

Timed play is optional throughout. Clock rules are covered briefly in Chapter 1 and time management in Chapter 27.

> COACHING JUDGMENT **Fewer, better games**
>
> Coaching judgment: three slow games that you analyze properly are worth more than thirty fast games you never look at again.

Companion products

The book is complete on its own. Three optional companions use the same positions and the same methods:

- **The Cheat Sheets.** One printable page per chapter with the chapter's decision procedure, the same steps as the summary box at the end of each chapter. Keep the current one next to your board while you play slow games, until the steps become automatic.
- **The 90-Day Accelerator workbook.** A day-by-day schedule that tells you which chapter to read, which exercises to solve, which games to play, and when to review your misses. Use it if you prefer to follow a plan rather than set your own pace.
- **The Interactive Trainer.** The book's exercises in an app that lets you play the moves on screen, checks your answer, shows the hint and explanation, and schedules your review dates for you.

None of these is required.

What is evidence, and what is coaching judgment

Chess instruction is full of confident advice, much of it never tested.

When a recommendation is supported by published research, it appears in an **evidence** box with the source. When it is the accumulated experience of chess coaching, it appears in a **coach** box, or is described as a guideline. Both are useful. Only the first has been measured. The research below underpins the way the exercises and review schedule are designed.

> EVIDENCE **Testing yourself beats rereading**
>
> Roediger and Karpicke (2006), "Test-enhanced learning: Taking memory tests improves long-term retention," *Psychological Science*. Students who practised recalling material remembered more of it a week later than students who spent the same time rereading it. This is why the book asks you to solve exercises and write down answers, not just play through solutions.

EVIDENCE **Spacing your review works better than cramming**

Cepeda, Pashler, Vul, Wixted and Rohrer (2006), "Distributed practice in verbal recall tasks: A review and quantitative synthesis," *Psychological Bulletin*. Across many studies, practice spread over time produced better long-term retention than the same amount of practice massed together. The 1-3-7-21 day review schedule applies this idea; the exact intervals are coaching judgment.

EVIDENCE **Serious study matters in chess**

Charness, Tuffiash, Krampe, Reingold and Vasyukova (2005), "The role of deliberate practice in chess expertise," *Applied Cognitive Psychology*. Among tournament players, the amount of time spent in serious solitary study, such as analyzing positions and games, was strongly associated with chess skill. Playing alone was a weaker predictor. This is why the loop includes analysis and fixing, not only playing.

EVIDENCE **Strong players see patterns, not single pieces**

Chase and Simon (1973), "Perception in chess," *Cognitive Psychology*. Stronger players reconstructed briefly shown game positions far better than weaker players, but the advantage largely disappeared with randomly placed pieces. Expertise depends on recognizing familiar groups of pieces, often called **chunks**. This is why the book repeats key patterns many times in realistic positions.

Everything else in this book, including the rating milestones, the 80% guideline and the order of the chapters, is coaching judgment: what usually works for improving players. Treat it as advice, not law, and adjust what clearly does not work for you.

A quick word on notation

The book uses standard algebraic notation, the same system every chess app and book uses. Every square has a name made of a letter (the file, a to h) and a number (the rank, 1 to 8): the square in the bottom-left corner from White's side is a1.

Moves are written with a capital letter for the piece and the square it moves to: K for king, Q for queen, R for rook, B for bishop and N for knight. Pawns have no letter. So `Nf3` means “a knight moves to f3”, and `e4` means “a pawn moves to e4”. An `x` means a capture (`Bxf7`), `+` means check, and `#` means checkmate. `O-O` is castling kingside and `O-O-O` is castling queenside.

When a move is numbered, `12.Nf3` is White’s 12th move and `12...Nf6` is Black’s reply. Throughout the book, **White** and **Black** with capital letters mean the players.

If notation is new to you, read Chapter 2 before going further. It takes about an hour to learn, and after that every diagram and line in this book will be easy to follow.

Your first week

To get started:

1. Read Chapter 1 with a board in front of you, even if you think you know the rules. Most players discover at least one rule they had wrong.
2. Solve the Chapter 1 exercises and start your review list.
3. Read Chapter 2 and practise writing down a short game.
4. Play two slow games and keep a scoresheet for each.
5. After each game, spend ten minutes finding the move where it turned, before using an engine.

Then turn the page and begin.

PART ONE

Stop Giving Things Away

Stage 1 · 400–600

At 400–600, games are decided by pieces left undefended and checkmates nobody saw coming. This part removes those losses first. You will learn the complete rules, read the board accurately, count exchanges, run a safety scan before every move, deliver the basic checkmates, and open the game without walking into traps. None of it is glamorous. All of it wins games immediately.

Chapter 1 · The Rules, Completely

Chapter 2 · Reading the Board: Notation and Board Vision

Chapter 3 · Piece Values and Counting

Chapter 4 · The Safety Scan: Checks, Captures, Threats

Chapter 5 · The Basic Checkmates

Chapter 6 · Opening Principles and Early Traps

CHAPTER 1 STAGE 400–600

The Rules, Completely

Every rule you need, including the ones most beginners get wrong

YOU WILL LEARN TO

- Set up the board correctly and know exactly how every piece moves and captures.
- Tell check, checkmate and stalemate apart at a glance.
- Know every condition for castling, en passant and promotion, including underpromotion.
- Recognize every way a game can be drawn, and avoid stalemating an opponent you are beating.
- Follow touch-move, resigning and clock etiquette in over-the-board play.

> IN YOUR NEXT GAME
>
> Before you castle, capture en passant or promote, run the rule checklist from this chapter; and when you are far ahead, check that your opponent still has a legal move before you play yours.

In a club game, a player with a queen against a lone king spends twenty moves chasing it around the board, then plays a natural-looking queen move. The opponent smiles, says “stalemate”, and offers a handshake. The game is a draw. In another game, someone wants to castle, sees that their rook is attacked, decides castling must be illegal, and plays a worse move instead.

These are not rare accidents. Players rated 400 to 800 lose a surprising number of half-points and points simply because they are unsure of a rule. This chapter removes that uncertainty. Even if you have played for years, read it with a board in front of you. Most players find at least one rule they had slightly wrong.

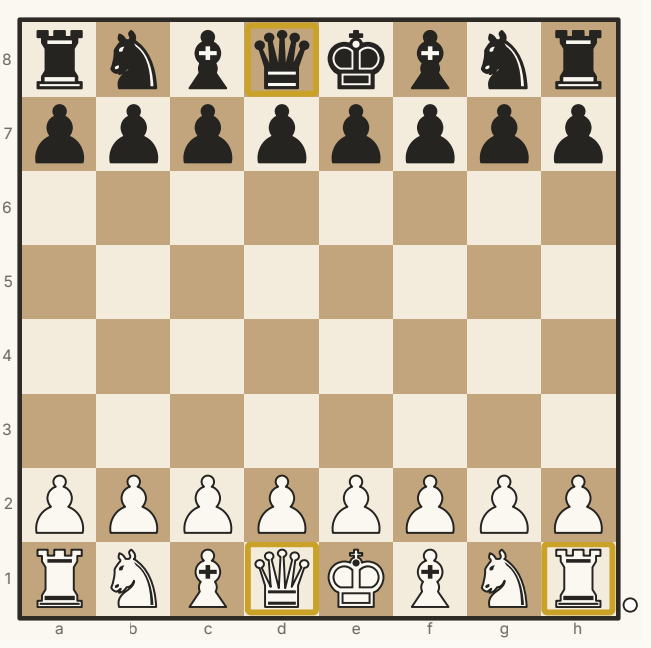
The board and the starting position

The board has 64 squares in eight rows and eight columns. The columns are called **files** and are lettered a to h from White's left to right. The rows are called **ranks** and are numbered 1 to 8 from White's side. Every square therefore has a name, such as e4 or g7. Chapter 2 builds on this.

Two rules decide how the board is placed:

- **Light square on the right.** Each player has a light square in the corner nearest their right hand (h1 for White, a8 for Black). If the corner on your right is dark, turn the board 90 degrees.
- **Queen on her colour.** The white queen starts on the light square d1; the black queen starts on the dark square d8. The kings stand next to them on the e-file.

The rest follows: rooks in the corners, knights next to the rooks, bishops next to the knights, and a row of eight pawns in front.



The starting position. h1 is a light square, the white queen stands on light d1, and the black queen on dark d8. White always moves first.

COMMON MISTAKE **The swapped king and queen**

The most common set-up error is placing the king and queen the wrong way round. Online this cannot happen, but over the board it happens often, and a game played from a wrong set-up can be invalid. Check "queen on her colour" before every game.

How the pieces move and capture

A piece **captures** by moving onto a square occupied by an enemy piece and removing it from the board. You can never capture your own pieces, and only the knight can jump over other pieces.

- **King (K):** one square in any direction. The king may never move onto a square attacked by an enemy piece. Kings can therefore never stand next to each other.
- **Queen (Q):** any number of squares along a file, rank or diagonal, as long as the path is clear.
- **Rook (R):** any number of squares along a file or rank.
- **Bishop (B):** any number of squares along a diagonal. A bishop stays on its starting colour for the whole game, so each side has a light-squared bishop and a dark-squared bishop.
- **Knight (N):** an L-shape, two squares in one direction and then one square at right angles. The knight jumps over anything in its way, and always lands on a square of the opposite colour.
- **Pawn:** moves straight forward one square, or two squares on its very first move if both squares are empty. A pawn captures differently from how it moves: one square diagonally forward. A pawn never moves or captures backwards, and a pawn blocked by any piece directly in front of it cannot move straight ahead.

The pawn is the piece beginners most often get wrong. It cannot capture straight ahead, and it cannot move diagonally unless it is capturing. Two pawns facing each other on the same file are both stuck until something changes.

Check, checkmate and stalemate

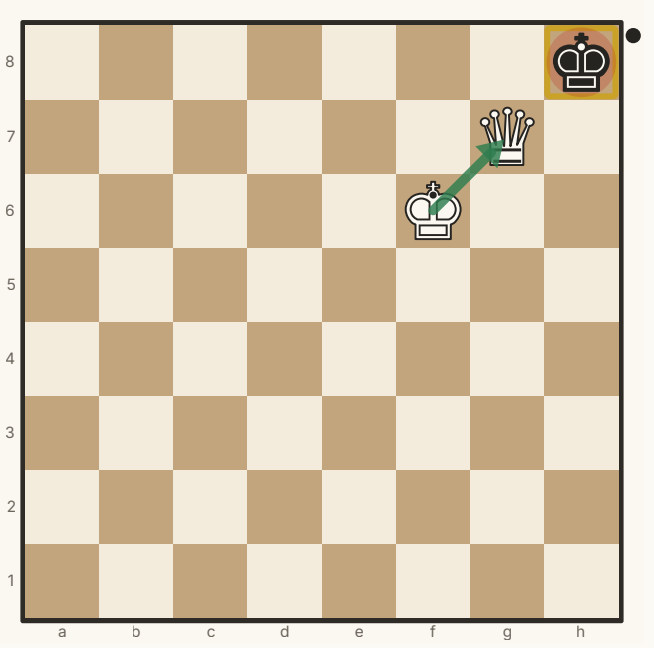
A king is in **check** when it is attacked by an enemy piece. You must get out of check immediately. There are only three ways:

1. **Move the king** to a square that is not attacked.
2. **Capture the checking piece.**
3. **Block** the check by putting a piece between the checker and the king (impossible against a knight or a pawn, or when the checker is adjacent).

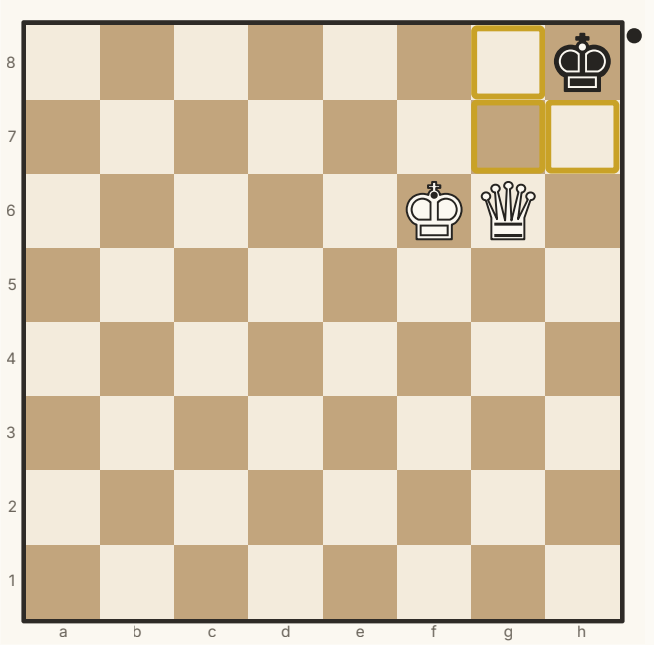
You are never allowed to make a move that leaves or puts your own king in check. Saying “check” aloud is not required by the rules.

Checkmate is check with no legal way out. The game ends immediately and the side delivering checkmate wins. The king is never actually captured.

Stalemate is the opposite situation: the side to move is *not* in check, but has no legal move at all. The game ends immediately as a **draw**. Stalemate is the reason a lone king can sometimes save a hopeless position.



Checkmate. The black king is attacked by the queen on g7, cannot capture it because the white king defends it, and has no safe square.



Stalemate, a draw. Black is not in check, but every square next to the black king is covered by the queen on g6. Black has no legal move.

These two positions differ by a single square. In the first, White played the queen to g7 with check. In the second, White played it to g6, and threw away a completely winning position.

> KEY IDEA **The "does my opponent have a move?" check**
>
> When you are far ahead, especially when your opponent has only a king or a king and blocked pawns, ask before every move: *after my move, does my opponent have at least one legal move, or am I giving check?* If the answer is "no legal move and no check", your move is stalemate.

Castling

Castling is the only move in which two pieces move at once. The king moves two squares towards one of its rooks, and that rook jumps over the king to the square on the king's other side.

- **Kingside castling** (written `O-O`): White's king goes from e1 to g1 and the rook from h1 to f1. For Black, e8 to g8 and h8 to f8.

- **Queenside castling** (written `O-O-O`): White's king goes from e1 to c1 and the rook from a1 to d1. For Black, e8 to c8 and a8 to d8.

Castling usually puts your king behind a wall of pawns and brings a rook towards the centre. It is one of the most important moves of the opening. Over the board, touch the king first and move it two squares; then move the rook.

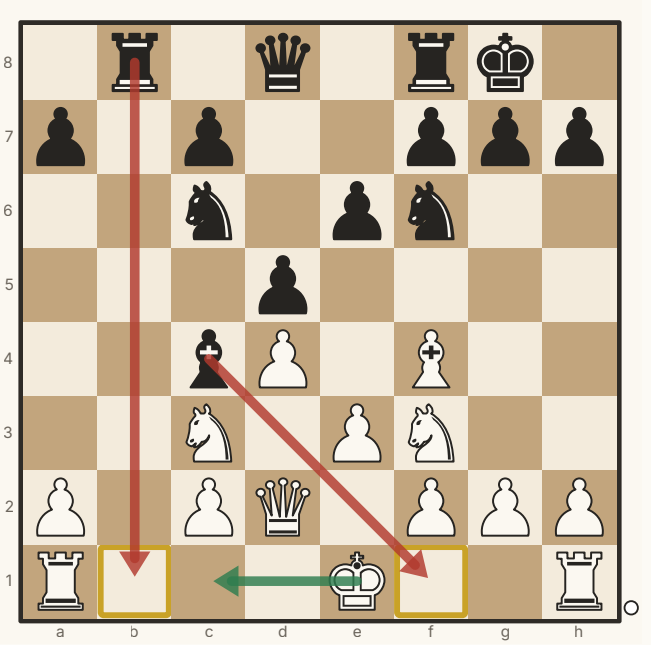
The full conditions

Castling is legal only if **all** of these are true:

1. **The king has never moved** during the game, not even once and back again.
2. **The rook you castle with has never moved.**
3. **Every square between the king and that rook is empty.** For queenside castling that includes the b-square (b1 or b8).
4. **The king is not in check** right now.
5. **The king does not pass through an attacked square.** For kingside castling that is f1 (or f8); for queenside castling it is d1 (or d8).
6. **The king does not land on an attacked square** (g1 or c1; g8 or c8). This is just the normal rule that you may not move into check.

Just as important are the things that do **not** stop you from castling:

- **The rook may be attacked**, and it may pass over an attacked square.
- **On the queenside, b1 (or b8) may be attacked.** Only the king's path matters, and the king never crosses the b-square.
- **Having been in check earlier** does not matter, as long as the king did not move to get out of it.



White cannot castle kingside, because the bishop on c4 attacks f1, a square the king must cross. Queenside castling is legal, even though the black rook attacks b1.

Worked example: can I castle?

Take the diagram above and run the checklist the way you would at the board.

First I check whether the king or the rooks have moved. Online, the app keeps track; over the board, you must remember, and a scoresheet helps. In this position neither has moved, so both castling options remain possible.

Next I look at the squares between the king and each rook. On the kingside, f1 and g1 are empty. On the queenside, b1, c1 and d1 are empty too.

Is my king in check now? I look along every line into e1: the e-file is blocked by my pawn on e3, the diagonals are blocked by my own pawns and queen, and no black knight is near. No check.

Now the key step: which squares will the king cross or land on, and are any of them attacked? For kingside castling, the king crosses f1 and lands on g1. The black bishop on c4 looks down the diagonal d3–e2–f1. Both d3 and e2 are empty, so f1 is attacked. Kingside castling is illegal.

For queenside castling, the king crosses d1 and lands on c1. The black queen on d8 points down the d-file, but her own pawn on d5 blocks her. Nothing else

reaches c1 or d1. The rook on b8 attacks b1, but the king never goes to b1. Queenside castling is legal.

That four-step check (moved? empty? in check? path attacked?) takes a few seconds once you are used to it. Do it every time you are unsure, rather than giving up a good castling move or attempting an illegal one.

> COMMON MISTAKE **"My rook is attacked, so I can't castle"**
>
> This is the commonest castling misunderstanding. The rule is about the king's safety only. You can castle with a rook that is under attack, and castling is often the best way to save it, because it moves the rook and the king in one go.

En passant

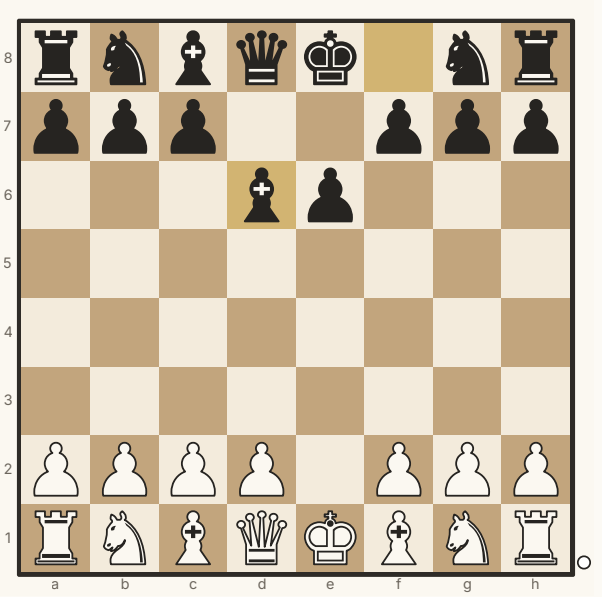
En passant (French for "in passing") is a special pawn capture. It exists because of the pawn's two-square first move: without it, a pawn could sneak past an enemy pawn that would have captured it if it had moved only one square.

The rule: if a pawn moves two squares forward from its starting square and lands directly beside an enemy pawn, that enemy pawn may capture it **as if it had moved only one square**. The capturing pawn moves diagonally to the square the other pawn passed over, and the pawn that moved two squares is removed.

Two conditions are easy to forget:

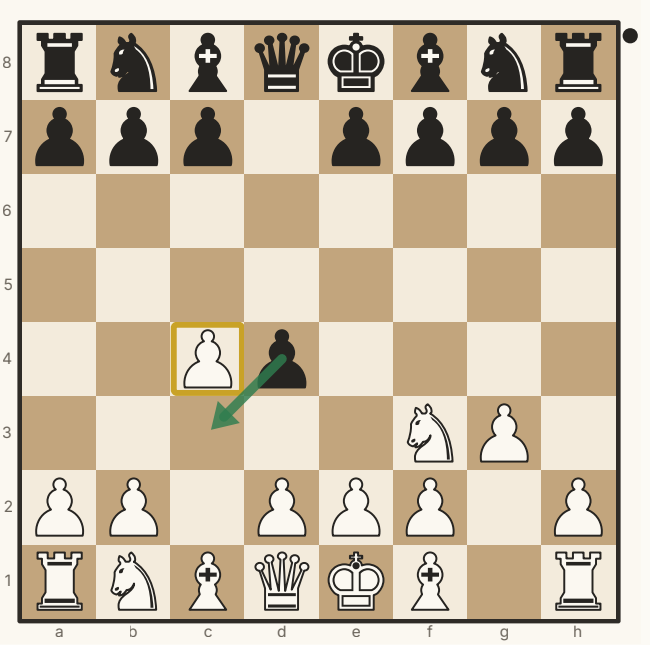
- **Only immediately.** En passant is allowed only on the very next move. If you play something else, the right is gone for that pawn.
- **Only from the fifth rank.** The capturing pawn must already stand on its fifth rank: rank 5 for a White pawn, rank 4 for a Black pawn. That is the only place where an enemy pawn can arrive next to it with a two-square move.

1. **e4 e6** 2. **e5 d5** 3. **exd6 Bxd6**



After 2...d5 the black pawn passed d6, next to White's pawn on e5. White captured en passant with 3.exd6, landing on d6 and removing the pawn from d5. Black recaptured with the bishop.

The same rule works for Black. In the next diagram White has just played c2-c4, passing the c3-square, which the black pawn on d4 attacks.



Black to move may capture en passant with 3...dxc3, landing on c3 and removing the pawn from c4. If Black plays any other move first, the chance is gone.

En passant is written like any pawn capture, using the square where the capturing pawn lands: `exd6`. Some books add "e.p.". Taking en passant is never compulsory, unless it is your only legal move.

> **BOARD DRILL** **Find the en passant squares**
>
> Set up a white pawn on e5 and black pawns on d7 and f7. Move the d-pawn to d5 and capture en passant. Reset, move the f-pawn to f5, and capture en passant on f6. Then try with a black pawn on d4 and a white pawn going from c2 to c4 or from e2 to e4.

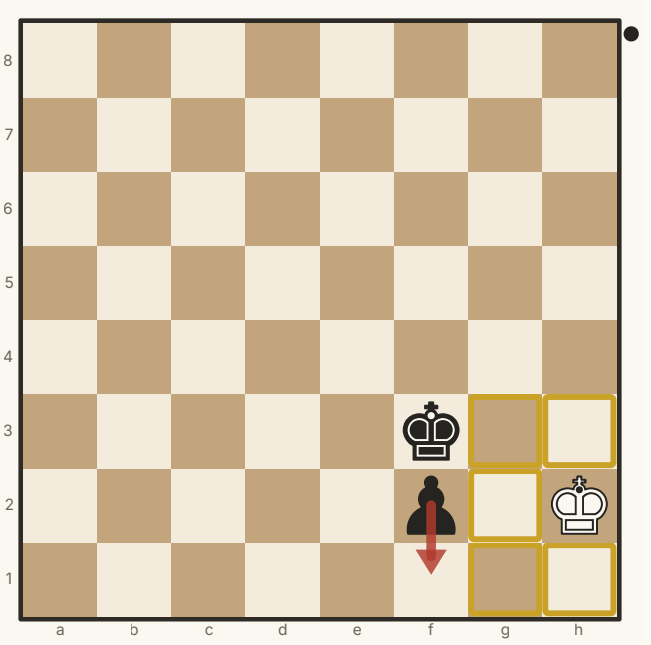
Promotion and underpromotion

When a pawn reaches the far end of the board (the eighth rank for White, the first rank for Black) it must immediately **promote**: you replace it with a queen, rook, bishop or knight of your own colour. It cannot stay a pawn, and it cannot become a king. You may promote to a second queen even if your first queen is still on the board. Over the board, if no spare queen is at hand, stop the clock and ask the arbiter for one; an upside-down rook counts as a rook, not a queen.

The promotion is part of the same move. A pawn that promotes with a capture or with check is written `e8=Q`, `dxc8=Q+`, or `g1=N+`.

Almost always you choose a queen. Choosing anything else is called **underpromotion**, and there are two main reasons for it:

- **To avoid stalemate.** A new queen can cover so many squares that the enemy king has no legal move. A rook covers fewer.
- **To win with a knight fork or check.** A knight attacks squares no queen can reach, so promoting to a knight with check can fork the king and another piece.



Black to move. 1...f1=Q?? is stalemate: the white king on h2 has no legal move. 1...f1=R leaves the h3 square free, and King and Rook against King is an easy win.

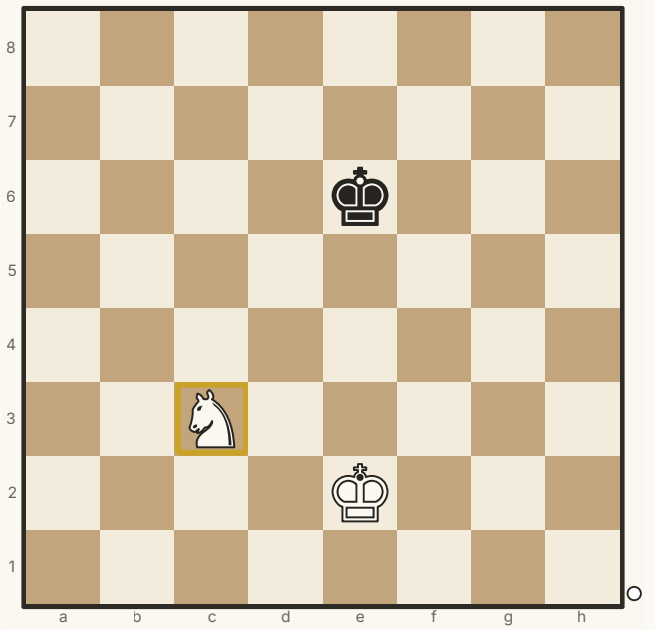
KEY IDEA **Promotion checklist**

Before you promote: (1) Is a queen stalemate? (2) Would a knight give a check or fork that wins more? (3) Otherwise, take the queen.

The ways a game can be drawn

A game is drawn in any of these ways:

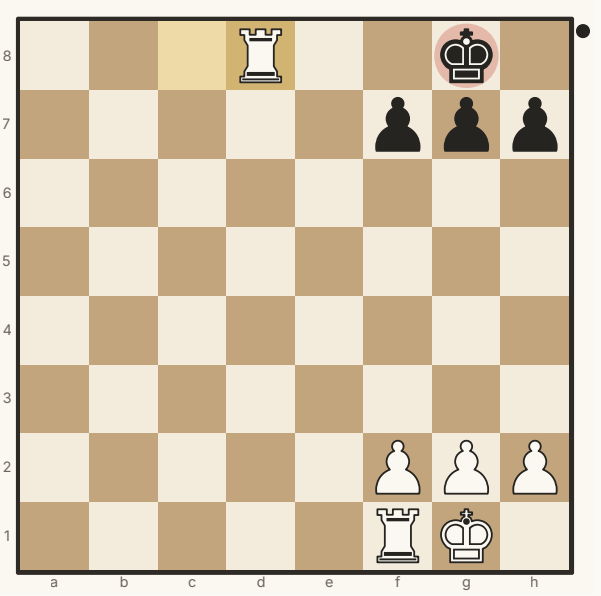
- **Stalemate:** the side to move has no legal move and is not in check.
- **Threefold repetition:** the same position occurs three times, with the same player to move and the same castling and en passant possibilities. The repetitions need not come in a row. Over the board, the player about to make the move that repeats the position, or who has just seen it repeat, claims the draw from the arbiter; most apps declare it automatically.
- **The fifty-move rule:** fifty moves by each player have been played without any capture and without any pawn move. Again, over the board you claim it; apps usually apply it automatically.
- **Insufficient material:** neither side can possibly checkmate. The standard cases are king against king, king and bishop against king, king and knight against king, and king and bishop against king and bishop with both bishops on the same colour.
- **Agreement:** one player offers a draw and the other accepts. Over the board, offer a draw just after making your move and before pressing the clock.



Insufficient material. A king and knight cannot force checkmate, and no checkmate is even possible against a lone king, so the game is drawn at once.

The most useful drawing weapon in practice is **perpetual check**: a series of checks that the opponent cannot escape. The same position soon repeats three times, and the game is drawn. When you are losing badly, look for it. When you are winning, keep a square for your king where the checks stop.

30... **Qb6??** 31. **Rc8+ Qd8** 32. **Rxd8#**



A reminder that checkmate ends everything. Black, with a queen against two rooks, ignored the back rank. A move such as 30...h6, giving the king a square, was necessary.

COACHING JUDGMENT **Do not offer draws out of fear**

Coaching judgment: at this level, positions that look dangerous are often still full of play, and opponents miss winning moves all the time. Offer or accept a draw only when the position is genuinely equal and simplified, or when a perpetual check is on the board.

Touch-move, resigning, and the clock

These rules matter mainly over the board. Online, the app enforces them for you.

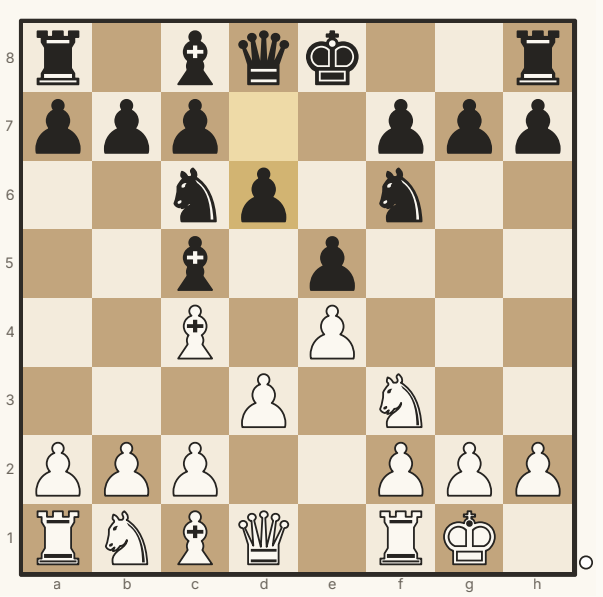
- **Touch-move.** If you deliberately touch one of your own pieces, you must move it, provided it has a legal move. If you deliberately touch an opponent's piece, you must capture it if you legally can. Once you have let go of a piece on a legal square, the move is made and cannot be changed.
- **Adjusting.** To straighten a piece without having to move it, say "I adjust" (or "j'adoube") *before* you touch it.
- **Resigning.** You may end the game at any time by resigning, which counts as a loss. Say "I resign", stop the clock, or offer your hand. At this level, coaching judgment is to resign rarely: opponents stalemate, blunder and run out of time far more often than you expect.
- **The clock.** In timed games, each player has a clock. Make your move with one hand and press your clock with the same hand. If your time runs out, you lose, unless your opponent has no possible way to checkmate you, in which case the game is drawn.

Timed play is optional in this book. When you do play with a clock, choose slow time limits (see "How to Use This Book"), and learn the time-management ideas in Chapter 27 before worrying about speed.

A short opening example

Here is how castling fits into a normal opening. Both sides develop a knight and a bishop, which clears the squares between king and rook, and White castles on move four.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bc4 Bc5** 4. **O-O Nf6** 5. **d3 d6**



White castled on move 4 after clearing f1 and g1. Black can castle next move for the same reason: the knight and bishop have left f8 and g8.

Notice what made castling possible: the knight left g1 and the bishop left f1. Developing your kingside pieces first is the fastest route to a safe king, which is one reason the opening principles in Chapter 6 put development first.

Practice at the board

You can do all of these with a physical set or any free analysis board. To load a position, use the "Set up position" or "Load FEN" option and paste the FEN.

1. **Castling checklist.** Load `1r1q1rk1/p1p2ppp/2n1pn2/3p4/2bP1B2/2N1PN2/P1PQ1PPP/R3K2R w KQ - 0 10`. Try to castle kingside in the app: it should refuse. Castle queenside: it should accept. Now move the black bishop from c4 to d7 (in the board editor) and check kingside castling again. Explain to yourself why it is now legal.
2. **En passant both ways.** From the starting position, play `1.e4 Nf6 2.e5 d5` and capture en passant. Then play `1.d4 e5 2.d5 c5` and check that `3.dxc6` is possible. Finally, in the same line, play a waiting move first and confirm the app no longer allows the capture.
3. **Stalemate traps.** Load `7k/8/5K2/8/8/8/8/6Q1 w - - 0 1`. Find the mate in one, and also find the queen move that stalemates. Then load `8/8/8/8/8/5k2/5p1K/`

`8 b - - 0 60` and play `f1=R`, then win the rook ending against a friend or an engine.

4. **Perpetual check.** Set up a position where you have only a queen and your opponent has queen and rook, with your opponent's king a little exposed. Try to give checks forever. Notice how often the king can hide, and which pawn moves would have given it shelter.

Check yourself by asking: could I explain each rule to a friend, including the exceptions? If not, re-read that section.

Chapter summary

- Set up with the light square on your right and the queen on her colour.
- You must escape check by moving the king, capturing the checker, or blocking. Checkmate wins; stalemate draws.
- Castling needs an unmoved king and rook, empty squares between them, and a king that is not in check and does not cross or land on an attacked square. The rook, and b1/b8, may be attacked.
- En passant is possible only immediately after the two-square pawn move, and only by a pawn on its fifth rank.
- A pawn on the last rank must promote; underpromote when a queen would stalemate or a knight wins more.
- Draws come from stalemate, threefold repetition, the fifty-move rule, insufficient material, and agreement. Perpetual check is the practical way to force a repetition.
- Over the board, touch-move applies, and you lose on time unless your opponent cannot possibly mate.

CHEAT CODE **Special-move checklist**

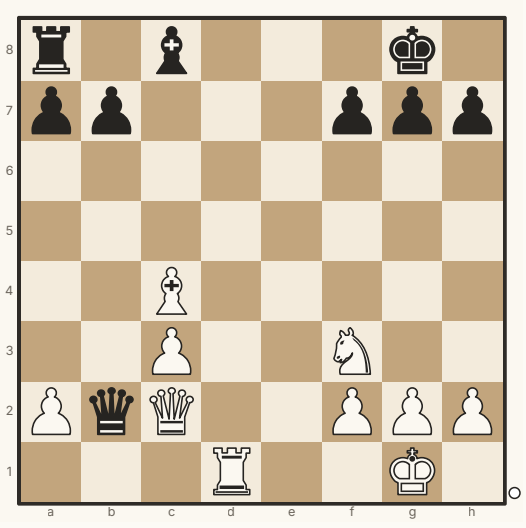
1. **Before castling:** have the king or this rook moved? Are the squares between them empty? Am I in check? Does the king cross or land on an attacked square?
2. **After an enemy pawn moves two squares next to your pawn on your fifth rank:** consider capturing en passant now, because next move it is too late.
3. **Before promoting:** would a queen be stalemate? Would a knight give a winning check or fork? Otherwise choose a queen.
4. **When far ahead:** after my move, does my opponent have a legal move or am I giving check? If neither, it is stalemate.
5. **When far behind:** look for perpetual check and stalemate tricks before resigning.

Now solve the exercises below. Then play two slow games, and in each one apply the checklist at least once, even if only to confirm that a castling move is legal.

Exercises

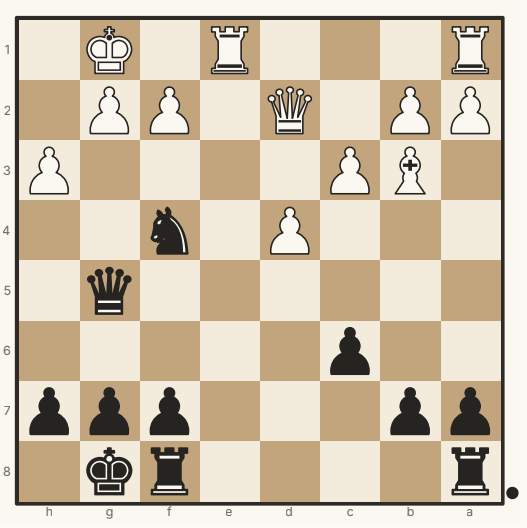
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

1.1 CHECKMATE



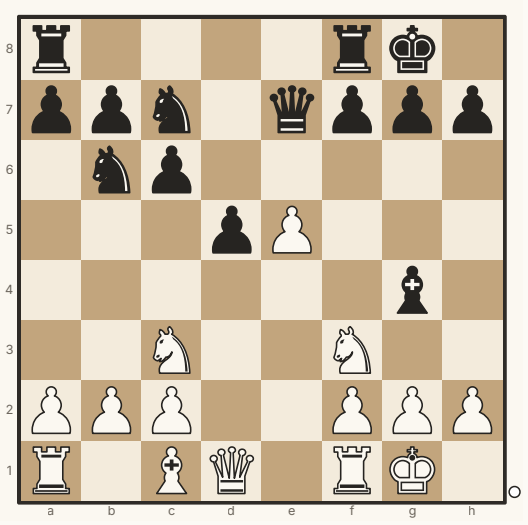
White to move. Black has just grabbed the pawn on b2 and attacks your queen. Find checkmate in one move. Solution p. 631

1.2 CHECKMATE



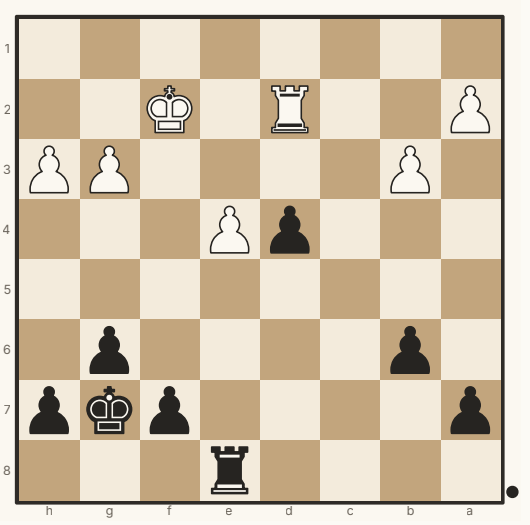
Black to move. Find checkmate in one move. Solution p. 631

1.3 RULES



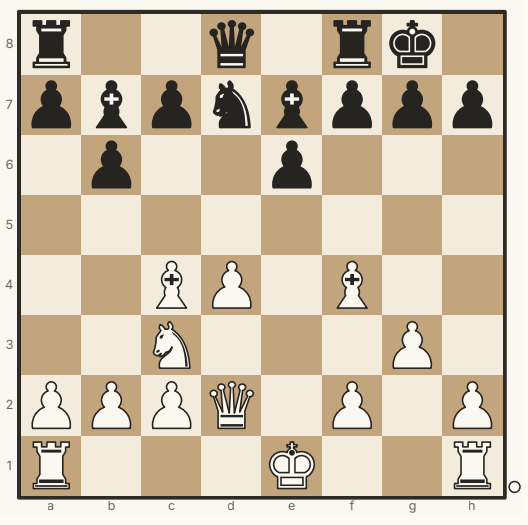
White to move. Black has just played ...d7-d5. Win material. Solution p. 631

1.4 RULES



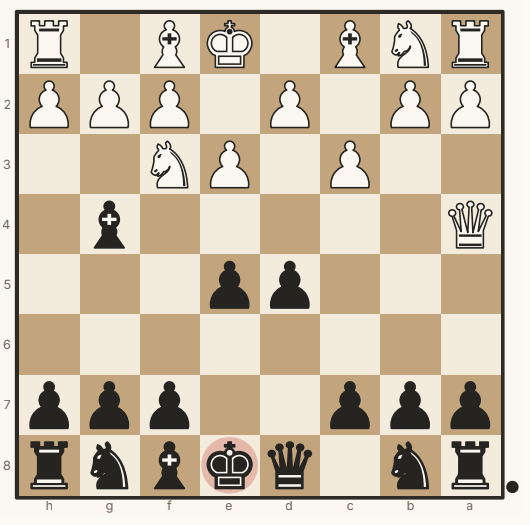
Black to move. White has just played e2-e4. What is the strongest move? Solution p. 631

1.5 RULES



White to move. Black's bishop on b7 attacks your rook on h1. Are you still allowed to castle? Find a move that looks after both the rook and the king. Solution p. 632

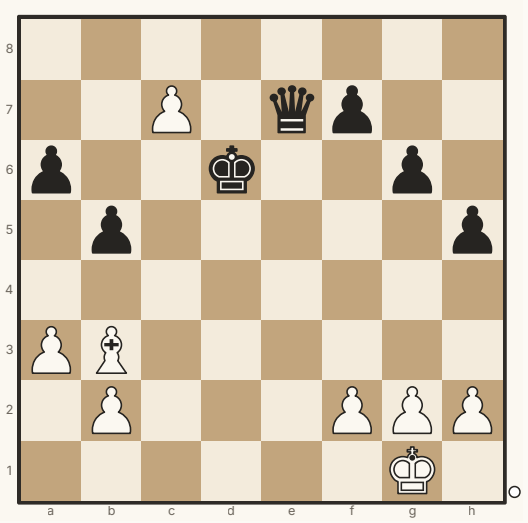
1.6 DEFENCE



Black to move. White has just played Qa4+, checking your king and attacking your bishop on g4. Castling is not allowed while you are in check. What is the best way out?

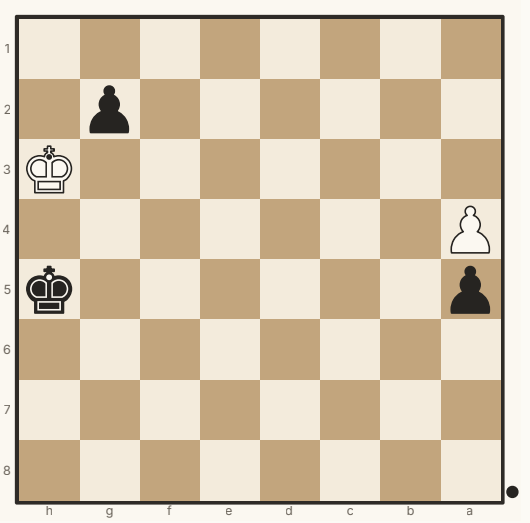
Solution p. 632

1.7 RULES



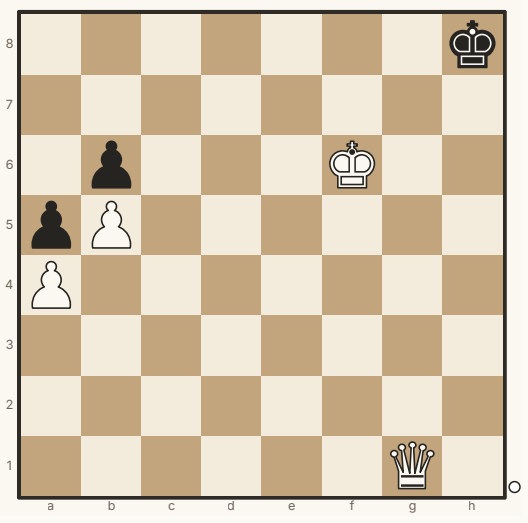
White to move. Black's queen threatens ... Qe1, which would be checkmate. You can promote your pawn. Which piece should it become? Solution p. 632

1.8 RULES



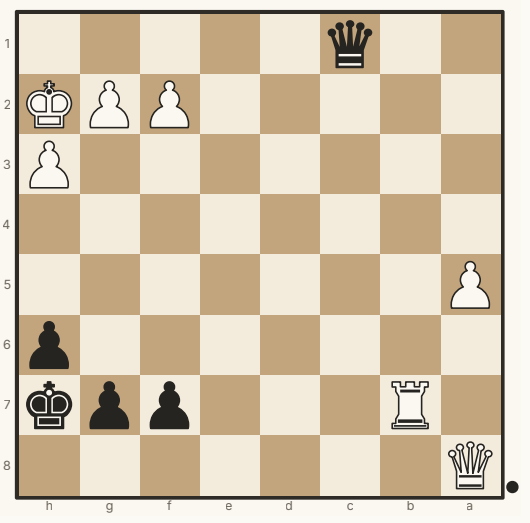
Black to move. White's king attacks your pawn on g2, so you must promote now. Which piece should it become? Solution p. 632

1.9 CHECKMATE



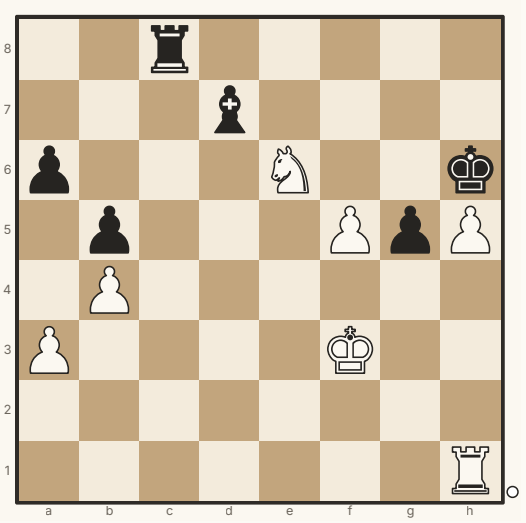
White to move. Black has only a king and blocked pawns. Finish the game, and be careful. Solution p. 633

1.10 DEFENCE



Black to move. You are a rook down and White is threatening to win more. Which move saves the game? Solution p. 633

1.11 CHECKMATE



White to move. Black has just played ...g7-g5. Find checkmate in one move.

Solution p. 633

HINTS

1.1 Which checks do you have? Look at the black king's back rank.

1.2 Can the king capture a piece that is defended?

1.3 Which pawn just moved two squares, and which of your pawns stands next to it?

1.4 En passant works for Black too, from the fourth rank.

1.5 Which piece moves when you castle kingside?

1.6 Which block also moves the attacked piece to safety?

1.7 Which new piece would give check?

1.8 After a queen promotion, where could the white king go?

1.9 Can Black move any pawn? Then look for a check.

1.10 When you are losing, look for checks that never run out.

1.11 Which en passant capture opens a line for your rook?

CHAPTER 2 STAGE 400–600

Reading the Board: Notation and Board Vision

Name every square, then see every line

YOU WILL LEARN TO

- Read and write standard algebraic notation, including captures, checks, castling, promotion and disambiguation.
- Keep a scoresheet and load any game (PGN) or position (FEN) into a free app.
- See the full reach of every piece, including long diagonals, sideways rook moves, backward moves and knight jumps.
- Scan the whole board, edges included, before every move.

> IN YOUR NEXT GAME
>
> Before each move, trace every line from your opponent's last-moved piece and from each of your long-range pieces to the edge of the board, and look backwards as well as forwards.

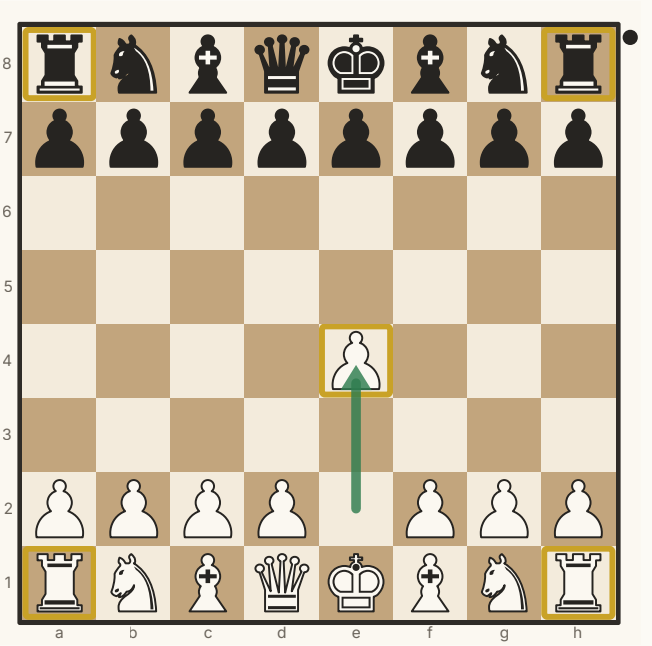
You are White. Your opponent plays a knight to the edge of the board, attacking your bishop. You look at the bishop, find a safe square for it, and move it. After the game, the engine shows that your queen could simply have taken the knight. It was undefended, and your queen had been looking at it the whole time from the other side of the board.

This is the most common way games are lost below 1000: not through a deep combination, but because a player did not see a line that was already there. This chapter teaches two linked skills. The first is notation, the language that lets you read this book, record your games and load positions into any app. The second is **board vision**: the habit of seeing the full reach of every piece on the board.

The language of the board

Squares, files and ranks

Every square has a two-character name. The letter gives the **file** (the column, a to h, from White's left to right) and the number gives the **rank** (the row, 1 to 8, from White's side). White's pieces start on ranks 1 and 2, Black's on ranks 7 and 8.



After 1.e4, the white pawn stands on e4: the e-file, fourth rank. The corner squares are a1 and h1 for White, a8 and h8 for Black.

Two tips make square names automatic. First, the squares are named from White's side even when you play Black; d4 is always the same square. Second, a1 is always a dark square, so the light-squared diagonal from h1 runs to a8, and the dark-squared diagonal from a1 runs to h8. Those two **long diagonals** come up again and again.

> **BOARD DRILL** **Name the squares**
>
> With a board in front of you, have someone call out ten squares, or use the coordinate trainer in any free app. Touch each square within two seconds. Then do it again from Black's side of the board. Ten minutes a day for a week is enough for most people.

Writing a move

A move is written as the piece letter followed by the square it lands on. The letters are **K** (king), **Q** (queen), **R** (rook), **B** (bishop) and **N** (knight, because K is taken). Pawns have no letter, so `e4` means a pawn moves to e4, and `Nf3` means a knight moves to f3.

The other symbols:

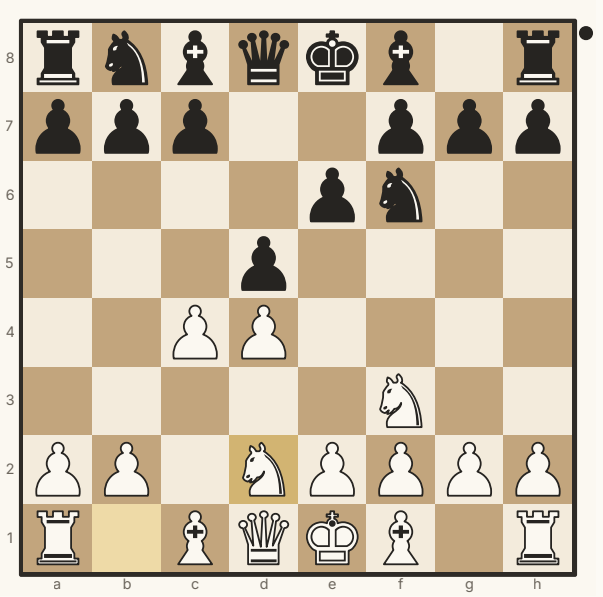
- **Capture:** an `x` before the square. `Bxf7` means a bishop captures on f7. A pawn capture gives the file the pawn came from: `exd5`.
- **Check:** a `+` after the move, as in `Bb5+`. **Checkmate:** a `#`, as in `Qxf7#`.
- **Castling:** `O-O` kingside, `O-O-O` queenside (some sources use zeros: 0-0).
- **Promotion:** the square, `=`, and the new piece: `e8=Q`, or `bxa8=N+` for a capture that promotes to a knight with check.
- **En passant:** written as an ordinary pawn capture to the square where the pawn lands, such as `exd6`.

Moves are numbered in pairs: `1.e4 e5` is White's first move and Black's reply. When a sentence starts from a Black move, it uses three dots: `5...Nf6` means "Black's fifth move was knight to f6".

Disambiguation: which piece moved?

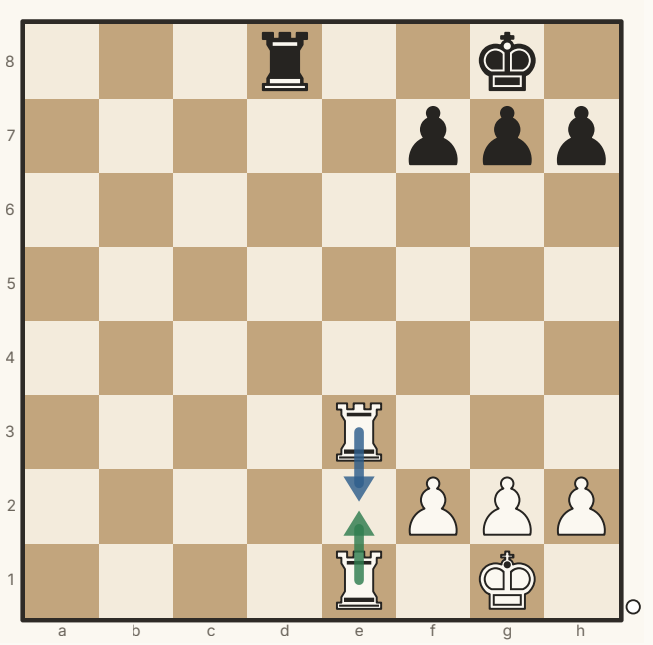
Sometimes two identical pieces could move to the same square. Then you add the file (or, if that is not enough, the rank) the piece came from. In the line below, both of White's knights could go to d2, so White writes which one did.

1. **d4 d5** 2. **c4 e6** 3. **Nf3 Nf6** 4. **Nbd2**



Both the b1-knight and the f3-knight could reach d2, so the move is written 4.Nbd2: the knight from the b-file.

When two pieces stand on the same file, you use the rank instead.



Both rooks can move to e2. If the rook on e1 goes there, write R1e2; if the rook on e3 goes there, write R3e2. Writing just Re2 would be ambiguous.

Annotation symbols

Books and apps add symbols after moves to give an opinion. This book uses only six: `!` (a good move), `!!` (a brilliant move), `?` (a mistake), `??` (a blunder), `!?` (interesting, worth trying) and `?!` (dubious). They are opinions, not part of the move.

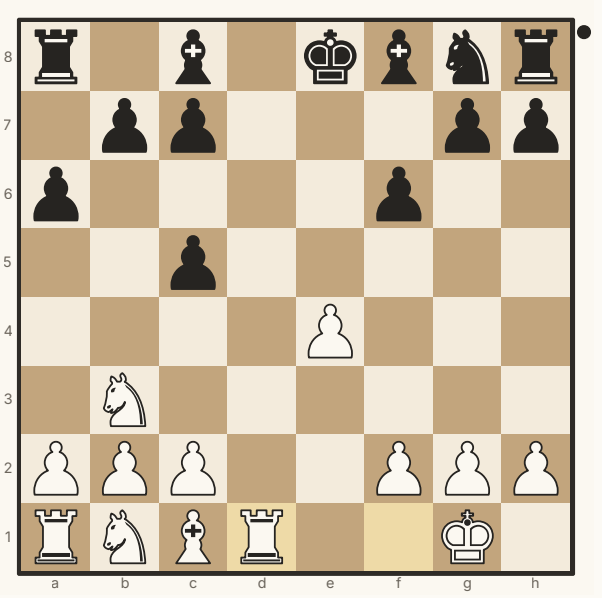
Scoresheets, PGN and FEN

Keeping a scoresheet

Over the board, in serious games you write down every move on a **scoresheet**, usually in two columns, White's moves on the left. Online, the app records the game for you, but writing it down for a few games is excellent practice, and you will need the skill at any club or tournament.

Here is a short, sound opening as you would write it.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bb5 a6** 4. **Bxc6 dxc6** 5. **O-O f6** 6. **d4 exd4** 7. **Nxd4 c5** 8. **Nb3 Qxd1** 9. **Rxd1**



The Exchange Variation of the Ruy Lopez. On a scoresheet: 1. e4 e5, 2. Nf3 Nc6, 3. Bb5 a6, and so on. Note the captures (4.Bxc6, 8...Qxd1) and the castling move 5.O-O.

Three habits help. Write your move before you play it, and look at it once more as a last check (many clubs allow this). Write the move number on every line so you can find your place again. And if you lose track, ask the arbiter or your opponent at a calm moment to compare scoresheets; do not guess.

> **COMMON MISTAKE** **Writing the square you came from**
>
> Beginners often write `Ng1f3` or `Ne5` when they meant the knight landed on f3. Standard notation records only the destination square (plus disambiguation when needed). The square you write is the square where the piece now stands.

PGN: a game in a text file

PGN (Portable Game Notation) is simply a game written in the notation you have just learned, with a few labelled lines on top such as the players' names, the date and the result. Every chess app can export your games as PGN and import a PGN you paste in. When you want to analyze a game you played over the board, type up your scoresheet in PGN form, paste it into a free analysis board, and step through it.

FEN: a position in one line

FEN (Forsyth–Edwards Notation) describes a single position. Every diagram and exercise in this book gives one, so that you can load it into an app. Here is the position after 1.e4:

```
rnbqkbnr/pppppppp/8/8/4P3/8/PPPP1PPP/RNBQKBNR b KQkq - 0 1
```

Read it in six parts, separated by spaces:

1. **The pieces**, rank by rank from the eighth rank down to the first, separated by `/`. Capital letters are White pieces, small letters Black, `P`/`p` are pawns, and a digit counts empty squares. `4P3` means four empty squares, a white pawn, three empty squares: the pawn on e4.
2. **Side to move:** `w` or `b`. Here Black is to move.
3. **Castling rights:** `K` and `Q` for White's kingside and queenside, `k` and `q` for Black's; `-` if nobody can castle.
4. **En passant square**, if a pawn has just moved two squares and can be captured; otherwise `-`. (Some apps write e3 here after 1.e4 even though no capture is possible.)

5. **Half-move clock** for the fifty-move rule.
6. **Move number.**

You will rarely need to write a FEN by hand. What matters is that you can paste one into an app's "Load FEN" or "Set up position" box, and check that the side to move is right before you start thinking.

> KEY IDEA **The two things to check after loading a FEN**
>
> Is the correct side to move? Is the board shown from that side? Many apps show the position from White's side by default; flip the board when you are solving as Black.

Board vision: seeing every line

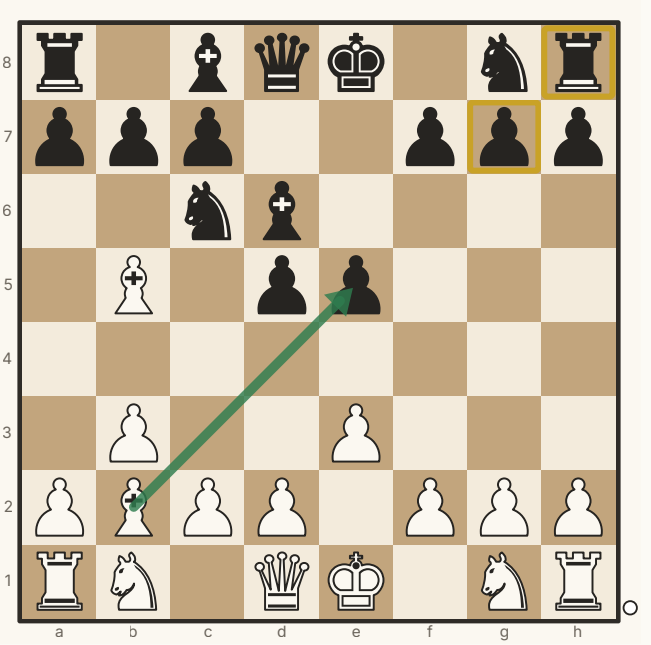
Knowing how the pieces move is not the same as seeing where they reach. **Board vision** is the ability to see, quickly and completely, every square each piece attacks, for both sides. Weak board vision produces the classic beginner losses: a queen captured from across the board, a knight fork nobody noticed, a backward bishop move that never entered the player's mind.

The good news is that board vision is trainable. It improves fast with deliberate practice, because the same geometric patterns appear in game after game.

Lines of force

Think of every queen, rook and bishop as sending out **lines of force**: rays that run along its files, ranks and diagonals until they hit a piece. The ray "sees" every empty square and the first piece it meets. If that first piece is an enemy piece, it is attacked. If it is your own piece, it is defended.

The rays of long-range pieces are easiest to underestimate when they are long. A bishop on b2 attacks h8 just as surely as it attacks c3. A queen on d1 attacks h5 even though it is four squares away and pointing at the "wrong" side of the board.

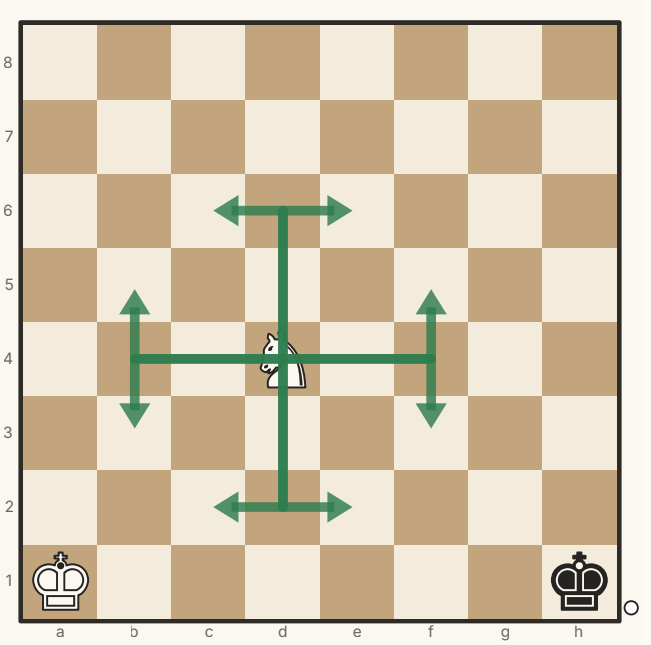


A standard position from Larsen's Opening (1.b3). White's bishop on b2 sends a ray down the long dark diagonal and presses on e5. If the e5-pawn ever moves, the ray reaches g7 and h8.

Knight geometry

The knight is the piece that most often surprises players, because it does not move in straight lines. Three facts make it easier to see:

- **It always changes colour.** A knight on a light square attacks only dark squares, and the other way round. If the enemy king and queen both stand on dark squares, a knight on a light square might fork them; a knight on a dark square cannot.
- **It reaches eight squares from the centre, but only two from a corner.** A knight on d4 controls b3, b5, c2, c6, e2, e6, f3 and f5. A knight on a1 controls only b3 and c2. That is the reason behind the saying "a knight on the rim is dim"; it is also why knights on the rim are often left undefended.
- **It is slow over distance.** To move one square diagonally, a knight needs two moves; to reach a square two diagonal steps away it may need four. When you plan a knight route, count the jumps instead of guessing.



The eight squares a knight on d4 attacks. All of them are light squares, while d4 is dark. Picture this star every time you look at a knight.

BOARD DRILL **The knight's tour of targets**

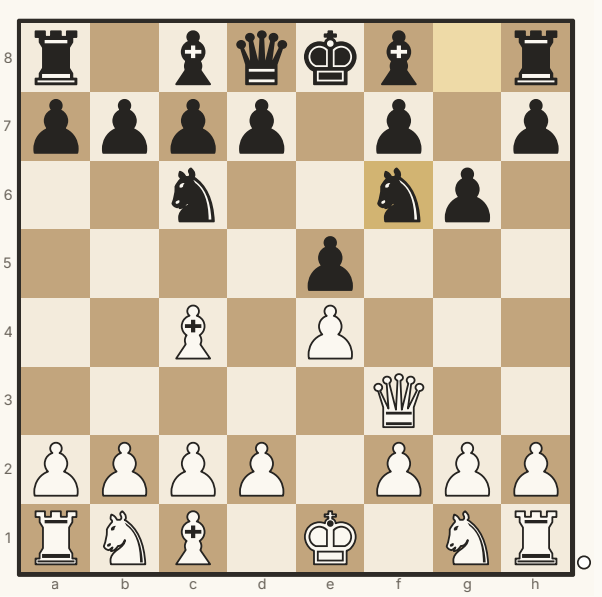
Place a knight on any square and two "targets" (coins work) on other squares. Find the shortest route by which the knight could attack both at once. Count the jumps aloud. Repeat ten times with different squares.

Ranks, files and diagonals

Rooks are usually pictured going up and down files. They are just as strong sideways along ranks, and players miss sideways captures surprisingly often, especially when the target is on the far edge. Bishops and queens have the same problem with diagonals that run backwards, towards their own side of the board.

A useful exercise: for each of your long-range pieces, trace all of its rays to the edge of the board, in every direction, including backwards. Backward moves are hard to see for a simple reason: in most games, pieces move forward, so your eyes learn to look forward.

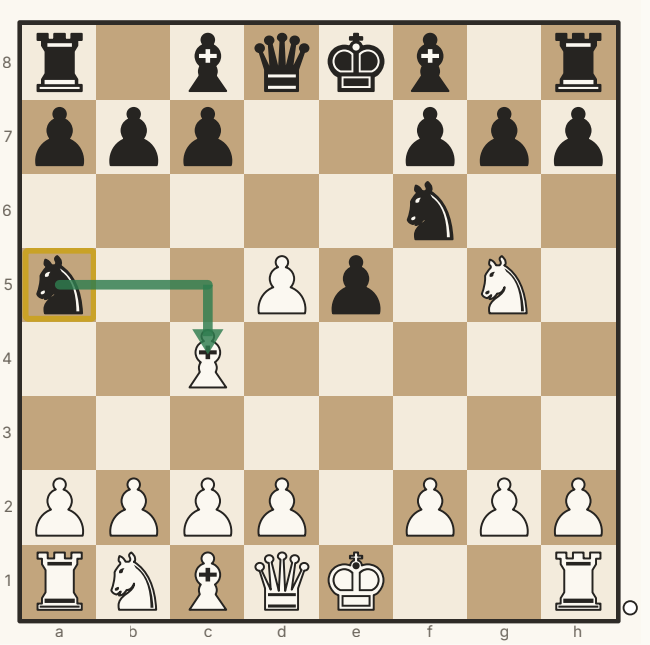
1. **e4 e5** 2. **Qh5?! Nc6** 3. **Bc4 g6** 4. **Qf3 Nf6**



A typical beginner line. After 3...g6 hits the queen, the backward move 4.Qf3 renews the threat on f7. Black must see the new line from f3 to f7, and 4...Nf6 blocks it.

Edge pieces

Pieces on the a-file, h-file, first rank and eighth rank are missed far more often than central ones. Your eyes are drawn to the middle of the board, and edge pieces sit at the end of the rays, where your attention fades. They also tend to be undefended, because few pieces can reach the edge to protect them.



A main line of the Two Knights Defence. Black's knight on a5 attacks the bishop on c4. Knights on the edge like this one are where players most often fail to look, both when attacking and when defending.

Five common blind spots

Most vision errors fall into a handful of families. Learn to look for them by name:

- **The piece behind the piece.** When a piece moves off a line, the piece behind it suddenly attacks along that line. A bishop on b2 behind a knight on c3, or a rook behind a pawn on an open file, can come to life in one move. Chapter 9 calls this a discovered attack.
- **The square just vacated.** When you move a piece, the square it leaves can become an entry point for the opponent, and the lines it was blocking open up. Look at the empty square, not only at the new one.
- **Long queen moves with check.** The queen can give check from far away, and checks that also attack something else are among the most common ways to lose a piece. Queen moves such as Qa4+ (along the a4-e8 diagonal) and Qh5+ (along the h5-e8 diagonal) catch players again and again in the opening.
- **The knight fork squares near the king.** Before your king has castled, c7 and f7 (for Black) and c2 and f2 (for White) are the classic knight targets, because a knight landing there often attacks the king and a rook together.

- **Your own pieces' defensive duties.** Vision is not only about attacking. If you do not see that your bishop is the only piece guarding a square, you will move it and leave that square undefended.

When you review a game, try to name the blind spot behind each missed move. Naming the pattern makes it far easier to recognize next time.

Visualizing without moving

In a game you cannot move the pieces to test an idea; you must see the position after the move in your head. Start small. Before each move in a training game, picture the board after your move and answer one question: *what does each of my opponent's pieces attack now?* The square your piece leaves can open a line, and the square it goes to can block one.

> COACHING JUDGMENT **Train vision in short daily doses**
>
> Coaching judgment: ten minutes a day of easy vision drills (naming squares, knight routes, "what does this piece attack?") does more for a beginner than an hour once a week. The skill is closer to reading speed than to knowledge.

Scanning the whole board

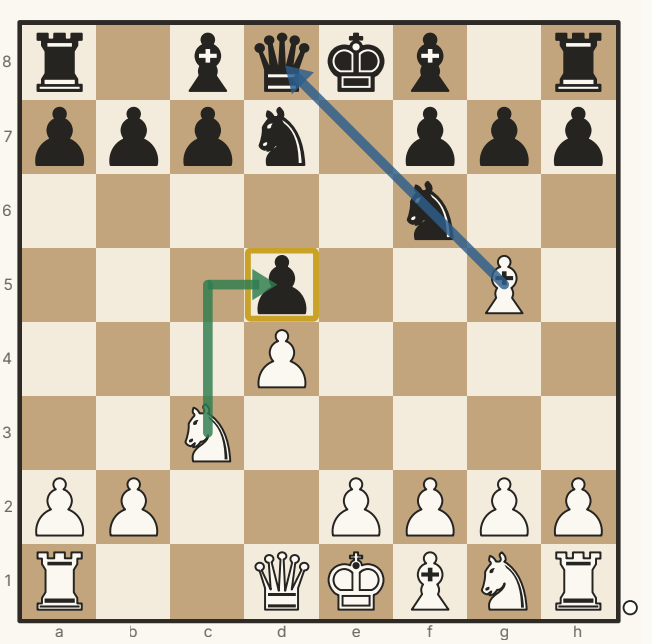
Board vision becomes useful when it turns into a routine. The **board scan** takes ten to twenty seconds:

1. **What did my opponent's last move change?** Trace every ray from the piece that moved, and every ray it opened by leaving its old square.
2. **What do the enemy long-range pieces see?** Queen, rooks, bishops: trace each one to the edges, including backwards.
3. **Where can the enemy knights jump?** Picture their stars.
4. **Now look at my own pieces the same way**, and especially at the edges and corners.

Chapter 4 turns this into the full safety scan of checks, captures and threats. For now, the goal is simply to see everything that is on the board.

Worked example: a hidden diagonal

The next position arises after a common Queen's Gambit Declined move order. White looks at the pawn on d5.



White to move. The d5-pawn is attacked by the knight on c3 and seems to be pinned: if the knight on f6 recaptures, White's bishop takes the queen. Is 6.Nxd5 safe?

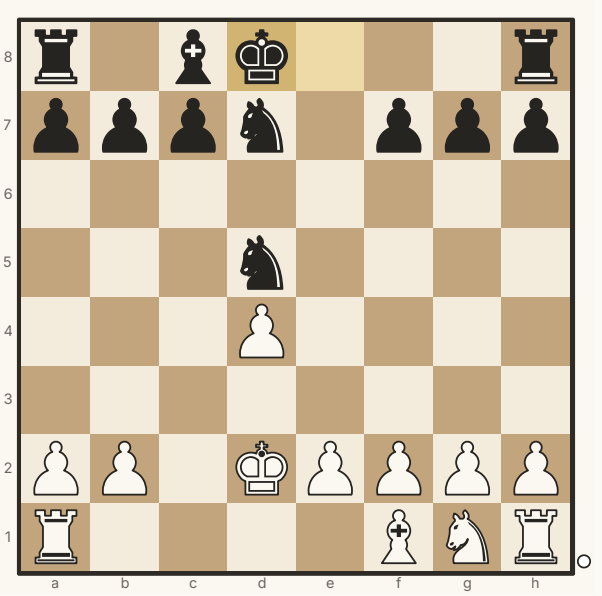
Here is the thinking process White should run before taking.

First I check the material: the d5-pawn is attacked once (by my knight) and defended once (by the knight on f6). The f6-knight appears unable to recapture, because if it moves my bishop on g5 takes the queen on d8.

Next I look at my opponent's best reply, not just the reply I want. After 6.Nxd5, Black can still play 6...Nxd5, accepting the loss of the queen. Then after 7.Bxd8, what does Black have? I scan Black's long-range pieces. The bishop on f8 looks along the diagonal e7, d6, c5, b4 to a3, and from b4 it would see c3, d2 and e1, where my king stands. The c3-square is now empty, because my knight left it.

So the line is 7...Bb4+, a check from the far side of the board, and after 8.Qd2 Bxd2+ 9.Kxd2 Kxd8, Black has won a piece. The move 6.Nxd5 is a blunder. This trap is so well known that it has a name, the Elephant Trap.

6. Nxd5?? Nxd5 7. Bxd8 Bb4+ 8. Qd2 Bxd2+ 9. Kxd2 Kxd8



The Elephant Trap. The hidden diagonal f8 to b4 to e1 decides: Black gave the queen and won it back with a check, ending a piece up.

The lesson is not the trap itself. It is the scan: when your piece leaves a square (here c3), ask what new lines that opens for your opponent, and trace every enemy ray towards your king, however far away the attacker stands.

> **COMMON MISTAKE** **Looking only where the action is**
>
> Most vision errors happen because the player looked only at the part of the board where the last few moves were played. The decisive piece is often sitting quietly on the other side: a bishop on its home square, a rook on the back rank, a queen that has not moved for ten moves.

How to find vision errors in your own games

When you go through a lost game, look for these signs:

- **The losing move was a capture or attack from far away**, a long diagonal or a whole rank or file, that you did not consider.
- **The capture came backwards**, towards the side that made it.

- **An edge piece**, yours or your opponent's, was hanging for more than one move.
- **A knight fork** appeared on a square you never pictured.

Mark these in your mistake journal under *missed tactic* or *hanging piece*, and note the geometry: "long diagonal", "backward move", "edge piece", "knight". After a few weeks, you will see which kind of line you miss most, and you can drill it.

Practice at the board

1. **Notation fluency.** Play through the Ruy Lopez line in this chapter on a real board, reading only the notation. Then write the moves on a sheet of paper from memory. Check yourself against the book.
2. **Load a position.** In a free app, load `r1bqkb1r/pppn1ppp/5n2/3p2B1/3P4/2N5/PP2PPPP/R2QKBNR w KQkq - 0 6` and play the Elephant Trap line. Then reload it and play the normal developing move 6.e3 instead, keeping the c3-knight where it blocks the diagonal from b4 to e1.
3. **Ray tracing.** Set up any middlegame position from one of your games. For each queen, rook and bishop on the board, both colours, place your finger on the piece and slide it along every ray to the first piece it meets. Say aloud what it attacks and what it defends.
4. **Knight stars.** Put a knight on each of these squares in turn and name its targets without moving it: a1, b1, d4, e5, h8. Check by moving it in an app.
5. **Edge sweep.** At the end of each training game, before resigning or agreeing a draw, look slowly along the a-file, the h-file, and both back ranks. Count the pieces you had not noticed.

Check yourself: if you can name every square a piece attacks in under five seconds, your vision is in good shape for the next chapters.

Chapter summary

- Every square has a fixed name: file letter, then rank number, always from White's side.
- Moves are piece letter plus destination: `x` for captures, `+` for check, `#` for mate, `O-O`/`O-O-O` for castling, `=Q` for promotion. Add the file or rank of departure when two identical pieces could make the move.

- PGN records a whole game; FEN records one position. Paste either into any free app, then check the side to move.
- Long-range pieces send rays to the edge of the board; trace them all, including backwards and sideways.
- Knights always change square colour, are weakest on the edge, and are slow over distance; picture their eight-square star.
- Edge pieces and backward moves are where most vision errors happen.
- Before each move, scan what your opponent's last move changed and what every long-range piece sees.

CHEAT CODE **The board scan**

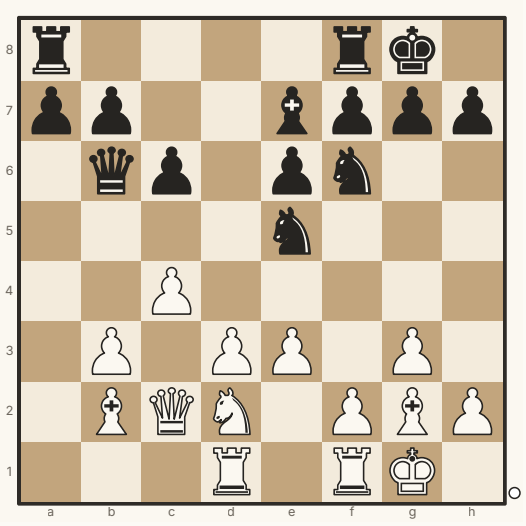
1. Trace every ray from the piece my opponent just moved, and from the square it left.
2. Trace every enemy queen, rook and bishop to the edge, forwards, sideways and backwards.
3. Picture every enemy knight's star of squares.
4. Look along both edges and both back ranks for loose pieces.
5. Repeat steps 2 to 4 for my own pieces to find my chances.

Now solve the exercises below. Then play two slow games, keep a scoresheet for both, and run the board scan before every move. Afterwards, load each game as PGN into a free app and check which lines you missed.

Exercises

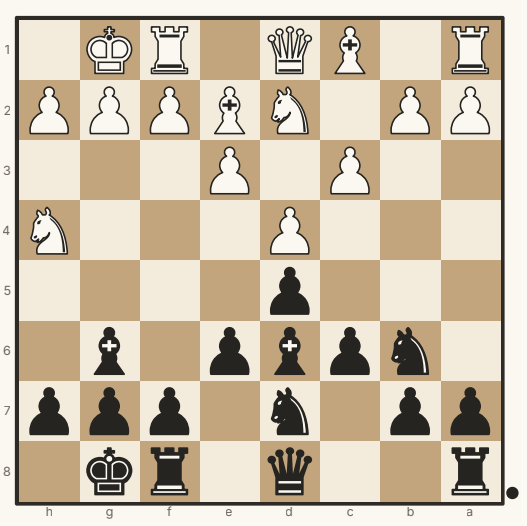
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

2.1 BOARD VISION



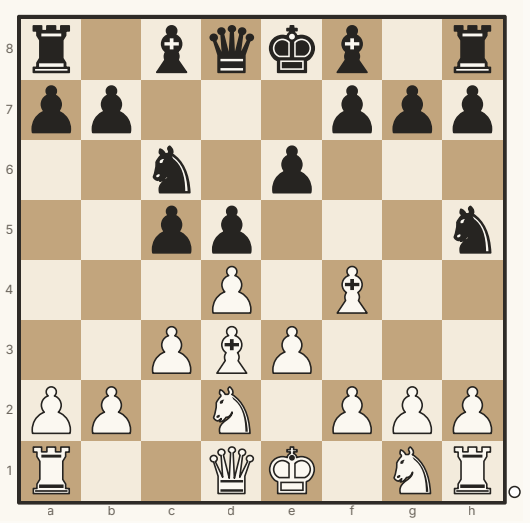
White to move. Black's knight has just jumped to e5. Win material. Solution p. 634

2.2 BOARD VISION



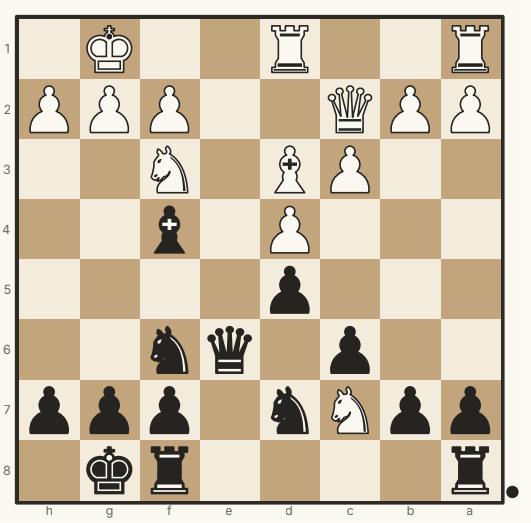
Black to move. White's knight has just gone to h4 to attack your bishop on g6. What is the best reply? Solution p. 634

2.3 BOARD VISION



White to move. Black has just played ...Nh5 to attack your bishop on f4. What is the best move? Solution p. 634

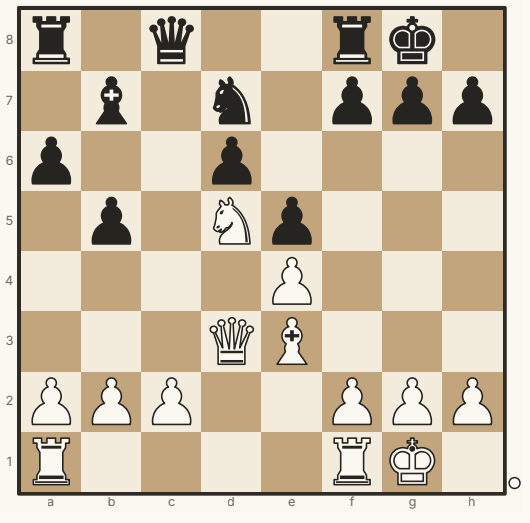
2.4 BOARD VISION



Black to move. White's knight has just captured on c7, attacking your queen and your rook on a8. What is the best move?

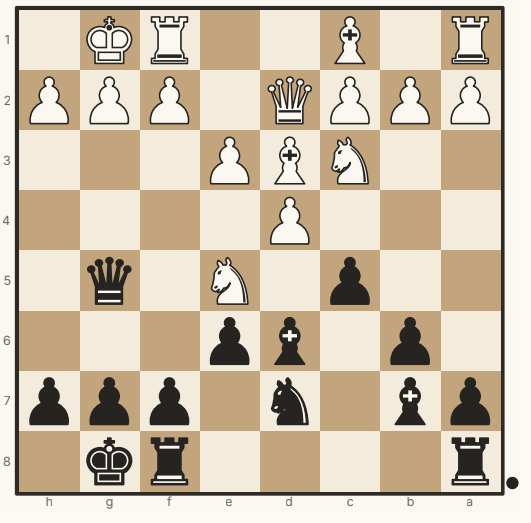
Solution p. 634

2.5 BOARD VISION



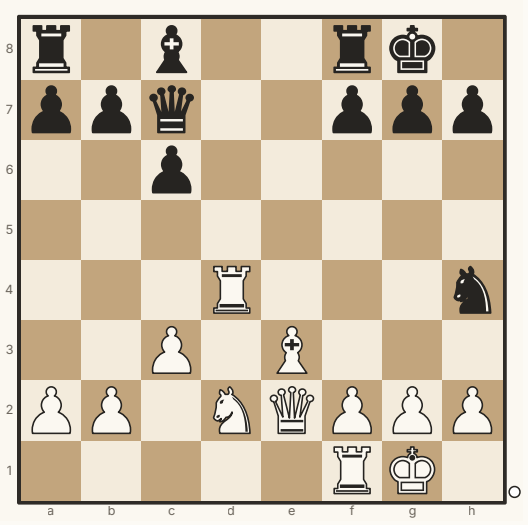
White to move. Find the knight move that wins material. Solution p. 635

2.6 CHECKMATE



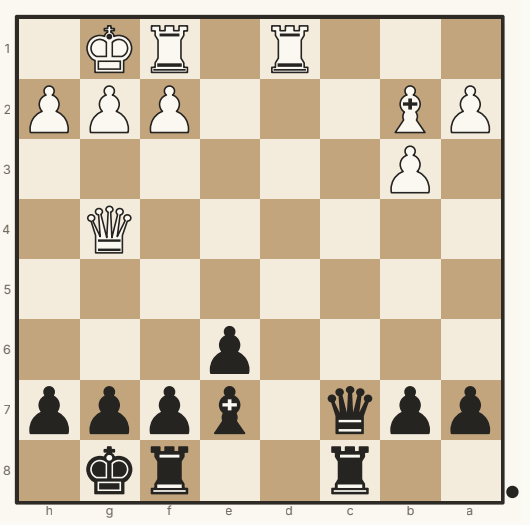
Black to move. Find checkmate in one move. Solution p. 635

2.7 BOARD VISION



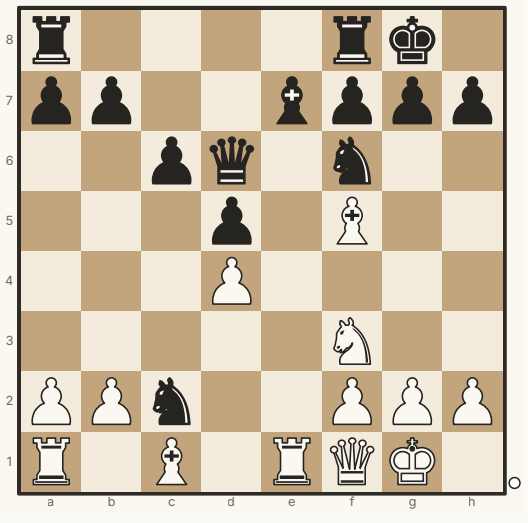
White to move. Black's knight has jumped to h4. Win material. Solution p. 635

2.8 DEFENCE



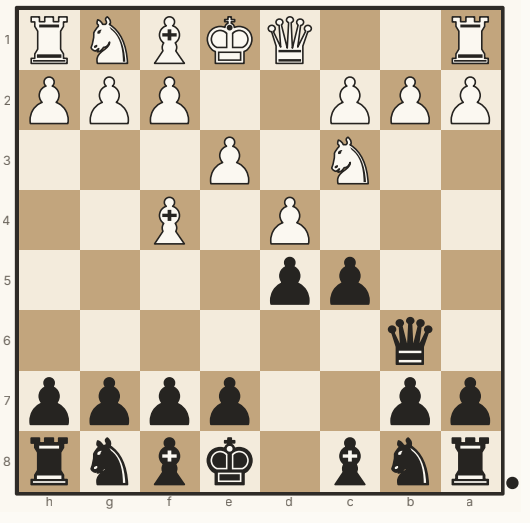
Black to move. What is White threatening, and how do you meet it? Solution p. 635

2.9 BOARD VISION



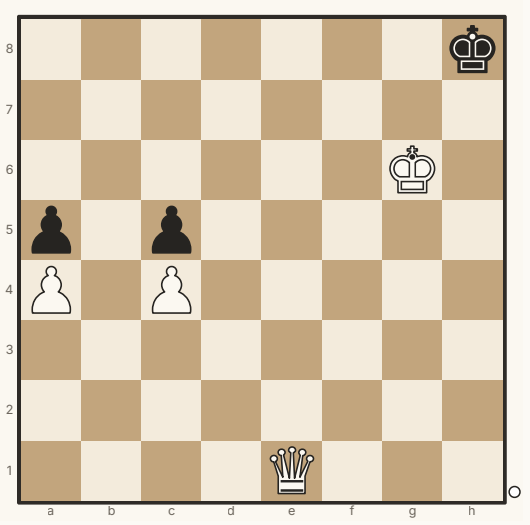
White to move. Black's knight has jumped to c2, forking both of your rooks. What is the best move? Solution p. 636

2.10 QUIET MOVE



Black to move. The pawn on b2 is attacked by your queen and has no defender. Is it safe to take? Play the best move. Solution p. 636

2.11 CHECKMATE



White to move. Black has a king and blocked pawns. Finish the game.

Solution p. 636

HINTS

2.1 Trace every line your bishop on b2 can see.

2.2 Look at the diagonal running from your queen towards the h-file.

2.3 What is your queen on d1 looking at?

2.4 Can any of your pieces reach c7?

2.5 Which square would let your knight attack the king and the queen at once?

2.6 What does your bishop on b7 see?

2.7 Look sideways along your rook's rank.

2.8 What does the bishop on b2 see?

2.9 Which of your pieces can reach c2?

2.10 Which of your pawns lost its defender when your queen left d8?

2.11 Use the full length of the e-file.

CHAPTER 3 STAGE 400–600

Piece Values and Counting

Know what everything is worth, then count before you capture

YOU WILL LEARN TO

- Use the standard piece values to judge trades, and know when they mislead.
- Count attackers and defenders on any square, including x-ray attackers.
- Capture in the right order, lowest-value attacker first.
- Decide whether a capture is safe with a short, reliable procedure.
- Choose between adding a defender, moving away and counterattacking when a piece is attacked.

> IN YOUR NEXT GAME
>
> Before every capture, count the attackers and defenders on that square and play out the exchange in your head, lowest-value piece first; before every move, check that the piece you move, and every piece it was defending, is still safe.

Your opponent's knight takes a pawn in the centre. You recapture with your queen because it is your strongest piece, and your opponent's bishop, which was defending that square all along, takes your queen. Or you see a "free" pawn, grab it with a knight, and lose the knight to a recapture you never counted.

Below 1000, a large share of games are decided by exactly this: material given away in exchanges that a few seconds of counting would have prevented. This chapter gives you two tools. The first is a simple scale of piece values that tells you whether a trade is good. The second is **counting**, the habit of adding up attackers and defenders before any capture. Together they will stop most one-move losses of material.

What the pieces are worth

The standard values, measured in pawns, are:

Piece	Value
Pawn	1
Knight	3
Bishop	3
Rook	5
Queen	9
King	cannot be traded

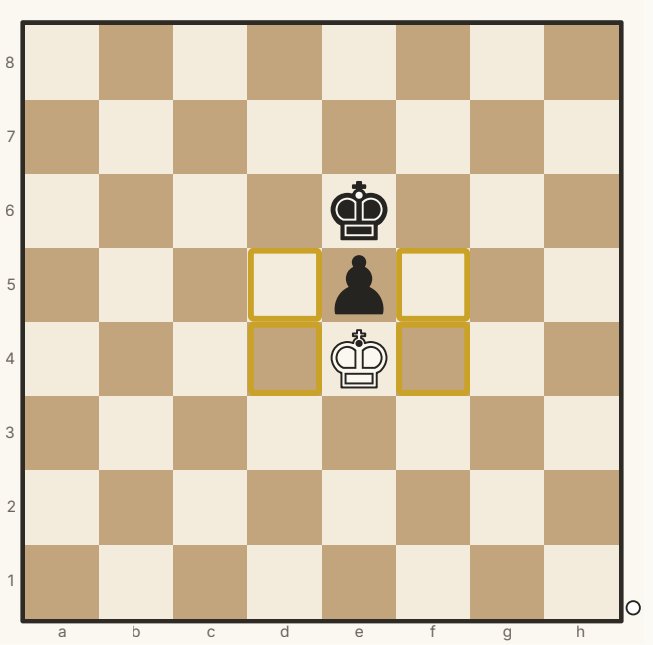
These numbers come from centuries of experience, and they work remarkably well as a first guide. They let you add up trades quickly: giving a knight (3) for a rook (5) gains two points, an advantage known as **winning the exchange**. Giving a bishop (3) for two pawns (2) loses a point.

A few practical consequences:

- **A minor piece for three pawns** is roughly even on paper, but the extra piece is usually stronger in the middlegame, and the pawns can become stronger in the endgame.
- **Two minor pieces for a rook and a pawn** is roughly even too; in the middlegame the two pieces are often better.
- **The queen** is worth about two rooks minus a little, or three minor pieces.

The king

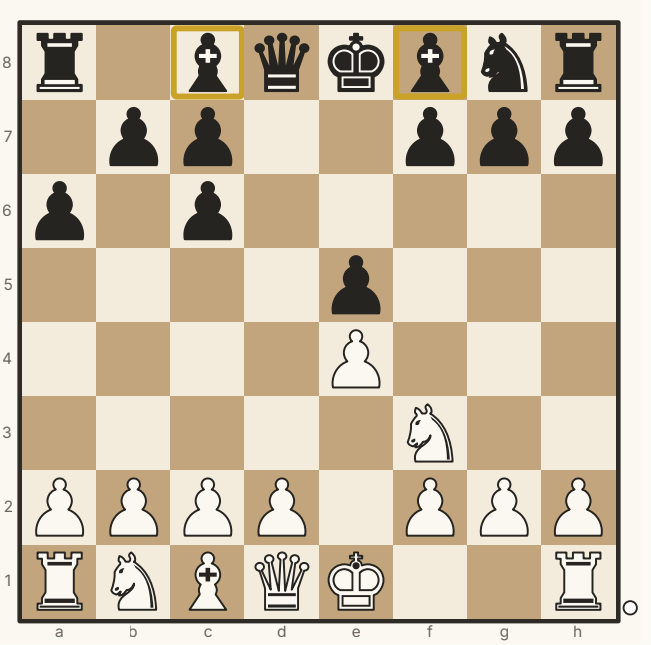
The king has no trading value, because losing it loses the game. But the king also has a fighting value. In the middlegame it is a liability that needs shelter. In the endgame, when the queens are off and mating attacks are unlikely, the king becomes an active piece, roughly as strong as a minor piece, that attacks pawns and escorts its own pawns forward.



In the endgame the kings are fighters. Both kings guard key squares around the e5-pawn. Chapter 18 shows how the side whose king is more active usually decides the game.

The bishop pair

Bishops and knights are both worth about three, but two bishops together are usually a little stronger than bishop and knight or two knights, especially when the position opens up. This advantage is called the **bishop pair**. Strong players often value it at around half a pawn; treat that figure as coaching judgment rather than law.



The Exchange Variation of the Ruy Lopez. White has given the bishop pair to damage Black's pawns. Material is level, but each side has a long-term trump.

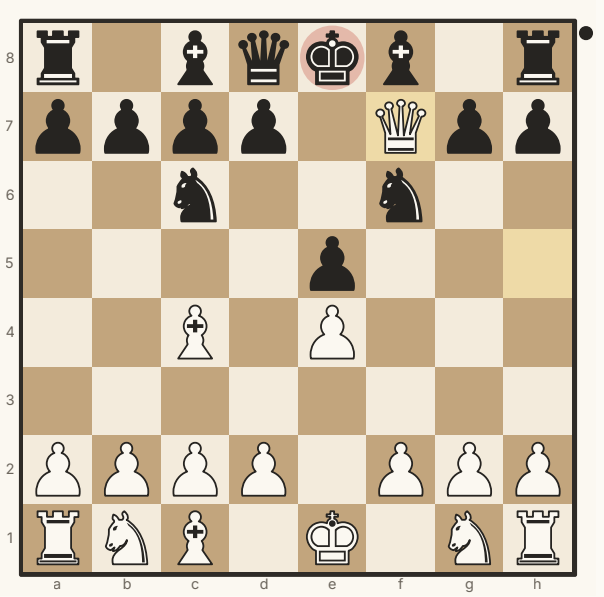
When the values mislead

The values describe pieces in a typical position. In a specific position, other factors can matter more:

- **Mate threats.** No amount of material matters if you are checkmated. A queen that captures a rook and allows mate in one has made a losing trade.
- **King safety.** A material advantage is often worth less than a safe king. An attack against an exposed king can be worth a piece or more.
- **Activity.** A rook that dominates an open file can be stronger than a passive rook buried in the corner, and a knight on a strong central square can outplay a bishop blocked by its own pawns.
- **Passed pawns.** A pawn one square from promotion is worth far more than one point.

The line below is the classic example of material blindness. Black "attacks" White's queen with a knight and forgets to look at the mate threat.

1. **e4 e5** 2. **Bc4 Nc6** 3. **Qh5?! Nf6??** 4. **Qxf7#**



Scholar's Mate. On move 3, Black needed a move that defended f7, such as 3...g6 or 3...Qe7. Attacking the queen with 3...Nf6 lost on the spot.

> **KEY IDEA** **Values are a guide, not the goal**
>
> Use the values to judge trades quickly. Then ask one more question before you decide: does this trade change the safety of either king, or allow a mate, a fork or a passed pawn?

Exchange sacrifices, briefly

Occasionally strong players deliberately give a rook for a minor piece, an **exchange sacrifice**, because the minor piece will be more active or the opponent's king will be exposed. You will meet the idea again in Stage 5. For now, treat your rooks as worth five and do not give the exchange unless you can see exactly what you get in return.

Trades: equal and unequal

A **trade** (or exchange) is any sequence where both sides capture. Keep two kinds apart:

- **Equal trades:** knight for knight, bishop for knight, rook for rook, queen for queen. They do not change the material balance, but they can change the pos-

ition: who gets the open file, whose pawns get damaged, whose king gets safer.

- **Unequal trades:** one side gives more value than it gets. These decide most games at this level.

A strong guideline: **when you are ahead in material, trade pieces, not pawns.** Each equal trade makes your extra material count for more, and leaves your opponent fewer pieces to attack with. When you are behind, avoid equal trades and keep pieces on the board to make threats.

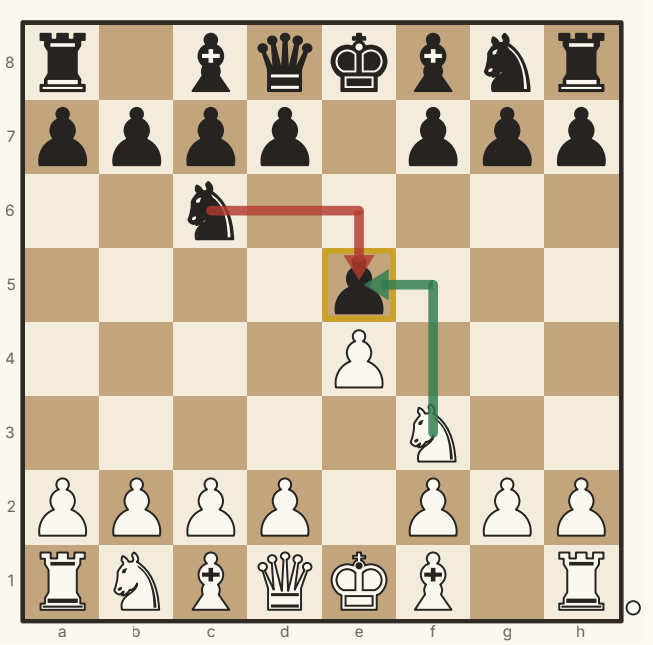
For example, suppose you have won a knight early in the game. With all the other pieces still on the board, your opponent can still create threats against your king, and your extra knight is one piece among many. Trade the queens and a pair of rooks, and your knight becomes one of only a few pieces left; it can now win pawns while your opponent has almost nothing to attack with. The same logic tells the defending side to avoid trades. Chapter 16 develops this into a full method for converting an advantage.

Counting attackers and defenders

The most important skill in this chapter is **counting**: before any capture, add up how many of your pieces attack the target square and how many of your opponent's pieces defend it.

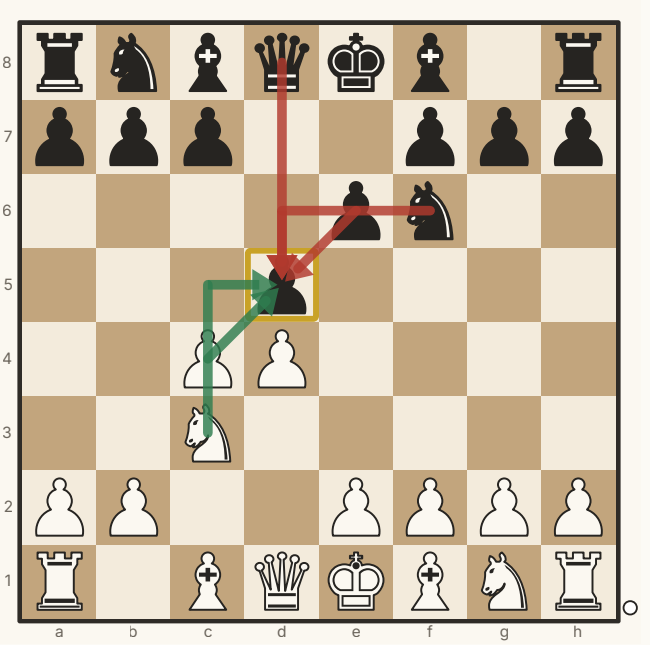
The basic rule:

- If a piece is attacked more times than it is defended, it is usually in danger.
- If it is defended at least as many times as it is attacked, it is usually safe, *as long as every attacker is worth more than or the same as the target.*



The simplest count. After 1.e4 e5 2.Nf3 Nc6, the e5-pawn is attacked once (by the knight on f3) and defended once (by the knight on c6). Taking it would give a knight for a pawn.

That last condition matters enormously. A knight defended twice can still be lost if it is attacked once by a pawn, because the pawn is cheaper. **When the attacker is worth less than the target, counting defenders does not help.**



The d5-pawn is attacked twice (by the c4-pawn and the c3-knight) and defended three times (by the e6-pawn, the f6-knight and the queen on d8). White can trade with cxd5, but Nxd5?? simply loses the knight to ...exd5.

Play out the exchange

Counting is the first step. The second is to play out the captures in your head, adding up the values as you go. Take the d5-square above and imagine 4.Nxd5. White has taken a pawn (+1). Black replies 4...exd5, taking a knight (White is now −3 + 1 = −2). White can recapture with 5.cxd5 (+1, total −1), and Black can take again with 5...Nxd5 or 5...Qxd5. Every time, White ends up behind. Counting told you the same thing in one second: three defenders, two attackers.

Capture order: lowest value first

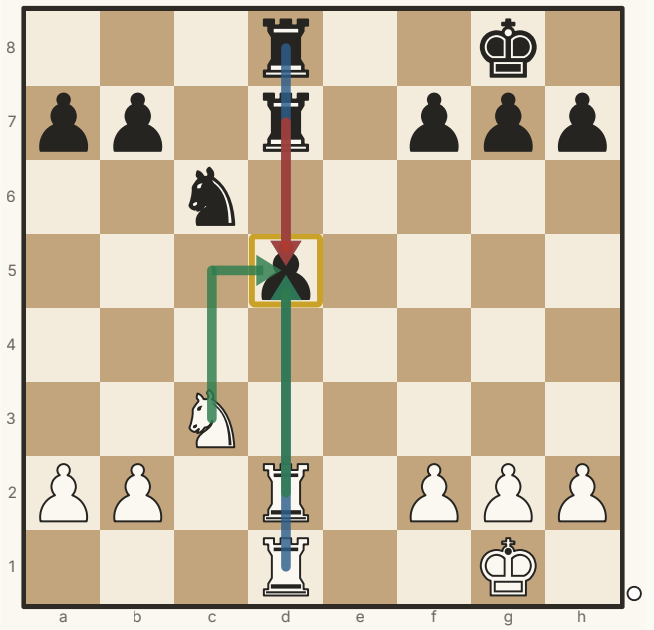
When you capture on a square several times, the order matters. The rule is simple: **capture with your lowest-value piece first, and keep your most valuable pieces for last.** The reason is that your opponent can stop recapturing at any moment. If your queen goes in first and is captured, you have lost nine points even if you later win the square back.

The same applies to your opponent. When you count, assume they will recapture with their cheapest piece first, and that they will stop as soon as further captures lose material.

> **COMMON MISTAKE** **Recapturing with the queen "because it's strongest"**
>
> Players often recapture with the queen because it is the most powerful piece. But the queen is also the piece you can least afford to have captured. If a pawn or a minor piece can recapture, it is usually the safer choice. Use the queen only when you have checked that nothing can take it on that square.

X-ray attackers and defenders

Pieces lined up on the same line count even when one stands behind the other. Two rooks doubled on a file, or a queen behind a bishop on a diagonal, form a **battery**. The rear piece is an **x-ray** attacker (or defender): it does not attack the square yet, but it will as soon as the front piece captures.



The isolated d5-pawn is defended by the rook on d7 and, behind it, the rook on d8. It is attacked by the knight on c3, the rook on d2 and, behind that, the rook on d1. Counting the x-rays, three attackers face two defenders.

When you count, always look behind the front piece on each line. X-rays are the most common reason a careful count still comes out wrong.

Is it safe to take? A procedure

Here is a short routine to run before any capture. It takes about ten seconds once you are used to it.

1. **Count** the attackers and defenders of the target, including x-rays.
2. **Compare values.** Is any defender worth less than my capturing piece? If a pawn defends the target, never take with a piece worth more than the target plus a pawn.
3. **Play out the exchange** in your head, lowest-value attacker first, and add up the values. Assume your opponent stops when it suits them.
4. **Check the final position.** After the last capture, is anything else of mine undefended? Did any of my pieces stop defending something when it moved? Does my opponent get a check, a fork or a mate threat?
5. **Only then decide.** If the count is in your favour and step 4 is clean, take. If not, look for a better move, often a quiet developing move that keeps the tension.

Worked example: the isolated pawn

Let us run the procedure on the x-ray position above. White is considering 25.Nxd5.

First I count. The d5-pawn is attacked by my knight on c3, my rook on d2 and, behind it, my rook on d1: three attackers. It is defended by the rook on d7 and, behind it, the rook on d8: two defenders. The knight on c6 does not help, because a knight on c6 jumps to b4, d4, e5, e7, d8, b8, a7 and a5, never to d5.

Next I compare values. Both defenders are rooks, worth more than my knight, so there is no cheap defender to worry about.

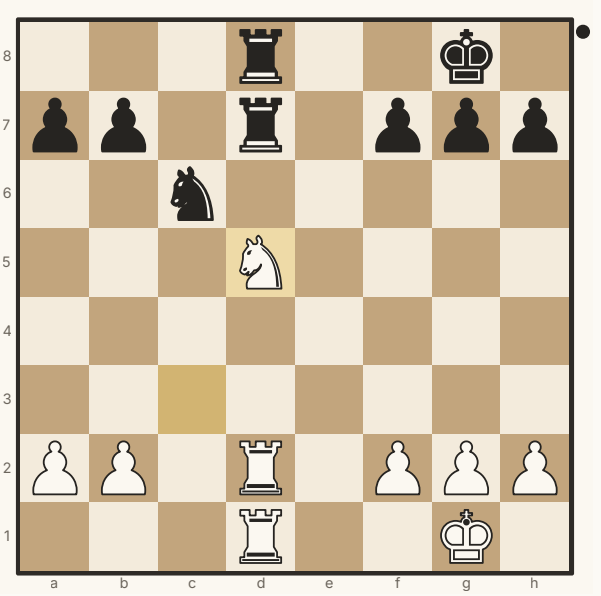
Now I play it out, lowest-value attacker first, adding up as I go. 25.Nxd5 wins a pawn (+1). If 25...Rxd5, then 26.Rxd5 (+1 −3 +5 = +3) Rxd5 27.Rxd5 (+3 −5 +5 = +3). Every way Black recaptures, I come out ahead, so Black will not recapture at all, and I have simply won the pawn.

Notice what happens if I forget the x-ray rook on d1. I would count two attackers against two defenders, play out 25.Nxd5 Rxd5 26.Rxd5 Rxd5, see that I have lost a knight and a rook for a pawn and a rook, and decide not to take. The rook behind the rook changes the answer.

Finally I check the new position. My knight on d5 is attacked only by the rook on d7, and taking it loses for Black. Did moving the knight from c3 leave anything undefended? Nothing of mine depended on it. Does Black have a check? No.

The decision: 25.Nxd5 wins the pawn, and the knight lands on a strong central square.

25. **Nxd5**



White has won the d5-pawn. Black cannot recapture: the third white attacker, the x-ray rook on d1, would win the last exchange.

> COACHING JUDGMENT **Count out loud at first**
>
> Coaching judgment: in your first few weeks of using this routine, whisper the count to yourself in training games: “two attackers, one defender, knight takes first.” It feels slow, but it builds the habit faster than counting silently, and the habit soon becomes automatic.

Four counting traps

Counting is simple in principle, but four situations regularly fool players who have learned the basic method.

The king as a defender. The king defends every square next to it, but it can only recapture on a square that is not attacked by another enemy piece. A pawn "defended" by the king alone is lost as soon as the square is attacked twice, because the king cannot take back into the second attacker's line.

The defender that moved away. A piece defends squares only from where it stands. When you move a piece, everything it was guarding loses a defender. Before every move, ask what the piece is protecting right now. This single question prevents a large share of hanging pieces at this level, and Chapter 4 builds it into the safety scan.

The defender that cannot really defend. A piece that is pinned against its king cannot recapture at all, and a piece that is already defending something else may not be able to do both jobs. Counting such a piece as a full defender gives the wrong answer. Chapters 8 and 10 look at these situations in detail; for now, notice when a defender is also busy elsewhere.

Captures with check. An exchange does not have to run in the order you expect. If one of the recaptures gives check, or if your opponent can insert a check or a bigger threat in the middle of the sequence, the simple count no longer applies. When a capture in the sequence comes with check, play that line out carefully before trusting your count.

> BOARD DRILL **Count ten squares**
>
> Take any position from one of your recent games. Pick ten squares that contain pieces or pawns, and for each one say aloud: attacked how many times, defended how many times, and what the cheapest attacker is worth. Check yourself by playing out the captures in an app. With practice, each square should take under five seconds.

When your piece is attacked

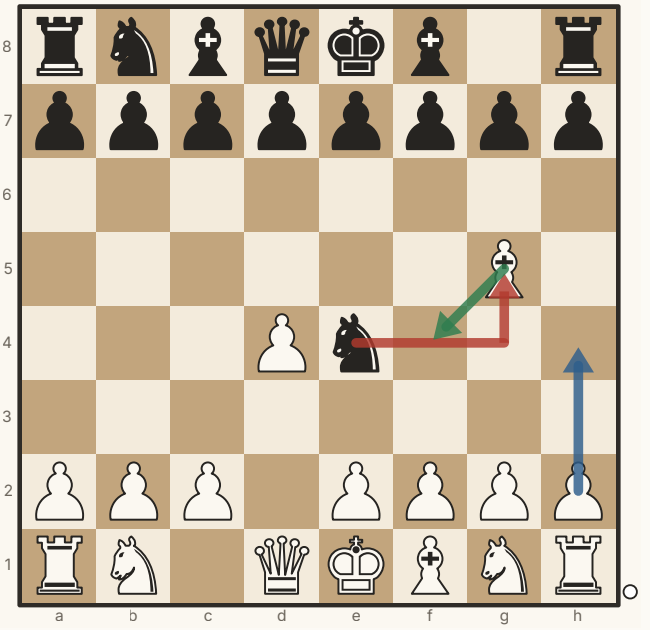
The other side of counting is defence. When one of your pieces is attacked, you have four main options:

1. **Move it away** to a safe square.
2. **Add a defender**, so that the capture would cost your opponent material.
3. **Block** the line of attack (only against queens, rooks and bishops).

4. **Counterattack**: make a threat of your own that is at least as big, or capture something.

Which option works depends on the values involved.

- **If the attacker is worth less than your piece** (a pawn attacking a knight, a knight attacking a rook), adding a defender does not help: your opponent will happily give the cheaper piece for yours. You must move it, block, or counterattack with something bigger.
- **If the attacker is worth the same or more**, adding a defender is often enough, and it can be the most useful option if it also develops a piece.



The Trompowsky Attack. Black's knight attacks the bishop on g5. Because the attacker is a knight, of equal value, White has real choices: move the bishop (3.Bf4) or defend it with a pawn (3.h4), and both are played by strong players.

Counterattack

A counterattack works only if your threat is at least as large as your opponent's and they cannot answer both at once. A typical example: your knight is attacked, and instead of moving it you attack your opponent's queen with a pawn. After the queen moves, you save the knight with a gain of time. But be strict: if your opponent can capture your piece *with check*, or can meet your threat while keeping theirs, the counterattack fails.

> **COMMON MISTAKE** **Defending a piece against a pawn attack**
>
> When a pawn attacks one of your pieces, beginners often add a defender. It does not help: the pawn captures your piece, you recapture the pawn, and you have traded a piece for a pawn. Against a pawn attack, move the piece, or use the time for a bigger threat.

How to find counting errors in your own games

When you analyze a game, mark every capture you made or allowed and ask:

- Did I count the attackers and defenders before capturing?
- Did I capture with the right piece, the lowest-value one?
- Did I miss an x-ray, a defender behind another piece?
- When my piece was attacked, did I choose the right defence for the values involved?
- After my move, did a piece I moved stop defending something?

Log mistakes of this type under *hanging piece* in your mistake journal. If they make up a large share of your losses, which is normal at this level, come back to this chapter after a month and solve the exercises again.

Practice at the board

1. **Count the centre.** Load `rnbqkb1r/ppp2ppp/4pn2/3p4/2PP4/2N5/PP2PPPP/R1BQKBNR w KQkq - 2 4`. Count attackers and defenders of d5 and of d4. Play out 4.cxd5 exd5 and recount. Then try 4.Nxd5?? and see the material swing.
2. **X-ray drill.** Load `3r2k1/pp1r1ppp/2n5/3p4/8/2N5/PP1R1PPP/3R2K1 w - - 0 25`. Play out 25.Nxd5 and every possible recapture. Then, in the board editor, remove the rook on d1 and count again: is 25.Nxd5 still safe? (It is not: after 25... Rxd5 26.Rxd5 Rxd5 you have given a knight and a rook for a pawn and a rook.)
3. **Capture order.** Set up a square attacked by your knight and your queen and defended by one enemy rook. Play out both orders, knight first and queen first, and add up the values each time.
4. **Pawn attacks.** In a free app, play the French Defence moves `1.e4 e6 2.d4 d5 3.Nc3 Nf6 4.e5`. Find Black's knight moves that save the knight, and explain why defending it does not work.

5. **Play it out.** Against a friend or an engine at a low level, play ten-minute games where you must say “count” before every capture. Afterwards, check how many captures you would have changed.

Chapter summary

- Pawn 1, knight 3, bishop 3, rook 5, queen 9. The king cannot be traded but is a strong fighter in the endgame.
- The bishop pair is usually worth a little more than bishop and knight; mate threats, king safety, activity and passed pawns can outweigh material.
- Count attackers and defenders before every capture, including x-ray pieces behind the front attacker.
- When the attacker is worth less than the target, the number of defenders does not matter.
- Capture with the lowest-value piece first; your opponent can stop recapturing whenever it suits them.
- Against a cheaper attacker, move, block or counterattack; adding a defender works mainly against attackers of equal or greater value.
- When ahead, trade pieces; when behind, keep pieces on and make threats.

CHEAT CODE **Is it safe to take?**

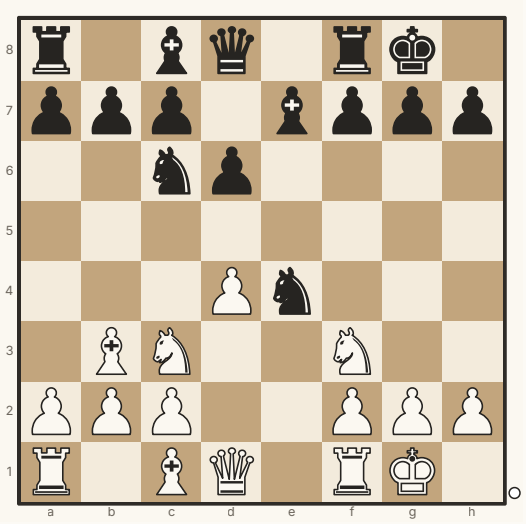
1. Count attackers and defenders of the target, including x-rays.
2. Is any defender cheaper than my capturing piece? If so, think twice.
3. Play out the exchange, lowest-value attacker first, adding up values.
4. Check the final position for loose pieces, checks, forks and mate threats.
5. Take only if the count favours me and step 4 is clean; otherwise play a solid developing move.

Now solve the exercises below. Then play two slow games in which you count out loud, or in your head, before every capture you make, and before every capture you allow.

Exercises

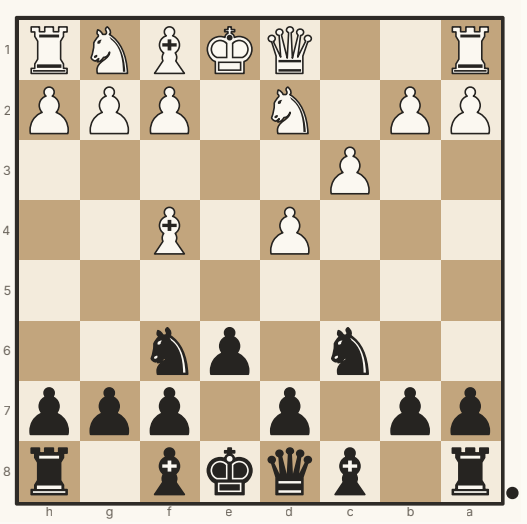
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

3.1 COUNTING



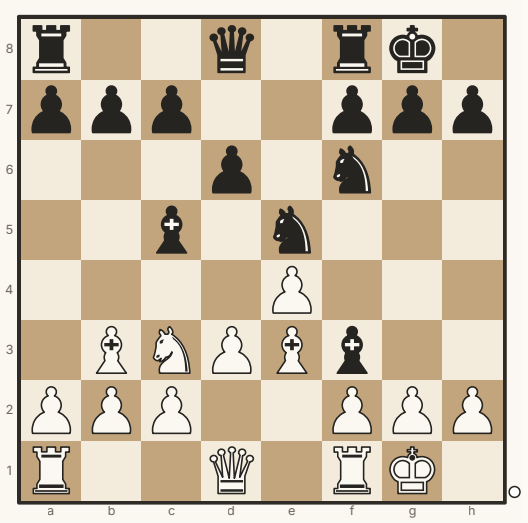
White to move. Black's knight has just taken your pawn on e4. Is the knight safe there? Play the right move. Solution p. 637

3.2 COUNTING



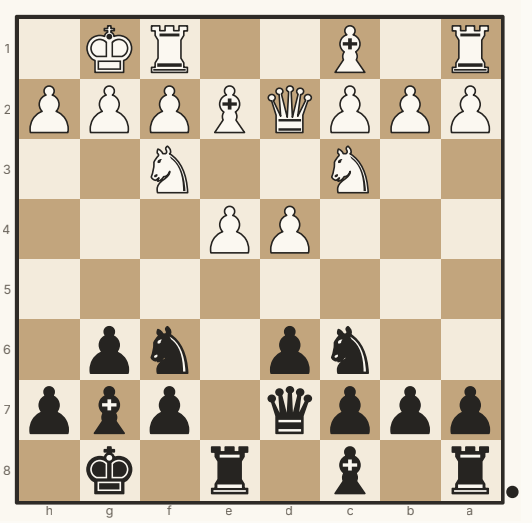
Black to move. Your knight on c6 attacks the pawn on d4. Is it safe to take? Play the best move. Solution p. 637

3.3 COUNTING



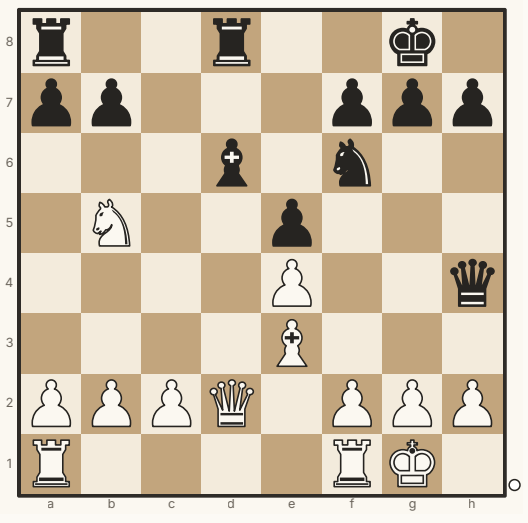
White to move. Black's bishop has just captured your knight on f3. How should you recapture? Solution p. 637

3.4 COUNTING



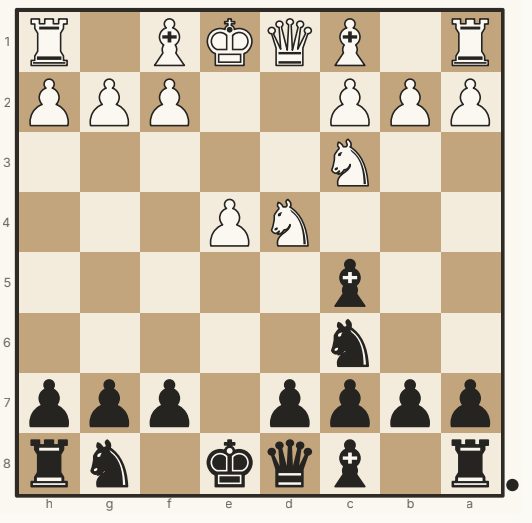
Black to move. The pawn on e4 is attacked by your knight and your rook. Can you win it, and with which piece should you capture first? Solution p. 637

3.5 COUNTING



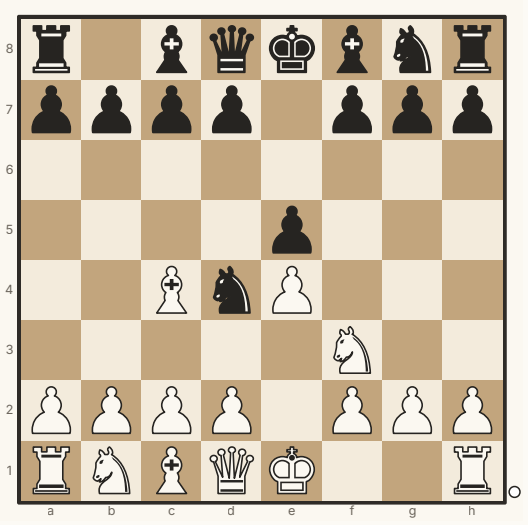
White to move. Black's bishop on d6 is attacked by your knight and your queen and defended only by the rook on d8. Win material. Solution p. 638

3.6 COUNTING



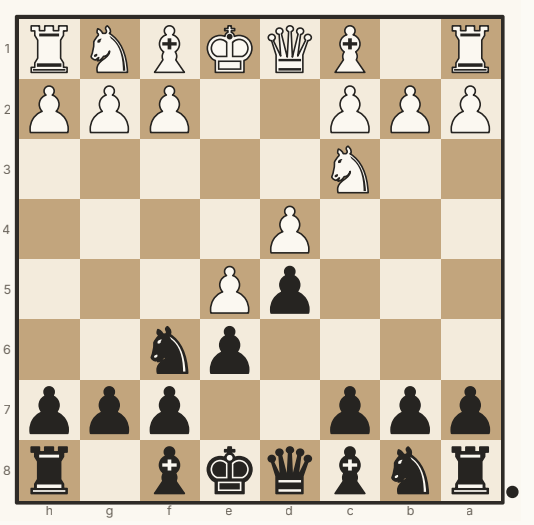
Black to move. White has just played Nc3 in the Scotch Game. Count the d4-square and win material. Solution p. 638

3.7 QUIET MOVE



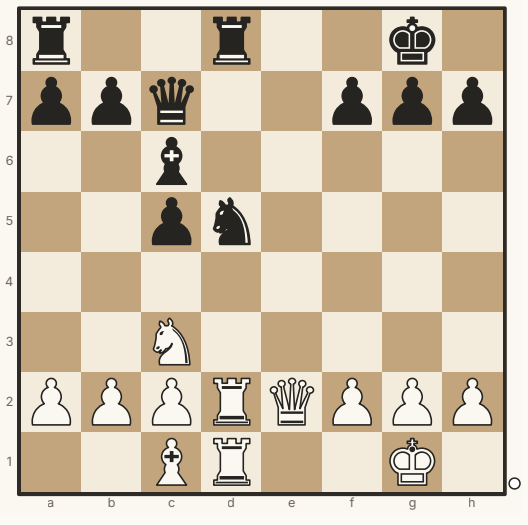
White to move. Black's last move, ...Nd4, left the e5-pawn with no defender. Is it safe to take? Play a good move. Solution p. 638

3.8 DEFENCE



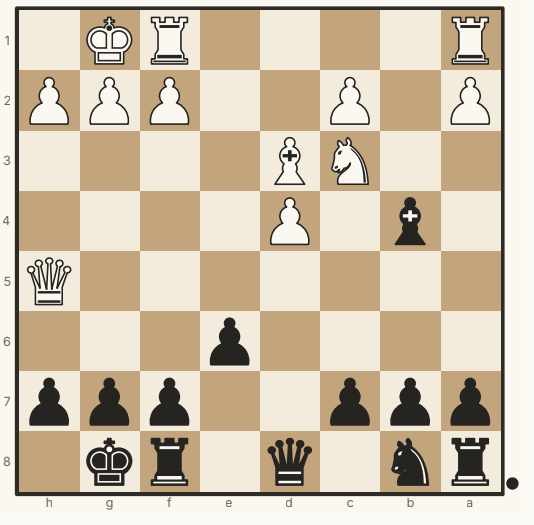
Black to move. White has just played e4-e5, attacking your knight on f6 with a pawn. What should you do? Solution p. 638

3.9 COUNTING



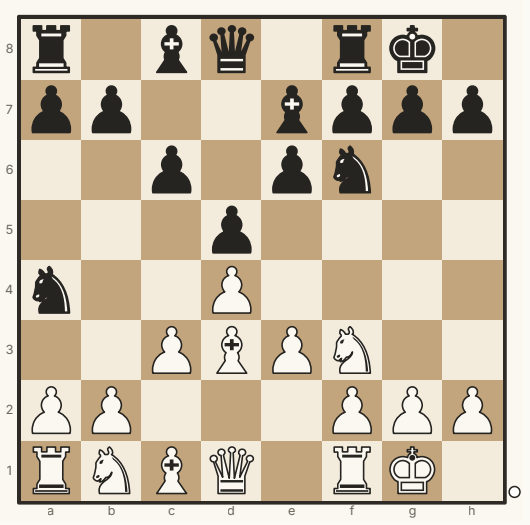
White to move. Count the d5-square carefully, including pieces standing behind other pieces, and win material. Solution p. 639

3.10 DEFENCE



Black to move. Your bishop can take White's undefended knight on c3. Should it? Find the best move. Solution p. 639

3.11 BOARD VISION



White to move. Black's knight has jumped to a4, attacking your b2-pawn. What is the best move? Solution p. 639

HINTS

3.1 How many black pieces defend e4?

3.2 Which piece defends d4, and what is it worth?

3.3 Which black piece still defends f3?

3.4 Which attacker is worth less?

3.5 Which of your two attackers should go in first?

3.6 Two attackers against one defender.

3.7 After taking on e5, which black moves would attack two things at once?

3.8 Does defending help when the attacker is a pawn?

3.9 Do not forget the rook on d1.

3.10 Before you count material, look at your king.

3.11 Edge pieces are easy to miss. What does your queen see?

CHAPTER 4 STAGE 400–600

The Safety Scan: Checks, Captures, Threats

The habit that stops you giving pieces away

YOU WILL LEARN TO

- Ask three questions about every move your opponent makes - what does it attack, what does it no longer defend, what line did it open.
- Find hanging pieces, both yours and your opponent's, before you choose a move.
- Scan checks, captures and threats for both sides in a fixed order.
- Run a blunder check on your chosen move before you let go of the piece.
- Turn the five-question pre-move checklist into a habit.

> IN YOUR NEXT GAME
>
> After every move your opponent makes, ask "What does it attack, what does it no longer defend, what line did it open?" and before you release your own piece, ask "After this move, what can my opponent capture or check?"

Look at almost any game between two players rated around 500 and you will find the same story. Someone leaves a piece where it can be taken for free. Sometimes the opponent takes it; often they do not notice, and two moves later the first player hangs something else. The game is decided not by a plan or a clever combination but by who stops giving material away first.

This chapter teaches the single most valuable habit in chess below 1000: the **safety scan**. It is a short, fixed routine you run every move. It does not require talent or memorised theory, only discipline. Players who run it honestly every move usually find that their results change faster than from any other single improvement. Learn it here, and every later chapter will build on it.

Why games are lost below 800

At this level most decisive mistakes fall into a few families. A piece is left **hanging** (it can be captured and nobody can recapture). A threat by the opponent is not seen until it is carried out. A move is played without asking what it allows. None of these is a lack of knowledge; you already know how the pieces move. They are lapses of attention.

That is good news, because attention can be trained with a routine. Strong players are not superhuman. They have simply asked the same few questions so many thousands of times that the questions run automatically. Your job in this chapter is to start asking them on purpose, slowly, until they become automatic for you too.

> COACHING JUDGMENT **A note on what to expect**
>
> This is coaching judgment, not a measured result: most players who start running a safety scan every move find that their games get longer and their blunders get rarer within a few weeks. The first games will feel slow and awkward. That is normal. You are building a habit, and habits are slow at first.

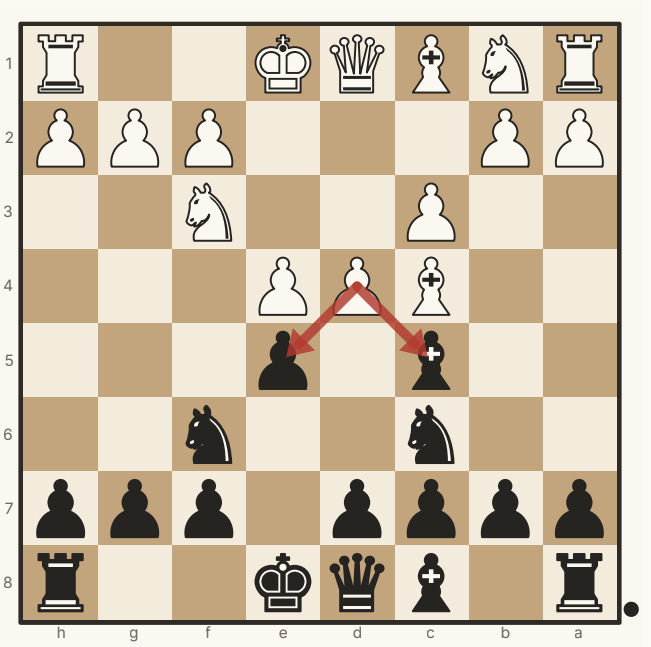
Step one: the three questions about your opponent's move

Every move your opponent makes changes the position. Before you think about your own ideas, find out what changed. Three questions cover almost everything:

1. **What does it attack?** Which of your pieces, pawns or squares can the moved piece now capture or hit?
2. **What does it no longer defend?** The piece left a square. What was it guarding there that is now unprotected?
3. **What line did it open?** When a piece or pawn moves, the file, rank or diagonal behind it may open for another piece. What can now see further than before?

Most beginners ask only the first question, and only sometimes. The second and third questions catch the threats that "come from nowhere".

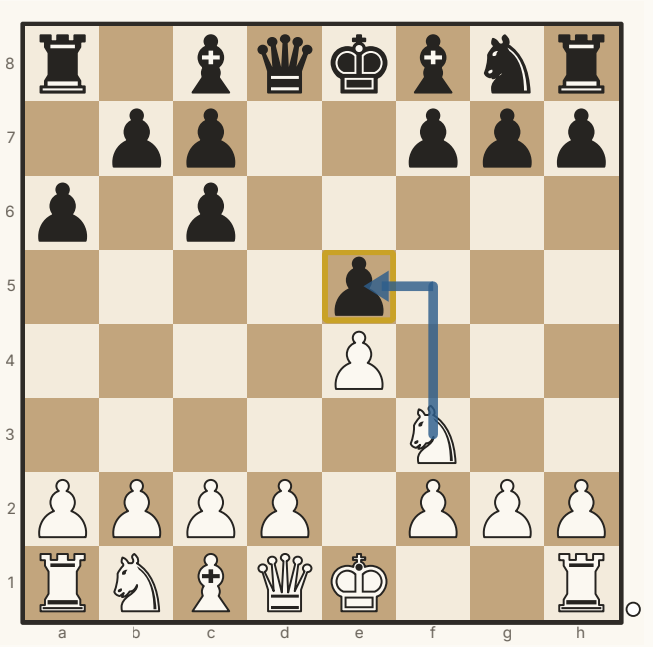
What does it attack?



After 1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d4, White's last move attacks the bishop on c5 and the pawn on e5 at the same time.

Here the answer to question one is easy to find if you look: the pawn on d4 attacks two things. The usual reply is 5...exd4, removing the attacker. A player who skips the question and plays a "plan" move such as 5...O-O simply loses a piece after 6.dxc5. The opponent's move is the first thing you look at, every time.

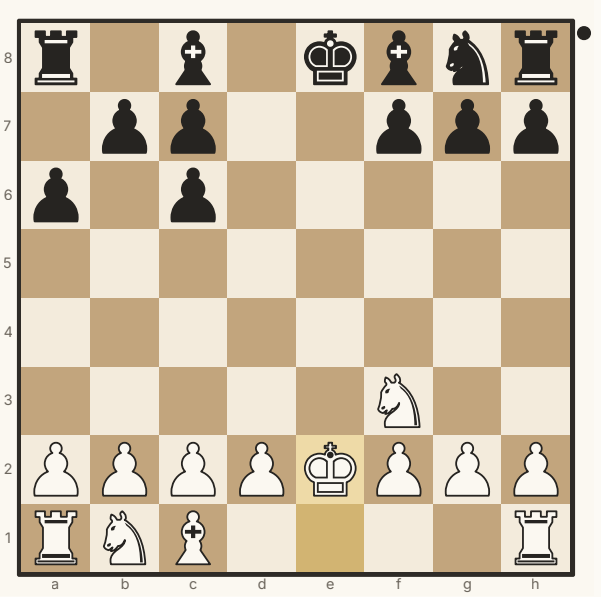
What does it no longer defend?



After 3.Bb5 a6 4.Bxc6 dxc6, the knight that defended e5 has gone. Is 5.Nxe5 a free pawn?

When White exchanged on c6, the black knight that guarded e5 disappeared. Asking "what is no longer defended?" reveals that the e5-pawn now hangs. But the scan does not stop there. Before taking, White must ask what Black can do in reply:

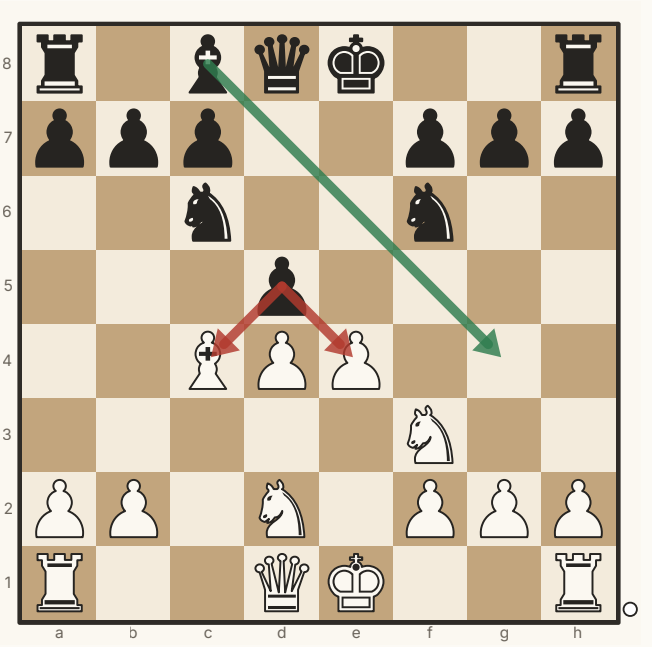
5. **Nxe5?! Qd4** 6. **Nf3 Qxe4+** 7. **Qe2 Qxe2+** 8. **Kxe2**



Black's queen attacks the knight and the e4-pawn at once, wins the pawn back, and White's king has lost the right to castle.

The same question works for your own pieces. When you move a piece, ask what it was guarding before. Many hanging pieces at club level were safe a move earlier, until their only defender wandered off.

What line did it open?



In the Giuoco Piano, 8...d5 attacks the bishop on c4 and the e4-pawn, and it also opens the c8-bishop's diagonal toward g4.

A pawn move can attack something directly and, at the same time, unblock a piece behind it. Here 8...d5 hits c4 and e4 and frees the bishop on c8. White's usual answer is 9.exd5 Nxd5, and both sides are fine. A beginner who only noticed "the pawn attacks my bishop" might retreat the bishop and leave e4 hanging.

Discovered lines are dangerous because the piece that does the attacking did not move. Your eyes follow the piece that moved, and the bishop or rook in the background slips past them. Train yourself to look *behind* the moved piece: along the file, rank or diagonal it was standing on.

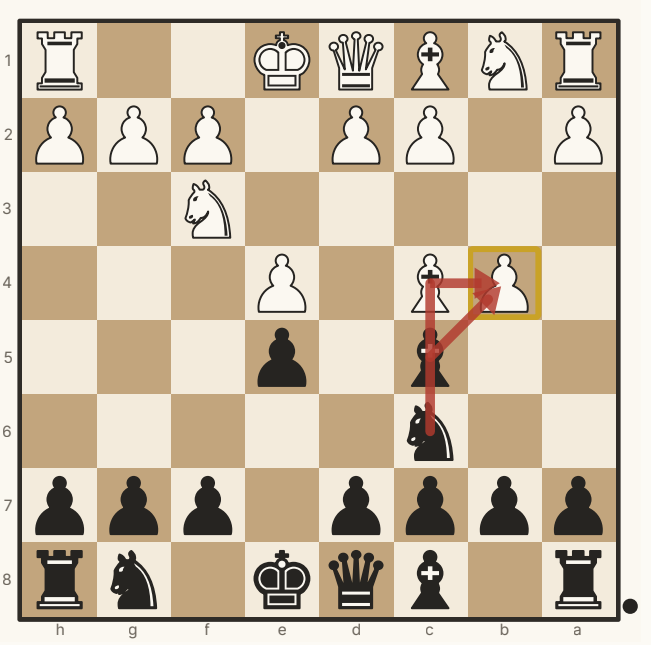
> **KEY IDEA** **The three questions**
>
> After every opponent move, ask in this order: 1. What does it attack? 2. What does it no longer defend? 3. What line did it open? Answer all three before you think about your own plans.

Hanging pieces: yours and theirs

A piece is **hanging** (also called *en prise*) when it can be captured and the capture wins material. The simplest case is a completely undefended piece under attack. A subtler case is a piece defended once but attacked twice, or a piece defended only by something more valuable than its attacker. Chapter 3 taught you to count attackers and defenders; this chapter makes you do it every move.

Undefended means target

A piece that has no defender at all is called **loose**. Loose pieces are not hanging yet, because nothing attacks them. But they are the first things to fall when a forcing move appears, because a single check or threat can attack them together with something else. Coaches use a short slogan for this: **loose pieces drop off**.



The Evans Gambit, 4.b4. The pawn on b4 is attacked by the bishop and the knight and defended by nothing. It is a deliberate gambit, but the scan must still notice it.

In this position White has offered a pawn on purpose, and Black may take it with 4...Bxb4. The point is not this opening; it is the method. Before you move, list your opponent's undefended pieces and pawns, and list your own. Loose pieces are targets for both sides.

A quick way to see them: run your eyes over each piece and ask "if this were captured, could I (or they) take back?" With practice this takes a few seconds per side.

Your own hanging pieces

Beginners spot their opponent's hanging pieces more easily than their own, because they are hunting. Reverse the habit: the first scan is defensive. After your opponent moves, look at each of your own pieces and ask whether it is attacked and how often it is defended. If something is hanging, deal with it now: move it, defend it, or make a bigger threat.

> COMMON MISTAKE **Reacting to the attack and missing the attacker**
>
> When your piece is attacked, first check whether you can simply take the attacker. In many club games a knight jumps forward to attack a bishop, the bishop's owner retreats it at once, and nobody notices that the knight was standing on a square defended by a pawn. Ask "can I capture the piece that is attacking me?" before you retreat.

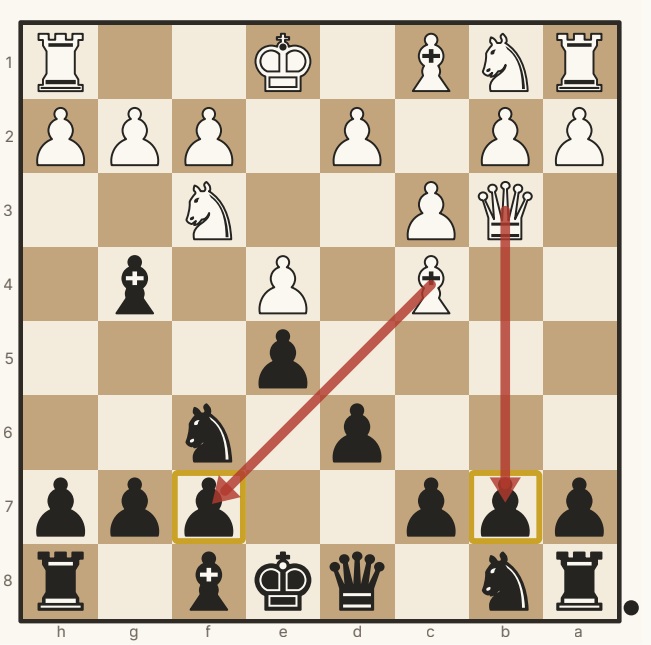
The forcing-move scan

A **forcing move** is a move that limits your opponent's replies: a **check** (they must answer it), a **capture** (they usually want to recapture), or a **threat** (a move that will win something next turn if ignored). Forcing moves are where tactics live. If there is a winning combination in a position, it almost always starts with one of them.

The forcing-move scan is simple to state: **list every check, every capture and every serious threat, for both sides.** In that order: checks first because they are the most forcing, then captures, then threats.

Your forcing moves

For your own side, look at each check, even silly ones. Most will be useless, but the one that also attacks a loose piece, or drives the king to a bad square, is the one you are looking for. Then look at every capture and count it out. Then look for moves that create a threat: attacking an undefended piece, threatening mate, or attacking a piece with something less valuable.



After 1.e4 e5 2.Nf3 d6 3.Bc4 Bg4 4.c3 Nf6 5.Qb3, White threatens both Qxb7 and Bxf7+. The bishop left c8, so b7 lost its only defender.

The threat scan finds two loose targets for one queen move. Black's best answer is the counter-attack 5...d5, which blocks the bishop's line to f7. A natural developing move fails:

5...**Nc6?** 6.**Bxf7+ Ke7** 7.**Qxb7**

Black loses two pawns and the king is stuck in the centre. Two threats at once are hard to meet with an ordinary move.

Your opponent's forcing moves

Do the same from your opponent's side of the board. What checks do they have right now? What captures? What are they threatening? This is the part of the scan that prevents disasters. If your opponent has a check that forks your king and rook, you need to know it *before* you start a slow plan.

KEY IDEA **Checks, captures, threats - both sides**

1. My checks, my captures, my threats.
2. Their checks, their captures, their threats. If a forcing move wins material for you, look at it first. If one wins material for them, deal with it first.

When there is nothing forcing

Often the scan finds nothing: no captures that win, no checks that matter, no threats. That is valuable information too. It means you are free to play a good normal move: develop, castle, improve your worst piece, or put a rook on an open file. Do not invent violence when the position is quiet. Many beginners lose because they launch an attack that is not there and leave pieces behind.

The blunder check: before you let go

You have scanned the position and chosen a move. One step remains, and it is the step that saves the most games: the **blunder check**. Before you release the piece (or click the mouse), look at the position *after* your move and ask:

> **After my move, what can my opponent capture or check?**

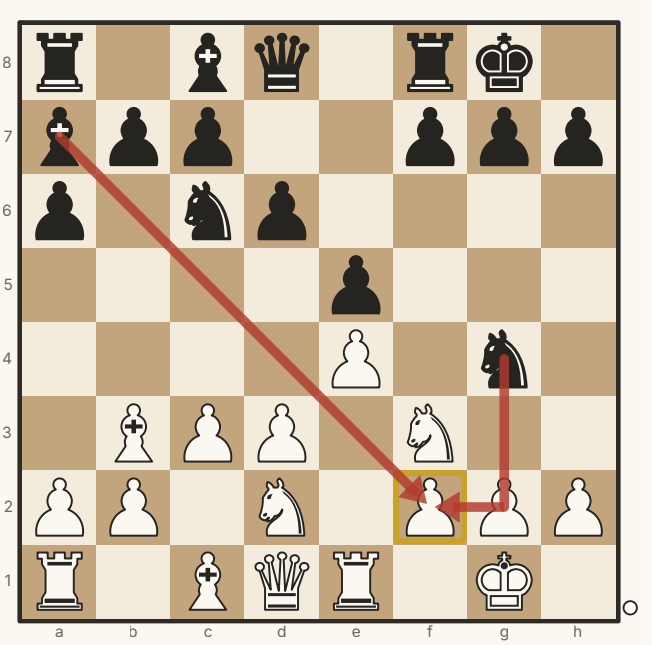
Your move changes the position. It may leave a piece undefended, open a line toward your king, or put a piece on a square where it can be attacked. The scan you ran a minute ago was for the old position. The blunder check is a fresh, short scan of the new one, from your opponent's side.

COMMON MISTAKE **The "obvious" move**

The most dangerous move is the one that feels obvious: an automatic recapture, a pawn push that "kicks" a piece, a capture of a free-looking pawn. Because it feels obvious, you skip the check. Make a rule for yourself: the more obvious the move looks, the more carefully you check it.

A worked example: the knight on g4

Here is the whole routine on one position from a typical Italian Game. Black has just played 9...Ng4. You are White.



Black's last move 9...Ng4 attacks f2, together with the bishop on a7. What should White do?

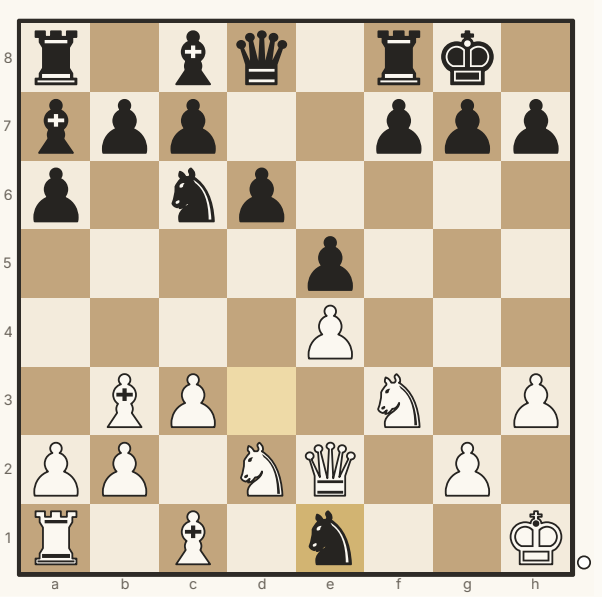
First, the three questions. What does 9...Ng4 attack? It hits f2 and h2. The bishop on a7 also aims at f2, so f2 is attacked twice. What does it no longer defend? The knight left f6, so e4 and h7 are less protected, but nothing hangs right now. What line did it open? The d8-queen now sees all the way to h4 along the diagonal; ...Qh4 or ...Qf6 may come later.

Second, count f2. Attacked twice (knight and bishop), defended once (the king). The rook on e1 does not defend f2. So Black is threatening 10...Nxf2 or 10...Bxf2+, winning a pawn and wrecking the king's cover. You must do something about it.

Third, candidate moves. The natural move is 10.h3, "kicking" the knight. Other ideas are 10.Re2, which defends f2, 10.Rf1, which also defends it, and 10.d4, which blocks the bishop's diagonal toward f2.

Fourth, the blunder check on 10.h3. After h3, what can Black capture or check? The knight can capture on f2 anyway. Now f2 is defended only by the king, and the knight on f2 attacks your queen on d1 and the pawns on d3 and h3:

10. **h3? Nxf2!** 11. **Qe2 Nxd3+** 12. **Kh1 Nxe1**



The "kick" fails. After White recaptures on e1, Black has won a rook and two pawns for the knight, and White's king is exposed.

So 10.h3 is a blunder. What about 10.d4? After 10...exd4 White's centre falls apart and f2 is still weak; the engine rates it clearly worse for White. Now check **10.Re2:**

10. **Re2 Kh8** 11. **h3 Nh6** 12. **d4**

The rook defends f2 a second time. Only then does h3 come, and the knight has to retreat to the edge.

Fifth, the opponent's strongest reply. After 10.Re2, Black's checks: none. Captures: the knight can still take on f2, but now the count has changed.

10... **Nxf2?** 11. **Rxf2 Bxf2+** 12. **Kxf2**

Black has given two minor pieces for a rook and a pawn, usually a poor trade with so many pieces still on the board.

Threats: nothing that works at once, and White's next move h3 will send the knight back. The move passes the check.

That is the whole process: read the opponent's move, count the attacked square, list candidates, and blunder-check the most natural one before playing it. It took a page to write, but at the board it becomes a minute or two.

The five-question pre-move checklist

Everything in this chapter fits into five questions. Ask them, in order, before every move in a slow game:

1. **What is my opponent threatening?** (The three questions about their last move, plus their checks, captures and threats.)
2. **Are any pieces undefended?** Mine and theirs.
3. **Do I have forcing moves?** Checks, captures and threats that win something.
4. **What changes after my candidate move?** What does my move attack, stop defending, and open?
5. **What is my opponent's strongest reply?** Especially every check and capture.

The first two questions keep you safe. The third finds your chances. The last two are the blunder check. If the answer to question five is "they win material" or "they mate", go back to question three and choose again.

> EVIDENCE **Why a routine helps**
>
> Psychology research on expertise, such as Adriaan de Groot's *Thought and Choice in Chess* (1965), found that strong players first spend time orienting themselves in the position before they calculate. A fixed checklist is a beginner's way to do the same orientation deliberately. The specific five questions here are coaching judgment, not a tested protocol.

Which moves deserve the full checklist?

In a slow game, run all five questions every move until it feels routine. Later you can lighten the load: when the position is quiet and nothing has changed, a quick glance at questions 1 and 5 is enough. When pieces are in contact, when the opponent has just captured or checked, or when you are about to capture, run the full list. Those are the moments when games are lost.

Making the scan a habit

Knowing the checklist is easy. Using it every move, in real games, is the hard part. Habits form through repetition in the situation where you need them. A few practical methods help.

Sit on your hands

Literally. When it is your move, put your hands under your legs or flat on the table away from the pieces. Only when you have finished the checklist and the blunder check do you reach for the piece. This simple physical rule stops the most common blunder of all: moving on impulse because you "saw" a move.

Over the board, the **touch-move rule** (Chapter 1) makes this even more important: if you touch a piece, you must move it. Online, set the app to require a confirmation or turn off "premoves" while you build the habit.

Play slow games

You cannot learn a five-question routine in a 3-minute game. Play games of at least 15 minutes per side plus a 10-second increment, and preferably longer: 30 minutes or more. Use the time. If you finish a 15-minute game with 12 minutes left, you did not use the checklist.

> COACHING JUDGMENT **Slow down on purpose**
>
> Coaching judgment: for the next month, make most of your games 15+10 or slower, and after each game check whether you lost material to something the scan would have found. Blitz is fine for fun, but it trains the opposite habit: moving fast without checking.

Say it quietly in your head

Give the routine words. Many players find it useful to say, silently, "Attack? Defence? Line?" after the opponent's move, and "Checks, captures?" before releasing their own. The words become a trigger. After a few hundred games you will notice that you ask the questions without deciding to.

Review your losses with one question

After each game, find the first moment you lost material or missed a free piece. Ask: which question of the checklist would have caught it? Write the answer down. Most players discover that the same one or two questions are the ones they skip. That tells you exactly which habit to work on.

IN YOUR NEXT GAME **Your next game**

Play one slow game (15+10 or longer). After each of your opponent's moves, ask the three questions. Before each of your moves, ask "What can my opponent capture or check after this?" Sit on your hands until you have answered. After the game, count how many moves you checked and how many you skipped.

Where the scan usually fails

Even players who run a scan miss things in the same few places. Knowing them helps you look there on purpose.

- **Backward moves.** Our eyes move forward, toward the opponent. A queen or bishop that can capture by moving backward, toward its own side, is easy to miss. Include backward captures in your scan.
- **Long-range pieces.** A bishop on b2 or a rook on a1 can strike across the whole board. When you list your opponent's captures, trace every open diagonal and file from their bishops, rooks and queen to the end.
- **The king as an attacker.** Kings capture too. A piece next to the enemy king that is not protected is hanging, even if no other piece attacks it.
- **Stopping after the first threat.** When you find one threat, it is tempting to answer it and stop looking. Many moves create two threats at once. Finish the list before you choose a reply.
- **Your own last move.** The line you opened or the piece you stopped defending with your own previous move is just as dangerous as anything your opponent did. Ask the three questions about your own move, too.

BOARD DRILL **Thirty seconds, every move**

In your next few slow games, give yourself a fixed budget: thirty seconds of scanning after every opponent move, even when the move looks harmless. Most moves will reveal nothing. The few that reveal a threat will win you games.

Not every free piece is free

One last warning before the exercises. The scan finds hanging pieces, but sometimes a piece is left hanging on purpose. A **poisoned** piece is one that loses if you take it: the capture walks into a mate, a fork, or a bigger loss.

The defence is the blunder check again. When you see a free piece, do not ask only "can I take it?" Ask "after I take it, what are my opponent's checks and captures?" If the capture puts your king on an open line, removes a defender from your back rank, or lands your queen on a square where it can be trapped, the free piece may be bait. Chapter 6 shows several famous opening traps that work exactly this way.

Practice at the board

Use a physical board or any free analysis board.

1. Set up `r1bqk2r/pppp1ppp/2n2n2/2b1p3/2BPP3/2P2N2/PP3PPP/RNBQK2R b KQkq - 0 5` (the 5.d4 position). Answer the three questions about White's last move out loud. Then list every Black capture and every Black check. Check yourself: d4 attacks c5 and e5; Black's captures are ...exd4, ...Bxd4 and ...Nxe4, and only ...exd4 is clearly sound.
2. Set up `r1bq1rk1/bpp2ppp/p1np4/4p3/4P1n1/1BPP1N2/PP1N1PPP/R1BQR1K1 w - - 4 10` (the worked example). Without looking at the text, play 10.h3 for White and find Black's best reply. Then play 10.Re2 and do the same. Compare.
3. Take any position from one of your own games. Put a coin on every piece that is undefended, for both sides. Count the coins. Then find one forcing move for each side that attacks a coin.
4. **Drill (10 minutes):** play a game against an engine or app set to a low level, but before every move say the five questions aloud. It is slow and that is the point. Stop after 20 moves.

Chapter summary

- Most games below 800 are decided by hanging pieces and unseen threats, not by deep ideas.
- After every opponent move ask: What does it attack? What does it no longer defend? What line did it open?

- Undefended pieces are targets; loose pieces drop off when a forcing move appears.
- Scan checks, captures and threats for both sides, in that order.
- When nothing forcing exists, play a sound normal move rather than inventing an attack.
- Before releasing a piece, blunder-check: after my move, what can my opponent capture or check?
- A "free" piece can be poisoned; check your opponent's replies before you take it.
- Build the habit with slow games, sitting on your hands, and reviewing which question you skipped.

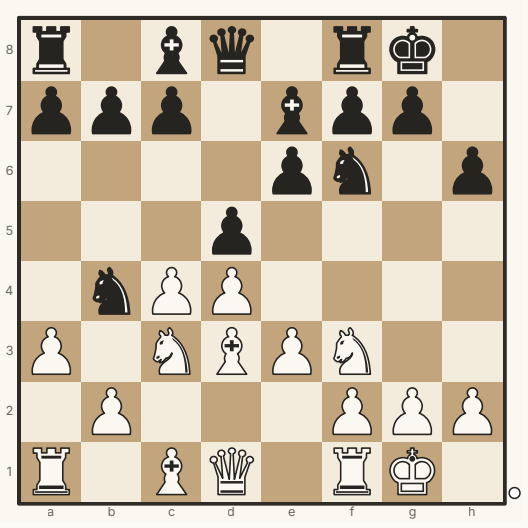
CHEAT CODE **The safety scan**

1. Opponent's move: What does it attack? What does it no longer defend? What line did it open?
2. Undefended pieces: list mine and theirs.
3. Forcing moves: my checks, captures, threats; then theirs.
4. Choose a candidate. What changes after it?
5. Blunder check: after my move, what can my opponent capture or check? If something bad, go back to step 3.

Exercises

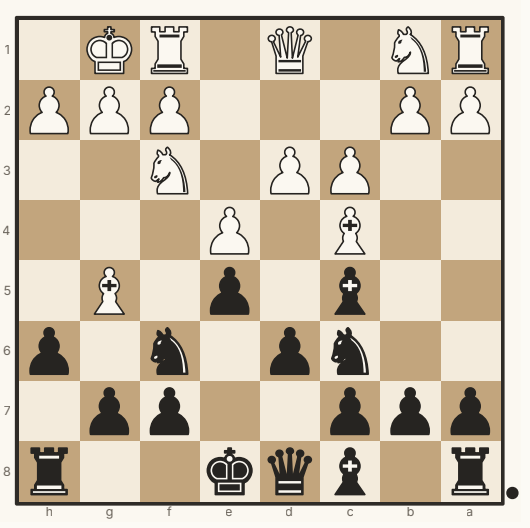
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

4.1 TACTIC



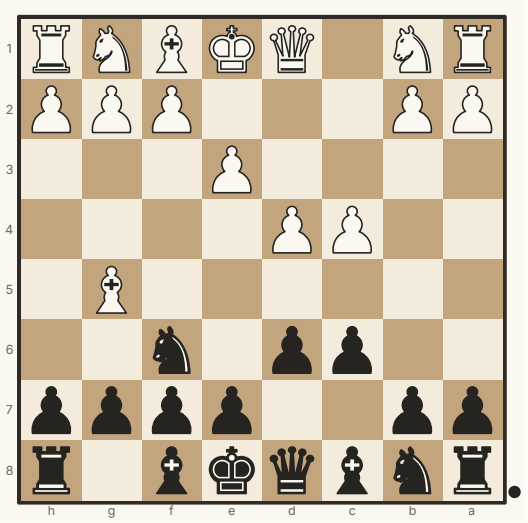
White to move. Black has just played 8... Nb4, attacking your bishop on d3. What does Black's last move allow? Solution p. 640

4.2 TACTIC



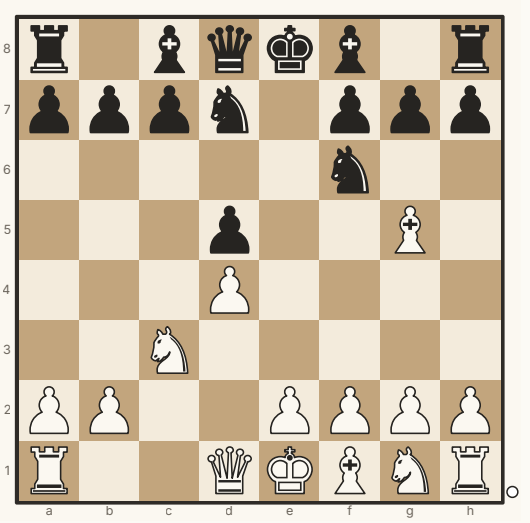
Black to move. White has just played 7.Bg5, pinning your knight. What is wrong with White's move? Solution p. 640

4.3 TACTIC



Black to move. White's bishop on g5 has no defender. Find the forcing move that wins it. Solution p. 640

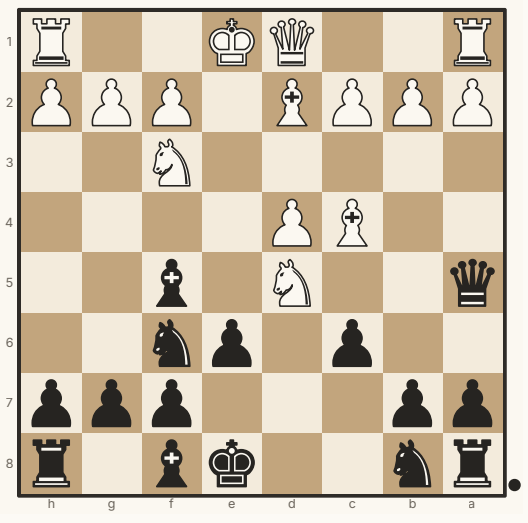
4.4 BLUNDER CHECK



White to move. The d5-pawn is attacked twice and the knight on f6 seems pinned. Is the pawn free? Play the right move.

Solution p. 640

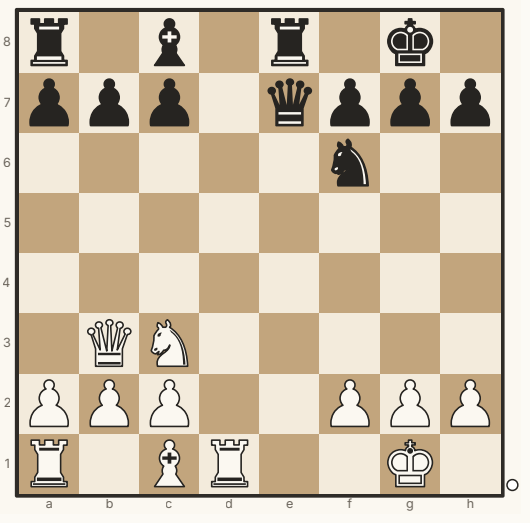
4.5 DEFENCE



Black to move. White has just played 8.Nd5. What does this move attack, and what line did it open? Find the only good reply.

Solution p. 641

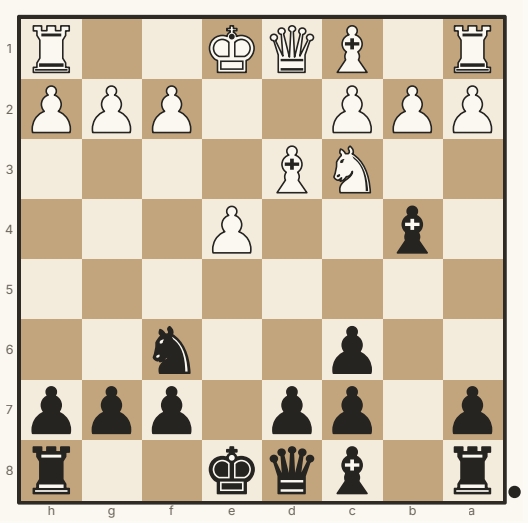
4.6 DEFENCE



White to move. The b7-pawn is undefended. Before you take it, run the safety scan: what is Black threatening? Make a safe move.

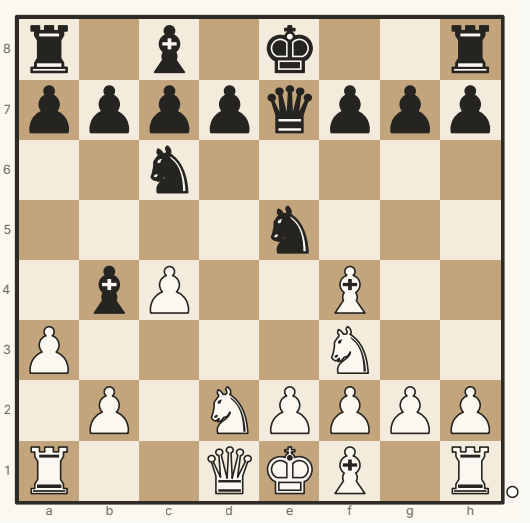
Solution p. 641

4.7 COUNTING



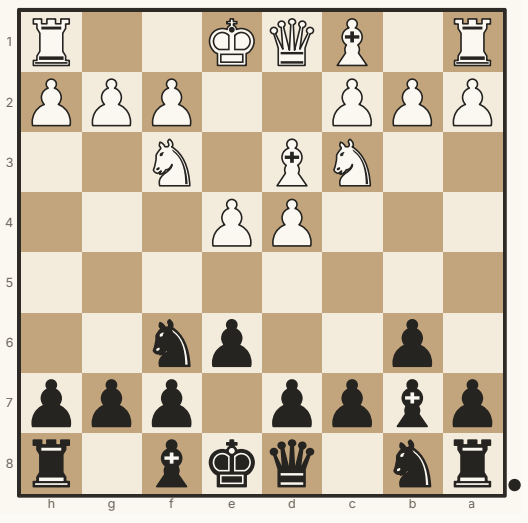
Black to move. Your bishop pins the knight on c3, so the e4-pawn looks free. Is it? Play the right move. Solution p. 641

4.8 BLUNDER CHECK



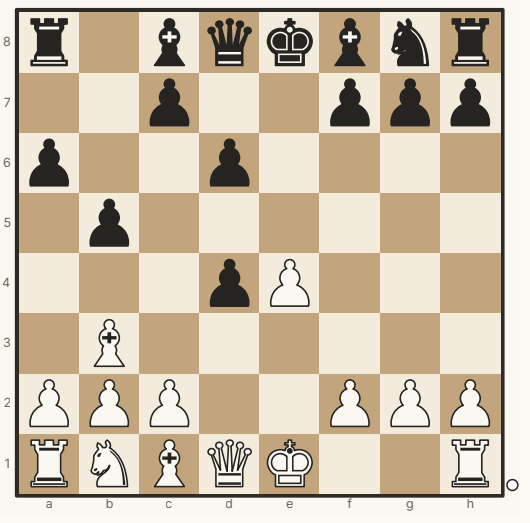
White to move. Black's bishop on b4 is attacked by your a3-pawn. Before you take it, ask: after my move, what can my opponent check or capture? Solution p. 641

4.9 BLUNDER CHECK



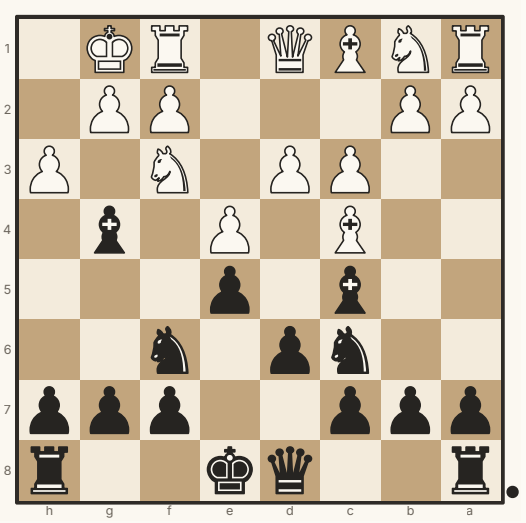
Black to move. You want to develop your f8-bishop. Find a safe square for it. Solution p. 642

4.10 BLUNDER CHECK



White to move. Black has just taken on d4. Recapturing with the queen looks natural. What is Black's strongest reply to it? Choose a safe move. Solution p. 642

4.11 DEFENCE



Black to move. White has just played 7.h3.
What does it attack? Make a safe move.

Solution p. 642

HINTS

4.1 Which of your pieces or pawns attack b4?

4.2 What guards g5?

4.3 Which check also attacks g5?

4.4 After Nxd5 Nxd5 Bxd8, what checks does Black have?

4.5 Look behind the knight: which piece was it blocking?

4.6 List Black's checks.

4.7 Count every defender of e4, not just the pinned one.

4.8 Look at every knight check Black would have if you took the bishop.

4.9 For each square, ask what a White pawn push would hit.

4.10 After Qxd4, which Black pawn moves come with tempo?

4.11 Which of your pieces does the h3-pawn attack?

CHAPTER 5 STAGE 400–600

The Basic Checkmates

Finishing the game when you are far ahead

YOU WILL LEARN TO

- Checkmate with king and queen against king, using the box and knight's-distance method, without stalemating.
- Checkmate with king and rook against king by shrinking the box, using opposition and waiting moves.
- Mate with two rooks (or rook and queen) using the ladder, keeping the pieces out of the king's reach.
- Recognize and avoid the three classic errors - stalemate, endless checks, and forgetting your own king.

> IN YOUR NEXT GAME
>
> When you reach a won ending with a queen or rook against a lone king, do not check; first build a box, bring your king, and before every move near the edge ask "Is it check, and if not, does my opponent still have a legal move?"

You have played well, won your opponent's pieces one by one, and now you have a queen and king against a lone king. Your opponent refuses to resign. Thirty moves later the game is a draw: either you stalemated them, or you ran out of time chasing their king around the board with checks.

This happens constantly at the 400 to 800 level, and every lost half-point hurts. The good news is that the basic checkmates are pure technique. There is no guesswork, just a method you can learn in an evening and use for the rest of your life. This chapter teaches the three mates you must know cold: queen, rook, and two rooks (or rook and queen). Master them on a board until you can do them without thinking.

Why technique beats talent here

When the opponent has only a king, there are no tactics to find and nothing to calculate deeply. The only question is whether you know the procedure. Players who do not know it try to mate by giving checks, because checks feel forcing. But a lone king can almost always step out of a check, and endless checks just push it around the board.

The method is the opposite: **restrict first, check last**. You build a cage around the enemy king, shrink it step by step, bring your own king close, and only then give the final check. Every mate in this chapter follows that pattern.

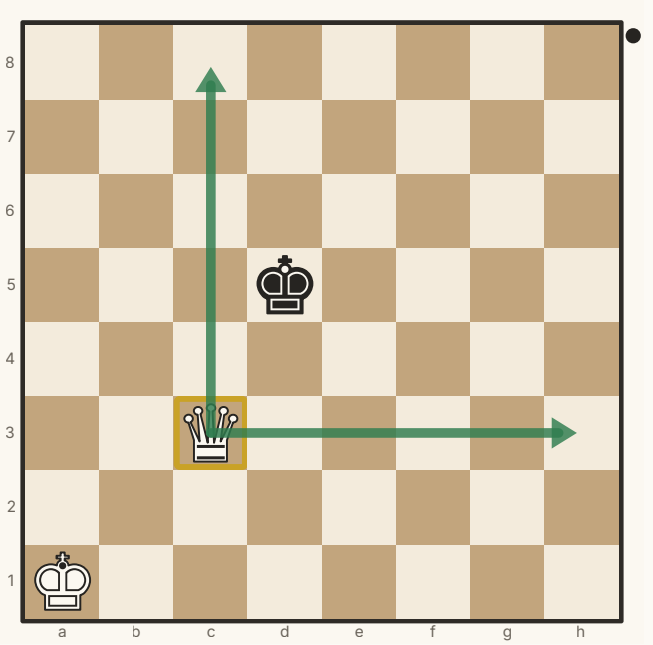
> KEY IDEA **Three rules for every basic mate**
>
> 1. Restrict the king; do not chase it.
> 2. Your king must help. A queen or rook alone cannot mate a king in the middle of the board.
> 3. Near the edge, before every move ask: is it check? If not, does my opponent still have a legal move?

King and queen against king

The queen is so powerful that this mate is fast once you know the method: with best play it never takes more than ten moves from any position. The danger is not failing to mate; it is stalemate.

Step 1: build the box

The queen controls a whole file and a whole rank at once, plus diagonals. Placed correctly, it cuts the board in two and keeps the enemy king inside a rectangle. That rectangle is called the **box**.



The queen on c3 keeps the king off the c-file and the third rank. The king is boxed into the area from d4 to h8, and the box will shrink.

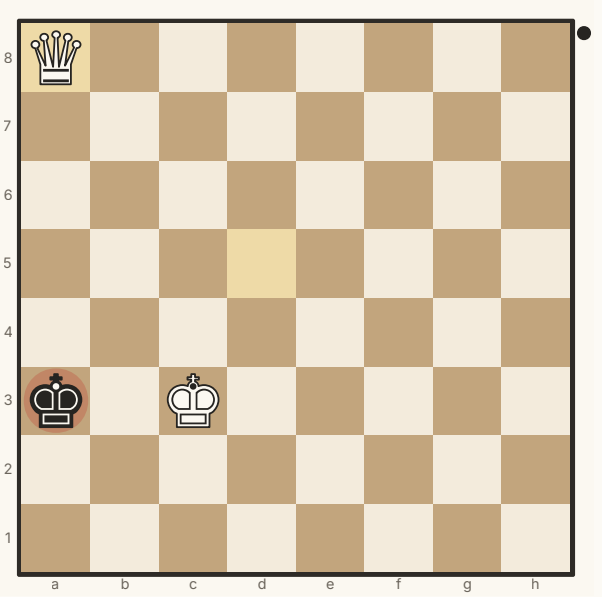
Step 2: shrink the box at a knight's distance

The simplest way to shrink the box is the **knight's-distance method**: after each king move, put your queen a knight's move away from the enemy king. From there the queen covers the squares around the king without standing next to it, and the king is pushed toward the edge. As soon as the king is on the edge and boxed into two files or two ranks, stop moving the queen and bring your king.

1... **Ke4** 2. **Qf6 Kd5** 3. **Qe7 Kd4** 4. **Qe6 Kc5** 5. **Qd7 Kc4** 6. **Qd6 Kc3** 7. **Qd5 Kb4**

After every king move, the queen goes to a square a knight's move away from it. The king is pushed step by step onto the a- and b-files.

8. **Kb2 Ka4** 9. **Kc3 Ka3** 10. **Qa8#**

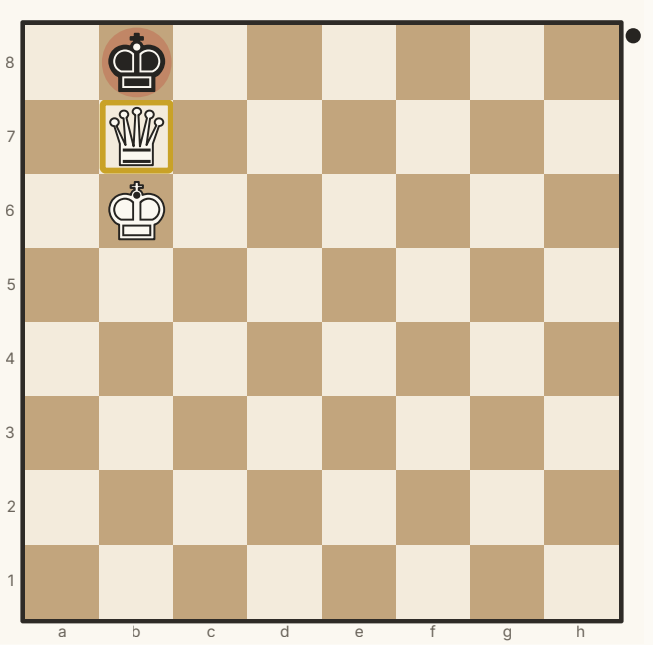


Once the king is trapped on the a- and b-files, the queen stays on d5 and White's king walks up; then the queen mates.

Notice what White did *not* do: no checks until the last move. Each queen move took squares away. When the king reached the a- and b-files, the queen stayed on d5, guarding the fifth rank and the c-file, and White's king came to c3 to cover b2, b3 and b4.

Step 3: the final patterns

There are two common mating patterns with king and queen. In the first, the queen stands directly in front of the enemy king, protected by your king. In the second, the queen checks along the edge while your king covers the escape squares from in front.

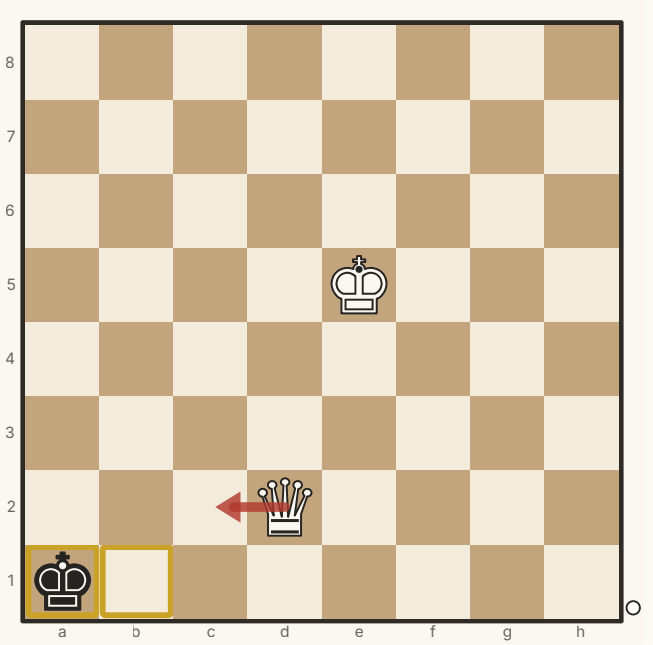


The "kiss": the queen on b7 checks and covers every square around the king, and White's king protects her. This is checkmate.

For the edge pattern, picture the black king on b8 and your king on b6: a queen check from anywhere on the eighth rank (for example Qd8 or Qe8) is mate, because your king covers a7, b7 and c7.

The stalemate trap

Stalemate happens most often at the moment you think the game is over. The enemy king is in the corner, your queen takes away its last squares, and you have forgotten to give check.



The black king is trapped on a1 and b1. 1.Qc2?? or 1.Qb3?? takes away every square without giving check: stalemate. The win requires the king.

The cure is a rule: once the king is on the edge and boxed in, **your queen stops moving** until your king has arrived. A queen that keeps "improving" near a cornered king is a stalemate waiting to happen.

Worked example: the corner

Let us think through the position in the last diagram move by move, the way you should at the board.

First I ask where the black king can go. It is on a1. The queen on d2 controls the whole second rank, so a2 and b2 are covered. The only free square is b1. On b1, the king could go back to a1, or to c1, but the queen covers c1 diagonally. So the king is stuck shuttling between a1 and b1. The box is as small as it gets.

Next I ask what the queen must not do. Any queen move that takes away b1 without giving check is stalemate. Qc2 covers b1 and a2; Qb3 covers a2, b1 and b2. Both leave Black with no legal move. So the queen stays where it is.

Now I bring the king. My king goes toward c3, where it will protect the queen on b2: 1.Kd4 Kb1 2.Kc3 Ka1. Before each king move I check that Black still has a

move. After 1.Kd4 Black has b1. After 2.Kc3, Black's king on b1 has only a1 (c2, b2 and a2 are covered, and c1 is covered by the queen), so it is not stalemate.

Finally I look for checks that mate. With the king on a1 and my king on c3, 3.Qb2# is the kiss mate: the queen on b2 is protected by the king and covers a1's neighbours. I also check: 3.Qd1+ Ka2 would let the king out, so b2 is the right square.

1. **Kd4 Kb1** 2. **Kc3 Ka1** 3. **Qb2#**

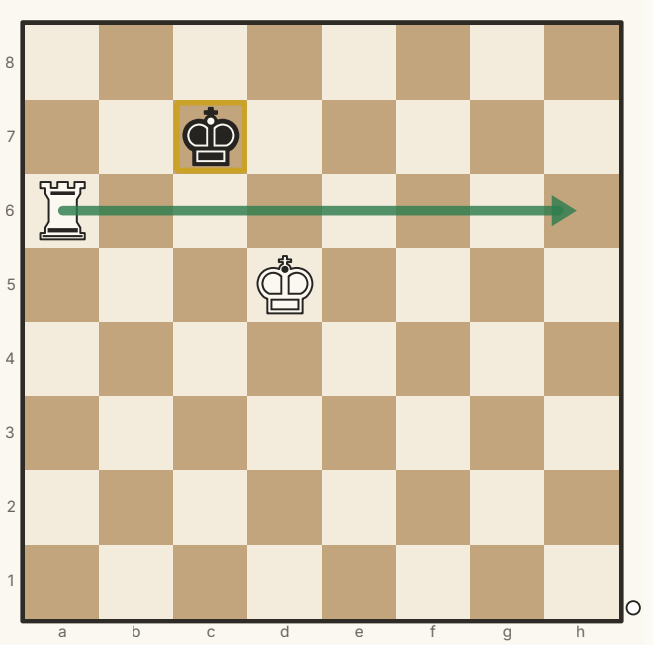
Queen stays still, king comes, then mate. Three moves, no risk.

King and rook against king

The rook mate takes longer, up to 16 moves with perfect play and usually more in practice, but the method is just as reliable. The rook controls only a file and a rank, so your king is needed much earlier, and the key idea is **opposition**.

The box and the cut-off

The rook cuts the enemy king off along a file or rank. In the next diagram the rook on a6 keeps the black king on the seventh and eighth ranks. That is the box. Your job is to make it smaller, one rank or file at a time, until the king is on the edge.



The rook on a6 cuts the black king off on the seventh and eighth ranks. White will shrink this box and bring the king up to face the black king.

Opposition: when to check

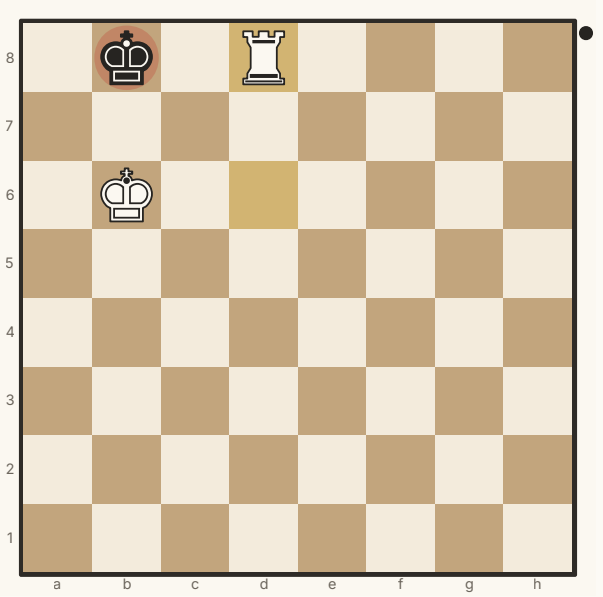
Two kings stand in **opposition** when they are on the same file (or rank) with exactly one square between them. When the enemy king is on the edge and the kings are in opposition, a rook check along the edge is mate, because your king covers every square in front. Before that, a rook check is useful only if it pushes the king back a rank.

If the enemy king refuses to stand opposite yours, you follow it with your king. If following would only repeat, you make a **waiting move** with the rook: a rook move along its line, far from the enemy king, that keeps the box exactly as it is and passes the move to your opponent. Then the king has to give way.

1. **Rd6 Kb7** 2. **Kc5 Kc7**

First the rook shrinks the box to the corner area a7-c8, then White's king walks up to face the black king.

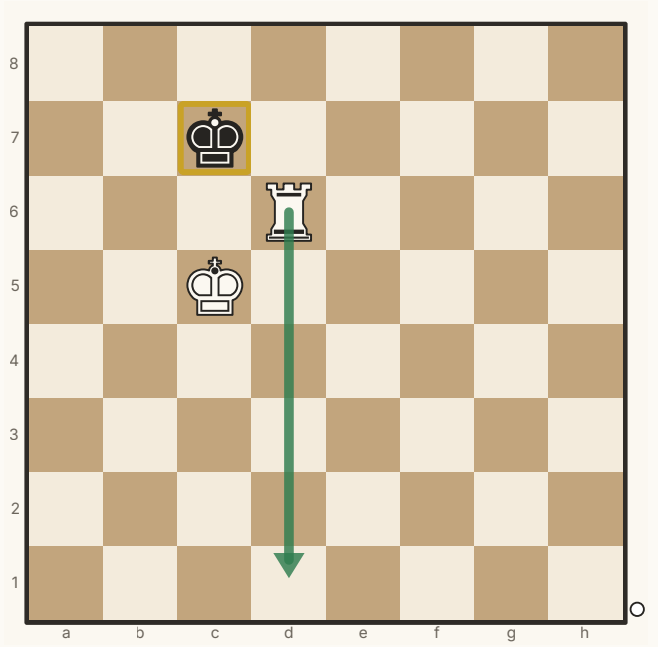
3. **Rd1 Kb7** 4. **Rd7+ Kc8** 5. **Kc6 Kb8** 6. **Kb6 Kc8** 7. **Rd6 Kb8** 8. **Rd8#**



Wait (3.Rd1), check to push the king to the edge (4.Rd7+), follow with the king, wait again (7.Rd6), and mate when the kings face each other.

Follow the plan in the line. 1.Rd6 shrinks the box to the corner area a7-c8. White's king walks up with 2.Kc5. When Black's king stands on c7, the rook simply waits with 3.Rd1, keeping the d-file closed, and Black must step back to b7. Now 4.Rd7+ drives the king to the eighth rank. White's king follows, and after 6.Kb6 Kc8 the second waiting move 7.Rd6 forces 7...Kb8, straight into opposition: 8.Rd8#.

The waiting move



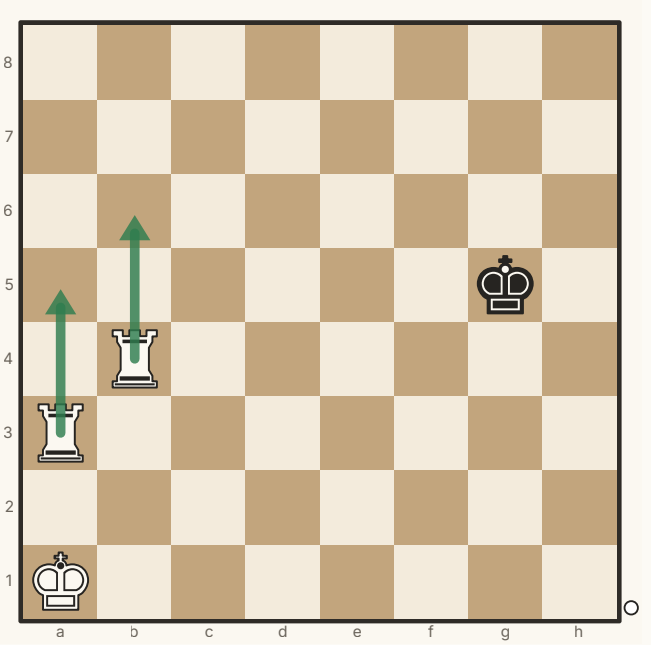
The black king has nowhere useful to go, but it is White's move. 3.Rd1 keeps the d-file closed and hands the move back to Black, who must retreat.

Waiting moves feel strange at first, because they do not seem to do anything. Their job is to change who is on move. In king-and-rook endings, the side that must move often has to give ground, so a quiet rook move far from the enemy king is one of the most useful tools you have.

> COMMON MISTAKE **Checking from too close**
>
> A rook check with the rook next to the enemy king, and no protection, just loses the rook. Keep the rook far away from the enemy king, at the other end of its rank or file. It works just as well from there.

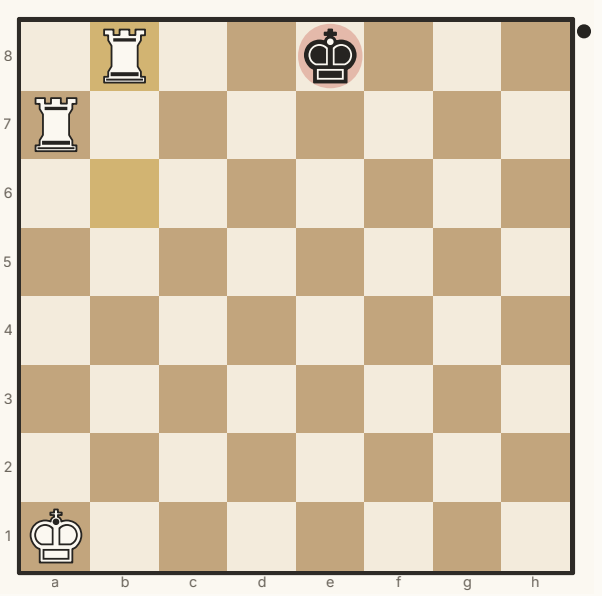
Two rooks, or rook and queen: the ladder

With two heavy pieces you do not need your king at all. The **ladder** (also called the lawnmower) uses the two pieces on neighbouring ranks or files. One guards a rank, the other gives check on the next rank, and they leapfrog each other until the king is on the edge.



Two rooks start the ladder. The rook on b4 guards the fourth rank; the a3-rook will check on the fifth, and so on up the board.

1. **Ra5+ Kg6** 2. **Rb6+ Kf7** 3. **Ra7+ Ke8** 4. **Rb8#**

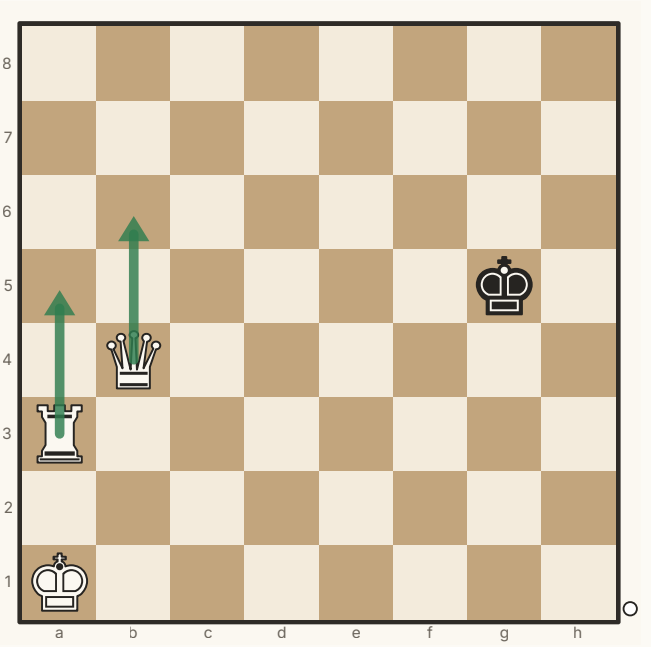


Each rook check pushes the king one rank back, while the other rook guards the rank it just left.

The only extra skill is the switch. If the enemy king walks toward your rooks and attacks one of them, do not check into its arms: move that rook to the far side of the board, along the same rank, and then carry on with the ladder. The same method works with a queen and a rook, or with two queens after a promotion.

Queen and rook together

The ladder works even faster with a queen and a rook, and it is the mate you will reach most often in real games, because players who are far ahead usually still have a rook when they promote a pawn.



Queen and rook on neighbouring files. The queen guards the fourth rank; the rook checks on the fifth.

1. **Ra5+ Kf6** 2. **Qb6+ Ke7** 3. **Ra7+ Ke8** 4. **Qb8#**

The same ladder: rook, queen, rook, queen. Because the queen also covers diagonals, the king has even fewer squares.

A queen and a rook can also mate with the rook guarding and the queen in front of the king, but the ladder needs no thought at all, so use it whenever the king is in the open.

Getting to the basic mate

The basic mates usually appear at the end of a game you have already won on material. How you get there matters too.

- **Trade pieces when you are ahead.** Every exchange brings you closer to a position where your extra material is decisive and your opponent has no counterplay. The opponent's last pieces are their only hope; take them off the board.
- **Promote with care.** When you promote a pawn, check for stalemate first, exactly as Chapter 1 explained. If promoting to a queen would stalemate, promote to a rook instead; king and rook still win.
- **Keep one piece that can mate.** If you have a choice, keep a rook or a queen rather than a lone minor piece. King and bishop, or king and knight, cannot mate a lone king.
- **Do not hurry.** Your opponent may play on in the hope that you stalemate them. Take your time, use the method, and say the stalemate question in your head before every move.

The rook mate as a plan in words

If you forget the details, remember this short plan for king and rook against king:

1. Put the rook a few squares away from the enemy king to cut it off along a file or rank.
2. Bring your king toward the enemy king, staying on the same side of the rook.
3. When the kings stand opposite each other, check along the rank or file behind the enemy king to push it back.
4. When the enemy king attacks the rook, move the rook to the far end of its line.
5. When you cannot make progress, make a waiting move with the rook.
6. On the edge, the check with the kings in opposition is mate.

Other mates: two bishops, bishop and knight

King with two bishops against a king is also a forced win, and king, bishop and knight against a king is a forced win too, though it can take more than 30 moves and needs a special technique. Both are much rarer in real games than the three

mates above. They are worth learning eventually, but not now. Two knights cannot force mate against a lone king, and a single minor piece cannot mate at all.

> COACHING JUDGMENT **What to study when**
>
> Coaching judgment: queen, rook and two-rook mates come up in real games every month at this level. The two-bishop mate is a good exercise later in Stage 2 or 3. Learn the bishop-and-knight mate only when you are above about 1500 and want a challenge; it appears so rarely that the time is better spent on tactics and rook endgames.

How the defender makes it hard

To mate quickly, it helps to know what the defending side is trying to do. A player with a lone king has only two hopes: stalemate, or running you out of moves or time. So the defender does two things.

First, the defending king **heads for the centre**, because a king in the middle of the board is hardest to box in. Every move you spend checking it without shrinking the box helps the defender. Second, near the edge, the defender **looks for stalemate squares**: corners and edge squares where one careless move takes away the last legal move.

That tells you how to play. Use your first moves to cut the king off from the centre, even if it means a quiet move instead of a check. Then, near the edge, slow down: the closer the king gets to the corner, the more carefully you count its legal moves.

Why the knight's distance works

A queen placed a knight's move away from the enemy king covers almost all of the squares around it, but cannot be attacked by it, because the king can only take something on a neighbouring square. Each time the king moves, you can usually find a new knight's-distance square that leaves it fewer choices. You do not need to calculate; you need only repeat the pattern until the king reaches the edge. When it does, stop: the next queen move toward the corner is the one that can stalemate.

When the opponent still has pawns

In real games the lone king often still has a pawn or two. Pawns change two things. A pawn that can move means stalemate is impossible for the moment, so the defender cannot use the stalemate trick while it can still push. But a blocked pawn (one with a piece directly in front of it) has no legal move and does not help the defender at all. Count it as if it were not there when you ask the stalemate question.

The other danger is a pawn that runs toward promotion while you are busy mating. Before you start the technique, check whether any enemy pawn can become a queen. If it can, stop it or take it first; then mate.

The three classic errors

Error 1: stalemate

You have seen it twice already. It happens in queen endings far more than in rook endings, because the queen covers so many squares. The defence is the question in the key box: before every move near a cornered king, ask "Is it check? If not, does my opponent have a legal move?" Also remember that blocked pawns do not help: if your opponent has a pawn that cannot move, it does not count as a legal move.

Error 2: chasing with checks

Checks feel active, but against a lone king they usually let it out of the box. A check is useful only when it pushes the king back (as with the rook in opposition) or mates. If you have given three checks in a row and the king is still in the middle, stop and build a box.

Error 3: forgetting the king

Neither the queen nor the rook can mate a king on its own in the middle of the board, and even on the edge the queen needs help in most positions. Bring your king early. In the rook mate, the king is half the method. In the queen mate, the king is the final step, and the moment to bring it is when the enemy king is trapped on the edge.

> **COMMON MISTAKE** **Running out of time**
>
> In a game with a clock, the basic mates must be fast. If your own time runs out while your opponent has only a king, the game is usually scored as a draw, because a lone king can never checkmate you. All your earlier good play is wasted. Practise until you can mate with king and queen in well under a minute and with king and rook in under two.

Practice at the board

Use a physical board or any analysis board, and play against yourself or a friend who defends as well as possible. An engine set to full strength makes the best defender.

1. Set up `8/8/8/3k4/8/2Q5/8/K7 b - - 0 1`. Play the queen mate with the knight's-distance method. Count your moves; aim for under 12. Then try again from `4k3/8/8/8/8/8/8/3QK3 w - - 0 1`.
2. Set up `8/8/8/4K3/8/8/3Q4/k7 w - - 0 1`. Find every queen move that stalemates. Then mate in three.
3. Set up `8/2k5/R7/3K4/8/8/8/8 w - - 0 1`. Mate with the rook. Say "opposition, check" or "follow" or "wait" out loud before each move.
4. Set up `4k3/8/8/8/8/8/8/R3K3 w - - 0 1` and mate with the rook from scratch. A defender who heads for the centre makes it take longer; aim for under 25 moves.
5. Set up `8/8/8/8/8/6k1/1R6/R6K w - - 0 1` and run the ladder. Then put the black king on d4 instead and see when you must switch the rooks.

Check yourself: if you ever stalemate or lose the rook, start the position again. Do each exercise until you can do it three times in a row without a mistake.

Chapter summary

- Against a lone king, restrict first and check last.
- King and queen: box the king, follow it at a knight's distance to the edge, then freeze the queen and bring your king.
- The final queen patterns are the protected "kiss" and the edge check with your king in front.

- King and rook: cut the king off, bring your king, check when the kings stand in opposition, and use waiting moves with the rook from far away.
- Two rooks (or queen and rook) mate with the ladder; switch the pieces to the far side when the king approaches.
- Before every move near a cornered king: is it check, and if not, does my opponent have a legal move?
- Two bishops and bishop-and-knight mates exist but are rare; learn them later.

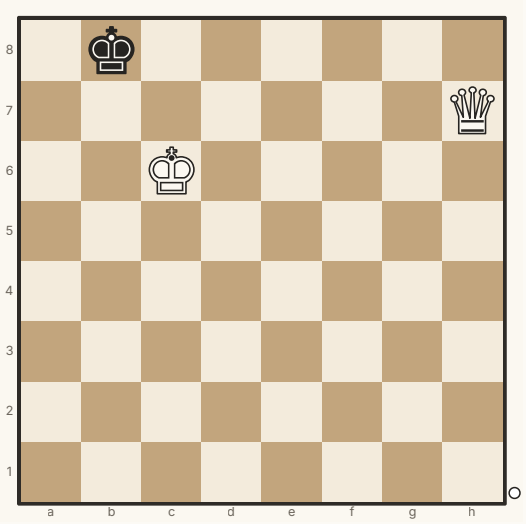
CHEAT CODE **Mating a lone king**

1. Box: cut the king off with your queen or rook.
2. Shrink: queen at a knight's distance; rook with checks only when the kings face each other.
3. Bring your king. Wait with the rook (far away) when needed.
4. Before every move near the edge: is it check? Does the opponent have a legal move?
5. Mate with the protected queen, the edge check, or the ladder.

Exercises

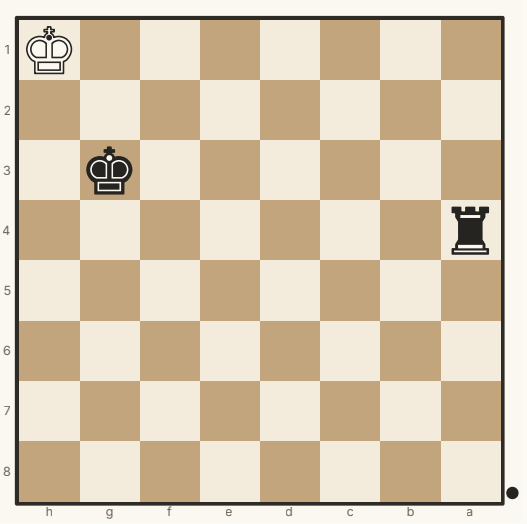
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

5.1 CHECKMATE



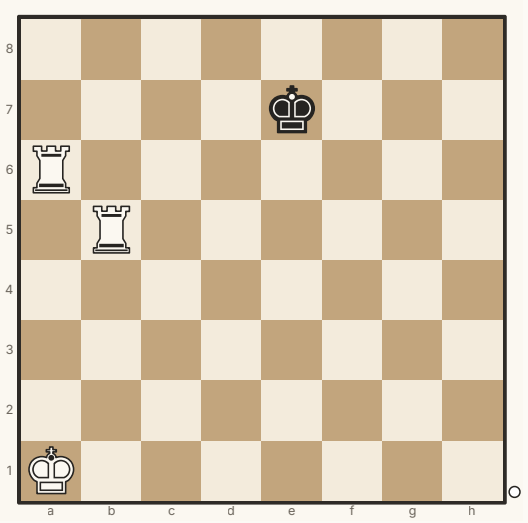
White to move. Mate in one. Solution p. 643

5.2 CHECKMATE



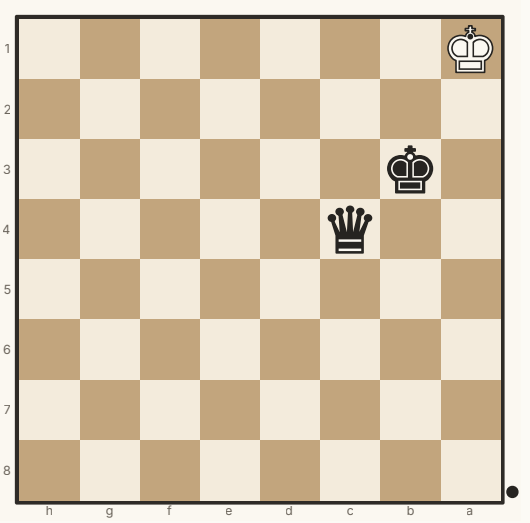
Black to move. Mate in one. Solution p. 643

5.3 CHECKMATE



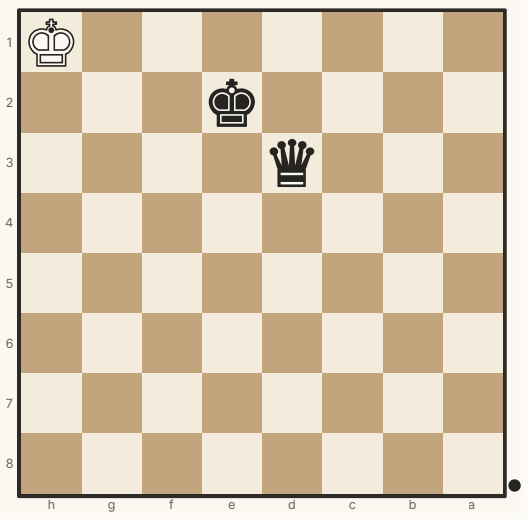
White to move. Use the two rooks to mate in two. Solution p. 643

5.4 CHECKMATE



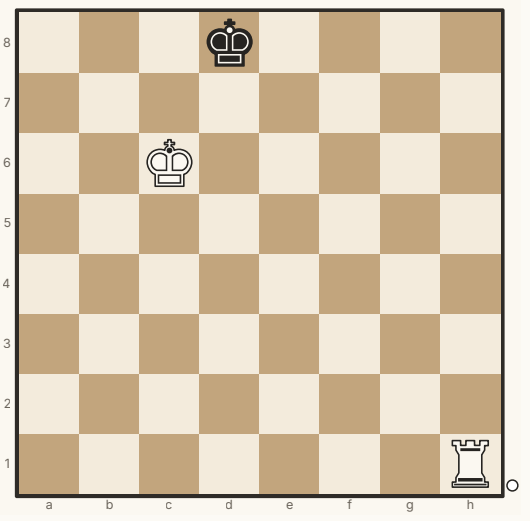
Black to move. Mate in one. Careful: one natural queen move only stalemates.
Solution p. 643

5.5 CHECKMATE



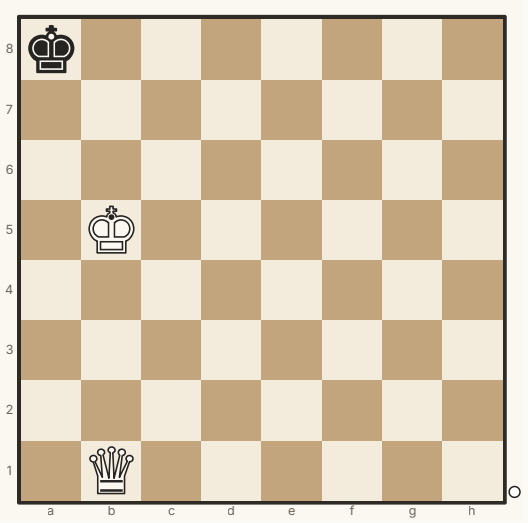
Black to move. Mate in two. Solution p. 643

5.6 CHECKMATE



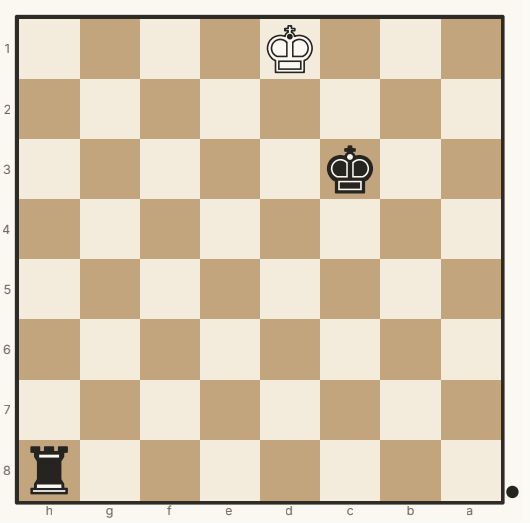
White to move. Mate in two with king and rook. Solution p. 644

5.7 CHECKMATE



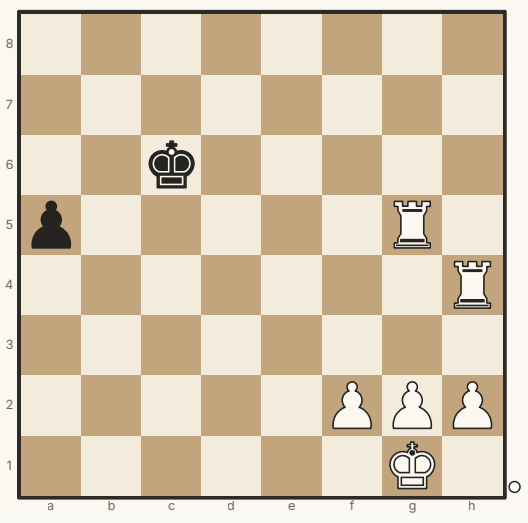
White to move. Mate in two. Do not stalemate. Solution p. 644

5.8 CHECKMATE



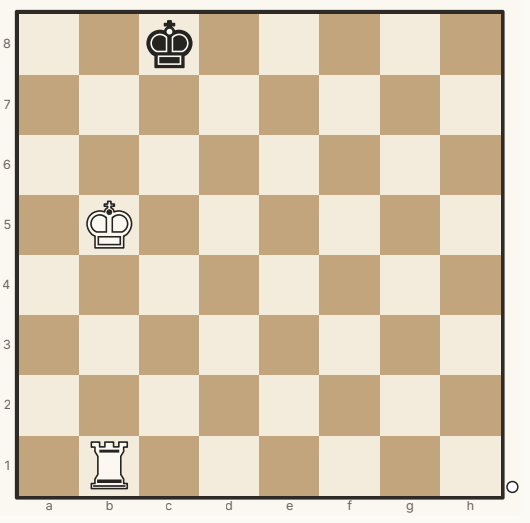
Black to move. Mate in two with king and rook. Solution p. 644

5.9 CHECKMATE



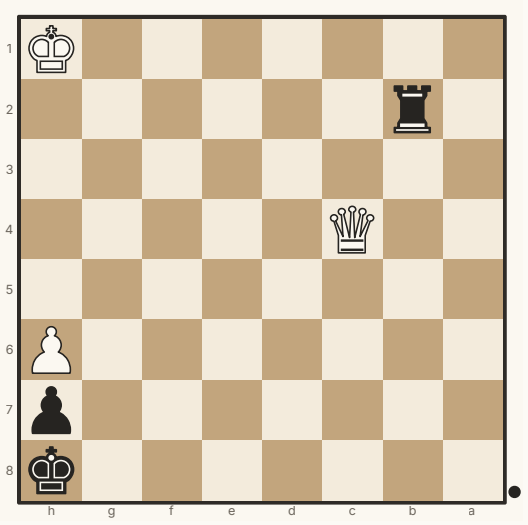
White to move. Black's king is in the open. Mate in three with the rook ladder.
Solution p. 644

5.10 CHECKMATE



White to move. Mate in three with king and rook. Solution p. 645

5.11 DEFENCE



Black to move. You are lost on material and White threatens Qc8#. Your king has no legal moves. Find the move that saves the game. Solution p. 645

HINTS

5.1 Which square next to the black king is protected by your king?

5.2 Check along the edge the king is standing on.

5.3 One rook guards a rank while the other checks on the next one.

5.4 Before you play a queen move next to a cornered king, ask: is it check?

5.5 Bring the king, not the queen.

5.6 Which file must the black king not reach?

5.7 Your king must protect the square in front of the cornered king, without taking away its last move.

5.8 Cut off the file the king wants to run to.

5.9 One rook guards, the other checks, one rank at a time.

5.10 First bring your king opposite the enemy king's file, one rank below.

5.11 Which piece is the only one keeping you from being stalemated?

CHAPTER 6 STAGE 400–600

Opening Principles and Early Traps

Get your pieces out, get your king safe, and do not fall for the classics

YOU WILL LEARN TO

- Apply the core opening principles - centre, development, castling, connected rooks - and know their common exceptions.
- Avoid the typical beginner errors - early queen moves, moving one piece repeatedly, pointless pawn moves, especially with the f-pawn.
- Defend against Scholar's Mate and punish the early queen with gains of tempo.
- Recognize Fool's Mate, Légal's Mate and the Fried Liver danger, and punish early f-pawn moves.

> **IN YOUR NEXT GAME**
>
> In the first ten moves of your next game, before each move ask "Does this move develop a piece, fight for the centre, or get my king safe?" and if your opponent's queen or bishop is aiming at f7 (or f2), check that it is defended before you do anything else.

A typical game at the 400 level begins like this. White plays 1.e4, Black answers 1...e5, and White brings the queen out on move two to attack f7. Black does not notice, develops a knight to the "natural" square, and is checkmated on move four. In the next game the roles are reversed: the player who has learned to attack f7 brings the queen out early, gets it chased around the board, and loses because no other piece has moved.

The opening is not about memorising long lines. At this level it is about a handful of principles that keep your position healthy, and a handful of traps you must know from both sides. This chapter gives you both. Chapter 12 and Chapter 13 then build your actual repertoire on these foundations.

The three jobs of the opening

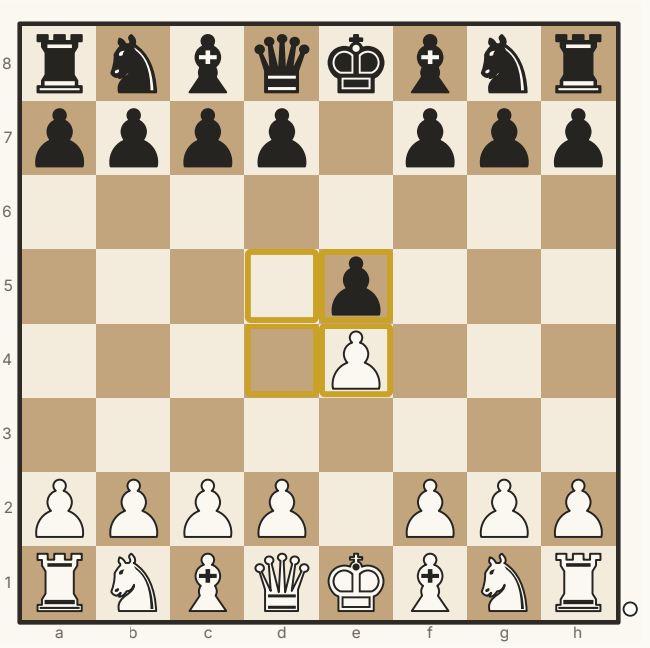
Every sound opening plan does the same three jobs in the first ten or so moves:

1. **Fight for the centre.** The four central squares, d4, d5, e4 and e5, are the most valuable squares on the board.
2. **Develop your pieces.** Bring the knights and bishops off the back rank to active squares.
3. **Make your king safe.** Castle, usually on the kingside, and connect your rooks.

If a move does one of these jobs, it is probably a reasonable opening move. If it does none, ask why you are playing it. That single question would save most beginners dozens of games.

Control the centre

Pieces in the centre reach more squares, and pieces that reach more squares are stronger. A knight on e4 attacks eight squares; a knight on a1 attacks two. The centre is also the crossroads of the board: pieces placed there can switch quickly to either wing.



After 1.e4 e5 each side has a pawn in the centre. The four central squares are the prize; most good opening moves fight for them.

You fight for the centre in two ways: by **occupying** it with pawns (1.e4, 1.d4) and by **controlling** it with pieces that aim at it (a knight on f3 hits e5 and d4). In the repertoire of this book you start with 1.e4 as White and answer 1.e4 with 1...e5 as Black, which puts a pawn in the centre at once.

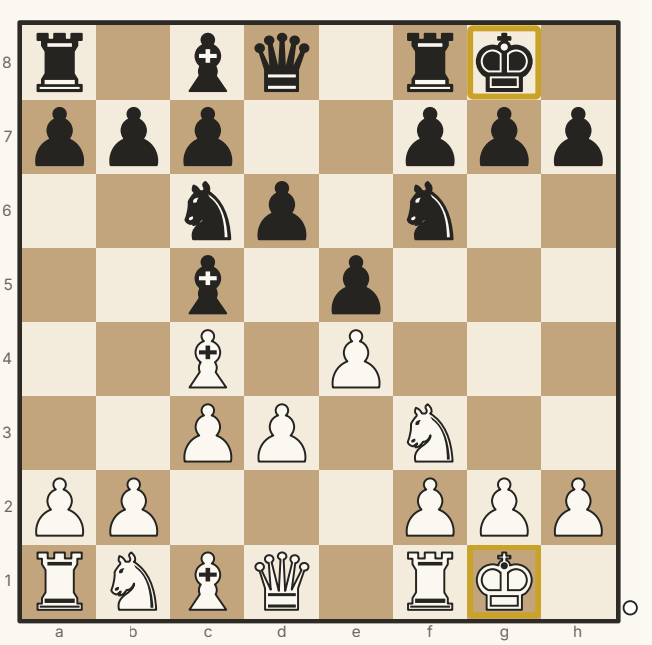
Develop: knights usually before bishops

Development means moving a piece from its starting square to a square where it does useful work. In the opening, try to move each minor piece once, to a good square, before you move any piece twice.

A strong guideline is **knights before bishops**. Knights have one or two natural squares (f3 and c3 for White, f6 and c6 for Black), and they are needed early to fight for the centre. Bishops have more choices, and it often pays to wait a move to see where the opponent's pieces go before deciding. This is a guideline, not a law: in the Italian Game, for example, White plays 3.Bc4 before the queen's knight comes out, because the bishop has an obvious good square.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bc4 Bc5** 4. **c3 Nf6** 5. **d3 d6** 6. **O-O O-O**

The Giuoco Pianissimo from your White repertoire. Both sides develop, support the centre, and castle within six moves.



After six moves each side has developed two minor pieces, put a pawn in the centre, and castled. The remaining job is to develop the queen's knight and bishop and connect the rooks.

Castle early and connect your rooks

The king in the centre is in danger as soon as the centre opens. **Castling** moves the king to the side, behind a wall of pawns, and brings a rook toward the centre in one move. Aim to castle within the first ten moves in most games. Kingside castling is usually faster because only two pieces need to move out of the way.

Once you have castled, leave the three pawns in front of your king alone for as long as you can. Each of those pawn moves creates a hole that enemy pieces can use later. A common beginner habit is to castle and then push the h-pawn or g-pawn "to make room"; usually this only weakens the shelter you just built. The exception is a quiet h3 (or ...h6) played to stop an annoying pin or to give the king a flight square against back-rank mates, which Chapter 4 showed can matter a great deal.

Queenside castling (O-O-O) is also legal and sometimes good, but it takes longer to prepare, because three pieces must leave the back rank, and the king on c1 (or c8) is often more exposed. Until you have more experience, castle kingside unless you have a clear reason not to.

After castling, finish development until your two rooks see each other along the back rank, with nothing between them. Then the rooks are **connected**: they defend each other and can move to the central files, where they will be needed when the position opens. Connecting the rooks is a good sign that your opening is over.

> KEY IDEA **A ten-move checklist**
>
> By about move 10, try to have: 1. At least one pawn in the centre, or pieces controlling it. 2. All four minor pieces developed. 3. Your king castled. 4. Your rooks connected. If you are far from this, your opponent probably is not, and that is a problem.

Why f7 and f2 are the weak points

At the start of the game, every pawn in front of the king is defended by at least one other piece, except one. The pawn on f7 is defended only by the king (and the same is true of f2 for White). That makes it the natural first target for an early attack: a queen on h5 or f3, a bishop on c4 and a knight on g5 can all aim at it.

Once you castle kingside, a rook stands on f8 (or f1) and defends the f-pawn as well, and the danger mostly disappears. This is one more reason to castle early: it fixes the weakest square in your position. Until you have castled, treat f7 like a piece that must always be defended at least as many times as it is attacked.

Counting development

A simple way to judge an opening position is to count developed pieces. Count each knight and bishop that has left its starting square, add one if the king has castled, and compare the totals. If you are two or more moves behind, avoid opening the position and finish developing first. If you are two or more moves ahead and the enemy king is still in the centre, look for ways to open lines: that is when attacks succeed.

This count is a rough tool, not an exact rule. A developed piece on a bad square, such as a knight on the rim at a3 or h6, is worth less than one on a good square. But at this level the count is usually right about who is better.

What not to do in the opening

The principles above tell you what to do. Most opening disasters at the 400–800 level come from breaking one of the "don'ts" below.

Don't move the same piece twice without a reason

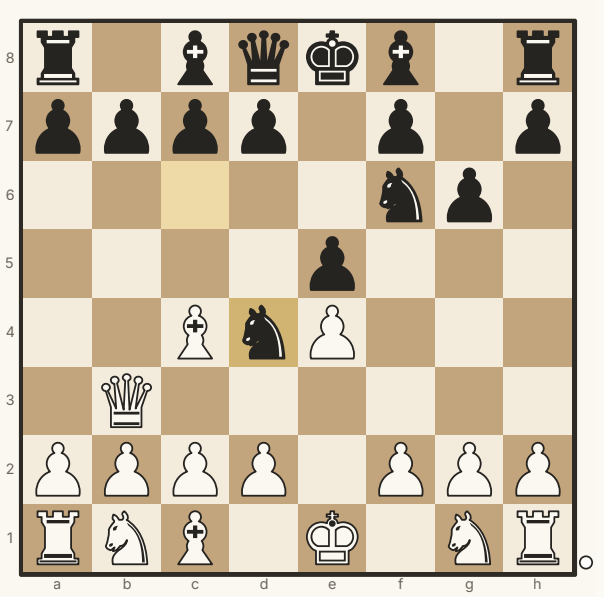
Every move spent moving a piece that is already developed is a move not spent developing another. Two or three wasted moves in the opening can leave you so far behind that your opponent's attack arrives before your pieces do.

There are good reasons to move a piece twice: to capture something, to escape an attack, or to answer a direct threat. "I want to attack something" is not usually a good reason in the first ten moves, because the attack is often easy to meet and the attacker then has to retreat, losing more time.

Don't bring the queen out early

The queen is your most valuable piece. If it comes out early, your opponent can attack it with pawns and minor pieces, developing with **tempo** (gaining time: making a useful move that also forces your opponent to react). Each time the queen has to move, your opponent gets a free developing move.

1. **e4 e5** 2. **Qh5?! Nc6** 3. **Bc4 g6** 4. **Qf3 Nf6** 5. **Qb3? Nd4**



White's queen has moved three times. Black has developed two knights with tempo, and 5...Nd4 attacks the queen and threatens ...Nxc2+.

Notice that each Black move developed a piece and, at the same time, attacked or blocked the queen. That is how you punish an early queen: not by hunting it with your own queen, but by developing with gains of tempo.

> COACHING JUDGMENT **When is an early queen move fine?**
>
> Coaching judgment: an early queen move is fine when it wins material safely, when it answers a direct threat, or when it is part of a known sound opening (such as the Scandinavian, 1.e4 d5 2.exd5 Qxd5). At the 400–800 level, the most common early queen move is the attempt at Scholar's Mate below, and it is much less dangerous once you know how to meet it.

You will meet the same idea in a more respectable form in the Centre Game, 1.e4 e5 2.d4 exd4 3.Qxd4. White's queen recaptures in the centre, and Black develops with tempo at once:

1. **e4 e5** 2. **d4 exd4** 3. **Qxd4 Nc6** 4. **Qe3 Nf6** 5. **Nc3 Bb4** 6. **Bd2 O-O** 7. **O-O-O Re8**

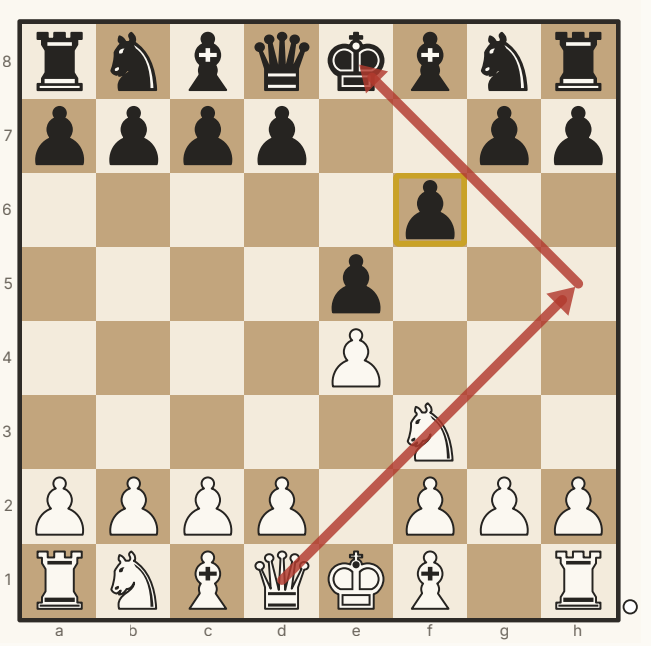
Black's knight hits the queen, then the other pieces come out with threats against e4. Black has castled and put a rook on the open e-file while White is still sorting out the queen.

Nothing dramatic has happened: no piece has been lost. But Black has gained a lead in development simply by attacking the queen with pieces that wanted to move anyway. That is the normal reward for punishing an early queen, and at this level it often turns into a winning attack a few moves later.

Don't make pointless pawn moves, especially with the f-pawn

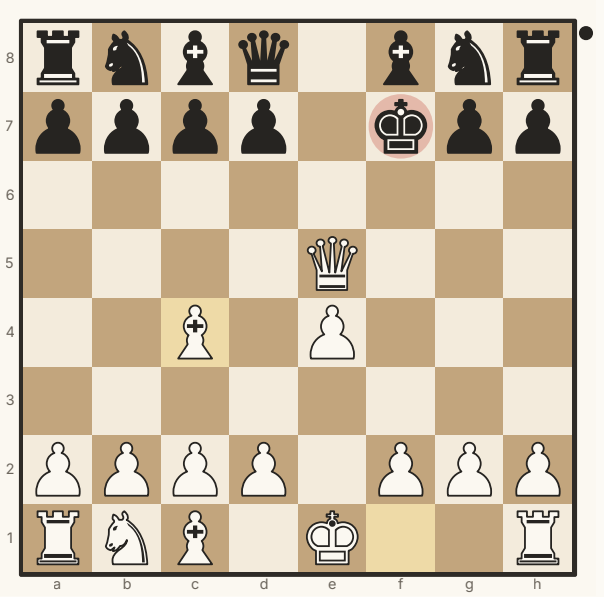
Pawn moves cannot be taken back. In the opening, move the centre pawns (usually the e- and d-pawns) to fight for the centre and open lines for the bishops. Moves like a3, h3, a6 or h6 do not develop anything, and moving the f- or g-pawn early can open lines to your king.

The f-pawn is a special case. On f7 (or f2) it shields the king along the e8-h5 diagonal (e1-h4 for White). Moving it early, especially to f6 or f3, opens that diagonal and takes away the knight's best square.



2...f6? defends e5, but it opens the diagonal from h5 to the black king and takes the f6-square from the knight. White can punish it at once.

1. **e4 e5** 2. **Nf3 f6?** 3. **Nxe5! fxe5?** 4. **Qh5+ Ke7** 5. **Qxe5+ Kf7** 6. **Bc4+**



After 3...fxe5? 4.Qh5+ the black king has to walk into the open, and White wins back the knight with a huge attack.

The best answer to 3.Nxe5 is 3...Qe7, attacking the knight and the e4-pawn:

3... **Qe7** 4. **Nf3** **Qxe4+** 5. **Be2**

Black has won the pawn back, but White is well ahead in development and Black's king and queen stand awkwardly in the centre.

Either way, 2...f6 has made Black's life hard.

Exceptions are real

Opening principles are strong guidelines, not laws. Some exceptions you will meet:

- **Capturing material** often justifies moving a piece twice or bringing the queen out, as long as you check the opponent's replies first (Chapter 4).
- **Useful small pawn moves** exist: h3 (or ...h6) to stop a pin by an enemy bishop on g4 (or g5), c3 to prepare d4, a6 to stop a piece coming to b5.
- **Established gambits** such as the King's Gambit (1.e4 e5 2.f4) move the f-pawn early on purpose, for concrete reasons. They are not for now.
- **Defending against a direct threat** always comes before any principle.

The principle is the default; concrete reasons override it. If you break a principle, you should be able to say exactly what you gain.

When your opponent leaves the book

Sooner or later your opponent will play a move you have never seen, often on move two or three. This is where the principles earn their keep. You do not need to know the "theory" of 1.e4 e5 2.Nf3 h6 or 1.e4 e5 2.Nc3 a6. Ask the same questions as always: what does the move attack, what does it no longer defend, and what line did it open? If nothing is attacked, answer with a good developing move toward the centre.

Strange opening moves are usually not traps. They are usually just slow. Punish slowness by developing quickly and castling, and your opponent will be the one with problems by move ten.

Traps every beginner must know

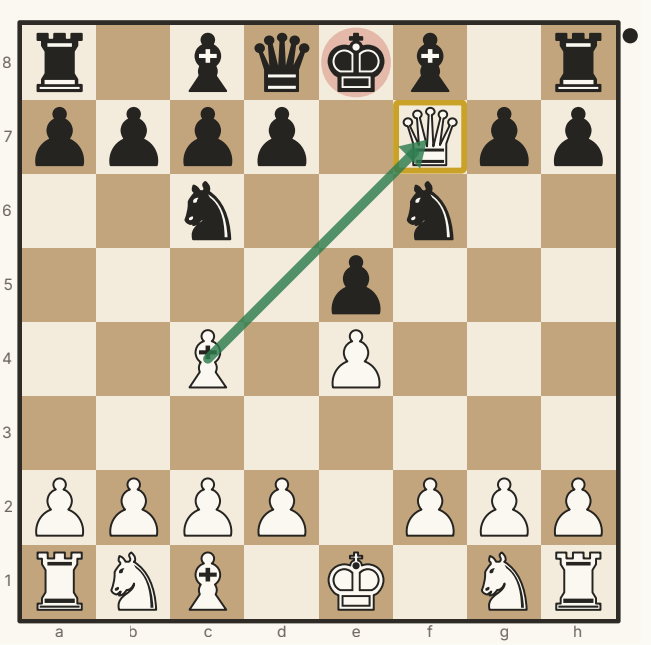
The traps below are the ones you will actually meet. Learn each from both sides: how to avoid it, and how to punish an opponent who falls into it.

Scholar's Mate

White brings the queen and the light-squared bishop together against f7, which at the start is defended only by the king.

1. **e4 e5** 2. **Qh5 Nc6** 3. **Bc4 Nf6??** 4. **Qxf7#**

Scholar's Mate. 3...Nf6?? attacked the queen but did nothing about f7.

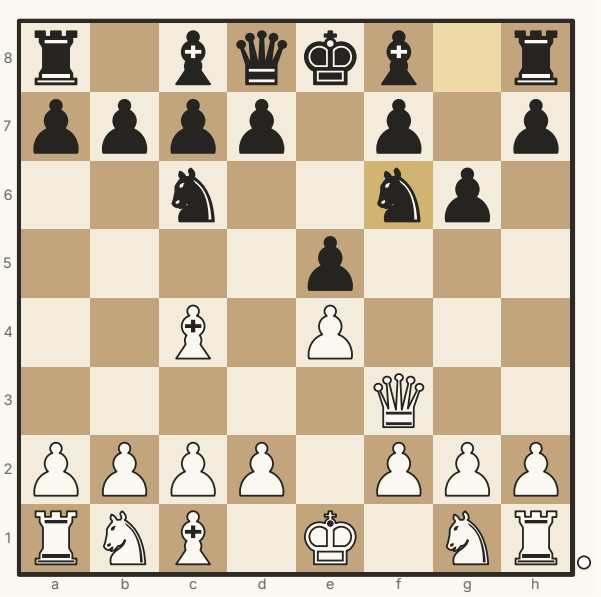


Checkmate. The queen on f7 is protected by the bishop on c4, and the king cannot take it or escape.

How to defend. The rule is simple: when an enemy queen and bishop aim at f7, defend f7 or block the line before you do anything else. Which defence is right depends on where the queen is:

- **Queen on h5** (attacking f7 along the diagonal and e5 along the rank): 3...g6! blocks the queen's line and attacks it. 3...Qe7 also defends both f7 and e5. Do not play 3...Nf6??, because the knight does not block the h5-f7 diagonal.
- **Queen on f3** (attacking f7 along the f-file): now 3...Nf6! is exactly right, because the knight on f6 blocks the f-file and develops to its best square.

1. **e4 e5** 2. **Qh5 Nc6** 3. **Bc4 g6** 4. **Qf3 Nf6**

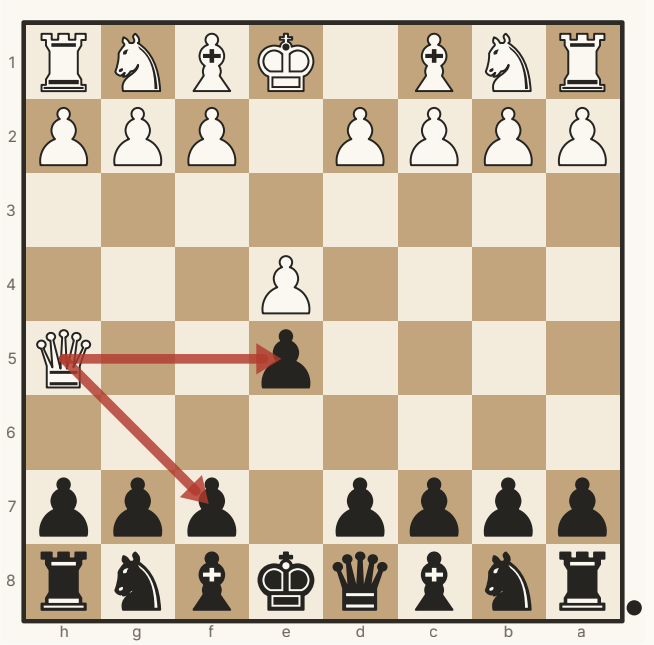


The correct defence. 3...g6 gains time on the queen, and when it goes to f3, 4...Nf6 blocks the f-file and develops. Black is already comfortable.

From here White's queen is a target. If White carries on with 5.Qb3, you have seen the answer: 5...Nd4!, developing with a threat.

Worked example: meeting 2.Qh5

Let us walk through Black's thinking after 1.e4 e5 2.Qh5, using the five-question checklist from Chapter 4.



White's queen attacks e5 and eyes f7. What does Black need to do?

What is my opponent threatening? I ask what the queen attacks. Along the fifth rank it hits my e5-pawn, which has no defender, so Qxe5+ is a threat: a capture with check. Along the diagonal it looks at f7, but f7 is defended by my king and only one White piece attacks it, so that is not yet a threat. If the bishop comes to c4, it will be.

Are any pieces undefended? My e5-pawn. White's queen on h5 is undefended too, but I cannot attack it with a pawn right now.

Do I have forcing moves? Nothing useful: no checks, no good captures.

Candidate moves. I want a move that defends e5 and develops. 2...Nc6 does both. 2...d6 defends e5 but blocks my f8-bishop. 2...Nf6?? develops but drops the pawn with check: 3.Qxe5+.

What is my opponent's strongest reply to 2...Nc6? Probably 3.Bc4, threatening Qxf7#. Then I will block with 3...g6, attacking the queen. After 4.Qf3, threatening mate again, 4...Nf6 blocks and develops. Every one of my moves develops and answers a threat, and White's queen has moved three times. I choose 2...Nc6.

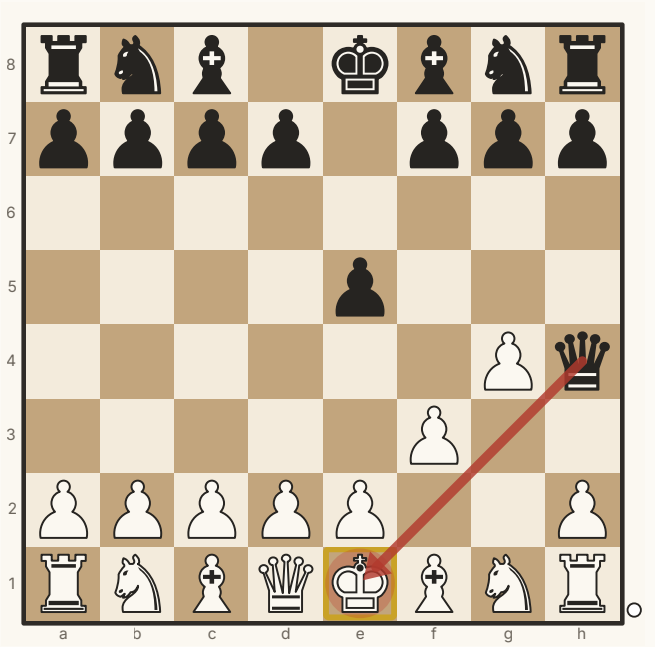
That is the whole skill: defend the real threats in time, and let your opponent's early queen give you free moves.

Fool's Mate

The fastest possible checkmate needs the loser's cooperation: two early pawn moves that open the diagonal to the king.

1. **f3?! e5** 2. **g4?? Qh4#**

Fool's Mate. The f- and g-pawns have moved, and the queen mates along the e1-h4 diagonal.



The final position of Fool's Mate. No White piece can block on g3 or capture the queen.

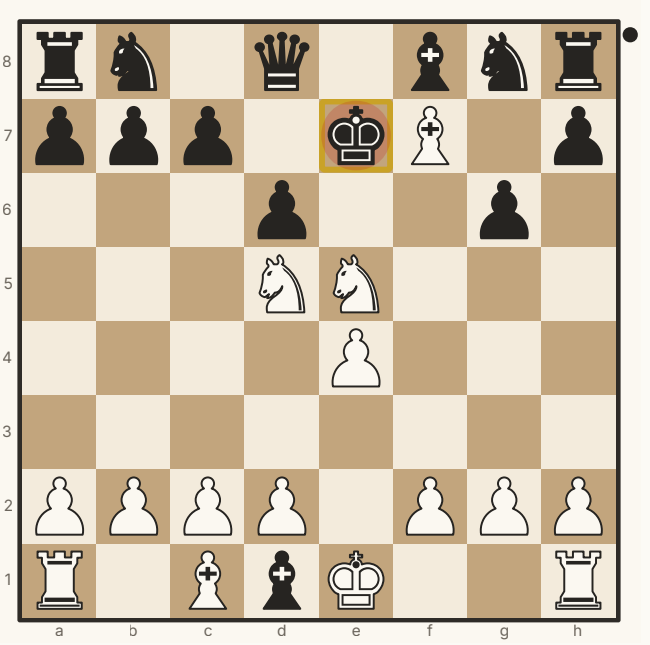
You will rarely see the exact Fool's Mate, but its pattern is common: if your f- and g-pawns have both moved and the diagonal to your king is open, an enemy queen or bishop can check you with no blocking square. The mirror version, with Black playing ...f6 and ...g5, is punished by Qh5#.

Légal's Mate

This trap, named after the 18th-century French player Légal, shows that a pinned piece is not always pinned. The bishop on g4 "pins" the knight on f3 to White's queen, but White can move the knight anyway, because the queen sacrifice leads to mate.

1. **e4 e5** 2. **Nf3 d6** 3. **Bc4 Bg4** 4. **Nc3 g6?** 5. **Nxe5! Bxd1??** 6. **Bxf7+ Ke7** 7. **Nd5#**

Légal's Mate. After 5...Bxd1?? three White minor pieces mate the king that has not castled. Black's best was 5...dxe5 6.Qxg4, only a pawn down.



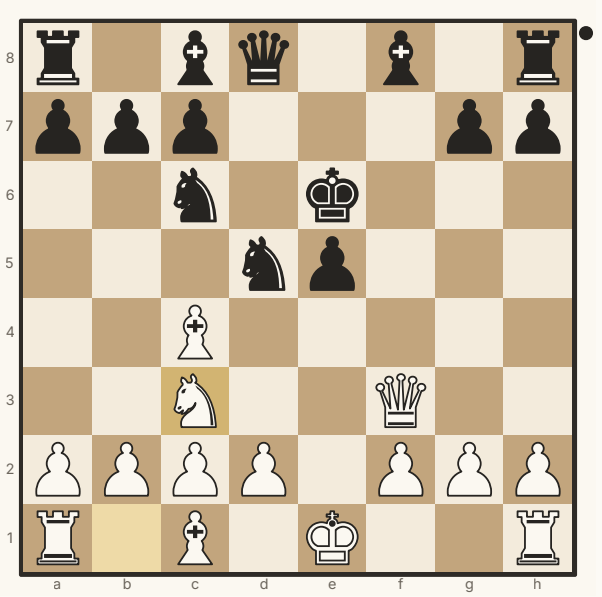
The final position. The d5-knight gives check and covers f6; the e5-knight protects the bishop and covers d7; the bishop covers e6 and e8.

The lessons go both ways. As Black, when an opponent "blunders" the queen, ask what their pieces can do to your king after you take it. As White, remember that a pin against the queen (not the king) is only a relative pin; the pinned piece can still move if the result is good enough.

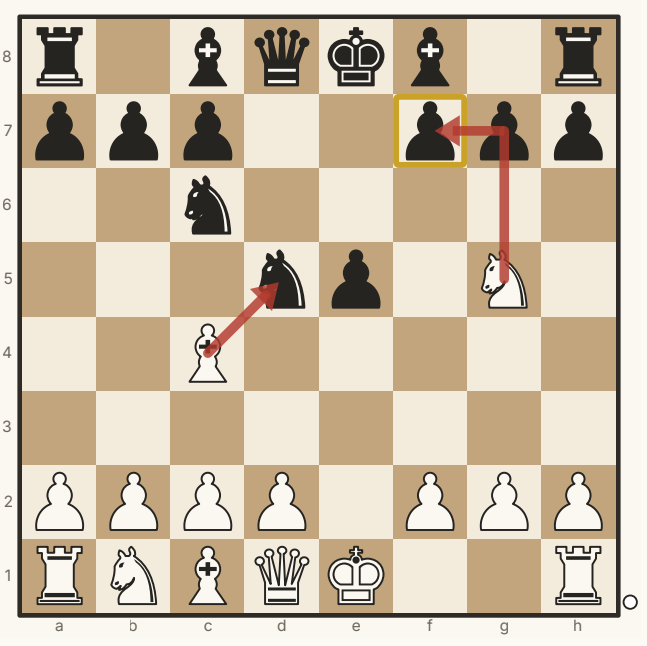
The Fried Liver: why 3...Nf6 4.Ng5 is dangerous

In the Two Knights Defence, 1.e4 e5 2.Nf3 Nc6 3.Bc4 Nf6, White can attack f7 at once with 4.Ng5. Black's best reply is 4...d5, blocking the bishop. After 5.exd5 the natural recapture is a mistake:

1.e4 e5 2.Nf3 Nc6 3.Bc4 Nf6 4.Ng5 d5 5.exd5 Nxd5? 6.Nxf7! Kxf7 7.Qf3+ Ke6 8.Nc3



The Fried Liver Attack. Black's king has been dragged to e6 in the middle of the board, and White's pieces pile up against the pinned knight on d5.



After 5...Nxd5? the knight blocks the bishop's diagonal only temporarily. 6.Nxf7! sacrifices the knight to pull the king into the open.

Theory says Black should play 5...Na5!, attacking the bishop instead of recapturing, and the resulting positions are sharp but fine for Black. That is a lot to learn and easy to get wrong at the board. For this reason the Black repertoire in this book avoids the whole line: after 1.e4 e5 2.Nf3 Nc6 3.Bc4 you will play 3...Bc5, the Giuoco Piano, instead of 3...Nf6. Chapter 13 explains it in detail. If you meet 4.Ng5 against 3...Bc5, simply castle: the rook on f8 then defends f7.

> **COMMON MISTAKE** **Copying the opponent's attack**
>
> Many players learn Scholar's Mate or the Fried Liver and then play them every game. Against careful defence these attacks leave the attacker's queen and knight out of play and the attacker worse. Learn them so you can defend against them, not as your main weapon.

Punishing early f-pawn moves

You have already seen the main idea: an early ...f6 or ...f5 by Black (or f3 or f4 by White) opens the diagonal toward the king. When your opponent plays such a move, look at once for a queen check on the opened diagonal, and for captures in the centre that the pawn no longer prevents. In 1.e4 e5 2.Nf3 f6?, 3.Nxe5! works precisely because the queen check on h5 is coming.

> **IN YOUR NEXT GAME** **Your next game**
>
> In the first ten moves, before each move ask "Does this develop, fight for the centre, or make my king safer?" Whenever your opponent's queen or bishop aims at f7 (or f2), check that it is defended before anything else.

Practice at the board

1. Set up `r1bqkbnr/pppp1ppp/2n5/4p2Q/2B1P3/8/PPPP1PPP/RNB1K1NR b KQkq - 3 3` and find every Black move that stops Qxf7#. Then play 3...g6 4.Qf3 and find the right defence again. Check yourself with an engine: 3...Nf6?? and 3...d6?? should both show mate.
2. Play the Légal's Mate line from the start position until you can do it from memory. Then play it again with Black's best reply 5...dxe5 and see what Black has lost.

3. Set up `rnbqkbnr/pppp2pp/5p2/4p3/4P3/5N2/PPPP1PPP/RNBQKB1R w KQkq - 0 3` and play 3.Nxe5 against a friend or engine from both sides. Try 3...fxe5 and 3...Qe7 and see which is better.
4. **Drill:** play five quick games (10+5) against an app at a low level, but with one rule: no queen moves before move 8 unless you are capturing or answering a threat. After each game, count how many moves each side spent on development.

Chapter summary

- The opening has three jobs: fight for the centre, develop the minor pieces, and castle; finish by connecting the rooks.
- Knights usually come out before bishops; move each piece once before moving one twice.
- An early queen gets chased; punish it by developing with gains of tempo.
- Avoid pointless pawn moves, especially early f-pawn moves that open lines to your king.
- Every principle has exceptions; break one only for a concrete reason you can name.
- Against Scholar's Mate, block or defend f7 according to where the queen stands: ...g6 or ...Qe7 against Qh5, ...Nf6 against Qf3.
- Know Fool's Mate, Légal's Mate and the Fried Liver from both sides; your Black repertoire avoids the Fried Liver with 3...Bc5.

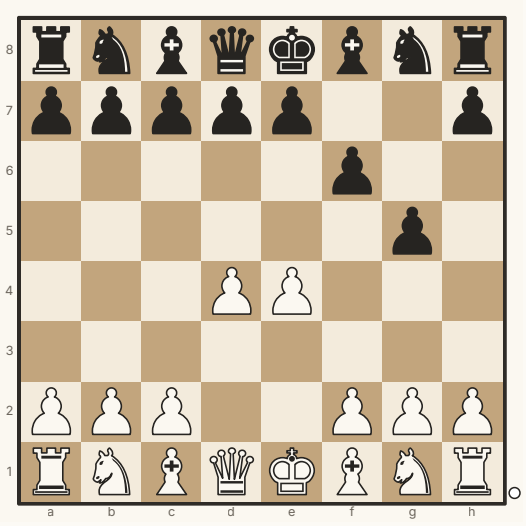
CHEAT CODE **Opening move check**

1. What is my opponent threatening? Is f7 (or f2) attacked twice?
2. Can I develop a piece toward the centre, preferably a knight?
3. Can I castle, or prepare to castle?
4. Am I moving a piece twice, moving the queen early, or making a pawn move that does not help? If so, can I name the concrete reason?
5. Blunder check: after my move, what can my opponent capture or check?

Exercises

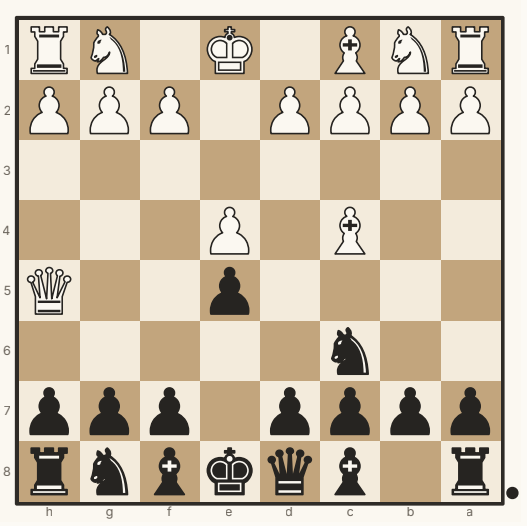
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

6.1 CHECKMATE



White to move. Black opened with 1...f6 and 2...g5. Punish it at once. Solution p. 645

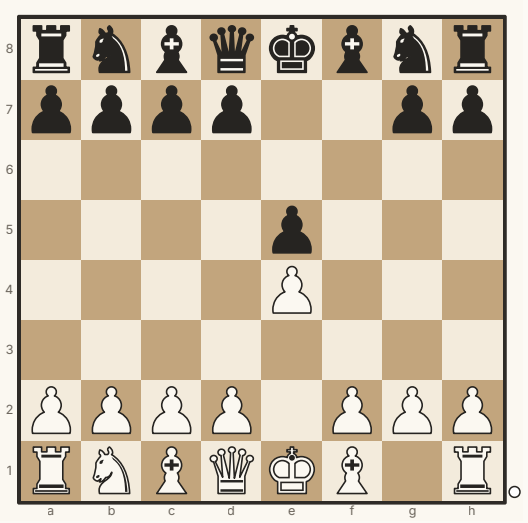
6.2 DEFENCE



Black to move. White's queen and bishop both aim at f7. Stop Scholar's Mate.

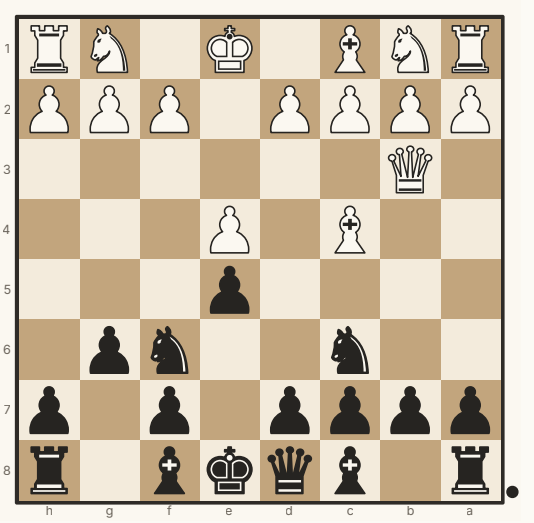
Solution p. 645

6.3 TACTIC



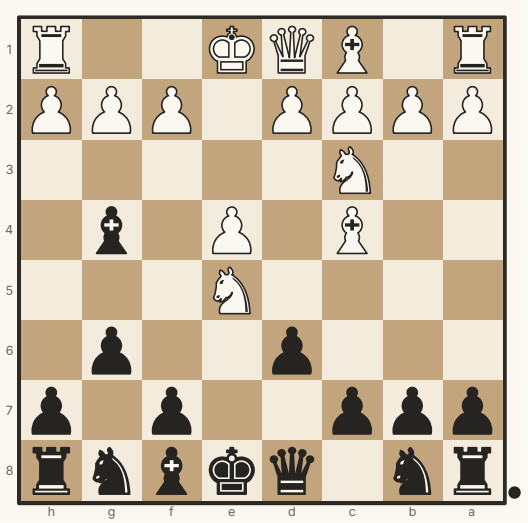
White to move. After 1.e4 e5 2.Nf3 f6? 3.Nxe5! Black took the knight with 3...fxe5. Punish the open king. Solution p. 646

6.4 OPENING



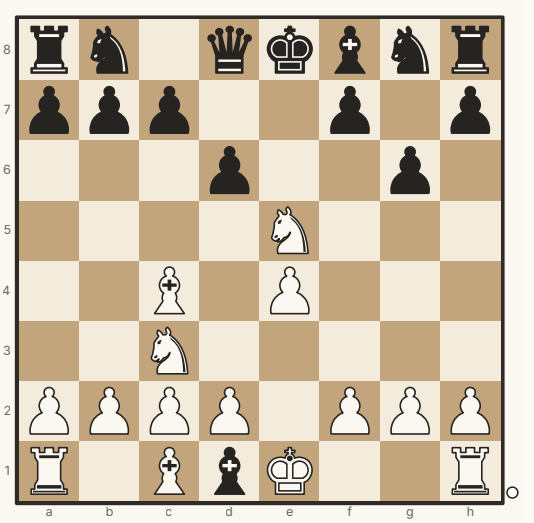
Black to move. White's queen has moved three times (Qh5, Qf3, Qb3) and now hits f7 again. Find the developing move that attacks the queen and creates a threat of its own. Solution p. 646

6.5 DEFENCE



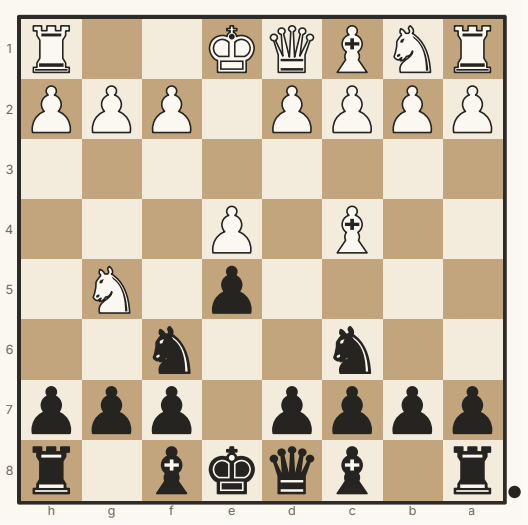
Black to move. White has just played 5.Nxe5!, leaving the queen on d1 en prise. Do you take it? Solution p. 646

6.6 CHECKMATE



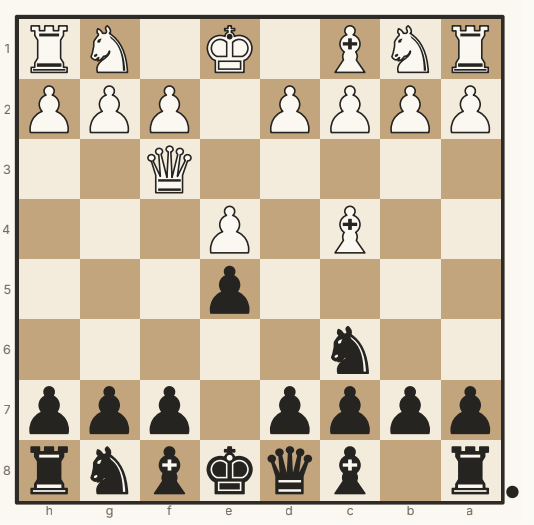
White to move. Black has just grabbed your queen with 5...Bxd1. Show why that was a mistake. Solution p. 646

6.7 OPENING



Black to move. White's knight and bishop both attack f7 (the Two Knights with 4.Ng5). What is Black's soundest reply? Solution p. 647

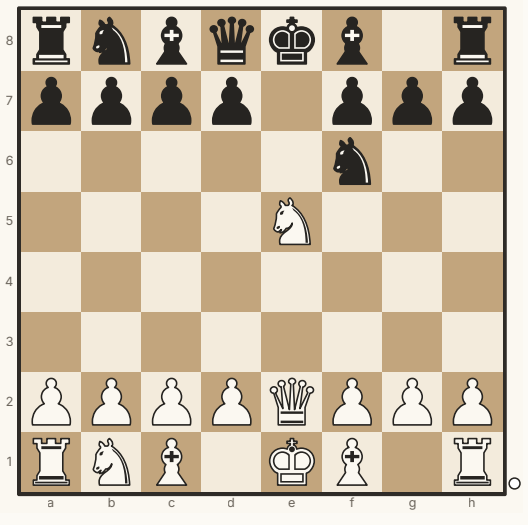
6.8 DEFENCE



Black to move. White threatens Qxf7#. Find the move that defends and develops.

Solution p. 647

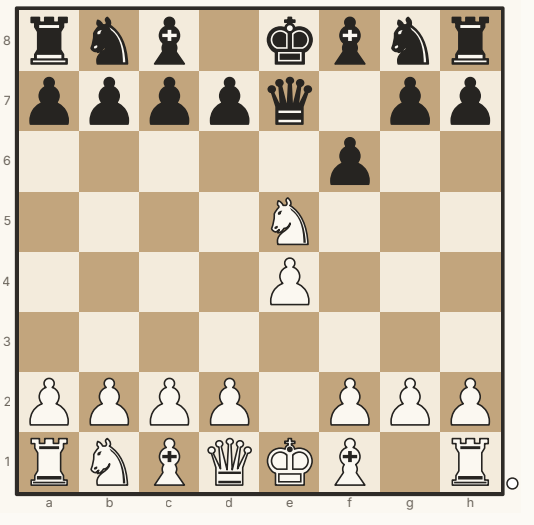
6.9 TACTIC



White to move. In the Petrov, Black copied White with 3...Nxe4?! and after 4.Qe2 retreated with 4...Nf6??. Win material.

Solution p. 647

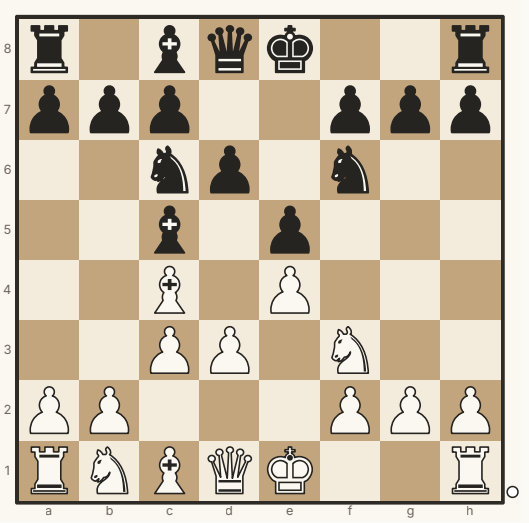
6.10 OPENING



White to move. After 1.e4 e5 2.Nf3 f6? 3.Nxe5 Qe7, your knight is attacked. Choose the retreat that keeps your advantage.

Solution p. 647

6.11 OPENING



White to move. This is the Giuoco Pianissimo from your repertoire. Choose a sound move. Solution p. 648

HINTS

6.1 Look along the diagonal from h5 to e8.

6.2 Ask what the queen threatens before you develop.

6.3 Which checks do you have?

6.4 Can you develop a piece and attack the queen at the same time?

6.5 After ...Bxd1, list every check White has.

6.6 Check first: which piece can give check with a capture?

6.7 Can you block the bishop's diagonal instead of defending f7 directly?

6.8 The queen's line to f7 runs along the f-file. Put a piece in the way.

6.9 Your queen and Black's king are on the same file. Move the knight with a threat.

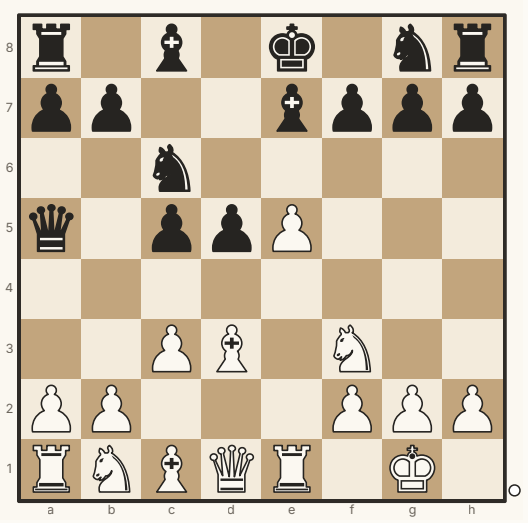
6.10 Which square keeps the knight safe without blocking your other pieces?

6.11 Is anything of yours attacked? If not, finish development.

Stage 1 Test

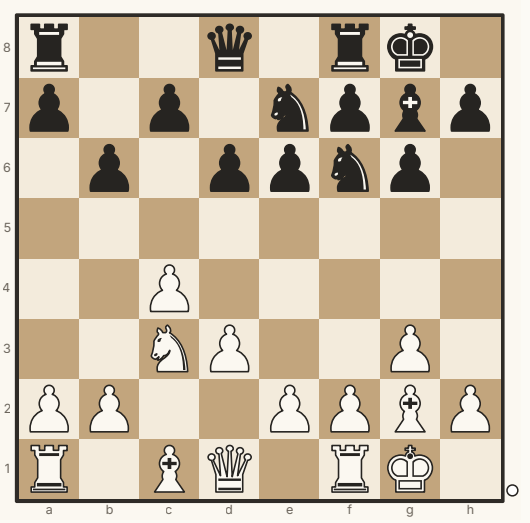
Fifteen mixed positions covering Part 1. Solve them without moving the pieces, write down your answers, then mark yourself. 12/15 or better: move on with confidence. 9–11: review the chapters behind your misses. Below 9: repeat this part's exercises before continuing.

T1.1 RULES



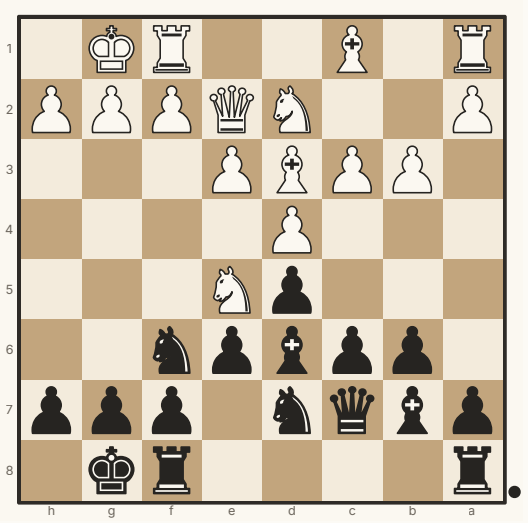
White to move. Black has just played ...d7-d5. Use the special pawn capture to win material. Solution p. 648

T1.2 BOARD VISION



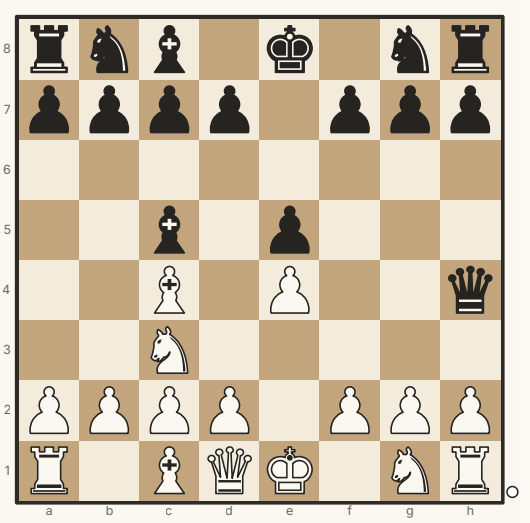
White to move. Black's knight has just retreated from c6 to e7. What did that open? Solution p. 648

T1.3 COUNTING



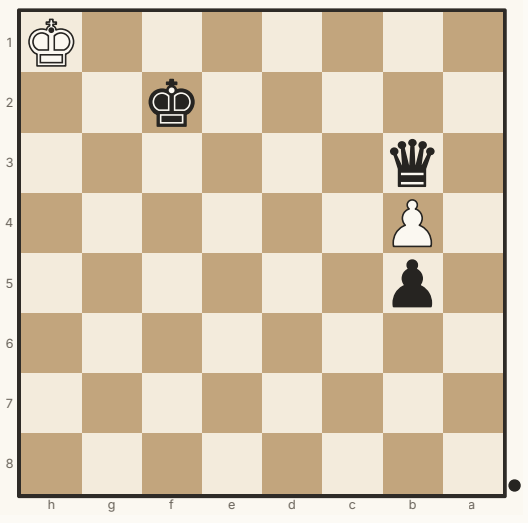
Black to move. White's knight on e5 is attacked twice and defended once. Win material. Solution p. 649

T1.4 DEFENCE



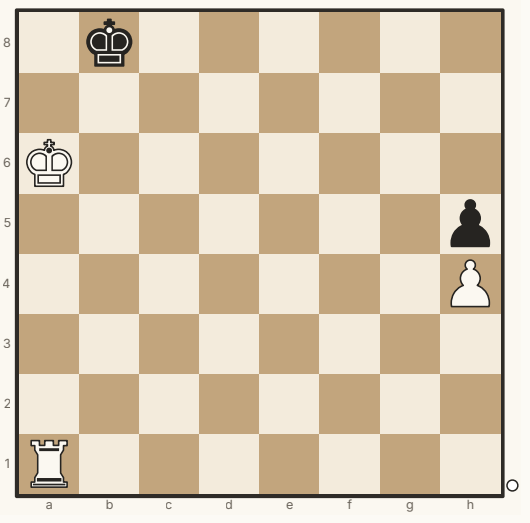
White to move. Black's queen and bishop both aim at f2. What does Black threaten? Make a move that stops it. Solution p. 649

T1.5 CHECKMATE



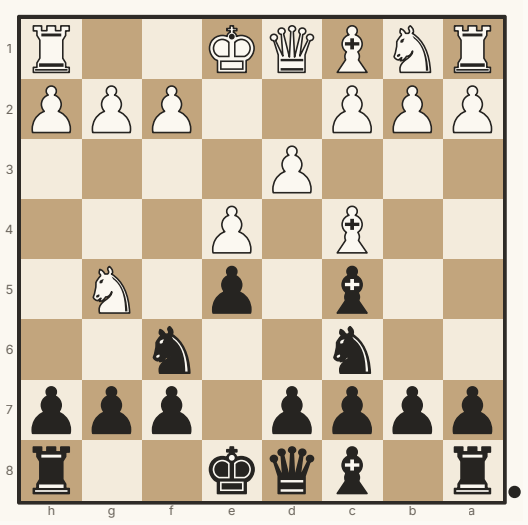
Black to move. Mate in one, and watch out for stalemate. Solution p. 649

T1.6 CHECKMATE



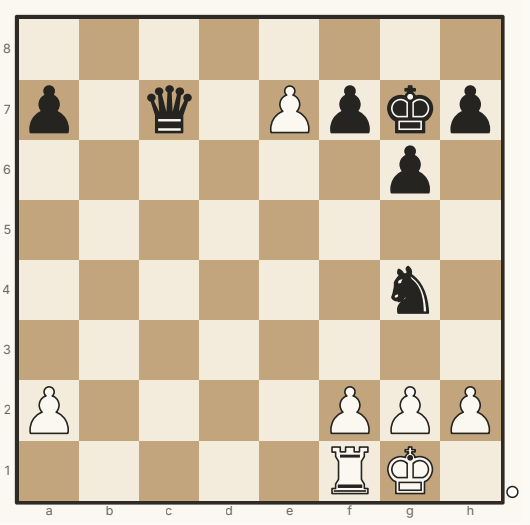
White to move. Mate in two with king and rook. Solution p. 649

T1.7 OPENING



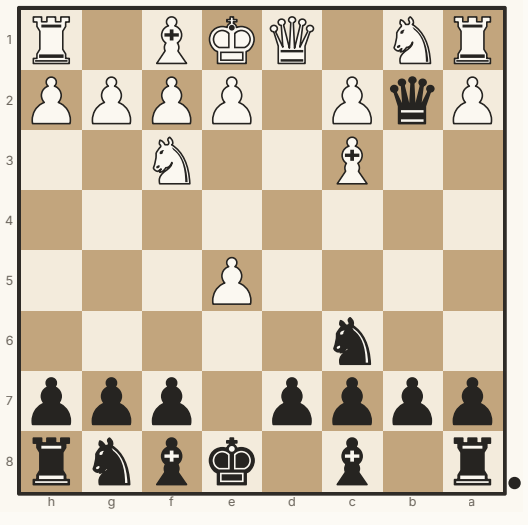
Black to move. White's knight and bishop both attack f7. Choose the move that solves the problem and follows opening principles. Solution p. 649

T1.8 RULES



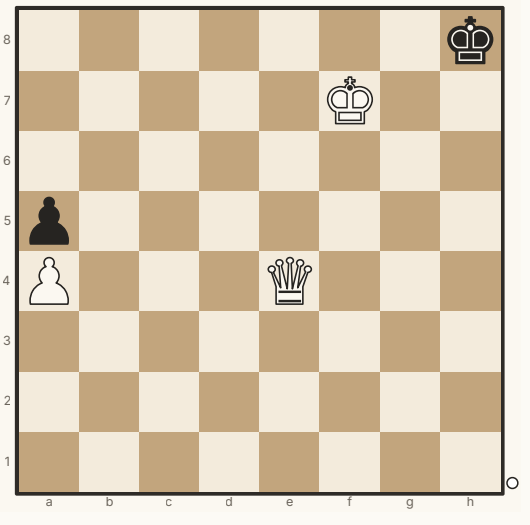
White to move. Black threatens ...Qxh2#. Promote your pawn, but choose the right piece. Solution p. 649

T1.9 OPENING



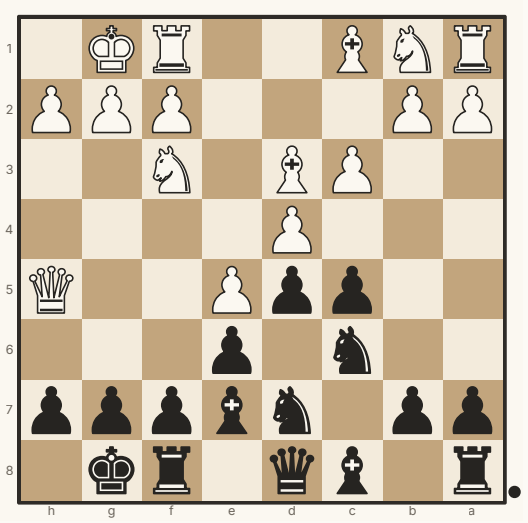
Black to move. White has just attacked your queen with 6.Bc3. Find the move that wins material. Solution p. 650

T1.10 CHECKMATE



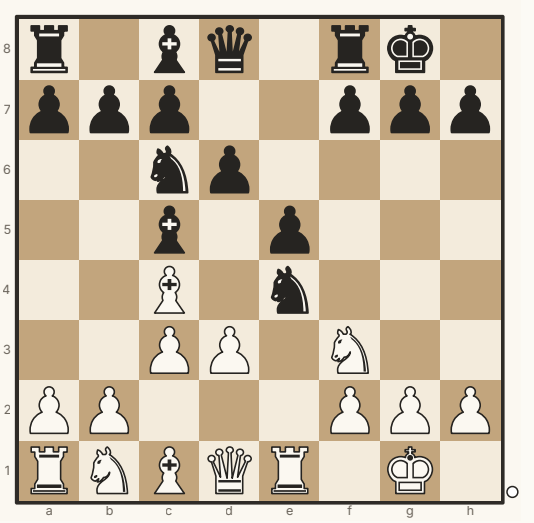
White to move. Mate in one. The a-pawns are blocked, so a stalemate is possible. Solution p. 650

T1.11 DEFENCE



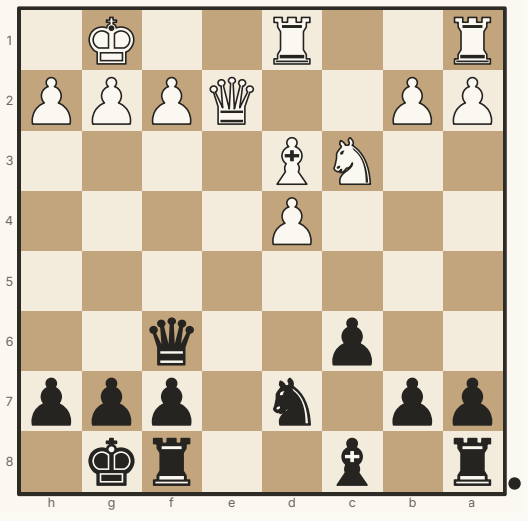
Black to move. White's queen and bishop point at your king. What is the threat, and how do you stop it? Solution p. 650

T1.12 TACTIC



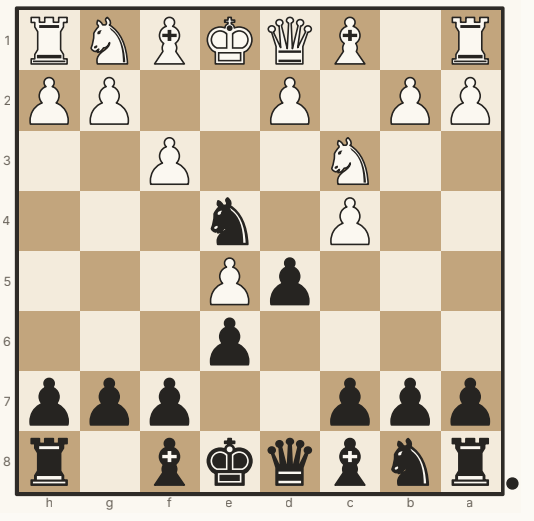
White to move. Black has just played 7... Nxe4, taking your pawn. Is the knight safe? Solution p. 650

T1.13 BLUNDER CHECK



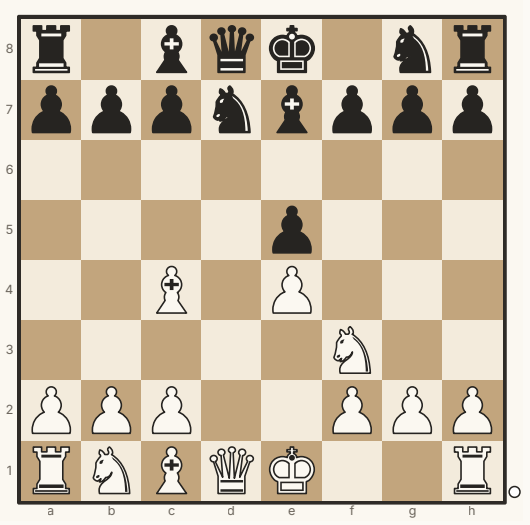
Black to move. White's d4-pawn has no pawn or knight defending it, and your queen attacks it. Is it safe to take? Play the right move. Solution p. 651

T1.14 TACTIC



Black to move. After 1.c4 Nf6 2.Nc3 e6 3.e4 d5 4.e5 Ne4, White has attacked your knight with 5.f3?. Before you retreat, run the safety scan: checks first. Solution p. 651

T1.15 TACTIC



White to move. In the Philidor Defence Black has just recaptured 5...dxe5, opening the d-file. Find the forcing move.

Solution p. 651

HINTS

T1.1 Which pawn capture is only legal immediately after a two-square pawn move?

T1.2 Trace the diagonal from your g2-bishop all the way to the corner.

T1.3 Attackers two, defenders one: capture, and decide which piece goes first.

T1.4 Which checks and captures does Black have next move?

T1.5 Give check from a square that also covers h2.

T1.6 Cut off the king's escape route before you check.

T1.7 Which move both protects f7 and gets your king out of the centre?

T1.8 Which new piece would give check?

T1.9 Can you make the attacking bishop unable to move?

T1.10 Which file is the king standing on?

T1.11 Block the bishop's diagonal or cover h7.

T1.12 Which of your pawns defends e4?

T1.13 After ...Qxd4, which white move is a check, and what would it uncover?

T1.14 Which diagonal did 5.f3 open next to White's king?

T1.15 Which queen move threatens mate on f7?

PART TWO

See the Tactics

Stage 2 · 600–800

Once you stop giving material away, the next rating jump comes from taking it. Tactics are short forcing sequences that win material or deliver mate, and they follow a small number of repeating patterns. This part teaches those patterns one at a time, then gives you a compact, sound opening repertoire for both colours so your games start in positions you understand.

Chapter 7 · Forks

Chapter 8 · Pins and Skewers

Chapter 9 · Discovered Attacks and Double Check

Chapter 10 · Removing the Defender, Overloading, Deflection and Decoys

Chapter 11 · Checkmate Patterns

Chapter 12 · Your Repertoire as White

Chapter 13 · Your Repertoire as Black

CHAPTER 7 STAGE 600–800

Forks

One move, two targets

YOU WILL LEARN TO

- Recognize the geometry of knight, pawn, queen, king, bishop and rook forks.
- Find fork squares by working backwards from two targets.
- Use forcing moves (checks and captures) that drive pieces onto forkable squares.
- Spot your opponent's fork threats and keep your own king and pieces off forkable squares.

> IN YOUR NEXT GAME
>
> Before every move, look for any square from which one of your pieces would attack two loose or more valuable targets at once, and check whether your opponent has such a square against you.

You are about to castle when your opponent's knight lands on c7. It checks your king and attacks your rook in the corner. You move the king, the knight takes the rook, and the game is effectively over. Or you are on the other side: your pawn steps forward and suddenly two enemy pieces are attacked, and only one can be saved.

That is a **fork**: one piece attacking two or more targets at the same time. Your opponent has only one move to answer, so usually only one target can be saved. Between 600 and 800, forks decide more games than almost any other tactic, because they are easy to miss and easy to set up. This chapter teaches you to see them, for both sides, before they happen.

What a fork is, and why it works

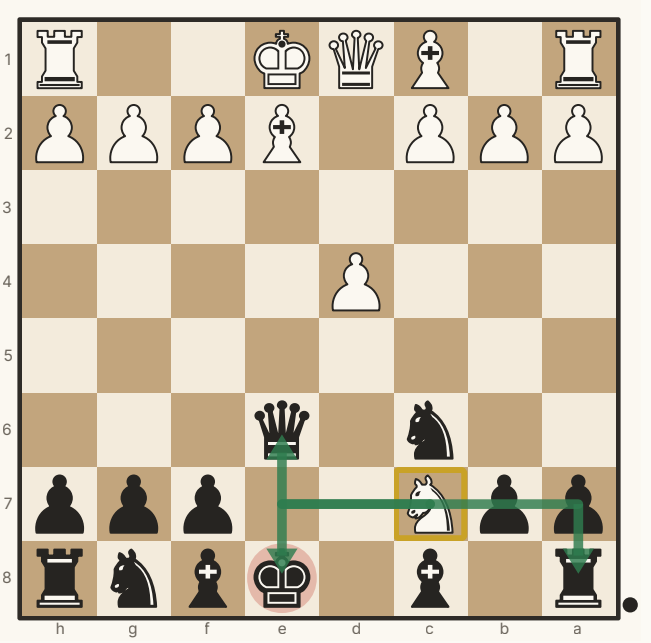
A **fork** (also called a **double attack**) is a move that creates two threats at once. The threats can be against pieces, against the king, or a mix such as "check plus

an attack on a rook". The defender gets one move. If no single move answers both threats, something is lost.

Three things make a fork work:

1. **Two targets.** Each target must be either *loose* (undefended) or *more valuable* than the forking piece. A knight attacking two defended pawns is not a fork in any useful sense.
2. **A safe forking piece.** If the opponent can simply capture the piece that forks, there is no fork. This is the most common reason "forks" fail at club level.
3. **No single answer.** The opponent must not have one move that saves both targets, for example by moving one target to a square where it protects the other, or by giving a check of their own.

The strongest forks include a **check**, because the check must be answered first. The king moves, and then you collect the second target.



A family fork. The knight on c7 checks the king and attacks the rook on a8 and the queen on e6. Black must move the king, and White takes the queen.

When a knight forks the king, queen and rooks together, coaches call it a **family fork**. When it hits king and queen, it is a **royal fork**. The names do not matter much. What matters is that the knight on c7 above is attacked by nothing, so Black is simply losing material.

Knight forks

The knight is the great forking piece for two reasons. It jumps, so it cannot be blocked. And its attack looks nothing like the attack of the other pieces, so a queen and a king standing "safely" can still be hit together by a knight.

The geometry

A knight always lands on a square of the opposite colour. That gives you a quick filter: **a knight can only fork pieces that stand on squares of the same colour** (both light or both dark), and the knight itself must stand on the other colour. If the king is on a dark square and the queen is on a light square, no knight can fork them.

The second filter is distance. A knight attacks squares two steps away in a "short L". Two targets it can fork are always within four squares of each other, with typical shapes such as:

- **Same rank or file, two squares apart with one square between** (king e8 and rook c8 can both be hit from d6).
- **Four squares apart on a rank** (king e8 and rook a8 can both be hit from c7).
- **A "long L" shape** (king g8 and queen d7 can both be hit from f6; king g8 and queen c6 from e7).

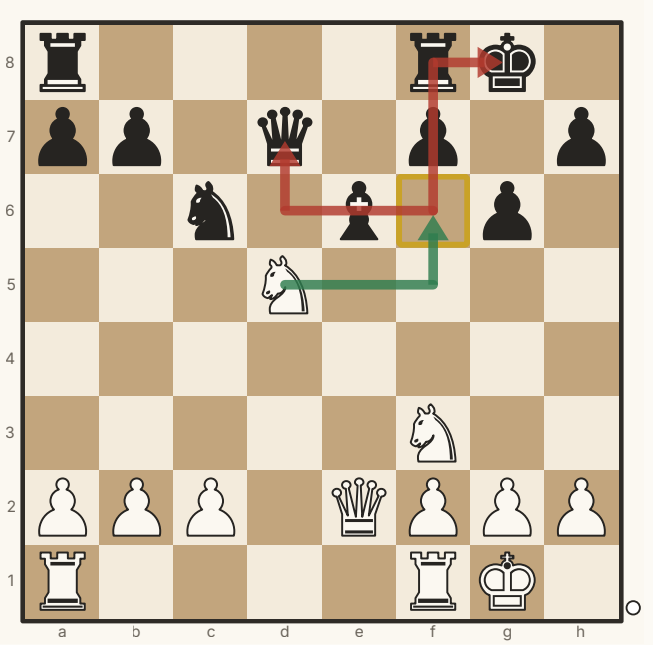
You do not need to memorize these shapes. Lists of shapes are hard to use at the board, and a method is better.

Working backwards from the targets

Instead of asking "where can my knight go?", ask **"which squares attack both targets?"** and only then "can my knight get there?"

The procedure:

1. Pick two targets: the king plus something valuable or loose.
2. List the squares from which a knight attacks the first target (at most eight).
3. Check which of those also attack the second target. These are the **fork squares**.
4. Ask whether your knight can reach a fork square *now*, and whether that square is guarded.



Targets: king on g8, queen on d7. Knight squares that attack g8 are e7, f6 and h6. Of those, only f6 also attacks d7. The knight on d5 can jump there, and with the pawn on g6 instead of g7, nothing guards f6.

White plays Nf6+, the king moves, and Nxd7 wins the queen for a knight. Notice how the defensive pawn move ...g6 changed everything: with a pawn on g7, the square f6 would be covered and the fork would not exist. Pawn moves create and remove fork squares all the time.

> KEY IDEA **The fork square is the whole story**
>
> A knight fork needs one square that touches both targets. Find that square first. Then ask two questions: can my knight get there (now, or after a check)? And is the square guarded? If it is guarded by a pawn, forget it. If it is guarded by a piece, count: the fork may still win if the target is worth more than the knight.

What commonly goes wrong

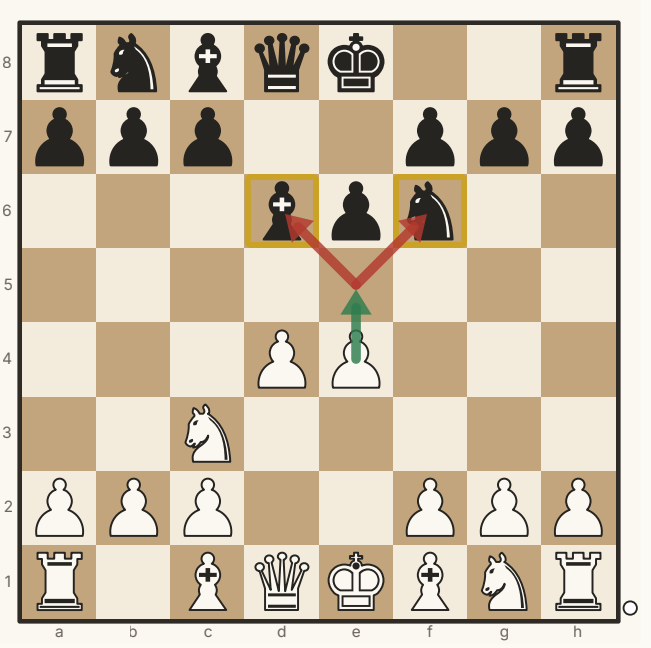
The most common knight-fork mistake at this level is **jumping onto a guarded fork square**. A knight lands on c7 "forking" king and rook, and the queen on d8 simply takes it. Exercise 07-11 is built on exactly this: count the defenders of the fork square before you jump.

The second mistake is **forgetting the forking piece itself**. Your opponent's knight lands on c7 attacking both of your rooks, you panic and move a rook, and only later notice that your queen could have captured the knight. Before you react to a fork, run the safety scan from Chapter 4: can I take the forking piece? Is it defended?

Pawn forks

A pawn fork is the cheapest fork of all. A pawn is worth about 1 point; if it forks two minor pieces, you win at least 2 points even if the pawn is captured afterwards.

A pawn attacks the two squares diagonally in front of it, so a pawn fork happens when **two enemy pieces stand side by side, one square apart, on the rank in front of a pawn that can advance safely**. The pawn needs to be safe on its new square, or at least it must not matter if it is taken.



The bishop on d6 and the knight on f6 stand on the same rank with one square between them. White plays e5, protected by the pawn on d4, and one of the two pieces is lost.

The pattern for Black is the mirror image: White pieces on d3 and f3, for example a bishop on d3 and a knight on f3, can be forked by ...e4. Beginners often develop

exactly like this with e3, Bd3 and Nf3 while Black still has pawns on d5 and e5. Exercise 07-02 shows the result.

When to look for pawn forks

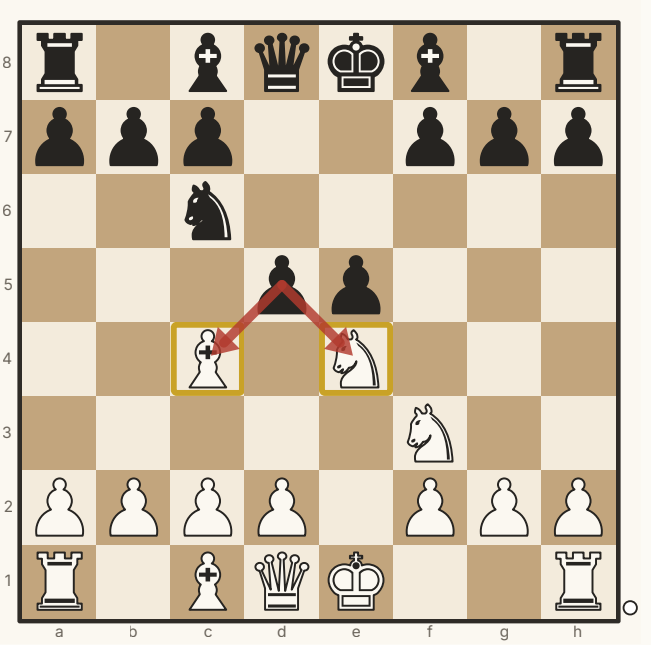
Pawn forks hide in plain sight because the pawn move looks quiet. Make it a habit: **whenever two enemy pieces stand on the same rank with exactly one square between them, look at the pawn that could step between and below them.** Check three things: can the pawn get there in one move, is the square safe (or does it not matter), and can the opponent answer with a check or a capture first?

The fork trick

The most famous pawn fork in the openings is called the **fork trick**. You give up a knight to remove a central pawn, then win the piece back with a pawn fork. Here it is in the Italian Four Knights, which can arise when White plays 4.Nc3 instead of the 4.d3 recommended in this book's repertoire.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bc4 Nf6** 4. **Nc3 Nxe4** 5. **Nxe4 d5** 6. **Bd3 dxe4** 7. **Bxe4 Bd6**

Black takes on e4, and after 5.Nxe4 the pawn fork 5...d5 hits the bishop on c4 and the knight on e4. White protects the knight with 6.Bd3, and after 6...dxe4 7.Bxe4 Black has regained the piece with equal chances.



The moment of the fork trick, after 5...d5. The pawn attacks c4 and e4 at once. White's best is 6.Bd3, protecting the knight, but after 6...dxe4 7.Bxe4 material is level again.

The fork trick is a fair trade, not a winning trap. It does show the principle clearly: a "sacrifice" of a piece is fine if a fork brings it straight back.

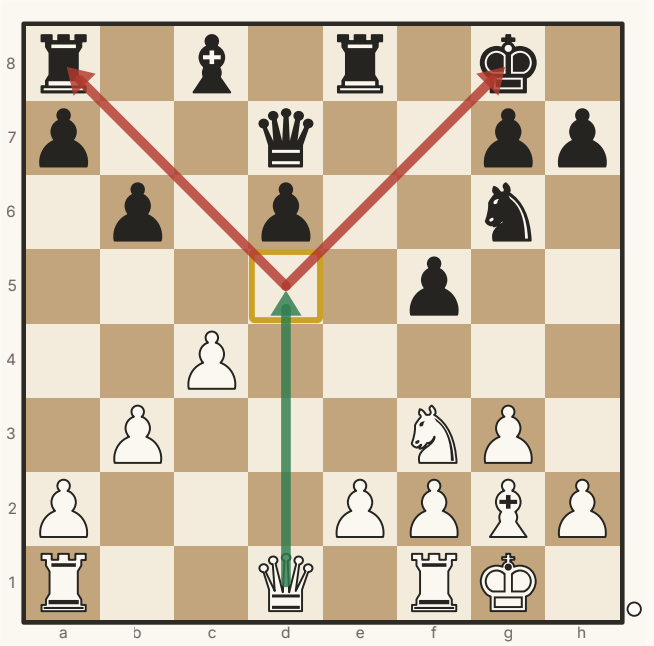
> **COMMON MISTAKE** **The fork that can be answered with a check**
>
> A fork only works if the opponent has no forcing answer. In the Giuoco Piano, 1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.Nxe5? Nxe5 5.d4 tries the fork trick for White: the d-pawn attacks the bishop on c5 and the knight on e5. But Black has 5...Bb4+!, moving one target away *with check*, and after 6.c3 Nxc4 Black keeps the extra piece. Before you play a fork, ask: "Can my opponent move one target with check, or capture something with check?"

1. e4 e5 2. Nf3 Nc6 3. Bc4 Bc5 4. Nxe5? Nxe5 5. d4 Bb4+ 6. c3 Nxc4 7. cxb4

The failed fork trick. Black escaped the pawn fork with a check and is a piece up for a pawn.

Queen forks

The queen moves in eight directions, so she can create a double attack from many squares. Her weakness is value: if either target is defended and can take her, the fork fails. That is why **queen forks almost always combine a check with an attack on a loose piece**, or attack two loose pieces at once.



Black's f-pawn has advanced to f5, opening the a2-g8 diagonal. Qd5+ checks the king through e6 and f7, and attacks the loose rook on a8 through c6 and b7. Whatever Black does, Qxa8 follows.

How to find queen forks:

1. **List the loose pieces** on both sides. A loose piece is one with no defender.
2. **Look at your possible checks.** For each check, ask: "From that square, does my queen also attack a loose piece?"
3. **Look along the lines from each loose piece back to the enemy king.** If a single queen square lies on a line to the king *and* a line to the loose piece, that square is a fork square.

The black version is just as common. In Exercise 07-03, White's bishop on g5 pins a knight but has no defender, and ...Qa5+ checks along the diagonal toward e1 while attacking g5 along the fifth rank.

> **KEY IDEA** **Loose pieces drop off**
>
> Coaches have a short slogan for this: **LPDO, "loose pieces drop off"**. A piece with no defender is not lost yet, but it is the raw material of almost every fork. Any time you see a loose piece (yours or your opponent's), stop and look for a double attack on it. Most queen forks in amateur games hit a loose bishop or knight that "was safe" one move earlier.

King, bishop and rook forks

Every piece can fork. The others simply do it less often.

- **King forks** appear in endgames, where the king is an active piece. A king that steps forward can attack two loose pawns or a loose pawn and a piece at once. The king cannot be blocked, and it can only be driven away by checks. In endgames, keep your pieces and pawns protected, or at least keep them where the enemy king cannot reach both.
- **Bishop forks** usually come with a check along a diagonal. A typical case: once Black's f-pawn has moved, a white bishop on d5 checks the king on g8 and hits a rook on a8 at the same time.
- **Rook forks** often happen on an open rank: a rook check on the back rank that also hits a loose piece on the same rank, or a rook on the seventh rank attacking two pawns.

The thinking is the same for all of them: find two targets that lie on lines (or, for the king, squares) one piece could reach in one move, and check whether that square is safe.

Setting up a fork

Often the fork is not there yet: one target is on the wrong square. Then you need a **forcing move that drives a piece onto a forkable square**. The usual tools:

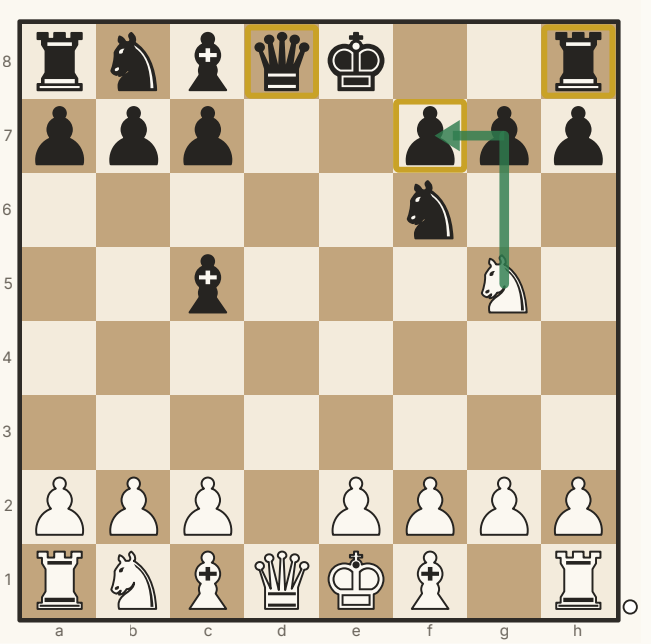
- **A capture that must be recaptured.** If only the king can recapture, you have decided where the king will stand.
- **A check that leaves the king one square.** A sacrifice with check can drag the king onto the fork square.

- **A trade of queens.** Trading on d8 or d1 often pulls the king to d8 or d1, where a knight on f7, c7, f2 or c2 is waiting.

The sequence matters. The forcing move must come *first*, because it limits your opponent's choices. If you play the fork first, the opponent gets a free move to escape or to make a forcing move of their own.

Worked example: getting the move order right

Here is a position of the kind that arises after an early queen's-pawn opening goes wrong for Black. Follow the thinking process, not just the moves.



White to move. The knight on g5 attacks f7, the d-file is open, and the queens face each other on d1 and d8.

Step 1: checks, captures and threats for both sides. White has one check, Qxd8+, which is also a capture. White's captures are Qxd8+ and Nxf7 (and Nxh7, which wins only a pawn). Black's threats matter too: Black can trade queens with ... Qxd1+, and the bishop on c5 eyes f2.

Step 2: look for targets. The black king is on e8 and the rook on h8. A knight on f7 would attack d8 and h8. So the fork square is f7, and the knight can capture there right now.

Step 3: candidate one, 6.Nxf7. It attacks the queen on d8 and the rook on h8. But it is not check, so Black gets a free move, and Black has forcing moves. The queen on d8 is attacked, but it can capture the white queen with check, and Black also has ...Bxf2+.

6. **Nxf7? Bxf2+** 7. **Kxf2 Qxd1**

The wrong move order. Black ignores the fork, answers with forcing moves, and wins the white queen for a bishop.

Step 4: candidate two, 6.Qxd8+ first. This capture is check, and the only recapture is 6...Kxd8, because no other black piece reaches d8. Now the king stands on d8, a knight's jump from f7, and the rook is still on h8.

6. **Qxd8+ Kxd8** 7. **Nxf7+ Ke8** 8. **Nxh8**

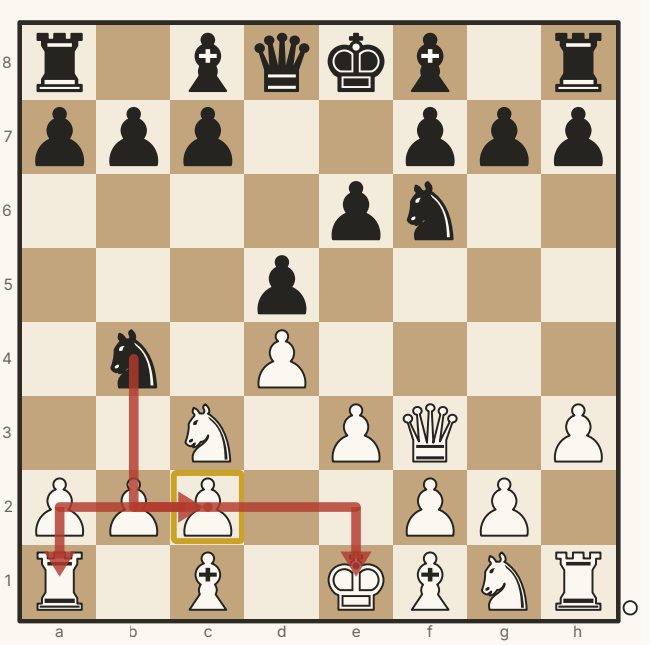
Trade first, fork second. The queen trade drives the king to d8, and Nxf7+ forks king and rook. White wins a rook and a pawn.

Step 5: blunder-check. After 8.Nxh8 the knight is in the corner and may be trapped, but White is already a full rook ahead, so even losing the knight later leaves White clearly winning.

The lesson: **the fork was the same in both lines, but only the forcing move order worked.** Check first, capture first, fork second.

Defending against forks

Every pattern in this chapter can be used against you. The defensive version of the safety scan from Chapter 4 adds one question: **"Where can my opponent's knights, pawns and queen go next move, and would they attack two things from there?"**



White's queen has left d1, so nothing guards c2. The knight on b4 threatens ...Nxc2+, forking the king and the rook on a1. White must deal with it now, for example with Qd1 or Bb5+.

If White ignores the threat with a routine developing move, the fork lands:

6. **Nge2? Nxc2+** 7. **Kd1 Nxa1**

The knight takes a pawn with check and then the rook on a1. Even if White later traps the knight in the corner, Black has won the exchange and a pawn, and the white king is stuck in the centre.

How to defend

When you spot a fork threat, you have four kinds of answer. Look for them in this order:

1. **Remove or block the forking piece.** Capture it, or attack it so it has to leave.
2. **Guard the fork square.** Put a piece or pawn where it covers the square, as ...Ba5 covers c7 in Exercise 07-06.
3. **Move one of the targets**, ideally to a square where it also protects the other, or with tempo (a check or an attack).
4. **Take the square away first**, as in Exercise 07-07, where White plays e4 before Black can play ...e4.

Prevention: habits that stop forks before they start

- **Castle early.** An uncastled king on e1 or e8 with rooks still in the corners is the classic target for knight forks on c2, c7, f2 and f7.
- **Watch your queen and king on the same colour.** A queen standing a knight's fork away from your king (king g8 with the queen on d7, fork square f6, or on c6, fork square e7) is a standing invitation. Check the fork squares between them, and whether a pawn or piece covers them.
- **Keep pieces protected.** Loose pieces are what queen forks feed on. When you finish a move, scan for any piece of yours with no defender.
- **Be careful with pawn moves near your king.** Moving a pawn can uncover a fork square, as ...g6 uncovered f6 in the earlier diagram.

> COMMON MISTAKE **Reacting to the fork instead of the board**
>
> When a fork appears, many players immediately save the more valuable target. Stop for ten seconds first. Can you capture the forking piece? Do you have a check or a bigger threat? Can one move save both targets? Exercise 07-10 shows a "fork" where the right answer is simply to take the knight.

How to spot forks in your own games

Forks are easy to find once you look for the right features. Train yourself to notice these signals during the game:

- **A loose piece.** Any undefended piece, yours or theirs, is a fork target.
- **An uncastled king with a rook still in the corner.** Look at c2/c7 and f2/f7.
- **King and queen on the same colour squares, close together.** Find the fork squares between them.
- **Two pieces side by side with one square between them** in front of an enemy pawn.
- **An open file or diagonal toward the king**, which allows queen checks that also hit something else.

After each game, go through it once looking only for forks: forks you played, forks you allowed, and forks that were available but missed. Write each one in your mistake journal under "missed-tactic" or "hanging-piece". After a few weeks

you will see the same shapes repeating, and that repetition is exactly what builds pattern recognition.

> **IN YOUR NEXT GAME** **Your fork checkpoint**
>
> In your next game, before each move, spend five seconds on two questions: "Is there a square where my knight, pawn or queen would attack two things?" and "Is there such a square for my opponent?" If the answer to the second is yes, deal with it before anything else.

Practice at the board

Set up each position on a board or in any free chess app's analysis mode. Try the task before moving anything, then check yourself.

1. **Fork squares by hand.** Set up `r4rk1/pp1q1p1p/2n1b1p1/3N4/8/5N2/PPP1QPPP/R4RK1 w - - 0 15`. Without moving a piece, name every square from which a white knight would attack both g8 and d7. (Answer: only f6.) Then put a black pawn on g7 instead of g6 and repeat: the fork disappears.
2. **Pawn fork scan.** Set up `rnbqk2r/ppp2ppp/3bpn2/8/3PP3/2N5/PPP2PPP/R1BQKBNR w KQkq - 0 5`. Find the pawn fork, then play out Black's best defence (...Bb4, attacking the knight on c3) and count the material after White captures on f6.
3. **Move order.** Set up `rnbqk2r/ppp2ppp/5n2/2b3N1/8/8/PPP1PPPP/RNBQKB1R w KQkq - 0 6`. Play 6.Nxf7 and find Black's best reply. Then reset and play 6.Qxd8+ first. Compare the results.
4. **Defensive drill.** Set up `r1bqkb1r/ppp2ppp/4pn2/3p4/1n1P4/2N1PQ1P/PPP2PP1/R1B1KBNR w KQkq - 1 6`. Find three different White moves that stop ...Nxc2+, and one natural move that does not.
5. **Random positions.** Open any of your recent games at move 15. For both sides, list every loose piece and every fork square against the kings. Do this for five positions.

To check yourself, use an engine only *after* you have written down your answer.

Chapter summary

- A fork is one move that creates two threats. It works when both targets are loose or more valuable than the forking piece, the forking piece is safe, and no single reply answers both threats.
- Knight forks are the most common. A knight forks only pieces on the same colour; find fork squares by working backwards from the two targets, then ask whether the square is guarded.
- Pawn forks hit two pieces standing side by side with one square between them. The fork trick (...Nxe4 and ...d5) uses one to regain a piece.
- Queen forks almost always combine a check with an attack on a loose piece. Loose pieces drop off.
- Kings, bishops and rooks fork too, especially in endgames and along open lines.
- Many forks must be set up: a capture, a check or a queen trade drives a piece onto the fork square. Forcing moves come first, the fork second.
- Defend by removing the forking piece, guarding the fork square, moving a target with tempo, or taking the square away before it is used. Castle early and keep pieces protected.

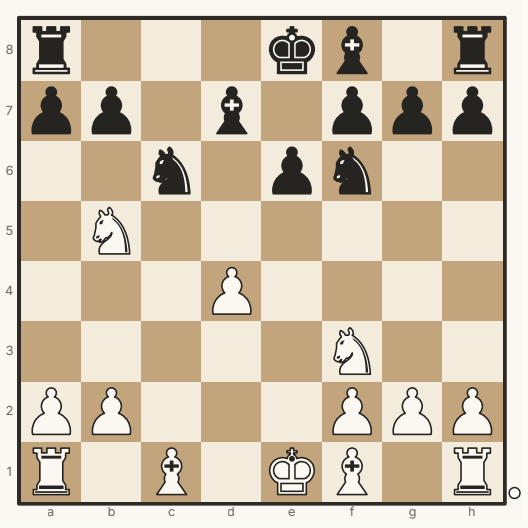
CHEAT CODE **The fork check**

1. List the targets: the enemy king, the queen, rooks, and every loose piece.
2. For each pair of targets, find the squares from which one of your pieces would attack both.
3. Ask: can a piece of mine reach that square now, or after a check or forced capture?
4. Check that the square is safe and that the opponent has no forcing answer (a check, a capture, or one move that saves both).
5. Play the forcing moves first and the fork second.
6. Then run the same scan for your opponent before you commit to your move.

Exercises

Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

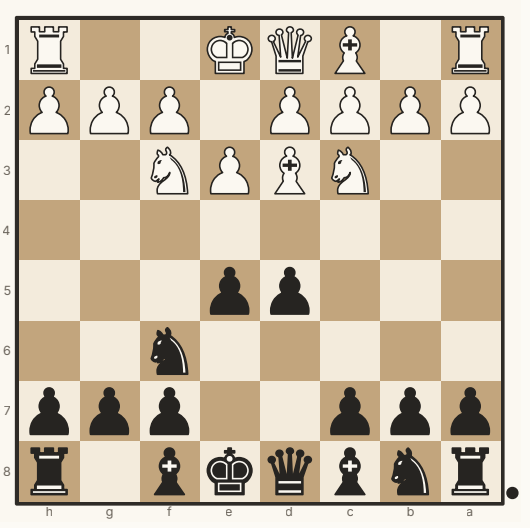
7.1 TACTIC



White to move. The queens are gone and Black has not castled. Win material.

Solution p. 652

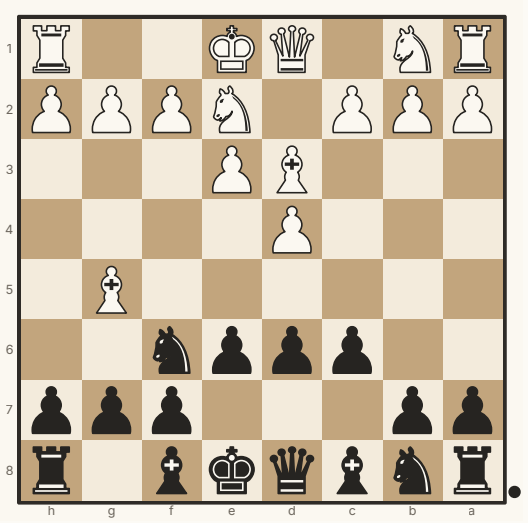
7.2 TACTIC



Black to move. White has developed with e3, Nc3, Bd3 and Nf3. Win material.

Solution p. 652

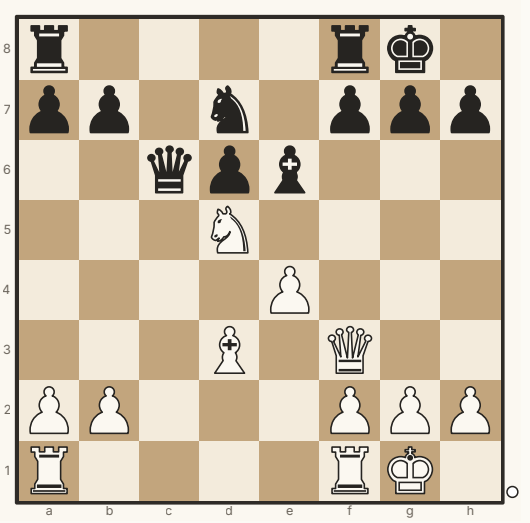
7.3 TACTIC



Black to move. White's bishop on g5 pins your knight to your queen. Is there something better than breaking the pin slowly?

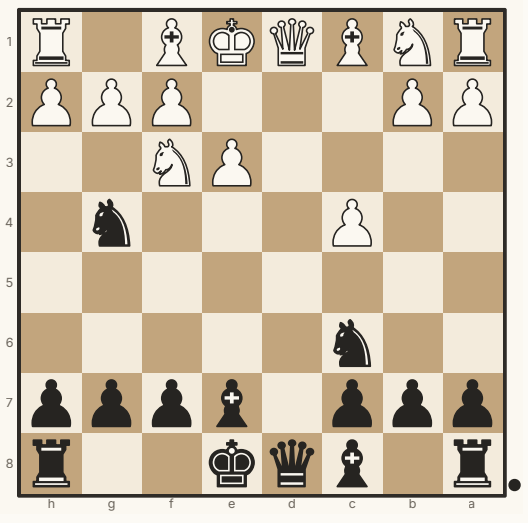
Solution p. 652

7.4 TACTIC



White to move. Win material. Solution p. 652

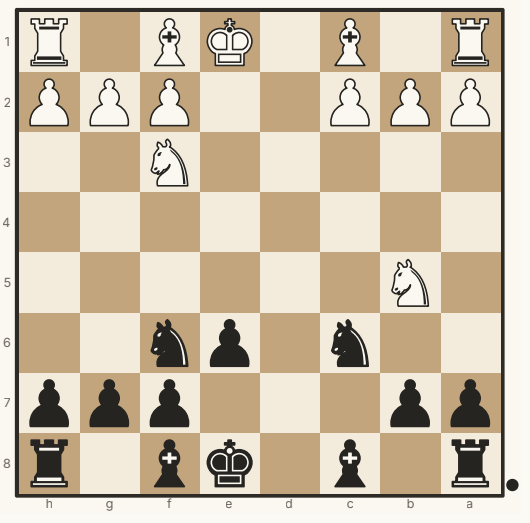
7.5 TACTIC



Black to move. The d-file is open and the queens face each other. Win material.

Solution p. 653

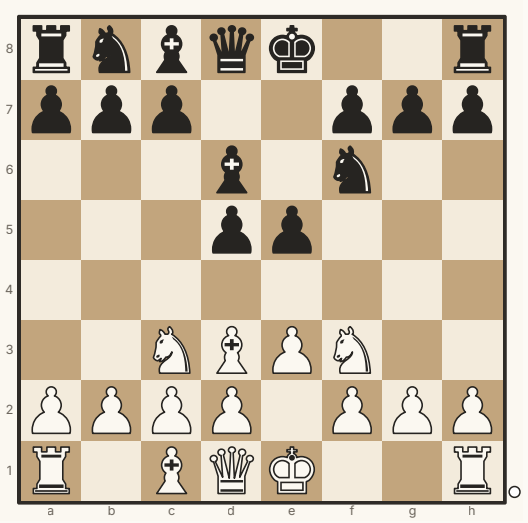
7.6 DEFENCE



Black to move. White's last move, Nb5, threatens Nc7+ forking your king and rook. Find the best way to meet the threat.

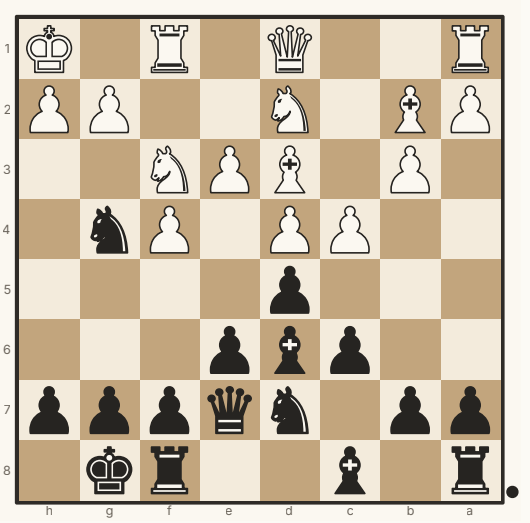
Solution p. 653

7.7 DEFENCE



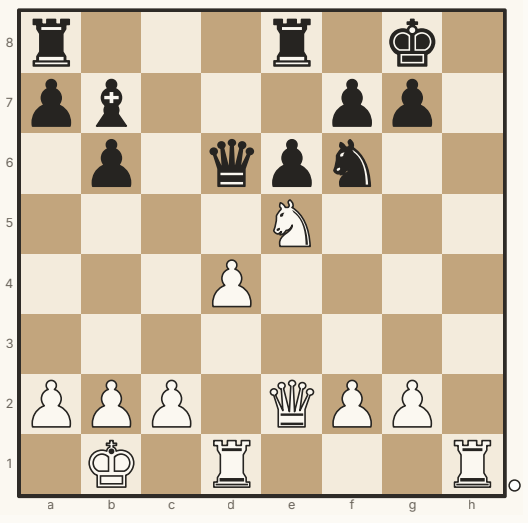
White to move. Black threatens ...e4, forking your bishop and knight. Find a good move. Solution p. 653

7.8 TACTIC



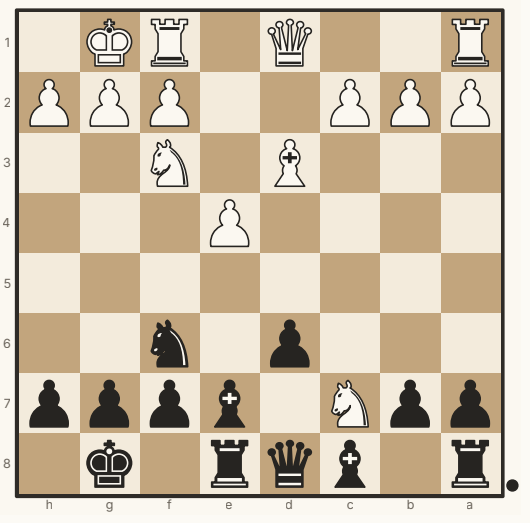
Black to move. White has set up a Stonewall with pawns on d4, e3 and f4. Win material. Solution p. 653

7.9 TACTIC



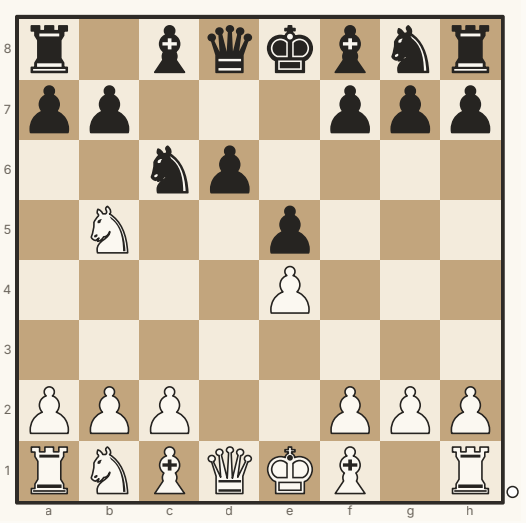
White to move. Your knight already attacks f7. Find the way to turn that into a winning fork. Solution p. 654

7.10 BLUNDER CHECK



Black to move. White's knight has jumped to c7, attacking both of your rooks. What should you do? Solution p. 654

7.11 QUIET MOVE



White to move. Your knight on b5 eyes c7, next to the king and the rook. Is there a fork? Play a good move. Solution p. 654

HINTS

7.1 Which squares does the knight on b5 attack? Is one of them next to the king and the rook at the same time?

7.2 Look at the two white pieces on the third rank. Which pawn can attack both?

7.3 Which white piece has no defender? Which queen move gives check and attacks it?

7.4 Find a square that touches both g8 and c6 with a knight's jump.

7.5 From f2 your knight would attack d1 and h1. How can you get the white king to d1?

7.6 Which square does White want? Can you guard it, or gain time first with a check?

7.7 Which of your pieces stand side by side in front of a black pawn?

7.8 Which white pawn is undefended, and what would a knight on that square attack?

7.9 From f7 the knight would attack d6 and h8. What check forces the king to h8?

7.10 Is the forking piece itself protected?

7.11 Count the defenders of each fork square before you jump.

CHAPTER 8 STAGE 600–800

Pins and Skewers

Attacking along the line

YOU WILL LEARN TO

- Tell an absolute pin from a relative pin, and know which pinned pieces may legally move.
- Exploit a pin by attacking the pinned piece again, especially with a pawn.
- Break a pin by interposing, moving the back piece, counter-attacking, or unpinning with tempo.
- Recognize skewers of king and rook or king and queen, including the rook-endgame skewer that promotes a pawn.

> IN YOUR NEXT GAME
>
> Whenever two enemy pieces, or an enemy piece and the king, stand on the same line, look for a bishop, rook or queen move onto that line; and before you count a piece as a defender, check whether it is pinned.

Your opponent's bishop lands on b5 and suddenly your knight on c6 cannot move: behind it, on the same diagonal, stands your king. A pawn walks up to attack the knight and there is nothing you can do. Or, the other way round, you "pin" a knight to your opponent's queen, grab a pawn, and the knight jumps away anyway, because your opponent has seen that moving it wins more than it loses.

Pins and skewers are attacks along a line: a rank, a file or a diagonal. They are the second big family of tactics after forks, and they work only with the pieces that move in lines: bishops, rooks and queens. This chapter teaches you to create them, exploit them, escape them, and, just as important, to know when a pin is not really a pin.

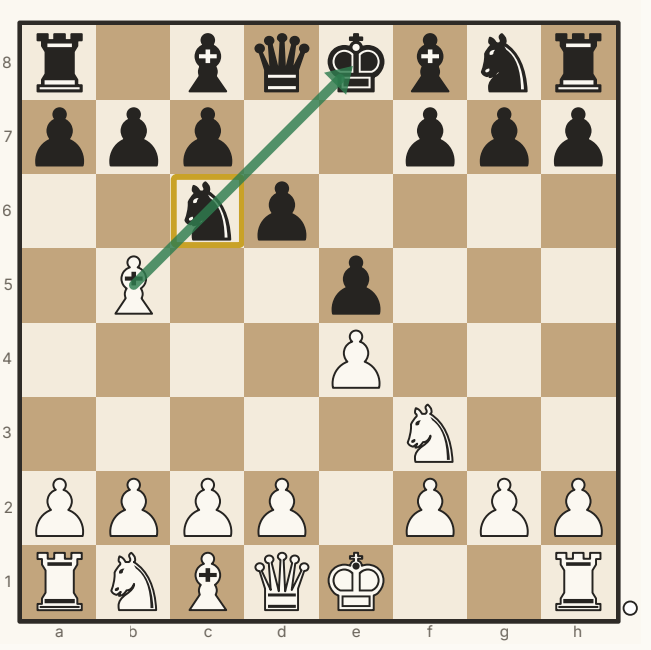
What a pin is

A **pin** is an attack on a piece that cannot move, or should not move, because a more valuable piece stands behind it on the same line. The attacking piece is the **pinner**, the piece in front is the **pinned piece**, and the piece behind is the one being protected.

There are two kinds, and the difference matters enormously.

Absolute pins

In an **absolute pin**, the piece behind is the king. Moving the pinned piece would expose the king to check, so the move is **illegal**. The pinned piece cannot move at all, except along the line of the pin (a pinned rook on a file may still slide along that file, and may even capture the pinner).

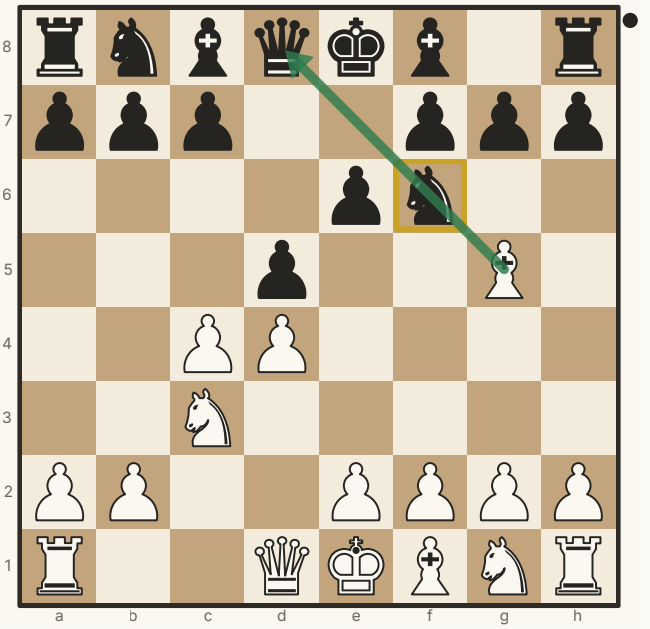


After 1.e4 e5 2.Nf3 Nc6 3.Bb5 d6, the knight on c6 is absolutely pinned. The bishop on b5, the knight and the king on e8 are on one diagonal, so the knight may not move until something changes.

A typical consequence: a piece that is absolutely pinned **does not defend anything**. In the diagram, if a black piece stood on d4 or e5 relying on the knight for protection, it would in fact be unprotected. Exercise 08-01 is built on this.

Relative pins

In a **relative pin**, the piece behind is not the king but something valuable, usually the queen or a rook. Moving the pinned piece is legal; it just loses the piece behind, *unless the move creates a bigger threat*.



The Queen's Gambit Declined after 1.d4 d5 2.c4 e6 3.Nc3 Nf6 4.Bg5. The bishop pins the knight on f6 to the queen on d8. The pin is relative: the knight may move, but if it does, Bxd8 wins the queen unless Black has something better.

This is one of the most common pins in chess, and you will meet it from both sides of your repertoire: Black plays this Queen's Gambit Declined set-up, and as White you will often see ...Bg4 pinning your knight on f3 to your queen.

> **COMMON MISTAKE** **Thinking a relatively pinned piece cannot move**
>
> The most expensive pin mistake at this level is treating a relative pin as if it were absolute. A piece pinned to the queen *can* move, and if it moves with a capture, a check or a mate threat, the "pin" may cost the pinner far more than the queen is worth. Before you rely on a pin, ask: "If the pinned piece moves anyway, what does it attack or threaten?"

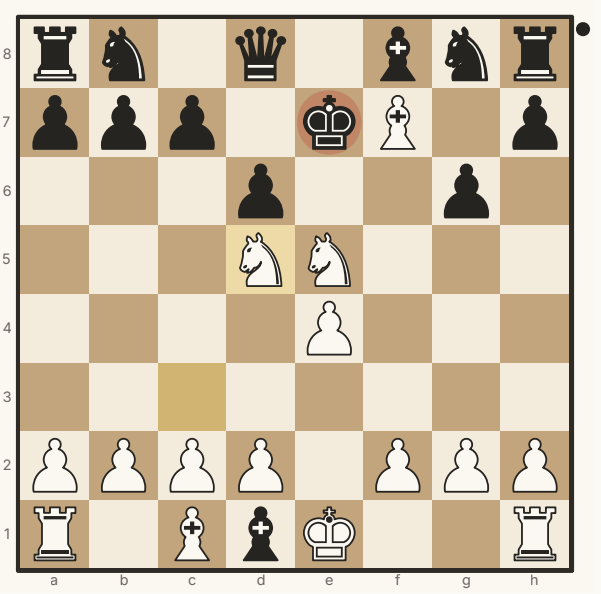
Two famous traps: the pinned piece that moved

The best way to remember the mistake above is to see it punished twice, once from each side.

The Légal trap

The trap is named after Légal, an eighteenth-century French master. Black pins White's knight to the queen, and White moves the knight anyway.

1. **e4 e5** 2. **Nf3 d6** 3. **Bc4 Bg4** 4. **Nc3 g6?** 5. **Nxe5! Bxd1??** 6. **Bxf7+ Ke7** 7. **Nd5#**



Black took the queen and was mated by three minor pieces. The knight on f3 was only pinned to the queen, so it was free to move.

Black's correct reply to 5.Nxe5 is 5...dxe5, and after 6.Qxg4 White has won a pawn. That is the price of the mistake 4...g6; it is a lot less than being mated. Exercise 08-05 asks you to find this defence.

The Elephant Trap

Now the same idea from Black's side, in the Queen's Gambit Declined. White thinks the knight on f6 is pinned and grabs a pawn on d5.

1. **d4 d5** 2. **c4 e6** 3. **Nc3 Nf6** 4. **Bg5 Nbd7** 5. **cxd5 exd5** 6. **Nxd5?? Nxd5!** 7. **Bxd8 Bb4+** 8. **Qd2 Bxd2+** 9. **Kxd2 Kxd8**

The Elephant Trap. Black's knight ignores the pin, and after the queen trade Black is a piece up.

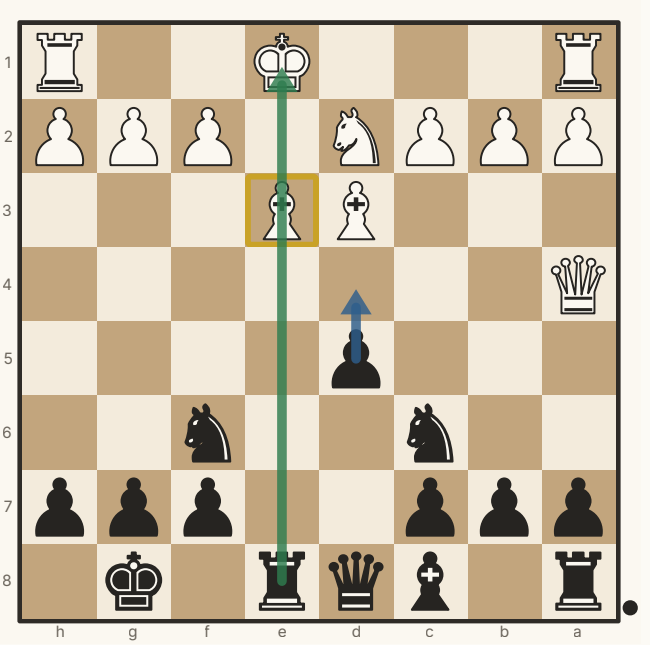
After 6...Nxd5 Black has won a knight, and the only thing hanging is Black's queen. If White takes it with 7.Bxd8, the check 7...Bb4+ forces 8.Qd2, Black wins White's queen back, and finishes a piece ahead. Exercises 08-06 and 08-11 test this trap from both sides.

Exploiting a pin

Once a piece is pinned, it is a sitting target. The pinned piece cannot run, so the usual plan is simple: **attack it again**.

Attack the pinned piece with a pawn

The most effective attacker is a pawn. A pawn is worth less than any piece, so if it attacks a pinned piece, the defender cannot solve the problem by "protecting" the piece: the pawn takes it anyway, and gives up only a pawn in return.



Black's rook on e8 pins the bishop on e3 to the white king. The pawn on d5 steps forward with ...d4, attacking the pinned bishop, and White cannot save it. Exercise 08-04 uses the same idea in a slightly different position.

How to recognize this chance:

1. Find every pinned piece on the board (for both sides).
2. For each one, ask: "Which of my pawns could attack it in one move?"
3. Check that the pawn's new square is not simply lost to a capture that also solves the problem, and that the opponent cannot break the pin with tempo first.

Pile up on the pinned piece

If no pawn is available, attack the pinned piece with more pieces than can defend it. A pinned knight on f6 attacked by a bishop on g5 can be hit again by a knight on d5 or e4, or by a rook on the f-file. Because the pinned piece cannot leave, every new attacker counts. A typical example is the Bg5 pin on a knight on f6: White adds Nd5, and if the pawn on e4 can advance to e5, the knight is attacked by a pawn as well. Black's usual defences are to interpose a piece (...Be7), to ask the bishop a question with ...h6, or to move the queen off the d8-h4 diagonal so that the knight is free again.

What goes wrong when you exploit a pin

Three errors waste most pins at this level.

- **Releasing the pin too early.** Capturing the pinned piece with the pinner (for example Bxc6+ or Bxf6) often just trades a bishop for a knight. The pin is worth most while it lasts, because it ties your opponent down. Capture only when it wins material or leaves real damage. In Exercise 08-09 the strong plan keeps the bishop on the pinning diagonal.
- **Giving the opponent time to castle.** An absolute pin against an uncastled king disappears the moment the king castles. If you have a pawn that can attack the pinned piece, use it *now*, before the king escapes. Exercise 08-03 is exactly this race.
- **Forgetting the counter-attack.** The pinned side may answer your attack with a threat of their own: a check, an attack on your queen, or a capture of the pinner. Before you push the pawn, look at your opponent's forcing moves just as you would for a fork.

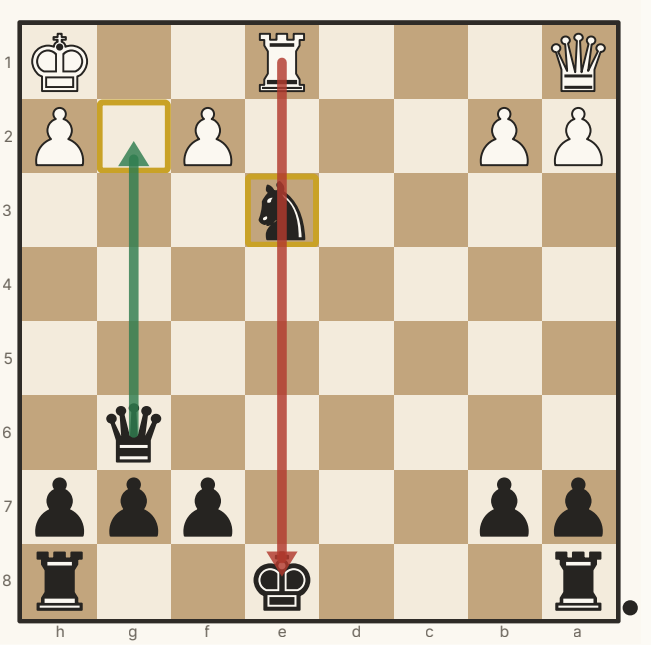
Pinned pieces do not defend

Whenever you count attackers and defenders (as in Chapter 3), **leave out every absolutely pinned defender**. A piece pinned to the king cannot capture anything off the line. A relatively pinned defender can capture, but only at the cost of the piece behind it, so count it with care.

The danger: a pinned piece is not a dead piece

Pinned pieces lose their freedom to move, but they keep some powers. Two of them catch players out again and again.

A pinned piece still controls squares for king moves. The king may never move onto a square attacked by an enemy piece, and it does not matter whether that enemy piece is pinned. A pinned piece can therefore still "defend" a square next to the enemy king.



The knight on e3 is pinned to the black king by the rook on e1. But it still guards g2, so ...Qg2 is checkmate: the white king cannot capture a queen that is protected, even by a pinned piece.

30... **Qg2#**

The pinned knight defends the mating square.

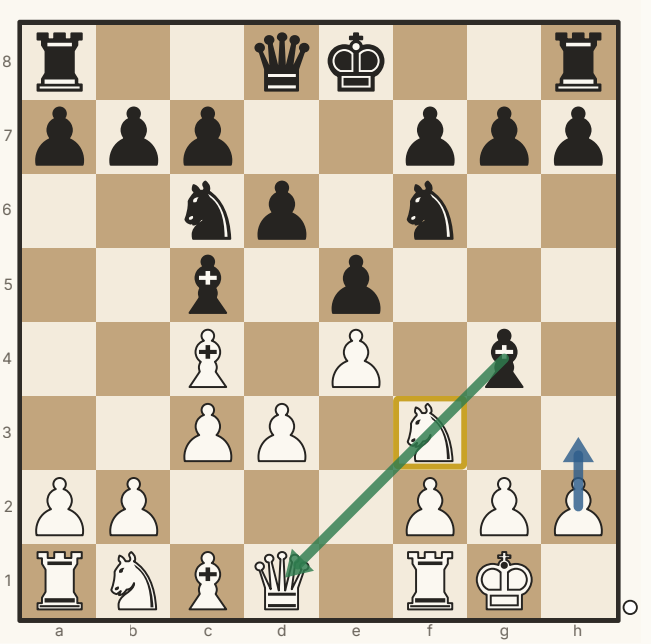
A pinned piece can still give check, and can move along the pin line. A rook pinned on a file can slide up and down that file, and may capture the pinner if it is unprotected. A pinned bishop can capture a pinning bishop on the same diagonal. Always check whether the pinned piece can simply take the pinner.

Breaking a pin

When one of your own pieces is pinned, you have four kinds of answer. Look for them in roughly this order, and look early: a pin is usually easy to break *before* the opponent has piled up on it.

1. **Counter-attack the pinner.** Ask the pinning piece a question: ...h6 against a bishop on g5, or a3 against a bishop on b4. The pinner must take, retreat, or keep the pin and allow a pawn advance like ...g5.

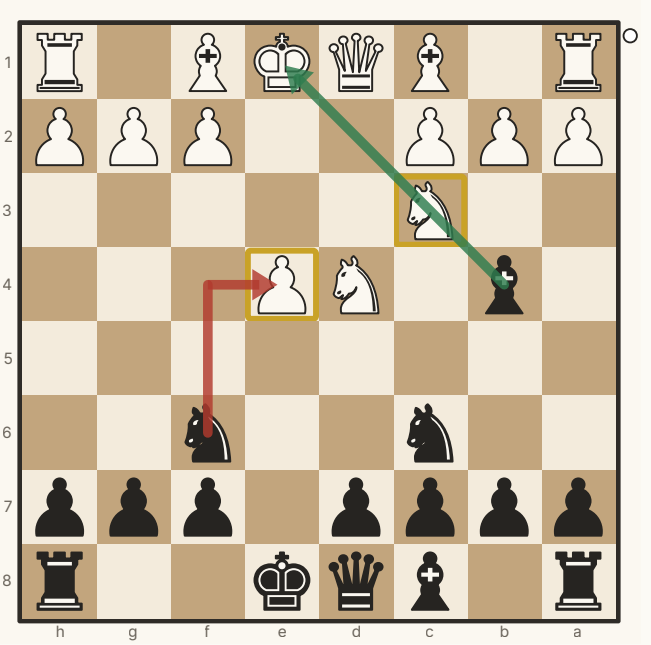
2. **Interpose a piece** between the pinner and the piece behind. Against Bg5 pinning a knight on f6, ...Be7 means the knight is now only "pinned" to a bishop.
3. **Move the piece behind.** Moving the queen off the line, or castling the king off the e-file, turns the pinned piece into a free piece. Castling is the easiest unpinning move in chess.
4. **Unpin with tempo.** The best way to move the back piece is with a threat of your own: a queen move that gives check, or attacks something, so the opponent has no time to exploit the pin.



A Giuoco Pianissimo position with ...Bg4 pinning White's knight on f3 to the queen. There is no immediate danger. White can ask the bishop a question with h3, or calmly continue with Re1 and Nbd2 and break the pin later.

The ...Bb4 pin

The mirror image of the Bg5 pin is the bishop pin on c3 (or ...Bb4 by Black against a knight on c3). When White's d-pawn has left d2 and the king is still on e1, the pin is absolute, and the knight stops defending the squares around it.



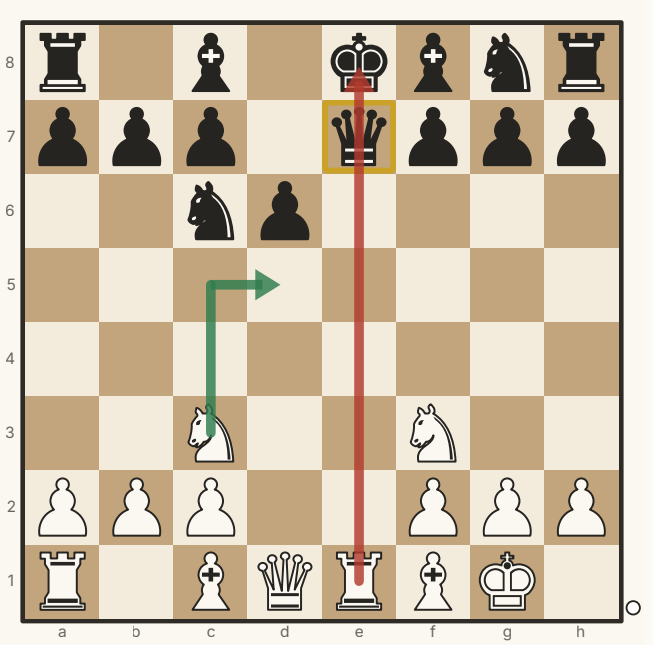
The Scotch Four Knights after 1.e4 e5 2.Nf3 Nc6 3.d4 exd4 4.Nxd4 Nf6 5.Nc3 Bb4. The knight on c3 is pinned to the king, so it no longer really defends e4: Black threatens ...Nxe4.

This is part of your repertoire as Black (see Chapter 13). White's best answer is 6.Nxc6 bxc6 7.Bd3, defending e4 with a piece that is not pinned. A careless developing move such as 6.Be2 allows 6...Nxe4, because the pinned knight cannot take back.

Pins along files

Rooks and queens pin along files and ranks. The two lines to watch most are:

- **The e-file toward an uncastled king.** Once the e-pawns have gone, any piece on the e-file in front of a king on e1 or e8 can be pinned by a rook or queen. Castle early.
- **The file or diagonal between a queen and a king.** A queen standing on the same line as its own king, with only one piece between them, is a pin waiting to happen.



Black's queen on e7 stands between the rook on e1 and the king on e8. It is absolutely pinned. Nd5 attacks it a second time, and Black must give the queen for a rook.

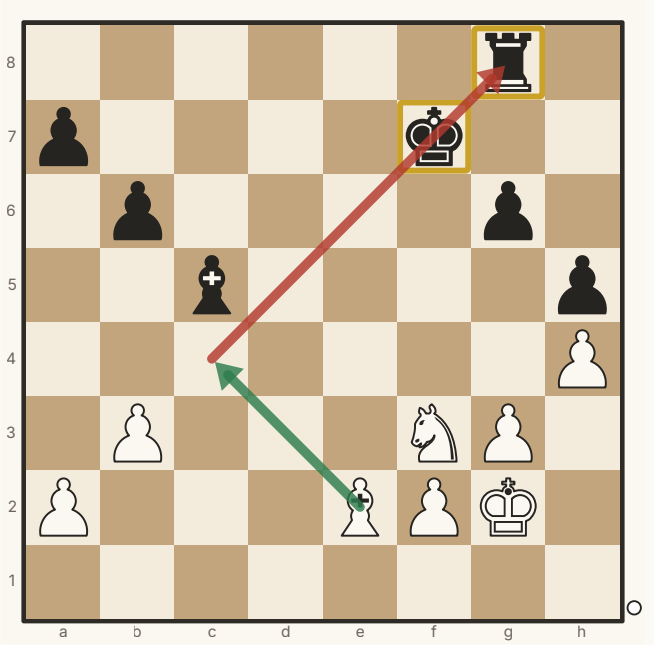
A queen pinned to a king is lost as soon as a cheaper piece attacks it. The defence is prevention: never put your queen on the same line as your king when an enemy rook or bishop can reach that line.

Skewers

A **skewer** is a pin in reverse. The more valuable piece stands *in front*. It is attacked, it must move (or must move because it is the king in check), and the piece behind it is captured.

Skewers are most deadly when the front piece is the king, because a check must be answered. The classic targets are:

- **King and rook** on the same rank, file or diagonal.
- **King and queen**, which usually wins the queen outright.
- **Queen and rook**, where the queen must move and the rook falls.



The king on f7 and the rook on g8 share the diagonal a2-g8. Bc4+ checks the king; when it steps away, Bxg8 wins the exchange.

Skewers that fail

A skewer needs the piece behind to be worth taking. If the rook behind the king is protected, "winning" it costs your skewering piece, and you have only traded. Check also that the front piece has no counter-blow: a king in check may be able to capture the checking piece, or block the check with a piece that defends the target behind it. And a queen attacked by a skewer may be able to move with check, which gives your opponent time to save the piece behind.

How to spot skewers

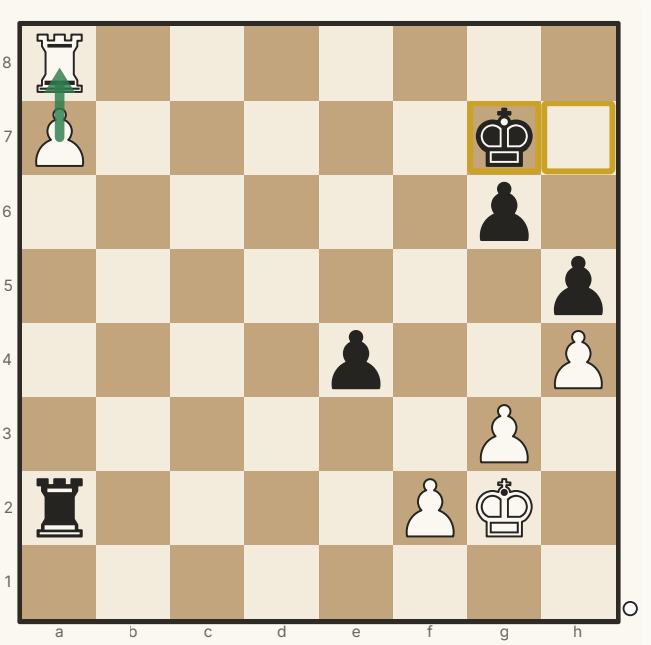
Skewers come from looking along lines *through* the king. For each enemy king:

1. Look along all eight lines from the king (two diagonals, the rank and the file, in both directions).
2. Note any valuable enemy piece standing on one of these lines beyond the king with nothing in between.
3. Ask whether one of your line pieces can give check on that line from the other side, on a safe square that cannot be blocked.

The same scan finds pins: if the piece on the line is *between* you and the king, it is a pin; if the king is between you and the piece, it is a skewer.

Rook-and-king skewers in endgames

In rook endgames, kings and rooks often stand on open ranks and files, and the skewer is one of the most important tactical ideas. The best-known version is the skewer that promotes a pawn.



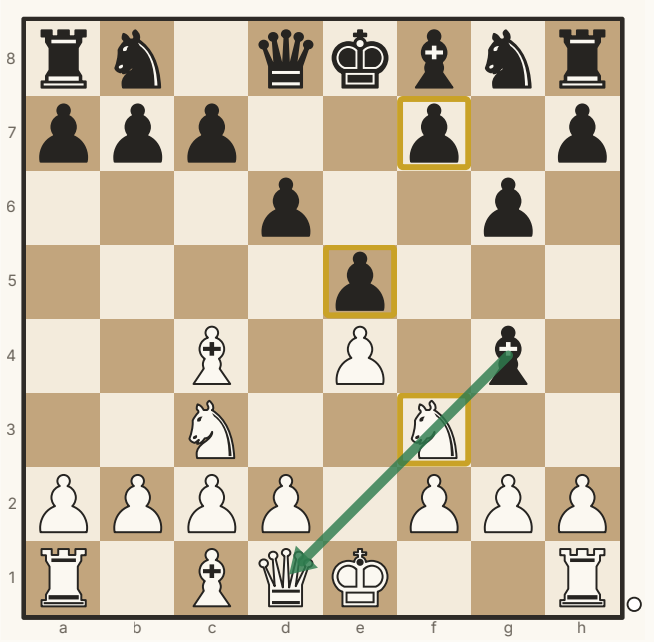
White's rook stands in front of the a-pawn, Black's rook behind it. With the black king on g7, Black has good drawing chances: the king covers h7 and g7, so the skewer trick does not work.

The defending king belongs on g7 or h7 in this kind of position. If it wanders onto another square of the seventh rank, such as f7 or e7, White has a trick: the rook leaves a8 sideways, for example Rh8, threatening to promote. If Black takes the pawn with the rook, Rh7+ skewers the king and the rook on a7. Exercise 08-07 shows the idea in action.

The general lesson for endgames: **whenever your king and rook stand on the same line, ask whether your opponent can check along it.** Many rook endgames are lost not by deep mistakes but by one careless king move onto the rook's rank or file.

Worked example: is the pinned knight really pinned?

Here is the position just before the Légal trap. Black has pinned White's knight to the queen with ...Bg4 and has just played 4...g6. White to move. Follow the thinking.



White to move. The knight on f3 is pinned to the queen on d1, but only relatively. The pawn on e5 is defended by the d6 pawn.

Step 1: what is pinned, and how? The knight on f3 is pinned to the queen. It is a relative pin, so the knight may legally move.

Step 2: checks, captures, threats. White's captures are Nxe5 and Bxf7+. Bxf7+ Kxf7 just loses a bishop for a pawn. Nxe5 takes a pawn and unpins the knight by moving it, but it hangs the queen on d1.

Step 3: what does the moving piece threaten? After Nxe5, the knight attacks f7. Together with the bishop on c4 and the knight on c3, which can jump to d5, White has a mating pattern if the black king is forced to e7.

Step 4: the opponent's best reply. If Black takes the queen with 5...Bxd1??, then 6.Bxf7+ Ke7 7.Nd5# is mate, as the line above showed. Black's best is 5...dxe5, and after 6.Qxg4 White has won a clean pawn and Black has a weakened kingside.

5. **Nxe5! dxe5** 6. **Qxg4**

With best defence, the "pinned" knight's move wins a pawn. If Black grabs the queen instead, it is mate.

Step 5: blunder-check. Is there anything Black can do that refutes 5.Nxe5? The checks for Black are none; the only big capture is the queen, which loses. So 5.Nxe5 is safe and strong.

The lesson works in both directions. As the pinner, never assume a relatively pinned piece will stay still. As the pinned side, look for the moment when moving the pinned piece wins more than it loses.

How to spot pins and skewers in your own games

- **Same-line scan.** Whenever two enemy pieces, or an enemy piece and the king, stand on the same rank, file or diagonal, look for your bishop, rook or queen reaching that line.
- **Pinned-defender check.** When you count the defenders of a square, ask of each defender: "Is it pinned?" Leave out pieces pinned to the king.
- **Uncastled king on an open file.** Look for pieces on the e-file in front of the king, for both sides.
- **King and rook in the endgame.** Before every king move in a rook ending, check whether the king will stand on the same line as your own rook.
- **Before you rely on a pin.** Ask what the pinned piece would attack if it moved anyway.

After each game, look for one moment where a pin or skewer was available or was allowed. Log it in your mistake journal under "missed-tactic" or, if a pinned piece fooled you, under "calculation".

> IN YOUR NEXT GAME **Your pin checkpoint**
>
> In your next game, every time your opponent puts a bishop, rook or queen on a line that runs toward your king or queen, stop and ask two questions: "What is between them?" and "Can I break the line now, before it costs me anything?"

Practice at the board

Set up each position on a board or in any free chess app. Decide your answer before you move anything.

1. **Absolute or relative?** Set up `r1bqkbnr/ppp2ppp/2np4/1B2p3/4P3/5N2/PPPP1PPP/RNBQK2R w KQkq - 0 4` and `rnbqkb1r/ppp2ppp/4pn2/3p2B1/2PP4/2N5/PP2PPPP/R2QKBNR b KQkq - 3 4`. For each, name the pinned piece and say whether it may legally move. Then find one way for the pinned side to break the pin.
2. **Attack with a pawn.** Set up `r1bqr1k1/ppp2ppp/2n2n2/3p4/Q7/3BB3/PPPN1PPP/R3K2R b KQ - 4 10`. Find Black's winning pawn move, then play every White reply you can think of and check that Black still wins a piece.
3. **The pinned defender.** Set up `r3k2r/pp3ppp/6q1/8/8/4n3/PP3P1P/Q3R2K b kq - 0 30`. Explain to yourself why ...Qg2 is mate even though the knight on e3 is pinned.
4. **Both traps.** Play through the Légal trap and the Elephant Trap from the start position three times each, once as the winner and once as the loser, until you can do it without the book.
5. **Rook-ending skewer.** Set up `R7/P5k1/6p1/7p/4p2P/6P1/r4PK1/8 w - - 0 45`. Move the black king from g7 to f7 and find White's win. Then put it back on g7 and convince yourself that the same trick fails.

Check yourself with an engine only after you have written down your answers.

Chapter summary

- A pin attacks a piece that cannot or should not move because a more valuable piece stands behind it. Only bishops, rooks and queens pin.
- In an absolute pin the piece behind is the king and the pinned piece may not leave the line; it defends nothing off the line. In a relative pin the pinned piece may move, and sometimes should.
- Exploit a pin by attacking the pinned piece again, ideally with a pawn, and by piling up attackers.
- Break a pin by counter-attacking the pinner, interposing, moving the piece behind (castling is the easiest), or unpinning with tempo. Do it early.
- A pinned piece still controls squares around the enemy king and can still move along the pin line or capture the pinner.

- The Légal trap and the Elephant Trap show the cost of assuming a relatively pinned piece will stay put.
- A skewer attacks a valuable piece in front, usually the king, to win the piece behind it. In rook endgames, watch every king move onto your own rook's line, and know the skewer trick that promotes a pawn.

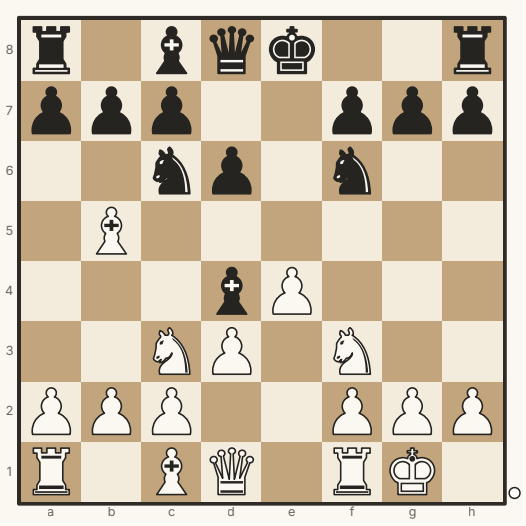
CHEAT CODE **The line check**

1. List every line (rank, file, diagonal) that runs through an enemy king or queen.
2. On each line, find what stands between, and what stands behind.
3. A piece between your line piece and the king or queen is pinned: attack it again, preferably with a pawn.
4. A king or queen in front of another piece can be skewered: find a safe check or attack along the line.
5. Before relying on a pin, ask what the pinned piece would do if it moved anyway.
6. Run the same scan for your own king and queen, and break dangerous pins early.

Exercises

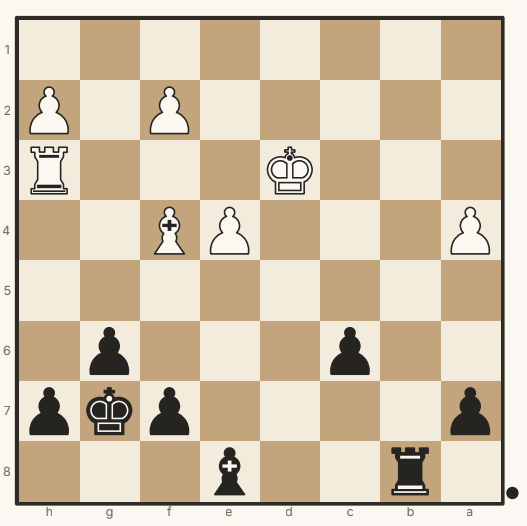
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

8.1 TACTIC



White to move. Black's bishop on d4 looks protected by the knight on c6. Is it really? Win material. Solution p. 655

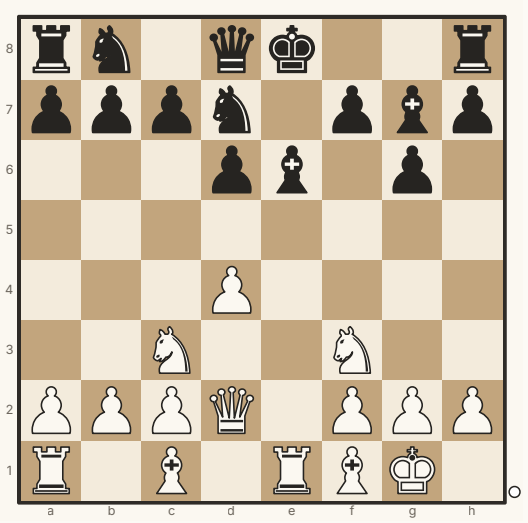
8.2 TACTIC



Black to move. The white king and rook stand on the same rank. Win material.

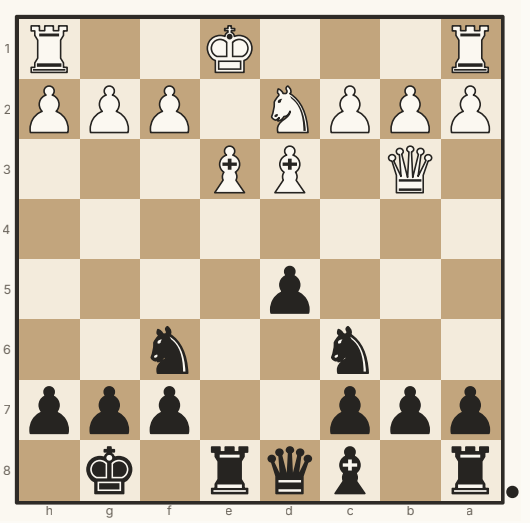
Solution p. 655

8.3 TACTIC



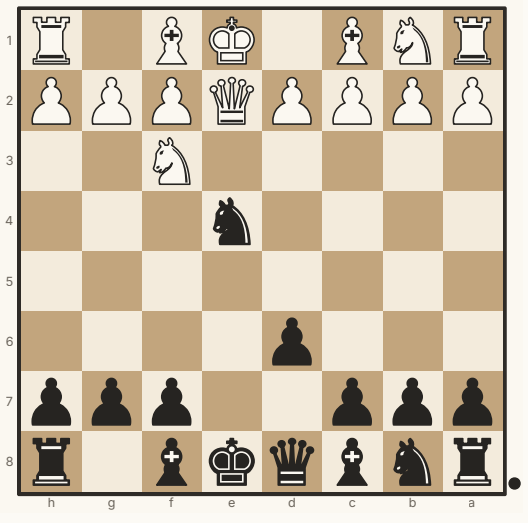
White to move. Black's bishop on e6 stands on the open e-file in front of the king. Exploit it. Solution p. 655

8.4 TACTIC



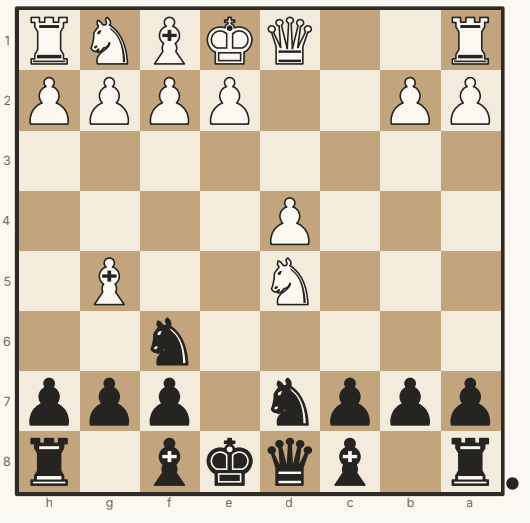
Black to move. White has not castled yet. Win material. Solution p. 655

8.5 DEFENCE



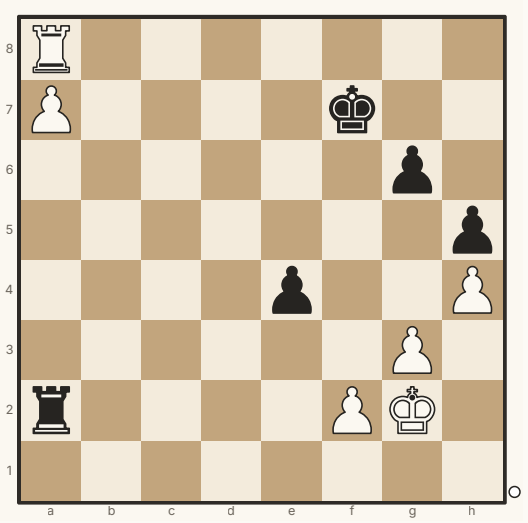
Black to move. In the Petrov (1.e4 e5 2.Nf3 Nf6 3.Nxe5 d6 4.Nf3 Nxe4), White has played 5.Qe2. Your knight on e4 is pinned to your king, and d2-d3 is coming. Find the only move that saves the knight. Solution p. 656

8.6 TACTIC



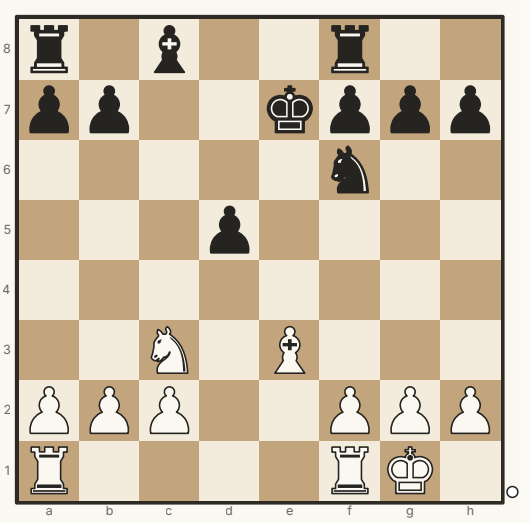
Black to move. White has just taken your d5 pawn with the knight, thinking your knight on f6 is pinned. Punish it. Solution p. 656

8.7 ENDGAME



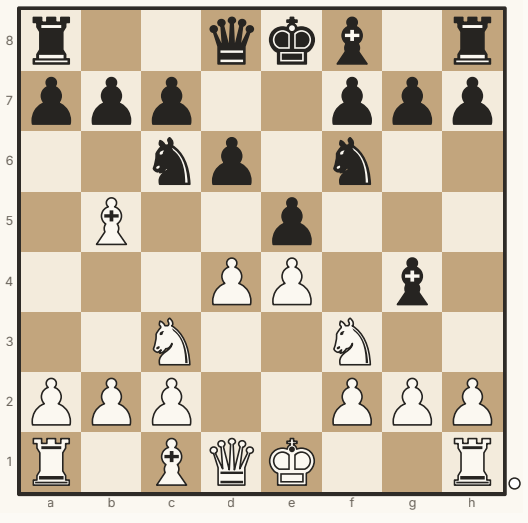
White to move. Your rook stands in front of your a-pawn, and Black's rook is behind it. Find the winning idea. Solution p. 656

8.8 TACTIC



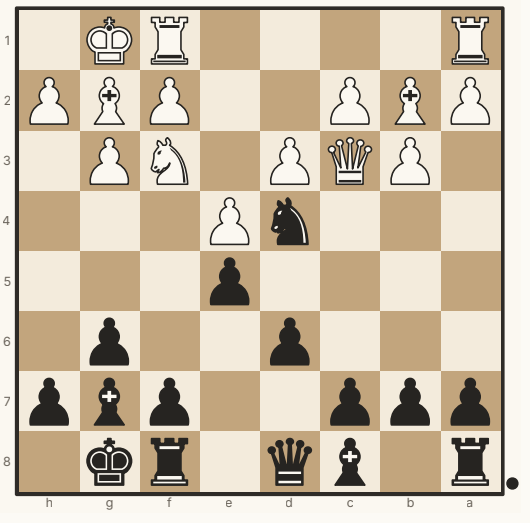
White to move. The queens are off and Black's king stands on e7. Win material.
Solution p. 656

8.9 TACTIC



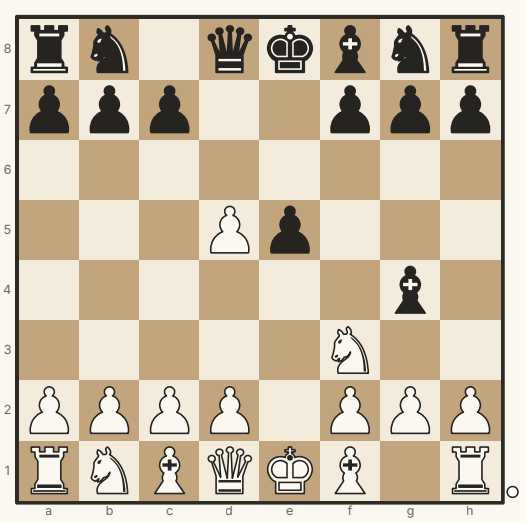
White to move. Your bishop on b5 pins the knight on c6 to the king. How do you exploit it? Solution p. 657

8.10 TACTIC



Black to move. Review from Chapter 7: find the move that wins the white queen.
Solution p. 657

8.11 BLUNDER CHECK



White to move. After 1.e4 e5 2.Nf3 d5 3.exd5, Black has pinned your knight to your queen with 3...Bg4. The e5-pawn has no defender. Is it safe to take? Play a good move. Solution p. 657

HINTS

8.1 Can the knight on c6 legally move?

8.2 Which check lines up the king with something behind it?

8.3 The pinned bishop cannot move. What is the cheapest piece you can attack it with?

8.4 Which white piece cannot move, and which pawn can attack it?

8.5 Defending a pinned piece is not enough when a pawn can attack it. Which piece can step onto e7?

8.6 Your knight is pinned to the queen, not to the king. What happens if you take on d5 and White takes your queen?

8.7 What happens if your rook leaves the a-file sideways and Black takes the pawn?

8.8 Which piece stands behind the black king on a diagonal?

8.9 Attack the pinned piece. If Black chases your bishop, how can you keep the pin?

8.10 Which square touches both g1 and c3 with a knight's jump?

8.11 If the knight on f3 moves, what can the bishop on g4 take?

CHAPTER 9 STAGE 600–800

Discovered Attacks and Double Check

The piece that moves and the piece that strikes

YOU WILL LEARN TO

- Recognize a battery (a front piece in front of a back piece on the same line) for both sides.
- Use discovered attacks and discovered checks, choosing the best square for the front piece.
- Understand why double check can only be answered by a king move, and use it to mate.
- Spot your opponent's batteries and defuse them before they fire.

> IN YOUR NEXT GAME
>
> Before every move, look along each of your bishops, rooks and queen to see whether one of your own pieces is standing in the way of a target; and look along your opponent's lines in the same way.

Your knight has been sitting on e5 for ten moves, blocking your own rook on e1. Your opponent's king is still on e8. One knight move, any knight move, and the rook gives check. Suddenly your knight has a free move: it can jump to a square where it attacks the queen, even a square where it could normally be captured, because your opponent must deal with the check first.

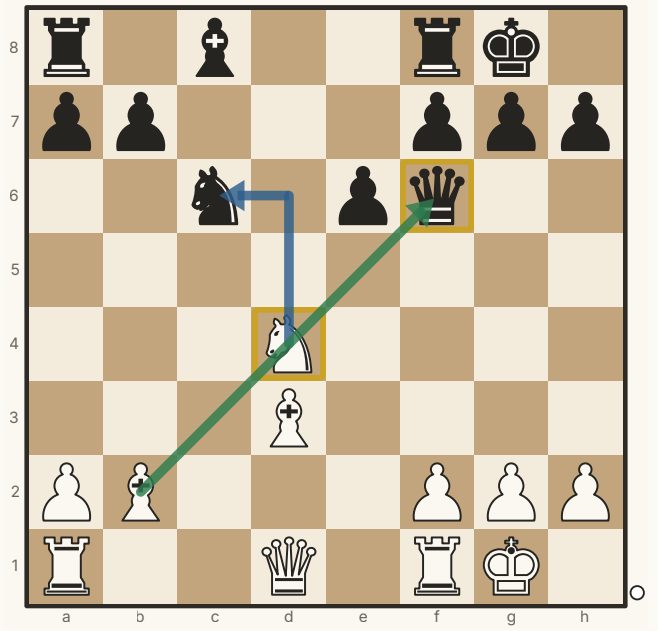
That is a **discovered attack**: you move one piece and uncover an attack by another. It is the most powerful of the basic tactics, because it creates two threats with one move, and one of them comes from a piece that did not move at all. This chapter shows you how discovered attacks work, why discovered check and double check are so strong, and how to see the hidden lines before your opponent uses them against you.

Anatomy of a discovered attack

A discovered attack needs two pieces of the same colour on one line:

- The **back piece** is a bishop, rook or queen. It is the piece whose attack will be uncovered.
- The **front piece** stands between the back piece and the target. It can be any piece, even a pawn or the king.

Together they form a **battery**: a line piece with a friendly piece in front of it, aimed at an enemy target. When the front piece moves, the back piece's attack is **discovered**. If the front piece also makes a threat with its move, the opponent faces two threats at once.



A battery on the long diagonal. The knight on d4 is the front piece and the bishop on b2 the back piece. Any knight move uncovers an attack on the queen on f6, so the knight can use its move to make a second threat.

The key idea is this: **the front piece gets a free move.** Your opponent must deal with the uncovered attack, so whatever the front piece does with its move, it does it "for free". If it captures something, threatens something, or gives check, your opponent cannot answer both.

What the front piece should do

Good uses of the front piece's move, roughly from best to worst:

1. **Give check.** Then the uncovered attack wins whatever it hits, because the check must be answered first.
2. **Capture something.** The front piece takes material, and the back piece's threat means the opponent has no time to recapture.
3. **Attack something valuable**, ideally a second piece that is loose or worth more than the front piece.
4. **Block or cut off the defence** of the target.

A front piece that moves to a "nice" square without a threat wastes the discovery. Your opponent just moves the attacked piece and nothing has been won.

> KEY IDEA **The two-threat test**
>
> A discovered attack only wins material if it creates **two** threats: the uncovered attack and the front piece's own threat. Before you play it, name both threats aloud. If you can only name one, your opponent will simply move the target.

Pawns and kings as front pieces

The front piece does not have to be a knight or a bishop. A **pawn** can be the front piece: a pawn advance that uncovers a rook or bishop, and at the same time attacks a piece, is a discovered attack with the cheapest possible unit. For example, with a white bishop on b2 behind a pawn on d4, the push d4-d5 can uncover an attack on the long diagonal while the pawn itself hits a knight on c6 or e6.

Even the **king** can be a front piece. In endgames a king standing between its own rook and the enemy king can step aside with a discovered check. The rule is the same as always: the move of the front piece should carry a threat of its own.

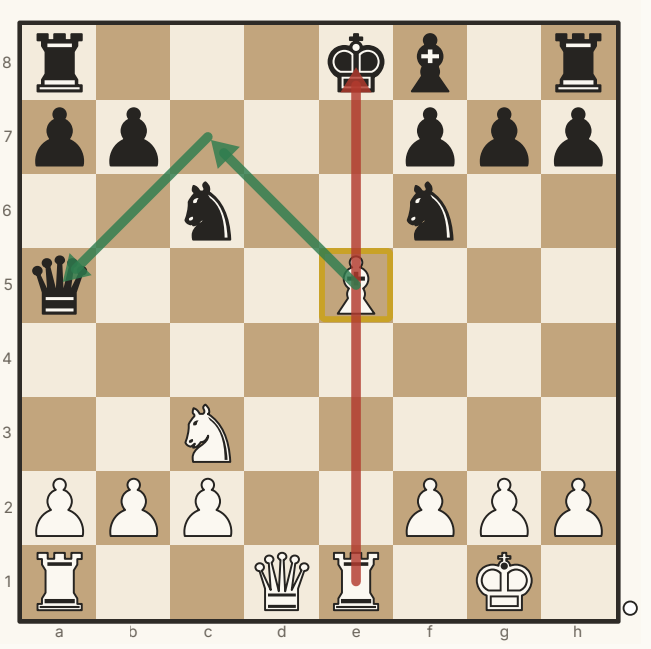
What commonly goes wrong

- **Firing too early.** A battery is often stronger as a threat than as a move. If the discovery wins nothing now, keep it loaded; your opponent must spend moves guarding against it, and later a better target may appear.
- **A front-piece move without a threat.** The opponent simply moves the target, and the battery has been used up for nothing.

- **Forgetting the opponent's reply.** If the uncovered attack is not a check, your opponent may answer with a check or a capture of their own before saving the target. Look at their forcing moves before you fire.
- **The target escapes with tempo.** If the attacked piece can move to a square where it attacks something of yours, it may gain the time it needs. Check where the target can go.

Discovered check

When the back piece's target is the **king**, the uncovered attack is a **discovered check**. This is the most dangerous version, for a simple reason: a check must be answered, so the front piece can go **anywhere**, even to a square where it is attacked, even to a square where it is defended twice. Your opponent cannot capture it while the king is in check, unless the capture itself also deals with the check.



The bishop on e5 blocks the rook's line to the king on e8. Every bishop move is check. Bc7+ attacks the queen on a5 through b6, and Black must answer the check first: after ...Be7, Bxa5 wins the queen.

When you have a discovered check, take your time. Look at **every square** the front piece can reach and ask what it attacks from there. Usually one square

stands out: the one that attacks the queen, or a loose piece, or that captures something valuable. Exercises 09-02 and 09-05 practise exactly this choice.

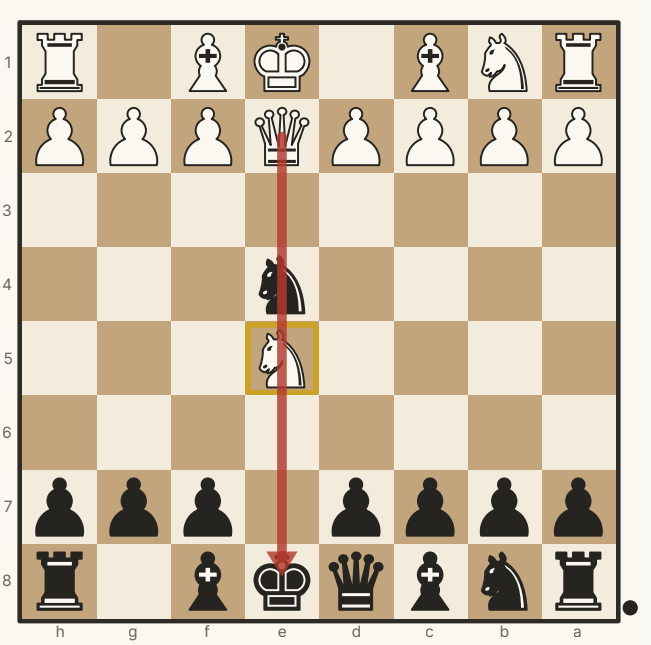
The ways a discovered check can fail

Discovered check is not automatically winning. Watch for three defences:

- **The king captures the front piece.** If the front piece moves next to the king onto an unprotected square, the king may take it and step off the line in the same move. In Exercise 09-02, ...Nxf2+ fails to Kxf2 for exactly this reason.
- **The checked side captures the back piece.** A check is answered by capturing the checking piece. If the back piece is loose and something can take it, the discovery fizzles.
- **A block that also defends.** The opponent may interpose a piece on the line that at the same time protects the target of the front piece.

A famous opening example

The Petrov Defence contains one of the best-known discovered-check traps. After 1.e4 e5 2.Nf3 Nf6 3.Nxe5, Black should play 3...d6 first. If Black copies White with 3...Nxe4?, then 4.Qe2 sets up a battery: queen behind knight on the e-file, aimed at the black king.



After 4.Qe2 the queen and the knight on e5 form a battery against the king on e8. Black must block the e-file at once with 4...Qe7. If the knight on e4 retreats, for example 4...Nf6??, then 5.Nc6+ is a discovered check that attacks the queen.

This book's White repertoire meets the Petrov with 3.Nxe5, so you will see this position. Remember the general lesson for both colours: do not grab pawns on the e-file while your king is still on it. Exercise 09-03 lets you play the winning discovered check.

Double check

A **double check** happens when the front piece gives check *and* uncovers a check from the back piece, so the king is in check from two pieces at once.

Double check is the most forcing move in chess. Against a single check you have three possible answers: move the king, capture the checking piece, or block the check. Against a double check, **only a king move works**. You cannot capture two pieces with one move, and you cannot block two lines with one piece.

That has two consequences:

1. **The front piece cannot be captured**, no matter how many times it is attacked. A knight can land on a square attacked by two pawns, and it does not matter.

2. **If the king has no safe square, it is checkmate**, even if every checking piece could otherwise be taken.

How to find double checks

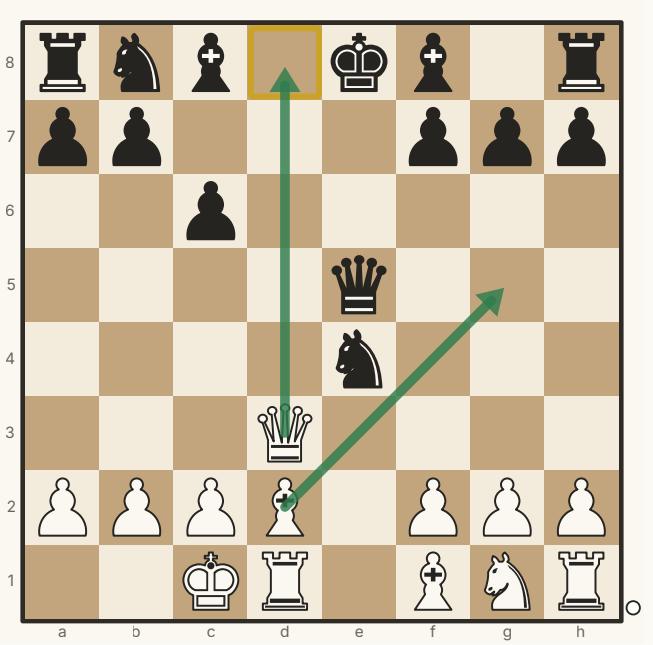
A double check needs a front piece that can give check *from its new square*. So when you have a discovered check available, ask one extra question: **"Can the front piece land on a square from which it also checks the king?"** For a knight in front of a rook on the e-file, those are the squares a knight's jump from the king; for a bishop in front of a rook, the diagonal squares next to the king's diagonals. Because the front piece cannot be captured, you can choose a square that is attacked two or three times. What matters is only whether the king has an escape square afterwards.

Double check to mate

The most famous double check in chess history is a short game played in Vienna in 1910.

1. **e4 c6** 2. **d4 d5** 3. **Nc3 dxe4** 4. **Nxe4 Nf6** 5. **Qd3 e5** 6. **dxe5 Qa5+** 7. **Bd2 Qxe5** 8. **O-O-O Nxe4??** 9. **Qd8+!! Kxd8** 10. **Bg5+ Kc7** 11. **Bd8#**

Réti vs. Tartakower, Vienna 1910. The queen sacrifice drags the king to d8, and 10.Bg5+ is a double check from the bishop and the rook on d1. After 10...Ke8 11.Rd8# is also mate.

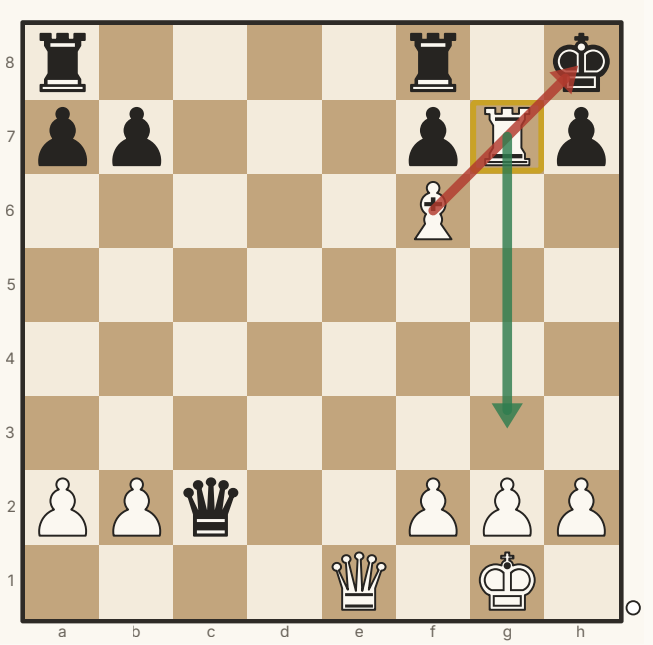


The position before 9.Qd8+. White's rook on d1 is behind the queen on the d-file, and the bishop on d2 is ready to swing to g5. The queen sacrifice puts the king on the square where both pieces can check it at once.

Notice the pattern: the back piece (the rook on d1) and the front piece (the bishop on d2) were already lined up. The queen sacrifice simply put the king where both could reach it. Exercises 09-07 and 09-09 contain double checks that mate in the same way.

The windmill: discovered check again and again

When a rook and a bishop form a battery against a king in the corner, something remarkable can happen. The rook gives check from g7 (or g2), the king steps into the corner, and then the rook moves *along the rank*, capturing a piece, while the bishop behind it gives discovered check. The king steps out, the rook returns with check, and the cycle repeats. Each turn of this **windmill** wins another piece.



The rook on g7 is the front piece and the bishop on f6 the back piece, aimed at the king on h8. Every rook move is discovered check. Here a rook move down the g-file, such as Rg3, is even checkmate: g8 and g7 are both covered.

You do not need to memorize windmill positions. Remember the shape: **a rook and a bishop (or queen) lined up on the enemy king, with the rook able to step out and back in.** When you see it, count how many pieces the front piece can collect with discovered checks before the king escapes.

Spotting batteries: yours and your opponent's

Discovered attacks are missed because the line is "blocked". Your eye sees a bishop behind your own knight and thinks the bishop is doing nothing. Train yourself to see through your own pieces.

The common batteries are:

- **Queen behind bishop** on a diagonal, for example a queen on c2 and a bishop on d3 aimed at h7.
- **Rook behind knight** on a file, for example a rook on e1 and a knight on e5 aimed at the king or queen on e8.
- **Rook or queen behind bishop** on a file, for example a rook on d1 behind a bishop on d3 aimed at a queen on d6.

- **Bishop behind knight** on a diagonal, for example a bishop on b7 behind a knight on e4 aimed at a king on h1.
- **Rook behind pawn**, where a pawn push uncovers the rook (less common, but it happens on open files).

A classic: queens facing each other on a file

One battery appears in game after game. The queens face each other on an open d-file, and one side has a bishop standing between them, for example a black bishop on d6 between the queens on d8 and d1. If that bishop moves with a threat, such as a capture with check on h2 or a capture of an unprotected piece, the queen behind it attacks the enemy queen, which is lost unless it is defended. The same is true with a rook behind the bishop instead of a queen. So whenever the queens or a queen and a rook share a file with one minor piece between them, both players must check: whose battery is it, and what can the front piece capture or check? Exercises 09-01 and 09-04 are built on this pattern.

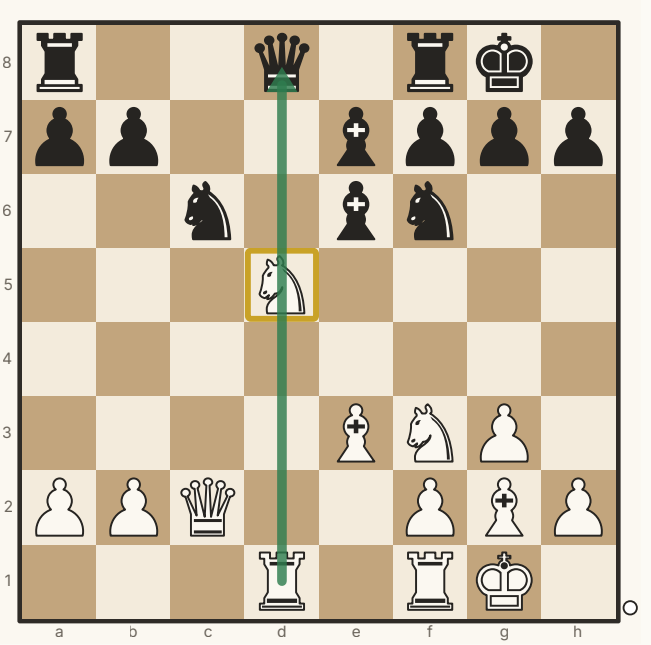
The battery scan

Once per move, run this short scan:

1. **For each of your bishops, rooks and queen**, follow its lines *through* your own pieces. Is there an enemy king, queen, rook or loose piece behind one of your pieces? If so, you have a battery.
2. **For that battery**, list every move of the front piece that comes with a threat: checks first, then captures, then attacks.
3. **Do the same for your opponent.** Follow each enemy line piece through *its* own pieces toward your king and queen.

Worked example: which knight move?

Here is a typical middlegame position with a battery on the d-file. Follow the thinking process.



White to move. The rook on d1 and the black queen on d8 are on the same file, and White's own knight on d5 stands between them.

Step 1: find the battery. Looking along the d-file from the rook on d1, the first piece is my own knight on d5, and behind it is the black queen on d8. So any knight move uncovers an attack on the queen.

Step 2: list the knight moves with threats. Checks first: Nxe7+ captures the bishop on e7 with check, and Nxf6+ captures the knight on f6 with check. Then captures and attacks: Nb6 attacks the rook on a8, Nc7 attacks the rook on a8 and the bishop on e6 (but is not check).

Step 3: what does my opponent do against each? After Nxe7+, Black recaptures with ...Qxe7, and the queen has left the d-file *with the recapture*. The discovery has fizzled: White keeps some initiative, but the queen is safe. After Nxf6+, Black must answer the check, and the only capture is ...Bxf6, which does not move the queen. Now Rxd8 takes the queen.

14. **Nxf6+ Bxf6** 15. **Rxd8 Rfxd8**

The right knight move: a capture with check that does not let the queen escape. White has won the queen for a rook and a knight.

Step 4: blunder-check. After 15...Rfxd8, White has queen against rook and knight. Can Black win anything back immediately? The bishop on e6 attacks a2 and the rooks share the d-file, but nothing of White's is loose enough to fall at once. White is clearly winning.

The lesson: **the best front-piece move is one that leaves the target no escape.** A capture with check is ideal, but only if your opponent's answer to the check does not also move the target off the line.

Defending against discovered attacks

Discovered attacks are hard to defend once they are played, because they come with two threats. The defence happens *before* they are played.

- **Move the target off the line.** If your queen stands on a file or diagonal behind an enemy piece that is aimed at it, move the queen. Exercise 09-06 asks you to do exactly that.
- **Capture or attack the front piece.** If you can take the front piece, the battery is gone. This is often the cleanest defence (Exercise 09-11).
- **Block the line with a protected piece** between the back piece and your target, so that even if the front piece moves, the line stays closed.
- **Keep your king off open lines.** An uncastled king on the e-file is the classic target for discovered checks. Castle early, and do not open the e-file while your king is still on it (the Petrov trap above).
- **Watch the "quiet" enemy piece.** A bishop or rook that seems to be blocked by its own piece is not asleep. Ask: "What happens if the piece in front of it moves?"

> COMMON MISTAKE **Only looking at pieces that can move**
>
> The most common reason players miss discovered attacks, for both sides, is that they scan only the moves a piece can make, not the lines a piece controls *behind* other pieces. A bishop on b2 with your own knight on d4 in front of it still "sees" f6. Always follow a line piece's lines through friendly pieces.

How to spot discovered attacks in your own games

- **Look for lined-up pieces.** Any time two of your pieces stand on one line with an enemy target beyond them, mark it. The same for your opponent.
- **Check the king's lines.** Before every move, look along the file, rank and diagonals of both kings. A friendly piece between a line piece and the enemy king is a discovered check waiting to happen.
- **Favour moves with tempo for the front piece.** When you have a battery, list the checks and captures of the front piece first.
- **After the game**, go through your moves and find every battery that existed for either side. Note which were used, which were missed, and which you allowed. Log missed ones in your mistake journal under "missed-tactic".

> IN YOUR NEXT GAME **Your battery checkpoint**
>
> In your next game, at least once every five moves, follow each of your bishops, rooks and queen to the edge of the board, *through* your own pieces. Then do the same for your opponent's pieces aimed at your king and queen.

Practice at the board

Set up each position on a board or in any free chess app. Work out your answer before you move a piece.

1. **The free move.** Set up `r3kb1r/pp3ppp/2n2n2/q3B3/8/2N5/PPP2PPP/R2QR1K1 w kq - 0 12`. List every bishop move. For each, say what it attacks in addition to the discovered check. Which is best?
2. **Which knight move?** Set up `r2q1rk1/pp2bppp/2n1bn2/3N4/8/4BNP1/PPQ2PBP/3R1RK1 w - - 0 14`. Play Nxe7+ and Nxf6+ in turn and compare what White wins.
3. **The Petrov trap.** From the start position, play 1.e4 e5 2.Nf3 Nf6 3.Nxe5 Nxe4 4.Qe2. Try Black's possible fourth moves and find the one that keeps the material balance (4...Qe7). Then play 4...Nf6 and find White's win.
4. **Double check.** Play through Réti vs. Tartakower until you can play it from memory. Then set up the position before 9.Qd8+ and try other White moves to see why only the queen sacrifice works so quickly.

5. **Battery hunting.** Take any of your recent games. At move 10, 15 and 20, find every battery on the board for both sides.

Check yourself with an engine only after you have written down your answers.

Chapter summary

- A discovered attack moves a front piece and uncovers an attack by a back piece (a bishop, rook or queen) on the same line. Together they form a battery.
- The front piece gets a free move: use it to give check, capture, or attack something, so your opponent faces two threats.
- In a discovered check the front piece can go anywhere, even to an attacked square, because the check must be answered first. Check every square it can reach.
- A double check can only be answered by a king move. The checking pieces cannot be captured, and if the king has no square, it is mate.
- The windmill is a repeated discovered check by a rook and bishop against a cornered king.
- Spot batteries by following each line piece through your own pieces. Do the same for your opponent.
- Defend by moving the target off the line, capturing the front piece, blocking with a protected piece, and keeping your king off open lines.

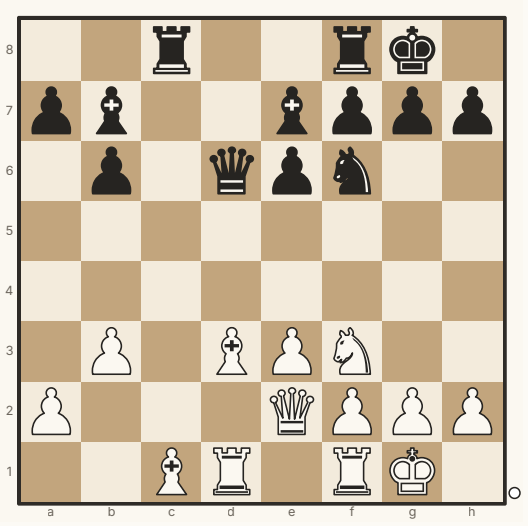
CHEAT CODE **The battery check**

1. Follow each of your bishops, rooks and queen through your own pieces: is an enemy king, queen, rook or loose piece behind?
2. If so, list the front piece's moves: checks first, then captures, then attacks.
3. Choose the move that leaves the target no escape and no counter-capture.
4. If the king is the target, check whether the front piece can also give check: a double check forces a king move.
5. Then run the same scan on your opponent's line pieces and defuse any battery aimed at your king or queen.

Exercises

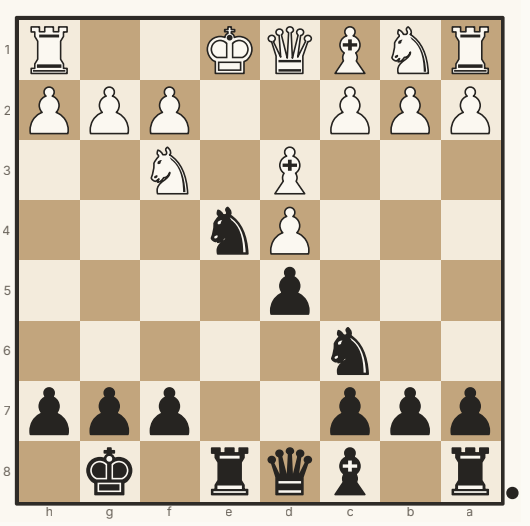
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

9.1 TACTIC



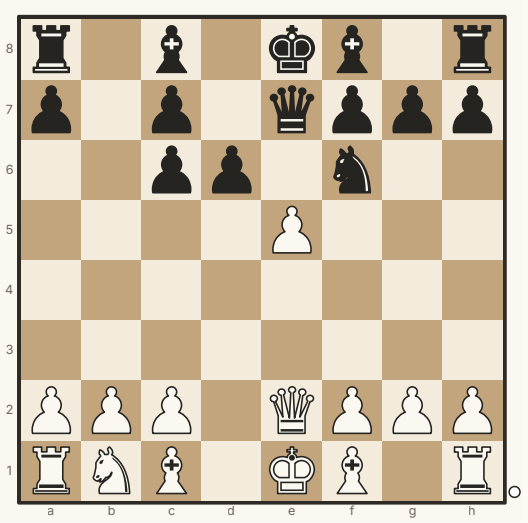
White to move. Your rook on d1 and the black queen on d6 are on the same file, with only your bishop between them. Win material. Solution p. 658

9.2 TACTIC



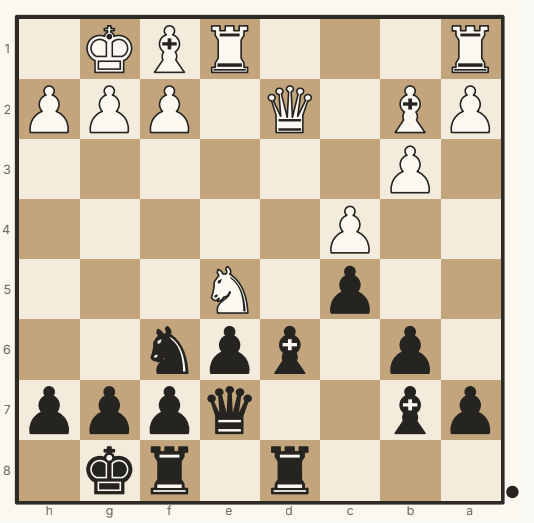
Black to move. White's king is still on e1 and your rook stands on the open e-file behind your knight. Win material. Solution p. 658

9.3 OPENING



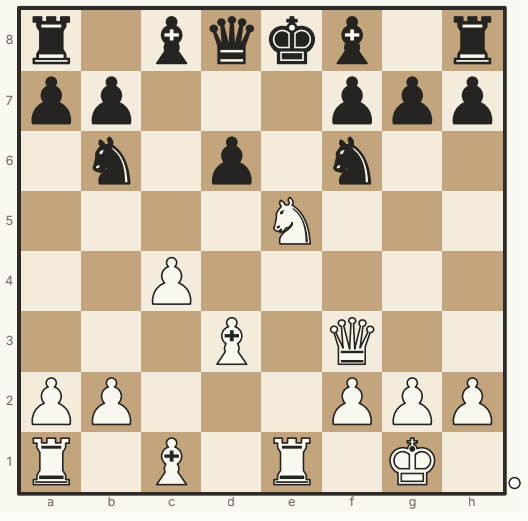
White to move. In the Scotch (1.e4 e5 2.Nf3 Nc6 3.d4 exd4 4.Nxd4 Nf6 5.Nxc6 bxc6 6.e5 Qe7 7.Qe2), Black attacked your e5-pawn with 7...d6?. Punish it. Solution p. 658

9.4 TACTIC



Black to move. Your rook on d8 and White's queen on d2 face each other on the d-file, with your bishop on d6 between them. Win material. Solution p. 658

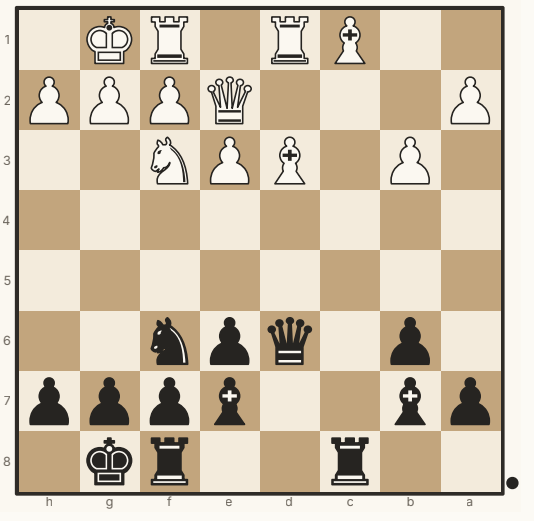
9.5 TACTIC



White to move. Black's king is still on e8 and your knight on e5 is attacked by the d6 pawn. Find the strongest knight move.

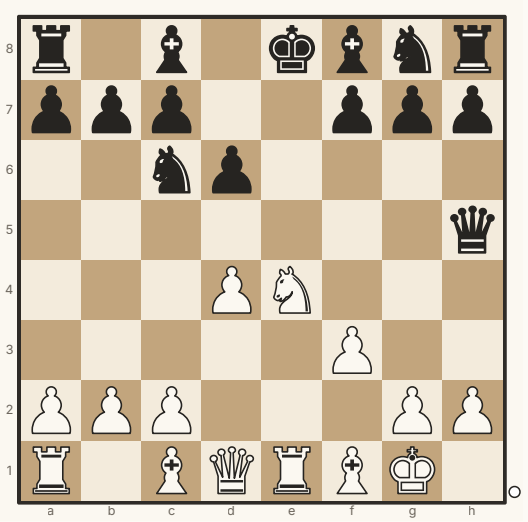
Solution p. 659

9.6 DEFENCE



Black to move. White's rook on d1 is aimed at your queen through the bishop on d3. Find a safe move. Solution p. 659

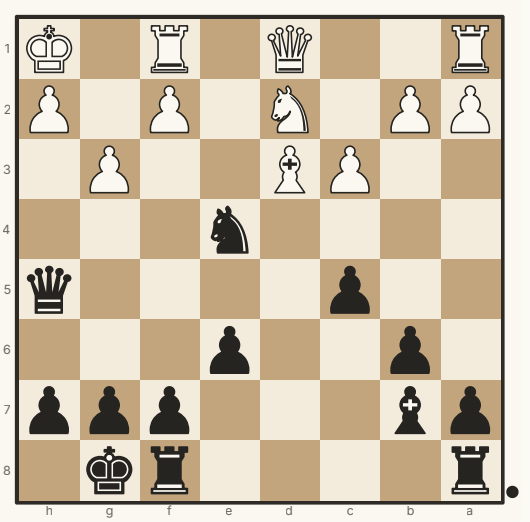
9.7 CHECKMATE



White to move. The black king is on the open e-file. Find checkmate in 2.

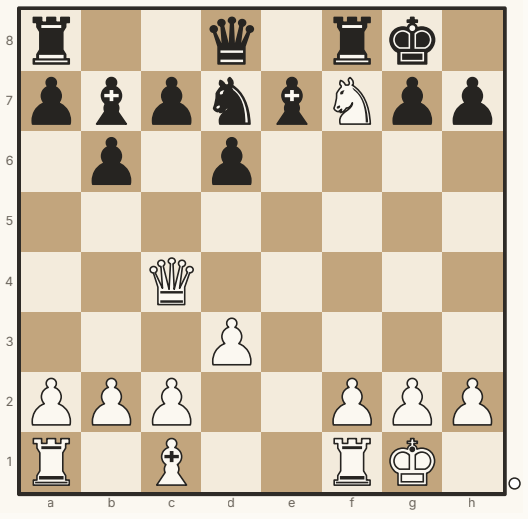
Solution p. 659

9.8 CHECKMATE



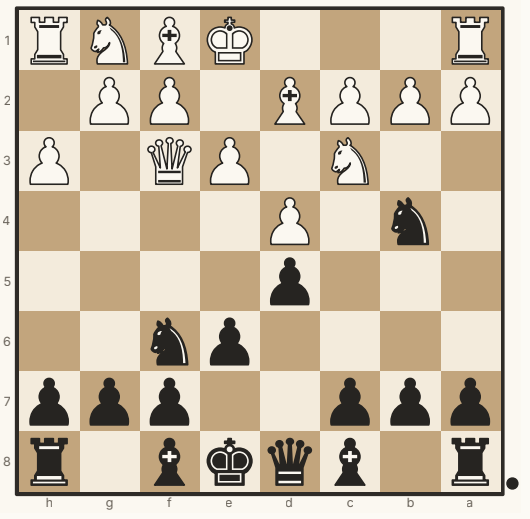
Black to move. Your bishop on b7 is aimed at the white king through your knight on e4. Find checkmate in 2. Solution p. 659

9.9 CHECKMATE



White to move. Your knight on f7 blocks your queen's diagonal to g8. Find checkmate in 3. Solution p. 660

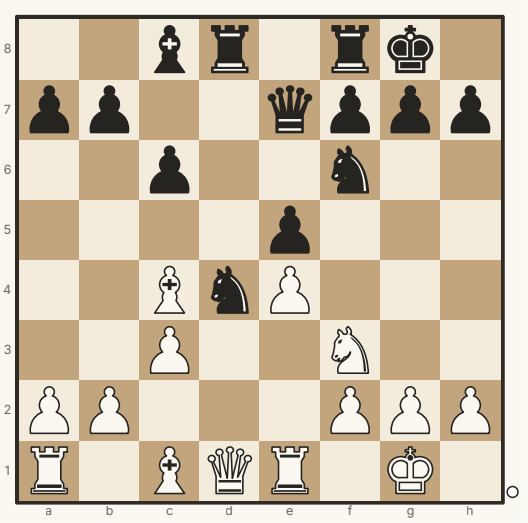
9.10 TACTIC



Black to move. Review from Chapter 7: White's queen has left d1. Win material.

Solution p. 660

9.11 BLUNDER CHECK



White to move. Black's knight has jumped to d4, attacking your knight on f3 and standing in front of the rook on d8, which is aimed at your queen. What do you play?

Solution p. 660

HINTS

9.1 What is behind your bishop on d3? Which bishop move comes with check?

9.2 Every knight move is check. Which one also attacks the white queen?

9.3 Your pawn stands between the two queens. What can it capture?

9.4 Which move of the d6 bishop captures something at the same time as it uncovers the rook?

9.5 Every knight move is check. Which one attacks the queen?

9.6 Which enemy piece is behind the bishop on d3, and what is it pointed at?

9.7 Which knight move gives two checks at once and leaves the king only one square?

9.8 All knight moves are check. Which one also stops the king from escaping?

9.9 Move the knight so that it gives check and uncovers the queen's check at the same time.

9.10 Which unprotected pawn stands a knight's jump from both the king and a rook?

9.11 What can capture the knight on d4, and what is it defended by?

CHAPTER 10 STAGE 600–800

Removing the Defender, Overloading, Deflection and Decoys

Every defender has a job. Make it fail.

YOU WILL LEARN TO

- List the duties of each enemy defender and spot the one that is doing too much.
- Win material by capturing, chasing or blocking a defender.
- Use deflection and decoy sacrifices to drag pieces off key squares or onto bad ones.
- Trap pieces by counting their escape squares before you attack them.
- Combine these ideas with forks, pins and discovered attacks from Chapters 7–9.

> IN YOUR NEXT GAME
>
> For each enemy piece that defends something, ask “what is its job?” — and if one piece has two jobs, look for a forcing move that makes it choose.

You have a fork in mind. Your knight could land on c7 and hit the king and the rook. Then you notice the bishop on d6 guarding c7, sigh, and play something else. A few moves later your opponent’s position has become solid and the chance has gone.

Stronger players don’t give up so quickly. When a defender stands in their way, they ask a simple question: *can I get rid of it?* They might capture it, chase it away, cut it off, or give it a second job it cannot handle. Tactics are seldom made of a single trick. Usually there is one move that prepares the trick by taking out a guard. This chapter teaches you to see those preparing moves. They turn “nearly a tactic” into a won game, and at 600–800 they are everywhere.

What is each piece doing?

Up to now you have mostly asked “what is attacked?” and “what is undefended?”. This chapter adds a deeper question: **what is each defender actually doing?**

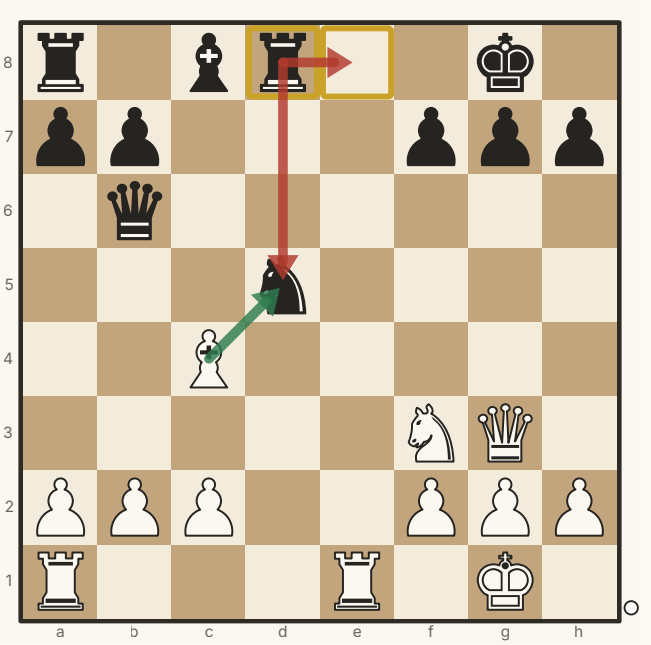
A **duty** is a job a piece performs for its side: guarding another piece, guarding a square (for example a square where mate could happen), or blocking a line. Most pieces have one or two duties at any moment. A piece’s duty can be more important than its value. A knight worth three points can be the only thing standing between its king and mate.

Here is the procedure we will use for the whole chapter. Pick the enemy piece that is defending something important and write down (in your head) its duty list:

1. Which of my targets does it defend?
2. Which squares does it guard: back rank, mating squares, forking squares, promotion squares?
3. Is it the *only* defender of any of these?

When a piece is the only defender of something valuable, it is a target. When a piece has two important duties, it is **overloaded**, and you are probably one forcing move away from winning material.

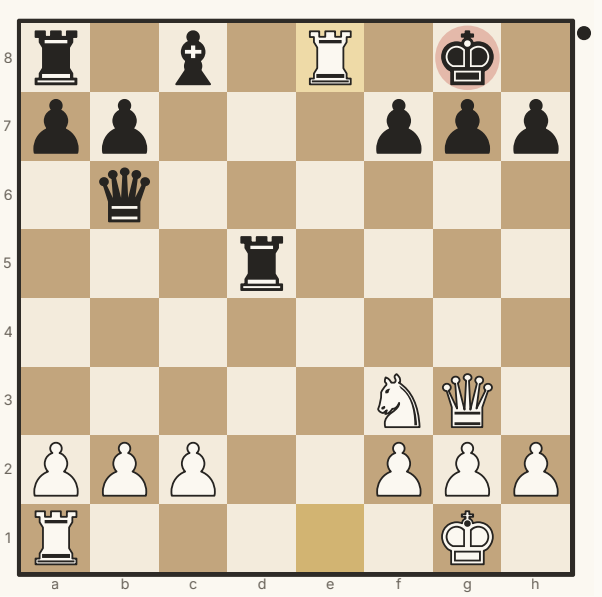
> KEY IDEA **Defenders are targets**
>
> If a piece is the only thing holding the enemy position together, attack that piece rather than the thing it protects. Capture it, chase it, block it, or overload it.



Duty list for the rook on d8: it guards the knight on d5 and the back rank. One rook, two jobs.

Look at Black's rook on d8. Its duties: it defends the knight on d5, and it covers e8, the only square through which White's rook could invade the back rank. Black's king has no **luft** (an escape square created by moving a pawn in front of the king). So the rook is overloaded, and White simply takes the knight:

18. **Bxd5 Rxd5??** 19. **Re8#**



The rook could defend the knight or the back rank, not both. Recapturing loses the game at once.

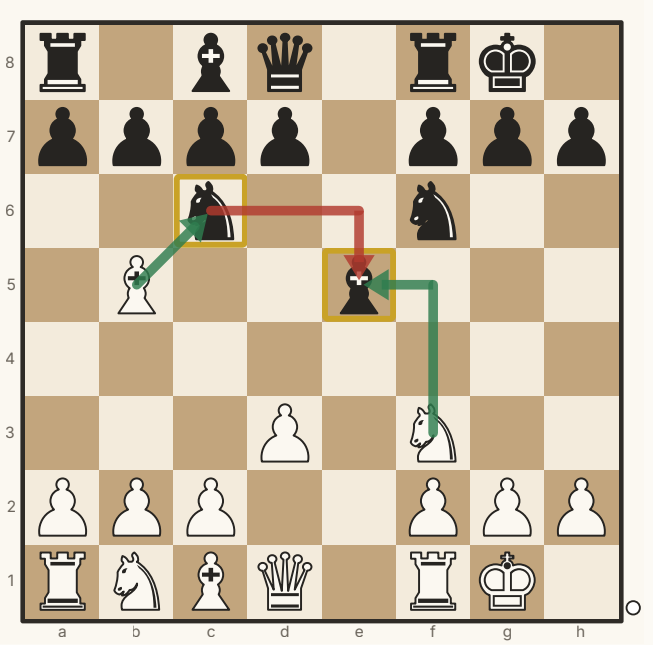
If Black does not recapture, White is a clean piece up. The duty list told you where to look before you calculated a single line.

Removing the defender

Removing the defender means getting rid of the piece that guards your real target. There are three main ways to do it: capture it, chase it, or block it.

Capture the defender

The most direct method is to take the defender, even if the trade looks equal on its own. A bishop for a knight is a fair exchange. A bishop for a knight that was the only guard of h7 or of a forking square is a winning exchange.



The bishop on e5 is attacked by the knight on f3 and defended only by the knight on c6. Capture the defender.

Black's bishop on e5 has exactly one guard, the knight on c6. White takes it: 8.Bxc6. If 8...dxc6, then 9.Nxe5 wins the bishop. Black's best try is to grab a pawn on the way out, but it doesn't save the material:

8. **Bxc6 Bxb2** 9. **Bxb2 dxc6**

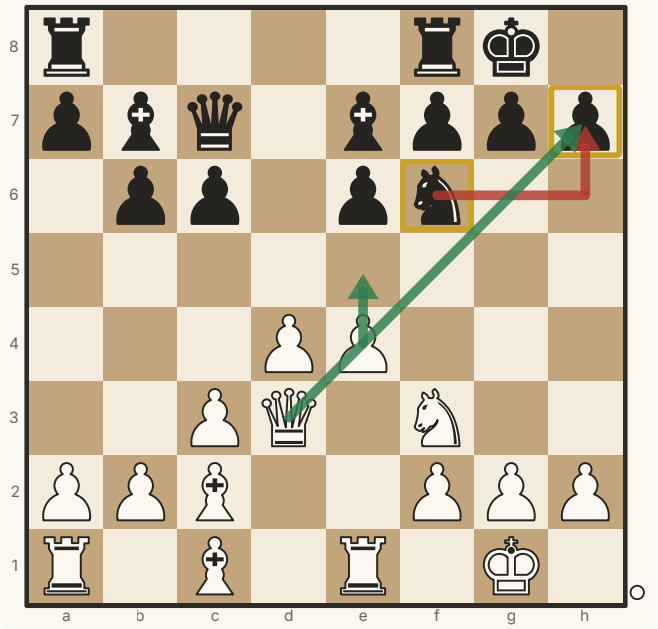
White has won a knight and bishop for a bishop and a pawn: a clean piece for a pawn.

This is why your counting from Chapter 3 must include a question about duties. If you count "bishop takes knight, pawn takes bishop, even trade", you miss the follow-up: once the knight is gone, something else falls, or a mate appears. The trade is only the first half of the combination.

> **COMMON MISTAKE** **Counting the trade, not the consequence**
>
> Beginners often reject Bxf6 or Bxc6 because "it gives up my good bishop". Ask what the captured piece was *guarding*. If its disappearance wins material or mates, the bishop has done its job.

Chase the defender

Sometimes you cannot capture the defender, but you can attack it with something cheaper, usually a pawn. A defender that has to run away is no longer defending.



White's queen and bishop aim at h7. Only the knight on f6 defends it. The e-pawn can chase the knight away.

Here White's queen on d3 and bishop on c2 form a **battery** (two pieces lined up on the same line, the front one supported by the one behind). The knight on f6 is the only guard of h7. The pawn push e4-e5 attacks the knight, and every square the knight can run to abandons h7.

12. **e5 g6** 13. **exf6 Bxf6**

Black has to block the diagonal with ...g6, and White collects the knight for a pawn.

If the knight moves, for example 12...Nd5, then 13.Qxh7# ends the game. Black's best is to give up the knight with 12...g6 and stay a piece down. The pawn push was decisive, but only because you had first noticed the knight's duty.

Block the defender

The third method is to cut the line between the defender and what it defends. If a rook on d8 protects a piece on d2 along the d-file, a piece of yours planted on d5

can break that connection. This is called **interference**. It is less common at this level than capturing or chasing, and you will meet it again in Chapter 24. For now, add it to your question: *can I capture it, chase it, or get in its way?*

Overloaded pieces

An **overloaded piece** is a defender with two duties that cannot both be carried out. You exploit it by forcing it to perform one duty, and then you take advantage of the other.

The rook on d8 in the first diagram was overloaded. Here are the most common overloaded defenders you will meet:

- **The back-rank rook.** It guards the first or eighth rank *and* a piece somewhere on its file or rank.
- **The queen guarding two pieces.** A queen often protects two loose pieces at once. Capture one, and if she recaptures, the other falls.
- **The pawn guarding two squares.** A pawn on g7 that guards both f6 and h6 cannot be in two places.
- **The king.** A king that defends a piece next to it and also has to stay on a particular square can be overloaded by a check.

To exploit an overload, you usually begin with a capture or a check on one of the two duties. That forces the defender to choose, and you take whatever it leaves behind.

> COACHING JUDGMENT **Look for the word "only"**
>
> In my coaching experience, the fastest way to find overloads is to say the word "only" out loud while you scan. "This rook is the only guard of the back rank. It is also the only guard of the knight." Two "only"s on the same piece means there is very likely a tactic.

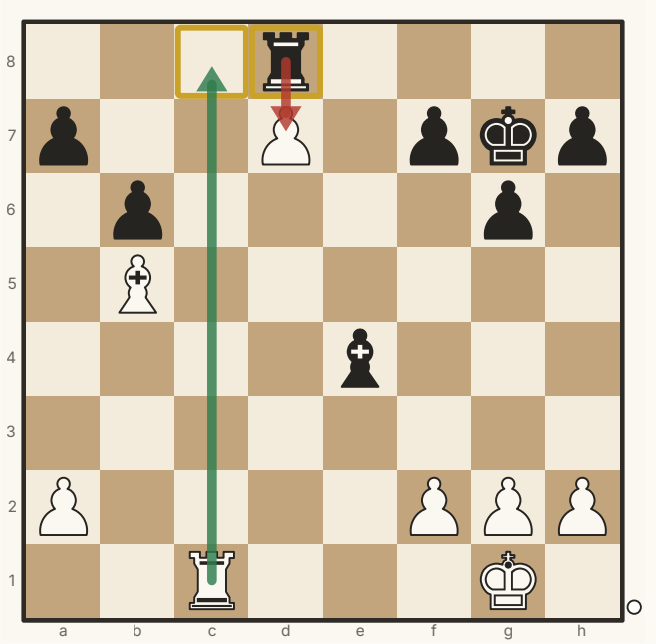
Deflection: force the defender away

Deflection is closely related to overloading. You force a defender to leave its post, usually with a check, a capture it must answer, or a sacrifice it must accept. The difference is mainly in how you think about it. With overloading, you notice the

defender has two jobs. With deflection, you look for a forcing move that pulls it away from its one critical job.

The defender you deflect can be any piece, including a pawn or the king. Kings are especially easy to deflect because a check must be answered. If the king is the only defender of a piece, a check that drags the king away usually wins that piece.

One of the most useful deflections at this level is against a defender that guards a promotion square.



Black's rook on d8 blocks the d-pawn. A rook offer on c8 deflects it.

The rook on d8 has one job: stop the pawn on d7 from becoming a queen. White's bishop on b5 protects the pawn, so Black cannot simply take it. Now 30.Rc8! attacks the rook. If 30...Rxc8, then 31.dxc8=Q. If the rook leaves the d-file, the pawn promotes on d8.

30. **Rc8 Rxd7** 31. **Bxd7**

The best Black can do is give up the rook for the pawn. White is a whole rook ahead.

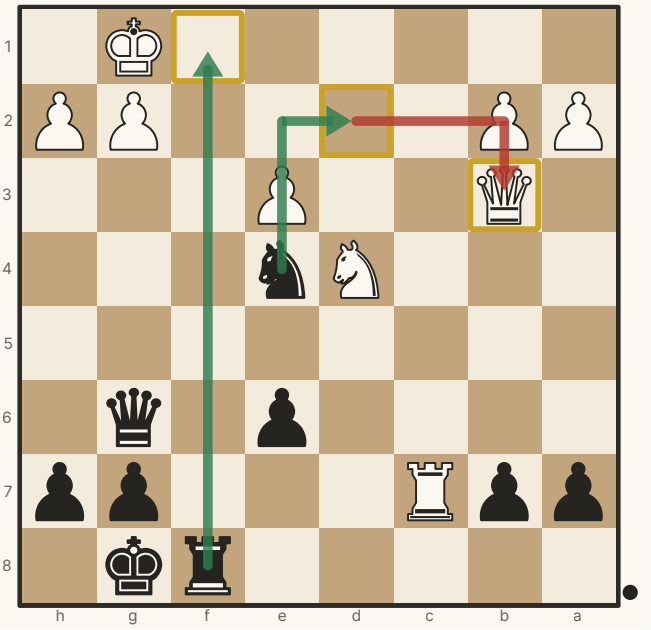
COMMON MISTAKE **Deflecting with a piece that can be ignored**

A deflecting move must be forcing. If your opponent can say "no thanks" and play something useful, nothing has been deflected. Checks, captures that must be answered, and threats to queen or mate are forcing. A quiet attack on a defended piece usually isn't.

Decoy: lure a piece onto a bad square

A **decoy** (also called **attraction**) is the mirror image of deflection. Instead of pulling a defender *away* from a good square, you lure an enemy piece, often the king, *onto* a bad square. Usually you do it with a sacrifice the opponent must accept.

Why would a square be bad? Most often because a fork, skewer or mate works there. A knight fork of the king and queen only works if the king stands on the right square. A decoy puts it there.



A knight on d2 would fork a king on f1 and the queen on b3. Black can force the king to f1.

From d2, Black's knight attacks b3 and f1. White's king is on g1, so it is safe from that knight, for now. The rook sacrifice 24...Rf1+ leaves the king only one legal reply, capturing the rook, which brings it to the forking square.

24...**Rf1+** 25.**Kxf1 Nd2+** 26.**Kf2 Nxb3** 27.**axb3**

Black gave a rook and won a queen. The decoy made the fork happen with check.

In this particular position 24...Nd2 at once is also strong, because the knight hits b3 and f1 together and the rook backs it up. But the pattern to remember is the decoy: *if the fork works only with the king on a certain square, look for a check or a sacrifice that forces the king there.* You will see it again with the king on h1 and h8, and in many mating attacks in Chapter 11.

> **KEY IDEA** **Deflection vs. decoy**
>
> Deflection: drag a defender *away from* a square it must guard. Decoy: drag a piece *onto* a square where it gets hit. Both use forcing moves, and both start from the same question: "Which square would I like this piece to be on — or not be on?"

Trapped pieces: remove the escape squares

So far we have taken away a piece's *defence*. The same thinking works on its *escape squares*. A **trapped piece** is one that is attacked and has no safe square to go to. You win it not by attacking it harder, but by checking that every exit is covered first.

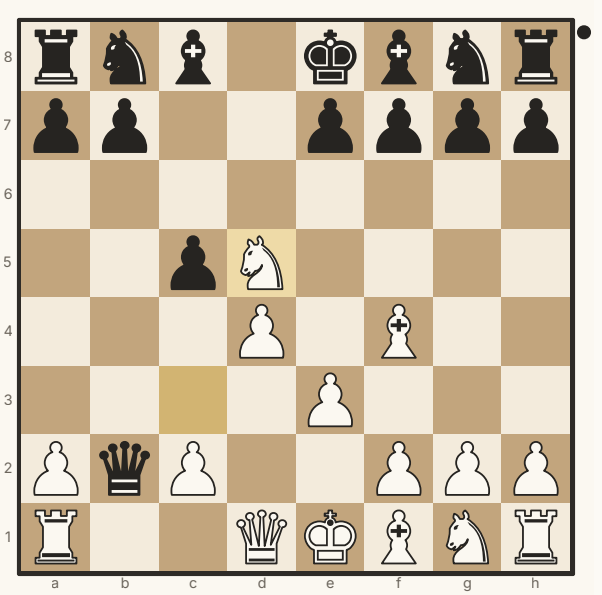
There are three classic situations at this level.

The greedy bishop. A bishop that captures a rook's pawn deep in enemy territory, typically on a2, a7, h2 or h7, can be shut in by a single pawn move. For example, a black bishop that takes on a2 relies on the diagonal a2-g8 to get out. If White plays b3, protected by the c2-pawn, the diagonal is closed and the bishop may have no way home.

The knight on the rim. A knight on the edge controls only four squares, or even two in the corner. If you cover those squares first, a single pawn attack wins it. (Exercise 10-05 is a perfect example.)

The queen hunt. A queen that grabs a pawn far from home, especially the b2- or b7-pawn, can run short of squares. Developing moves with gain of time, like Nb5, Rb1 or Nc3-a4, can trap her outright.

1. **d4 d5** 2. **Bf4 c5** 3. **e3 Qb6** 4. **Nc3 Qxb2?** 5. **Nxd5**



Black's queen has grabbed b2 and is far from home. White threatens Nc7+ and Rb1, and the queen soon runs out of squares.

CHEAT CODE **Before you attack a piece, count its exits**

1. Name every square the piece can move to.
2. Cross out the squares your pieces or pawns cover.
3. Cross out the squares its own pieces block.
4. If none are left, find the cheapest attacker — usually a pawn — and attack it.
5. Check your attacker is protected and that the opponent has no strong counter-attack in the meantime.

The same procedure protects your own pieces. Before your bishop takes on a7 or your queen takes on b7, count its way back. If you cannot name a safe escape route, the "free" pawn is probably bait.

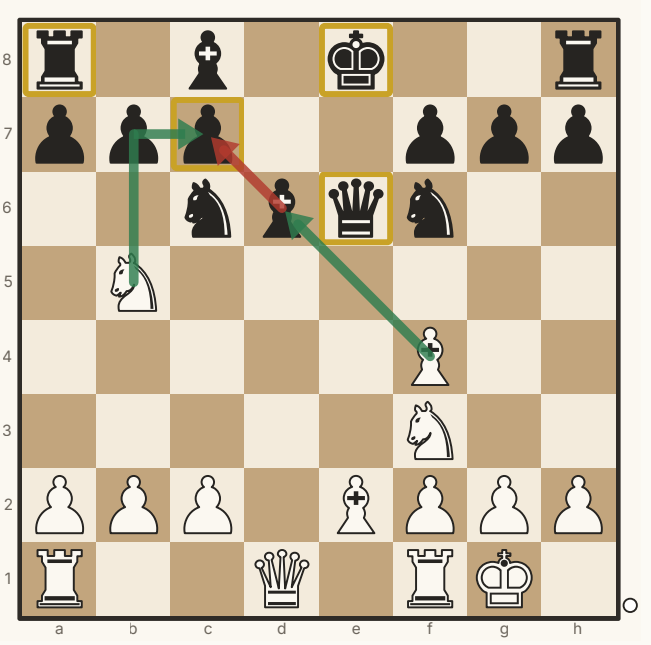
Combining the ideas

Removing the defender rarely stands alone. It is the preparation; the payoff is usually a motif you already know from Chapters 7–9: a fork, a pin, a discovered attack, or mate. The skill is to see the two-step plan: "This fork would work *if* that piece were gone."

A useful habit is to work backwards. Look for the tactic you *wish* you had: a fork square, a mating square, a skewer line. Then ask what stops it. The answer is the defender, and the defender is your new target.

Worked example: the fork that needs help

Let's walk through a position the way you should think at the board.



White to move. A knight on c7 would fork the king, queen and rook. What stops it?

Step 1: checks, captures, threats. I look at my forcing moves first. Nxd6+ is a check and a capture. Bxd6 is a capture. Nc7+ is a check. Nxa7 is a capture.

Step 2: what would I like to happen? Nc7+ catches my eye immediately. From c7 the knight would attack e8, e6 and a8: the king, the queen and a rook. That is the tactic I *wish* I had.

Step 3: what stops it? If I play Nc7+ now, the bishop on d6 just takes my knight: Bxc7. So the bishop on d6 is the guard of c7. I check the rest of Black's pieces: the queen on e6 does not cover c7, nor do the knights. The bishop is the *only* defender of the forking square.

Step 4: can I remove the defender? I have two ways to capture it: Nxd6+ and Bxd6. After 9.Nxd6+ cxd6 my knight is gone, so there is no fork any more. After 9.Bxd6 the knight stays on b5, ready to jump.

Step 5: what is Black's best reply? If 9...cxd6, the c7-square is unguarded and 10.Nc7+ forks king, queen and rook. If Black does not recapture, I am a piece up. Black's most stubborn try is to recapture with the queen and give it up for my knight:

9. **Bxd6 Qxd6** 10. **Nxd6+ cxd6**

Black has given the queen for a bishop and knight. White is close to winning.

Step 6: blunder-check. Before playing 9.Bxd6, I make sure Black has no check or capture that changes things, for example a capture of my queen or a mate threat. There isn't one. I play it.

Notice what happened. The fork was the idea, but removing the defender was the move. Without the duty list, it would have been easy to see Nc7+ fail and move on.

How to spot these ideas in your games

These motifs hide in plain sight because the decisive move often looks like an ordinary exchange or a pointless sacrifice. Here is how to train your eye.

When you are attacking. After you have looked at checks, captures and threats (Chapter 4), go through the enemy defenders one by one and make a duty list for the important ones. Pay special attention to:

- the piece guarding the back rank;
- the piece guarding the square in front of the castled king (h7, g7, h2, g2);
- any piece that defends two other pieces;
- any piece guarding a square where you have a fork.

When you are defending. Do the same for your own pieces. Ask "Is any of my pieces overloaded? Is my back rank guarded by one piece that is also doing something else?" Making luft with h3 or g3 at a quiet moment removes a duty from your rooks forever.

When you analyse your games. Whenever you lose material to a tactic, ask which of your pieces was overloaded or deflected. Record it in your mistake journal under "missed-tactic" or "hanging-piece" and note the defender that failed. After a few games you will see the same patterns repeating, usually the back rank and the overloaded queen.

> IN YOUR NEXT GAME **The "two jobs" scan**
>
> In every game this week, at least once per phase (opening, middlegame, endgame), stop and name the duties of your opponent's most important defender. If it has two jobs, spend a minute looking for a forcing move that makes it choose.

Practice at the board

Set up each position on a board or in any free analysis app. Try to solve it without moving the pieces, then play out your answer against the engine (or a friend) to check it.

1. **Overloaded rook.** `r1br2k1/pp3ppp/1q6/3n4/2B5/5NQ1/PPP2PPP/R3R1K1 w - - 0 18`. Name the two duties of the rook on d8, then win a piece. Check: Black must not be able to recapture without allowing mate.
2. **Chasing the defender.** `r4rk1/pbq1bppp/1pp1pn2/8/3PP3/2PQ1N2/PPB2PPP/R1B1R1K1 w - - 0 12`. Find the pawn move. Then try every knight move for Black and prove each one loses to Qxh7#.
3. **Deflecting a blockader.** `3r4/p2P1pkp/1p4p1/1B6/4b3/8/P4PPP/2R3K1 w - - 0 30`. Find the rook move that wins. Play through all three Black replies: take, run, or capture the pawn.
4. **Remove the guard of the fork.** `r1b1k2r/ppp2ppp/2nbqn2/1N6/5B2/5N2/PPP1BPPP/R2Q1RK1 w kq - 0 9`. Play the worked example from the Black side against a friend or engine and try every defence. Does any of them hold?
5. **Your own games.** Take your last three losses. For each tactic you suffered, mark the defender that failed and write its two duties next to the move.

BOARD DRILL **Duty cards**

Take any middlegame position from your games. For each piece on both sides, write one short line: "Nf6: guards h7 and d5." Pieces with two important lines are your tactical targets — or your weaknesses.

Chapter summary

- Every defender has duties: guarding pieces, guarding squares, blocking lines. Its duty can matter more than its point value.
- Removing the defender means capturing, chasing or blocking the piece that guards your real target.
- An overloaded piece has two duties it cannot fulfil at once. Force it to perform one, then exploit the other.
- Deflection drags a defender away from its post with a forcing move; a decoy lures a piece onto a square where a fork, skewer or mate works.
- Trapped pieces are won by covering every escape square first, then attacking with the cheapest piece.
- These ideas are usually the preparing move; the payoff is a fork, pin, discovered attack or mate from Chapters 7–9 and 11.
- Work backwards: find the tactic you wish you had, then ask what stops it.

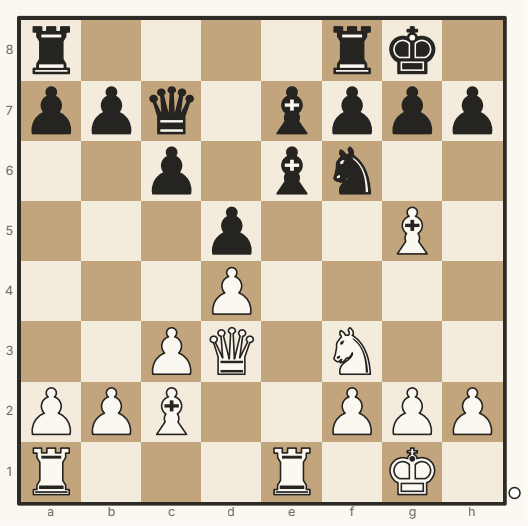
CHEAT CODE **The duty-list procedure**

1. After checks, captures and threats, pick the enemy pieces that defend something important.
2. For each, list its duties: which pieces and which squares it guards.
3. Mark any piece that is the *only* defender of something valuable, and any piece with *two* duties.
4. Find a forcing move (capture, check, threat) that removes, chases, blocks, deflects or overloads that piece.
5. Calculate the opponent's best reply, then blunder-check before you play.

Exercises

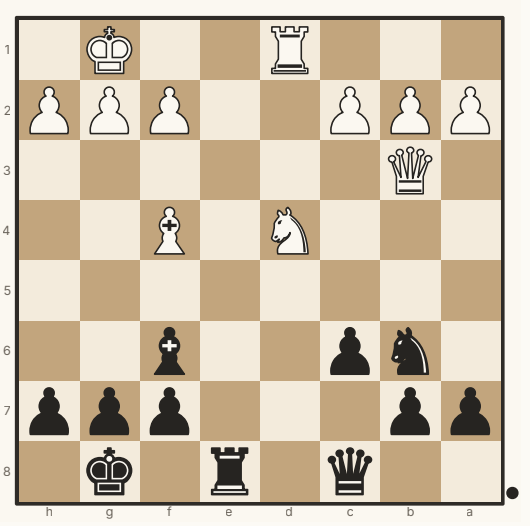
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

10.1 TACTIC



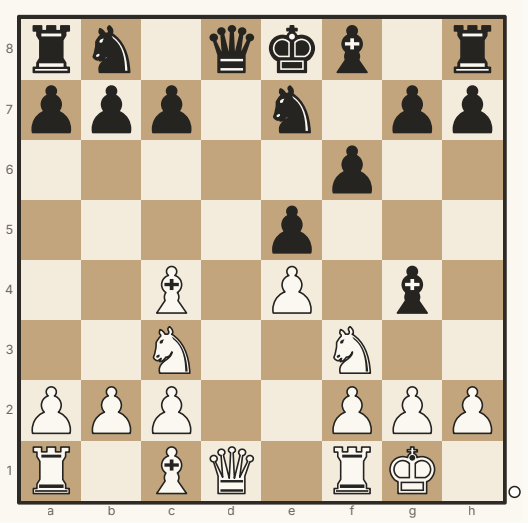
White to move. The knight on f6 has one very important job. Remove it and win material. Solution p. 661

10.2 TACTIC



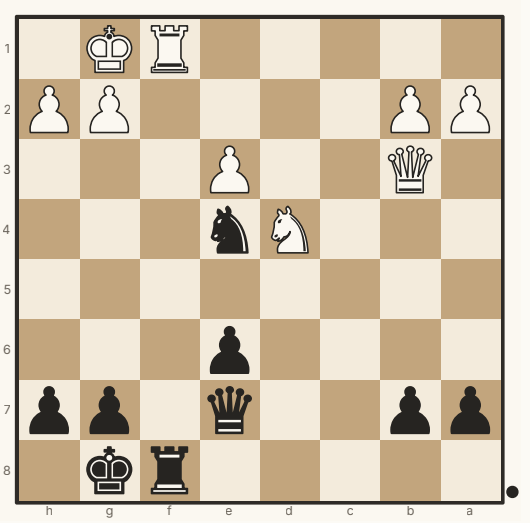
Black to move. White's rook on d1 is doing two jobs. Punish it. Solution p. 661

10.3 TACTIC



White to move. Black's queen on d8 is defended only by the king. Deflect the king and win the queen. Solution p. 661

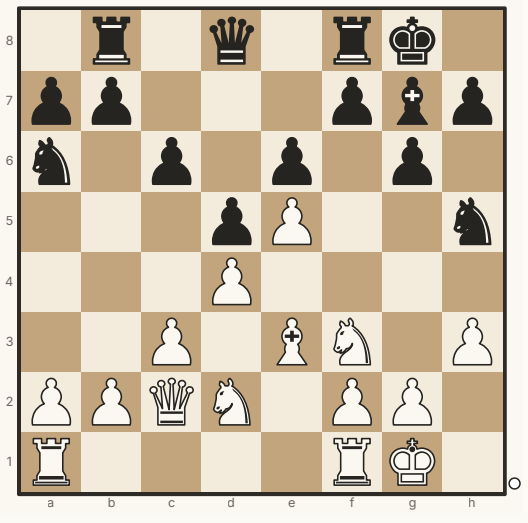
10.4 TACTIC



Black to move. Lure White's king onto a square where your knight can fork it.

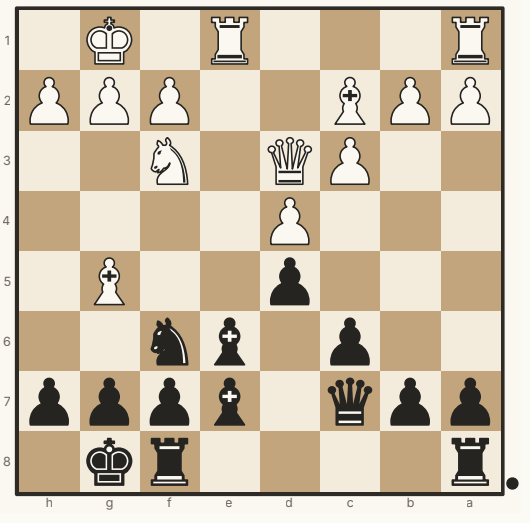
Solution p. 661

10.5 TACTIC



White to move. The knight on h5 has wandered to the edge. Trap it. Solution p. 662

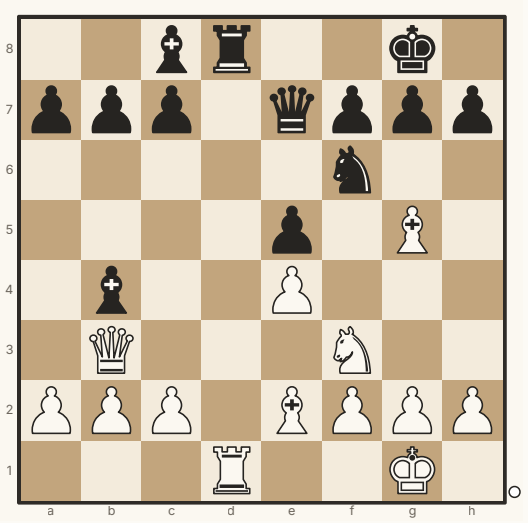
10.6 DEFENCE



Black to move. White threatens Bxf6 followed by Qxh7#. Find a move that meets the threat without losing material.

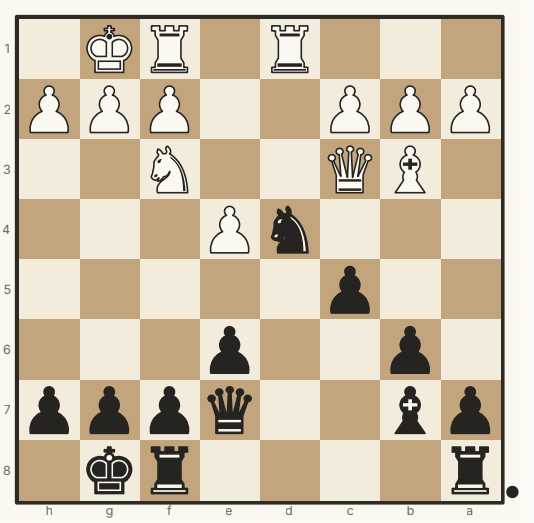
Solution p. 662

10.7 TACTIC



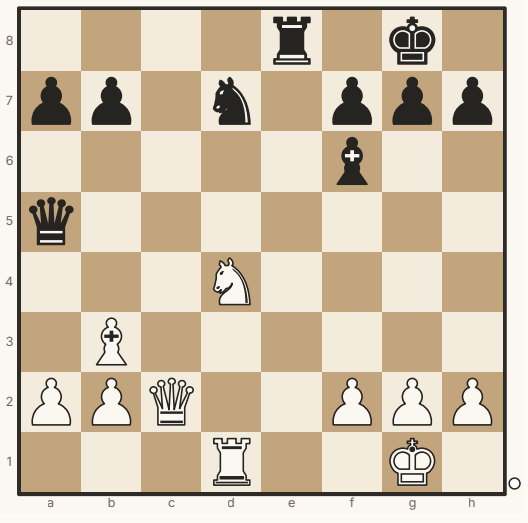
White to move. Black's queen on e7 protects the rook on d8 and the bishop on b4. Overload it. Solution p. 662

10.8 TACTIC



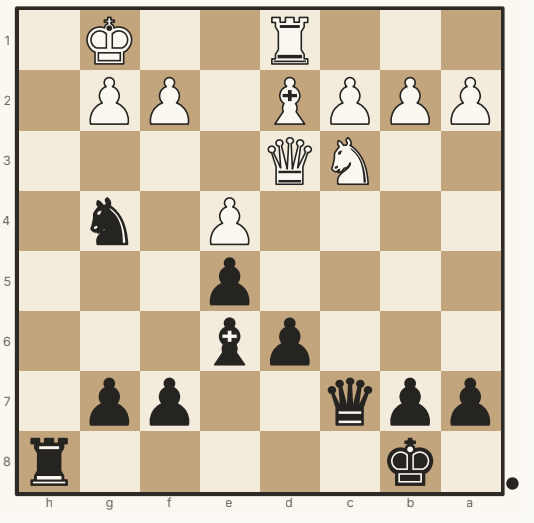
Black to move. Review from Chapter 7: win White's queen. Solution p. 662

10.9 QUIET MOVE



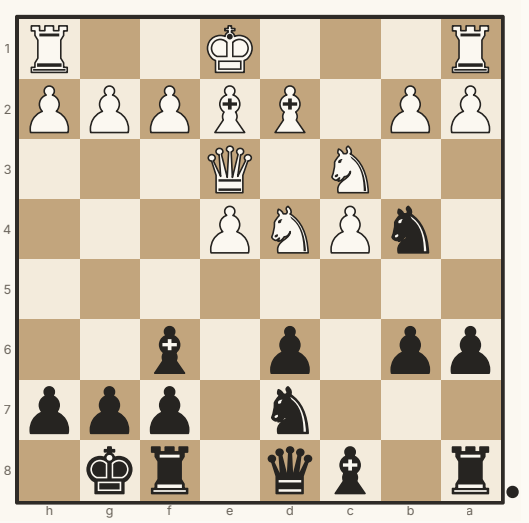
White to move. Your rook on d1 has two jobs and Black threatens both ...Qe1+ and ...Bxd4. Find a calm move that solves the problem. Solution p. 663

10.10 TACTIC



Black to move. Use the open h-file to decoy White's king, then fork. Solution p. 663

10.11 TACTIC



Black to move. The knight on d4 guards an important square. Remove it. Solution p. 663

HINTS

10.1 What would happen on h7 if the knight on f6 disappeared?

10.2 List the duties of the rook on d1.

10.3 Which forcing move drags the king away from e8?

10.4 From which square does a knight attack both f1 and b3?

10.5 Which squares can the knight on h5 move to? Are any of them safe?

10.6 Instead of defending the defender, can you block the attacking line?

10.7 What happens if the queen has to recapture on d8?

10.8 Which knight check also attacks the queen?

10.9 Can one move relieve the rook of one of its duties?

10.10 On which square would the white king be forked by a knight landing on f2?

10.11 Which white piece stops ...Nc2+? Can you capture it?

CHAPTER 11 STAGE 600–800

Checkmate Patterns

Recognize the picture, then play the moves

YOU WILL LEARN TO

- Recognize the most common mating patterns from their key ingredients.
- Use sacrifices (deflection and decoy) to force a known pattern onto the board.
- Judge when the Greek gift sacrifice Bxh7+ is likely to work.
- Spot a weakened king — your opponent's and your own — before the mate happens.
- Defend against these patterns with luft, extra defenders and counter-checks.

> **IN YOUR NEXT GAME**
>
> When the enemy king is short of squares, picture the mating patterns from this chapter and ask which one is "nearly there" — and do the same for your own king.

At 600–800, many games are decided by a mate that one player never saw coming. A rook slides to the back rank. A knight hops to f7 and the king, buried behind its own pieces, cannot move. The loser often had plenty of material. What they lacked was the picture.

Strong players do not calculate every mate from scratch. They recognize **mating patterns**: small arrangements of a few pieces that occur again and again. When they see two or three of the ingredients on the board, they know which moves to look for. In this chapter you will learn the most important patterns, the sacrifices that force them, and the warning signs that a king, yours or your opponent's, is about to be caught.

How to use mating patterns

A pattern is not a position to memorize. It is a short list of ingredients:

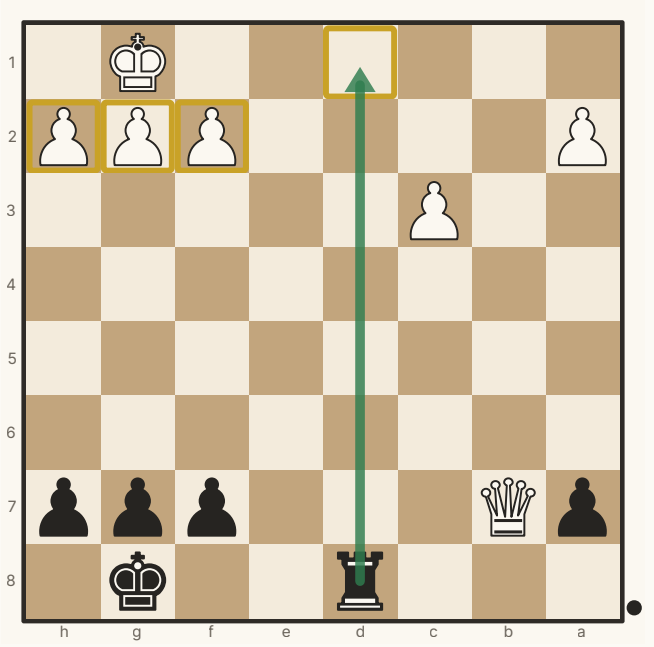
- where the king is,
- which of its flight squares are blocked by its own pieces,
- which squares the attacker controls,
- and which piece gives the final check.

At the board you look for two or three of the ingredients. Then you ask: "Can I force the rest?" Usually the answer involves a check or a sacrifice that drags the king or a defender into place. That is where Chapter 10 comes in: deflection and decoy are the engines that drive most mating combinations.

> **KEY IDEA** **A mate needs two things**
>
> Every checkmate has a **checking piece** and a **net**: the squares around the king that are covered or blocked. When you study a pattern, notice who gives check and who builds the net. Often the king's own pieces build half of it.

The back-rank mate

The **back-rank mate** happens when a rook or queen checks the king along its first rank (the eighth rank for Black) and the king cannot step forward because its own pawns are in the way. It is the most common mate at club level, by a long way.



White's queen has gone pawn-hunting on b7. Nothing guards the first rank and the king has no luft: 30...Rd1# ends the game.

The ingredients are simple: an open file for the rook, a king boxed in by its own pawns, and no defender of the back rank. When you see a king castled behind three unmoved pawns, start checking which of its pieces guard the back rank. If the answer is "one rook", that rook is overloaded the moment it is also asked to guard something else (see Chapter 10).

The attacker's most common tool is a **deflecting sacrifice** on the back rank: a queen or rook offers itself to drag the last defender away, and the second heavy piece mates. You will find several in the exercises.

Prevention: luft

The cure is cheap. **Luft** (German for "air") is an escape square for the king, made by moving one of the pawns in front of it, usually h3 (or ...h6 for Black). In the diagram above, if White's h-pawn stood on h3, 30...Rd1+ would just be a check and the king would walk to h2.

COACHING JUDGMENT **When to make luft**

Coaching judgment: in most middlegames, a quiet moment once the heavy pieces have come out is the right time for h3 or g3. You don't have to do it on move 8, but once the rooks are fighting over open files, a king without luft is a liability.

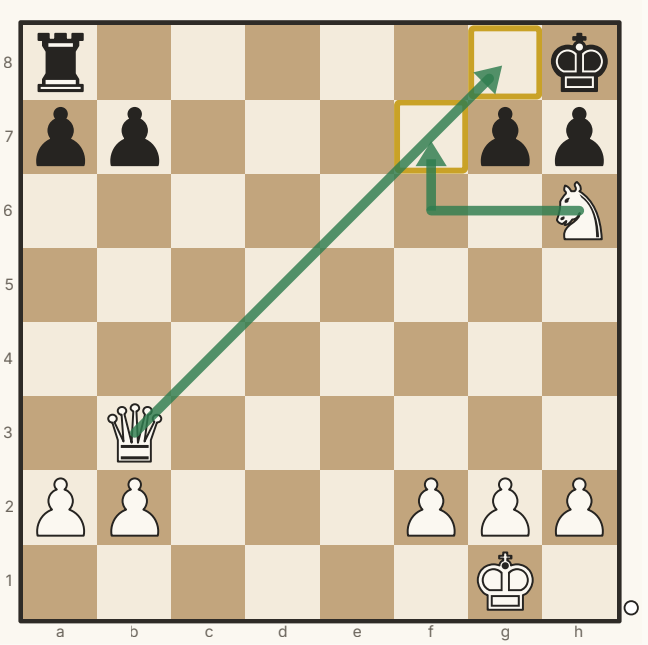
COMMON MISTAKE **The greedy queen**

The most common way to lose to a back-rank mate is to send your queen off to grab a pawn on the other side of the board, as White did with Qxb7 above. Before you take a far-away pawn, ask what your queen was guarding.

Smothered mate

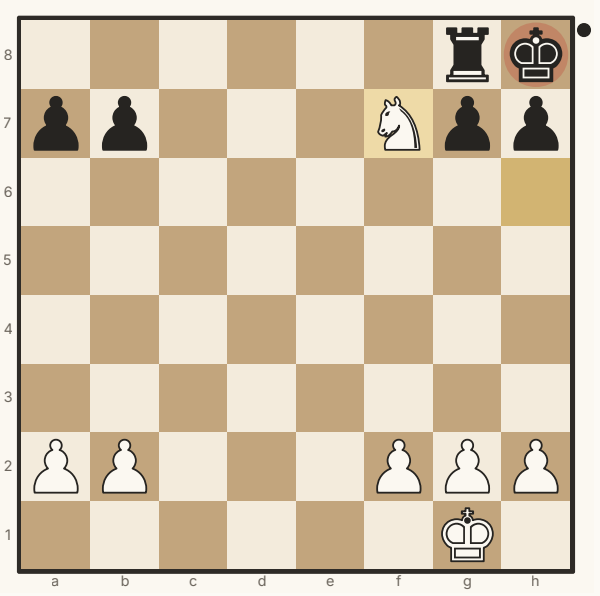
A **smothered mate** is a knight check against a king that is completely surrounded by its own pieces. Since knights jump, the check cannot be blocked; since the king has no free squares, it cannot run. The classic case is a king in the corner on h8 with a rook on g8 and pawns on g7 and h7.

The famous route to it is sometimes called **Philidor's legacy**: knight check, double check to drive the king into the corner, a queen sacrifice on g8, and the knight mates on f7. Here is the final phase.



The knight on h6 guards g8. The queen sacrifice on g8 forces the rook to block the king's last square.

25. **Qg8+ Rxg8** 26. **Nf7#**



Smothered mate. Black's own rook and pawns fill every square around the king.

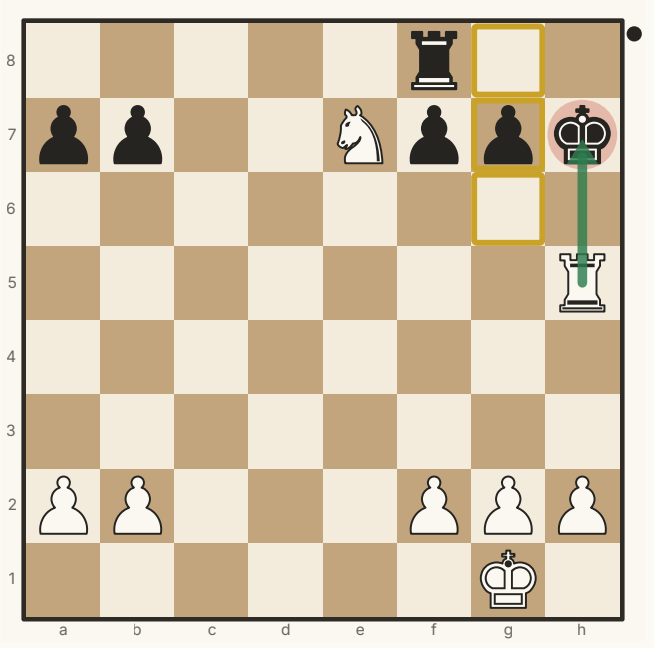
Remember the ingredients: the king in the corner, its neighbours occupied, a knight that can reach f7 (or f2 for White's king), and a queen on the a2-g8 diagonal (or a7-g1). If you see a king on h8 with a rook on g8, look for knight checks on f7. The same picture works for Black against a white king on h1: ...Qg1+, Rxg1, ... Nf2#.

Knight-and-rook mates: Anastasia and Arabian

Rooks and knights work well together near the edge of the board. Two patterns are worth knowing by name.

Anastasia's mate

In **Anastasia's mate**, a knight on e7 (or e2 for Black) covers g8 and g6, the king is on h7 hemmed in by its own g7-pawn, and a rook or queen mates on the h-file.

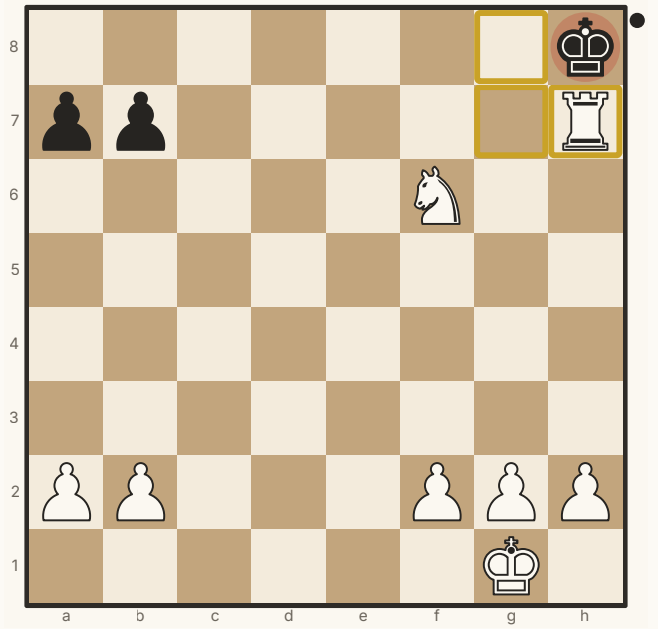


Anastasia's mate. The knight takes g8 and g6, the g7-pawn blocks, and the rook checks along the h-file.

To get there, the attacker often sacrifices on h7 to open the file and decoy the king: Qxh7+ Kxh7, then a rook swings to the h-file with mate. The key question is "Can my rook reach the h-file in one move?"

The Arabian mate

The **Arabian mate** is the oldest known pattern: rook and knight against a king in the corner. The rook checks from the square next to the king, and the knight both protects the rook and covers the remaining flight square.

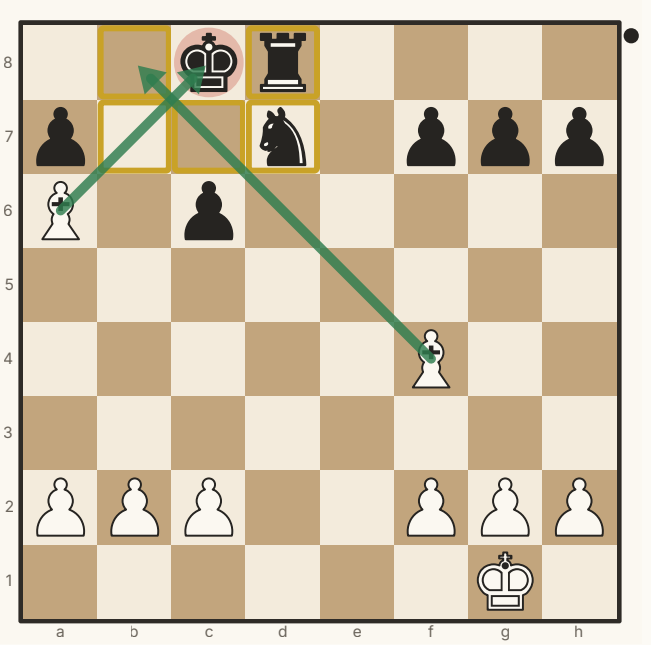


Arabian mate. The rook on h7 checks, the knight on f6 protects it and covers g8; the rook covers g7.

A related idea is the **hook mate**: a rook gives check next to the king, protected by a knight, which is itself protected by a pawn, while the king's own pawn blocks its escape. You will not need its exact geometry often, but remember the lesson. A rook protected by a knight is very hard to drive away from the king.

Bishop mates: Boden's mate and the two-bishop net

Bishops attack from far away, so their mates often look surprising. The best-known is **Boden's mate**: two bishops on criss-crossing diagonals against a king that has castled queenside.



Boden's mate. The bishop on a6 checks along a6-c8, the bishop on f4 covers b8 and c7, and Black's own rook and knight block d8 and d7.

Typically the attacker sacrifices the queen on c6 (or c3 for Black) to force ...bxc6 and open the diagonal to the king. The ingredients: a king on c8 (or c1), its own pieces on d7 and d8, one bishop able to control b8 and c7, and the other able to reach a6.

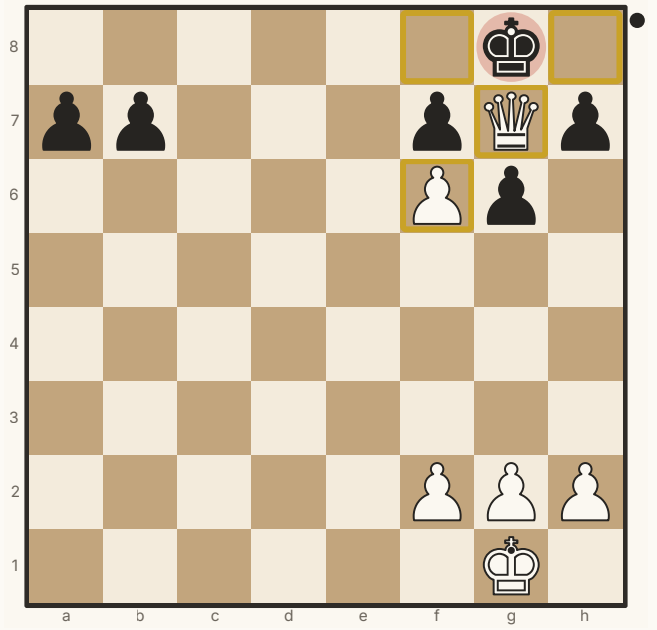
More generally, a **mating net** with minor pieces is any position where bishops and knights between them cover all the king's squares. These nets are easy to overlook because no single piece seems dangerous. When the enemy king is short of squares, count which squares *each* of your minor pieces covers. Sometimes the net is already complete and you only need a check.

Queen mates near the king: h7, g7 and the supporting pawn

Most mating attacks against a castled king end with the queen on h7 or g7 (h2 or g2 against White). The queen cannot do it alone: she needs a partner that protects her on the mating square.

The most common partners are:

- **A bishop on the b1-h7 diagonal**, supporting Qxh7#. This is the "Q + B battery" you met in Chapter 10.
- **A knight on g5**, supporting Qxh7# or Qh7#.
- **A pawn on f6 (or h6)**, supporting Qg7#. This picture, with a pawn wedged deep in front of the king and the queen landing next to it, is a version of what older books call **Damiano's mate**.

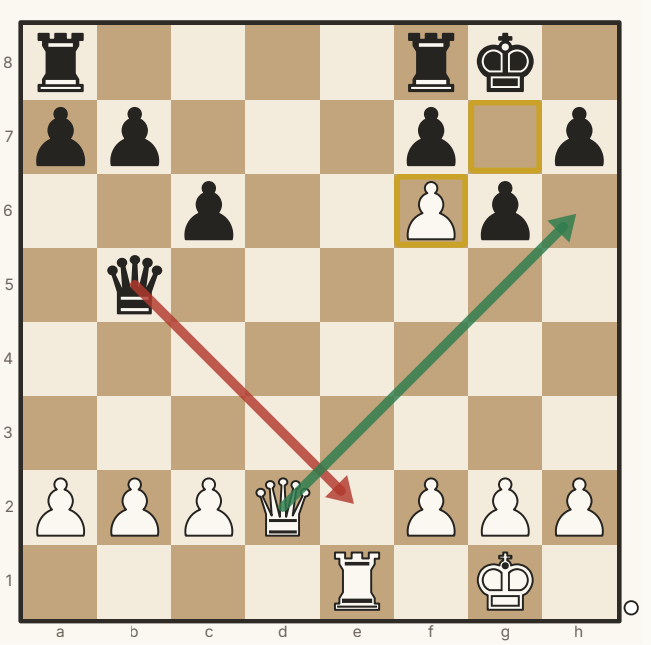


The pawn on f6 protects the queen on g7. The king cannot take, and f8 and h8 are covered by the queen.

> KEY IDEA **The advanced pawn is a dagger**
>
> A pawn on f6 (or f3 against White) next to a castled king is worth far more than one point. It covers g7 and e7 and prepares Qg7# or Qh6 with mate on g7. When you can safely plant a pawn there, the enemy king is usually in serious trouble.

Worked example: the quiet queen move

Let's see how to think when one of these patterns is "nearly there".



White to move. The pawn on f6 is the dagger. Which square does the queen need?

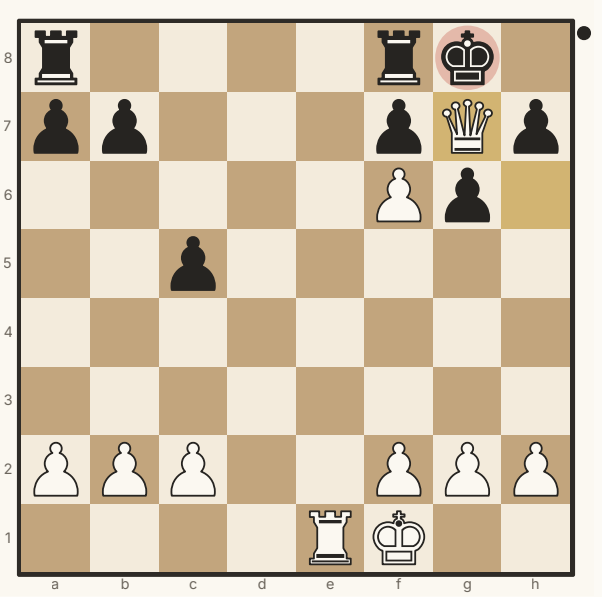
Step 1: checks, captures, threats. I have no useful checks. Qd8 is not a check (the rook on f8 is in the way). There are no good captures. Black's queen on b5 attacks b2 and e2, but nothing is hanging that matters.

Step 2: which pattern is nearly there? My pawn on f6 already covers g7. If my queen reaches g7, it will be protected by the pawn, and the king on g8 cannot go to f8 (its own rook) or h8 (covered by the queen). That is the f6-pawn pattern.

Step 3: can the queen get there? From d2, the queen reaches h6 in one move. From h6 she threatens Qg7#.

Step 4: what is Black's best defence? I look for every way Black could stop Qg7#: something guarding g7, something blocking, or a check that changes everything. Black's pieces are far away. Black's only "defence" is a desperate check, 22...Qf1+, which my king simply captures. Then nothing can stop the mate.

22. **Qh6 Qf1+** 23. **Kxf1 c5** 24. **Qg7#**



A quiet queen move that mates. The pawn on f6 did half the work.

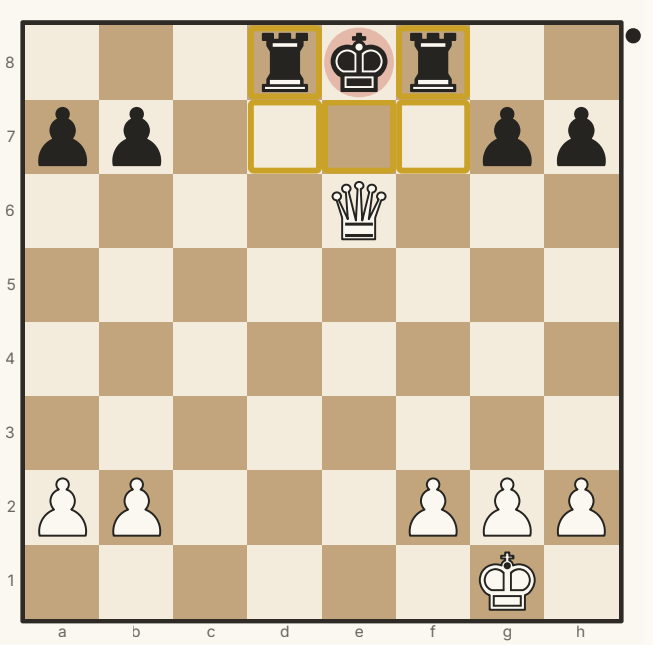
Step 5: blunder-check. Before playing 22.Qh6, I check that Black has no check that wins my queen or gives perpetual. There is only the queen sacrifice on f1. I play it.

The lesson: a quiet move can be stronger than any check when it creates a mate threat the opponent cannot meet. Recognizing the pattern told me which square to aim for.

The corridor and epaulette mates

When a king is stuck in a "corridor", it can only move along one line, and a single long-range piece can cut it off. The back-rank mate is the most common corridor mate, but the same thing happens on files and diagonals, for example a king on the edge of the board with its own pieces in front of it.

The **epaulette mate** is a special case: the king's own rooks (or other pieces) stand on both sides of it, like the shoulder decorations on a uniform. A queen check from two squares in front is then mate.



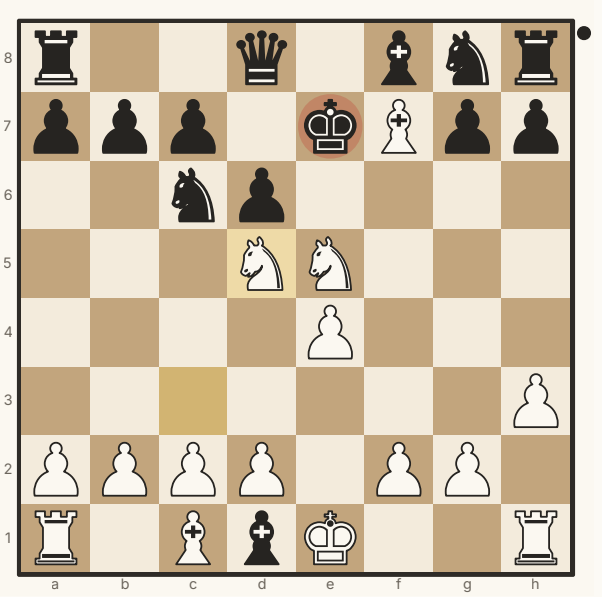
Epaulette mate. Black's rooks on d8 and f8 block the king's sideways escape; the queen on e6 covers d7, e7 and f7.

You will not see the exact epaulette often, but the idea, that the king's own pieces form part of the net, appears in almost every mate in this chapter.

The Légal pattern: a recap

Several of the best-known opening traps, the kind discussed in Chapter 6, are based on a pin to the queen rather than the king. The most famous is **Légal's pattern**: White ignores a pin on the knight, gives up the queen, and mates with minor pieces. The typical finish is a bishop check on f7 followed by a knight mate on d5.

1. **e4 e5** 2. **Nf3 d6** 3. **Bc4 Nc6** 4. **Nc3 Bg4** 5. **h3 Bh5?** 6. **Nxe5 Bxd1??** 7. **Bxf7+ Ke7** 8. **Nd5#**

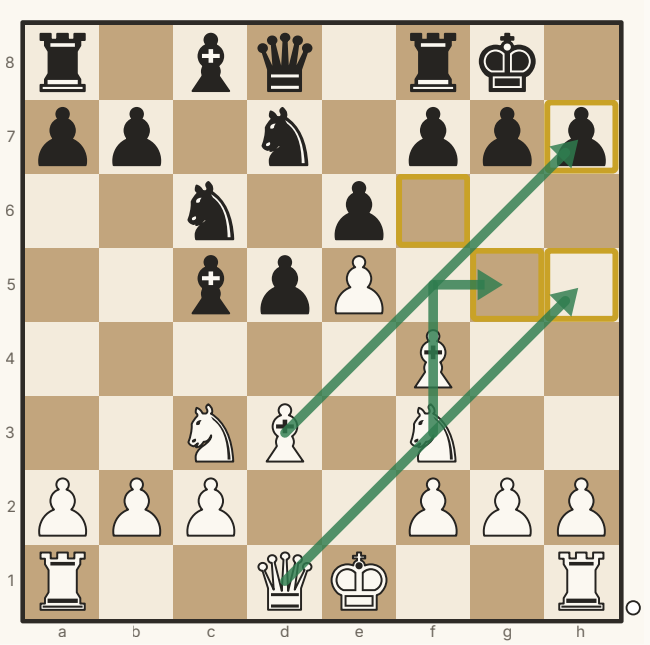


Three minor pieces against a king that left home. Black should have played 6...Nxe5, losing only a pawn.

The pattern to remember: a pin to the *queen* is not absolute. If breaking it lets you deliver mate or win more than the queen, the pin is only an illusion.

The Greek gift: Bxh7+

The **Greek gift** is the classic bishop sacrifice on h7 (or ...Bxh2+ for Black) against a castled king. The usual follow-up is Ng5+ and Qh5, bringing the knight and queen to the king with gain of time. It is one of the most important attacking ideas in chess, and also one of the easiest to get wrong. Treat this section as an introduction; you will study attacks on the king in depth in Chapter 21.



The typical Greek-gift set-up: bishop on d3, knight on f3, queen ready for h5, pawn on e5 keeping a black knight off f6.

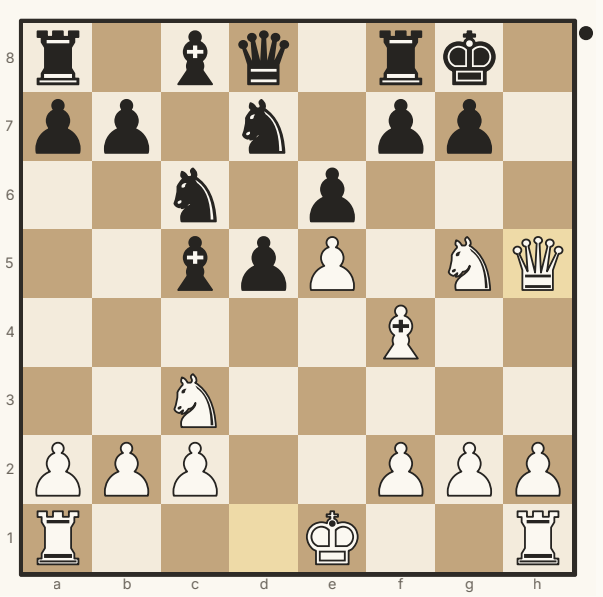
The conditions

Coaching judgment, based on the ingredients that appear in most successful examples: check these before you sacrifice.

1. **A bishop on the b1-h7 diagonal** that can capture on h7 with check.
2. **A knight that can reach g5** with check after ...Kxh7, and that cannot simply be captured there for free.
3. **A queen that can reach h5** (or the h-file) quickly, usually from d1 via an open d1-h5 diagonal.
4. **No black knight on f6** (or one that can be driven away, typically by a pawn on e5) and no black bishop or queen that can defend h7 or cover h5.
5. **A plan against each king move**: ...Kg8, ...Kg6 and ...Kh6, and also against declining the sacrifice with ...Kh8.

If several of these are missing, the sacrifice is usually unsound. If all are present, it often wins outright.

1. **e4 e6** 2. **d4 d5** 3. **Nc3 Nf6** 4. **e5 Nfd7** 5. **Nf3 c5** 6. **dxc5 Nc6** 7. **Bf4 Bxc5** 8. **Bd3 O-O?** 9. **Bxh7+ Kxh7** 10. **Ng5+ Kg8** 11. **Qh5**



White threatens Qh7#. Black has to give up the queen with 11...Qxg5 to stop the mate.

COMMON MISTAKE **Sacrificing on hope**

The most common Greek-gift mistake is to play Bxh7+ because "it looks like the textbook" when one ingredient is missing, for example a black knight that can come back to f6, or a bishop on e7 that captures on g5. Before you sacrifice, play through ...Kg8, ...Kg6 and declining with ...Kh8 in your head. If you can't find a strong answer to one of them, keep the bishop.

Recognizing a weakened king

Patterns tell you *how* to mate. The first step is to notice *when* a king is vulnerable. Run this checklist on both kings, yours and your opponent's, whenever the position changes:

- **Flight squares.** How many safe squares does the king have? Zero or one is a red flag.
- **Pawn shield.** Have the pawns in front of the king moved? Each move creates holes (squares no pawn can protect), especially f6, g6, h6 for Black and f3, g3, h3 for White.
- **Missing defenders.** Is the f6-knight (or f3-knight) gone? It is the most important defender of a castled king.

- **Attackers nearby.** Count your pieces that can reach the king's area in one or two moves. Three attackers against one defender usually means a tactic is close.
- **Open lines.** Is there an open file toward the king, or a long diagonal pointed at it?
- **Back rank.** Is the back rank guarded by more than one piece, and does the king have luft?

> IN YOUR NEXT GAME **The two-king scan**
>
> Once every few moves, run the weakened-king checklist on *both* kings. If the enemy king scores badly, look for one of the patterns from this chapter. If yours scores badly, spend a move on defence before you attack.

Defending against mating patterns

Knowing the patterns also makes you a better defender. Most mates can be prevented a move or two in advance, cheaply, if you recognize what is coming. Your main tools:

- **Make luft** before the back rank becomes a problem.
- **Add a defender** to the mating square, ideally with gain of time, for example a knight to f6 attacking a queen on h5.
- **Block the attacking line**, for instance ...g6 against a bishop on the b1-h7 diagonal.
- **Trade off a key attacker**, such as the bishop or knight that supports the queen.
- **Look for counter-checks.** A check against the attacking king can sometimes break the whole combination.

A defender's question: which pattern is my opponent building?

Every pattern in this chapter can also be read from the defender's side. If your king is on h8 with a rook on g8, an enemy knight that can reach f7 is a smothered-mate warning. If your king has castled queenside and your c-pawn has moved, look at the a6-c8 diagonal for a Boden-style bishop check. If an enemy pawn has arrived on f6 (or f3 against your king), the g7 square is under permanent pressure and you will often need a piece dedicated to guarding it. A

knight that lands on e7 next to your king on h8 or h7 means the h-file must stay closed.

The idea is to recognize the picture one or two moves *before* it is complete, while a cheap defensive move still works. Once the final sacrifice is on the board, it is usually too late. Many club games are lost not because the defender missed the final combination, but because they made a careless pawn move or traded off their best defender a few moves earlier.

The danger in defence is the "natural move" that ignores the threat: a developing move or a pawn grab while the queen and bishop line up against h7. Every time your opponent's move creates a battery aimed at your king, stop and count attackers and defenders of the target square.

Practice at the board

Set up each position on a board or in an analysis app. Solve it first without moving the pieces, then check with the engine.

1. **Back rank with and without luft.** `3r2k1/pQ3ppp/8/8/8/2P5/P4PPP/6K1 b - - 0 30`. Play 30...Rd1#. Then move White's h-pawn to h3 and see why the same move is only a check.
2. **Smothered mate.** `r6k/pp4pp/7N/8/8/1Q6/PP3PPP/6K1 w - - 0 25`. Find the two-move mate. Then put the knight on f7 and the black king on g8 and find Philidor's full sequence (Nh6+, Kh8, Qg8+, Rxg8, Nf7#), with the queen on c4.
3. **The quiet queen move.** `r4rk1/pp3p1p/2p2Pp1/1q6/8/8/PPPQ1PPP/4R1K1 w - - 0 22`. Find the mate. Try every Black defence you can think of.
4. **Greek gift test.** Play through the Greek gift line above from move 8. At move 9 try declining with 9...Kh8 instead of taking. Is White still better? Ask the engine and compare with your own judgment.
5. **Build the nets.** On an empty board, place only the pieces needed for Anastasia's, the Arabian, Boden's and the epaulette mate. If you can set each one up from memory, you own the pattern.

> **BOARD DRILL** **Pattern flashcards**
>
> Make a card for each pattern in this chapter: draw the final position and list its ingredients on the back. Review the cards before each tournament game for a week.

Chapter summary

- Mating patterns are short lists of ingredients: the king's position, its blocked squares, the attacker's controlled squares, and the checking piece.
- The back-rank mate is the most common club mate; luft and a second back-rank defender prevent it.
- Smothered mate (with Philidor's queen sacrifice on g8) and the knight-and-rook mates (Anastasia's, Arabian, hook) show how knights exploit kings blocked by their own pieces.
- Bishops mate from long range: Boden's mate and minor-piece nets are easy to miss, so count every square your minor pieces cover.
- Queen mates on h7 and g7 need a partner: a bishop, a knight on g5, or a pawn on f6.
- The Greek gift Bxh7+ works only when its ingredients are present; check every king move and the option of declining before you sacrifice.
- Run the weakened-king checklist on both kings and defend early: luft, extra defenders, blocked lines and counter-checks.

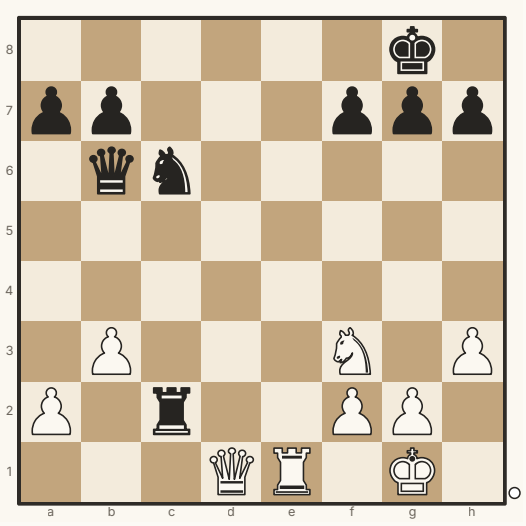
CHEAT CODE **Finding (and stopping) a mate**

1. Count the enemy king's flight squares and note which are blocked by its own pieces.
2. Name the pattern that is "nearly there" and the ingredient that is missing.
3. Look for a check, sacrifice or quiet threat that supplies the missing ingredient — usually a deflection or decoy.
4. Calculate the opponent's best defence, including counter-checks and declining a sacrifice.
5. Run the same scan on your own king before you commit to the attack.

Exercises

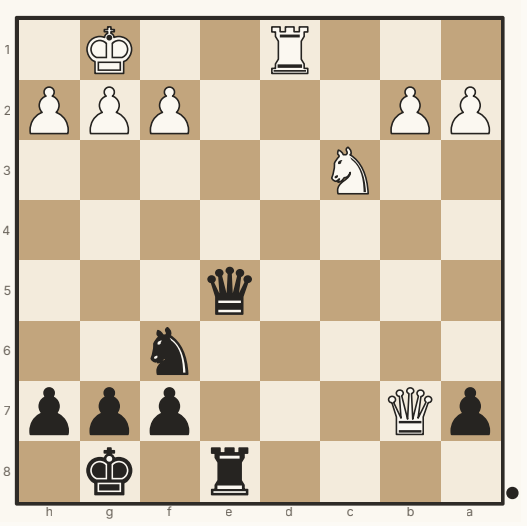
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

11.1 CHECKMATE



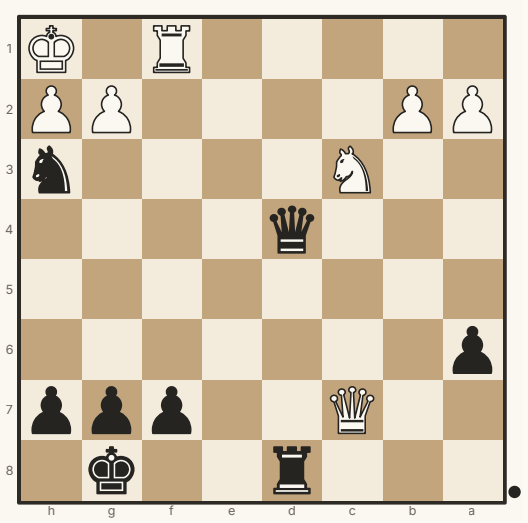
White to move. Black's rook on c2 is hanging, but there is something better. Mate in 1. Solution p. 664

11.2 CHECKMATE



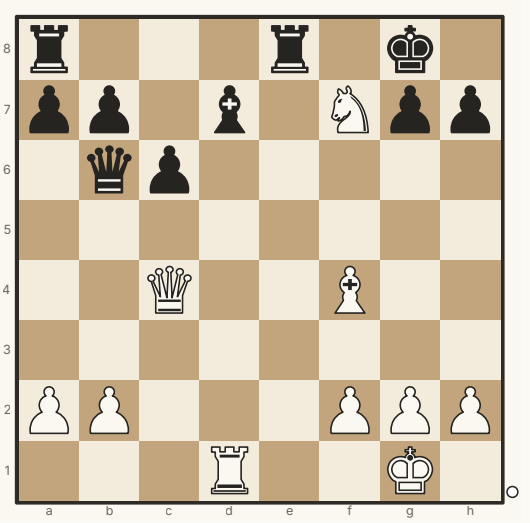
Black to move. White has just grabbed the b7-pawn. Mate in 2. Solution p. 664

11.3 CHECKMATE



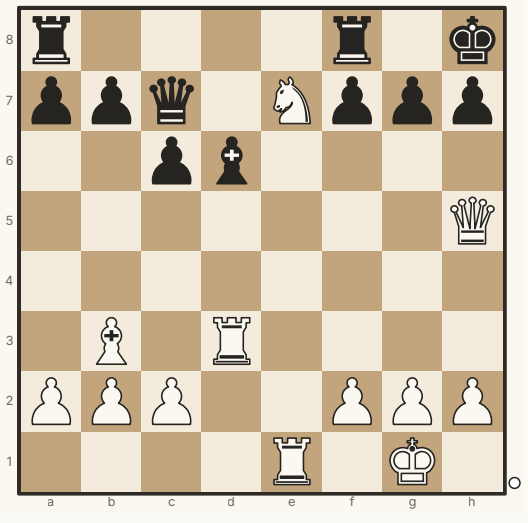
Black to move. White's king is surrounded by its own pieces. Mate in 2. Solution p. 664

11.4 CHECKMATE



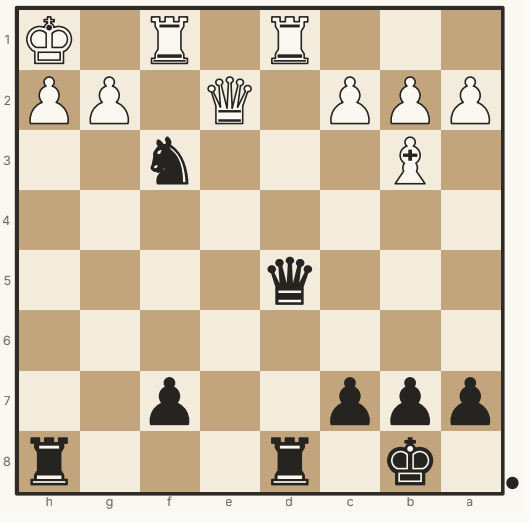
White to move. Play Philidor's smothered mate. Mate in 3. Solution p. 664

11.5 CHECKMATE



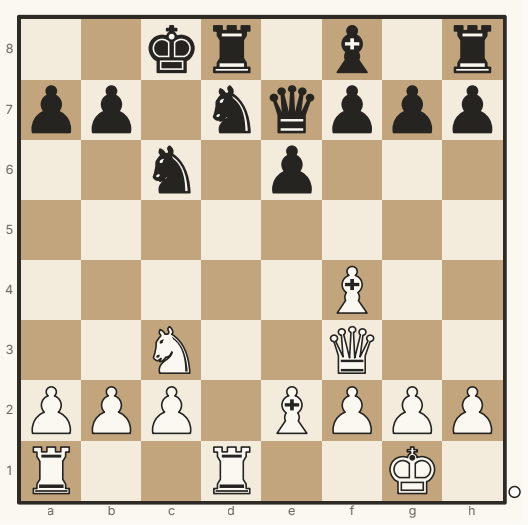
White to move. Your knight on e7 already controls g8 and g6. Mate in 2. Solution p. 664

11.6 CHECKMATE



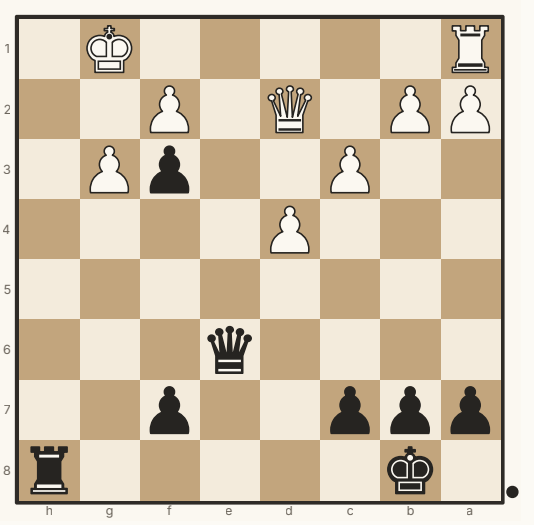
Black to move. Your knight on f3 is attacked twice. Mate in 1. Solution p. 665

11.7 CHECKMATE



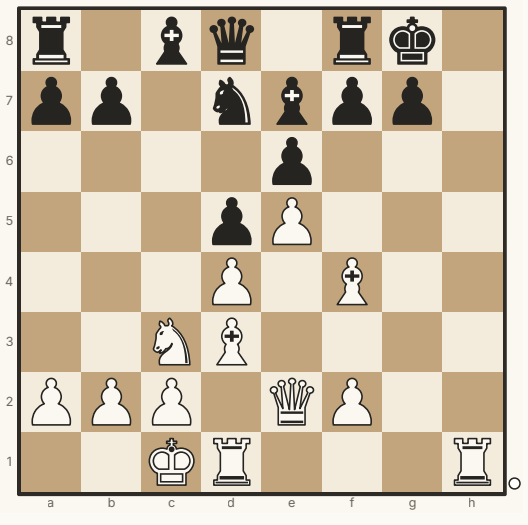
White to move. Black castled long. Use your two bishops. Mate in 2. Solution p. 665

11.8 CHECKMATE



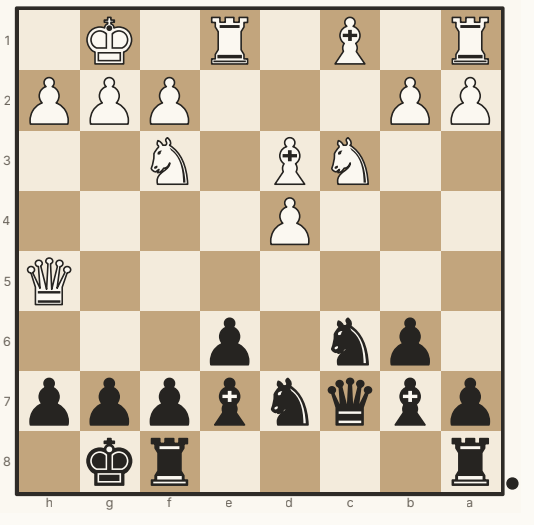
Black to move. Your pawn on f3 is a dagger next to White's king. Find the quiet move that mates in 2. Solution p. 665

11.9 CHECKMATE



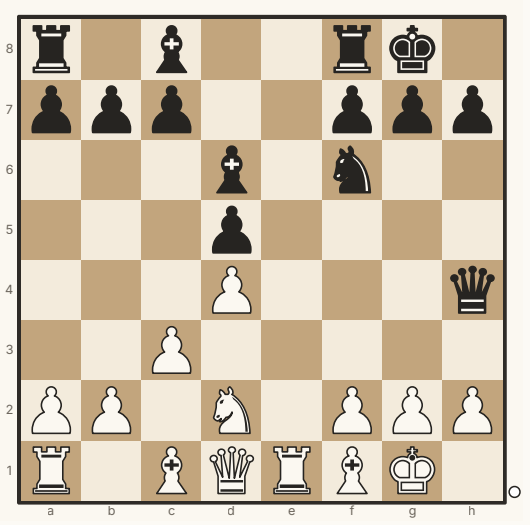
White to move. The h-file is open and your bishop aims at h7. Mate in 3. Solution p. 665

11.10 DEFENCE



Black to move. White threatens Qxh7#. Find the best defence. Solution p. 666

11.11 DEFENCE



White to move. Black's queen and bishop are aiming at h2. Find a move that meets the threat. Solution p. 666

HINTS

11.1 Which black piece could stop a rook on the eighth rank?

11.2 Which white piece is the only guard of e1?

11.3 Fill g1 with a white piece, then check with the knight.

11.4 Start with a double check that forces the king into the corner.

11.5 Which white piece can reach the h-file in one move once h7 has been opened?

11.6 Rook in front of the king, knight guarding the rook.

11.7 Which pawn must move away to open a diagonal to c8?

11.8 A queen supported by the f3-pawn wants to land on g2. Which square aims at g2 and h1 together?

11.9 Sacrifice to bring the king to h8, then check on the h-file.

11.10 Which piece can defend h7 and attack the queen at the same time?

11.11 Count attackers and defenders of h2.

CHAPTER 12 STAGE 600–800

Your Repertoire as White

1.e4, the Italian Game, and a plan against everything else

YOU WILL LEARN TO

- Play the Italian Game with c3 and d3 (the Giuoco Pianissimo) from memory of its set-up, not of long move lists.
- Know the one or two critical moves against every common Black reply to 1.e4.
- Recognize the standard middlegame plans of each structure (Re1, Nbd2-f1-g3, h3, a4/b4, the d3-d4 break).
- Punish the dubious replies (2...f6, 2...d5, 2...Qf6, early ...Nf6xe4 in the Petrov) with short, safe ideas.
- Notice when the standard plan no longer fits and switch to a different one.

> IN YOUR NEXT GAME
>
> Play 1.e4 in every White game this week; after each one, check the first ten moves against this chapter and write down the first move where you or your opponent left the repertoire.

You sit down as White, play 1.e4, and by move eight you are already thinking for five minutes because your opponent played something you have never seen. Worse, you spend that time hunting for a trap you half remember instead of simply developing. At club level, most games are not decided by opening theory. They are decided by whether you reach a normal middlegame with a safe king, developed pieces and a plan you understand.

This chapter gives you exactly that. You will learn one main system, the **Italian Game** with c3 and d3, and short, practical answers to everything else Black can play after 1.e4. The goal is not to memorize twenty moves. The goal is to know *where your pieces go*, *what you are trying to do*, and the handful of concrete moves where a mistake would cost you the game.

How to use a repertoire

A **repertoire** is the set of openings you have decided to play, so that you are not reinventing the wheel every game. A good repertoire for a player rated 600–1200 has three properties: it follows the opening principles from Chapter 6, it leads to positions with clear plans, and it does not collapse if you forget a move.

Every line in this chapter has been checked with a strong engine. That does not mean you must memorize every move. Treat the move lists as examples of the *ideas*, and learn the parts that are marked as critical.

> COACHING JUDGMENT **How much to memorize**
>
> Coaching judgment: below about 1200, learn your first five or six moves by heart in each line, and learn the *plan* that comes after. Beyond that, play by principles. If you find yourself memorizing move fifteen of a line, stop and spend the time on tactics instead.

The three questions after your opponent leaves the book

Sooner or later your opponent plays a move this chapter does not mention. When that happens, ask:

1. **Is something hanging or threatened?** Run the safety scan from Chapter 4: checks, captures, threats, for both sides.
2. **Did the move break an opening principle?** An early queen sortie, a pawn move that weakens the king, a piece moved twice. If so, look for a move that develops *with tempo* (attacking something).
3. **If nothing is wrong, what is my set-up?** Put your pieces on the squares this chapter recommends and continue with the standard plan.

The Italian Game: your main weapon

After 1.e4 e5 2.Nf3 Nc6 3.Bc4, your bishop points at f7, the weakest square in Black's camp, and you are ready to castle on move four or five. That is the **Italian Game**. We will play it in its calm form, called the **Giuoco Pianissimo** ("very quiet game"), with the pawns on c3 and d3.

The set-up

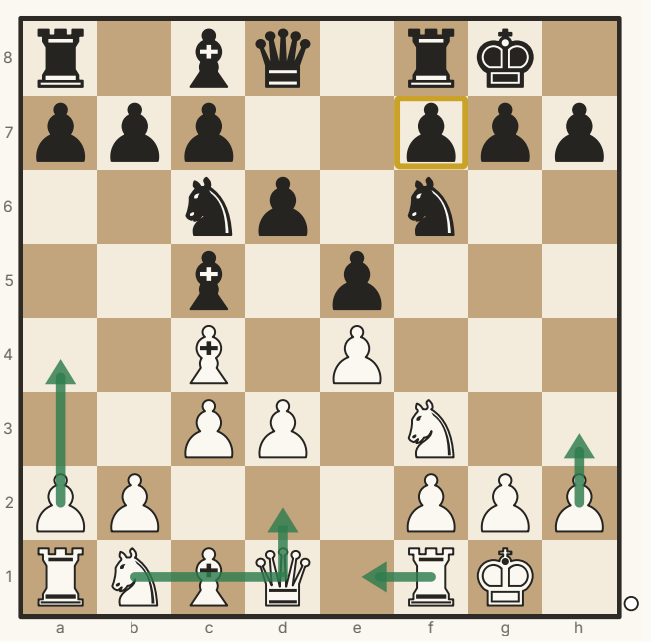
1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6 6.O-O O-O

The basic Giuoco Pianissimo. Both sides have castled and the centre is solid.

Learn the *jobs* of the White moves, not just the moves:

- **c3** prepares a later d4, takes the d4 square away from Black's knight, and gives the bishop a retreat square on c2 if it is ever attacked by ...Na5.
- **d3** supports the e4 pawn so that it can never hang, and opens the c1 bishop's diagonal.
- **O-O** puts the king safe before anything sharp happens.

Notice what White does *not* do: no early Ng5, no early queen move, no grabbing pawns. You are building a house before you invite anyone inside.



The standard plan from here: Re1 (protect e4, prepare Nbd2), Nbd2-f1-g3 (the knight reroute), h3 (stop ...Bg4 and ...Ng4), and a4 to gain queenside space.

The move-order choices

You may play 4.c3 and 5.d3, or 4.O-O and 5.d3 and 6.c3. Both usually reach the same position. The order in this book (4.c3 Nf6 5.d3) is the most common and

avoids an important early pitfall: with d3 already played, your e4 pawn is always protected.

Against 3...Nf6 (the **Two Knights Defence**) you play 4.d3 first, and then c3 and O-O follow:

1.e4 e5 2.Nf3 Nc6 3.Bc4 Nf6 4.d3 Bc5 5.c3 d6 6.O-O O-O

Against 3...Nf6, 4.d3 protects e4 and transposes to the same calm position.

Why not the famous 4.Ng5, attacking f7? Because it leads to very sharp lines in which Black sacrifices a pawn for a fast attack, and you would need to memorize a lot of theory. 4.d3 is sound, simple and keeps the game in the structures you know. If Black plays 4...Be7 instead of 4...Bc5, the plan is identical:

1.e4 e5 2.Nf3 Nc6 3.Bc4 Nf6 4.d3 Be7 5.O-O O-O 6.Re1 d6 7.c3

Same set-up, same plan. Next come h3, Nbd2-f1-g3 and a4 or the d3-d4 break.

Against 3...Be7, the **Hungarian Defence**, Black is very solid but passive. Here you can claim the centre at once with 4.d4:

1.e4 e5 2.Nf3 Nc6 3.Bc4 Be7 4.d4 exd4 5.Nxd4 Nf6 6.Nc3 O-O 7.O-O

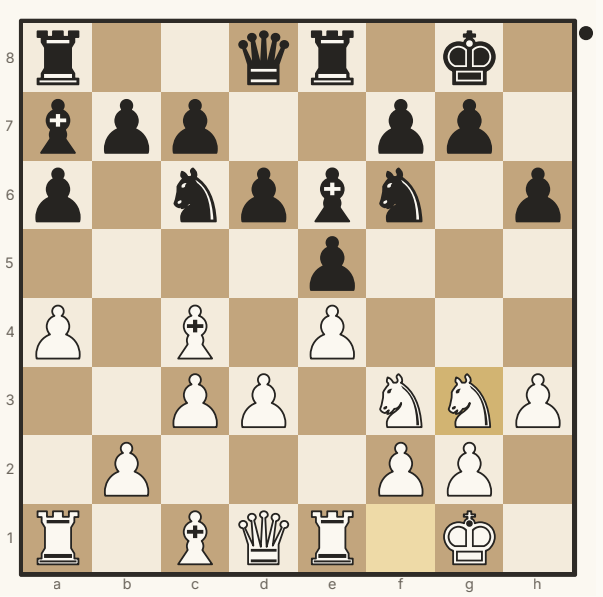
Against the Hungarian, White takes the centre with d4 and develops freely.

If Black keeps the pawn on e5 with 4...d6, push past it with 5.d5, gain space and develop: 5...Nb8 6.Nc3 Nf6 7.h3 O-O 8.O-O. If you prefer, the usual c3/d3 set-up is also perfectly fine against 3...Be7.

The middlegame plan: Re1, h3 and the knight tour

The c3/d3 structure has one awkward piece: the knight on b1. It cannot go to c3, because the pawn is there. Its standard route is **b1-d2-f1-g3**: on g3 it eyes f5 and h5, supports an attack on the kingside, and the f1 square it passes through also protects the king. For the knight to go via f1, the rook must first leave f1, which is one more reason to play Re1.

1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6 6.O-O O-O 7.Re1 a6 8.a4 Ba7 9.h3 h6 10.Nbd2 Re8 11.Nf1 Be6 12.Ng3



A model Pianissimo position. White's knight has arrived on g3, h3 has stopped ...Bg4 and ...Ng4, and a4 has gained queenside space.

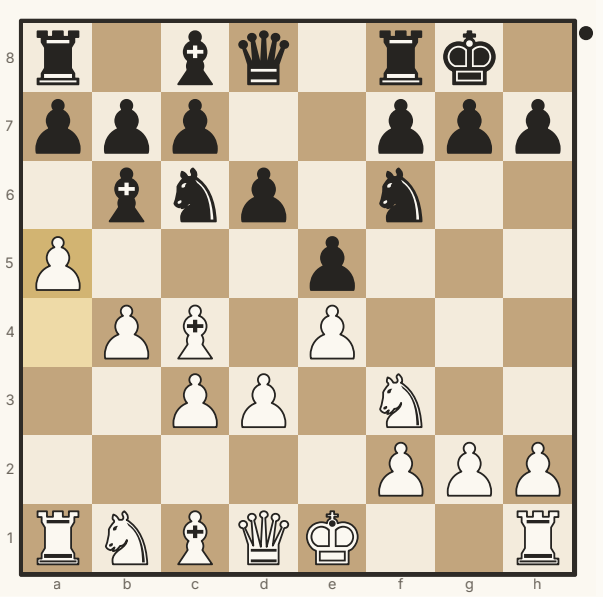
Look at the moves 8.a4 and 9.h3. They look like slow pawn moves, but each one does a job:

- **h3** takes the g4 square away from Black's bishop and knight. Without it, ...Bg4 pins your f3 knight to the queen, and ...Ng4 can join with ...Bc5 against f2.
- **a4** gains space and prepares a4-a5 or b4-b5, pushing Black's queenside pieces back.

The a4 idea against the bishop on c5

Black's bishop on c5 is a strong piece. White often plays b4 to kick it, and then a4 threatens to trap it with a5. Black must answer with ...a6 or ...a5 to give the bishop room. Watch what happens if Black castles automatically:

1.**e4 e5** 2.**Nf3 Nc6** 3.**Bc4 Bc5** 4.**c3 Nf6** 5.**d3 d6** 6.**b4 Bb6** 7.**a4 O-O?** 8.**a5**



The bishop on b6 has no safe square: c5 is covered by the b4 pawn and a7 is blocked by Black's own pawn. Black loses a piece for a pawn or two.

This is the trap. The correct reply to 7.a4 is 7...a6 or 7...a5, after which the position is balanced. The lesson for White: when Black's bishop is on c5 or b6 and Black has not played ...a6, *count its retreat squares*. For Black (see Chapter 13), the lesson is the reverse.

1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6 6.b4 Bb6 7.a4 a5 8.b5 Ne7 9.O-O

After the correct 7...a5, White plays b5 to push the knight away from the centre and castles. White has queenside space; the game is level.

When ...Bg4 pins: play h3

If Black plays ...Bg4 before you have played h3, the knight on f3 is **pinned** (see Chapter 8): it cannot move without exposing your queen. The standard reaction is to ask the bishop at once with h3.

1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6 6.O-O O-O 7.Re1 Bg4 8.h3 Bh5 9.Nbd2

8.h3 forces Black to decide. After 8...Bh5 White simply develops; after 8...Bxf3 9.Qxf3 White gets the bishop pair.

There are two possible answers for Black, and both are fine for you:

- **8...Bxf3 9.Qxf3**: you have the two bishops, and your queen eyes f7 and the kingside.
- **8...Bh5**: keep developing with 9.Nbd2. Later, g4 can break the pin, but be careful.

> COMMON MISTAKE **The early g4 lunge**
>
> After ...Bh5, many players push g4 at once to chase the bishop. With Black's knight on f6 and your king on g1, that can allow ...Nxg4! hxg4 Bxg4, and Black gets two pawns and an attack on your bare king. Only play g4 when Black cannot sacrifice on it, typically when the knight has left f6 or when your king is not castled on that side.

The Bg5 pin, used by White

Your own c1 bishop can pin Black's f6 knight with Bg5. This is most useful once Black has castled, because the pin then also weakens the defence of the black king.

1. e4 e5 2. Nf3 Nc6 3. Bc4 Bc5 4. c3 Nf6 5. d3 d6 6. O-O O-O 7. Bg5 h6 8. Bh4 Be6 9. Nbd2 Bxc4 10. Nxc4

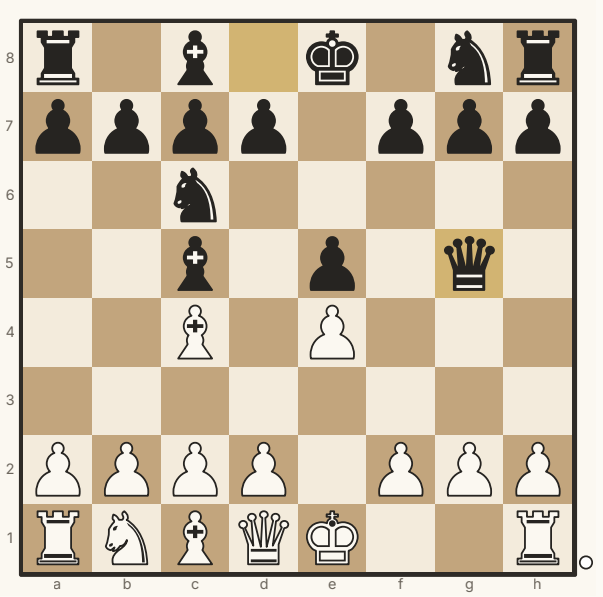
White pins the knight; after ...h6 the bishop keeps the pin from h4. A typical exchange of light-squared bishops follows.

If Black chases the bishop with ...h6 and ...g5, retreat with Bg3. The pawn advance has weakened the squares around Black's king (especially f5 and h5), which your knight can later use. Do not automatically sacrifice a piece on g5 for two pawns; check first whether you have follow-up attackers.

Traps to avoid: the early Ng5 adventure

Beginners love Ng5 in the Italian because it attacks f7 twice with knight and bishop. With Black's bishop on c5 and nothing on f6, it can lose a piece instantly:

1. e4 e5 2. Nf3 Nc6 3. Bc4 Bc5 4. Ng5?? Qxg5



The g5 square is defended by nothing; the queen simply takes the knight along the d8-g5 diagonal.

Even when Ng5 does not lose a piece, a knight raid followed by Nxf7 usually trades two minor pieces for a rook and pawn, which at club level is almost always bad for the side doing it:

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bc4 Bc5** 4. **c3 Nf6** 5. **d3 O-O** 6. **Ng5?! h6** 7. **Nxf7? Rxf7** 8. **Bxf7+ Kxf7**

White gave a knight and bishop for a rook and a pawn. Black's two minor pieces will dominate the middlegame.

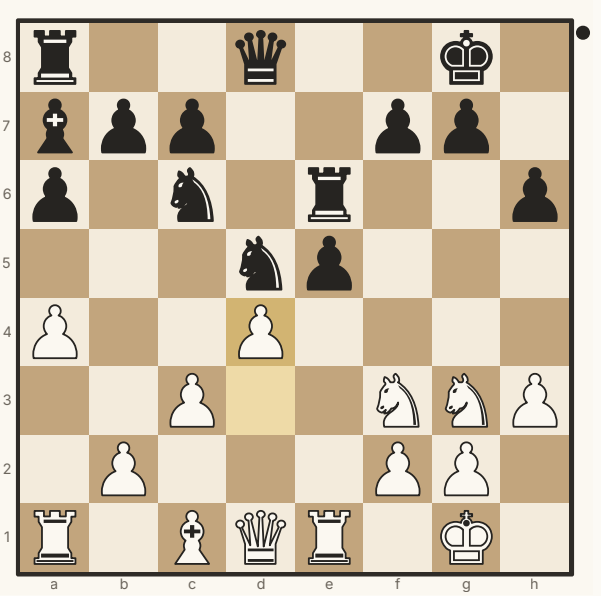
> KEY IDEA **Pieces before adventures**
>
> In the Italian, every early attacking move (Ng5, Qb3, Bxf7+) should be checked against one question: *what happens after the most natural defending move?* If the answer is "I have to retreat and I've lost time", it is not an attack, it is a waste of tempi.

The d3-d4 break

The pawn on d3 is not meant to stay there forever. When your pieces are developed and the centre is stable, the **d3-d4 break** (a pawn move that challenges the opponent's pawn chain) opens lines for your rooks and bishops. The best mo-

ment is usually when Black has spent a tempo, or when Black has opened the centre first.

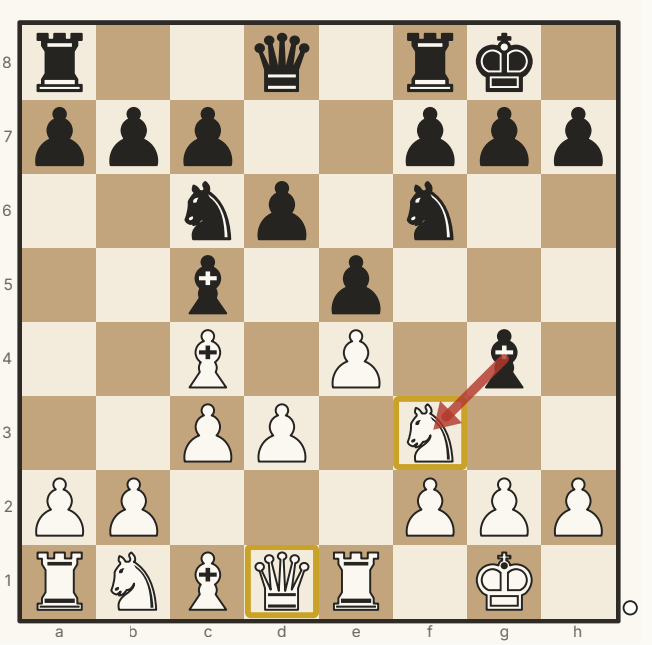
1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6 6.O-O O-O 7.Re1 a6 8.a4 Ba7 9.h3 h6 10.Nbd2 Re8 11.Nf1 Be6 12.Bxe6 Rxe6 13.Ng3 d5 14.exd5 Nxd5 15.d4



Black opened the centre with ...d5, so White strikes back with d4. The e5 pawn is attacked, the rook on e1 eyes the e-file, and White's pieces are ready.

A worked example: what do you play after ...Bg4?

Here is the full thinking process in a position you will meet often. White has just played Re1; Black answers ...Bg4.



White to move. Black's bishop pins the knight on f3 to the queen. What do you play?

Step 1 — Safety scan. Checks: none for either side that make sense. Captures: Bxf7+ Rxf7 gives a bishop for a pawn, no. Threats: Black's pin means that if the knight on f3 moves, Black's bishop takes the queen. Is anything hanging right now? No. Could Black increase the pressure? Yes: ...Nd4 attacks f3 a second time, and ...Nxf3+ gxf3 would wreck the pawns in front of your king. So there *is* a slow threat.

Step 2 — Candidate moves. h3 (ask the bishop), Nbd2 (defend f3 again), Be3 (trade the c5 bishop), and Bb5 (pin the c6 knight).

Step 3 — Compare. h3 forces a decision at once and costs nothing: if 8...Bxf3 9.Qxf3, you have the two bishops; if 8...Bh5, the pin remains but you can now meet ...Nd4 calmly, and Nbd2 follows next move. Nbd2 is also fine (the engine rates it almost equal), but it allows Black to keep all options. Be3 invites ...Bxe3, and after Rxe3 your rook stands on an awkward square.

Step 4 — Decision. 8.h3. Then, against 8...Bxf3 9.Qxf3 Na5, the bishop simply steps aside with 10.Bb5 (or Bb3), and White is comfortable.

1.**e4 e5** 2.**Nf3 Nc6** 3.**Bc4 Bc5** 4.**c3 Nf6** 5.**d3 d6** 6.**O-O O-O** 7.**Re1 Bg4** 8.**h3 Bxf3** 9.**Qxf3 Na5** 10.**Bb5**

After the bishop exchange White has two bishops, and the queen on f3 supports kingside play.

Watch your f2 square

Once you play Re1, the f2 pawn is protected only by your king. Black's typical counterplay is ...Ng4, aiming at f2 together with the bishop on c5. The standard antidote is h3 before Black can use g4, or a calm Re2 or Re1-f1 if the knight arrives. Here, after 7...Ng4 8.Re2, Black's knight has no follow-up and soon retreats:

1.**e4 e5** 2.**Nf3 Nc6** 3.**Bc4 Bc5** 4.**c3 Nf6** 5.**d3 d6** 6.**O-O O-O** 7.**Re1 Ng4** 8.**Re2 Kh8** 9.**h3 Nf6**

Re2 defends f2 a second time. Black's knight raid achieved nothing and it returns home.

When the Italian plan should change

The plan "Re1, Nbd2-f1-g3, h3, a4" is a default, not a law. Change it when:

- **Black opens the centre.** If Black plays ...d5 and exchanges pawns, your pieces need open lines quickly: play d4, put rooks on e1 and d1, and stop slow manoeuvring.
- **Black castles queenside.** Then your a- and b-pawns become attackers: b4, a4, b5 against the black king.
- **Black's bishop leaves c5 for good.** If the dangerous bishop goes to e7, f2 is no longer a target, and you may play d4 earlier.
- **Black gives up the centre with ...exd4.** Recapture with cxd4 and you have the ideal centre (pawns on e4 and d4). Then your plan becomes e4-e5 or d4-d5 to gain space.
- **Black plays ...Na5 to hunt your bishop.** Do not lose time; retreat to b3 or c2 (that is why you played c3) and use the time Black spends with the knight.

Black's other second moves

Not everyone plays 2...Nc6. Here is how to deal with the alternatives you will actually meet.

The Petrov: 2...Nf6

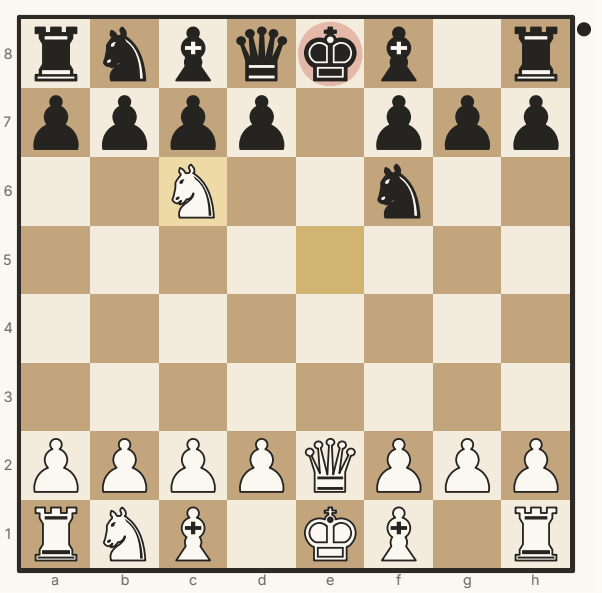
In the Petrov Defence, Black counterattacks e4 instead of defending e5. Take the pawn first, then retreat:

1.e4 e5 2.Nf3 Nf6 3.Nxe5 d6 4.Nf3 Nxe4 5.d4 d5 6.Bd3 Bd6 7.O-O O-O 8.c4 c6 9.Nc3 Nxc3 10.bxc3

The main line in simple form. White develops, castles, and challenges Black's centre with c4.

The critical point is move three. Black must play 3...d6 *before* taking on e4. Against the copycat 3...Nxe4?!, play **4.Qe2**. The queen and the knight on e5 line up on the e-file against Black's king, and if Black moves the attacked knight carelessly, a discovered check wins the queen:

1.e4 e5 2.Nf3 Nf6 3.Nxe5 Nxe4?! 4.Qe2 Nf6?? 5.Nc6+



The knight on c6 gives a discovered check from the queen on e2 and attacks the queen on d8 at the same time. White wins the queen.

The best defence after 4.Qe2 is 4...Qe7 5.Qxe4 d6 6.d4 dxe5 7.dxe5, and White keeps a small but lasting edge thanks to the better development; you do not need to memorize more than that.

Two more Petrov sidelines:

- **3...Qe7?!** (attacking the knight and the e4 pawn): 4.Nf3 Qxe4+ 5.Be2. You have given back the pawn, but you are ahead in development and Black's queen is exposed.

- **3...Nc6?** (the "Stafford Gambit", popular online, which simply gives up a pawn): 4.Nxc6 dxc6 5.d3 and develop with Be2, O-O. Most of Black's tricks rely on White playing f3 or Bc4 too early; d3 and Be2 keep everything protected.

1. **e4 e5** 2. **Nf3 Nf6** 3. **Nxe5 Nc6?** 4. **Nxc6 dxc6** 5. **d3 Bc5** 6. **Be2**

Against the Stafford, a piece up, White simply develops with d3 and Be2 and castles.

The Philidor: 2...d6

Play 3.d4 and develop normally. Black's position is solid but cramped.

1. **e4 e5** 2. **Nf3 d6** 3. **d4 exd4** 4. **Nxd4 Nf6** 5. **Nc3 Be7** 6. **Be2 O-O** 7. **O-O**

Black gave up the centre. White has more space and easy development.

If Black pins with 3...Bg4?!, the game often follows a famous pattern: 4.dxe5 Bxf3 5.Qxf3 dxe5 6.Bc4. Now 6...Nf6? allows 7.Qb3, attacking f7 and b7 at once:

1. **e4 e5** 2. **Nf3 d6** 3. **d4 Bg4?!** 4. **dxe5 Bxf3** 5. **Qxf3 dxe5** 6. **Bc4 Nf6** 7. **Qb3 Qe7** 8. **Nc3 c6** 9. **Bg5**

White's lead in development is huge. This is the pattern of Morphy's famous "Opera Game" (Paris 1858).

Dubious defences: 2...f6, 2...f5, 2...Qf6 and 2...d5

These are moves you will see at the 400–800 level. Each one breaks an opening principle. You do not need deep knowledge; you need one good idea against each.

2...f6? (the Damiano Defence) weakens the e8-h5 diagonal. Take the pawn: 3.Nxe5!. If Black takes back, the queen check is crushing:

1. **e4 e5** 2. **Nf3 f6?** 3. **Nxe5 fxe5?** 4. **Qh5+ Ke7** 5. **Qxe5+ Kf7** 6. **Bc4+**

Black's king is dragged into the open. White has a winning attack (after 4...g6 5.Qxe5+ wins the rook on h8).

The better defence for Black is 3...Qe7 4.Nf3 Qxe4+ 5.Be2, and White is ahead in development while Black's king and queen are awkwardly placed.

2...f5? (the Latvian Gambit) is sharper. A simple, safe answer is 3.Nxe5 Qf6 4.d4 d6 5.Nc4 fxe4 6.Nc3. Your knight is safe on c4, you have developed, and Black's queen is exposed.

1. **e4 e5** 2. **Nf3 f5?** 3. **Nxe5 Qf6** 4. **d4 d6** 5. **Nc4 fxe4** 6. **Nc3**

Against the Latvian, simple development is enough. Next White can play Ne3, Be2, O-O, or attack the e4 pawn.

2...Qf6?! defends e5 with the queen. Develop with gain of time: 3.Nc3 (threatening Nd5 to hit the queen) and then d4.

2...d5? (the Elephant Gambit) gives a pawn for little. Take it: 3.exd5. If 3...e4, 4.Qe2 pins the pawn to the king and wins time: 4...Nf6 5.d3. If 3...Qxd5, 4.Nc3 attacks the queen and you develop with tempo.

1. **e4 e5** 2. **Nf3 d5?** 3. **exd5 e4** 4. **Qe2 Nf6** 5. **d3**

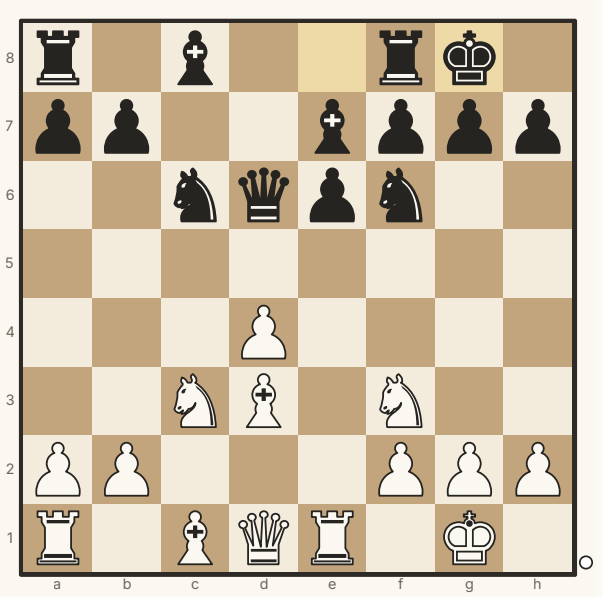
White keeps the extra pawn. The e4 pawn will fall or be exchanged, and White develops normally.

Against the Sicilian: 2.c3 (the Alapin)

After 1.e4 c5, the main lines of the Sicilian are a jungle of theory. The **Alapin**, 2.c3, prepares d4 so that after ...cxd4 you can recapture with the c-pawn and keep two pawns in the centre. It leads to clear positions with easy development.

2...d5: take, then build the centre

1. **e4 c5** 2. **c3 d5** 3. **exd5 Qxd5** 4. **d4 Nf6** 5. **Nf3 e6** 6. **Bd3 Nc6** 7. **O-O cxd4** 8. **cxd4 Be7** 9. **Nc3 Qd6** 10. **Re1 O-O**



White has an isolated d-pawn but active pieces: Nc3 came with tempo against the queen, and the Bd3, Re1 and Nf3 aim at Black's kingside.

On move three, Black's queen comes to d5 and cannot be chased by Nc3 at once, because c3 is occupied by your pawn. That is fine: 4.d4 builds the centre, Nf3 develops, and when Black takes on d4 and you recapture with the c-pawn, Nc3 *does* come with tempo.

The resulting pawn on d4 with no neighbouring c- or e-pawn is called an **isolated queen's pawn** (IQP). It can become weak in an endgame, but in the middlegame it gives your pieces open lines and the e5 square. Your plan: pieces to active squares (Bd3 or Bc4, Re1, Qe2, Ne5), and look for an attack on the black king. Avoid exchanging pieces without a reason; exchanges help the side playing against the isolated pawn.

2...Nf6: push past the knight

1.**e4 c5** 2.**c3 Nf6** 3.**e5 Nd5** 4.**d4 cxd4** 5.**Nf3 Nc6** 6.**cxd4 d6** 7.**Bc4 Nb6** 8.**Bb5 dxe5** 9.**Nxe5 Bd7** 10.**Nxd7 Qxd7** 11.**Nc3**

The knight on f6 was attacked; White gained space with e5 and d4 and develops quickly. Typical continuations are O-O, Qg4 or Qf3, and Re1.

The key ideas: 3.e5 chases the knight, 4.d4 builds the centre, and Nf3 and Bc4 develop while hitting the knight on d5. If Black takes on d4 at move four, you can

recapture with the queen (4...cxd4 5.Qxd4) or play 5.Nf3 first and take back with the pawn. The pawn recapture is the simpler choice.

Other Sicilian moves

- **2...e6**: 3.d4 d5 4.exd5 exd5 5.Nf3, and you develop with Be2 or Bd3, O-O and dxc5 at a good moment.
- **2...d6**: 3.d4 Nf6 4.Bd3, keeping e4 protected. If Black pushes ...Bg4 and ...Nc6, d5 gains space.
- **2...Nc6** and **2...g6**: 3.d4 cxd4 4.cxd4, and you have the ideal centre. Against 2...g6, push e5 when Black's knight wants f6.

1. **e4 c5** 2. **c3 Nc6** 3. **d4 cxd4** 4. **cxd4 d5** 5. **exd5 Qxd5** 6. **Nf3 e5** 7. **Nc3 Bb4** 8. **Bd2 Bxc3** 9. **Bxc3 e4** 10. **Ne5**

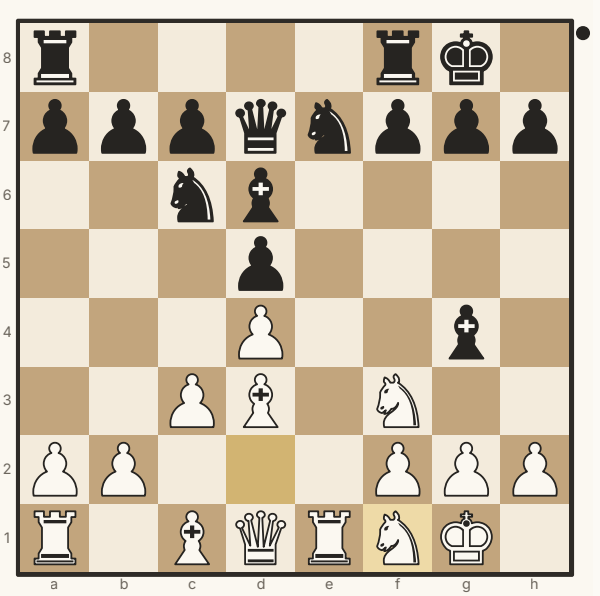
Against 2...Nc6, White recaptures with the c-pawn, develops with tempo on the queen, and the knight lands on a strong central square.

> COMMON MISTAKE **The pawn-grabbing 4...e5?**
>
> In the 2...d5 line, after 3.exd5 Qxd5 4.d4 Black sometimes plays 4...e5? to fight for the centre. Then 5.dxe5 Qxd1+ 6.Kxd1 leaves White a healthy pawn up. You do not lose castling rights in any important way, because with queens off the board your king belongs in the centre anyway.

Against the French: the Exchange Variation

After 1.e4 e6 2.d4 d5, the French Defence normally leads to a locked centre that many club players find uncomfortable. With **3.exd5 exd5** you create a symmetrical, open position where your easy development and the half-open e-file count. You keep the extra tempo that comes with moving first.

1. **e4 e6** 2. **d4 d5** 3. **exd5 exd5** 4. **Bd3 Nc6** 5. **c3 Bd6** 6. **Nf3 Nge7** 7. **O-O Bg4** 8. **Re1 Qd7** 9. **Nbd2 O-O** 10. **Nf1**



A typical Exchange French. White has the same knight manoeuvre as in the Italian (Nbd2-f1-g3) and the e-file for the rook.

The set-up: Bd3, Nf3, O-O, c3 (supporting d4), Re1, and then Nbd2-f1 or Bg5 to pin a knight on f6. The plan: occupy the e-file and the e5 square, and only then think about a kingside attack with Ne5, Qf3 or Qc2.

> **KEY IDEA** **Symmetry is not a draw by itself**
>
> In symmetrical positions the side that controls the open file first usually gets the better of it. Put a rook on e1 early, and if Black challenges with ... Re8, be ready to exchange on e8 when that leaves you in control of the file.

If Black plays an early 4...c5 to create an isolated pawn for you, the simplest antidote is 5.Nf3 Nc6 6.Qe2+, forcing Black to block with the bishop, and then dxc5 at a good moment. You take on c5 and Black recovers the pawn with the bishop, but Black's d5 pawn will be isolated, not yours.

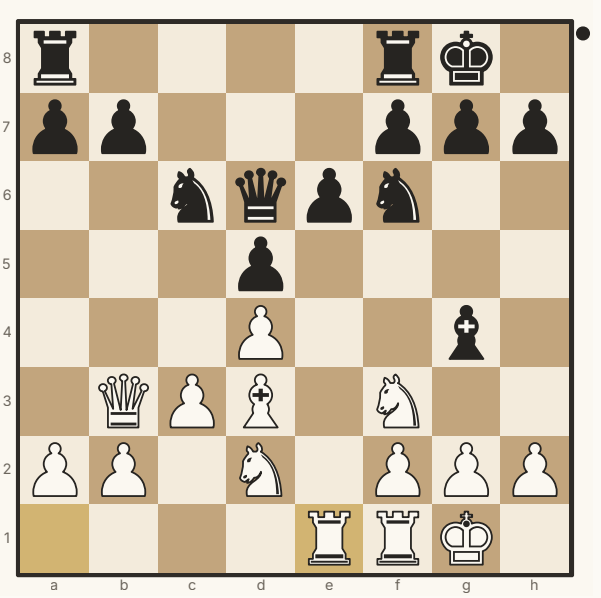
1.e4 e6 2.d4 d5 3.exd5 exd5 4.Bd3 c5 5.Nf3 Nc6 6.Qe2+ Be7 7.dxc5 Nf6 8.O-O O-O 9.h3 Bxc5 10.c3

Now Black has the isolated d-pawn. White blockades with pieces and keeps the queen on the e-file.

Against the Caro-Kann: the Exchange Variation

After 1.e4 c6 2.d4 d5 3.exd5 cxd5 the pawn structure resembles a Queen's Gambit with colours reversed. Play **4.Bd3**, controlling f5 so Black's light-squared bishop cannot develop easily.

1. **e4 c6** 2. **d4 d5** 3. **exd5 cxd5** 4. **Bd3 Nc6** 5. **c3 Nf6** 6. **Bf4 Bg4** 7. **Qb3 Qd7** 8. **Nd2 e6** 9. **Ngf3 Bd6** 10. **Bxd6 Qxd6** 11. **O-O O-O** 12. **Rae1**



The Exchange Caro-Kann. Note 7.Qb3, hitting b7 and d5, which is the classic answer once Black's bishop has left c8.

The set-up: Bd3, c3 (supports d4 and gives the queen the b3 square), Bf4, Nd2, Ngf3, O-O. Two ideas to remember:

- **Qb3** as soon as Black's light-squared bishop leaves c8: the b7 pawn is then defended only by the queen or a rook.
- **h3 and Qc2**, aiming the queen and bishop at h7, if Black castles kingside while your bishop is on d3.

When the Caro plan changes

If Black plays an early ...e5 to free the position, the centre opens. Take on e5 and target the isolated d5 pawn that remains:

1. **e4 c6** 2. **d4 d5** 3. **exd5 cxd5** 4. **Bd3 Nf6** 5. **c3 Nc6** 6. **h3 e5** 7. **dxe5 Nxe5** 8. **Bc2**

Black freed the position with ...e5, but the d5 pawn is now isolated. White develops with Nf3, O-O and puts pieces in front of it.

Against the Scandinavian: 2.exd5 Qxd5 3.Nc3

Take the pawn and attack the queen at once. This is the simplest example of the opening principle "develop with tempo".

1. **e4 d5** 2. **exd5 Qxd5** 3. **Nc3 Qa5** 4. **d4 Nf6** 5. **Nf3 c6** 6. **Bc4 Bf5** 7. **Bd2 e6** 8. **Qe2 Bb4** 9. **O-O-O**

White develops every piece with ideas of Nd5 or Ne5. Castling long is one sharp option; castling short with O-O and Re1 is equally good.

The queen must move after 3.Nc3. It has three main squares:

- **3...Qa5**: 4.d4, then Nf3, Bc4, Bd2 (watch for Nd5 or Nb5 tricks against the queen on a5), O-O.
- **3...Qd6**: 4.d4 Nf6 5.Nf3, then Be2 or g3/Bg2, O-O and Be3. Beware of Black's ... Nb4 or ...Bg4 pressure on d4; keep it protected.
- **3...Qd8**: Black has lost time. 4.d4 Nf6 5.Nf3, and if 5...Bg4 6.h3 Bxf3 7.Qxf3 you have the bishop pair.

1. **e4 d5** 2. **exd5 Qxd5** 3. **Nc3 Qd6** 4. **d4 Nf6** 5. **Nf3 a6** 6. **Be2 Nc6** 7. **O-O Bf5** 8. **Be3**

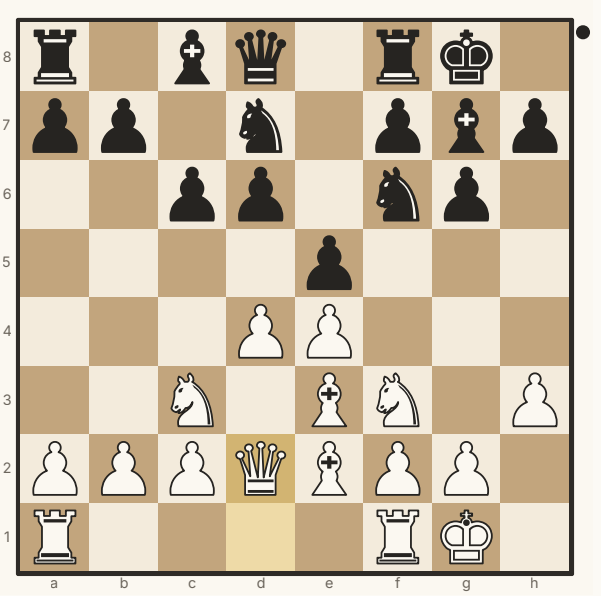
Against 3...Qd6, a calm development: Be2, O-O and Be3 protect d4. Later the queen's rook goes to d1 or e1.

If Black answers 2.exd5 with 2...Nf6, simply play 3.d4 Nxd5 4.Nf3 and develop with Be2, O-O and c4 to chase the knight.

Against the Pirc and Modern

In the **Pirc** (1...d6 2.d4 Nf6) and the **Modern** (1...g6), Black lets you build a big centre and plans to attack it later. Do not over-press. A solid, flexible set-up keeps everything protected:

1.e4 d6 2.d4 Nf6 3.Nc3 g6 4.Nf3 Bg7 5.Be2 O-O 6.O-O c6 7.h3 Nbd7 8.Be3 e5 9.Qd2



White's "Classical" set-up: knights on c3 and f3, bishops on e2 and e3, castled king, and h3 to keep Black's pieces off g4.

The plan: keep the pawns on e4 and d4 as long as possible. When Black challenges with ...e5, you can either exchange on e5 or push **d5** to gain space and play on the queenside with a4 and later c4 (after moving the knight). Against ...c6 and ...b5 (Black gaining queenside space), a4 is the standard answer.

If Black pins with ...Bg4 before you play h3, Be3 and Qd2 are fine; after ...Bxf3 Bxf3 you get the bishop pair:

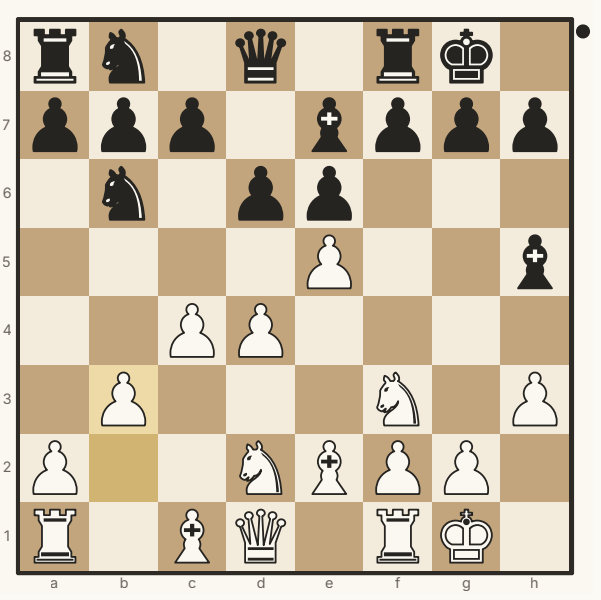
1.e4 d6 2.d4 Nf6 3.Nc3 g6 4.Nf3 Bg7 5.Be2 O-O 6.O-O Bg4 7.Be3 Nc6 8.Qd2 e5 9.d5 Bxf3 10.Bxf3 Nd4 11.Bxd4 exd4 12.Qxd4

Black's queenside play went wrong: White wins a pawn because the d4 pawn could not be held.

Against the Alekhine: 2.e5 Nd5 3.d4 d6 4.Nf3

In the Alekhine Defence (1...Nf6), Black invites you to chase the knight. Do so once, then develop. Do not try to chase it all over the board with c4, f4 and other pawn moves; those big centres need precise theory.

1.**e4** **Nf6** 2.**e5** **Nd5** 3.**d4** **d6** 4.**Nf3** **Bg4** 5.**Be2** **e6** 6.**O-O** **Be7** 7.**c4** **Nb6** 8.**Nbd2** **O-O** 9.**h3** **Bh5** 10.**b3**



The modern main line in simple form. White has a space advantage, every piece is safe, and Black's bishop on h5 is a little offside.

The set-up: e5 and d4 pawns, Nf3, Be2, O-O, then c4 to kick the knight once more and h3. If Black plays ...dxe5, take back with the knight (Nxe5) and use your lead in development:

1.**e4** **Nf6** 2.**e5** **Nd5** 3.**d4** **d6** 4.**Nf3** **dxe5** 5.**Nxe5** **g6** 6.**Bc4** **c6** 7.**O-O** **Bg7** 8.**Re1** **O-O** 9.**Bb3**

White's knight on e5 and bishop on b3 put pressure on f7, and the rook supports the knight.

When plans should change: a general guide

Every set-up in this chapter has a "default plan". Here are the most common signals that it is time to switch to a different one.

- **The opponent opens the centre.** Slow manoeuvres (Nbd2-f1-g3, a4) belong to closed centres. In an open centre, development, rooks on open files and king safety come first.
- **The opponent's king goes to the other side.** Pawn storms only work against a king you are not sheltering behind the same pawns. If kings are castled on opposite sides, advance the pawns in front of *their* king.
- **A piece of theirs is stranded.** A bishop with no retreat (the a4-a5 trap), a queen chased early (the Scandinavian), a knight on the rim. Stop your plan and attack it.
- **They exchange your best piece.** If your bishop on c4 or d3 is traded, look for a different target: the kingside plan usually needs that bishop.
- **They make a pawn move that creates a hole.** ...g5, ...f6 or ...h6 followed by ...g5 creates weak squares like f5 or h5 that your knights can occupy. Re-route a knight there.

> IN YOUR NEXT GAME **Track where the game left the book**
>
> After each game, note the move where your opponent (or you) deviated from this chapter. If the same deviation appears twice, it is time to study it: look up what this chapter suggests and play through the line on a board.

Practice at the board

You learn a repertoire by playing it, not by reading it. Use a physical board or any free chess app with an "analysis board" or "play vs computer from position" function.

1. **Set up the key positions and play them out.** Set up each of these and play at least three games from it against a friend or the computer at a moderate level (as White). Try to follow the plan from this chapter in each game. * Italian, after 6...O-O: `r1bq1rk1/ppp2ppp/2np1n2/2b1p3/2B1P3/2PP1N2/PP3PPP/RNBQ1RK1 w - - 2 7` * Alapin, after 10...O-O in the 2...d5 line: `r1b2rk1/pp2bppp/2n-qpn2/8/3P4/2NB1N2/PP3PPP/R1BQR1K1 w - - 5 11` * Exchange French, after

10.Nf1: `r4rk1/pppqnppp/2nb4/3p4/3P2b1/2PB1N2/PP3PPP/R1BQRNK1 b - - 10 10` (here Black is to move; you play White). * Exchange Caro-Kann, after 12.Rae1: `r4rk1/pp3ppp/2nqpn2/3p4/3P2b1/1QPB1N2/PP1N1PPP/4RRK1 b - - 3 12`

2. **Drill the first six moves.** On an empty board, play out each main line in this chapter from move one, saying the *purpose* of each White move aloud ("c3: prepares d4 and gives the bishop c2"). If you cannot say why a move is played, reread that section.
3. **Play the traps from both sides.** Set up the position after 7.a4 in the b4/a4 line (`r1bqk2r/ppp2ppp/1bnp1n2/4p3/PPB1P3/2PP1N2/5PPP/RNBQK2R b KQkq - 0 7`) and play Black's 7...O-O? to see the bishop trapped; then replay with 7...a5. Do the same for 3...Nxe4?! 4.Qe2 in the Petrov.
4. **Check yourself.** After each training game, compare your first ten moves with this chapter. Mark every move where you deviated and ask whether your move followed the same plan.

> **BOARD DRILL** **Five-minute set-up drill**
>
> Set up the starting position. Have a friend or the app play a random reasonable first move for Black. You have thirty seconds to name your second and third moves from this chapter, and one sentence for the plan. Repeat ten times.

Chapter summary

- Your White repertoire starts with 1.e4 and aims for clear, principled positions rather than sharp theory.
- Against 1...e5 2.Nf3 Nc6 3.Bc4, play the Giuoco Pianissimo: c3, d3, O-O, Re1, h3, Nbd2-f1-g3, and later a4 or the d3-d4 break. Against 3...Nf6 play 4.d3; against 3...Be7 play 4.d4.
- Count the retreat squares of Black's c5 bishop: b4 and a4-a5 can trap it if Black forgets ...a6 or ...a5.
- Avoid early Ng5 and Nxf7 adventures; answer ...Bg4 pins with h3, and keep an eye on f2 once the rook leaves f1.
- Against the Petrov, play 3.Nxe5, and after 3...Nxe4?! use 4.Qe2 with its discovered-check trick. Against 2...d6, 2...f6, 2...f5, 2...Qf6 and 2...d5, develop with tempo and take what is offered when it is safe.

- Against the Sicilian play 2.c3 and d4; against the French and Caro-Kann exchange on d5 and develop with Bd3, c3, Nf3, O-O and Re1 (plus Qb3 in the Caro); against the Scandinavian hit the queen with 3.Nc3; against the Pirc and Alekhine build a solid centre with Nf3, Be2 and O-O.
- Change plans when the centre opens, when kings are on opposite sides, or when an enemy piece is stranded.

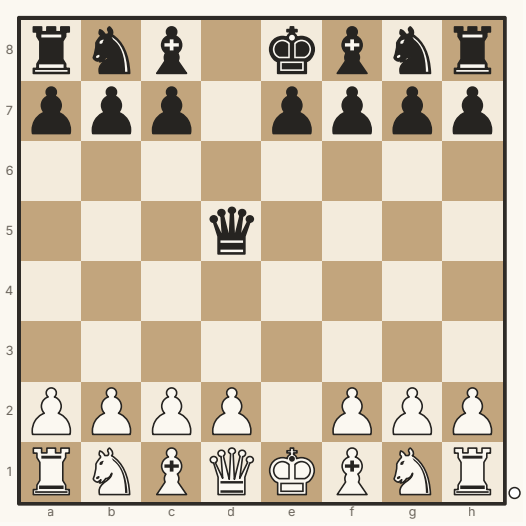
CHEAT CODE **Opening checklist as White**

1. After every opponent move, run the safety scan: checks, captures, threats.
2. If the move broke a principle (early queen, weakening pawn move, undefended piece), look for development with tempo or a safe capture.
3. Otherwise, place your pieces on this chapter's squares: centre pawns first, knights, bishops, castle, then Re1.
4. Play the prophylactic moves at the right time: h3 against ...Bg4/...Ng4; c3 so the bishop has a retreat.
5. Choose the plan that fits the centre: closed → manoeuvre (Nbd2-f1-g3, a4); open → rooks to open files, d3-d4 or attack.
6. Before any early attack (Ng5, Nxf7, g4), check the opponent's most natural defence.

Exercises

Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

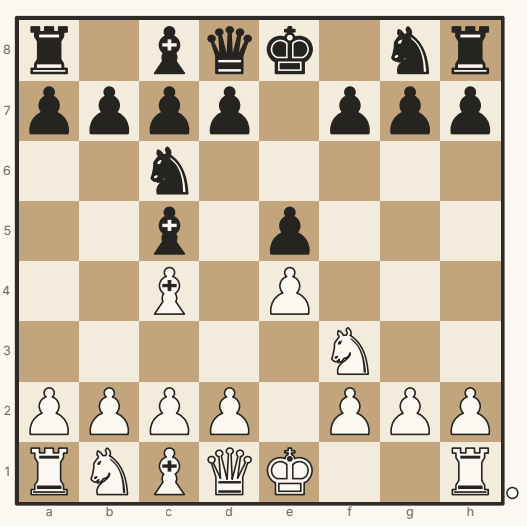
12.1 OPENING



White to move. Your opponent answered 1.e4 with the Scandinavian and has just recaptured 2...Qxd5. Play the repertoire move.

Solution p. 666

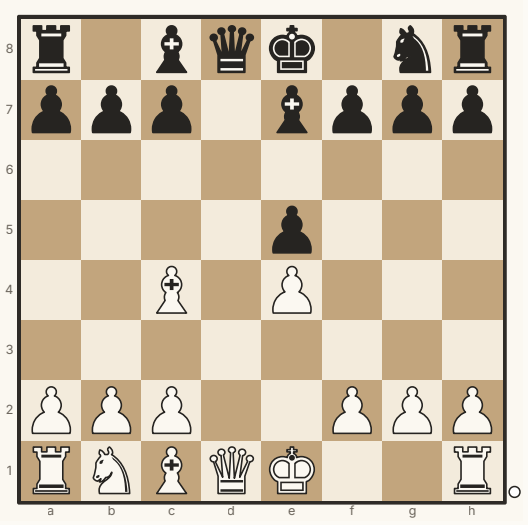
12.2 OPENING



White to move. You reached the Giuoco Piano after 3...Bc5. Choose a sound repertoire move and avoid the beginner's trap.

Solution p. 666

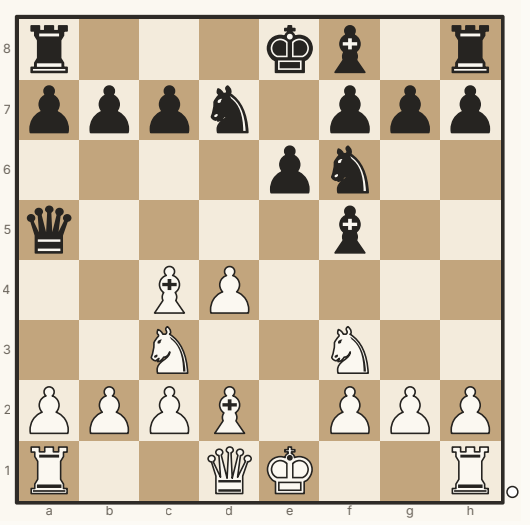
12.3 TACTIC



White to move. Against the Philidor (2...d6 3.d4) Black played 3...Nd7 4.Bc4 Be7?, and after 5.dxe5 Nxe5? 6.Nxe5 dxe5 both f7 and the e5-pawn are weak. Punish it.

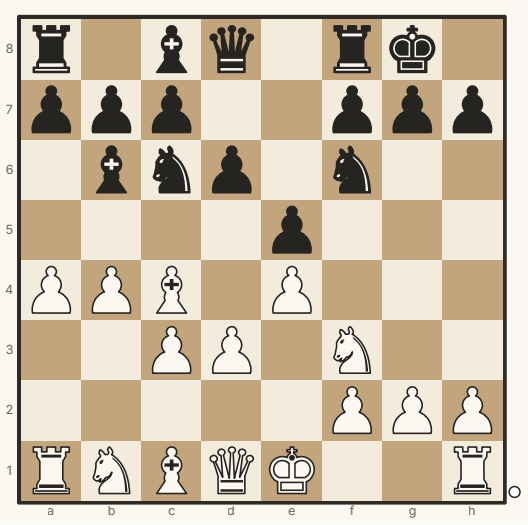
Solution p. 667

12.4 TACTIC



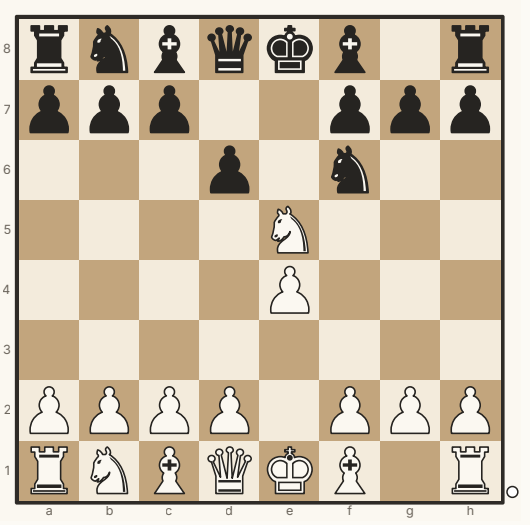
White to move. Against the Scandinavian (2...Qxd5 3.Nc3 Qa5 4.d4 Nf6 5.Nf3 Bf5 6.Bc4 e6 7.Bd2) Black has played the natural 7...Nbd7?! instead of the careful 7...c6. Win material. Solution p. 667

12.5 TACTIC



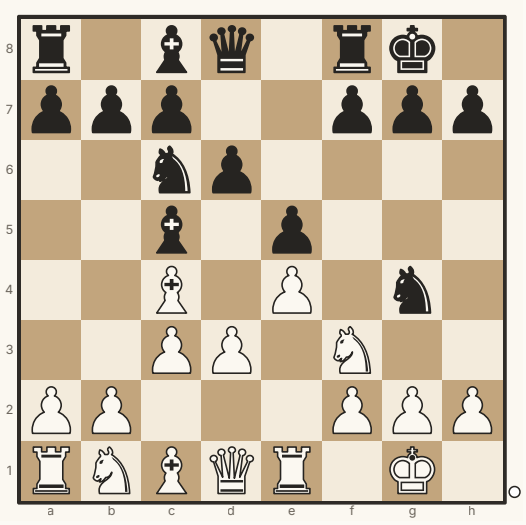
White to move. You played b4 and a4 against the bishop; Black castled automatically with 7...O-O?. Win material. Solution p. 667

12.6 OPENING



White to move. Against the Petrov (2...Nf6) you took on e5, and Black has correctly answered 3...d6, attacking your knight. Where should it go? Solution p. 667

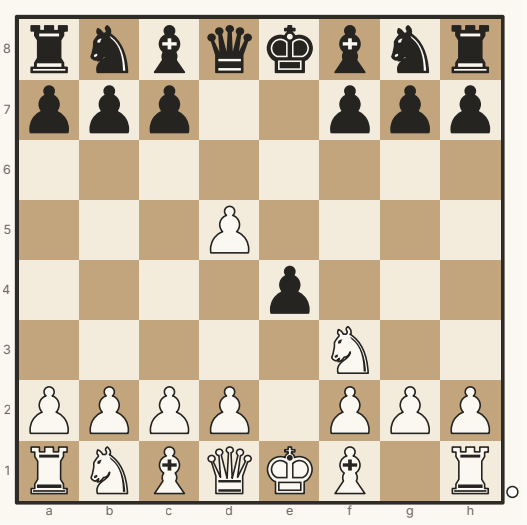
12.7 DEFENCE



White to move. Black's knight jumped to g4, and together with the bishop on c5 it aims at f2. Find a safe move that defends f2.

Solution p. 668

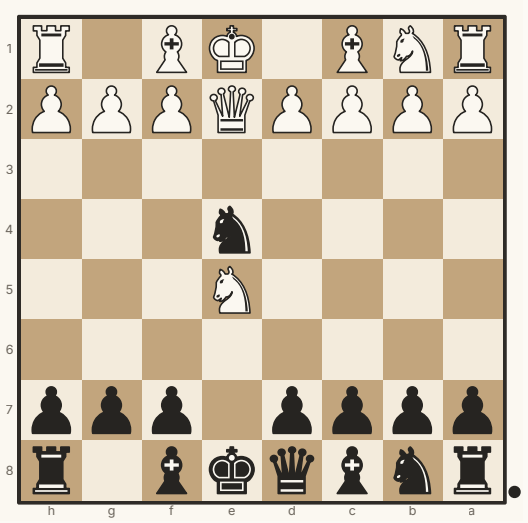
12.8 OPENING



White to move. Black played the Elephant Gambit 2...d5?, you took on d5, and Black pushed 3...e4, attacking your knight. Find the move that keeps an advantage.

Solution p. 668

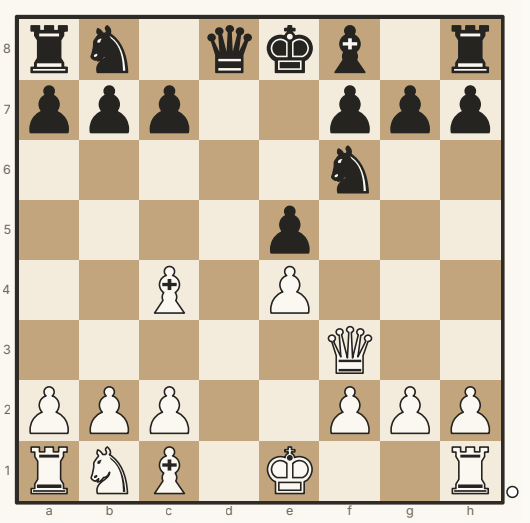
12.9 DEFENCE



Black to move. See the Petrov trap from your opponent's side: Black played the copycat 3...Nxe4?! and you answered 4.Qe2. What is Black's only good defence?

Solution p. 668

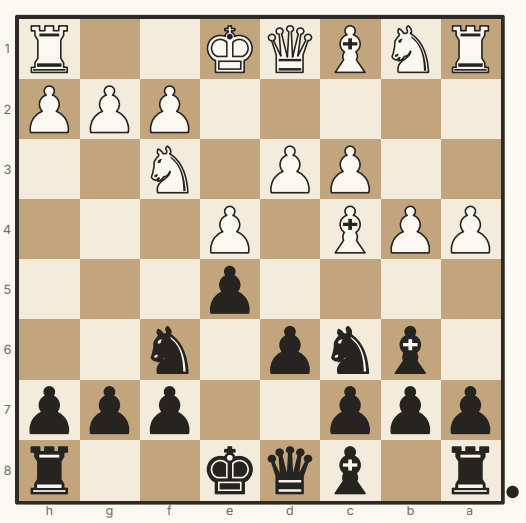
12.10 OPENING



White to move. Against the Philidor, Black played ...Bg4 and ...Bxf3 and has just developed 6...Nf6. Find the strongest move.

Solution p. 668

12.11 DEFENCE



Black to move. See the a4 trap from your opponent's side: you have played b4 and a4 against Black's bishop. What should Black play? Solution p. 669

HINTS

12.1 Which developing move attacks the queen?

12.2 Before you attack f7, ask who defends your attacking piece.

12.3 Find one queen move that attacks two weak points.

12.4 What happens to the queen on a5 if the knight on c3 moves away with a threat?

12.5 How many safe squares does the bishop on b6 have?

12.6 The knight is attacked by a pawn. Which retreat keeps it safe and pointed at the centre?

12.7 Count attackers and defenders on f2 before you chase the knight.

12.8 Is the e4 pawn free to move?

12.9 What is lined up on the e-file?

12.10 Find a queen move that attacks two undefended pawns.

12.11 Count the bishop's retreat squares after White plays a5.

CHAPTER 13 STAGE 600–800

Your Repertoire as Black

1…e5 against 1.e4, the Queen's Gambit Declined against 1.d4, and one set-up for everything else

YOU WILL LEARN TO

- Answer 1.e4 with 1…e5 and 2…Nc6, and know your reply to every common White third move.
- Build the Queen's Gambit Declined set-up against 1.d4 and know its two freeing ideas, …c5 and …e5.
- Use one universal set-up (…Nf6, …d5, …e6, …Be7, …O-O) against the London, 1.c4, 1.Nf3 and other systems.
- Defend against the early attacks club players love (Scholar's Mate attempts, Ng5 raids, the King's Gambit) without panic.
- Punish the typical White opening errors in these lines.

> IN YOUR NEXT GAME
>
> In every game as Black this week, play the set-up from this chapter and, before each of your first ten moves, check whether White's last move created a threat against f7 (after 1.e4) or a pin on your f6 knight (after 1.d4).

Playing Black feels different. White moves first, chooses the battlefield, and at club level often comes straight at your king: a queen on h5, a bishop on c4, a knight jumping to g5. Many games as Black are lost in the first ten moves, not because the defender lacked knowledge, but because they had no clear picture of where their pieces belong.

This chapter gives you that picture. Against 1.e4 you play **1…e5** and **2…Nc6**, the most classical defence there is. Against 1.d4 you play **1…d5** with a **Queen's Gambit Declined** set-up. Against everything else you use one simple scheme. As in Chapter 12, the aim is to understand plans and the few critical moves, not to memorize long lines.

How to defend an opening as Black

Your job as Black in the opening is slightly different from White's. White wants an advantage; you want **equality with a safe king and a plan.** If you reach move fifteen with all your pieces developed, your king castled and no pawns lost, you have done your job, whatever the engine says.

> KEY IDEA **The Black opening routine**
>
> 1. Before every move, ask: *what does White's last move threaten?* (Especially f7 after 1.e4 and the f6 knight after 1.d4.)
> 2. Develop knights before bishops, and get castled.
> 3. Fight for the centre with a pawn (...e5 or ...d5), and later challenge White's centre with a second pawn (...d5, ...c5 or ...e5).

Every line in this chapter is checked by an engine. Mistakes made by White on purpose, so that you see how to punish them, are marked with ? or ??.

Against 1.e4: 1...e5 2.Nf3 Nc6

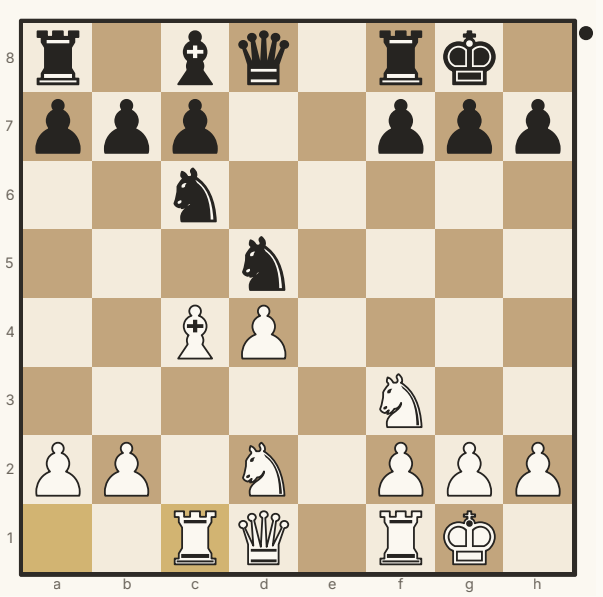
After 1.e4 e5 2.Nf3, the knight attacks your e5 pawn. 2...Nc6 defends it with a developing move and controls d4. Now White has four main third moves: 3.Bc4 (Italian), 3.Bb5 (Ruy Lopez), 3.d4 (Scotch), and the quieter 3.Nc3 or 3.c3.

3.Bc4: the Giuoco Piano with 3...Bc5

Put your bishop on the same kind of square as White's: 3...Bc5 aims at f2, just as White's bishop aims at f7. Now White has two main plans: the sharp central push with 4.c3 and 5.d4, or the slow 4.c3 and 5.d3.

The sharp line: 4.c3 Nf6 5.d4

1.**e4 e5** 2.**Nf3 Nc6** 3.**Bc4 Bc5** 4.**c3 Nf6** 5.**d4 exd4** 6.**cxd4 Bb4+** 7.**Bd2 Bxd2+** 8.**Nbxd2 d5** 9.**exd5 Nxd5** 10.**O-O O-O** 11.**Rc1**



The critical main line. Black has challenged the centre with ...d5, White has an isolated d-pawn, and Black's knight sits on the strong blockading square d5.

The main ideas:

- **5...exd4** takes White's centre pawn. Do not leave your bishop and pawn hanging in front of d4.
- **6...Bb4+** is the key move. After 6.cxd4 your bishop is attacked; instead of retreating to b6 you give check, gain time, and prepare to trade off a pair of pieces.
- **8...d5!** strikes at the e4 pawn immediately. This is the move that makes Black's position work: White's big centre is destroyed before it can advance.

If White blocks the check with 7.Nc3 instead of 7.Bd2, take the e4 pawn: 7...Nxe4. The play becomes sharp, but the engine confirms that Black is fine if you know the next few moves:

1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d4 exd4 6.cxd4 Bb4+ 7.Nc3 Nxe4 8.O-O Bxc3 9.d5 Bf6 10.Re1 Ne7 11.Rxe4 d6

White has sacrificed a pawn for activity. Black castles next, plays ...Bf5 or ...Ng6, and stays solid.

If this feels too sharp, coaching judgment: memorize just the first moves, 7...Nxe4 and ...Bxc3 followed by ...Bf6, ...Ne7 and ...d6. The tempting 7...d5 is clearly

weaker here: with White's knight on c3, opening the centre helps White's development.

White sometimes plays 6.e5 instead of 6.cxd4, attacking your knight. Counterattack in the centre: 6...d5! 7.Bb5 Ne4, and your knight is active on e4.

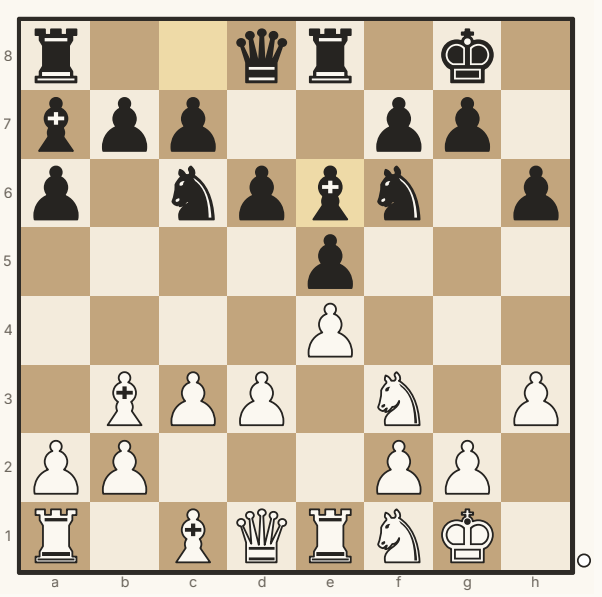
1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d4 exd4 6.e5 d5 7.Bb5 Ne4 8.cxd4 Bb6

Black's knight is strong on e4 and the bishop has retreated to safety on b6. Next comes ...O-O and ...Bg4 or ...f6.

The slow line: 4.c3 Nf6 5.d3

Many club players prefer the quiet 5.d3 (the Giuoco Pianissimo, White's choice in Chapter 12). Mirror White's development: ...d6, ...O-O, ...a6, ...Ba7, ...h6, ...Re8 and ...Be6.

1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6 6.O-O O-O 7.Re1 a6 8.Bb3 Ba7 9.h3 h6 10.Nbd2 Re8 11.Nf1 Be6



A model Black set-up. The bishop on a7 is safe from b4/a4 tricks, ...h6 stops Bg5, and ...Be6 challenges White's strongest bishop.

Three moves do real work here:

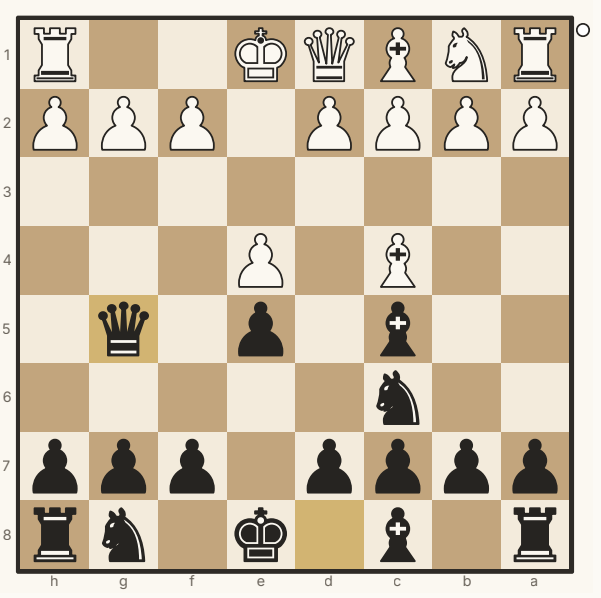
- **...a6 and ...Ba7.** The bishop on c5 is your best piece, and White would like to chase it with b4 and a4-a5. With the bishop on a7 it can never be trapped. Watch out: after 6.b4 Bb6 7.a4, you must play 7...a6 or 7...a5. Castling automatically with 7...O-O? loses the bishop to 8.a5.
- **...h6.** It stops Bg5, which would pin your knight on f6.
- **...Be6.** It offers to trade bishops. White's light-squared bishop on c4 or b3 is White's best attacking piece; exchanging it reduces the pressure on f7.

Your middlegame plan: if White plays slowly, you can strike with ...d5 (when it is well prepared) or regroup with ...Ne7-g6.

Traps and adventures against your Italian

The early Ng5. With your bishop on c5 and no knight on f6, 4.Ng5?? simply hangs the knight: 4...Qxg5. White's c1 bishop is blocked by the d2 pawn and cannot protect g5.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bc4 Bc5** 4. **Ng5?? Qxg5**



The queen wins the knight. Always check whether an attacking piece is defended before you defend against it.

If your knight is already on f6 (say after 4.d3 Nf6 5.Ng5), f7 really is attacked twice. Castle: 5...O-O, and if 6.O-O, then ...h6 sends the knight home. Your king now protects f7 together with the rook.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bc4 Bc5** 4. **d3 Nf6** 5. **Ng5 O-O** 6. **O-O h6** 7. **Nf3 d6**

White's knight raid achieved nothing. Black has castled and gained a useful ...h6 for free.

The fork trick 4.Nxe5?. Sometimes White tries 4.Nxe5? Nxe5 5.d4, hoping to fork bishop and knight. Answer with check: 5...Bb4+! 6.c3 Nxc4, and you keep the extra piece.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bc4 Bc5** 4. **Nxe5? Nxe5** 5. **d4 Bb4+** 6. **c3 Nxc4**

The check saves the bishop and the knight escapes by taking the bishop on c4. Black is a piece up.

The Evans Gambit: 4.b4

White sacrifices a pawn to gain time for c3 and d4. Take it and put the bishop on a5. From there it is hard to attack, and once White's d-pawn has moved it pins the c3 pawn to the white king:

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bc4 Bc5** 4. **b4 Bxb4** 5. **c3 Ba5** 6. **d4 d6** 7. **Qb3 Qd7** 8. **dxe5 Bb6** 9. **Nbd2 Na5**

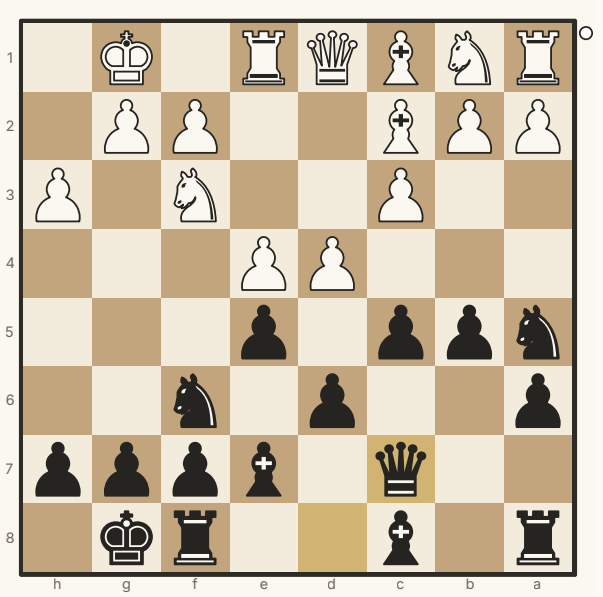
White has regained the pawn on e5, but Black has defended f7 safely, and the knight on a5 hits White's light-squared bishop and queen. Black has a comfortable game.

The two key defensive moves to know are **...d6** (supporting e5 and opening the c8 bishop) and **...Qd7** or **...Qe7** against the queen attack on f7. If you do not want to learn a gambit, 4...Bb6 (declining the pawn) is also perfectly playable.

3.Bb5: the Ruy Lopez

The Ruy Lopez (or **Spanish**) is White's most respected weapon. Its idea is long-term pressure on your knight on c6, which defends e5. You play the classical main line, but you only need the ideas.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bb5 a6** 4. **Ba4 Nf6** 5. **O-O Be7** 6. **Re1 b5** 7. **Bb3 d6** 8. **c3 O-O** 9. **h3 Na5** 10. **Bc2 c5** 11. **d4 Qc7**



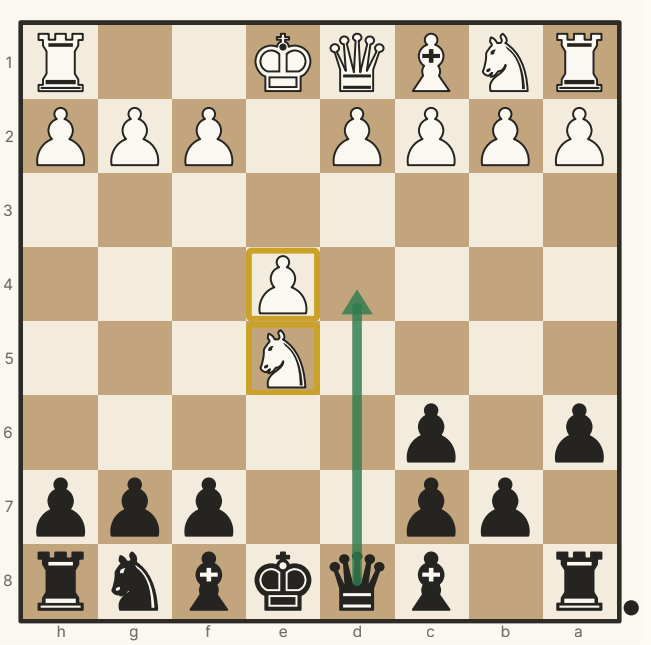
The Closed Ruy Lopez. Black has gained queenside space with ...b5 and ...c5, holds the e5 pawn with ...d6 and ...Qc7, and has castled safely.

Learn what each Black move does:

- **3...a6** asks the bishop a question at once.
- **4...Nf6** develops and attacks e4.
- **5...Be7** prepares castling and keeps things simple.
- **6...b5 and 7...d6** drive the bishop away and secure the e5 pawn. Now the bishop on b3 is no longer pinning anything.
- **...Na5 and ...c5** chase White's best bishop and gain space on the queenside. Another plan is ...Bb7, ...Re8 and ...Bf8, putting pressure on e4.

Is e5 hanging? The Exchange Variation

After 3...a6 White may take the knight: 4.Bxc6 dxc6. Now nothing defends e5, so it looks as if 5.Nxe5 wins a pawn. It does not: 5...Qd4! attacks the knight and the e4 pawn at the same time, and Black regains the pawn.



Black to move. White has just taken on e5. The queen move ... Qd4 forks the knight and the e4 pawn.

1.e4 e5 2.Nf3 Nc6 3.Bb5 a6 4.Bxc6 dxc6 5.Nxe5? Qd4 6.Nf3 Qxe4+ 7.Qe2 Qxe2+ 8.Kxe2

Material is level again, and Black has the bishop pair and an easy game. The pawn on e5 was never really hanging.

The best try for White is 5.O-O, which really does threaten d4 and Nxe5. Then defend the pawn with 5...f6 and develop. If the queens come off early, as in 6.d4 exd4 7.Nxd4 c5 8.Nb3 Qxd1 9.Rxd1 Bg4, your bishop pair makes the endgame comfortable.

1.e4 e5 2.Nf3 Nc6 3.Bb5 a6 4.Bxc6 dxc6 5.O-O f6 6.d4 exd4 7.Nxd4 c5 8.Nb3 Qxd1 9.Rxd1 Bg4 10.f3 Be6

The endgame after the Exchange Variation. Black's two bishops compensate for the doubled c-pawns.

Quiet Ruy Lopez lines

Many club players avoid the main line with 5.d3 or 6.d3. Your set-up stays the same: ...b5, ...Be7, ...d6, ...O-O. If White plays a4 against your b5 pawn, the simplest answer is ...b4 or ...Bd7.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bb5 a6** 4. **Ba4 Nf6** 5. **d3 b5** 6. **Bb3 Be7** 7. **O-O d6** 8. **c3 O-O**

Against the slow 5.d3, the same set-up. Black's next moves are ...Na5, ...c5 or ...Bb7 and ...Re8.

3.d4: the Scotch

White opens the centre at once. Take the pawn and develop with an attack on e4: 3...exd4 4.Nxd4 Nf6. Now your knight on f6 hits e4, and White must decide how to protect it.

The main line is 5.Nxc6 bxc6 6.e5, chasing the knight. The key moves are **6...Qe7!** (pinning the e5 pawn against the white king, so it cannot be defended easily) and **7...Nd5:**

1. **e4 e5** 2. **Nf3 Nc6** 3. **d4 exd4** 4. **Nxd4 Nf6** 5. **Nxc6 bxc6** 6. **e5 Qe7** 7. **Qe2 Nd5** 8. **c4 Nb6** 9. **Nd2 Qe6** 10. **b3 a5**

Black's queen blockades in front of the e5 pawn and the a-pawn begins to annoy White's queenside. Both sides have chances.

If you prefer something calmer, White's other moves let you develop simply:

- **5.Nc3** (the Scotch Four Knights): 5...Bb4 6.Nxc6 bxc6 7.Bd3 d5. Then castle and put a rook on e8.
- **5.Nxc6 bxc6 6.Bd3:** 6...d5, and your centre is excellent.

1. **e4 e5** 2. **Nf3 Nc6** 3. **d4 exd4** 4. **Nxd4 Nf6** 5. **Nc3 Bb4** 6. **Nxc6 bxc6** 7. **Bd3 d5** 8. **exd5 cxd5** 9. **O-O O-O** 10. **Bg5 c6**

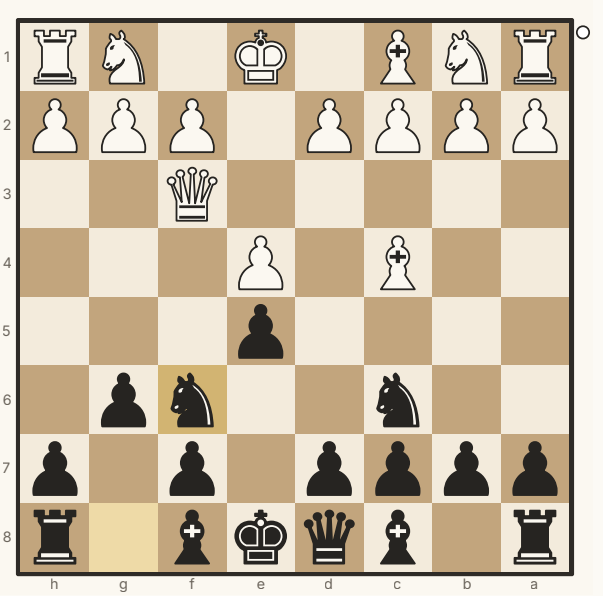
The Scotch Four Knights. Black has a solid centre and active bishops; the c6 pawn holds d5.

If White plays a gambit with 4.c3 or 4.Bc4, the same rule applies: develop with an attack on e4 (4...Nf6) and strike back with ...d5 or ...d6 when White pushes e5.

Early queen attacks and Scholar's Mate attempts

At 400–800 you will often meet 2.Qh5 or 2.Bc4 and 3.Qh5, aiming at **Scholar's Mate** (Qxf7#). The defence is simple if you *look at f7 before every move*.

1. **e4 e5** 2. **Qh5 Nc6** 3. **Bc4 g6** 4. **Qf3 Nf6**



Black has blocked the queen with ...g6 and, when it returned to f3, covered f7 with ...Nf6 at the right moment. The knight move develops and defends.

The two key moments:

1. **After 3.Bc4**, f7 is attacked twice. Do *not* play 3...Nf6??, which allows 4.Qxf7#. Block with 3...g6 (or defend with 3...Qe7).
2. **After 4.Qf3**, f7 is attacked twice again. Now 4...Nf6 blocks the queen's path.

Sometimes White then tries 5.Qb3?, attacking f7 a third time from a different angle. Here the engine shows a strong answer: 5...Nd4!, which attacks the queen and the c2 pawn. (The solid 5...Qe7 also defends, but it gives White time to recover.)

1. **e4 e5** 2. **Qh5 Nc6** 3. **Bc4 g6** 4. **Qf3 Nf6** 5. **Qb3? Nd4** 6. **Qc3? Nxe4**

The knight on d4 hit the queen and c2; now the other knight takes e4 and attacks the queen yet again. White's queen has made five moves and Black has won a pawn with a huge lead in development. (White's best was 6.Qd3, but Black is clearly better there too.)

> **COMMON MISTAKE** **Chasing the queen with pawns**
>
> After a premature Qh5, many players try to "punish" the queen by chasing it around with pawn moves. Usually a calm developing move that defends f7 (...g6 or ...Nf6 at the right moment, ...Qe7) is better. Develop *with* tempo against the queen (...Nf6, ...Nd4) when you can, and never with a weakening pawn move when a piece move is available.

If White plays the queen raid with 2.Bc4 first, the same ideas apply: 2...Nc6 3.Qh5 g6 4.Qf3 Nf6.

White's other second moves

2.Bc4 (the Bishop's Opening)

Play 2...Nf6, attacking e4. After 3.d3 you can play 3...Bc5 and the game resembles a Giuoco Pianissimo; after 3.Nf3 Nc6 it is a real Italian or Two Knights.

1. **e4 e5** 2. **Bc4 Nf6** 3. **d3 Bc5** 4. **Nf3 d6** 5. **c3 O-O** 6. **O-O**

A familiar position: the same set-up as the Pianissimo.

2.Nc3 (the Vienna)

Play 2...Nf6. Against 3.Bc4 develop with 3...Nc6, and against the sharper 3.f4 strike in the centre with 3...d5!:

1. **e4 e5** 2. **Nc3 Nf6** 3. **f4 d5** 4. **fxe5 Nxe4** 5. **Nf3 Be7** 6. **d4 O-O** 7. **Bd3 f5**

Against the Vienna Gambit, ...d5 and ...Nxe4 give Black an active knight; ...f5 supports it.

After 3.Nf3 Nc6 you reach the Four Knights Game, and the Scotch ideas apply: after 4.d4 exd4 5.Nxd4 Bb4, develop and strike with ...d5.

2.d4 (the Center Game)

Take the pawn. After 2...exd4 3.Qxd4, White's queen is in the centre early. Develop with tempo: 3...Nc6 4.Qe3 Nf6, then ...Bb4, ...O-O and ...Re8 against the e4 pawn.

1. **e4 e5** 2. **d4 exd4** 3. **Qxd4 Nc6** 4. **Qe3 Nf6** 5. **Nc3 Bb4** 6. **Bd2 O-O** 7. **O-O-O Re8**

Every Black move develops a piece. The rook on e8 and knight on f6 both attack e4.

If White gambits with 3.c3 (the Danish), the simplest answer is 3...d5, giving back the pawn to free your game: 4.exd5 Qxd5 5.cxd4 Nc6, or 4...Nf6.

2.f4 (the King's Gambit)

The King's Gambit is aggressive and full of traps. The simplest reply is the counterattack **2...d5**, striking at e4 before White can use the f-file. After 3.exd5 exf4 4.Nf3 Nf6, develop quickly and take back on d5.

1. **e4 e5** 2. **f4 d5** 3. **exd5 exf4** 4. **Nf3 Nf6** 5. **Bc4 Nxd5** 6. **O-O Be7** 7. **Bxd5 Qxd5** 8. **Nc3 Qd8** 9. **d4 O-O** 10. **Bxf4**

Material is level and both kings are safe. Black has no weaknesses; White's extra activity is small.

Watch for the beginner's error 3.fxe5??: then 3...Qh4+ 4.g3 Qxe4+ forks the king and the rook on h1.

1. **e4 e5** 2. **f4 d5** 3. **fxe5?? Qh4+** 4. **g3 Qxe4+** 5. **Qe2 Qxh1**

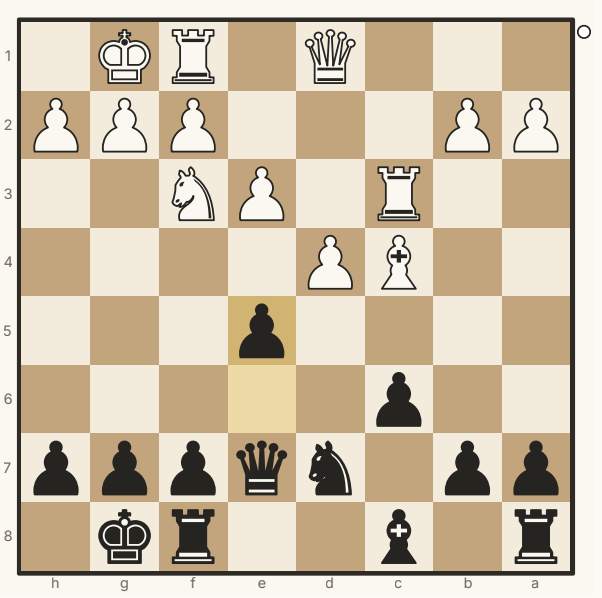
The check on h4 exploits the weakened e1-h4 diagonal. Black wins a whole rook.

Against 1.d4: the Queen's Gambit Declined

After 1.d4 d5 2.c4, White offers a pawn to pull your d5 pawn away from the centre. Decline with **2...e6**: you keep a pawn on d5 and open a diagonal for your f8 bishop. This is one of the most solid openings in chess.

The set-up

1. **d4 d5** 2. **c4 e6** 3. **Nc3 Nf6** 4. **Bg5 Be7** 5. **e3 O-O** 6. **Nf3 Nbd7** 7. **Rc1 c6** 8. **Bd3 dxc4** 9. **Bxc4 Nd5** 10. **Bxe7 Qxe7** 11. **O-O Nxc3** 12. **Rxc3 e5**



The classical freeing manoeuvre. Black traded pieces with ...Nd5 and then broke out with ...e5. The c8 bishop is free at last.

Learn the pieces' squares: **...e6, ...Nf6, ...Be7, ...O-O, ...Nbd7, ...c6**. The pin with Bg5 is not dangerous once your bishop is on e7. The only real problem of the QGD is your **light-squared bishop on c8**, which is blocked by the e6 pawn. Every Black plan aims to free it.

The two freeing ideas:

1. **...dxc4 and ...e5** (as above): trade on c4 when White's bishop has moved, exchange a pair of minor pieces with ...Nd5, then push ...e5. The bishop on c8 is free.
2. **...dxc4 and ...c5**: attack White's d4 pawn directly with the c-pawn.

1. **d4 d5** 2. **c4 e6** 3. **Nc3 Nf6** 4. **Bg5 Be7** 5. **e3 O-O** 6. **Nf3 h6** 7. **Bh4 Nbd7** 8. **Bd3 dxc4** 9. **Bxc4 c5** 10. **O-O a6**

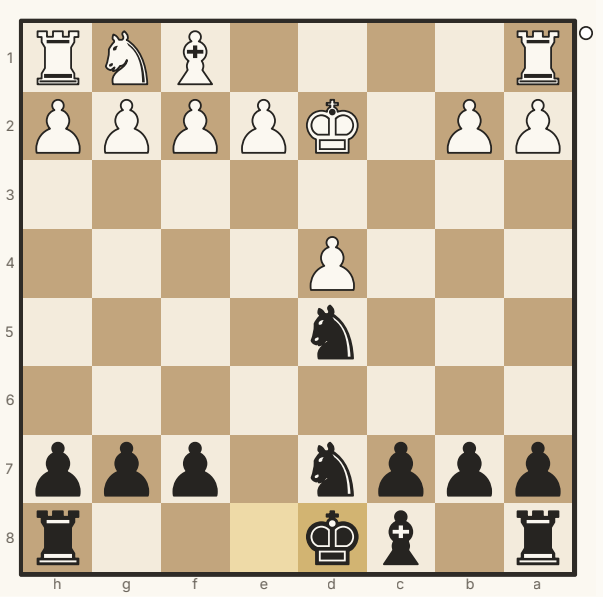
The ...c5 plan. Next comes ...b5 and ...Bb7, and the c8 bishop finds a long diagonal.

The move **...h6** is useful: it asks White's bishop whether it wants to take on f6 (giving you the bishop pair) or retreat to h4.

The Elephant Trap pattern

The QGD has one famous trap that you can set and must understand. With White's bishop on g5 pinning your knight on f6, the pin is *not* absolute: your queen is on d8, not your king. If White tries to win the d5 pawn by taking with the knight, you can break the pin:

1. **d4 d5** 2. **c4 e6** 3. **Nc3 Nf6** 4. **Bg5 Nbd7** 5. **cxd5 exd5** 6. **Nxd5?? Nxd5** 7. **Bxd8 Bb4+** 8. **Qd2 Bxd2+** 9. **Kxd2 Kxd8**



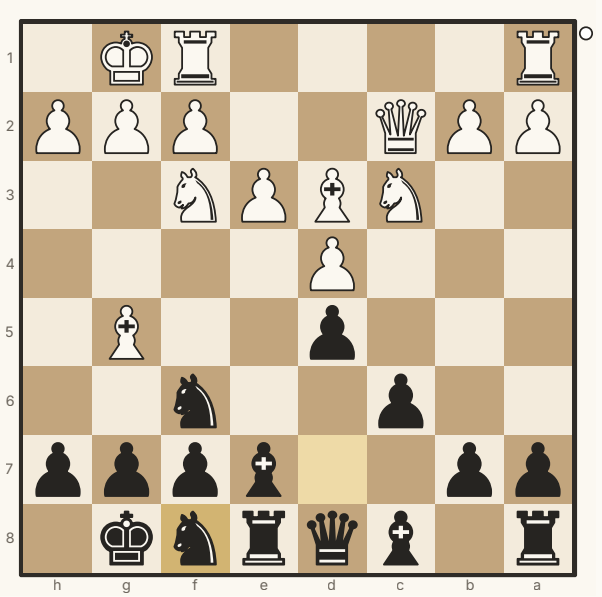
The queens have been traded and Black has won a knight for a pawn. The trap has netted a whole piece.

After 6.Nxd5?? Nxd5!, if White takes the queen with 7.Bxd8, Black plays 7...Bb4+ and wins White's queen back. When the dust settles, Black is a knight for a pawn ahead. (If White does not take the queen, Black is simply a piece up.) This is called the **Elephant Trap**. It works only because White's queen is on d1 and can be forced to block the check on d2. If White's queen has already left d1, check before you rely on it.

The Exchange Variation: the Carlsbad structure

If White takes on d5 early with cxd5 exd5, you get the **Carlsbad structure**: your pawns on d5 and c6 (or still c7) with no black e-pawn, against White's pawns on d4 and e3 with no white c-pawn. This structure appears so often that it is worth knowing its main ideas.

1. **d4 d5** 2. **c4 e6** 3. **Nc3 Nf6** 4. **cxd5 exd5** 5. **Bg5 Be7** 6. **e3 O-O** 7. **Bd3 Nbd7** 8. **Qc2 Re8** 9. **Nf3 c6** 10. **O-O Nf8**



The Carlsbad structure. Black's bishop on c8 is free now, the knight goes ...Nf8-e6 or -g6, and the rook on e8 controls the half-open e-file.

Your plans in the Carlsbad:

- **Develop the c8 bishop** to g4 or f5 (the diagonal is open now). Trading it for White's knight on f3 or bishop on d3 is often good.
- **Use the e-file**: rook to e8, knight to e4 when it can be supported (...Ne4 often follows ...Nf8-e6-g5 or ...Nd7).
- **Watch for White's minority attack** (b4-b5 to create a weakness on c6). Answer it with ...a6 or ...a5 to slow it down, and by keeping pieces active on the kingside.

Other White systems in the Queen's Gambit

- **4.Nf3 Be7 5.Bf4**: same set-up, with ...c5 as the main freeing break.
- **3.Nf3 and g3 (Catalan)**: you can take on c4 with ...dxc4 and keep the pawn with ...a6 and ...b5 when it is safe, or simply play ...Be7, ...O-O and ...c6.
- **4.e3**: the quietest move. Play ...Be7, ...O-O, ...dxc4 and ...c5.

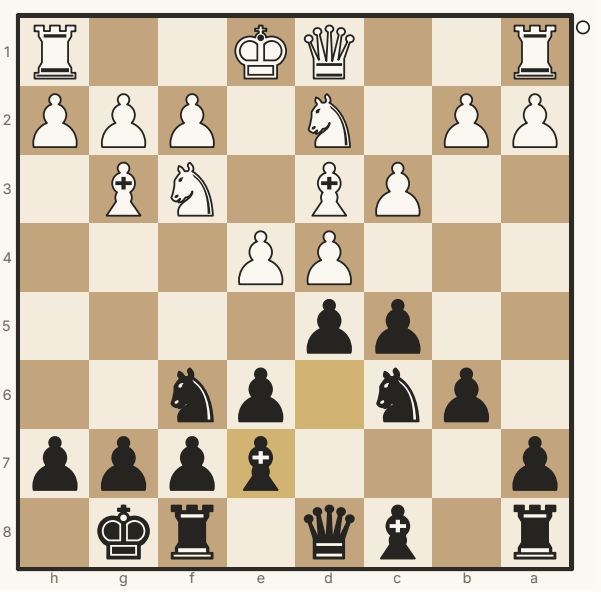
1.**d4 d5** 2.**c4 e6** 3.**Nc3 Nf6** 4.**e3 Be7** 5.**Nf3 O-O** 6.**Bd3 dxc4** 7.**Bxc4 c5** 8.**O-O a6**

Against the quiet 4.e3, ...dxc4 and ...c5 free the game immediately. Next comes ...b5 and ...Bb7.

Against the London: 2.Bf4

The **London System** (1.d4 d5 2.Bf4 or 2.Nf3 and 3.Bf4) is one of the most common openings at club level. White builds a solid pyramid with e3, c3 and Nbd2 and often attacks with Ne5, Bd3 and Qf3 or Qh5. Your antidote: hit the centre with **...c5** and challenge the bishop with **...Bd6**.

1.**d4 d5** 2.**Bf4 Nf6** 3.**e3 e6** 4.**Nf3 c5** 5.**c3 Nc6** 6.**Nbd2 Bd6** 7.**Bg3 O-O** 8.**Bd3 b6** 9.**e4 Be7**



Black's set-up against the London: ...e6, ...c5, ...Nc6, ...Bd6 to challenge White's bishop, and ...b6/...Bb7 to develop the last minor piece.

The ideas:

- **...c5** puts pressure on d4 and starts queenside play.
- **...Bd6** offers to trade White's best bishop. If White takes, 7.Bxd6 Qxd6, your queen is well placed and the e5 break follows.
- **...Qb6** at the right moment attacks b2, which White's bishop has left behind. After 5.c3 Qb6 6.Qb3 c4 7.Qc2, you gain queenside space.

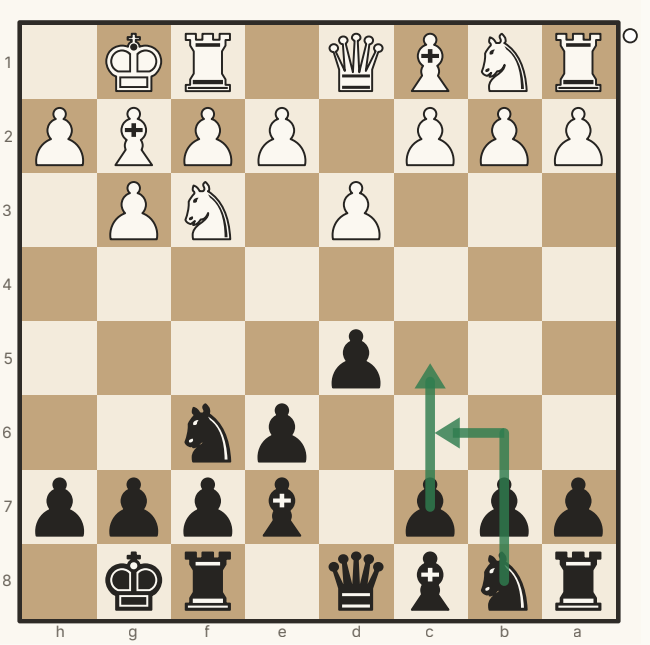
1. **d4 d5** 2. **Bf4 Nf6** 3. **e3 e6** 4. **Nf3 c5** 5. **c3 Nc6** 6. **Nbd2 Bd6** 7. **Bxd6 Qxd6** 8. **Bd3 e5** 9. **dxe5 Nxe5** 10. **Nxe5 Qxe5**

After the bishop trade, ...e5 frees Black's game completely. Black has a fine, active position.

> COACHING JUDGMENT **The Ne5 attack**
>
> London players love to put a knight on e5 and attack with Qf3, Qh3 or g4. Coaching judgment: answer an unsupported knight on e5 with ...Nd7 to challenge it, and do not castle into an attack you have not counted. In the line 7.Bg3 O-O 8.Bd3 Qe7 9.Ne5 Nd7, Black challenges the knight at once and is fine.

Against everything else: one universal set-up

Against 1.c4, 1.Nf3, 1.g3, 1.b3, the Colle (d4, Nf3, e3, Bd3) and most other quiet openings, you do not need separate preparation. Build the same set-up you use in the QGD: **...Nf6, ...d5, ...e6, ...Be7, ...O-O**, followed by **...c5** and **...Nc6**.



The universal set-up against 1.Nf3 and 1.g3. Black has a pawn in the centre, a safe king and two natural plans: ...c5 and ...Nc6, or ...b6 and ...Bb7.

Examples:

- **Against the Colle:** 1.d4 d5 2.Nf3 Nf6 3.e3 e6 4.Bd3 c5 5.c3 Nc6 6.Nbd2 Bd6 7.O-O O-O. White plans e3-e4; if 8.dxc5 Bxc5 9.e4, answer with ...Qc7 to keep an eye on e5 and e4.
- **Against 1.c4:** 1...Nf6 2.Nc3 e6 and ...d5 next. If White plays 3.e4, push past the knight with 3...d5 4.e5 d4!.
- **Against 1.b3 or 1.f4:** ...d5, ...Nf6, ...e6 (or ...g6 and ...Bg7 against 1.f4) and castle.

1.**d4 d5** 2.**Nf3 Nf6** 3.**e3 e6** 4.**Bd3 c5** 5.**c3 Nc6** 6.**Nbd2 Bd6** 7.**O-O O-O** 8.**e4 cxd4** 9.**cxd4 dxe4** 10.**Nxe4 Be7**

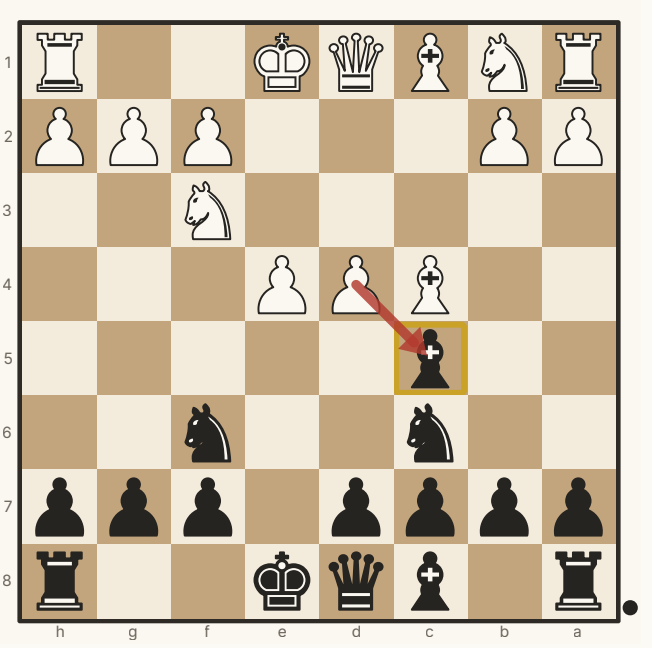
Against the Colle, White achieved e4, but Black traded the centre pawns and White's d4 pawn is now isolated.

1.**c4 Nf6** 2.**Nc3 e6** 3.**e4 d5** 4.**e5 d4** 5.**exf6 dxc3** 6.**bxc3 Qxf6** 7.**d4 c5**

Against 1.c4 and 3.e4, the counterattack ...d5 and ...d4 leads to an equal, open game with active pieces for Black.

A worked example: after 6.cxd4 in the Giuoco Piano

Here is the full thinking process in the critical position of your 1.e4 e5 repertoire.



Black to move. White has built the big centre with c3 and d4. Your bishop on c5 is attacked by the d4 pawn.

Step 1 — What does White threaten? The d4 pawn attacks my bishop on c5. Also, if White gets in d5 or e5, the pawns will push my knights back. So I must move the bishop *and* do something against the centre.

Step 2 — Checks, captures, threats for me. Checks: ...Bb4+. Captures: ...Nxe4 takes a pawn but leaves the bishop on c5 hanging (dxc5). ...Bxd4 gives the bishop for a pawn. Threats: ...d5 attacks e4, but it also leaves the bishop on c5 en prise.

Step 3 — Candidates. 6...Bb4+, 6...Bb6, 6...d5?!.

- 6...Bb6 is safe, but passive: after 7.d5 or 7.e5 White's centre rolls forward with tempo.
- 6...d5?! looks active, but after 7.exd5 Nxd5 the bishop is still attacked and White keeps the initiative; the engine rates it about a pawn worse than the check.
- 6...Bb4+ forces White to spend a move dealing with the check. If 7.Bd2 Bxd2+ 8.Nbxd2, I have traded one pair of pieces and can play 8...d5, smashing the centre.

Step 4 — Decide. 6...Bb4+, with ...d5 as the follow-up. Notice the general lesson: when White's centre is about to roll forward, use a **check** to gain the time you need for your own central pawn break.

When plans should change

Your default Black plans are: equalize in the centre, castle, then challenge with a second pawn break. Change them when:

- **White's king is still in the centre and lines are open.** Sometimes development is enough to start an attack. Look for ...Re8+ or ...Bb4+ ideas.
- **White's queen has wandered.** Every queen move in the opening is a chance to develop with tempo (...Nc6, ...Nf6, ...Nd4, ...Nb4).
- **White castles queenside.** Then your queenside pawns (...a5, ...b5, ...c5) become attackers, as long as your own king is safe.
- **The pawn break isn't available.** In the QGD, if ...c5 or ...e5 would lose a pawn, prepare it with one more move (...Re8, ...Qc7, ...dxc4) rather than forcing it.
- **White's pieces abandon the queenside.** In the London and Colle, ...Qb6 against b2 is often stronger than any kingside plan.

Practice at the board

1. **Play out the key positions.** Use a physical board or any free app that lets you play the computer from a position. Play each position at least three times as Black: * Giuoco Piano after 8...d5: `r1bqk2r/ppp2ppp/2n2n2/3p4/2BP4/5N2/PP1N1PPP/R2QK2R w KQkq - 0 9` (you are Black; White moves first). * Closed Ruy Lopez after 8...O-O: `r1bq1rk1/2p1bppp/p1np1n2/1p2p3/4P3/1BP2N2/PP1P1PPP/RNBQR1K1 w - - 1 9` * QGD after 6.Nf3: `rnbq1rk1/ppp1bppp/4pn2/3p2B1/2PP4/2N1PN2/PP3PPP/R2QKB1R b KQ - 1 6` * London after 5.c3: `rnbqkb1r/pp3ppp/4pn2/2pp4/3P1B2/2P1PN2/PP3PPP/RN1QKB1R b KQkq - 0 5`
2. **The f7 drill.** Set up the starting position and have a friend (or the computer at a low level) play 1.e4 e5 2.Qh5 and 2.Bc4 lines. Before each of your moves, say aloud how many times f7 is attacked and how many times it is defended.
3. **The trap rehearsal.** Set up the Elephant Trap position after 6.Nxd5?? (`r1bqkb1r/pppn1ppp/5n2/3N2B1/3P4/8/PP2PPPP/R2QKBNR b KQkq - 0 6`) and the Exchange Ruy position after 5.Nxe5? (`r1bqkbnr/1pp2ppp/p1p5/4N3/4P3/8/`

`PPPP1PPP/RNBQK2R b KQkq - 0 5`). Play the winning or equalizing moves from memory until you can do it without hesitation.

4. **Check yourself after each game.** Find the move where you left this chapter. If you lost because you did not know a line, add it to your notebook with the correct move.

> **BOARD DRILL** **The set-up race**
>
> On an empty board, set up Black's pieces in the QGD and the universal set-up (...Nf6, ...d5, ...e6, ...Be7, ...O-O, ...Nbd7 or ...Nc6, ...c5 or ...c6) as fast as you can, saying each move's purpose. Then do the same for the Italian (...e5, ...Nc6, ...Bc5, ...Nf6, ...d6, ...O-O, ...a6, ...Ba7, ...h6).

Chapter summary

- Against 1.e4, play 1...e5 2...Nc6, and answer 3.Bc4 with 3...Bc5, 3.Bb5 with 3...a6 and the Closed Ruy set-up, and 3.d4 with 3...exd4 4.Nxd4 Nf6.
- In the Giuoco Piano, meet 5.d4 with ...exd4, ...Bb4+ and ...d5; meet 5.d3 with ...d6, ...O-O, ...a6, ...Ba7, ...h6 and ...Be6.
- Against the Evans Gambit, take the pawn and retreat the bishop to a5; in the Exchange Ruy, remember that 5.Nxe5? is met by 5...Qd4.
- Against early queen attacks, look at f7 before every move: ...g6, ...Nf6 at the right moment, ...Qe7, and develop with tempo.
- Against 1.d4, play the QGD: ...d5, ...e6, ...Nf6, ...Be7, ...O-O, ...Nbd7, ...c6, then free the c8 bishop with ...dxc4 and ...e5 or ...c5. Know the Elephant Trap.
- Against the London, use ...c5, ...Nc6, ...Bd6 and ...Qb6 ideas; against everything else, use the universal set-up.
- Change plans when White's queen wanders, when White's king stays in the centre, or when the usual pawn break is not yet possible.

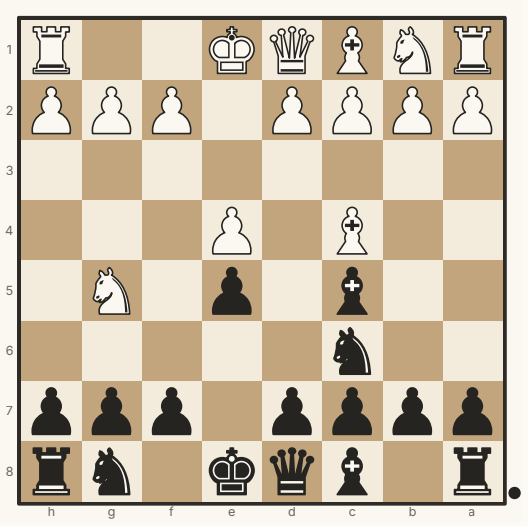
CHEAT CODE **Opening checklist as Black**

1. What did White's last move threaten? Look at f7 (after 1.e4) and pins on your f6 knight (after 1.d4).
2. Can you develop with tempo against an exposed white piece? Do it.
3. Otherwise, put your pieces on their set-up squares: knights, the bishop that is free, castle.
4. Prepare your second pawn break: ...d5 against 1.e4 structures; ...c5 or ...e5 in the QGD and against the London.
5. Before every pawn break, check that it does not hang a piece or a pawn, and that your king is safe.

Exercises

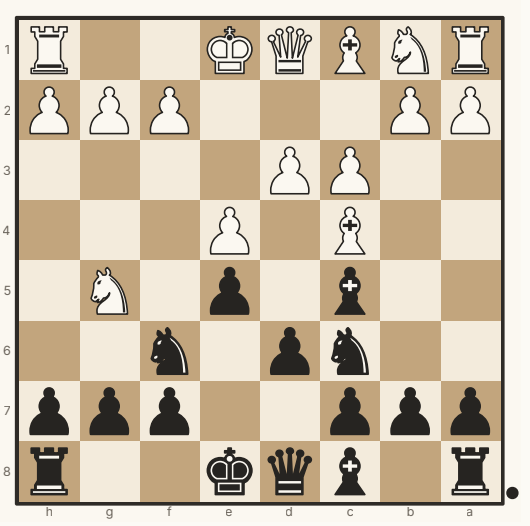
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

13.1 TACTIC



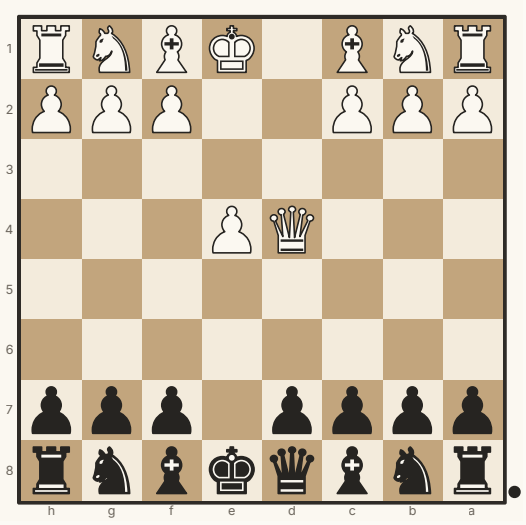
Black to move. In your Giuoco Piano, White has just played 4.Ng5, attacking f7 twice. Do you have to defend? Solution p. 669

13.2 DEFENCE



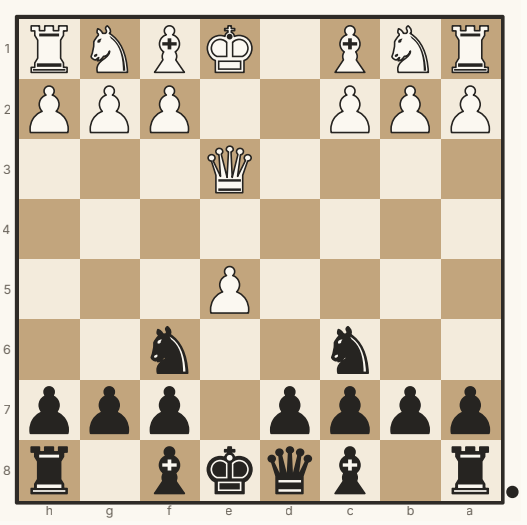
Black to move. In your Giuoco Piano (1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6), White has just played 6.Ng5, so the knight and the bishop on c4 both attack f7. Find a simple, safe move. Solution p. 669

13.3 OPENING



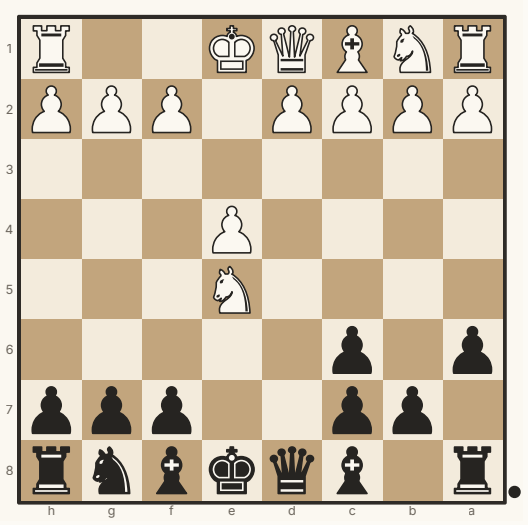
Black to move. In the Center Game, White has recaptured with the queen on d4. Play the repertoire move. Solution p. 670

13.4 OPENING



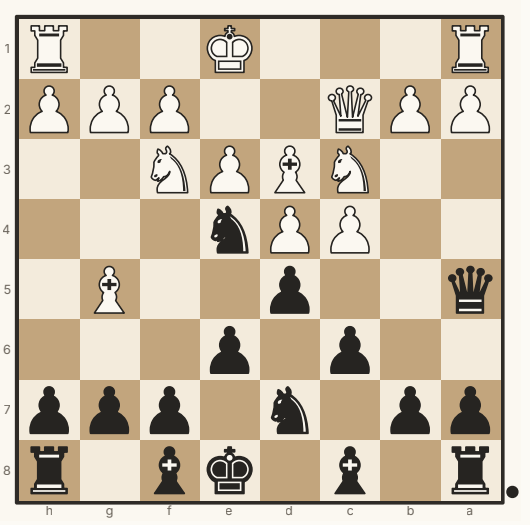
Black to move. In the Center Game (1.e4 e5 2.d4 exd4 3.Qxd4 Nc6 4.Qe3 Nf6), White has pushed 5.e5, attacking your knight. Find the move that saves the knight and gains another tempo against the queen. Solution p. 670

13.5 OPENING



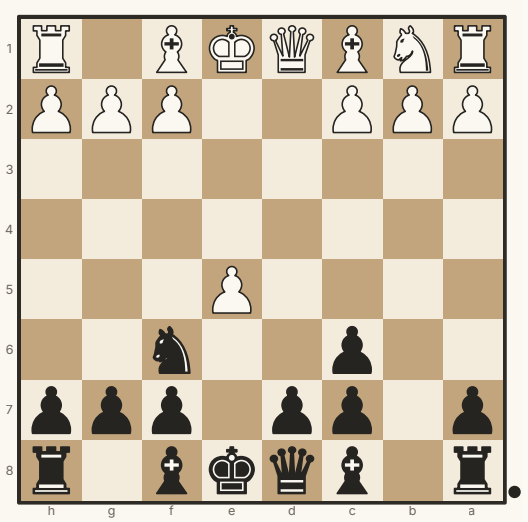
Black to move. In the Exchange Ruy Lopez, White has just grabbed the e5 pawn with 5.Nxe5. Win back the pawn. Solution p. 670

13.6 TACTIC



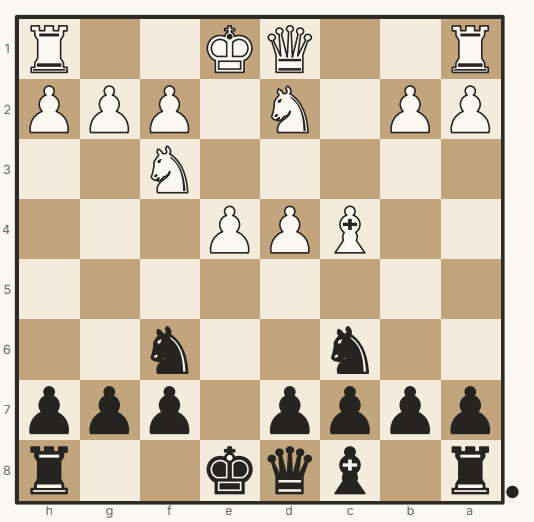
Black to move. In your Queen's Gambit Declined you chose the Cambridge Springs set-up (4.Bg5 Nbd7 5.e3 c6 6.Nf3 Qa5), pinning the knight on c3. White played 7.Bd3 Ne4 8.Qc2?, guarding c3 but forgetting the bishop on g5. Win material. Solution p. 670

13.7 OPENING



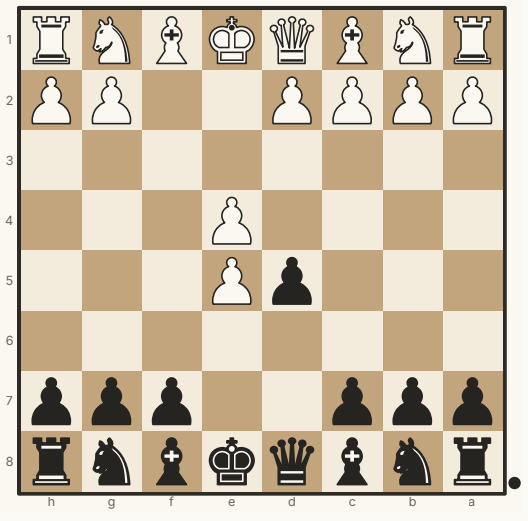
Black to move. In the Scotch, White has played 6.e5, attacking your knight on f6. Find the move that keeps your position together. Solution p. 671

13.8 OPENING



Black to move. In the Giuoco Piano main line you played ...Bb4+ and traded bishops on d2. Now strike at White's centre. Solution p. 671

13.9 TACTIC



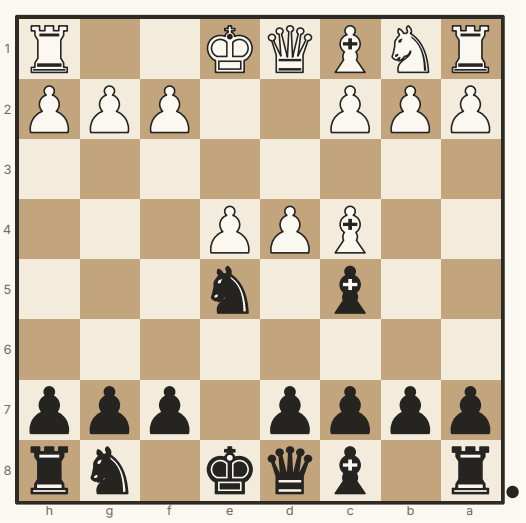
Black to move. You met the King's Gambit with 2...d5, and White has just taken on e5 with 3.fxe5??. Punish it. Solution p. 671

13.10 OPENING



Black to move. In the Giuoco Piano main line, White blocked your check with 7.Nc3 instead of 7.Bd2. Find the move that grabs the initiative. Solution p. 671

13.11 TACTIC



Black to move. White tried the fork trick 4.Nxe5? Nxe5 5.d4, attacking your bishop and knight with one pawn. Keep your extra piece. Solution p. 672

HINTS

13.1 Who protects the knight on g5?

13.2 Count the attackers and defenders of f7. Which move adds a defender and makes your king safe at the same time?

13.3 Which developing move attacks the queen?

13.4 Where can the knight go and attack the queen at the same time?

13.5 Find a queen move that attacks two things at once.

13.6 Take the bishop on g5. If White recaptures, what would your queen see if the d5-pawn moved?

13.7 Which move puts pressure on the e5 pawn while saving the knight's future?

13.8 Which pawn move hits e4?

13.9 Which diagonal did White's f-pawn leave open?

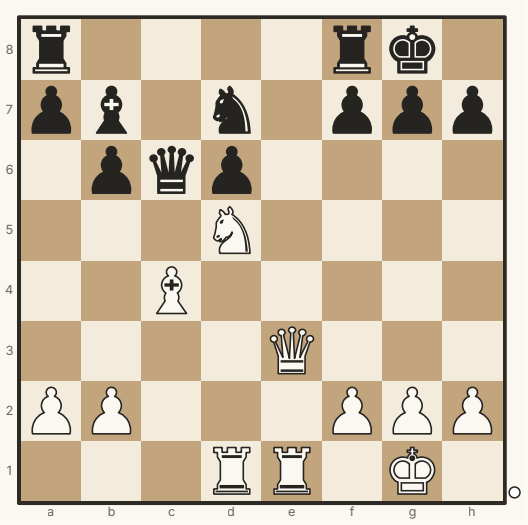
13.10 White's knight on c3 is pinned. What does that mean for the e4 pawn?

13.11 Two pieces are attacked. Can one of them move with check?

Stage 2 Test

Fifteen mixed positions covering Part 2. Solve them without moving the pieces, write down your answers, then mark yourself. 12/15 or better: move on with confidence. 9–11: review the chapters behind your misses. Below 9: repeat this part's exercises before continuing.

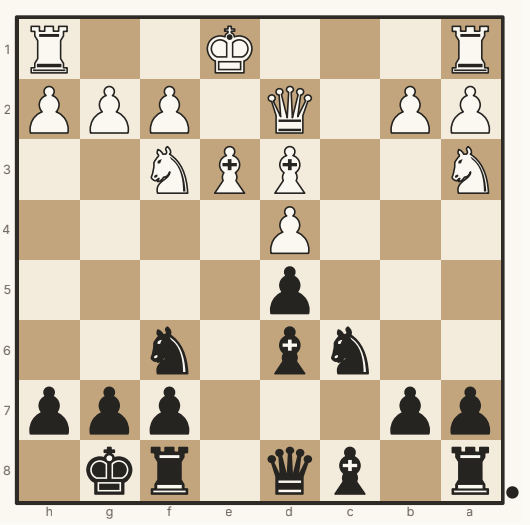
T2.1 TACTIC



White to move. Black threatens Qxg2#. Answer the threat and win material.

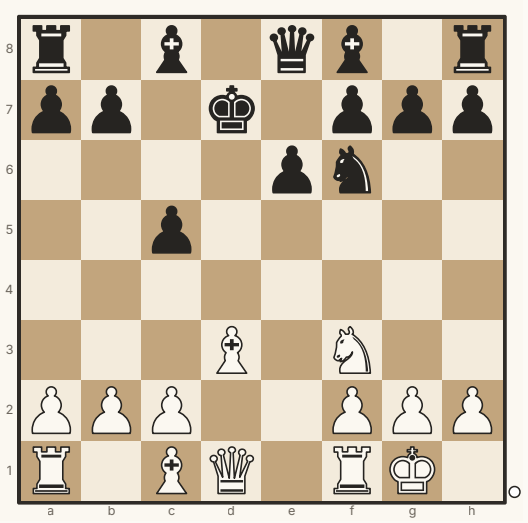
Solution p. 672

T2.2 TACTIC



Black to move. White has not castled yet. Win the queen. Solution p. 672

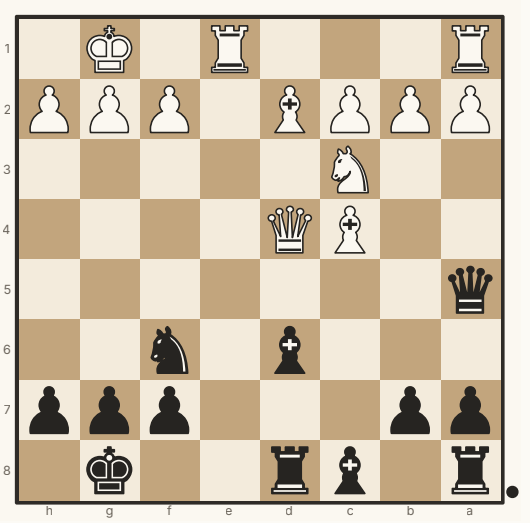
T2.3 TACTIC



White to move. Black's king and queen are on the same diagonal. Win the queen.

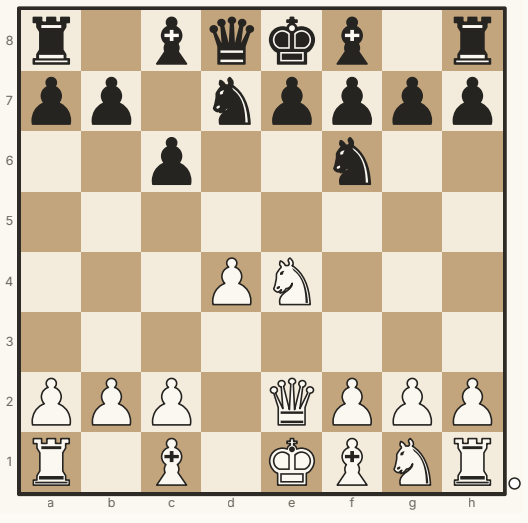
Solution p. 673

T2.4 TACTIC



Black to move. Your bishop on d6 stands between your rook and White's queen. Move it with maximum force. Solution p. 673

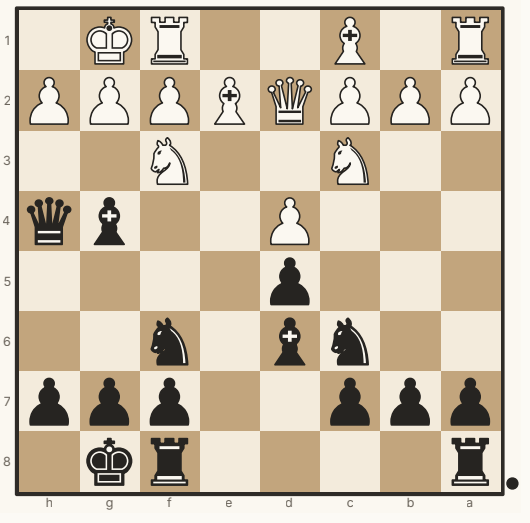
T2.5 CHECKMATE



White to move. Black has just played 5... Ngf6?? in the Caro-Kann. Mate in 1.

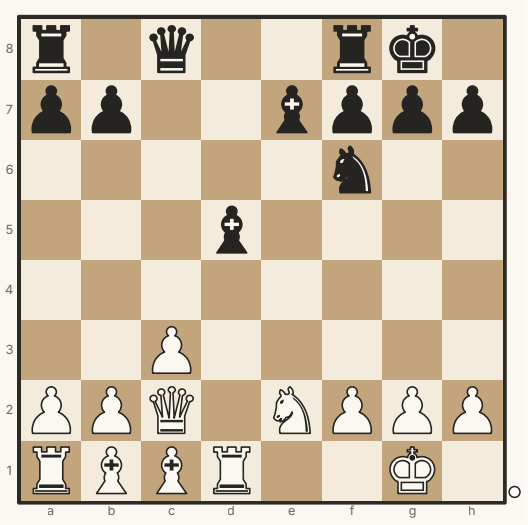
Solution p. 673

T2.6 TACTIC



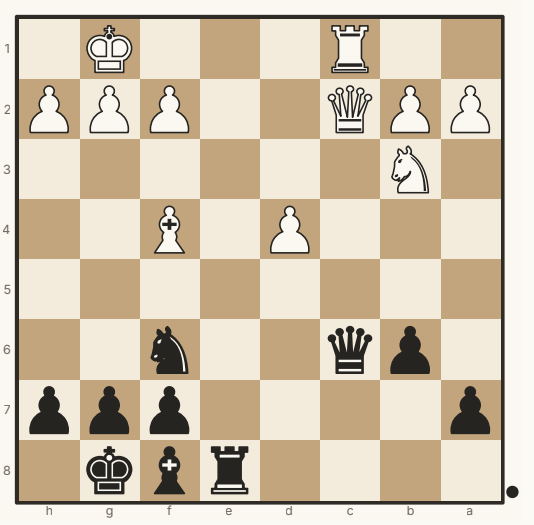
Black to move. White has just played Nf3, attacking your queen. Find the strongest reply. Solution p. 673

T2.7 TACTIC



White to move. Black's knight on f6 is defending two things. Overload it. Solution p. 673

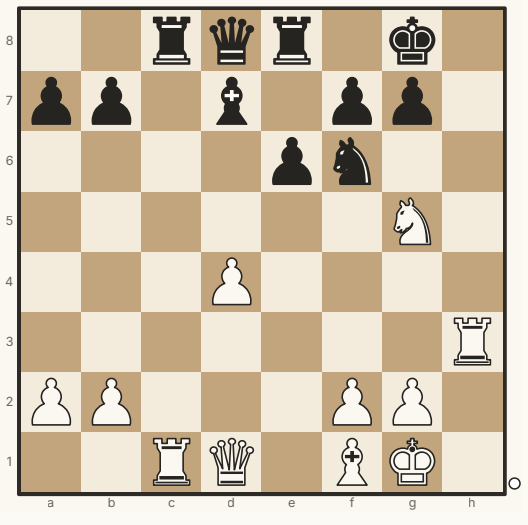
T2.8 TACTIC



Black to move. White's rook on c1 guards the queen and the back rank. Win material.

Solution p. 674

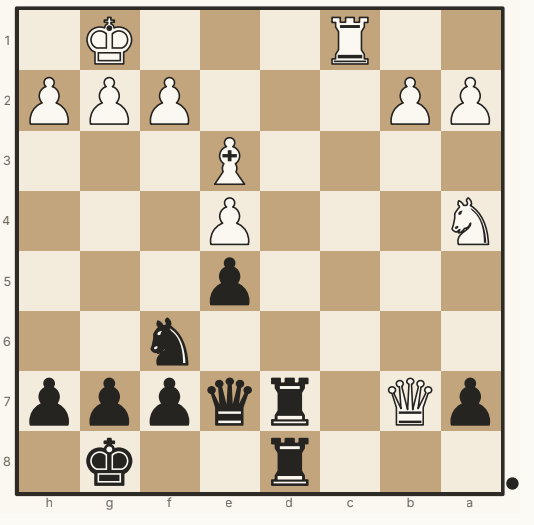
T2.9 TACTIC



White to move. Use a sacrifice to bring Black's king onto a forkable square.

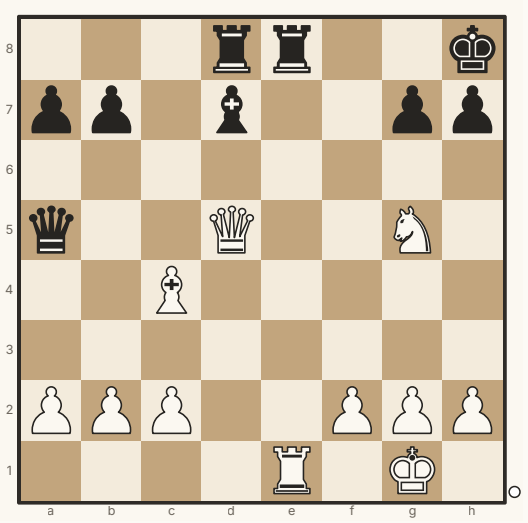
Solution p. 674

T2.10 CHECKMATE



Black to move. Your rooks are doubled on the open d-file. Mate in 2. Solution p. 674

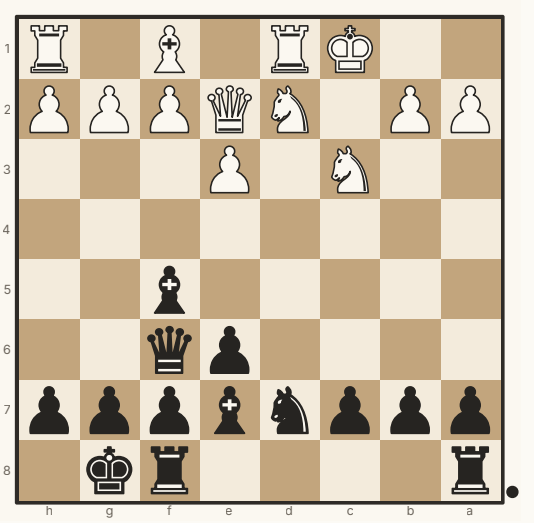
T2.11 CHECKMATE



White to move. Your queen, bishop and knight all point at Black's king. Mate in 2.

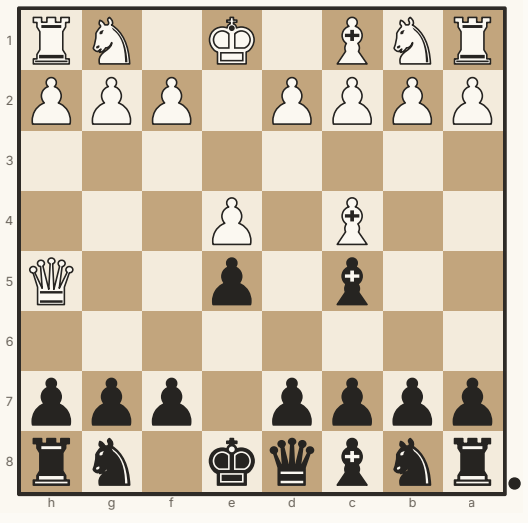
Solution p. 674

T2.12 CHECKMATE



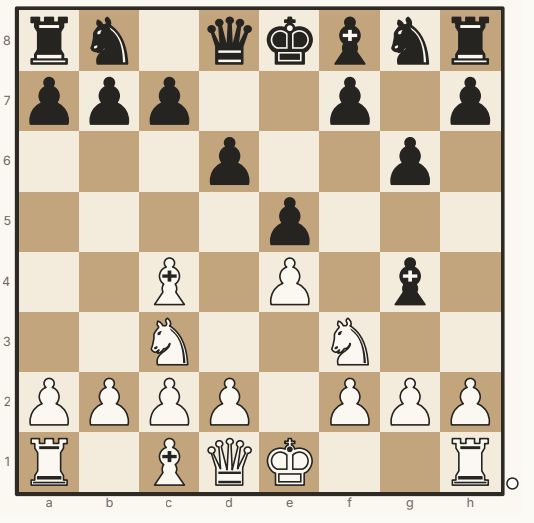
Black to move. White castled queenside and the c2-pawn has gone. Mate in 2. Solution p. 675

T2.13 OPENING



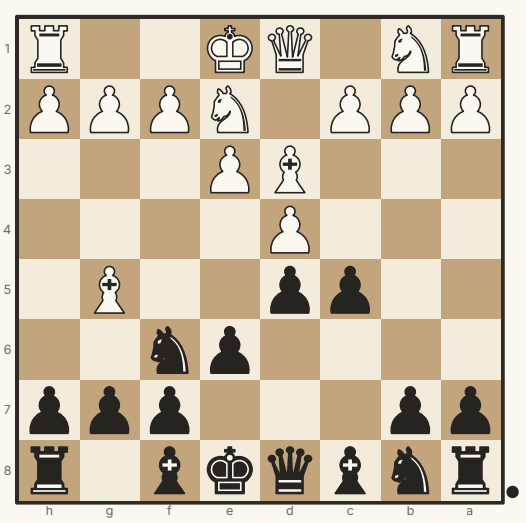
Black to move. After 1.e4 e5 2.Bc4 Bc5 3.Qh5, White's queen attacks f7 (together with the bishop on c4) and the e5-pawn at the same time. Find the move that meets both threats. Solution p. 675

T2.14 OPENING



White to move. Black's bishop pins your knight to your queen. Is the e5-pawn safe to take? Play the best move. Solution p. 675

T2.15 TACTIC



Black to move. White's bishop on d3 has very few squares. Win it. Solution p. 675

HINTS

T2.1 Which check also attacks the queen?

T2.2 Look for a check-line between the queen and the king.

T2.3 Which check hits the king with something valuable behind it?

T2.4 A discovered attack is strongest when the moving piece gives check.

T2.5 Which knight move gives two checks at once?

T2.6 What does the knight on f3 defend?

T2.7 If the knight recaptures on d5, what happens on h7?

T2.8 If the rook recaptures on c2, what is left guarding e1?

T2.9 Which square would put Black's king and queen a knight's jump apart?

T2.10 Two attackers against one defender on d1.

T2.11 Make Black's rook take away its own king's escape square.

T2.12 Which white pawn guards b2? Make it move.

T2.13 List both of White's threats before you choose a move.

T2.14 Is the knight on f3 pinned to the king or only to the queen?

T2.15 Which pawn move attacks the bishop on d3?

PART THREE

Calculate and Convert

Stage 3 · 800–1000

Seeing a pattern is not the same as proving it works. This part builds a reliable move-selection routine, accurate short calculation, the technique to turn winning positions into wins, the resourcefulness to save bad ones, and the king-and-pawn endgame knowledge that every later endgame rests on.

Chapter 14 · How to Think: Candidate Moves and the Move-Selection Routine

Chapter 15 · Calculation Fundamentals

Chapter 16 · Converting a Winning Position

Chapter 17 · Defense and Resourcefulness

Chapter 18 · King and Pawn Endgames

CHAPTER 14 STAGE 800–1000

How to Think: Candidate Moves and the Move-Selection Routine

A short routine for every move, and a longer one for the moves that matter

YOU WILL LEARN TO

- Run a fixed move-selection routine on every move, from the opponent's idea to the final blunder check.
- Build a list of two to four candidate moves instead of falling in love with the first idea.
- Compare candidates by finding the opponent's best reply, not the reply you hope for.
- Recognize critical moments that deserve extra thinking time.

> **IN YOUR NEXT GAME**
>
> On every move, ask "what does my opponent want?" first, and before you touch a piece, name at least two candidate moves and the opponent's best reply to the one you choose.

You know the tactics now. You can spot a fork, see a pin, and mate with a rook. And yet you still lose games to moves you "would have seen" in a puzzle. The reason is almost never missing knowledge. It is missing *procedure*: you looked at your own idea, liked it, and played it, without asking what your opponent wanted or whether a better move was available.

Between 800 and 1000, games are decided less by who knows more and more by who thinks in a more orderly way. This chapter gives you a routine — a short checklist you run on every move — and shows it working, move by move, inside a player's head. It is not complicated. The hard part is doing it *every time*, especially when you are sure you already know the answer.

The move-selection routine

Here is the routine in full. The rest of the chapter explains each step and shows how it sounds in practice.

1. **What does my opponent want?** Look at the move just played. What does it attack, what does it threaten, what does it prepare?
2. **Safety scan.** Are any of my pieces loose (undefended) or attacked more often than they are defended? Is my king safe? This is the scan from Chapter 4.
3. **Forcing moves for both sides.** List the checks, captures and threats available to *me* — and to *my opponent*, as if it were his move.
4. **Candidate moves.** Choose two to four moves worth considering seriously.
5. **Compare.** For each candidate, find the opponent's best reply. Pick the candidate whose worst case is best.
6. **Blunder check.** With the chosen move "played" in your head, run the safety scan once more on the new position.
7. **Play.** Then, and only then, touch the piece.

A **candidate move** is simply a move you decide to examine seriously. The term was made famous by the Soviet grandmaster and trainer Alexander Kotov, whose book *Think Like a Grandmaster* taught a generation to list candidates *before* calculating any of them. The point is to avoid analysing the first idea for ten minutes and then playing something else in a panic.

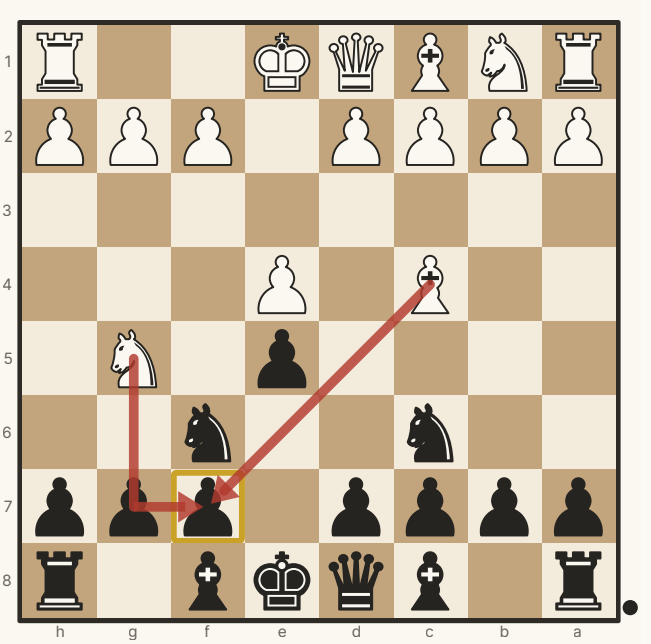
> KEY IDEA **The routine in one sentence**
>
> His idea, my safety, everyone's forcing moves, a short list, his best reply to each, one last check — then move.

Step 1: What does my opponent want?

Every move your opponent plays has a purpose. Sometimes it is a direct threat; sometimes it prepares a pawn break or brings a piece closer to your king. At this level the most common cause of lost games is simply not asking. A player sees his own plan so clearly that the opponent's last move becomes background noise.

Ask three short questions about the move just played: *What does the moved piece attack now? What did it stop defending? What line did it open?* The third question

catches discovered attacks, which are invisible if you only look at the piece that moved.



White has just played 4.Ng5. Knight and bishop both hit f7, and Nxf7 would fork the queen and the rook on h8.

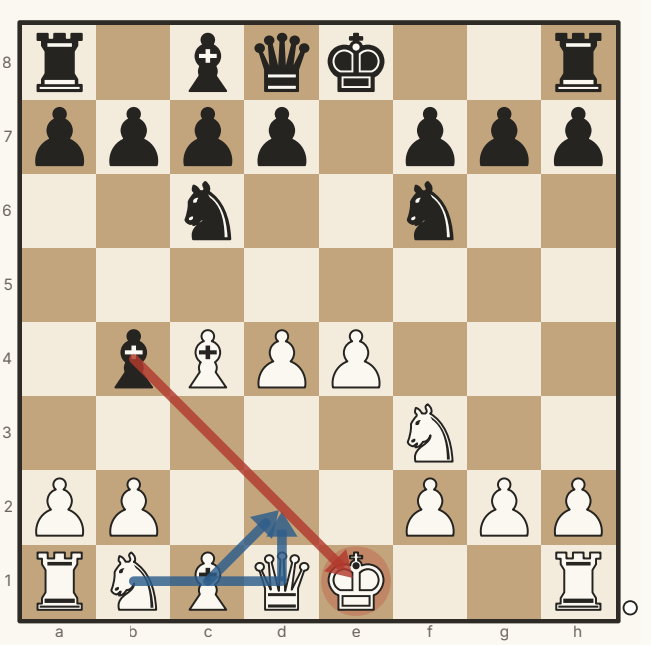
Here the question "what does he want?" has a clear answer: Nxf7, forking queen and rook. Only now is it worth listing candidates. The routine move 4...Be7, played without asking, loses material. The standard answer is to block the bishop's diagonal with a counter-attack.

4...**d5** 5.**exd5 Na5** 6.**Bb5+ c6** 7.**dxc6 bxc6**

4...d5 blocks the c4-bishop's line to f7 and fights back; Black gives a pawn but gains time against the bishop.

Step 2: The safety scan

You learned the safety scan in Chapter 4: checks, captures and threats against *you*. Inside the routine it has one extra job — it prevents the most painful kind of loss, the one where you never looked at your own loose piece. Run it after step 1, before you think about anything ambitious. It takes ten seconds in a quiet position.



An Italian Game after 6...Bb4+. The safety scan starts with the check: White can block with the bishop or a knight, or give up castling with Kf1. Blocking keeps the king safe and develops a piece.

Two quick tests catch most problems. First, **loose pieces**: any piece of yours that is not defended can be hit by a fork or a double attack. Second, **counting**: for each attacked piece, count attackers and defenders. If the attackers outnumber the defenders, or if the cheapest attacker is worth less than your piece, you have a problem to solve before anything else.

Step 3: Forcing moves for both sides

A **forcing move** is a check, a capture, or a threat that the opponent must answer. Forcing moves deserve to be looked at first because they limit the opponent's replies, which makes them easy to calculate and very often decisive.

Most players at this level do half of this step: they look at their own checks and captures. The other half is to look at *your opponent's* checks and captures as if it were his turn. That is how you notice that a back-rank mate is already threatened, or that the knight you want to move is the only defender of a key square. You will practise this in the exercises: several of them contain a capture that looks free until you ask what checks the opponent has afterwards.

> **COMMON MISTAKE** **The free piece**
>
> When something is offered for nothing, ask one question before you take it: "After I capture, what are my opponent's checks?" Many opening traps, including Légal's mate, work only because the victim never asks.

Step 4: Build a short candidate list

Now choose the moves you will seriously consider. Two to four is the right number. One is too few — it means you are not comparing anything. More than four is usually a sign that you have not used the first three steps to filter.

Good candidates come from three sources:

- **Forcing moves** that survive a first look (step 3).
- **Answers to the opponent's idea** (step 1) — defending, blocking, counter-attacking or getting out of the way.
- **Plan moves** when nothing forcing is happening: develop a piece, improve the worst-placed piece, prepare a pawn break, put a rook on an open file.

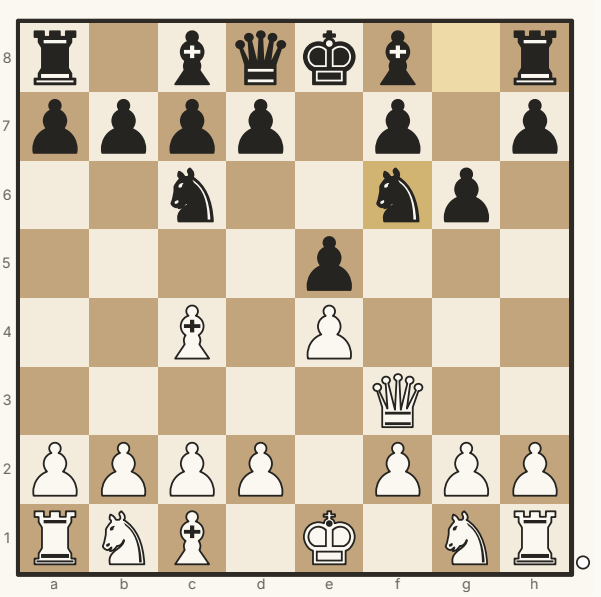
The difference between a candidate list and **single-line fixation** is the whole point of this chapter. Single-line fixation means you see one idea, calculate it deeper and deeper, and never ask whether another move is better. The calculation may even be correct — but if you never compared it to anything, you have no idea whether it was the best use of your move.

Step 5: Compare by the opponent's best reply

For each candidate, ask: *what is his best answer?* Not his most likely answer, not the answer you would like, but the one that hurts you most. Then compare the candidates by those worst cases.

This step is where "hope chess" turns into "real chess". The chess teacher Dan Heisman popularized the term **hope chess** for the habit of making a move and *hoping* the opponent does not have a good reply, instead of checking. In **real chess**, you check the opponent's forcing replies to every candidate before you play it. Hope chess sometimes wins against weaker players, which is exactly why it is so hard to give up. It loses to anyone who looks.

1. **e4 e5** 2. **Qh5?! Nc6** 3. **Bc4 g6** 4. **Qf3 Nf6**



Hope chess in its purest form. White "hopes" for Qxf7#; after Black's best replies the queen has been chased around and Black is developing with gain of time.

The early queen attack is the classic example. It sets a threat, and if Black misses it the game ends at once. But against the best reply — here 3...g6 and 4...Nf6 — White has spent three moves on the queen and Black is happily ahead in development. A player doing real chess would ask "what if he defends correctly?" before 2.Qh5 and see that the answer is "I am worse."

Step 6: The blunder check

You have chosen a move. Before you play it, imagine it on the board and run the safety scan again on the *new* position. This is the **blunder check**. It catches a very specific kind of error: the move you calculated so carefully also leaves something else undefended, or opens a line to your own king.

The blunder check is quick — ten to twenty seconds — and it is the single most valuable habit in this book. Make it physical if that helps: sit on your hands, or keep the hand off the board until you have said "blunder check done" to yourself.

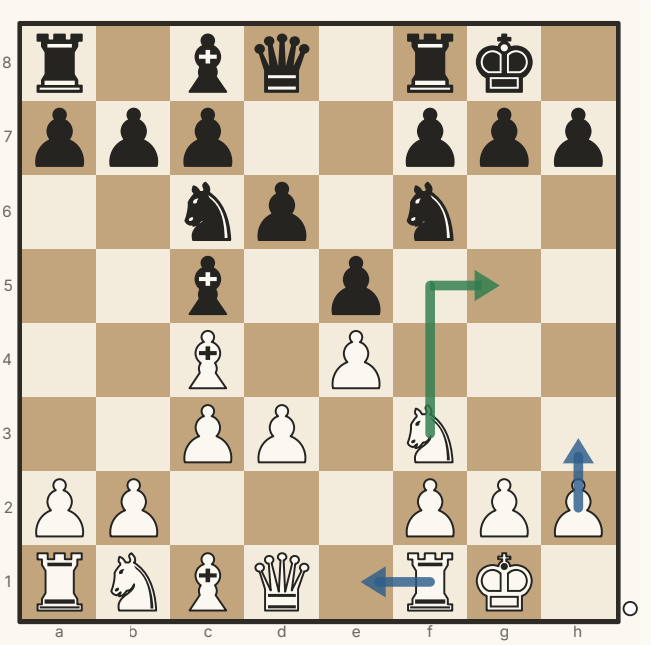
Step 7: Play — and use your opponent's time

Then play the move. Do not change your mind at the last second for a move you have not checked; that is how a careful routine produces a careless blunder.

While your opponent thinks, start step 1 for his likely replies. You will often find you have already done half the work when he moves.

A worked example: the full monologue

Here is the routine in action in a quiet opening position from your Italian Game repertoire (Chapter 12). Nothing dramatic is happening — which is exactly when players drift into hope chess.



The Giuoco Pianissimo after 6...O-O. White to move. Nothing is attacked, which is exactly when the routine matters most.

WORKED EXAMPLE **Thinking it through**

"Step 1 — what did he want with 6...O-O? He tucked the king away and connected his rook to f8. No new threat. But I should remember his usual ideas: ...Bg4 pinning my knight, and ...Na5 to swap off my good bishop.

Step 2 — safety. Everything of mine is defended. My king is castled. Fine.

Step 3 — forcing moves. Mine: Bxf7+ Rxf7 gives a bishop for a pawn — no. Ng5 threatens Nxf7. His: no checks, no captures that win anything.

Step 4 — candidates. Ng5, because it is aggressive. h3, to stop ...Bg4. Re1, to support e4 and prepare Nbd2-f1. Bb3, to take the bishop off the c4-square before ...Na5 arrives.

Step 5 — compare. Ng5: his best reply is ...h6. Then my knight must go back to f3 and I have lost two tempi for nothing. That is hope chess — it only works if he blunders. h3, Re1 and Bb3 are all calm plan moves; his best replies don't hurt me. I'll play h3, because ...Bg4 is the most annoying idea he has.

Step 6 — blunder check. After h3, does anything hang? No. Does it weaken my king? The g3-square a little, but nothing can reach it. OK."

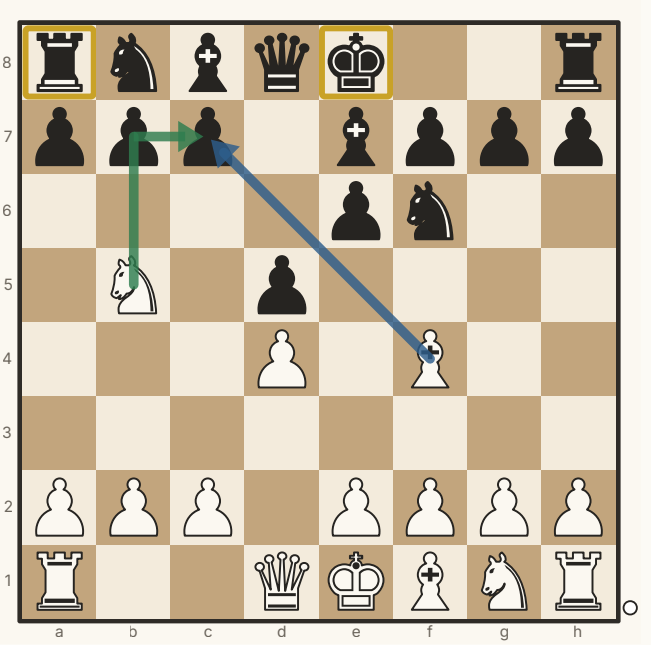
7. **Ng5?! h6** 8. **Nf3**

The "attacking" candidate fails to the best reply. After 7...h6 the knight goes home, and Black has gained a useful move for free.

Notice how short each step is. The whole monologue takes perhaps a minute. The only step that took real thought was step 5, and it took thought precisely because one candidate (Ng5) looked attractive until the opponent's best reply was checked.

"When you see a good move, look for a better one"

There is a saying often attributed to Emanuel Lasker: *when you see a good move, look for a better one.* It describes a trap that catches strong and weak players alike: stopping the search the moment you find something that works. Psychologists call this **satisficing** — accepting the first option that is "good enough" instead of looking for the best.



Black has just played 4...Be7?, ignoring the threat to c7. White to move. Bxc7 wins a pawn with an attack on the queen — but is there something better?

The first good move most players see is 5.Bxc7, winning a pawn and hitting the queen. It is good. But it is not the best. The check 5.Nxc7+ forks king and rook, and after the king moves White takes the rook on a8. Checks come first on the candidate list for exactly this reason.

5. **Nxc7+ Kf8** 6. **Nxa8**

The fork wins the rook on a8. Even if the knight is later lost in the corner, White has won the exchange and a pawn, far more than the single pawn won by 5.Bxc7.

There is a practical limit, of course. You do not need to look for a better move than a forced mate, and in a calm position you do not need to compare twelve decent plan moves. A useful rule: if your move wins material or creates a big threat, spend one more minute specifically looking for checks and captures you have not yet examined. That one minute pays for itself many times over.

COACHING JUDGMENT **Coaching judgment: how many candidates?**

In my experience, two candidates are usually enough in a quiet position and three or four in a sharp one. The goal is not to find more moves, but to make sure the move you play has beaten at least one real rival.

EVIDENCE **What strong players actually do**

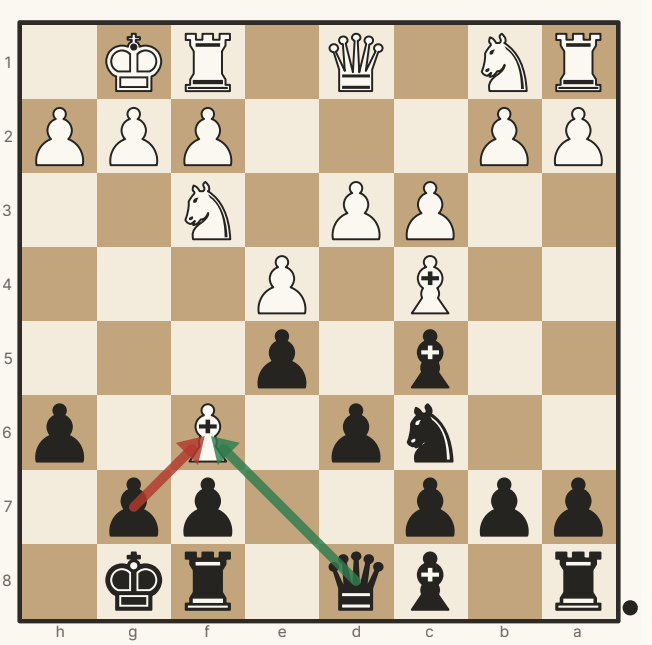
In his classic study of chess thinking, the Dutch psychologist Adriaan de Groot (*Thought and Choice in Chess*, English edition 1965) asked players of different strengths to think aloud. The strongest players did not examine dramatically more moves than weaker ones; the main difference was that the good moves showed up in their thinking earlier. A candidate list built from forcing moves, threats and plans is a way of steering your attention toward those moves.

When to think long and when to move quickly

You cannot spend ten minutes on every move; in a 30-minute game you would lose on time by move 20. The routine has a *short version* for routine moves — opponent's idea, safety scan, blunder check — and a *long version* for **critical moments**. Learning to tell them apart is a skill in itself.

A critical moment is a point where the character of the game can change. The most common ones are:

- **Captures and recaptures.** Every exchange changes the pawn structure or the balance of pieces. "Which way do I recapture?" is a classic critical decision.
- **Checks** — yours and your opponent's — because they force replies and can change the position quickly.
- **Pawn breaks and pawn moves near your king.** Pawns cannot move back; a careless pawn move is permanent.
- **Changes in the structure**, such as opening a file or giving up the bishop pair.
- **Transitions to the endgame**, especially trading the last pair of pieces into a king and pawn ending.



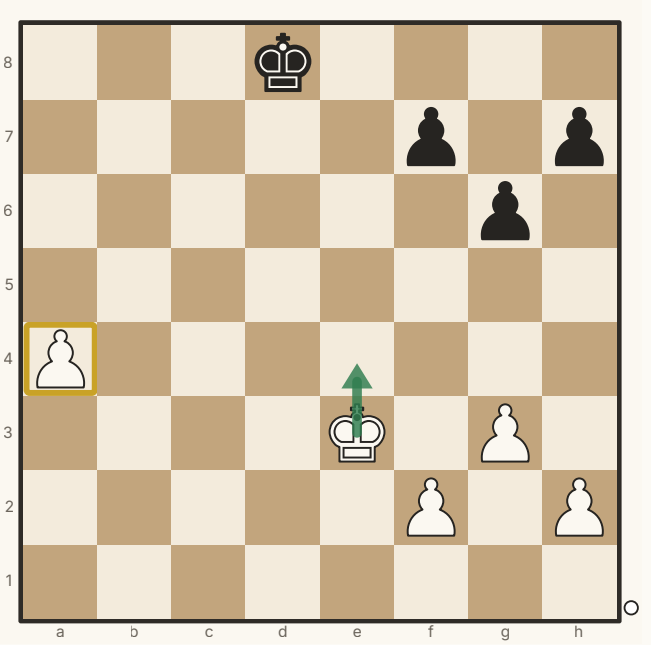
White has just captured on f6. Recapturing with the queen keeps Black's kingside intact; recapturing with the g-pawn opens Black's own king. A recapture is a critical moment.

Here Black has two recaptures. 8...Qxf6 is natural and good: the queen is active and the kingside pawns stay together. 8...gxf6? is not a blunder in the tactical sense, but it gives White a long-term target in front of Black's king — and the engine judges it more than a pawn and a half worse. A five-second "recapture with the queen, obviously" is fine here; but you should at least *notice* that the choice exists.

> **IN YOUR NEXT GAME** **Flag your critical moments**
>
> In your next game, every time a capture, check or pawn break is possible for either side, say to yourself "critical" and use the long routine. For other moves, use the short routine and save your clock.

Endgame transitions

The moment when you can trade the last pieces is one of the most important critical moments of all, because the evaluation can jump from "slightly better" to "completely winning" — or to "dead draw". Chapter 16 is about this kind of decision. For now, simply notice it: before any trade that leaves only kings and pawns, stop and count.



After a rook trade, White's outside passed a-pawn decides the king and pawn ending. Before trading, you must be sure that this is the position you are getting.

Common thinking errors and how to spot them in your games

When you analyse your games (see Chapter 28), you will find that most of your mistakes fall into a small number of thinking errors. Naming them makes them easier to fix.

- **Skipping step 1.** You played your plan without asking what his last move did. *Symptom in your games:* "I didn't see his threat."
- **One-candidate thinking.** You played the first move you liked. *Symptom:* the engine's best move was something you never considered at all.
- **Hope chess.** You calculated your idea assuming a convenient reply. *Symptom:* "I thought he would take."
- **Ignoring his forcing moves.** You checked your own checks and captures but not his. *Symptom:* a back-rank mate or fork "out of nowhere".
- **No blunder check.** You calculated a line and then played a different move, or the right move with a hanging piece elsewhere. *Symptom:* a one-move blunder in a position you understood.

- **Clock panic.** You spent ten minutes on a quiet move and then rushed the critical one. *Symptom:* the big mistake comes right after a long think.

> **BOARD DRILL** **The two-candidate drill**
>
> Play a training game (any time control of 15 minutes or more) in which you are not allowed to move until you have said two candidate moves and the opponent's best reply to each, out loud or on your score sheet. It will feel slow. After ten such games the habit starts to run by itself.

Practice at the board

Set up each position on a board or in any free chess app's analysis mode. Do not move pieces while you think. Write your answers to the routine on paper: his idea, your safety, forcing moves for both sides, your candidates, his best reply to each, your choice. Then check with an engine.

1. **The Fried Liver threat.** FEN `r1bqkb1r/pppp1ppp/2n2n2/4p1N1/2B1P3/8/PPPP1PPP/RNBQK2R b KQkq - 5 4`. Black to move. Name White's threat in one sentence, then list three candidates. Check that 4...d5 is one of them and compare it with 4...Nxe4.
2. **A quiet Italian.** FEN `r1bq1rk1/ppp2ppp/2np1n2/2b1p3/2B1P3/2PP1N2/PP3PPP/RNBQ1RK1 w - - 2 7`. White to move. Write a full monologue in the style of the worked example. Then let the engine show you its top three moves; were your candidates among them?
3. **Look for the better move.** FEN `rnbqk2r/ppp1bppp/4pn2/1N1p4/3P1B2/8/PPP1PPPP/R2QKBNR w KQkq - 2 5`. White to move. Find two moves that win material, and decide which is better *before* you check.
4. **Recapture choices.** FEN `r1bq1rk1/ppp2pp1/2np1B1p/2b1p3/2B1P3/2PP1N2/PP3PPP/RN1Q1RK1 b - - 0 8`. Black to move. Play out 8...Qxf6 and 8...gxf6 for a few moves each against the engine and notice where White's pieces go in the second case.

How to check yourself: the routine is working when your written candidate list usually contains the engine's best move, even if you did not choose it. If the best move is regularly *not on your list*, slow down at step 3 and step 4.

Chapter summary

- Use the same routine on every move: his idea, safety scan, forcing moves for both sides, candidates, compare by his best reply, blunder check, play.
- Step 1 — "what does my opponent want?" — prevents more lost games than any tactic you know.
- A candidate list of two to four moves protects you from single-line fixation.
- Compare candidates by the opponent's *best* reply. Hope chess assumes a convenient reply; real chess checks.
- When you find a good move, spend one more minute looking for a better one, starting with checks.
- Spend your time on critical moments: captures, checks, pawn breaks, structural changes and endgame transitions.
- The blunder check — one more safety scan with your move "played" — is the most valuable ten seconds of every move.

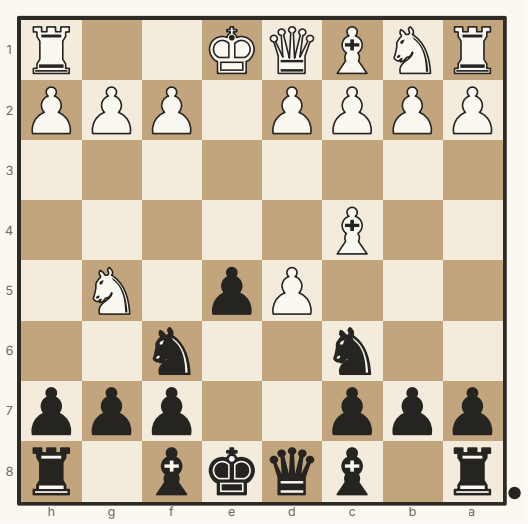
CHEAT CODE **The move-selection routine**

1. What does my opponent's last move attack, threaten or prepare?
2. Safety scan: loose pieces, counted attacks, my king.
3. Forcing moves — checks, captures, threats — for me *and* for him.
4. Choose 2–4 candidate moves (forcing moves, answers to his idea, plan moves).
5. For each candidate, find his best reply; pick the move whose worst case is best.
6. Blunder check the new position, then play — and think on his time.

Exercises

Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

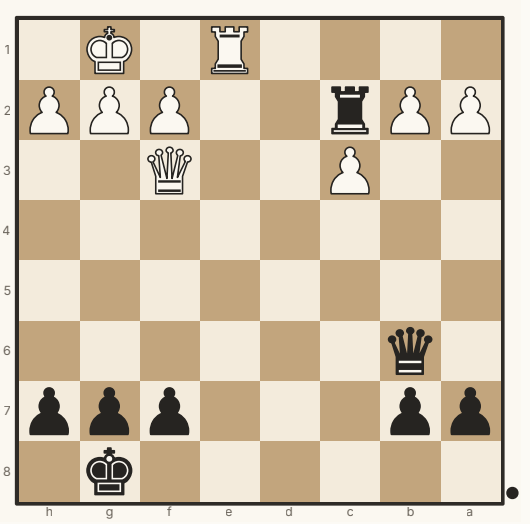
14.1 BLUNDER CHECK



Black to move. In the Two Knights Defence (1.e4 e5 2.Nf3 Nc6 3.Bc4 Nf6 4.Ng5 d5 5.exd5), taking back on d5 with the knight looks completely natural. Run your blunder check first, then play the best move.

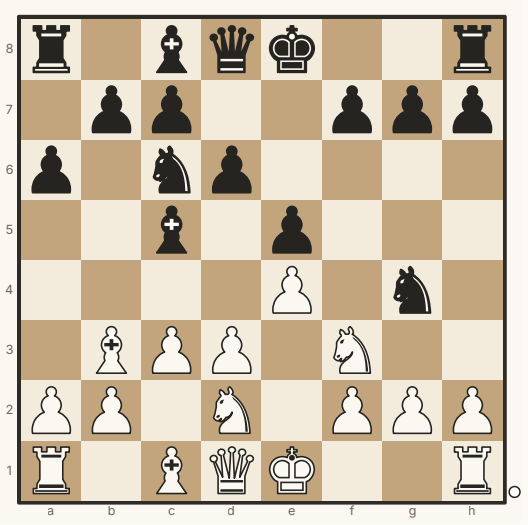
Solution p. 676

14.2 DEFENCE



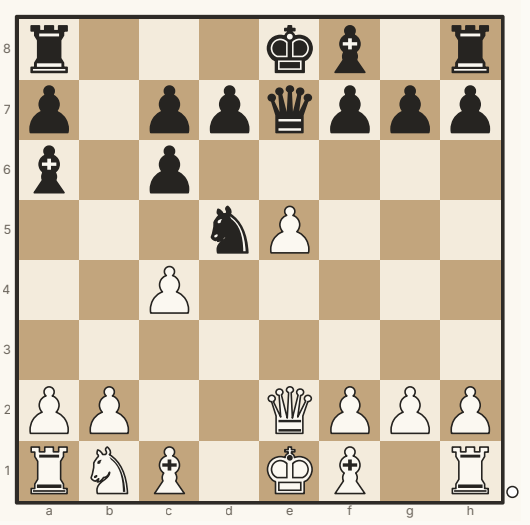
Black to move. Your rook has reached the second rank, and both b2 and f2 look tempting. First ask what White is threatening, then play the best move. Solution p. 676

14.3 DEFENCE



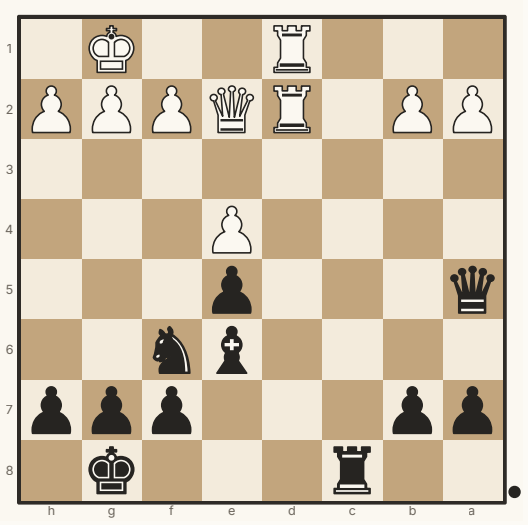
White to move. Black has just played 7... Ng4. Count the attackers and defenders of f2, then choose the best move. Solution p. 676

14.4 STRATEGY



White to move. In the Scotch (1.e4 e5 2.Nf3 Nc6 3.d4 exd4 4.Nxd4 Nf6 5.Nxc6 bxc6 6.e5 Qe7 7.Qe2 Nd5 8.c4) Black has played 8... Ba6, and the knight on d5 is attacked by your c-pawn. Your first idea is 9.cxd5. Compare it with at least one other candidate and play the best move. Solution p. 676

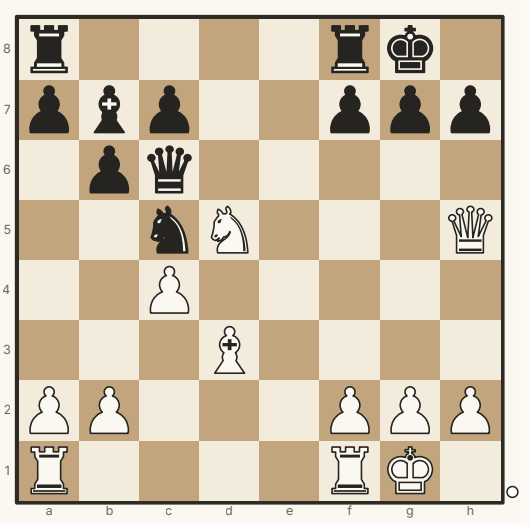
14.5 BLUNDER CHECK



Black to move. The pawn on a2 is loose. Is it safe to take? Play the right move.

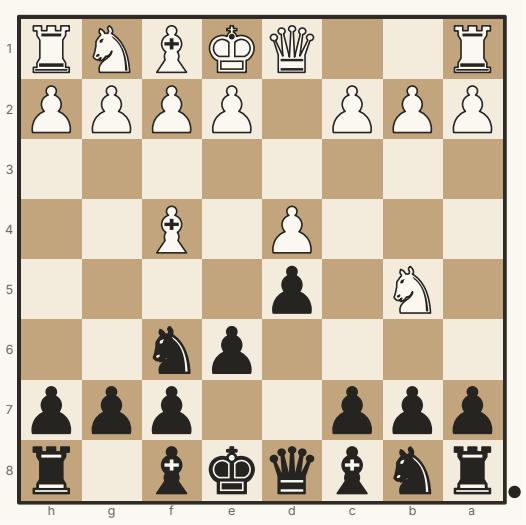
Solution p. 677

14.6 CHECKMATE



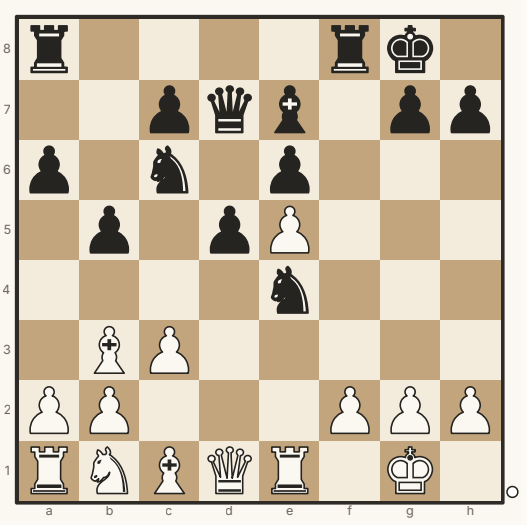
White to move. Ne7+ wins Black's queen. When you see a good move, look for a better one. Solution p. 677

14.7 DEFENCE



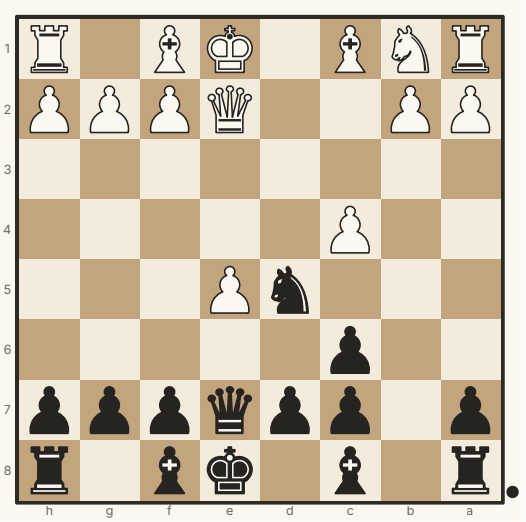
Black to move. White's last move was 4.Nb5. What is the threat, and what is the best way to stop it? Solution p. 677

14.8 STRATEGY



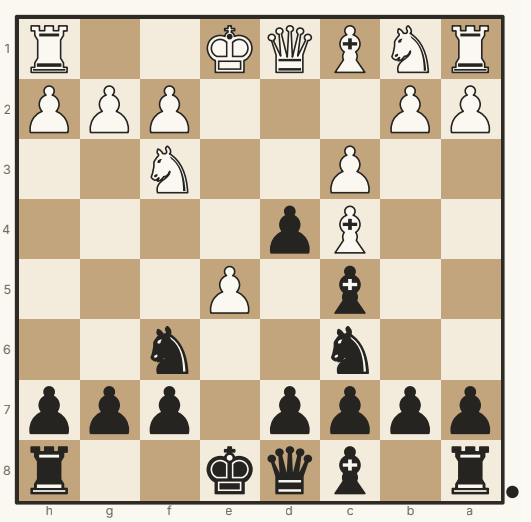
White to move. In the Open Ruy Lopez Black played 11...Qd7?, and after 12.Nxe6 fxe6 the knight on e4 looks safely protected by the d5-pawn. The natural move is 13.f3, kicking it. Compare candidates and play the best move. Solution p. 677

14.9 OPENING



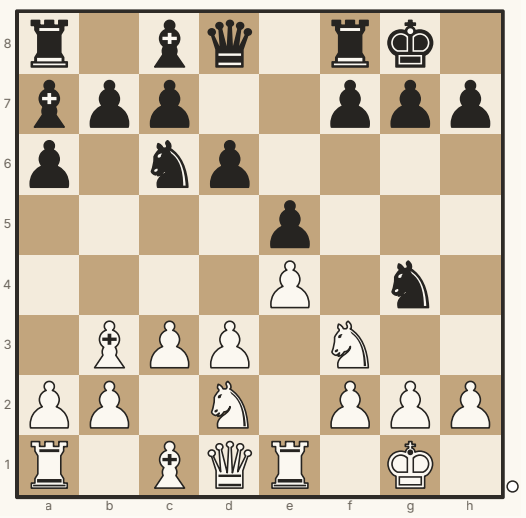
Black to move. Your knight on d5 is attacked. List the candidate squares and play the best move. Solution p. 678

14.10 OPENING



Black to move. White's pawn attacks your knight on f6. One move keeps the game balanced — find it. Solution p. 678

14.11 DEFENCE



White to move. Black's last move, 9...Ng4, attacks f2 for the second time. Your rook has left f1. Choose the best defence.

Solution p. 678

HINTS

14.1 If you take on d5, which white sacrifice on f7 becomes possible?

14.2 What would Re8+ do?

14.3 Which move adds a defender to f2 and improves the king at the same time?

14.4 What stands behind the c4-pawn on the diagonal from a6?

14.5 What are White's checks? Look at d8.

14.6 List every check, not just the first one you notice.

14.7 Which square do the knight and bishop both attack?

14.8 Which black pawn has two jobs at once?

14.9 Can you make the attacking pawn unable to capture?

14.10 Is there a move that creates a bigger threat than White's?

14.11 Count: two attackers on f2. How many defenders does your move add?

CHAPTER 15 STAGE 800–1000

Calculation Fundamentals

Seeing a few moves ahead, accurately

YOU WILL LEARN TO

- Calculate forcing lines (checks, captures, threats) two to four moves deep without moving the pieces.
- Visualise the final position of a line and know where every piece stands.
- Count capture sequences and pawn races correctly.
- Check the opponent's best defence and know when to stop calculating.

> IN YOUR NEXT GAME
>
> Before you play any capture or check, calculate the line until the position is quiet, and at the end ask "where is every piece now, and what are his checks and captures?"

You see the idea. The knight jumps in, the queen follows, the king is caught — and then, three moves later, the position on the board looks nothing like the one in your head. A bishop you "removed" is still guarding the key square. Your opponent did not take the bait but played a quiet move you never considered. The combination was real; your calculation was not.

Calculation means working out a sequence of moves in your head, without moving the pieces, and judging the position at the end. It is a skill like any other: it improves with a clear method and regular practice. This chapter gives you the method — what to calculate, in what order, how far, and how to check yourself — and the most common ways calculation goes wrong.

Forcing moves first

When you start calculating, the order in which you look at moves matters enormously. Look first at **forcing moves**: checks, captures and direct threats. There are three good reasons.

First, forcing moves limit the opponent's replies. After a check he may have two or three legal moves; after a quiet move he may have thirty. A line made of forcing moves is a narrow path you can follow to the end; a line made of quiet moves branches like a tree and quickly becomes impossible to hold in your head.

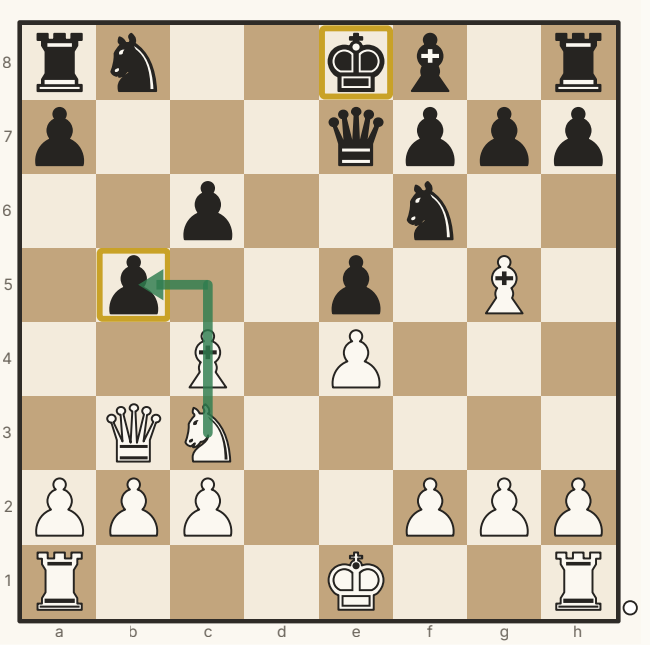
Second, forcing moves are where the decisive tactics live. Almost every combination you met in Chapters 7–11 started with a check or a capture.

Third, forcing moves are where *your opponent's* resources live too. A line that wins a queen is worthless if it allows a mate in two. So when you calculate, list the forcing moves for both sides at every step: "I check, he has three replies; if he blocks, I capture; now what are *his* checks?"

> **KEY IDEA** **Checks, captures, threats — in that order**
>
> At each step of a line, look at checks first, then captures, then threats. Do this for whichever side is to move in your imagined position, not only for yourself.

A worked example: Morphy's Opera game

The best way to learn the method is to watch it work. The position below comes from one of the most famous games ever played: Paul Morphy against the Duke of Brunswick and Count Isouard, who consulted together, at the Paris opera in 1858.



Morphy's Opera game after 9...b5. White to move. Black's king is still in the centre and Black is behind in development.

1.e4 e5 2.Nf3 d6 3.d4 Bg4 4.dxe5 Bxf3 5.Qxf3 dxe5 6.Bc4 Nf6 7.Qb3 Qe7 8.Nc3 c6 9.Bg5 b5

How the position arose. Black gave up the bishop pair early and is several moves behind in development.

WORKED EXAMPLE **Thinking it through**

"What did Black's last move want? 9...b5 attacks my bishop on c4. So my bishop must move, or I must do something more forcing.

Forcing moves for me: Nxb5 (a capture), Bxb5 (a capture), Bxf6 (a capture), Bxf7+ (a check, but Qxf7 and I have lost a bishop). Forcing moves for him if I do nothing: ...bxc4, winning my bishop.

Candidates: Nxb5, Bxb5, and the retreat Be2.

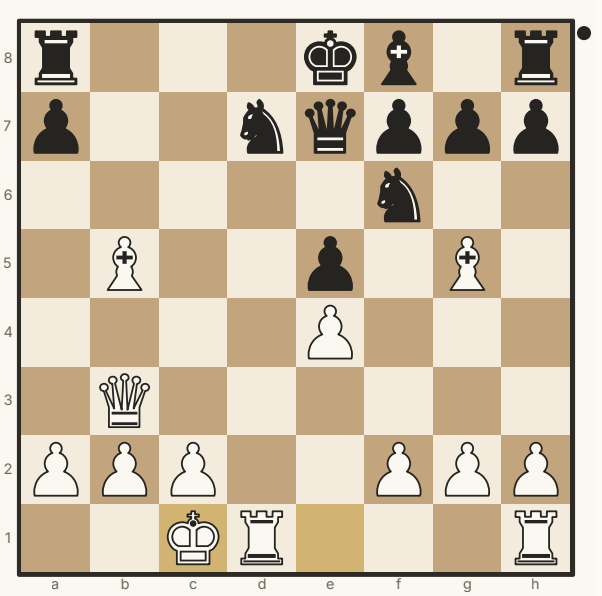
Nxb5 first. If 10...cxb5, then 11.Bxb5+ is a check: Black must block with ...Nbd7 or move the king to d8. My bishop on b5 pins the knight on d7, my rook can come to d1 after castling queenside, and Black's king has nowhere to hide. That looks crushing.

But that is the line I hope for. What is his best defence? Checks first — does Black have a check after 10.Nxb5? Yes: 10...Qb4+! It offers a queen trade, and after 11.Nc3 Qxb3 the attack is gone. I am still a pawn up, with the better position, but it is not a quick win.

Compare: after the retreat 10.Be2 or the capture 10.Bxb5, I win nothing. So Nxb5 is still the best candidate — I just have to remember that its best answer is the queen check, not the recapture."

The game went the way Morphy hoped. The allies recaptured, and Morphy's attack crashed through.

10. **Nxb5 cxb5?** 11. **Bxb5+ Nbd7** 12. **O-O-O**



The game continuation. The knight on d7 is pinned twice, Black's king is stuck in the centre, and White's rooks are coming. You will meet Morphy's finish in the exercises.

But the engine's verdict is instructive. Black's best defence was the queen check, and while White is still clearly better after it, the game is far from over.

10. **Nxb5 Qb4+** 11. **Nc3 Qxb3** 12. **axb3**

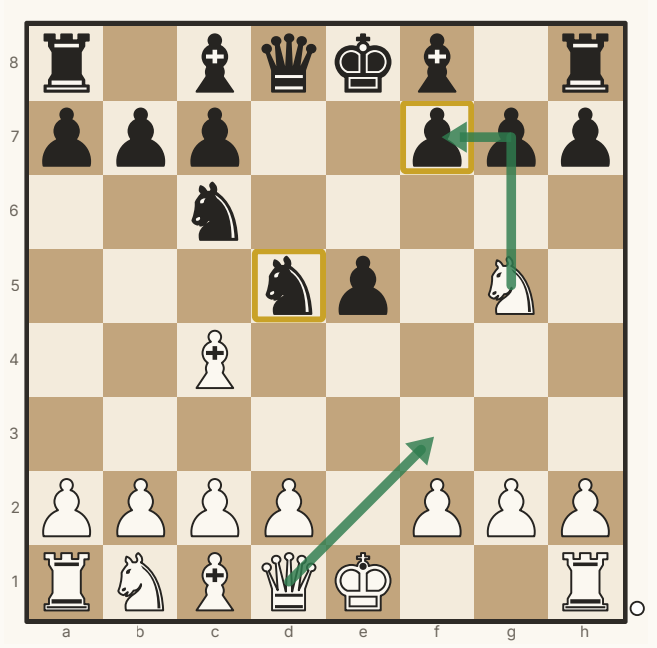
The best defence: a check that forces queens off. White is a pawn up in an endgame — good, but much less than the game continuation.

> **COMMON MISTAKE** **The hoped-for reply**
>
> The most common calculation error at every level is calculating the reply you want instead of the reply that hurts you. At every branching point, ask: "What would I play here if I were him?" Then look at his checks first.

Visualise the final position

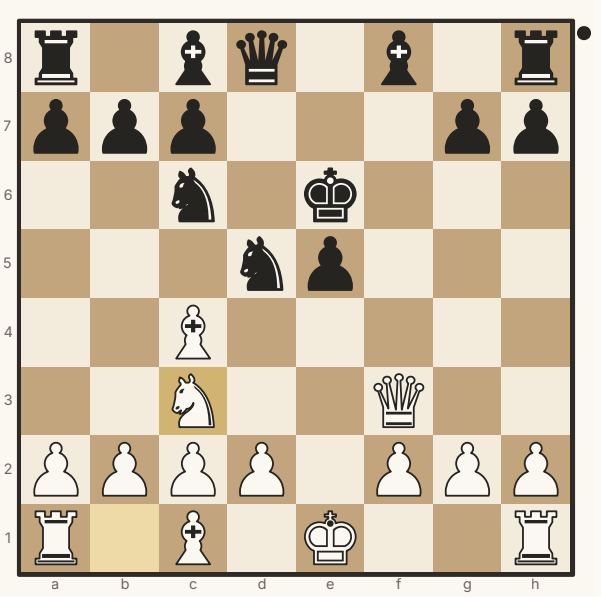
At the end of every line you calculate, stop and build a clear picture of the new position. Ask: *where is every piece now?* Which pieces have been captured, which have moved, which squares are now empty? Then run a quick safety scan on that imagined position, for both sides.

This sounds obvious, but it is exactly where calculation breaks down. Your mind keeps the old position as a background image, and the pieces that moved in your line tend to “stay” on their old squares in your head.



A sharp Two Knights position after 5...Nxd5. White to move. Can you see the position after Nxf7, Kxf7 and Qf3+ without moving a piece?

6. **Nxf7 Kxf7** 7. **Qf3+ Ke6** 8. **Nc3**



The Fried Liver. After 7...Ke6 the knight on d5 is pinned and attacked by queen and bishop; the quiet 8.Nc3 adds a third attacker.

Try it: from the diagram, play the line in your head and then answer without looking — where is the black king? What defends the knight on d5? (The king on e6 and the queen on d8 — and the knight is pinned by the bishop on c4.) What are Black's checks? (None.) If you can answer these three questions correctly, you have really calculated the line.

Notice that 8.Nc3 is not a check or a capture. That brings us to an important point.

Quiet moves inside forcing lines

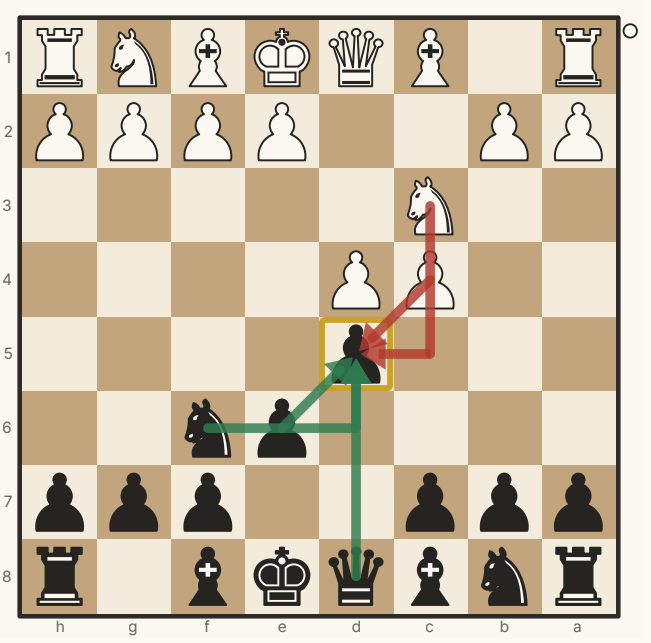
A **quiet move** is one that is not a check or a capture. Forcing lines sometimes contain a quiet move in the middle: a move that does not attack the king but creates a threat so strong that the opponent cannot meet it. Because we look at forcing moves first, these moves are easy to miss — both your own and your opponent's.

Look for a quiet move when your forcing moves have "run out" but the opponent's position is clearly creaking: the king is exposed, a piece is pinned, or one more attacker would win. Ask: "Is there a move that adds an attacker, or cuts off the only defence, even without check?" In the Fried Liver, 8.Nc3 is exactly that.

Counting capture sequences

When several pieces attack and defend one square, you need to **count**: will the final capture leave you ahead or behind? You met counting in Chapter 3. At this stage, add two refinements.

1. **The order of captures matters.** Each side normally captures with its least valuable piece first. If you capture with a queen while a pawn defends the square, you lose the queen at once, whatever the count says.
2. **Either side may stop.** The defender does not have to recapture if recapturing loses more material. Count the sequence step by step, and at each step ask whether the side to move would rather stop.



The Queen's Gambit Declined after 3...Nf6. The d5-pawn is attacked twice (c4, Nc3) and defended three times (e6, Nf6, Qd8). White cannot win it.

In this position from your Black repertoire (Chapter 13), White's two attackers on d5 face three defenders, and the first defender is a pawn. Any capture on d5 simply trades. That is why Black can calmly continue with ...Be7 and ...O-O. If White adds a third attacker — say Qb3 or Qa4 — count again.

BOARD DRILL **Counting without moving**

Take any position from one of your games where many pieces bear on one square. Before touching the pieces, write down: attackers, defenders, the value of each, and the order you expect. Then play the sequence out and check.

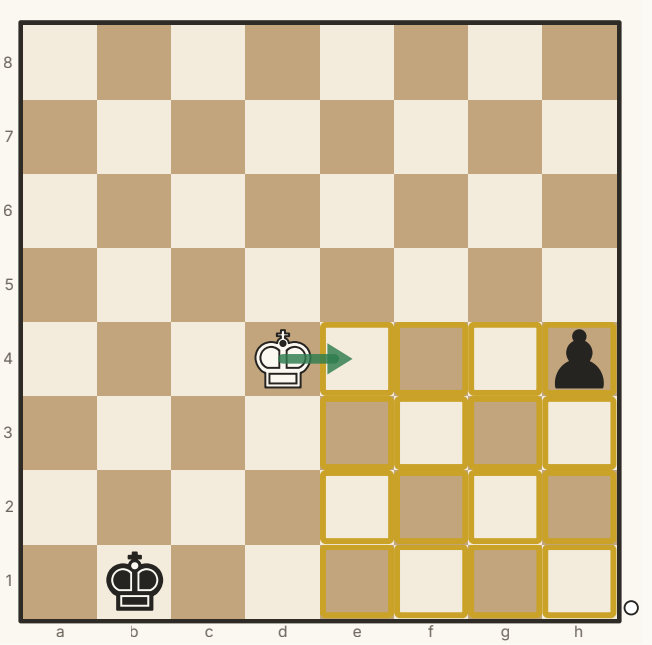
Tempo counting in races

In endgames, calculation is often pure counting. A **tempo** is a single move. In a **pawn race**, both sides push passed pawns, and you need to know who queens first — and what happens after that.

Three questions decide most races:

1. **How many moves does each pawn need to queen?** Count from the pawn's square to the last rank; a pawn on its starting square can use the double step.
2. **Who is to move?** A one-tempo difference changes everything.
3. **What happens when the first queen appears?** Does it come with check? Does it control the other pawn's queening square? Can the other side queen next move?

For races between a king and a pawn, use the **rule of the square**: draw a square with one side running from the pawn to its queening square. If the defending king can step into that square, it catches the pawn.

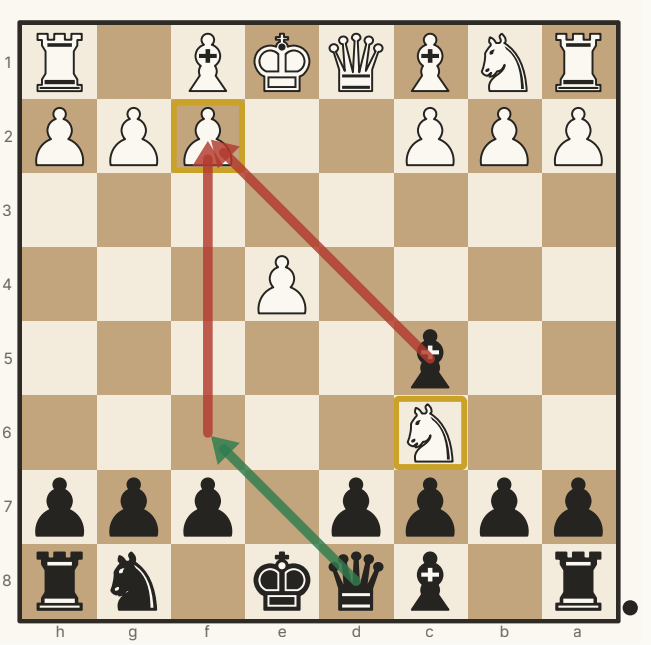


The square of the h4-pawn runs from h4 to e4, e1 and h1. With White to move, Ke4 enters it and catches the pawn; with Black to move, ...h3 shrinks the square and the pawn queens.

Races also teach the most important lesson of calculation in miniature: count the opponent's tempi as carefully as your own. Several of this chapter's exercises look like "just push the pawn" until you count what the other side does with the moves you give him.

Intermediate moves

An **intermediate move** — often called by its German name **zwischenzug** ("in-between move") — is a forcing move played in the middle of a sequence the opponent expects to be automatic. The classic case is a recapture: your opponent captures, expecting you to recapture, and instead you first play a check or a bigger threat, and recapture afterwards.



An illustrative Scotch Game line. White has just captured on c6. Before recapturing, Black can attack f2 with the queen, threatening mate.

5... **Qf6** 6. **Qf3 bxc6**

5...Qf6 threatens Qxf2#. White must deal with the threat first, and only then does Black recapture on c6 — with the queen already on an active square.

Intermediate moves matter in two ways. In attack, they let you insert a check or a capture that improves the final position. In defence, they are the main reason "forced" lines are not forced: when you calculate a capture sequence, always ask whether the opponent must recapture at once or has something better first. The exercises include a position in which an intermediate check wins a whole rook.

Check the opponent's best defence

We saw it in the Opera game: the line you calculate is only as good as the defence you assume. When you check a line, look at the opponent's replies in this order:

1. **His checks.** They are the easiest to overlook and the most dangerous.
2. **His captures**, especially captures of the piece you just moved or of a piece that the moved piece stopped defending.

3. **His threats**, including counter-threats against your king.
4. **Quiet defensive moves** that cover the key square or get the king out of the way.

If the line survives all four, trust it. If not, either repair the line or choose another candidate. This is real chess, as Chapter 14 described it: you do not play a combination because it works against the obvious reply, but because it works against the best one.

> COACHING JUDGMENT **Coaching judgment: assume he is strong**
>
> When you calculate, imagine your opponent is rated 300 points higher than he really is. You will sometimes be pleasantly surprised at the board — but you will rarely be shocked.

The stopping rule

How far should you calculate? The answer is not "as deep as possible". It is: **stop when the position is quiet and you can evaluate it.** A position is quiet when there are no checks, no captures that change the material balance, and no immediate threats for either side. At that point, count the material, look at the kings and the activity of the pieces, and decide whether the line is good for you.

If the position at the end of your line is *not* quiet — a piece is still hanging, a check is coming — you have not finished. Keep going until it is, or until you can see that one side wins too much material for the rest to matter.

There is a second stopping rule: **stop when you have found what you need.** If one candidate wins a rook by force, you do not need to calculate whether the others win a pawn. And if a candidate fails against the best defence, you can drop it; you do not need to calculate it further.

> IN YOUR NEXT GAME **The quiet-position test**
>
> In your next game, every time you calculate a line, say the final position to yourself in one sentence before you play: "I am a knight up, no checks for him, my king is safe." If you cannot say such a sentence, the position is not quiet yet — or you have not visualised it.

Evaluating the end of the line

Once the position at the end of your line is quiet, you still have to decide whether you like it. Use a short checklist, in this order:

1. **Material.** Count it. Who is ahead, and by how much?
2. **King safety.** Is either king exposed, or stuck in the centre with open files nearby?
3. **Threats.** Does either side have an immediate threat that will cost material next move?
4. **Activity.** Whose pieces are better placed? Is there a passed pawn?

Most of the time the first point decides: if you are a piece up with a safe king, the line is good. The other points matter when material is level, or when the side that is "ahead" has a king in danger. Do not try to be too precise. At this stage, "clearly better for me", "about equal" and "clearly worse" are enough to choose between candidates.

How deep is deep enough?

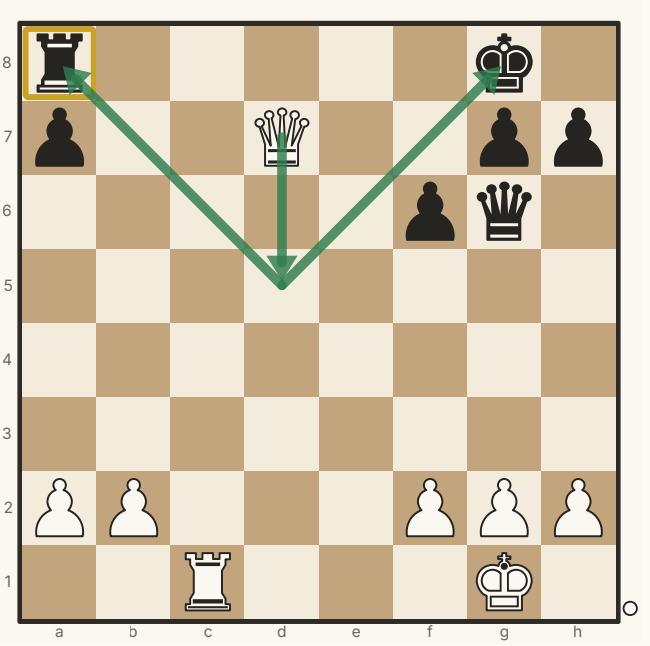
At 800–1000, most games are decided by lines of two to four moves for each side. You do not need to calculate ten moves deep; you need to calculate three moves *accurately*, including the opponent's best defence and the final check of the position. Depth comes with practice. Accuracy comes from method: forcing moves first, best defence, visualise the end, evaluate.

Common calculation errors

Most calculation mistakes at this level fall into a few families. Knowing their names helps you catch them.

- **Backward moves are missed.** Our eyes look forward: toward the opponent's king, toward the centre. Moves that go *back* toward your own side — a queen retreating to fork, a bishop dropping back to defend — are invisible to many players.
- **Pieces "disappear".** In a long line a piece is captured or moves away, but you keep calculating as if it were still there. Typical symptom: you think a square is guarded by a piece that was exchanged two moves ago.
- **Ghost pieces.** The reverse: a piece has moved in your line, but in your mind it still controls its old squares. A knight that went to f7 no longer guards e4.

- **Stopping one move too early.** You end your line at the moment you win something, without checking the opponent's next check or capture.
- **Calculating the hoped-for reply.** The Opera game mistake from the other side.



White to move. The winning move goes backwards: Qd5+ checks the king and attacks the loose rook on a8 along the long diagonal.

25. **Qd5+ Kf8** 26. **Qxa8+ Qe8**

A backward queen move forks king and rook. White wins the rook on a8 and is a whole rook ahead.

Players who look only "forward" from d7 see Qe6+, Qe7 and Qb7 and miss Qd5+. When you list your forcing moves, scan in all eight directions for each queen, rook and bishop, and all eight jumps for each knight — including the ones that go home.

How to train calculation

Calculation improves with deliberate practice. These methods work at any level.

- **Solve without moving the pieces.** Set up a position, then calculate with your hands off the board. Moving pieces while you calculate trains the wrong skill.

- **Write the lines down.** Before you check the answer, write the main line and the most important side line, ending with a one-sentence evaluation of the final position. Writing forces precision: you cannot write "and then something wins".
- **Then check — honestly.** Compare your written line with the solution or an engine. Did you miss a defence? A check? A backward move? Note the *type* of error in your training diary; patterns will appear.
- **Prefer slightly-too-hard problems.** Puzzles you solve in two seconds train pattern recognition, which is valuable, but not calculation. For calculation, choose problems that take you two to five minutes.
- **Play long games.** Calculation needs time. Games of at least 30 minutes per side, ideally longer, are where you practise it under real conditions.

> EVIDENCE **Pattern knowledge supports calculation**
>
> Classic research on chess expertise, starting with Adriaan de Groot's *Thought and Choice in Chess* (English edition 1965) and the memory experiments of William Chase and Herbert Simon ("Perception in Chess", 1973), suggests that strong players rely heavily on a large store of familiar patterns. Recognising patterns tells you *which* lines to calculate; calculation then checks them. That is why this book trains both.

Practice at the board

Use a physical board or any free chess app in analysis mode, but do not move the pieces until you have written your answer.

1. **The Fried Liver.** FEN `r1bqkb1r/ppp2ppp/2n5/3np1N1/2B5/8/PPPP1PPP/RNBQK2R w KQkq - 0 6`. White to move. Without moving pieces, play 6.Nxf7 Kxf7 7.Qf3+ Ke6 in your head and write down where all pieces stand. Then find a quiet move for White. Check with the line in this chapter.
2. **The square.** FEN `8/8/8/8/3K3p/8/8/1k6 w - - 0 1`. Set up the position twice, once with White to move and once with Black to move. Predict the result of each before playing it out against a friend or engine.
3. **Counting.** FEN `rnbqkb1r/ppp2ppp/4pn2/3p4/2PP4/2N5/PP2PPPP/R1BQKBNR w KQkq - 2 4`. Count attackers and defenders of d5. Then imagine White's queen on b3 and count again.

4. **Backward moves.** FEN `r5k1/p2Q2pp/5pq1/8/8/8/PP3PPP/2R3K1 w - - 0 25`. Before reading the solution again, list every queen move that gives check. How many did you find on the first try?

How to check yourself: compare your written lines with an engine's top line. Your calculation is improving when your errors move from "missed his check" toward "evaluated the final position slightly wrongly".

Chapter summary

- Look at forcing moves first — checks, captures, threats — for both sides at every step of a line.
- At the end of a line, visualise the final position: where every piece stands, and what each side's checks and captures are.
- Watch for quiet moves inside forcing lines, both yours and your opponent's.
- Count capture sequences by value and order, and remember that either side may stop capturing.
- In races, count tempi for both sides; use the rule of the square for king against pawn.
- Before every automatic recapture, ask whether an intermediate move is available — for you or for him.
- Check the opponent's best defence (checks, captures, threats, quiet defences) and stop when the position is quiet and evaluable.

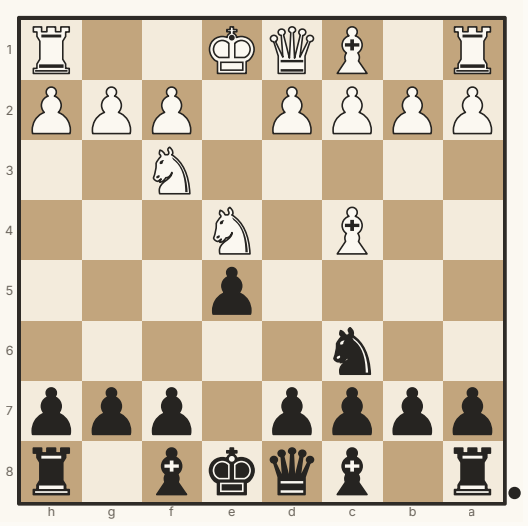
CHEAT CODE **Calculating a line**

1. List forcing moves — checks, captures, threats — for the side to move.
2. Follow the most forcing line; at each step, find the opponent's best reply, starting with his checks.
3. Look for intermediate moves and quiet moves that add an attacker.
4. Stop when the position is quiet; picture where every piece stands.
5. Evaluate in one sentence (material, king safety, activity) and compare with your other candidates.

Exercises

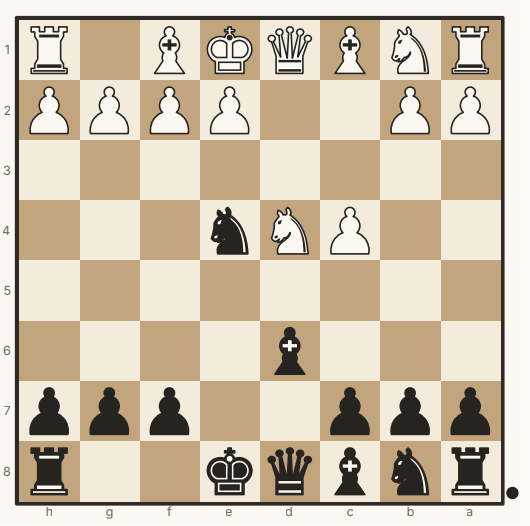
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

15.1 CALCULATION



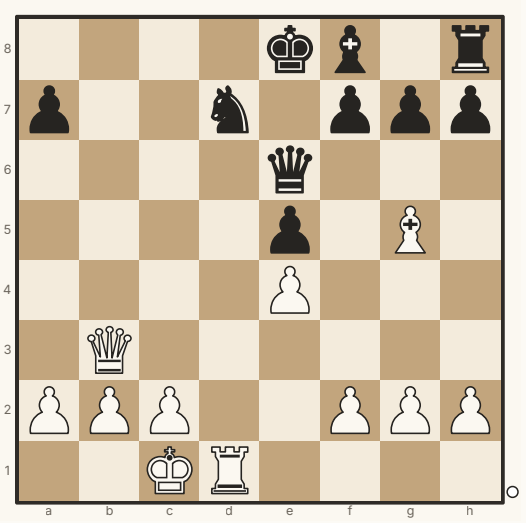
Black to move. You played 4...Nxe4 and White answered 5.Nxe4, so you are a piece down for a pawn. Find the forcing move that wins the piece back. Solution p. 679

15.2 CALCULATION



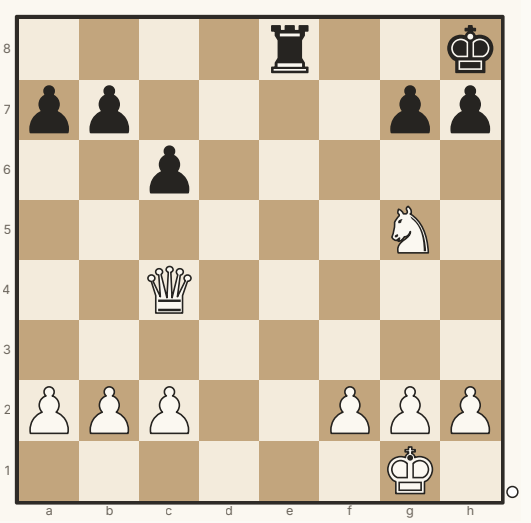
Black to move. In the Fajarowicz Gambit (1.d4 Nf6 2.c4 e5 3.dxe5 Ne4 4.Nf3 d6 5.exd6 Bxd6), White has just centralised the knight with 6.Nd4?. It is defended by the queen. Calculate, and win material. Solution p. 679

15.3 CHECKMATE



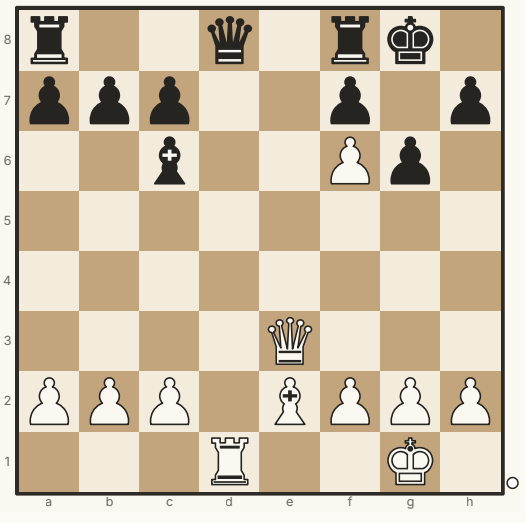
White to move and mate in two. Solution p. 679

15.4 CHECKMATE



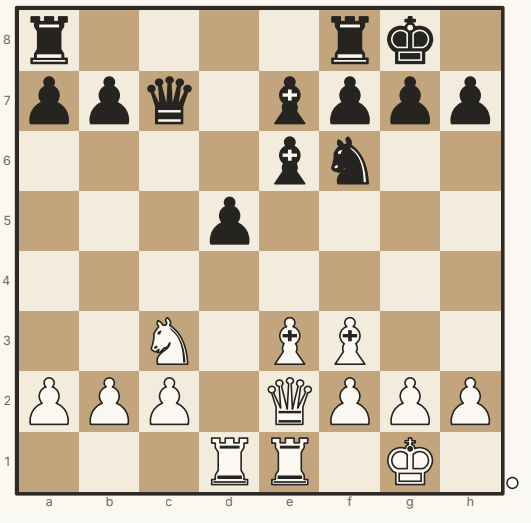
White to move and mate in four. Visualise the whole line before you move. Solution p. 679

15.5 CALCULATION



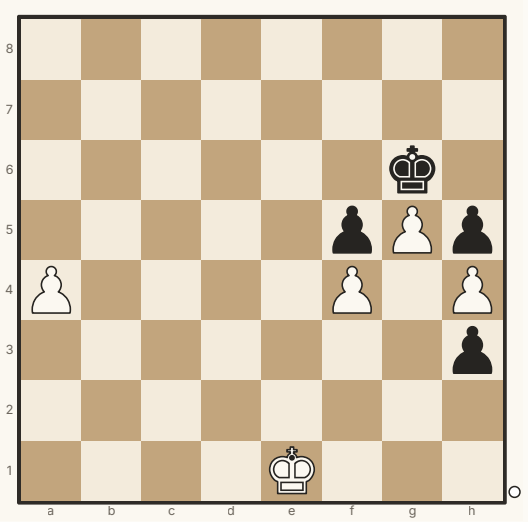
White to move. The pawn on f6 and a queen on h6 would mate on g7. Find the forcing way to make the idea work. Solution p. 680

15.6 COUNTING



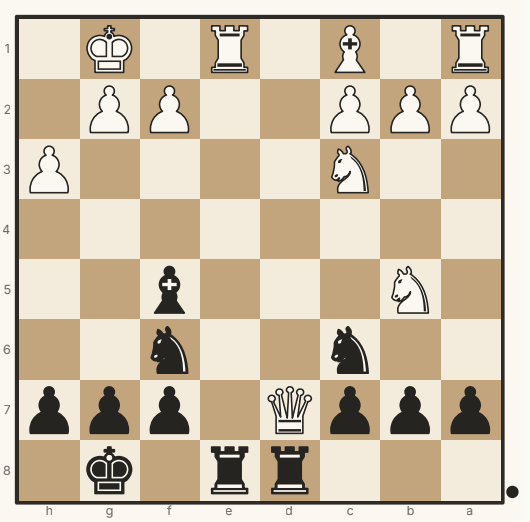
White to move. Count the attackers and defenders of the pawn on d5. Can you win it? Play the right capture. Solution p. 680

15.7 ENDGAME



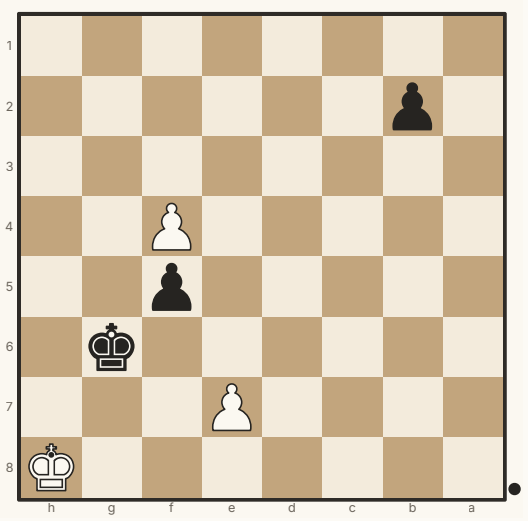
White to move. Both sides have a passed pawn. Count the tempi: should you race with the a-pawn or stop the h-pawn first? Solution p. 680

15.8 CALCULATION



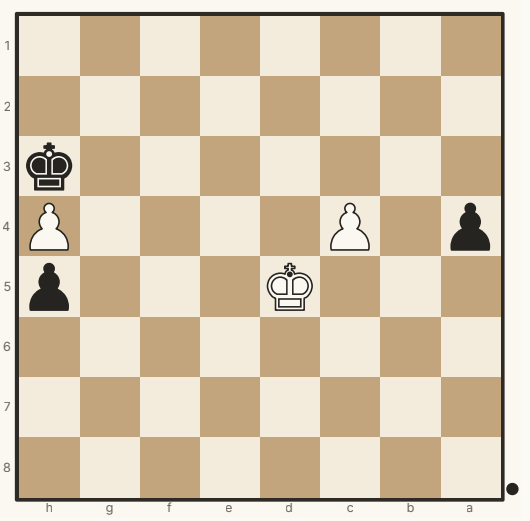
Black to move. White's queen has just captured on d7. Recapture is natural — but is there something better first? Solution p. 680

15.9 ENDGAME



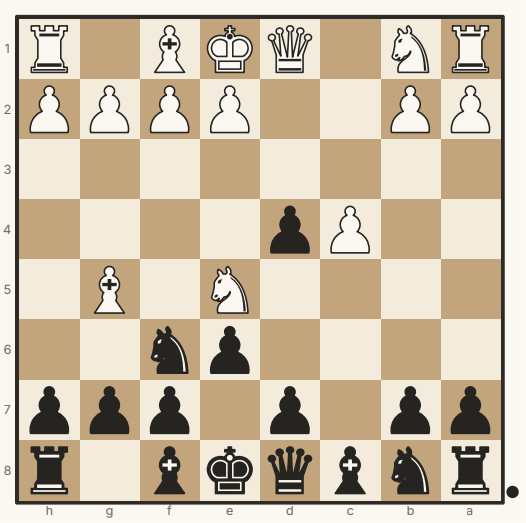
Black to move. You can queen at once with 52...b1=Q. Count what happens next, then find the move that wins. Solution p. 681

15.10 ENDGAME



Black to move. Taking the h4-pawn looks natural. Count the race first, then play the winning move. Solution p. 681

15.11 CALCULATION



Black to move. After 1.d4 Nf6 2.c4 e6 3.Nf3 c5 4.Bg5 cxd4, White has jumped into the centre with 5.Ne5?. Find the forcing move that wins material. Solution p. 681

HINTS

15.1 Which pawn move attacks two white pieces at once?

15.2 Your bishop is in front of your queen. Can it move with check?

15.3 Which black piece guards d8? Can you force it away with check?

15.4 Use knight checks to bring the king back to h8, then sacrifice on g8.

15.5 Which black piece can stop Qg7#? Remove it with a forcing move first.

15.6 Count, then capture with the least valuable attacker first.

15.7 How many moves does each pawn need? Can your king reach h1 or g2 in time?

15.8 Before recapturing, look for a capture that comes with check.

15.9 What is White threatening to do on the very next move?

15.10 How many moves does each passed pawn need to queen?

15.11 Which queen move gives check and attacks the knight at the same time?

CHAPTER 16 STAGE 800–1000

Converting a Winning Position

How to win a won game

YOU WILL LEARN TO

- Turn a material advantage into a win by reducing counterplay before grabbing more.
- Decide which trades help you (pieces, especially active ones) and which do not (usually pawns).
- Simplify into won king and pawn endings, and check them by counting first.
- Avoid stalemate tricks and survive the opponent's desperate counterattack.
- When you are the one behind, avoid the simplifications that help your opponent.

> IN YOUR NEXT GAME
>
> When you are ahead in material, ask before every move "what is his counterplay, and how do I take it away?" — and before every trade, ask "does this trade make his counterplay smaller or my win easier?"

You are a piece up. The hard part is over — or so it feels. You relax, you grab another pawn, and your opponent, who has nothing to lose, throws everything at your king. Three moves later you are fighting for a draw, or worse. Every player at this level has a game like that. Many have dozens.

Conversion is the process of turning an advantage into a win. It is a different skill from winning material, and it has its own rules. The good news is that those rules are simpler than tactics: take away counterplay, trade the right pieces, keep your king safe, and use your king in the endgame. This chapter shows how, and also what to do when you are on the other side of the board.

The converting mindset

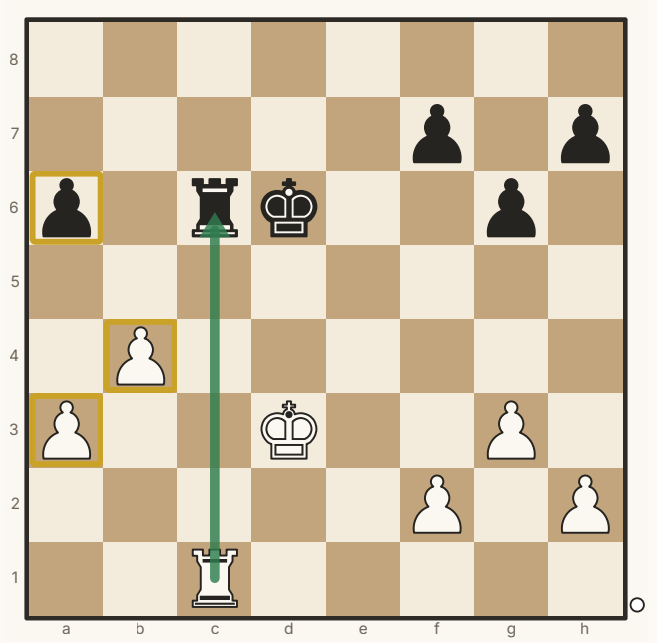
When you are winning, your opponent's only hope is **counterplay**: threats against your king, an active piece, a dangerous passed pawn, or a trick such as stalemate or perpetual check. Your job changes. You no longer need to find the strongest attacking move; you need to find the move that leaves him the fewest chances.

In practice this means three habits:

- **Ask what he wants, every move.** The routine from Chapter 14 does not stop when you are winning; it matters more, because your opponent will now look for tricks.
- **Prefer safe to spectacular.** A clean line that wins in thirty moves is better than a brilliant one that wins in ten if you have miscounted.
- **Don't grab more if consolidation wins.** An extra piece wins on its own. A second extra pawn is nice, but not if it costs you a tempo that lets his attack start.

> KEY IDEA **The converter's question**
>
> "What is his best counterplay, and which move takes it away?" Ask it before every move when you are ahead.

A worked example: choosing the cleanest path



White to move, a pawn up. Black's rook is active on the c-file. Several moves keep the advantage; one converts at once.

WORKED EXAMPLE **Thinking it through**

"What does Black want? His rook on c6 offers a trade, and if I avoid it, he will play ...Rb6 or ...Rc4 to attack my queenside pawns. His king is close to the centre.

Safety scan: my b4-pawn is defended by the a3-pawn, my king is fine — but my rook on c1 is attacked by his rook and defended by nothing! So whatever I do, I must deal with ...Rxc1.

Forcing moves: Rxc6+ is a capture with check. His forcing move is ...Rxc1, which is exactly the problem I just found.

Candidates: (a) Rxc6+ Kxc6, going into the king and pawn ending; (b) Rc2 or Rc4, keeping the rooks.

Compare. After (b), Black's rook stays active and I have to defend my pawns; my advantage is real but slow. After (a), count the pawn ending: I have two queenside pawns against one, so I can make a passed pawn there. My king gets to c4 first and keeps his king from advancing. Black's king will have to deal with my passed pawn, and then my king wins his pawns on the other side. That is a clean win.

Blunder check: after Rxc6+ Kxc6, does Black have anything? No pieces left, no checks. I play Rxc6+."

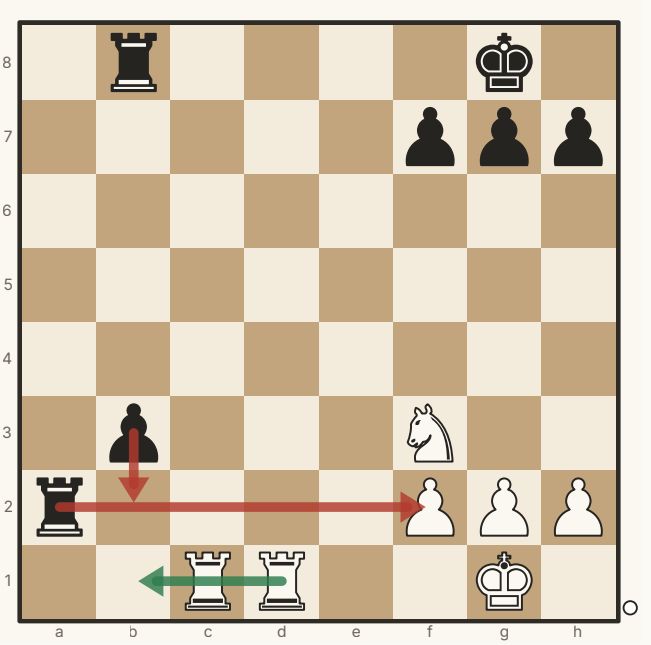
Notice that the safety scan found something the first impression missed: the white rook was hanging. That is normal — the routine is there precisely to catch it. And the forcing move that solves the problem turns out to be the cleanest conversion as well.

35. **Rxc6+ Kxc6** 36. **Kc4**

Trading into the pawn ending. White's king heads for the queenside, and the extra pawn there becomes a winning passed pawn.

Reduce counterplay before you grab

Most failed conversions follow the same script: the stronger side makes a "free" capture or an ambitious move, and the weaker side uses that tempo to start something. The fix is **prophylaxis** — a move that prevents the opponent's idea before it happens.



White is a knight up, but Black's b-pawn and the rook on the second rank are real counterplay. First stop the pawn (Rb1) and give the king air (g3 or h3); only then think about attacking.

Here White's extra knight should win, but Black's b-pawn is two steps from queening and his rook attacks f2. The engine's top choices are calm: Rb1, blockading the pawn, or a luft move such as g3. A greedy or ambitious move — say Rc7 to attack f7 — gives Black time for ...b2, and suddenly White is the one defending.

Typical counterplay to look for and remove:

- **Mate threats and checks** against your king, especially on the back rank. Make luft (an escape square for your king) at a quiet moment.
- **An opponent's passed pawn.** Blockade it, or put a rook behind it.
- **His most active piece.** Trade it off, or take away its best squares.
- **Perpetual check.** If his queen is near your king, consider offering a queen trade or tucking your king away before you do anything else.

> COMMON MISTAKE **Greed with a threat on the board**
>
> The most common conversion blunder at this level is capturing a pawn while the opponent has a mate threat or a perpetual. Before any capture when you are ahead, run the safety scan *for your own king* first.

Keep your king safe

A material advantage is only worth something if your king survives to use it. When you are ahead, your opponent's pieces will head for your king, because that is where a lost game can still be saved. So the moment you win material, look at your own king's shelter. Are the pawns in front of it still intact? Is there an escape square in case of a back-rank check? Are your pieces close enough to defend?

Often the best move right after winning material is a quiet king-safety move: luft with h3 or g3, a rook back to the first rank, a knight back to f3 to cover h2. These moves look modest, but they take away the opponent's only real chance. A safe king also lets you trade queens on your own terms instead of being forced to.

Don't grab more when consolidation wins

Consolidation means bringing your pieces back to safe, coordinated squares after winning material, so that the opponent has nothing to attack. It is the opposite of greed. A typical situation: you have won a piece, and now a pawn on b7 or a2 is hanging. Before taking it, ask three questions. Does the capture take my queen or rook away from my king? Does it give him a tempo for a check or a threat? Would the position be easily winning even without the pawn? If the answers are "yes, yes, yes", leave the pawn and consolidate. You can collect it later, when it is safe.

Trade pieces, not pawns

The best-known conversion guideline is: **when ahead in material, trade pieces, not pawns.** Each piece trade makes your extra material count for more. A knight up with all the pieces on the board is an advantage; a knight up with no other pieces is usually a simple win. And with fewer pieces left, your opponent has fewer ways to create counterplay.

Pawns are different. Your winning plan in the endgame usually involves making a passed pawn and queening it. Every pawn trade reduces your chances of doing that, and if all the pawns disappear, even an extra piece may not be enough: king and knight, or king and bishop, against a bare king is a draw.

This guideline has important nuances:

- **Trade his active pieces, keep your good ones.** The piece that creates his counterplay is the one to trade first.
- **Not every trade is good.** Before trading, look at the position you get. Trading into a lost or drawn pawn ending because "trades are good when ahead" is a classic error.
- **Some pawn trades help.** If a pawn trade opens a file for your rook, or removes his last pawn on one wing so that your extra pawn becomes passed, it can be the fastest way to win.

> COACHING JUDGMENT **Coaching judgment: which piece to trade?**
>
> When you are ahead, look at your opponent's pieces and ask, "Which one is giving me the most trouble?" That piece is usually the one to trade. Then look at your own and ask, "Which one would I miss most?" Keep that one.

Trading queens when ahead

The queen is the piece that creates the most counterplay: it gives perpetual checks, finds forks and attacks two weaknesses at once. That is why a queen trade usually helps the stronger side. A practical method is to offer the trade *with a threat or a check*, or on a square where refusing it costs your opponent something — for example, blocking a check with your queen on a square protected by a pawn. Then he must either trade or let your queen take over the centre.

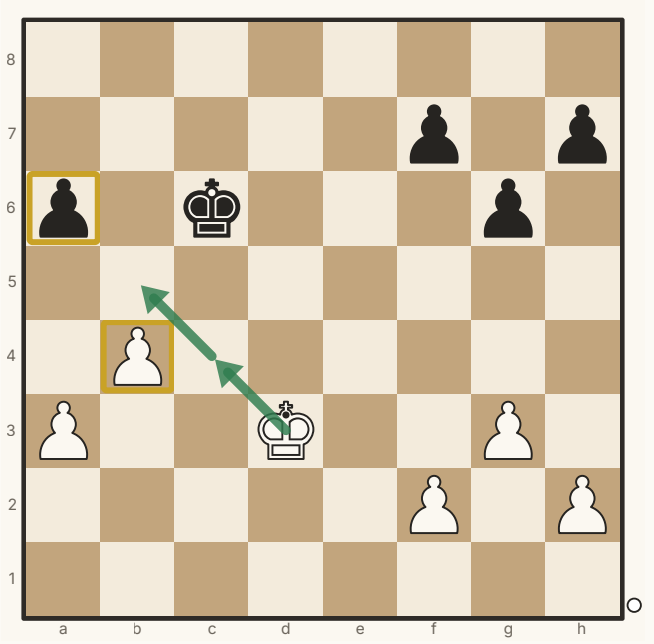
Two cautions. First, the square on which you offer the trade must be protected: blocking a check with an unprotected queen simply loses it. Second, check the ending after the trade, exactly as you would before any other simplification.

Simplify into a won king and pawn ending

The cleanest conversion is often to trade everything into a **king and pawn ending** that you have checked is winning. With no pieces left, there are no tactics, no perpetual checks, and no back-rank tricks: just counting. But it is also the most unforgiving decision in chess, because pawn endings are either won or drawn (or lost), and there is no way back.

Before such a trade, count:

1. **Can I make a passed pawn?** An extra pawn on a wing where you have more pawns usually can.
2. **Is it an outside passed pawn?** A passed pawn far from the kings drags the enemy king away, and your king then wins the pawns on the other wing.
3. **Whose king is more active?** The side whose king gets to the centre or to the enemy pawns first usually wins.
4. **Is there a breakthrough or race to check?** Count the tempi as in Chapter 15.



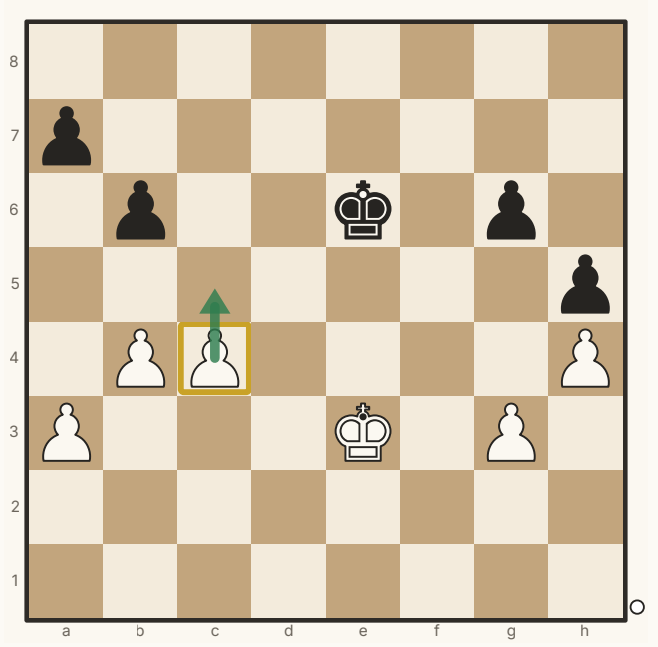
The pawn ending from the worked example. White to move. The king goes forward to c4 and b5, and the two-against-one majority on the queenside produces a passed pawn.

Activate the king

In the endgame the king is a strong piece. In the middlegame you kept it safe; once the queens are gone and the danger of mate is small, bring it toward the centre and the pawns. In a pawn ending, a king move is often worth more than any pawn move, because pawn moves cannot be taken back and the king decides who wins the race to the important squares. A useful guideline: in a pawn ending, ask “can my king improve?” before you consider a pawn move.

Create a passed pawn

A **passed pawn** is one with no enemy pawns in front of it or on the neighbouring files. Your extra material almost always wins through a passed pawn in the end. Where you have more pawns than your opponent — a **majority** — push the pawn that has no opponent in front of it first (the **candidate** pawn), supported by the others.



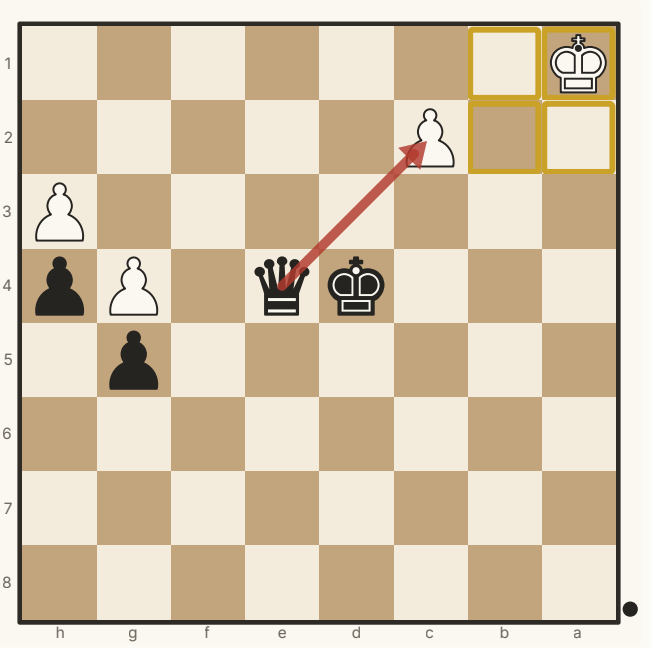
White's three queenside pawns face two. With the king supporting, the c-pawn — the one with no enemy pawn directly in front — leads the advance and the majority produces a passed pawn.

Two weaknesses, briefly

Sometimes one passed pawn is not enough: the defender blockades it and holds. Then the stronger side creates a **second weakness** on the other side of the board — another passed pawn, or a target pawn to attack. The defender cannot cover both at once. This is often called the **principle of two weaknesses**. At this stage, the practical version is simple: if your first plan is stopped, look for a second target far from the first, and switch to it.

Avoid stalemate tricks

Stalemate — the side to move has no legal move and is not in check — is a draw (Chapter 1). When you are far ahead, especially with a queen, it is your opponent's last hope. He may even sacrifice his remaining pieces to reach it, a trick called a **desperado** sacrifice.



Black to move. The "natural" 55...Qxc2?? takes White's last mobile pawn and covers a2, b1 and b2 without check — stalemate.

The rule is easy: **before every move when the opponent has few pieces left, count his legal moves after your move.** If the answer is zero and you are not giving check, stop. Here the white king can move only to b1 or b2 and the pawns on g4 and h3 are blocked, so the c2-pawn is White's last free unit. Bring the king closer first, and the same capture becomes mate.

55...**Kc3** 56.**Kb1 Qxc2+** 57.**Ka1 Qb2#**

With the king on c3, the capture on c2 comes with check, and the king covers b2 for the mate.

> **BOARD DRILL** **Count his moves**
>
> In any won ending, before every move, say aloud how many legal moves your opponent will have afterwards. Do this for ten games and the stalemate trap will never surprise you again.

Handling the desperate counterattack

When your opponent is lost, he has nothing to lose. Expect sacrifices, wild pawn storms and offers of material designed to open your king. Three guidelines help:

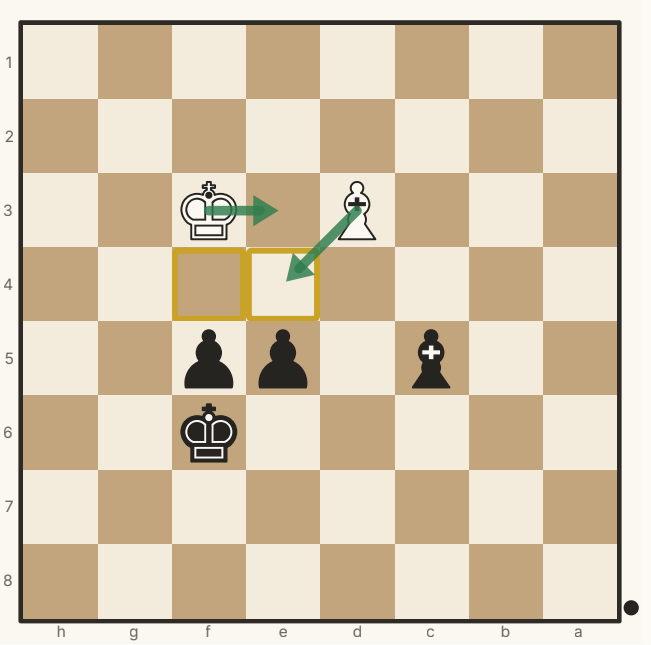
- **Remove the key attacker.** If one piece makes his attack work, take it or trade it, even if that means giving back some material.
- **Give back material when it kills the attack.** Returning a pawn or the exchange to reach a calm position with a big advantage is excellent technique, not cowardice.
- **Check every check.** Desperate attacks usually end in a perpetual check or a back-rank mate. Before each move, list his checks after your move.

The exercises include positions in which one move kills a desperate attack and the "natural" defence loses on the spot.

When your opponent is ahead

Everything in this chapter can be turned around. If *you* are the one behind, you want exactly what the converting side wants to avoid:

- **Keep pieces on** — especially your active ones — and trade pawns.
- **Avoid the pawn ending** unless you have counted it as a draw. Trading the last pair of rooks when a pawn down very often turns a drawable rook ending into a lost pawn ending.
- **Aim for drawish endings:** rook endings with all the pawns on one side, and endings with **opposite-coloured bishops** (one bishop moves on light squares, the other on dark).
- **Create counterplay and complications.** A passed pawn, a target near his king, a stalemate idea — anything that gives him a chance to go wrong.



Black is two pawns up, but with opposite-coloured bishops White holds easily: the king and bishop control the blockade squares e4 and f4, and the black bishop can never challenge the light squares.

> IN YOUR NEXT GAME **Before you trade, count**
>
> In your next game, before any trade that leaves only kings and pawns — whether you are ahead or behind — stop the clock in your head and count: passed pawns, outside passers, king activity, races. Then decide.

A phase-by-phase checklist

The technique changes with the stage of the game, so it helps to have a slightly different focus in each.

- **Right after winning material:** look at your king and at his forcing moves. Make luft, defend loose pieces, bring back a piece that has wandered off to make the capture.
- **In the middlegame:** trade his active pieces and his queen if you can; keep your pawns; avoid opening lines toward your own king.
- **At the transition to the endgame:** count before every trade that leaves few pieces, and above all before trading into a pawn ending.

- **In the endgame:** activate your king, create a passed pawn, and switch to a second target if the first one is blockaded. Count his legal moves to stay clear of stalemate.

This is not a fixed recipe — some positions demand that you keep attacking — but it is the right default. Strong players win most won games not with brilliant moves but by following these steps patiently.

How to identify conversion problems in your own games

When you analyse a game you were winning but did not win, look for these patterns:

- **The greedy capture.** You took material, and the opponent's next move was a check or a threat you had not looked at.
- **The missing prophylaxis.** His counterplay (a passed pawn, an active rook) grew for several moves while you pursued your own plan.
- **The wrong trade.** You exchanged your good piece for his bad one, or traded into an ending you had not counted.
- **The king that stayed home.** In the endgame your king stood on g1 while his marched into your pawns.
- **The stalemate.** You did not count his legal moves.
- **Time trouble.** You relaxed in the middlegame and spent your clock; the conversion then happened in a rush.

Tag these games in your mistake journal with the category *conversion*, and note which of the patterns occurred. Most players find that one or two of them repeat again and again — and those are the ones to work on first.

Practice at the board

Play each position out against a friend or a computer set to full strength. The goal is not to find one move but to *win the position*.

1. **The rook trade.** FEN `8/5p1p/p1rk2p1/8/1P6/P2K2P1/5P1P/2R5 w - - 0 35`. Play 35.Rxc6+ Kxc6 and win the pawn ending against the engine. If you fail, find the moment your king or pawns went wrong.

2. **Stop the pawn.** FEN `1r4k1/5ppp/8/8/8/1p3N2/r4PPP/2RR2K1 w - - 0 28`. White to move. Win with the extra knight. Count how many moves you spend on prophylaxis before you start your own plan.
3. **Stalemate awareness.** FEN `8/8/8/6p1/3kq1Pp/7P/2P5/K7 b - - 0 55`. Black to move. Mate as quickly as you can, saying White's number of legal moves out loud before each of your moves.
4. **The defender's view.** FEN `8/8/5k2/2b1pp2/8/3B1K2/8/8 b - - 0 50`. Take White and hold the draw against the engine. Notice which squares your bishop must watch.

How to check yourself: the engine's evaluation should not drop by more than about a pawn while you convert. If it does, go back to that move and ask what counterplay you allowed.

Chapter summary

- When ahead, your opponent needs counterplay; your job is to take it away, every move.
- Don't grab more if consolidation wins. Deal with threats, checks and passed pawns first.
- Trade pieces — especially his active ones — and usually keep pawns; but always look at the position you are trading into.
- A won king and pawn ending is the cleanest conversion; count passed pawns, king activity and races before you trade.
- In endgames, activate the king and create a passed pawn; if one target is not enough, find a second.
- Count your opponent's legal moves to avoid stalemate, and check his checks to avoid perpetuals.
- When behind, do the opposite: keep pieces, trade pawns, avoid lost pawn endings, and look for drawish endings and counterplay.

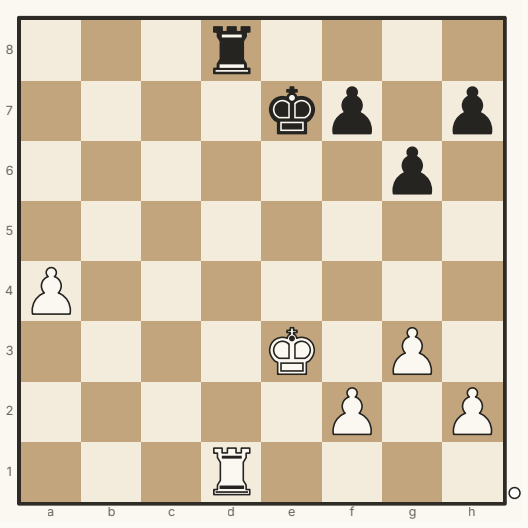
CHEAT CODE **Converting an advantage**

1. Ask: what is his counterplay (threats, checks, passed pawns, active pieces, stalemate)?
2. Remove it first: prophylaxis, luft, blockade, or trade his most active piece.
3. Trade pieces, keep pawns — but check the ending you are trading into.
4. Before trading into a pawn ending, count passers, kings and races.
5. In the endgame, activate the king and create a passed pawn; switch to a second target if needed.
6. Before every move in a won ending, count his legal moves and his checks.

Exercises

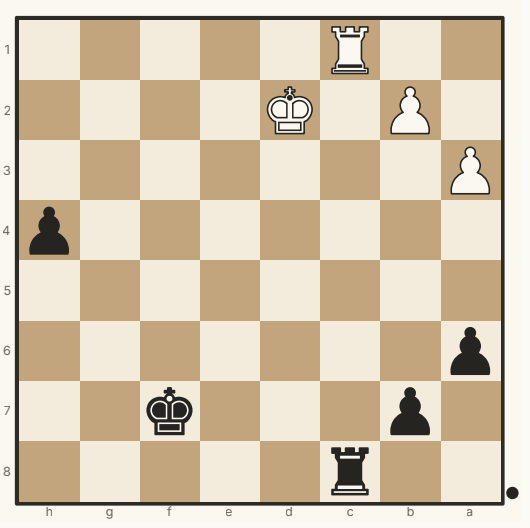
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

16.1 CONVERSION



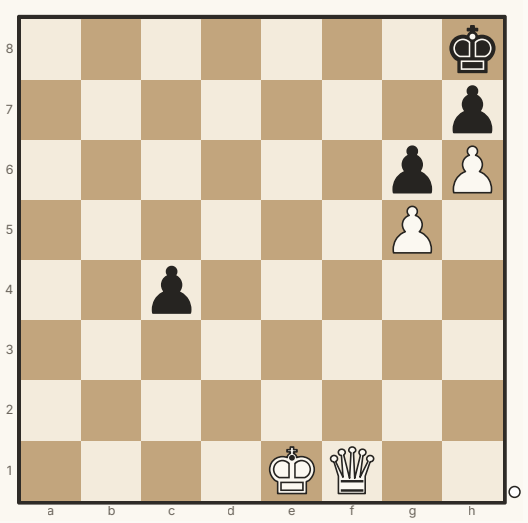
White to move. You are a pawn up with an outside passed a-pawn. Choose the move that converts most safely. Solution p. 682

16.2 CONVERSION



Black to move. Can you trade rooks? Count whether the white king can catch your h-pawn afterwards. Solution p. 682

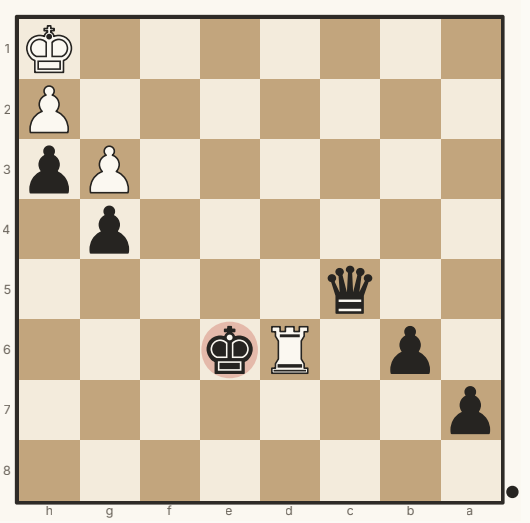
16.3 CHECKMATE



White to move. You are a queen up. Finish the game — and check that your move does not leave Black without legal moves.

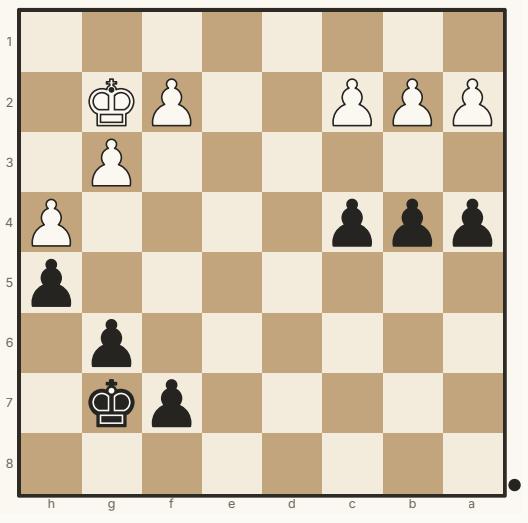
Solution p. 682

16.4 CONVERSION



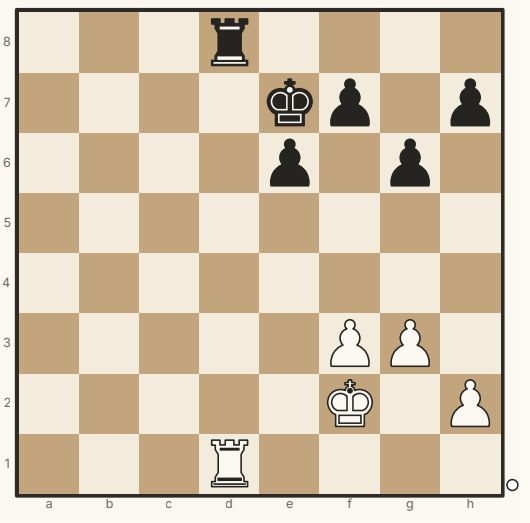
Black to move. White's rook gives check and offers itself. Which way of capturing is correct? Solution p. 682

16.5 ENDGAME



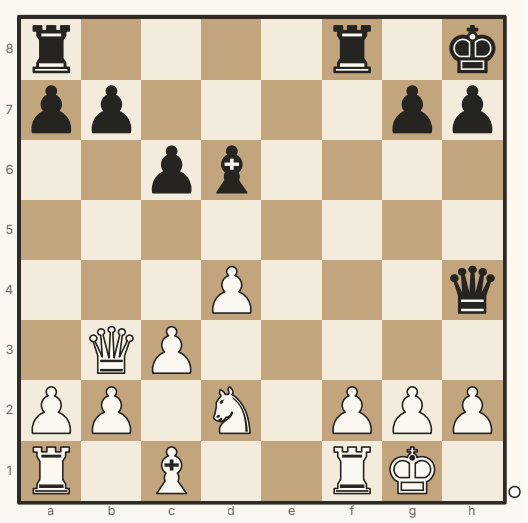
Black to move. Your three queenside pawns face three white pawns. Create a passed pawn that queens. Solution p. 683

16.6 DEFENCE



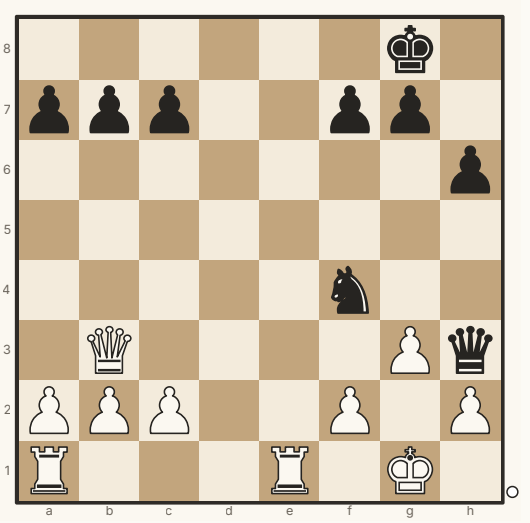
White to move. You are a pawn down. Black offers a rook trade on the d-file. Should you take? Solution p. 683

16.7 CONVERSION



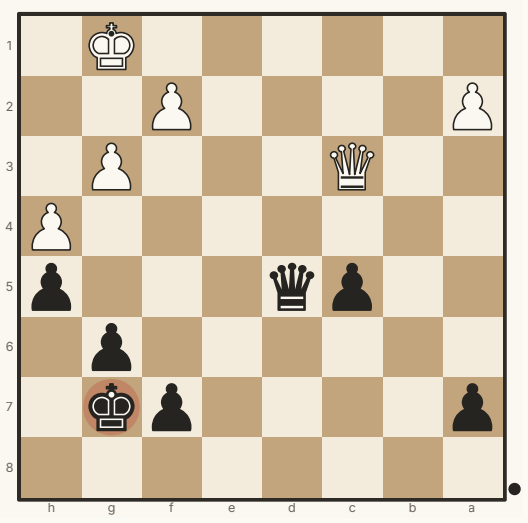
White to move. You are a knight and two pawns up, but Black threatens ...Qxh2#. Choose the safest way to meet the threat.
Solution p. 683

16.8 CONVERSION



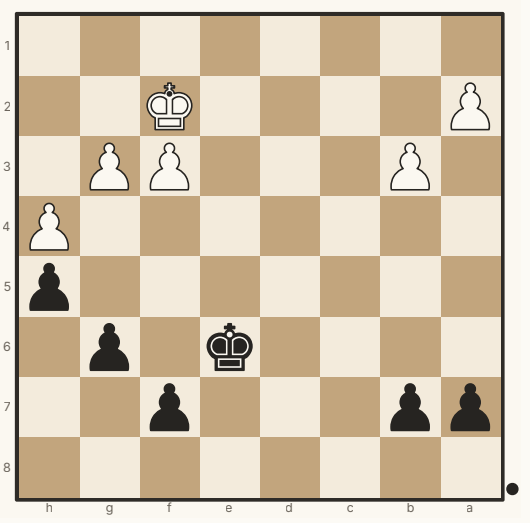
White to move. You are two rooks up for a knight, but Black has thrown everything at your king and threatens ...Qg2#. Find the safest way to win. Solution p. 683

16.9 CONVERSION



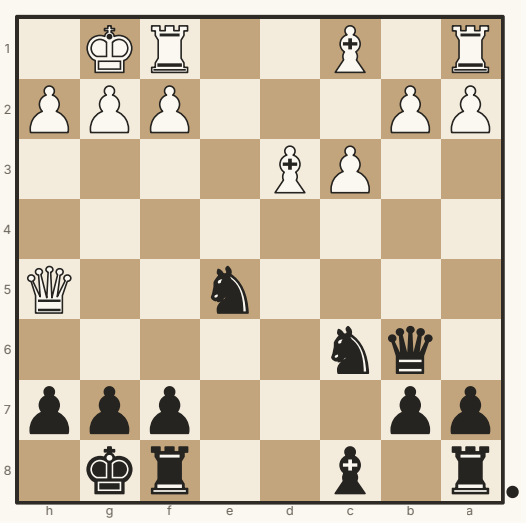
Black to move. White checks on the long diagonal. You are a pawn up. Which reply converts best? Solution p. 684

16.10 ENDGAME



Black to move. Material is level, but your king is closer to the centre. Choose the move that makes the most of it. Solution p. 684

16.11 CONVERSION



Black to move. You are a piece up. White threatens Qxh7#, and the pawn on b2 is loose. Choose the move that converts best.

Solution p. 684

HINTS

16.1 Which ending would you most like to have with an outside passed pawn?

16.2 After the trade, draw the square of the h-pawn.

16.3 Which black moves are legal right now? Do not take them all away without check.

16.4 If you take with the king, what can White move?

16.5 Push the middle pawn first.

16.6 Which ending is easier to hold a pawn down: with rooks, or without?

16.7 Which defence cannot be broken by a sacrifice?

16.8 Which black piece makes the mate threat possible?

16.9 Block the check with a protected piece that also offers a trade.

16.10 Which piece should go forward first in a pawn ending?

16.11 Which white piece makes the mate threat work? Can you trade it off?

CHAPTER 17 STAGE 800–1000

Defense and Resourcefulness

Making your opponent earn the point

YOU WILL LEARN TO

- Name the opponent's threat before choosing a move, every time.
- Choose from the full defensive menu - capture, move, defend, block, counter-attack, interpose with tempo.
- Prefer active defence and avoid trades that help the opponent, especially into lost pawn endings.
- Recognize saving resources - perpetual check, stalemate and simple fortresses.
- Keep creating practical chances when behind, and decide honestly when to resign.

> IN YOUR NEXT GAME
>
> Every time your opponent moves, ask "What does this threaten?" and, when you are worse, look for checks, counter-threats and drawing resources before you defend passively.

You are a piece down in move 25. Your opponent's rooks are coming, and your first instinct is to trade everything and hope, or to resign. Five moves later the game is over.

At the 800–1000 level, games are rarely won by perfect play. They are won by the player who makes the second-to-last mistake. That means defence matters as much as attack: every move you make your opponent find is another chance for them to go wrong. Good defenders are not passive. They spot threats early, answer them with the most useful move, avoid trades that help the opponent, and know the saving tricks of chess: perpetual check, stalemate and the fortress. This chapter teaches that toolkit and the attitude that goes with it.

Step one: name the threat

You cannot defend against a threat you have not seen. The safety scan from Chapter 4 and the move-selection routine from Chapter 14 both start with the same question, and it is the heart of defence: **after my opponent's move, what does it threaten?**

Ask it in a fixed order:

1. Does the move give check, or prepare a check?
2. Does it capture, or threaten to capture, something of mine?
3. Does it threaten mate or a tactic (fork, pin, discovered attack)?
4. What did the moved piece stop defending, and what new lines did it open?

A **threat** is a move your opponent wants to play next that would hurt you. Some threats are loud (an attacked queen), some are quiet (a rook that will land on your back rank). Say the threat out loud in your head, in words: "He threatens Qxf7 mate." Players who name the threat almost never blunder into it; players who only "feel" it often do.

> **IN YOUR NEXT GAME** **Say the threat**
>
> In your next game, after every move by your opponent, finish the sentence "This threatens ..." before you look at your own ideas. If the answer is "nothing", say that too.

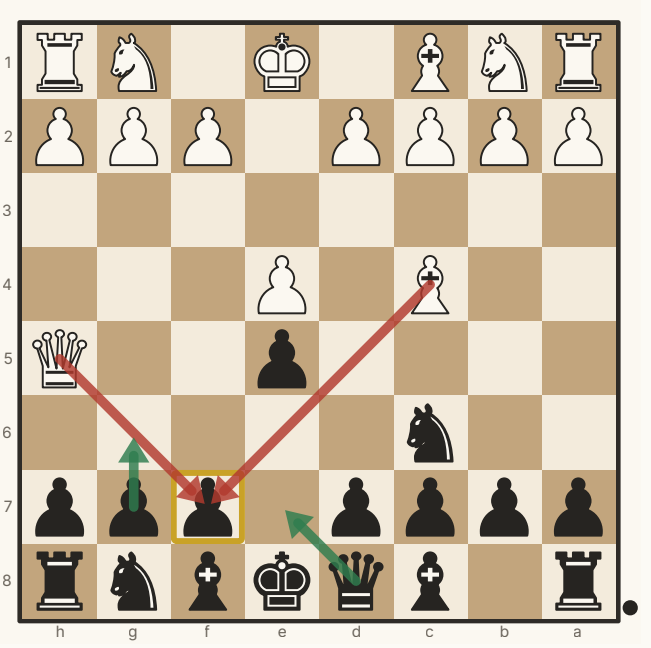
The defensive menu

Once the threat is named, you have a menu of possible answers. Beginners usually see only one item on it (move the attacked piece away). Strong defenders check all of them and pick the best.

1. **Capture the attacker.** If the attacking piece can be taken safely, that is often the strongest answer, because it removes the threat and wins material.
2. **Move the target.** Step the attacked piece to a safe and, ideally, useful square.
3. **Defend the target.** Add a defender so that the capture loses material for your opponent.
4. **Block the line.** Put something between a long-range attacker and its target.
5. **Counter-attack.** Make a threat of your own that is at least as big as theirs.

6. **Interpose with tempo.** Block or defend with a move that also attacks something, so your opponent has to spend a move answering you.

Check them in roughly this order, captures and forcing moves first, as you would in any candidate-move search. Then compare the candidates by asking which one leaves your position best.



White threatens Qxf7#. Black has several good answers: ...g6 blocks the queen's line and attacks it, ...Qe7 or ...Qf6 defends f7, ...Nh6 also defends. The natural developing move ...Nf6?? ignores the threat.

Each menu item is available here. The best is the one that does two jobs: 3...g6 blocks the h5–f7 line *and* attacks the queen, so White must move it again. That is interposing with tempo.

1. **e4 e5** 2. **Bc4 Nc6** 3. **Qh5 g6** 4. **Qf3 Nf6**

After 3...g6 4.Qf3 the queen renews the threat on f7, and 4...Nf6 blocks again while developing. Black is already comfortable.

1. **e4 e5** 2. **Bc4 Nc6** 3. **Qh5 Nf6??** 4. **Qxf7#**

The classic Scholar's Mate. 3...Nf6 attacks the queen but ignores the bigger threat.

Interposing with tempo

The last item on the menu deserves extra attention because it is so efficient. When a long-range piece (queen, rook or bishop) attacks along a line, look for a blocking move that also creates a threat. A pawn that blocks and attacks the checking piece, a knight that interposes and forks, a bishop that blocks a check and pins: each of these defends and wins a tempo in the same move. The same idea works against checks. When you are checked, the three answers are to move the king, capture the checking piece, or interpose. Before you move the king, always look for a capture or a blocking move that also attacks something.

The size of a counter-attack

A counter-attack only works if your threat is at least as big as the one you are ignoring. A threat to win a queen does not help if your opponent can mate you first. A threat to win a knight does not help if your opponent wins your queen. Before you choose a counter-attack, compare the two threats honestly: whose threat lands first, and what is it worth?

> COMMON MISTAKE **The irrelevant counter-threat**
>
> A common pattern at this level: your opponent threatens mate, you see a nice attack on their queen, and you play it. They give mate. Always ask "Is my threat bigger, or does theirs come first?" Mate is bigger than everything; check is faster than everything.

When two pieces are attacked

Sometimes the threat is a double attack: a fork or a pawn hitting two of your pieces at once. The menu still applies, but two items become especially important.

- **The desperado.** If one of the attacked pieces is lost anyway, let it capture something on its way out, ideally with check. Your opponent must recapture, and you gain the time to save the other piece.
- **The in-between move.** Before you recapture or retreat, ask whether you have a check or a bigger threat first (a **zwischenzug**, German for "in-between move"). Forcing moves change the order of events, and the order of events decides who wins material.

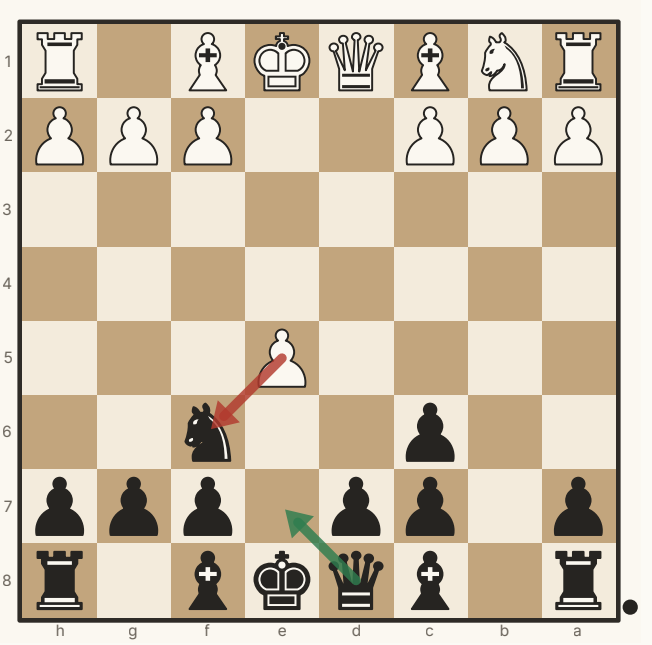
Both ideas come from the same habit: before you accept that you must lose something, look at every forcing move you have.

Defend actively

When two defences both hold, prefer the active one: the move that keeps your pieces working, creates counterplay or gains time. Passive defence (pieces huddled around a weakness) tends to get worse with every move, because the attacker can keep adding pressure while you have nothing to do.

Worked example: a knight under attack

This position comes from the Scotch Game, which you may play as Black (Chapter 13). White has just played 6.e5, attacking the knight on f6.



White's e-pawn attacks the knight. Retreat it, or find something more active? Notice the open e-file and the white king on e1.

Here is how to think it through.

First I name the threat. The pawn attacks my knight; next move it will take it. There is no check or mate threat.

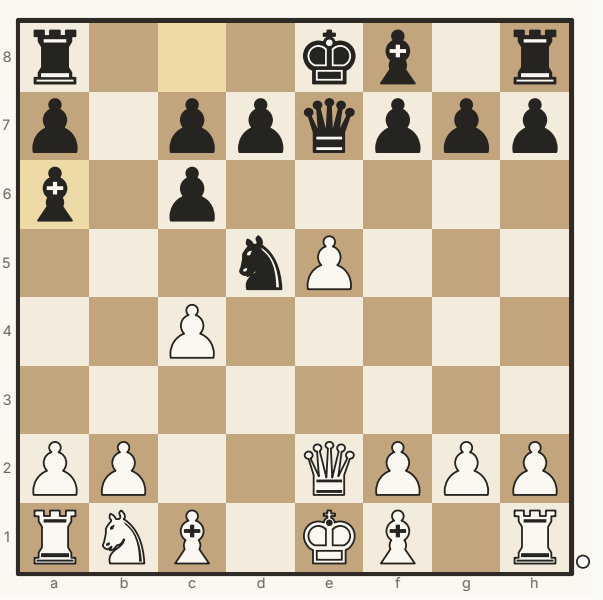
Then I list the menu. I cannot capture the attacker. Moving the knight gives four candidates: ...Nd5, ...Ne4, ...Ng8, ...Nh5. Defending the knight does not help, because the attacker is a pawn: if it takes, I lose a knight for a pawn anyway. Blocking is impossible. That leaves counter-attack.

Next I look for a counter-attack. The e5-pawn stands on the e-file in front of the white king. If I put my queen on e7, the pawn is pinned: exf6 would expose the king to check, so it is illegal. The queen also attacks e5 itself.

Now I check White's best reply. White must defend the pawn and the king's file. The usual answer is 7.Qe2, blocking the file. Then my knight can retreat to d5 at leisure, and my queen already stands on an active square.

Finally I compare. The retreat 6...Nd5 is also playable, but 6...Qe7 gains a tempo and keeps more options. The passive 6...Ng8 just loses time.

1. **e4 e5** 2. **Nf3 Nc6** 3. **d4 exd4** 4. **Nxd4 Nf6** 5. **Nxc6 bxc6** 6. **e5 Qe7** 7. **Qe2 Nd5** 8. **c4 Ba6**



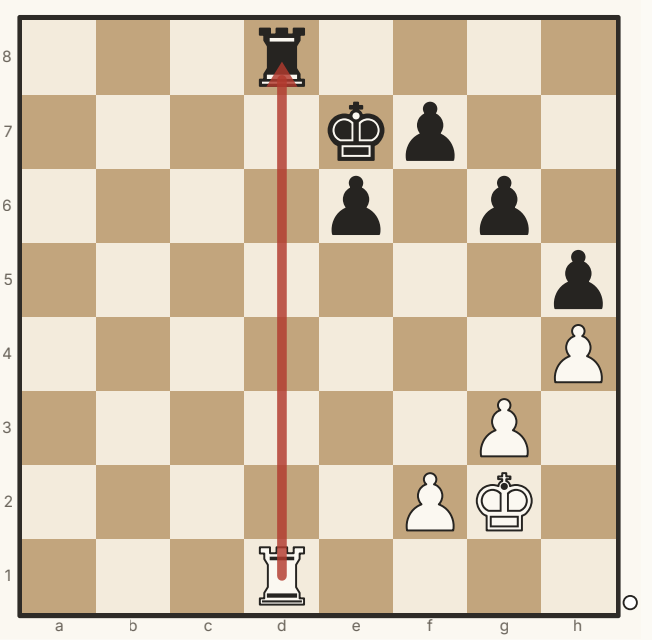
Black's pieces are active: the knight is safe, the bishop pins the c4-pawn to the queen, and White must keep watching the e-file.

The lesson is not this particular line. It is the routine: name the threat, run the whole menu, look for a counter-attack, check the opponent's best reply, and then compare.

Do not panic-trade

When you are under pressure, trading pieces feels like relief. Sometimes it is: trading your opponent's most dangerous attacking piece is often the best defence of all. But trades are not automatically good for the defender. Ask what each trade does.

- **Trades that help you:** removing an attacker near your king, removing the piece that creates a threat, or reaching an ending your opponent cannot win (for example a known draw).
- **Trades that help your opponent:** any trade when you are behind in material (the side ahead wants simplification, as you saw in Chapter 16), trading your active pieces for their passive ones, and above all trading into a lost pawn ending.



White is a pawn down. With rooks on, an active rook gives White a comfortable draw. After 1.Rxd8? Kxd8 the pawn ending is lost: Black's king and extra pawn decide.

In rook endings, the defender's rook is usually the reason the position holds. A pawn down with rooks on, you have checks, counterattacks on pawns and the chance to cut the enemy king off. In the pawn ending you have none of these. Chapter 18 shows how to calculate pawn endings; the defensive rule is simple:

never trade into a pawn ending you have not calculated, and especially not when you are behind.

The same logic applies in the middlegame. If your opponent is attacking your king, trading their most active attacker (often the queen) is excellent defence, even at a small cost. If instead your opponent is a piece up and offers to trade queens, that trade helps them, because your best chance was counterplay with the queen. So before every trade, ask one question: "After this exchange, whose pieces are left, and who is happier?"

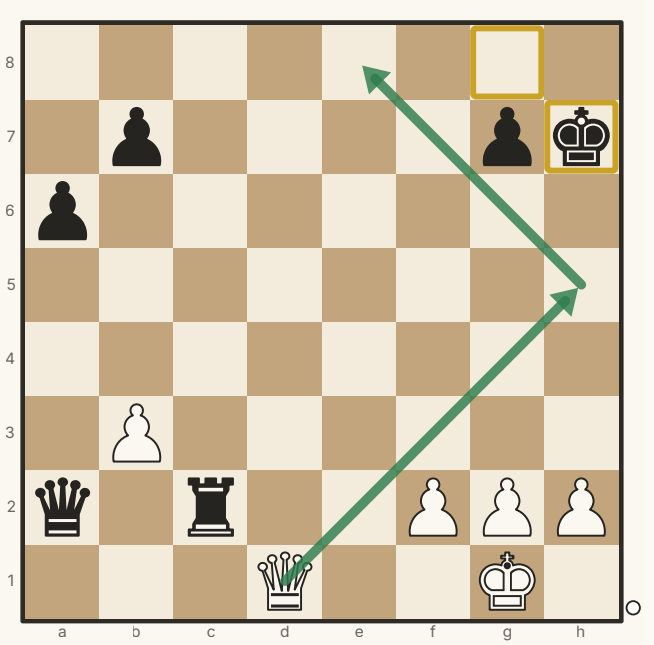
> KEY IDEA **When simplification helps the opponent**
>
> Trades help the side with more material, the side with the better pawn structure, and the side whose king is more active. If that is your opponent, keep pieces on and look for activity.

Saving resources

Sometimes the position is objectively lost, but chess has a few escape hatches. They come up in real games far more often than you might think, mostly because the winning side relaxes.

Perpetual check

Perpetual check is a series of checks the opponent cannot escape. The game is drawn, usually by threefold repetition. You look for it when the enemy king has few safe squares and your queen (sometimes with a rook or knight) can reach checking squares that cover them all.



White is a rook down and Black threatens ...Rc1. But the black king has only h7 and g8. 1.Qh5+ Kg8 2.Qe8+ Kh7 3.Qh5+ is perpetual check.

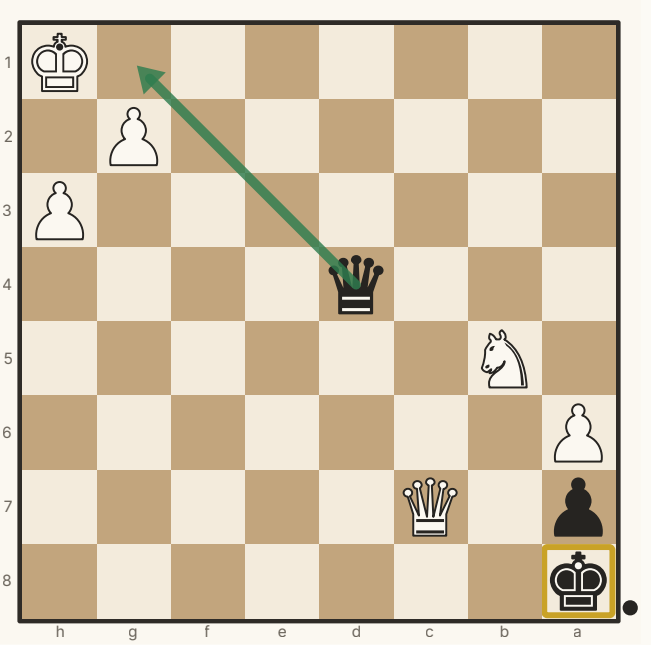
1. **Qh5+ Kg8** 2. **Qe8+ Kh7** 3. **Qh5+ Kg8** 4. **Qe8+**

Black cannot block on h6 or f8 and cannot run. The position repeats and the game is drawn.

How do you find perpetuals? When you are worse, scan for checks every move, especially against a king whose pawn shield has been broken. List the enemy king's flight squares and ask whether two or three checking squares cover them all. Then check that the opponent cannot block a check with a piece. In the diagram, the tempting 1.Qd3+ also checks, but after 1...g6 the black king walks out via h6 and g5, and the perpetual is gone.

Stalemate

Stalemate (no legal move, not in check) is a draw. When your king has no moves and your pawns are blocked, every remaining piece you own is a possible stalemate weapon: if you can force your opponent to capture it, the game is drawn.

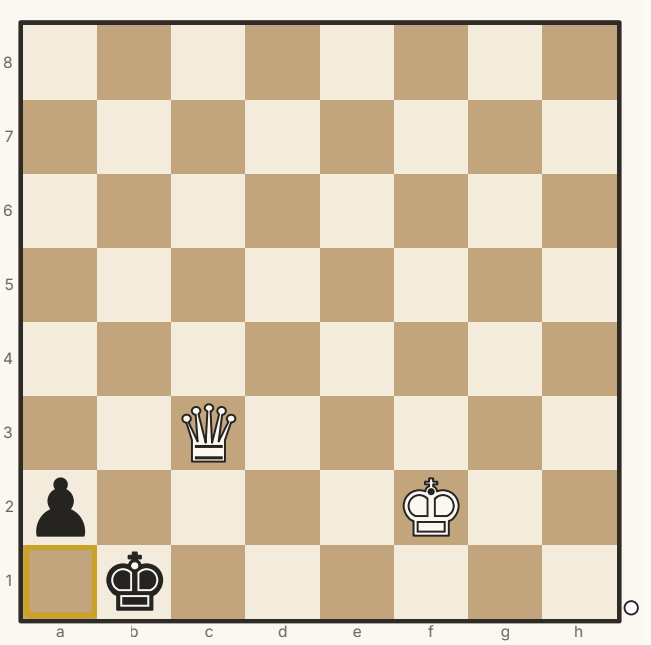


Black is lost: White threatens Qb7# or Qb8#. But the black king has no legal move and the a7-pawn is blocked. 1...Qg1+! forces 2.Kxg1, and Black is stalemated.

1... **Qg1+** 2. **Kxg1**

Stalemate. Black also has other checks that draw here, but the queen sacrifice ends the game at once.

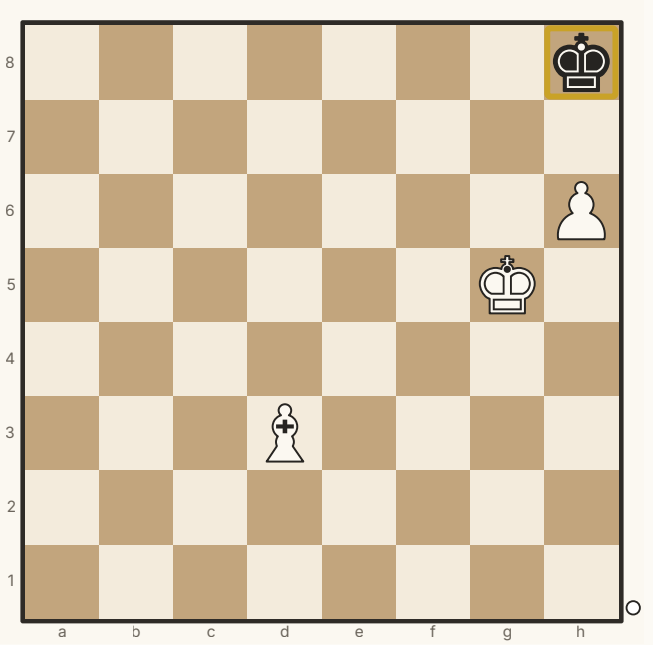
Stalemate is also the main defensive idea in many endings. One classic case: a queen against a pawn on the seventh rank on the a-, c-, f- or h-file is often a draw, because the defending king can hide in front of or beside the pawn where the attacker can only stalemate it.



Queen against a rook pawn on the seventh. If White checks, Black goes to a1; then any quiet approach by the white king is answered by stalemate. With the white king this far away, it is a draw.

The fortress

A **fortress** is a defensive set-up the opponent cannot break, even with extra material. At this level, learn one simple fortress: a rook pawn with the "wrong" bishop.



White has an extra bishop and pawn, but the h-pawn queens on h8, a dark square, and the bishop is light-squared. The black king sits in the corner and can never be driven out. It is a draw.

The defender's job is simply to get the king to the corner square (or next to it) in time. If you are ever a piece down with only a rook pawn left for your opponent, check the colour of their bishop and the queening square: you may be able to draw by running to the corner.

> **EVIDENCE** **Theory note**
>
> The wrong-coloured bishop and rook pawn draw, and the queen against rook or bishop pawn on the seventh, are standard results covered in every endgame manual, for example Jeremy Silman, *Silman's Complete Endgame Course* (2007).

When you are behind

Being worse is not the same as being lost. The practical question is: how do I make winning as hard as possible for my opponent?

- **Seek activity.** An active piece is worth more than a pawn when you are defending. Put rooks on open files and behind passed pawns, and centralize your king in the endgame.
- **Create threats.** Every threat you make forces your opponent to stop and think, and gives them a chance to miscalculate. A passed pawn, an attack on a loose piece, or a check that drives the king to a worse square all count.
- **Keep pieces on.** As we saw, simplification helps the side that is ahead. Avoid trades unless they kill a direct attack or reach a known draw.
- **Look for your own tactics.** Run the safety scan for *your* side too: checks, captures and threats. Players who are winning often stop defending, and a resource such as a fork, a back-rank trick or a perpetual appears.
- **Avoid passive resignation of the mind.** Many games are lost not when the position becomes hopeless but when the defender stops looking. Keep using the full routine on every move.
- **Use the clock wisely.** Spend time on the critical defensive moments, not on obvious recaptures.

A practical example of the mindset: you are a rook down in a queen ending. Instead of trading queens (which ends your chances), keep your queen as active as possible near the enemy king, check whenever the king is exposed, and look for stalemate ideas if your own king is boxed in. Your opponent has to avoid all of that while also making progress. Many players with an extra rook cannot do both, and the game ends in a draw by perpetual check or a blunder.

> COACHING JUDGMENT **Coaching judgment**
>
> In my experience at club level, a determined defender saves a large share of "lost" positions, because the opponent relaxes, rushes or tries to win too fast. Your job is to keep asking questions they must answer correctly.

Finding defensive moments in your own games

When you analyze your games (the full method comes in Chapter 28), mark three kinds of defensive moments.

- **Missed threats.** The move where your opponent's threat appeared and you did not name it. Write down what the threat was and which item on the menu would have met it.
- **Wrong choice from the menu.** You saw the threat but chose a passive answer when an active one existed, or a counter-attack that was too small.
- **Missed resources.** In games you lost, go back to the moment the position became bad and ask an engine whether a perpetual, a stalemate trick, a fortress or simply a more stubborn defence was available.

Record these under "king-safety", "missed-tactic" or "endgame" in your mistake journal. After a dozen games you will see your personal pattern, and that pattern tells you what to drill. Many players discover that they defend well when they are calm but stop scanning for threats the moment they get ahead, which is exactly when opponents start to swindle them.

When to resign

Resigning is part of chess etiquette, and there is no shame in it. But at the 800–1000 level the honest advice is: **resign rarely.** Positions that look hopeless to you are often not yet won for your opponent, and the experience of defending a bad position is itself valuable training.

Some guidelines (coaching judgment, not rules):

- If your opponent still has to show technique (for example, convert a piece-up middlegame), play on and make them prove it.
- If you are down a queen or more with no counterplay, and your opponent has shown they can deliver checkmate, resigning is reasonable and polite.
- Do not play on purely to waste your opponent's time, and do not wait for an obvious mate in one to be played. Resigning a clearly finished game is a courtesy.
- In casual or online games, use lost positions as training: try to find the most stubborn defence, then resign once nothing is left.

Whatever you decide, decide with a clear head. Never resign because you are annoyed by a mistake you just made; take a breath, look for resources, and then judge the position.

Practice at the board

Defence is trained by defending. Set up these positions against a friend or an engine and play the defending side until you can hold or swindle consistently.

- **The menu.** Set up `r1bqkbnr/pppp1ppp/2n5/4p2Q/2B1P3/8/PPPP1PPP/RNB1K1NR b KQkq - 3 3`. Write down every legal defence of f7 you can find, then check them with an engine. Which ones also develop or gain a tempo?
- **Perpetual.** Set up `8/1p4pk/p7/8/8/1P6/q1r2PPP/3Q2K1 w - - 0 1` and play White against an engine. Find the perpetual; then try other first moves and see how quickly you lose.
- **Rook ending vs pawn ending.** Set up `3r4/4kp2/4p1p1/7p/7P/6P1/5PK1/3R4 w - - 0 1`. Play it twice as White against an engine: once trading rooks, once keeping them. Compare the results.
- **Fortress.** Set up `7k/8/7P/6K1/8/3B4/8/8 w - - 0 1` and try to win as White. Then move the black king to e7 and see whether it can still reach the corner in time.
- **Defend a lost position.** Take a game where you were winning, and play the losing side from the critical moment against an engine. Count how many moves you can survive and note every resource you find.

Check yourself afterwards with the engine: for each defensive move you chose, did a better item on the menu exist?

Chapter summary

- Defence starts with naming the threat in words after every opponent move.
- Run the whole defensive menu: capture the attacker, move, defend, block, counter-attack, interpose with tempo.
- A counter-attack must be at least as big and as fast as the threat it ignores.
- Prefer active defence; passive positions usually get worse.
- Trades help the side that is ahead; never trade into a pawn ending you have not calculated.

- Perpetual check, stalemate and simple fortresses save many lost positions; look for them whenever you are worse.
- When behind, seek activity, create threats and keep pieces on; resign rarely, but courteously when nothing is left.

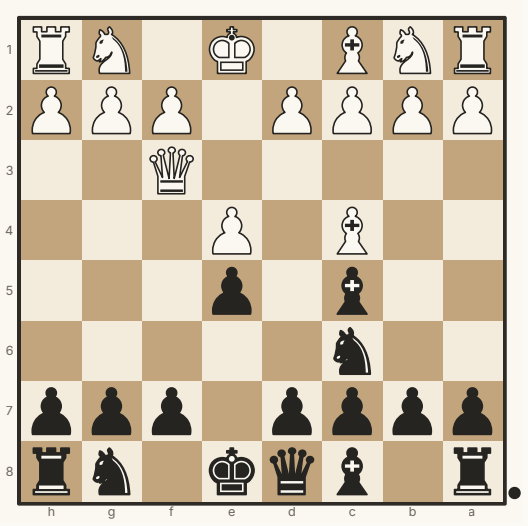
CHEAT CODE **Defensive routine**

1. Name the threat: checks, captures, mate or tactics, and what the last move stopped defending.
2. List the menu: capture the attacker, move, defend, block, counter-attack, interpose with tempo.
3. Compare: is my counter-threat bigger and faster? Which defence also improves my position?
4. Before any trade, ask who the resulting position helps; calculate pawn endings first.
5. If you are losing, look for checks (perpetual), stalemate ideas and fortresses before playing on passively.

Exercises

Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

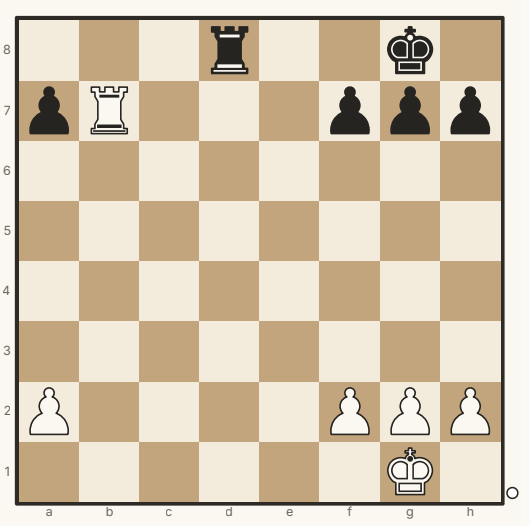
17.1 DEFENCE



Black to move. White's queen and bishop both aim at f7. What is the best response?

Solution p. 685

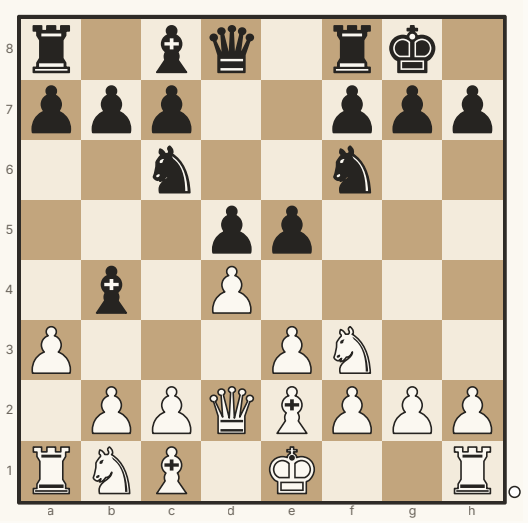
17.2 DEFENCE



White to move. The a7-pawn is hanging. Is it safe to take? Play the right move.

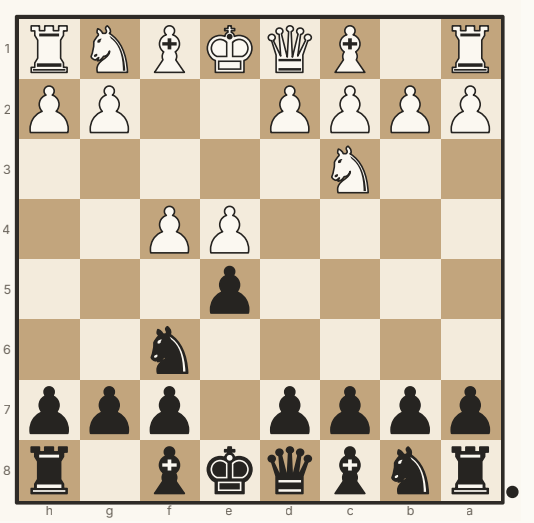
Solution p. 685

17.3 DEFENCE



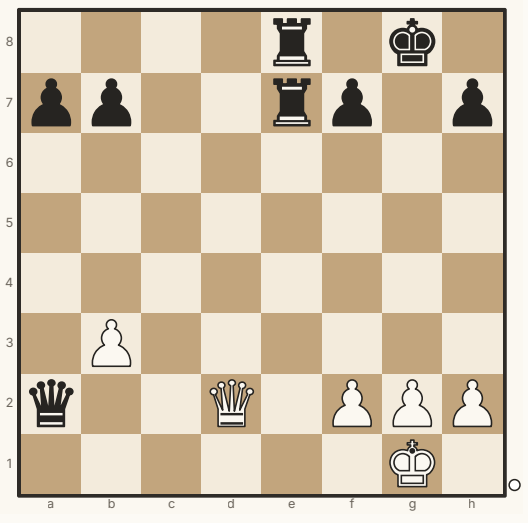
White to move. Your opponent just played ...Bb4, attacking your queen. What is the best response? Solution p. 685

17.4 DEFENCE



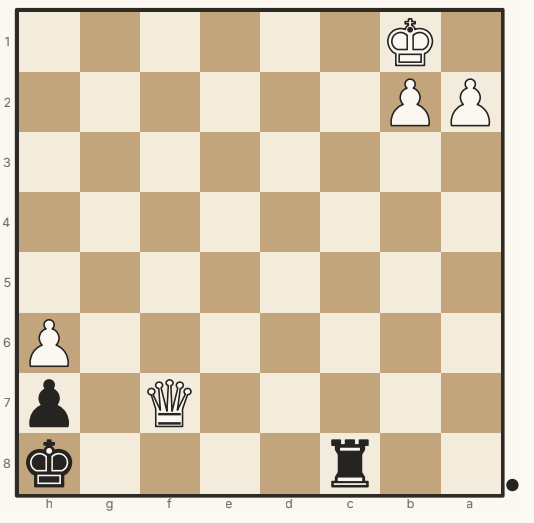
Black to move. In the Vienna Gambit (1.e4 e5 2.Nc3 Nf6 3.f4), White's f-pawn attacks e5, and fxe5 would also hit your knight. What is the best response? Solution p. 685

17.5 DEFENCE



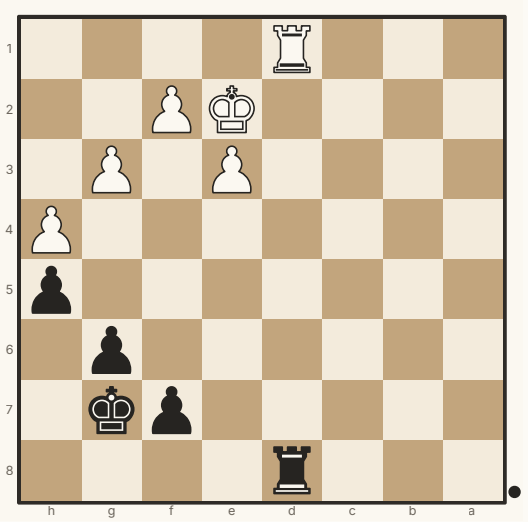
White to move. You are two rooks down and Black threatens to finish you off. Which move saves the game? Solution p. 686

17.6 DEFENCE



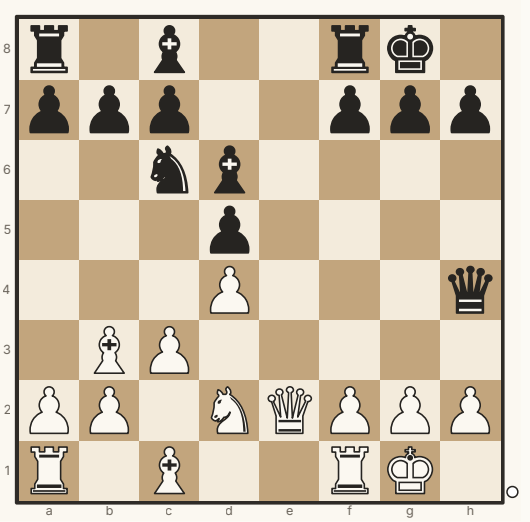
Black to move. White threatens Qg7#. Which move saves the game? Solution p. 686

17.7 DEFENCE



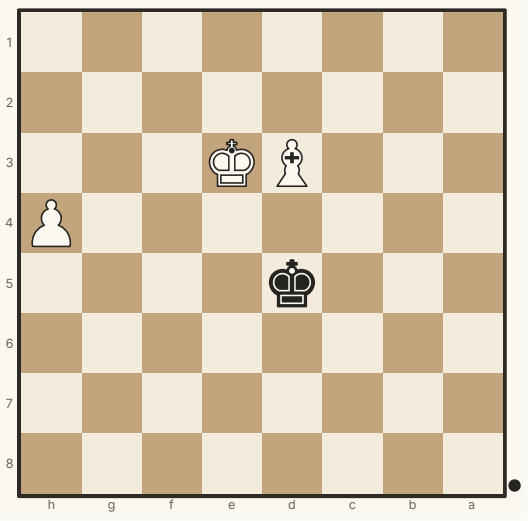
Black to move. You are a pawn down. White offers a rook trade on the d-file. What is the best move? Solution p. 686

17.8 DEFENCE



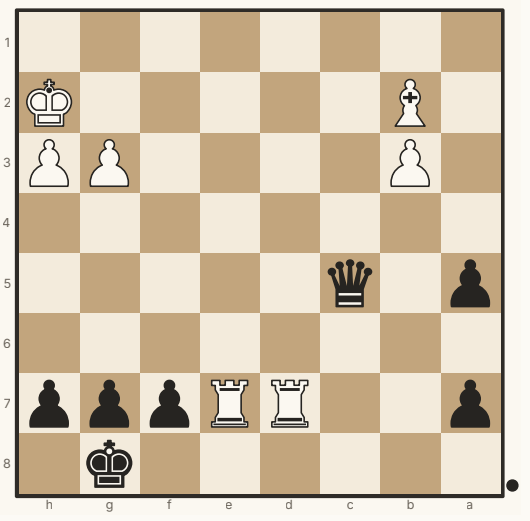
White to move. Black threatens Qxh2#. Several moves defend. Find the strongest.
Solution p. 686

17.9 DEFENCE



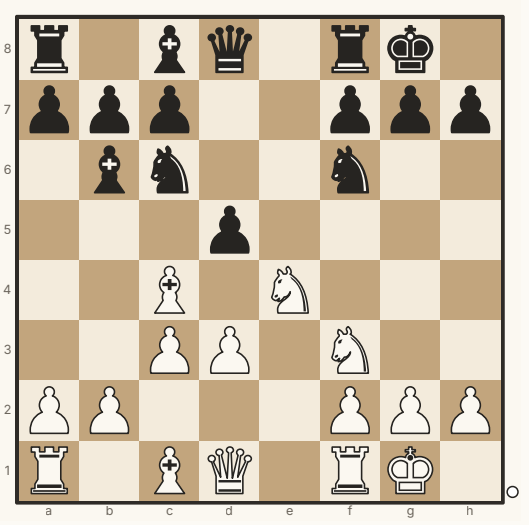
Black to move. White has a bishop and an h-pawn. Find the only move that draws.
Solution p. 687

17.10 DEFENCE



Black to move. White's rooks on the seventh rank are about to win your f7-pawn and more. Which move saves the game?
Solution p. 687

17.11 DEFENCE



White to move. Black's ...d5 attacks both your bishop and your knight. Find the only good move. Solution p. 687

HINTS

17.1 Name White's threat before you look at your own ideas.

17.2 What does Black's rook threaten?

17.3 Captures first: what can take the attacker?

17.4 Which white pawn can you attack in return?

17.5 Checks first. Where can the black king go?

17.6 Your king cannot move. Can you make your rook disappear?

17.7 Which ending is easier to defend: rook ending or pawn ending?

17.8 Defend with a move that also attacks.

17.9 Which corner square can the bishop never attack?

17.10 The white king has very few squares. Checks!

17.11 Can one attacked piece take something with check?

CHAPTER 18 STAGE 800–1000

King and Pawn Endgames

Every tempo counts

YOU WILL LEARN TO

- Use the rule of the square to judge pawn races instantly.
- Win and defend king and pawn against king with key squares and the opposition.
- Recognize the rook-pawn draw, outflanking, and the idea of losing a move.
- Use outside and protected passed pawns, and execute the classic pawn breakthrough.
- Count tempi in pawn races and calculate before trading into a pawn ending.

> IN YOUR NEXT GAME
>
> Before you trade the last pieces, count the pawn ending that follows; when you are in one, put your king in front of your pawns and ask who has the opposition.

The rooks come off, the last minor pieces disappear, and suddenly only kings and pawns are left. Many players at this level relax here: “It’s just pawns.” Then they push the wrong pawn, walk the king to the wrong square, and a winning position turns into a draw, or a draw into a loss.

King and pawn endings are the most concrete part of chess. There are no hidden tactics with knights and no long-range bishops, and the number of moves is small enough that you can often calculate to the end. That is exactly why they are unforgiving: a single wasted tempo usually decides the game. The good news is that a handful of ideas (the square, the opposition, key squares, and counting) cover most of what you will meet in your games. This chapter teaches those ideas and the habits to use them.

Why pawn endings are different

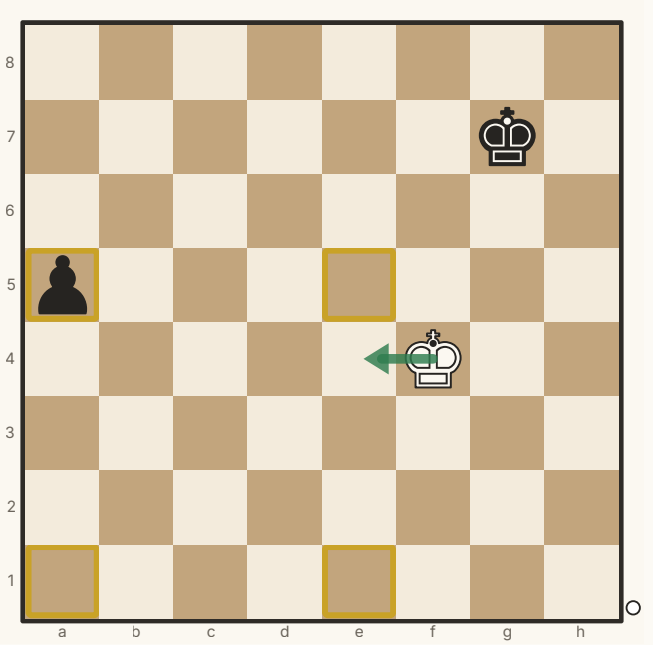
In a pawn ending the king is your strongest piece. It attacks pawns, escorts its own pawns and blocks enemy ones. The king that gets to the centre first, or into the enemy pawns first, usually wins.

The second difference is that **tempo** (a single move of time) matters more than anywhere else. In a middlegame, wasting a move is a small inaccuracy. In a pawn ending, having to move can be fatal. The situation where any move worsens your position is called **zugzwang**, a German word meaning "compulsion to move". Many pawn endings are won by putting the opponent in zugzwang.

The third difference is permanence. Once you trade into a pawn ending, there is no way back: you cannot bring a rook back to save you. That is why strong players treat the decision to enter a pawn ending as a calculation, not a reflex.

The rule of the square

The first skill is judging whether a king can catch a passed pawn. Counting moves works, but there is a faster tool. Draw an imaginary square: one side runs from the pawn to its promotion square, and the square extends sideways towards the defending king. If the defending king is inside the square, or can step into it with the move, it catches the pawn. If not, the pawn queens.



The square of the a5-pawn is a5-a1-e1-e5. White to move steps into it with Ke4 and catches the pawn; with Black to move, ... a4 shrinks the square and the white king is too late.

With White to move, the king walks straight into the pawn's path:

1. **Ke4 a4** 2. **Kd3 a3** 3. **Kc2 a2** 4. **Kb2**

The king stays inside the shrinking square all the way and captures the pawn on a2 next.

Three details cause most mistakes with the square:

- **A pawn on its starting rank** can move two squares at once. Draw the square as if the pawn were already one rank further forward.
- **Obstacles.** If your own pawns stand in the king's path, the geometry lies. The king has to walk around them.
- **Support.** The square only applies to a pawn running alone. If the enemy king can escort its pawn, you are no longer in a race but in a king-and-pawn fight, which the rest of this chapter covers.

> KEY IDEA **The square in one sentence**
>
> If your king is inside the square of the pawn (counting the move), it catches the pawn; if it is outside, it does not. Check it before every pawn race.

King and pawn against king

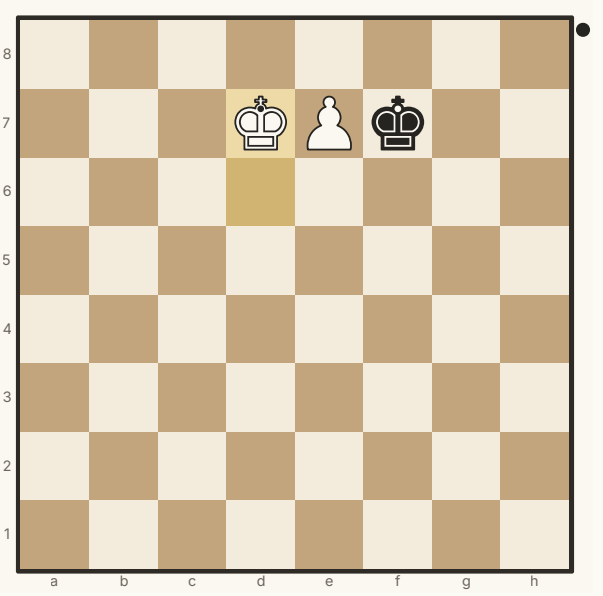
This is the basic ending that all others reduce to. One side has a king and a single pawn; the other has only a king. The result depends on where the kings stand and whose move it is. Master it and you will know, many moves ahead, whether trading into a pawn ending wins.

The king leads, the pawn follows

The most common beginner mistake is pushing the pawn and hoping. Pawns cannot protect themselves, and a pawn that runs ahead of its king is usually blockaded and lost. The right method is the opposite: your king goes in front of the pawn and clears the way.

A useful guideline: with any pawn except a rook pawn, if your king stands on the sixth rank directly in front of its pawn, you win no matter who is to move.

1. **Kd6 Kd8** 2. **e6 Ke8** 3. **e7 Kf7** 4. **Kd7**

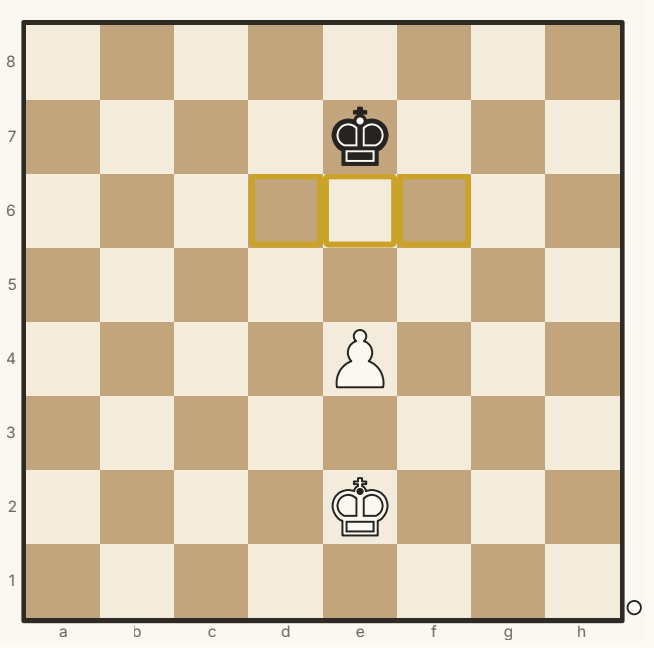


The king on the sixth controls the promotion path. After 4.Kd7 nothing stops e8=Q.

Key squares

Key squares are squares that, once your king reaches one of them, guarantee the win regardless of who moves. For a pawn on the second, third or fourth rank

(not a rook pawn), the key squares are the three squares two ranks directly in front of the pawn. For a pawn on e4 they are d6, e6 and f6.



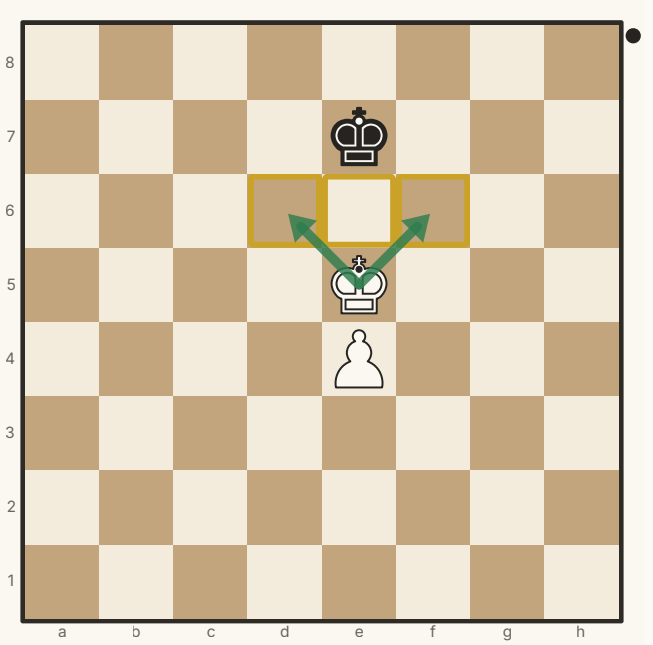
The key squares of the e4-pawn are d6, e6 and f6. White's task is to get the king to one of them; Black's task is to prevent it.

The rule above covers pawns on their second, third and fourth ranks, which is where most of your real endings start. As the pawn advances further, the key squares move forward with it and become easier for the attacker to reach, but the method stays the same: name the squares your king needs, then fight for them.

Thinking in key squares simplifies everything. You stop asking "can I push the pawn?" and start asking "can my king reach a key square?" The defender asks the mirror question: "can I keep the enemy king away from those squares?"

The opposition

The main tool for fighting over key squares is the **opposition**. Two kings stand in opposition when they face each other on the same file (or rank) with one square between them. The side that does *not* have to move "has the opposition": the other king must give way.



The kings are in opposition. With Black to move, Black must step aside and White's king reaches a key square. With White to move, the same position is only a draw.

With Black to move, every king move gives up one of the key squares:

1...**Kd7** 2.**Kf6 Kd8** 3.**Kf7 Kc7** 4.**e5**

Black stepped left, so White's king went right to f6 and then f7, controlling the pawn's path.

With White to move in the diagram, White can only shuffle; if 1.Kd5 Kd7 or 1.Kf5 Kf7, Black keeps facing the white king and the pawn never gets through. Always notice whose move it is: in pawn endings the same position can be a win or a draw depending on that alone.

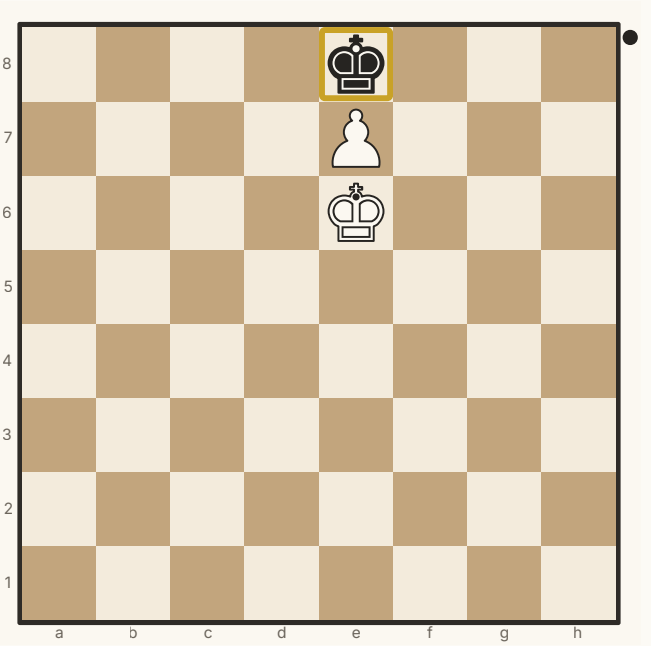
Defending: stay in front

The defender's recipe is simple to say: keep your king directly in front of the pawn, and when the enemy king approaches, take the opposition. When you are pushed back, retreat straight backwards (for example from e7 to e8), not to the side, so that you can take the opposition again when the white king advances.

Here is the drawn version of the opposition diagram, with White to move. Watch how Black always answers the white king by facing it.

1. **Kd5 Kd7** 2. **e5 Ke7** 3. **e6 Ke8** 4. **Kd6 Kd8** 5. **e7+ Ke8** 6. **Ke6**

Black kept the opposition every time the white king stepped forward. The final position is stalemate.



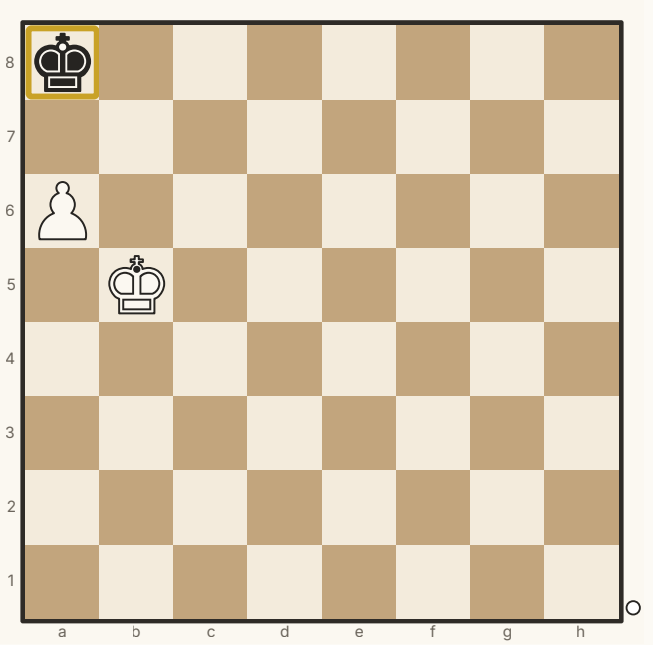
The defender's goal. Black to move has no legal move: stalemate. If White had played anything else on move 6, the black king would have captured the pawn.

This final picture is worth remembering. If you can reach it as the defender, you have drawn. If you are the attacker and you see it coming, you went wrong earlier: usually you pushed the pawn before the king had reached a key square.

> **COMMON MISTAKE** **Pushing the pawn too early**
>
> When the attacking king is beside or behind its pawn, pushing usually throws away the spare tempo you need later for the opposition. Bring the king forward first. A pawn on its second or third rank is a reserve of waiting moves; do not spend them until you need them.

The rook-pawn exception

Rook pawns (a- and h-pawns) are special. If the defending king reaches the corner in front of the pawn, the attacker cannot drive it out: the only way to force it out of the corner would stalemate it.



Black's king sits on a8. White can try Kb6, but after ...Kb8 and a7+ Ka8, any further king move either releases Black or gives stalemate. It is a draw.

So when you are down a pawn in a pawn ending, it is often worth trading so that your opponent is left with only a rook pawn. And when you are the one ahead, keep the pawn that is not on the edge.

Winning tricks: outflanking and losing a move

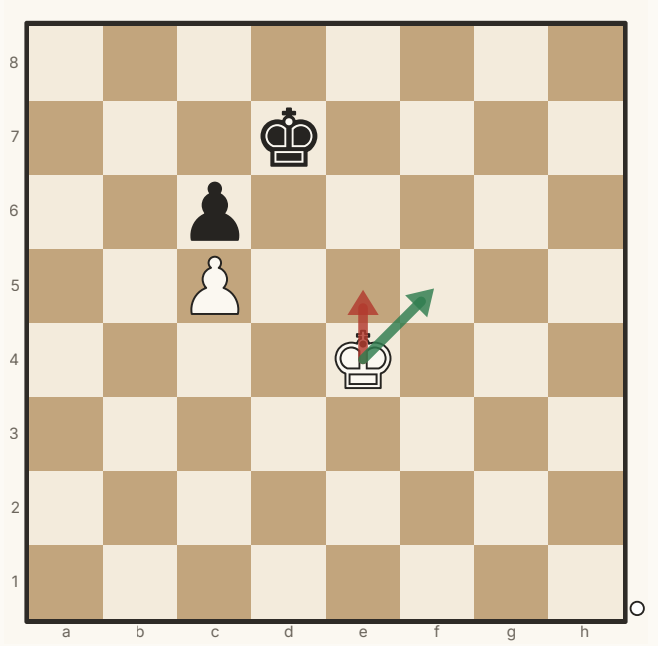
Real pawn endings usually have more than one pawn, and the kings fight over a target. Two ideas appear again and again.

Distant opposition and outflanking

Opposition also works at a distance: kings on the same file with three or five squares between them, with the other side to move, are in **distant opposition**. If you hold the distant opposition, you can usually turn it into the direct opposition as the kings approach.

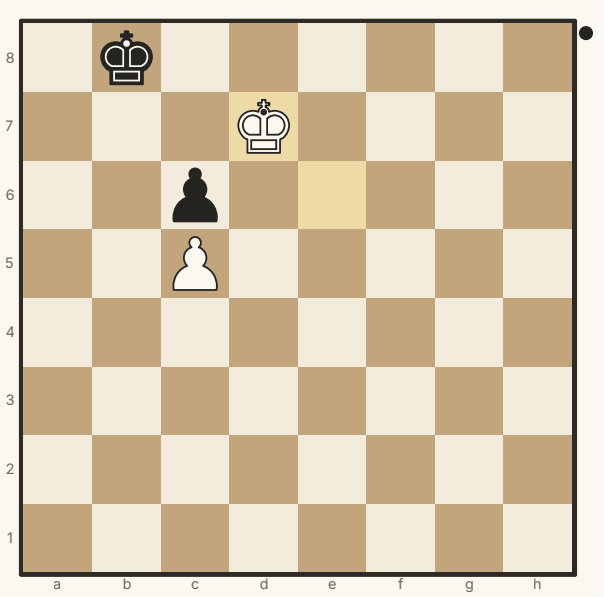
A quick way to check distant opposition on a file: count the squares between the kings. If the number is odd (one, three or five) and it is your opponent's move, you have the opposition. If it is even, or it is your move, you do not. The same counting works on ranks.

When direct confrontation does not work, the attacker goes around. **Outflanking** means using the opposition on one side to step past the enemy king on the other side. Here is a typical case with blocked pawns.



White wants the c6-pawn. The c5-pawn takes d6 and b6 away from Black's king. The direct 1.Ke5 Ke7 meets the opposition, but 1.Kf5! goes around.

1. **Kf5 Kc8** 2. **Ke5 Kc7** 3. **Ke6 Kb8** 4. **Kd7**



The white king reached e6 and d7. Next comes Kd6 or Kxc6, and the c-pawn falls.

If Black answers 1.Kf5 with 1...Ke7, then 2.Ke5! puts Black in zugzwang: the black king must leave, and the white king walks to d6. Notice the trick: White's king went e4–f5–e5, taking two moves to reach e5, so that it arrived there with Black to move.

Triangulation, simply

That last idea has a name. **Triangulation** means making a small detour (often around three squares in a triangle) to lose a move on purpose, so that a position you want arises with the *opponent* to move. It works when your king has room to manoeuvre and the enemy king does not, because it must stay on a small set of defending squares.

Why can one king lose a move and the other not? Usually because the defending king has only two good squares, and it must answer every attacking move by going to the matching square. The attacking king, with more room, can take a three-move route back to the same square while the defender can only shuttle between two. After the detour, the same position stands on the board, but with the defender to move.

You do not need to memorize triangulation positions at this stage. Remember the question it answers: "This position wins if it were my opponent's move. Can my

king waste a move, while theirs cannot?" When the answer is yes, look for a detour.

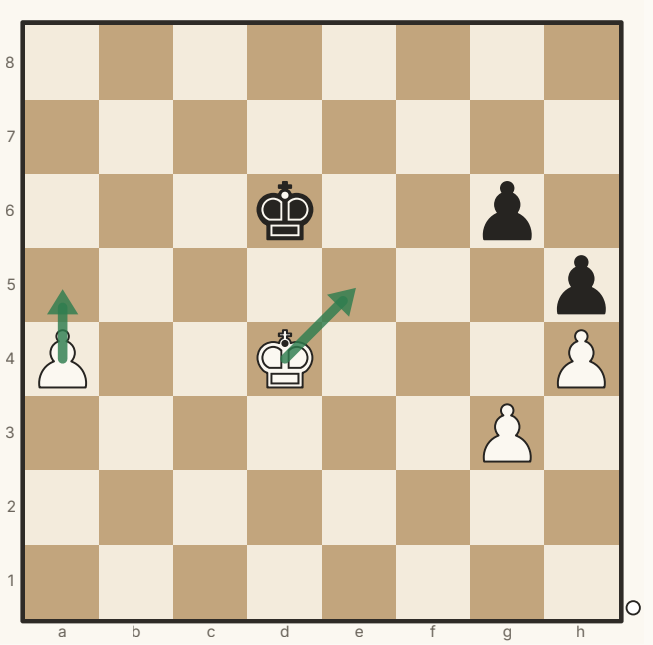
> COACHING JUDGMENT **Coaching judgment**
>
> Club players lose far more pawn endings by ignoring whose move it is than by any lack of theory. Before every move in a pawn ending, ask: "If I play this, who has the opposition?"

Passed pawns that win pawn endings

When both sides have several pawns, the endgame is decided by the quality of passed pawns. A **passed pawn** has no enemy pawns in front of it on its own or the adjacent files.

The outside passed pawn

An **outside passed pawn** is a passed pawn far away from the other pawns. Its strength is not that it queens by itself, but that it lures the enemy king away. While the defender's king runs to stop it, your king eats the pawns on the other side.



White's a-pawn is an outside passed pawn. White pushes it, Black's king must go after it, and the white king feasts on the kingside.

Worked example: thinking through the outside passer

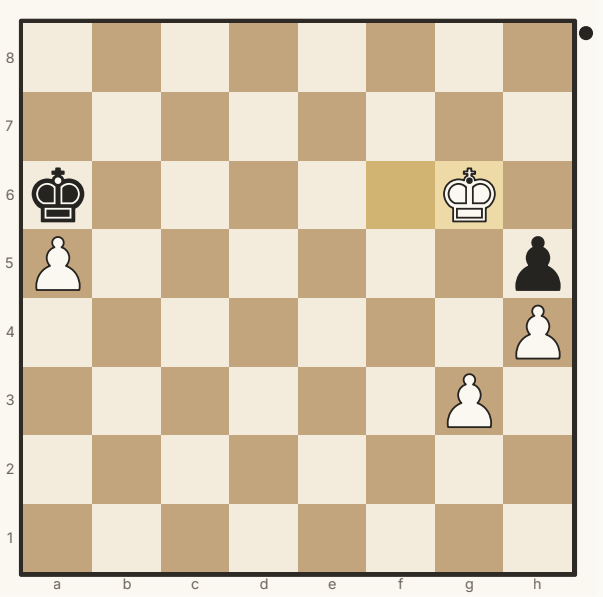
Let us run the thinking process on the diagram.

First I count material: White has three pawns, Black two. Next I look for the decisive plan, not a single move. The black king is the only defender for both the a-pawn and the kingside pawns, so it cannot do both jobs.

My candidate moves are 1.a5 (push the decoy) and 1.Ke4 (approach the kingside first). Both ideas look good, so I calculate the most forcing one, 1.a5. The natural reply is to go after the pawn: 1...Kc6. Now my king heads for the kingside: 2.Ke5 Kb5 3.Kf6. Black can grab the a-pawn, but it takes time, and the black king is far from the action.

Then I check the race at the end. After my king takes on g6, the h5-pawn falls next, and my h-pawn is much closer to queening than anything Black has. The plan works.

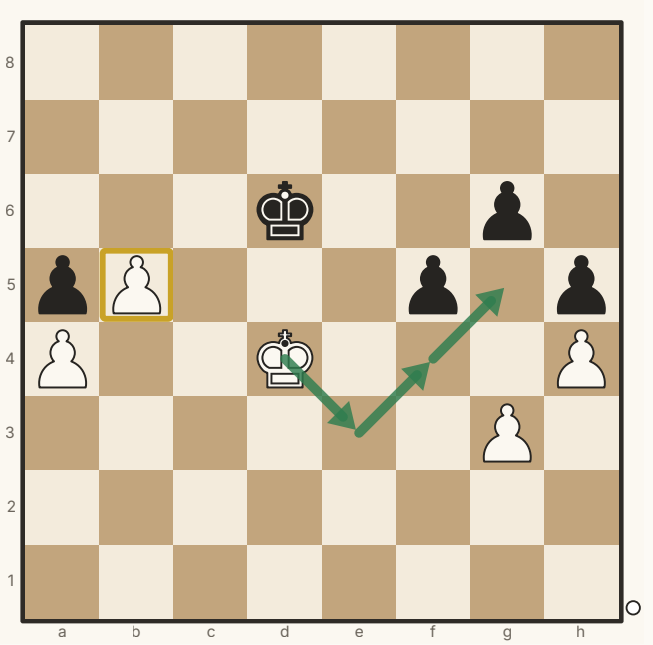
1.**a5 Kc6** 2.**Ke5 Kb5** 3.**Kf6 Ka6** 4.**Kxg6**



The decoy worked. White's king has entered Black's pawns and the h5-pawn falls next.

The protected passed pawn

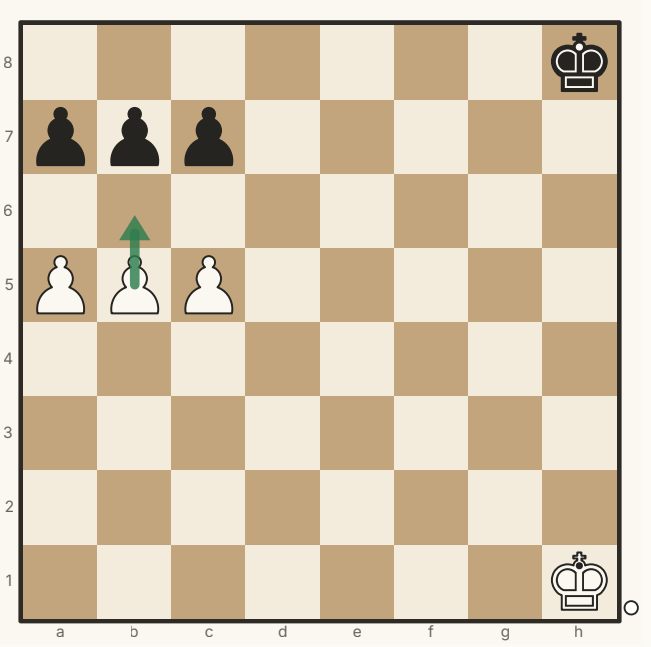
A **protected passed pawn** is a passed pawn defended by another pawn. The enemy king can never take it, so it must keep watching it forever. That ties the defender's king down while your king is free to go anywhere.



The b5-pawn is passed and protected by a4. Black's king must stay near it, so White's king walks to the kingside (Ke3, Kf4, Kg5) and wins the black pawns.

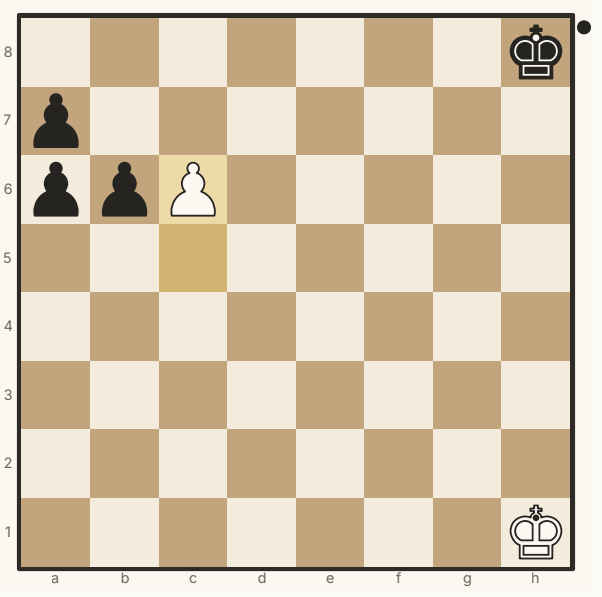
The pawn breakthrough

Sometimes a group of pawns can force a passed pawn by sacrifice. The classic case is three pawns against three, facing each other with one rank between them.



White to move wins with 1.b6!, a pawn sacrifice that forces open a path for one of White's pawns. The kings are too far away to interfere.

1. **b6 cxb6** 2. **a6 bxa6** 3. **c6**

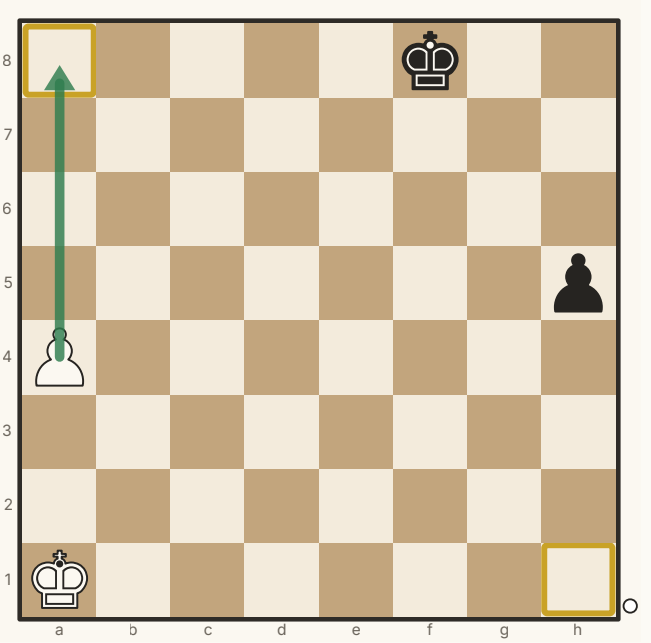


After 3.c6 the c-pawn runs to c8. If 1...axb6, then 2.c6! bxc6 3.a6 works the same way.

The pattern to remember: the middle pawn goes first, then the pawn on the side where the capture happened. Breakthroughs also work for the defender's opponent, so when your pawns face a pawn majority, check whether the enemy has one before you let the kings wander away.

Pawn races: counting tempi

When both sides have passed pawns and the kings cannot catch them, you are in a **pawn race**. Count, do not guess. For each pawn, count the moves to queen. Remember who moves first. Then look one step further: does the new queen give check, or control the other pawn's queening square?



Both pawns need four moves. White moves first, queens on a8 with check, and the new queen also covers h1 along the long diagonal.

1. **a5 h4?!** 2. **a6 h3** 3. **a7 h2** 4. **a8=Q+**

White queens with check, and after the king moves, Qxh1 or simply covering h1 stops the black pawn. Black's best was to give up the race, but the result is the same.

Counting races is also how you decide *whether* to race at all. Sometimes the right move is not to push your own pawn but to bring your king one step closer to the

enemy pawn first, entering its square, and only then run. Compare the two plans by counting both: how many moves until each side queens, and can either king catch the other pawn along the way? If both pawns queen, ask what the resulting queen ending looks like; an extra pawn or a queen that arrives with check usually decides it.

> **COMMON MISTAKE** **Forgetting the second step**
>
> Counting "who queens first" is not enough. Check whether the first new queen gives check or controls the other pawn's queening square, and whether the other pawn can queen with check. One extra tempo there flips the result.

Trading into a pawn ending

Everything in this chapter leads to one practical decision. You are in a rook or minor-piece ending and you have the chance to trade the last pieces. Should you?

Pieces give counterplay; pawn endings are pure. If you are winning the pawn ending, trading is usually the cleanest way to convert, and it removes all your opponent's tricks. If the pawn ending is a draw or a loss, trading throws away every chance you had. So the rule is: **trade into a pawn ending only when you have calculated it**, at least to the point where you can name the plan that wins (outside passer, protected passer, better king, opposition).

Your quick checklist before the trade:

1. After the trade, whose king is closer to the centre and to the pawns?
2. Does anyone get an outside or protected passed pawn?
3. Count any pawn race with the square and with tempi.
4. Who will have the opposition when the kings meet?

If you cannot answer these, keep the pieces and look for another way. Defending players should apply the same test in reverse: do not trade into a pawn ending just because you are tired of defending. You can see that in action in Chapter 17.

Finding these ideas in your own games

Pawn endings in your own games rarely look like textbook diagrams at first. They usually arise from a rook or minor-piece ending a few moves earlier, which is exactly where the important decision was made. When you analyze a game that reached a pawn ending, go back to the moment of the last trade and ask three questions.

- **Was the trade calculated?** Did you (or your opponent) know the pawn ending was won, or was it a hopeful swap?
- **Where did the tempo go?** Find the first move where a king went to the wrong side, or a pawn was pushed while the king was still behind it. In most lost pawn endings there is exactly one such moment.
- **Which pattern was hiding?** Square, opposition, rook pawn, outside passer, breakthrough. Name it and record it in your mistake journal under "endgame".

A useful habit during the game: whenever only kings and pawns are left, slow down. The position is simple enough that a few minutes of counting will usually tell you the truth, and one careless move can never be repaired.

Practice at the board

Pawn endings are ideal for practice because a friend, a coach or a free engine can play the other side. Set up each position, play it out several times with both colours, and check your result.

- **Opposition drill.** Set up `8/4k3/8/4K3/4P3/8/8/8 w - - 0 1`. Play it with Black to move (White should win) and with White to move (Black should draw). Switch sides. Do it until you win and draw it without thinking.
- **Square drill.** Set up `8/6k1/8/p7/5K2/8/8/8 w - - 0 1`. Before moving, say aloud whether the king catches the pawn. Then move the white king one square further away and test again.
- **Rook pawn.** Set up `k7/8/P7/1K6/8/8/8/8 w - - 0 1` and try to win as White against an engine. You will not, and you will see why.
- **Outflanking.** Set up `8/3k4/2p5/2P5/4K3/8/8/8 w - - 0 1` and win the c6-pawn against a defender. Then try it after 1.Ke5 to feel the difference.

- **Breakthrough.** Set up `7k/ppp5/8/PPP5/8/8/8/7K w - - 0 1` and play 1.b6. Try every Black reply.
- **Outside passer.** Set up `8/8/3k2p1/7p/P2K3P/6P1/8/8 w - - 0 1` and win it against an engine set to full strength.

To check yourself, use an engine or an endgame tablebase after you finish, not during. Write down any position where your result differed from the correct one and replay it the next day.

Chapter summary

- In pawn endings the king is a fighting piece: centralize it and lead with it.
- The rule of the square tells you instantly whether a king catches a lone passed pawn.
- In king and pawn against king, the attacker wins by reaching a key square; the defender draws by staying in front and taking the opposition.
- Rook pawns are usually drawn if the defending king reaches the corner.
- When direct opposition fails, outflank; when you need the opponent to move, look for a way to lose a move (triangulation).
- Outside and protected passed pawns win pawn endings by tying the enemy king down; three against three can be broken through with the middle pawn.
- In races, count the moves, remember who moves first, and check whether a queen arrives with check or covers the other queening square.
- Trade into a pawn ending only when you have calculated it.

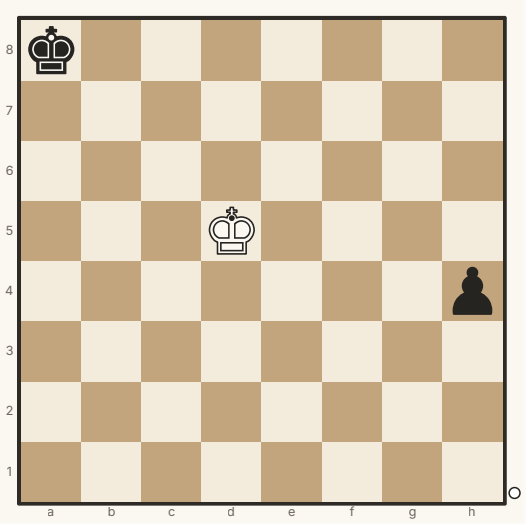
CHEAT CODE **Pawn ending procedure**

1. Count: material, passed pawns, and the square of every passed pawn.
2. Kings first: bring your king towards the centre and in front of your pawns.
3. Name the key squares and ask who has the opposition with the move.
4. Look for a decoy (outside passer), a breakthrough, or a protected passer.
5. In a race, count tempi including checks and control of the queening squares.
6. Before trading into a pawn ending, run steps 1-5 on the position after the trade.

Exercises

Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

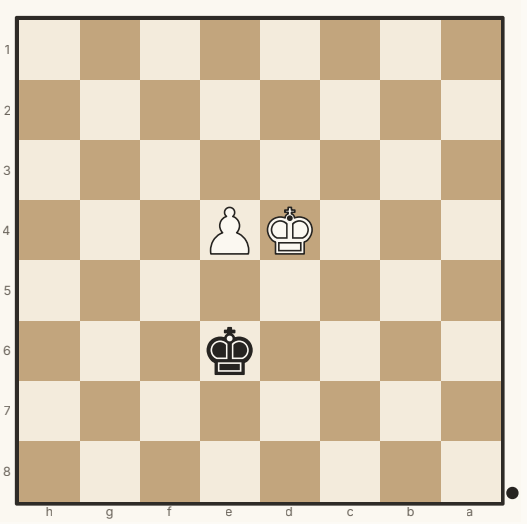
18.1 ENDGAME



White to move. Black's h-pawn is running. Which king move saves the game?

Solution p. 688

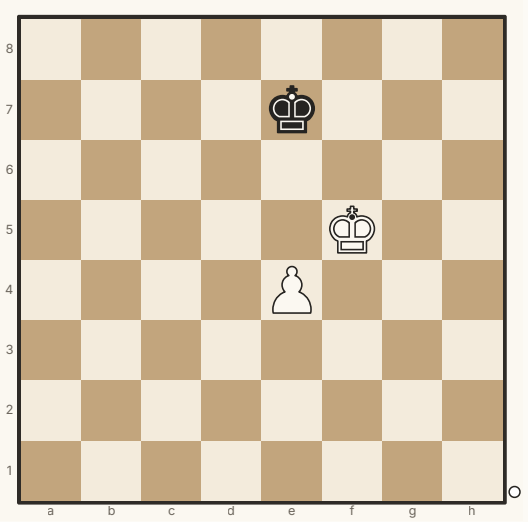
18.2 ENDGAME



Black to move. White wants to escort the e-pawn. Find the move that draws.

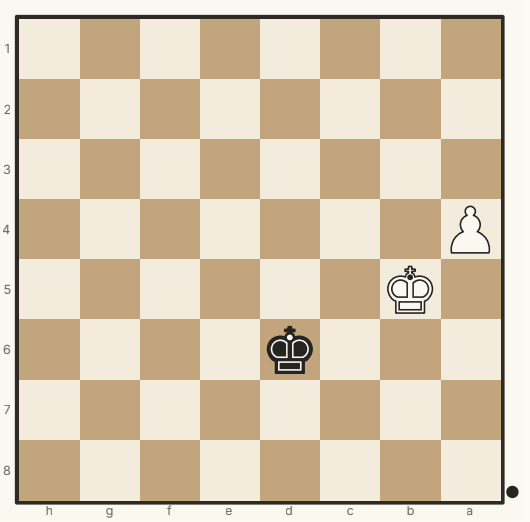
Solution p. 688

18.3 ENDGAME



White to move and win. Solution p. 688

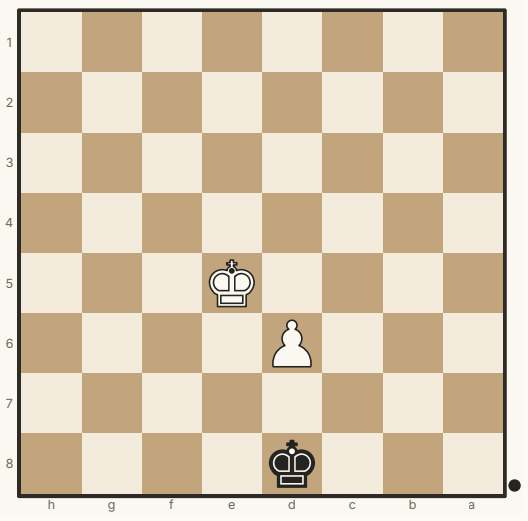
18.4 ENDGAME



Black to move. White has a rook pawn. Where must your king go to draw?

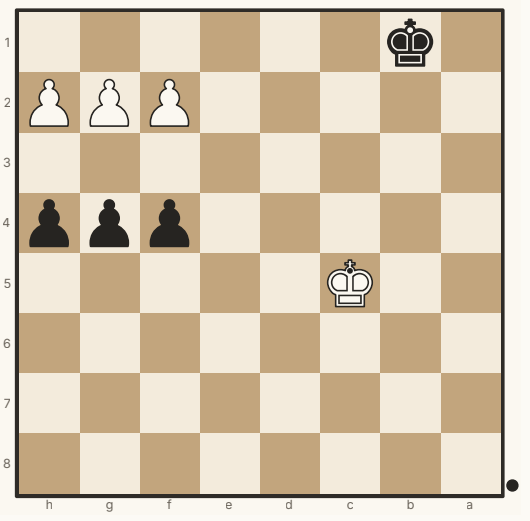
Solution p. 688

18.5 ENDGAME



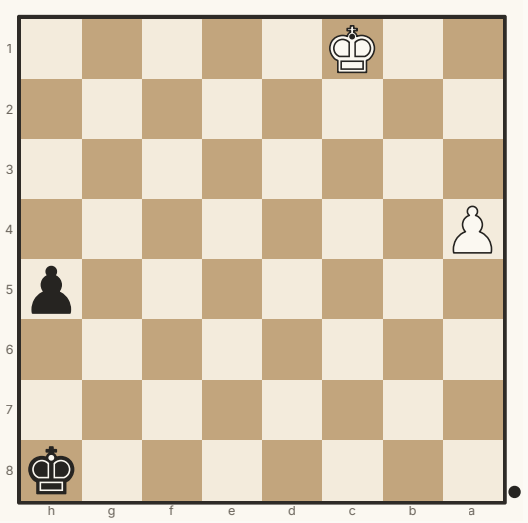
Black to move. Only one move holds the draw. Find it. Solution p. 689

18.6 ENDGAME



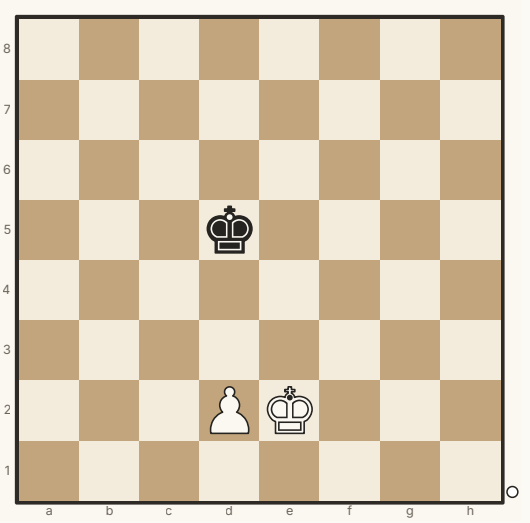
Black to move. Both kings are far from the kingside. Win the game. Solution p. 689

18.7 ENDGAME



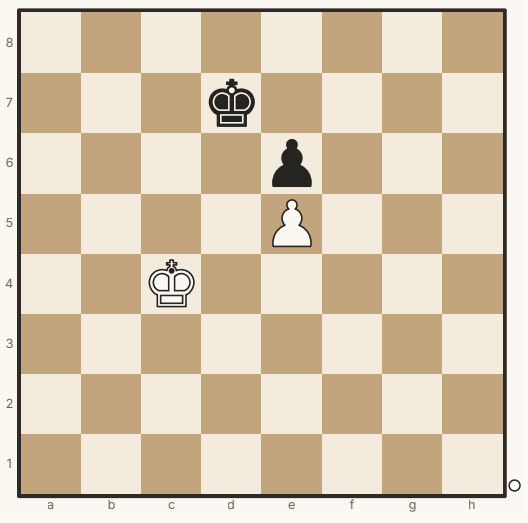
Black to move. Should you run with the pawn or chase White's pawn? Count, then play the right move. Solution p. 689

18.8 ENDGAME



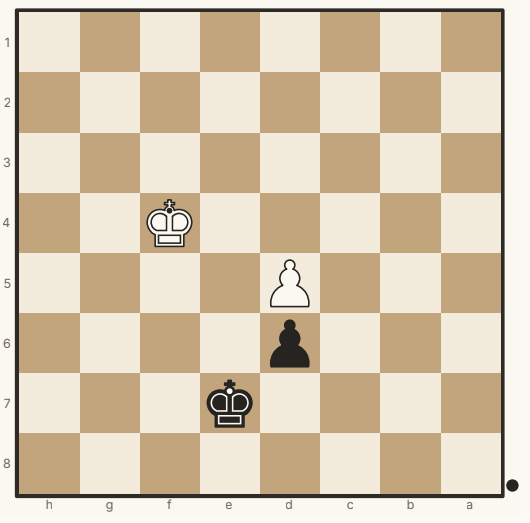
White to move. The pawn is still on d2. Which move wins? Solution p. 689

18.9 ENDGAME



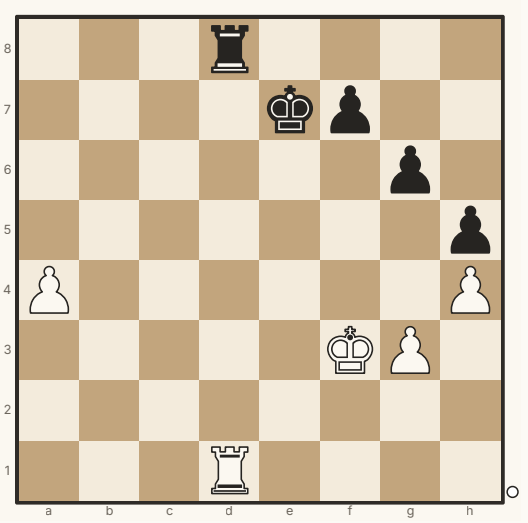
White to move. The black pawn on e6 is the target. Find the winning plan move.
Solution p. 690

18.10 ENDGAME



Black to move. White's king wants to go round to e6. How do you stop it? Solution p. 690

18.11 ENDGAME



White to move. Material is equal. Should you trade rooks? Play the best move.

Solution p. 690

HINTS

18.1 Draw the square from the pawn to its promotion square.

18.2 Face the white king with one square between you.

18.3 King in front of the pawn, and make the opponent move.

18.4 With a rook pawn, the corner is your fortress.

18.5 Stand in the pawn's path.

18.6 Sacrifice the middle pawn.

18.7 Count the moves to queen, and check what the new queen controls.

18.8 King first, pawn later.

18.9 Go around, not straight ahead.

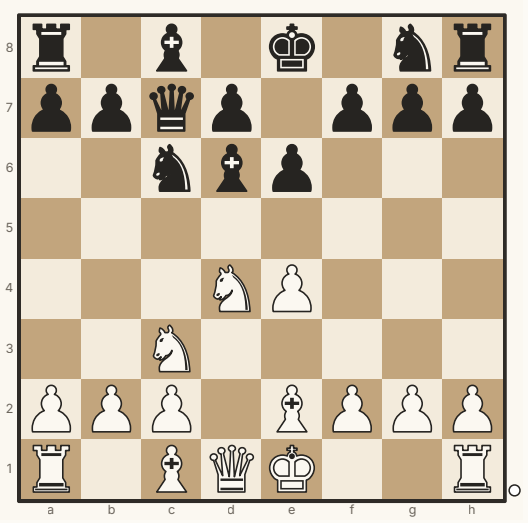
18.10 Mirror the white king on its own file.

18.11 Which pawn ending would you get, and who wins it?

Stage 3 Test

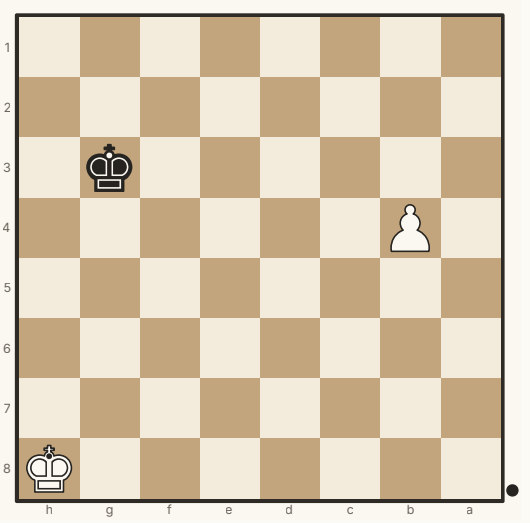
Fifteen mixed positions covering Part 3. Solve them without moving the pieces, write down your answers, then mark yourself. 12/15 or better: move on with confidence. 9–11: review the chapters behind your misses. Below 9: repeat this part's exercises before continuing.

T3.1 TACTIC



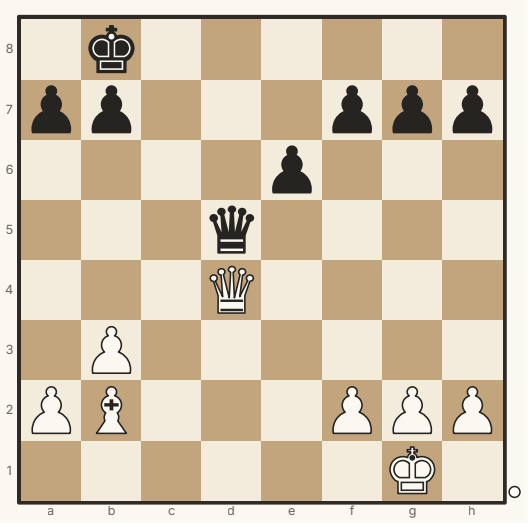
White to move. In a Sicilian (1.e4 c5 2.Nf3 Nc6 3.d4 cxd4 4.Nxd4 e6 5.Nc3 Qc7 6.Be2), Black has just developed with 6...Bd6?. Win material. Solution p. 691

T3.2 ENDGAME



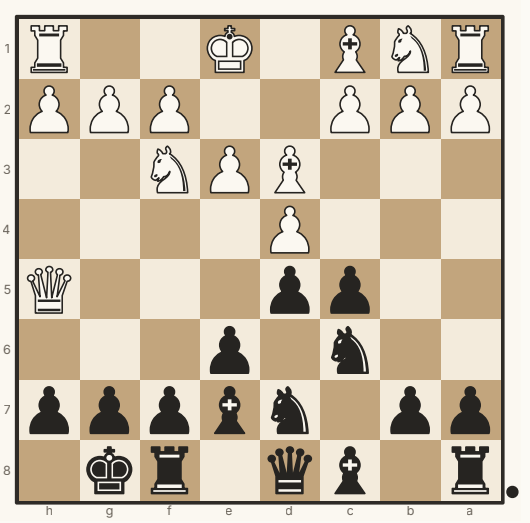
Black to move. White's king is far away, but the b-pawn is running. Which move saves the game? Solution p. 691

T3.3 CONVERSION



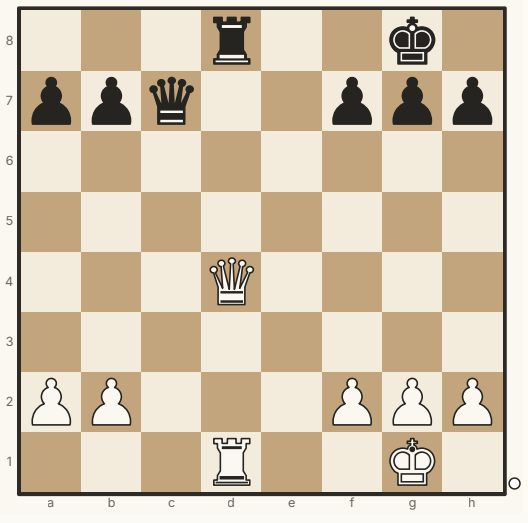
White to move. You are a bishop up. Black's queen offers a trade. What is the safest winning move? Solution p. 691

T3.4 DEFENCE



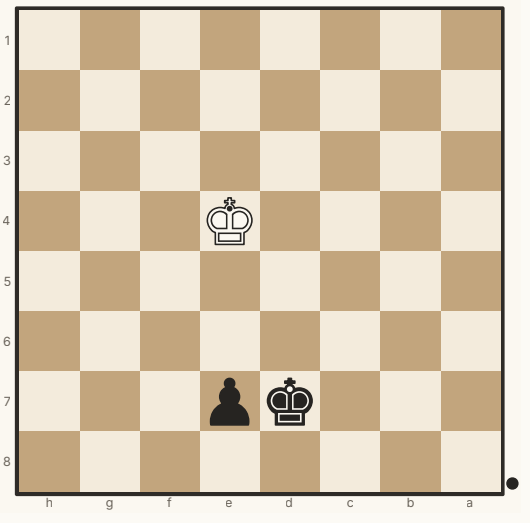
Black to move. White's queen and bishop aim at h7. Find a good defensive move. Solution p. 691

T3.5 CHECKMATE



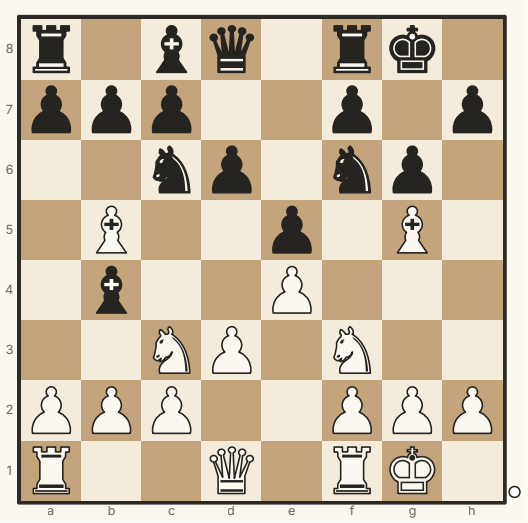
White to move. Black's queen guards the back rank. Find mate in two. Solution p. 692

T3.6 ENDGAME



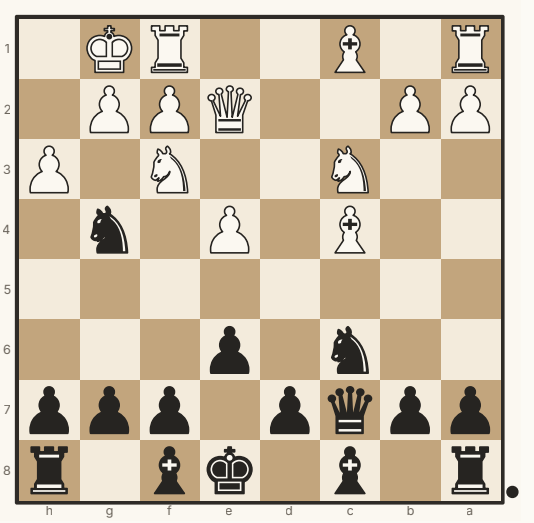
Black to move. Your pawn is still on e7. Which move wins? Solution p. 692

T3.7 CALCULATION



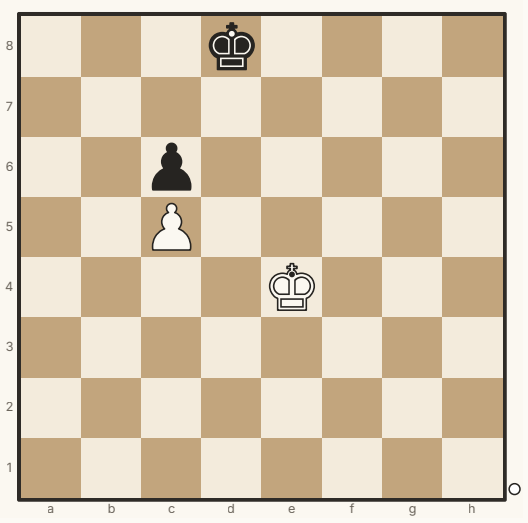
White to move. In the Symmetrical Four Knights (1.e4 e5 2.Nf3 Nc6 3.Nc3 Nf6 4.Bb5 Bb4 5.O-O O-O 6.d3 d6 7.Bg5) Black played 7...g6?. Your bishop pins the knight on f6 to the queen. Calculate, and play the strongest move. Solution p. 692

T3.8 CALCULATION



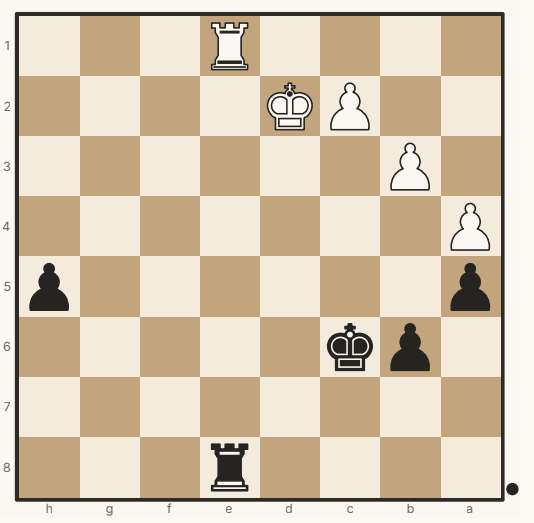
Black to move. In the Smith-Morra Gambit (1.e4 c5 2.d4 cxd4 3.c3 dxc3 4.Nxc3 Nc6 5.Nf3 e6 6.Bc4 Qc7 7.O-O Nf6 8.Qe2 Ng4), White has just attacked your knight with 9.h3. Retreating looks forced. Calculate the alternatives and play the best move. Solution p. 692

T3.9 ENDGAME



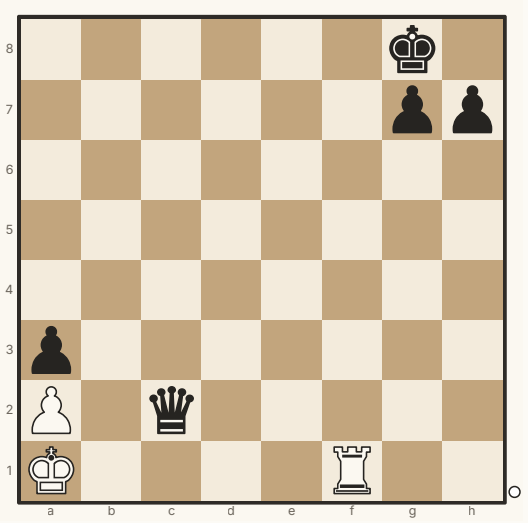
White to move. Win the c6-pawn and the game. Solution p. 693

T3.10 CONVERSION



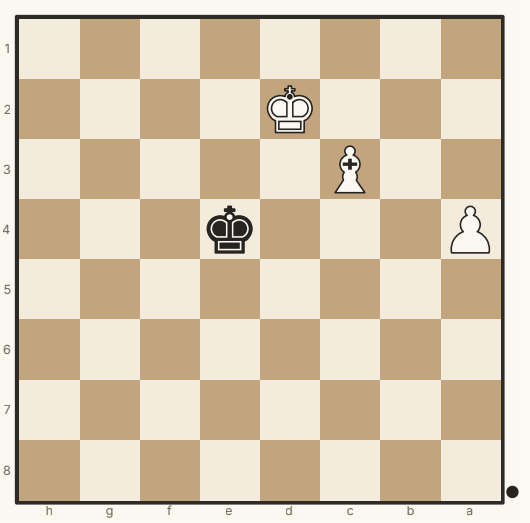
Black to move. Should you trade rooks? Play the best move. Solution p. 693

T3.11 DEFENCE



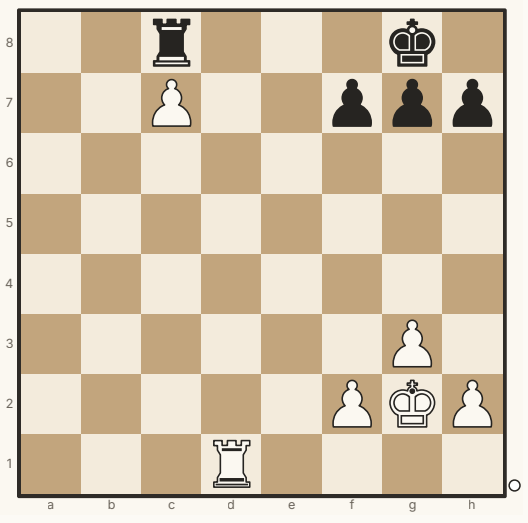
White to move. Black threatens Qb2#. Which move saves the game? Solution p. 693

T3.12 DEFENCE



Black to move. White has a dark-squared bishop and an a-pawn. Find the only move that draws. Solution p. 693

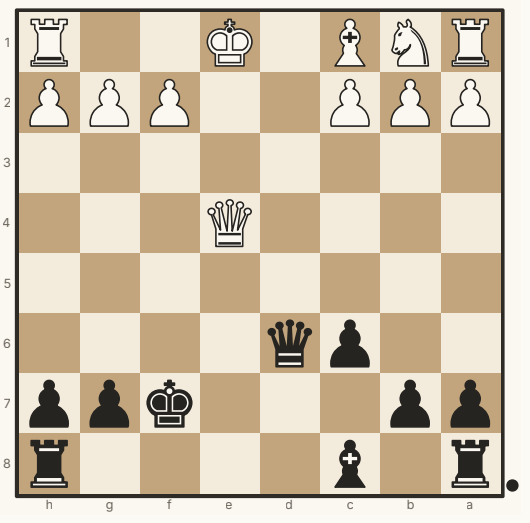
T3.13 CHECKMATE



White to move. The c7-pawn is one step from queening. Find mate in two.

Solution p. 693

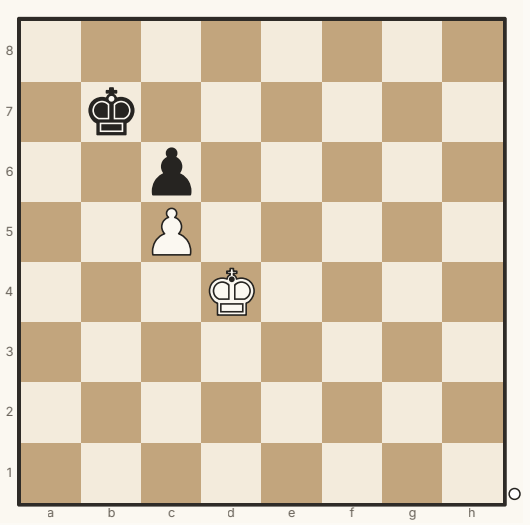
T3.14 TACTIC



Black to move. You are behind in material. Find the move that turns the game around.

Solution p. 694

T3.15 ENDGAME



White to move. Find the only winning move. Solution p. 694

HINTS

T3.1 Which knight jump attacks the queen and the bishop at the same time?

T3.2 Draw the square.

T3.3 What is Black's only active piece?

T3.4 What does White threaten on h7?

T3.5 Forcing moves first: what happens after the biggest capture?

T3.6 King in front of the pawn, and make the opponent move.

T3.7 Which move attacks the pinned knight a second time and something else as well?

T3.8 Can you attack the queen in a way that makes capturing your knight fatal?

T3.9 Approach from the side, not head-on.

T3.10 Which pawn ending would you reach, and who wins it?

T3.11 Your king cannot move. Can you make your rook vanish?

T3.12 Which corner can the bishop never control?

T3.13 Checks first.

T3.14 Look at the e-file.

T3.15 Head for d6 by the shortest route that avoids the opposition.

PART FOUR

Positional Judgment

Stage 4 · 1000–1200

When there is no tactic, what do you do? This part answers the question with piece activity, pawn structure, king safety, and simple planning, then adds the rook endgames that decide more games than any other ending.

Chapter 19 · Piece Activity

Chapter 20 · Pawn Structure Essentials

Chapter 21 · King Safety and Attacking the King

Chapter 22 · Making a Plan

Chapter 23 · Rook Endgames

CHAPTER 19 STAGE 1000–1200

Piece Activity

Count the squares, find the worst piece, improve it

YOU WILL LEARN TO

- Judge whether a piece is active or passive by counting the squares and targets it controls.
- Find your worst-placed piece and a concrete route to improve it.
- Recognize outposts, good and bad bishops, and knights stranded on the rim.
- Use rooks on open and half-open files, double them, and invade the seventh rank.
- Centralize the king in the endgame and trade your bad pieces for your opponent's good ones.

> **IN YOUR NEXT GAME**
>
> When there is no tactic on the board, ask "which of my pieces is doing the least?" and spend your move improving that piece — after a blunder check.

You have reached the position every improving player knows. No checks, no captures, no threats. Nothing is hanging. Your opponent has made a quiet move and you stare at the board, wanting to "do something". So you push a pawn, or shuffle a rook, and ten moves later you are worse without having blundered anything.

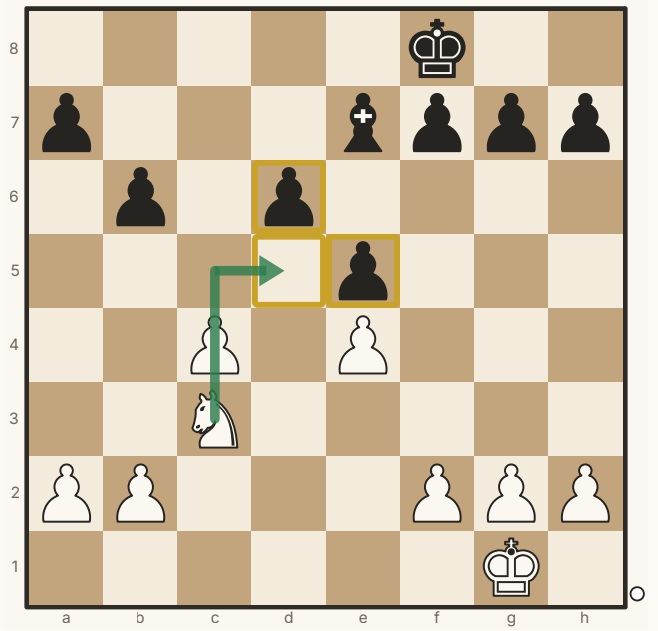
Between 1000 and 1200, more games are lost this way than you might think. The cure is not a secret plan. It is one question that always has an answer: **which of my pieces is doing the least, and where would it do more?** This chapter teaches you to measure activity, spot the pieces that matter most, and make the quiet improving move that strong players seem to find by instinct.

Active and passive pieces

A piece is **active** when it controls many squares, attacks targets, or restricts the enemy. It is **passive** when it only defends, is blocked by its own pawns, or has few safe squares to go to. Piece values from Chapter 3 are averages: a knight is "worth three" only when it has squares. A trapped knight is worth much less, a dominant one much more.

The simplest way to judge activity at the board is to count. For each piece ask:

- **Squares:** how many safe squares can it move to?
- **Targets:** does it attack anything — weak pawns, key squares, the king's zone?
- **Future:** can it get to a better square in one or two moves?



Equal material, unequal pieces. White's knight is heading for d5, where no black pawn can ever attack it. Black's bishop on e7 is hemmed in by its own pawns on d6 and e5.

Count in this position. White's knight on c3 has six squares now, and it can jump to d5, from where it would attack b6, c7, e7 and f6. Black's bishop on e7 has only four squares, all on the edge of the action, and its own pawns on d6 and e5 sit on the colour it needs. On paper material is equal. In practice White is playing with a better piece, and the engine agrees that White is clearly more comfortable.

> KEY IDEA **Activity is a number you can count**
>
> Count squares and targets for each piece. The side whose pieces have more of both is usually the side with the better position, even when material is equal.

The worst-placed piece

Strong players use a simple trick in quiet positions: they look for their **worst-placed piece** and ask how to improve it. This works because a position is only as strong as its weakest link. One sleeping rook or a bishop staring at its own pawns can be the difference between pressure and passivity.

The procedure is short. Count each of your pieces' squares and targets. Pick the one doing the least. Then find a route — usually two or three moves — to a square where it would do more. Before you play the first move of the route, run your blunder check: is the route safe, and what is your opponent threatening meanwhile?

The worst-placed piece is often a knight on the edge of the board, a bishop behind its own pawns, or a rook still on its starting square while the game happens elsewhere. In the endgame it is very often the king, still sheltering on g1 when it should be walking to the centre.

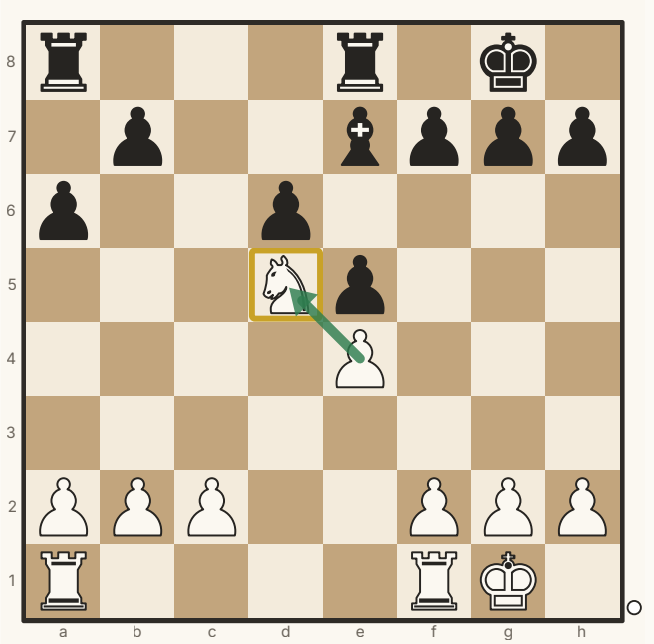
> COMMON MISTAKE **Improving a piece that is already fine**
>
> A common error is to move your best piece again because it is the one you like looking at. If your queen is already active, moving it to an "even better" square while a rook sleeps on a1 wastes a tempo. Improve the worst piece, not the favourite.

Knights: outposts and the rim

Outposts

An **outpost** is a square in or near the opponent's half of the board that **cannot be attacked by an enemy pawn**. Ideally it is also **supported by one of your own pawns**, so that if the opponent trades a piece for the occupant, you can recapture

with a pawn. Knights love outposts, because a knight needs to be close to the action and hates being chased by pawns.



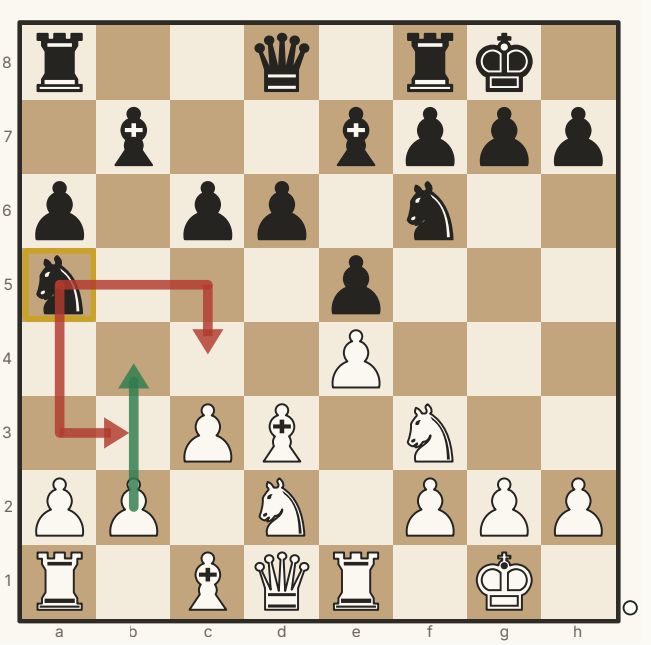
d5 is a model outpost. Black has no c-pawn and the e-pawn has already passed e6, so no black pawn can attack d5; White's e4 pawn supports the knight.

To check whether a square is an outpost, look at the enemy pawns on the two **adjacent files**. For d5 that means Black's c- and e-pawns. If neither can ever reach c6 or e6 to attack d5 — because they have advanced past, been exchanged, or are blocked — the square is a permanent home. Black's only way to remove the knight is to give up a piece for it.

This is why the pawn move ...e5 in the Sicilian or the Italian has a cost: it gains space and controls d4, but it gives up control of d5 for ever. Pawns cannot move backwards, a theme we return to in Chapter 20.

Knights on the rim

"A knight on the rim is dim" is one of the oldest chess sayings, and it is usually right. A knight in the centre controls eight squares; on the edge it controls four, and in the corner only two. Worse, a knight on the edge has few retreats, which makes it easy to trap.



Black's knight on a5 has no safe square. b3 and c4 are covered, b7 and c6 are blocked by Black's own pieces. White to move traps it with a pawn.

Look at the knight's possible squares: b3 (covered by the a2 pawn and the knight on d2), c4 (covered by the bishop and the knight), b7 (Black's own bishop) and c6 (Black's own pawn). After b2-b4 it has nowhere to go.

12. **b4** **Qc7** 13. **bxa5**

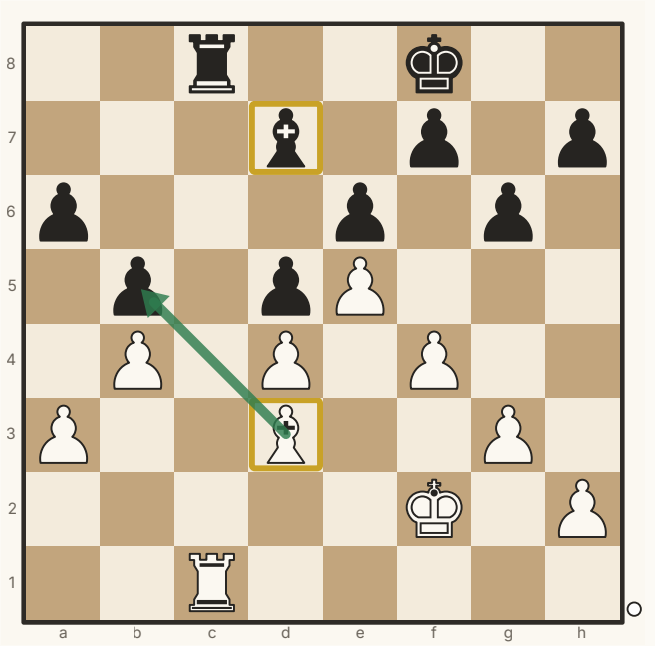
The knight is lost; Black's best is to accept it and look for counterplay a piece down.

The lesson for your own games has two sides. Before you put a knight on the edge, check its way back. And when your opponent's knight goes to the rim, count its retreats — a pawn move may win it.

> COACHING JUDGMENT **When the rim is fine**
>
> Coaching judgment: a knight on the edge is acceptable when it has a specific job and a clear way back — for example ...Na5 to trade off a bishop on b3, or Nh4 to jump to f5. It is a temporary visit, not a home. If you cannot name the job and the route back, do not go.

Bishops: good, bad and blocked

A bishop moves on one colour only. Your own pawns therefore decide much of its future. A **good bishop** has its own centre pawns on the *other* colour, so its diagonals are open. A **bad bishop** has its own centre pawns fixed on its *own* colour, so it bites on granite and mostly defends.



Same-coloured bishops, very different futures. All of Black's pawns stand on light squares, so the d7-bishop can only defend them. White's pawns are on dark squares and the d3-bishop attacks b5 and g6.

"Bad" does not mean useless. A bad bishop often defends its pawns perfectly well. The problem is that it cannot attack anything, so its owner must sit and wait while the other side improves. In an endgame that difference often decides the game.

What can you do about a bad bishop?

- **Trade it.** Offer an exchange for your opponent's good bishop (a French Defence player plays ...b6 and ...Ba6 to trade the light-squared bishops).
- **Put it outside the pawn chain.** Bring it out *before* the pawns lock it in — in the French, some players develop the bishop to b7 or via d7-e8-h5.
- **Change the pawn structure.** A pawn break such as ...f6 or ...c5 can open lines for it.

And from the other side: if your opponent has a bad bishop, keep the pawns where they are, avoid trading your good minor piece for it, and play on the colour it cannot cover.

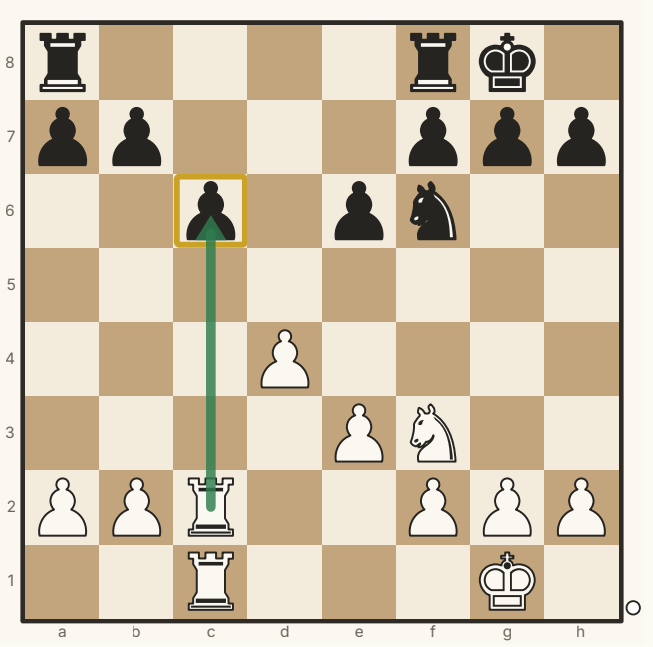
> KEY IDEA **Trade bad pieces, keep good ones**
>
> When you can exchange your worst piece for your opponent's best piece, the position usually improves for you even though "material is equal". Ask before every exchange: which piece leaves the board, and which one stays?

Rooks: files, doubling and the seventh rank

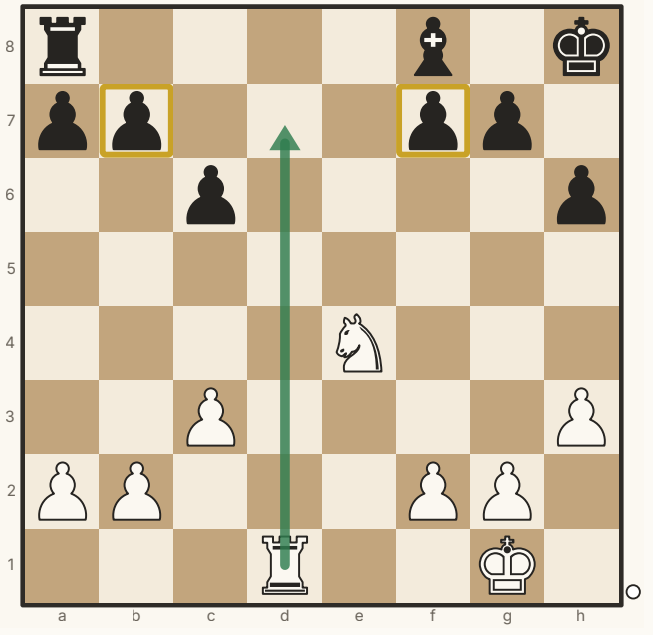
Rooks are often the last pieces to join the game. They need open lines. An **open file** has no pawns of either colour on it; a **half-open file** has only the opponent's pawn on it, which gives your rook a target.

Three rules of thumb cover most rook decisions at this level:

1. **Put rooks on open files** — or on the file you expect to open after a pawn exchange. The first rook to arrive usually owns the file.
2. **Double them.** Two rooks on the same file are much stronger than one each; the front rook can invade while the rear rook supports it.
3. **Aim for the seventh rank.** A rook on the seventh (the second, for Black) attacks pawns that cannot be defended by other pawns, and it cuts the enemy king off on the back rank.



White's rooks are doubled on the half-open c-file and aim at c6. Black's rooks are still looking for a job.



White owns the only open file. Rd7 puts the rook on the seventh rank, where it attacks b7 and f7 and ties Black's pieces to their defence.

24.**Rd7 Re8** 25.**f3**

Black's rook finally becomes active, but White calmly covers e2 and keeps the rook on the seventh. Black must spend the next moves defending b7 and f7.

Two more ideas belong here briefly. A **rook lift** brings a rook from the first rank up the board and sideways — for example Re1-e3-g3 or Rf1-f3-h3 — to attack the king when the files are closed; Chapter 21 shows it in action. And in the endgame, an **active rook** behind enemy lines is often worth more than a pawn. If you must choose between passively guarding a pawn and counter-attacking with the rook, look hard at the counter-attack first. Chapter 23 develops this.

> COMMON MISTAKE **The rook that never moves**
>
> Count how many games you finish with a rook still on a1 or h1. Before move 20, ask once: "Have both my rooks found a file?" If not, the next quiet move should probably be a rook move.

The king and queen in the endgame

In the middlegame the king hides. When the queens come off and only a few pieces remain, mating attacks become unlikely and the king turns into a strong fighting piece. It should walk to the centre, where it supports its own pawns, attacks the opponent's, and stops passed pawns.

Count king moves in such endings. If your king needs four moves to reach the centre and your opponent's needs three, a "useful" pawn move on the wing usually just hands him the race. The exercises at the end of this chapter include two king races, one for each colour.

The same principle applies to the queen in a queen ending: a queen in the centre covers both wings, gives fewer free checks to the opponent and can switch to attack in one move. A queen in the corner usually ends up defending.

Space

Space is the territory behind your pawn chain where your pieces can move freely. If your pawns stand on the fourth and fifth ranks and your opponent's on

the second and third, you have more room to manoeuvre, and he may find his pieces stepping on each other's toes.

Two practical rules follow:

- **With more space, avoid unnecessary exchanges.** Every trade gives the cramped side more room to breathe.
- **With less space, look for exchanges and pawn breaks.** Trading a pair of minor pieces, or striking at the opponent's pawn chain, frees your position.

Space is an advantage only if you use it. Pawns pushed too far can become targets, and space is not worth weakening your own king for.

Improving a piece: three typical routes

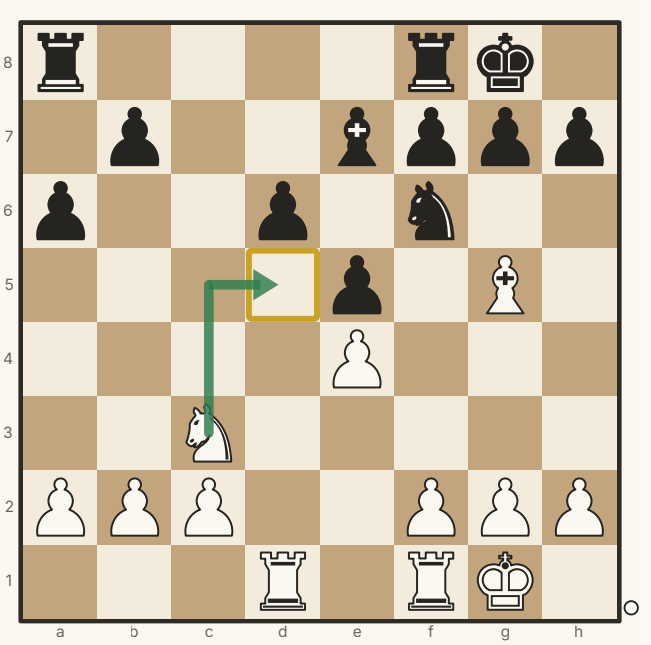
Knowing that a piece is badly placed is half the job; the other half is finding the route. Three routes come up again and again, and it pays to know them by heart.

- **The knight tour to an outpost.** A knight needs two or three moves to travel anywhere useful, so plan the whole trip. Nb1-d2-f1-g3-f5 in the Ruy Lopez, or Nc3-e2-g3 to reach f5, are classic examples. Ask before you start: will the square still be available when I arrive, and what does my opponent do with the three tempi?
- **The bishop outside the chain.** A bishop behind its own pawns can often be rerouted: Bc1-d2-e1-h4 in some closed positions, or ...Bc8-d7-e8-h5 for Black in the French. If no route exists, look for a trade.
- **The rook to the open file — or the file that will open.** If no file is open yet, place the rook behind the pawn you intend to advance, so that the file opens with your rook already on it.

The common thread is patience. A quiet three-move plan that turns your worst piece into a good one is often worth more than a one-move threat that your opponent parries easily. Your opponent at this level will rarely stop such a plan, because he is usually busy with his own threats — or with none at all.

Worked example: getting the knight to its outpost

Let us put the chapter together in one position. Follow the thinking, not just the moves.



White to move. Queens are off. White's knight wants d5; Black's knight on f6 guards that square.

Evaluate. First the safety scan: no white piece is attacked, and Black threatens nothing immediate. Now count activity. White's rook on d1 already presses on d6, the bishop pins nothing but annoys the f6-knight, and the knight on c3 would be magnificent on d5 — an outpost, since Black has no c-pawn and the e-pawn has passed e6. Black's worst piece is the bishop on e7, which is hemmed in by d6 and e5. Black's weakness is the backward pawn on d6, defended only by that bishop.

Plan. The plan is obvious: knight to d5. The question is how. The square d5 is guarded by one black piece, the knight on f6. If White simply plays Nd5, that knight can take.

Candidate moves. 15.Nd5, 15.Bxf6, and a quiet developing move such as 15.Rfe1.

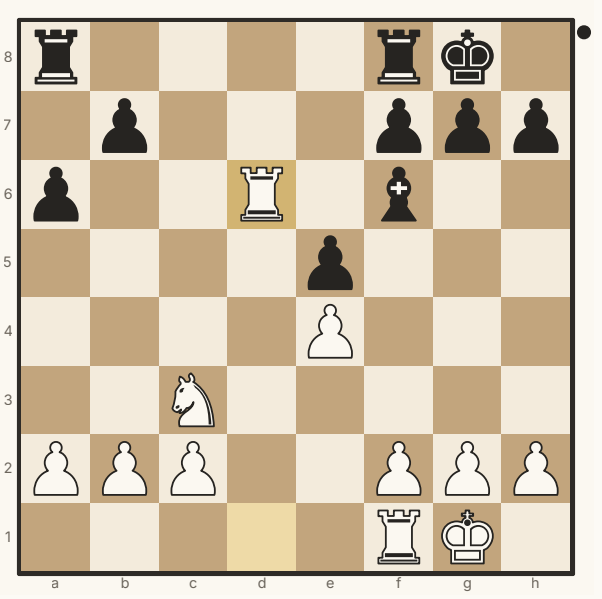
Opponent's best reply to each. Check 15.Nd5 first, because it is the move you want to play:

15. **Nd5? Nxd5** 16. **Bxe7 Nxe7**

The routine move loses a piece. The knight on d5 was not yet on an outpost: an enemy piece could still take it, and the exchange on e7 does not come back with check.

Now 15.Bxf6. Giving the bishop for the knight looks like a concession, but it removes the only defender of d5 *and* deflects the e7-bishop, which is also the only defender of d6.

15. **Bxf6 Bxf6** 16. **Rxd6**



White wins the d6 pawn, and the knight still has d5 waiting. Black's remaining bishop cannot challenge a knight on a light square.

Move. 15.Bxf6. The blunder check is short: after 15...Bxf6 nothing of White's hangs, and if Black does not recapture he is simply a piece down. The thinking pattern is the one to copy: you wanted a square, you counted who guards it, and you removed the guard first. The bishop-for-knight trade also made sense in the language of this chapter — White traded a bishop that had no outpost for the piece that defended the key square, leaving his best piece against Black's worst.

WORKED EXAMPLE **What the worked example teaches**

- An outpost is only safe when no enemy *piece* can simply take the occupant for free.
- "Remove the guard, then occupy" is a two-step plan you can use with knights, rooks and even the king.
- Trading a good-looking piece is correct when it leaves your best piece facing their worst.

Finding activity in your own games

When you review a game (see Chapter 28), stop at every quiet move and answer two questions: *which of my pieces had the fewest squares?* and *did my move improve it?* You will usually find the same culprits game after game — a bishop locked behind your pawns, a rook that waited until move 30, a king that stayed home in the endgame. That pattern is your training plan.

A second habit: before every exchange, name the pieces. "My bad bishop for his good knight" is a trade you want. "My active knight for his passive bishop" is one you should refuse unless it wins something concrete.

IN YOUR NEXT GAME **In your next game**

On every quiet move, point (mentally) at your worst piece before choosing your move. If you can improve it safely, do it. In the endgame, move your king toward the centre at the first safe moment.

Practice at the board

Set up each position on a board or in a free analysis app and play it out against a friend or the engine at a modest level.

1. **The knight race.** FEN `5k2/p3bppp/1p1p4/4p3/2P1P3/2N5/PP3PPP/6K1 w - - 0 25`. Play White. Get your knight to d5 and your king to d3, then try to win the b6 or a7 pawn. Then swap sides and try to hold with Black. Notice how little the bad bishop can do.
2. **The seventh rank.** FEN `r4b1k/pp3pp1/2p4p/8/4N3/2P4P/PP3PP1/3R2K1 w - - 0 24`. Play White: 24.Rd7, and keep the rook on the seventh as long as possible.

Check yourself: after ten moves, how many black pawns have you won, and how many black pieces are stuck defending?

3. **King march.** FEN `5k2/pp3ppp/2n5/8/8/2N1K3/PP3PPP/8 b - - 0 30`. Play Black twice against a friend or the engine: once starting with 30...Ke7, once with 30...a5. Compare where the kings stand five moves later.
4. **Square count drill.** Take any middlegame from your recent games. For each piece, write down how many safe squares it has. Circle the lowest number. Ask the engine whether improving that piece was a good idea.

Chapter summary

- A piece is active when it has squares, targets and a future; count them.
- In quiet positions, find your worst-placed piece and improve it — after a blunder check.
- An outpost is a square no enemy pawn can attack, ideally supported by your own pawn. Remove any piece that guards it before you occupy it.
- Knights on the rim have few squares and few retreats; count the retreats before you go there, and look for traps when your opponent does.
- A bad bishop is blocked by its own pawns on its colour. Trade it, get it outside the chain, or change the structure.
- Rooks belong on open and half-open files; double them and aim for the seventh rank.
- In the endgame the king is a fighting piece: centralize it early.
- Trade your bad pieces for your opponent's good ones; with more space, avoid unnecessary trades.

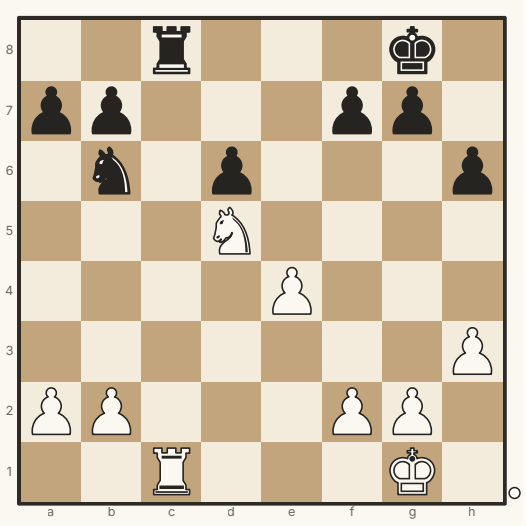
CHEAT CODE **Piece activity — the quiet-move procedure**

1. Safety scan first: checks, captures and threats for both sides.
2. Count squares and targets for each of your pieces; name your worst piece.
3. Find a two- or three-move route that improves it (outpost, open file, seventh rank, centre).
4. If an enemy piece guards the target square, plan to remove or exchange it first.
5. Before any exchange, ask which piece leaves and which stays — trade bad for good.
6. Blunder-check the first move of the route, then play it.

Exercises

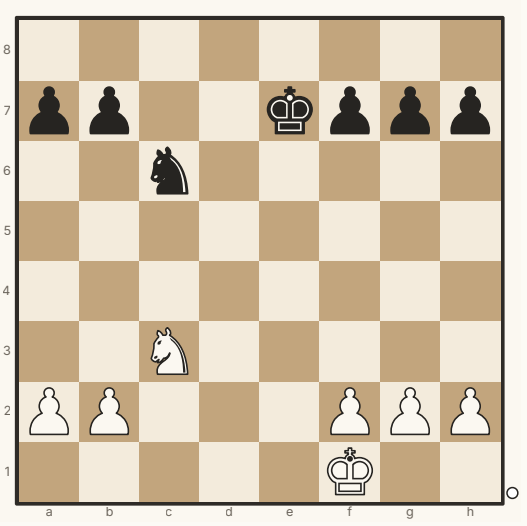
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

19.1 TACTIC



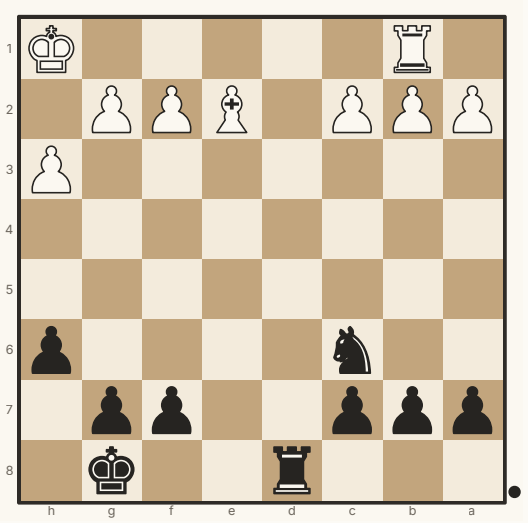
White to move. Black has just played ...Nb6, attacking your knight on its outpost. Punish it. Solution p. 694

19.2 ENDGAME



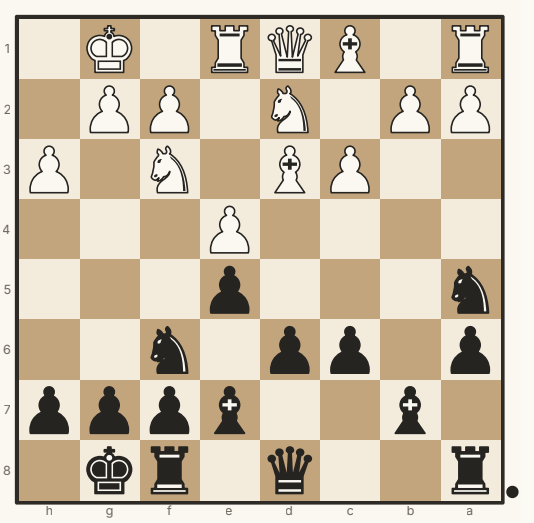
White to move. Queens and rooks are gone. Choose the best plan move. Solution p. 694

19.3 STRATEGY



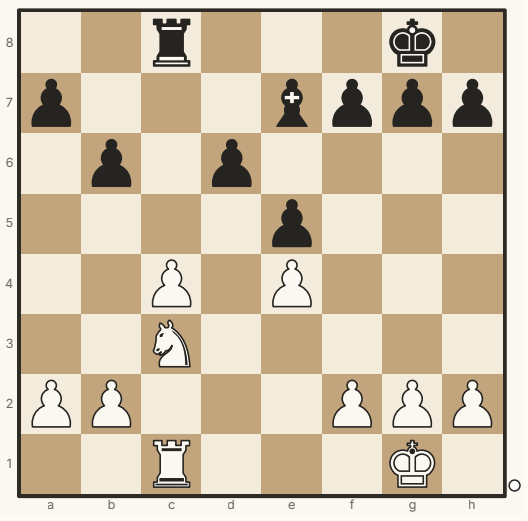
Black to move. The d-file is open and White's rook is passive. Find the most active move. Solution p. 695

19.4 DEFENCE



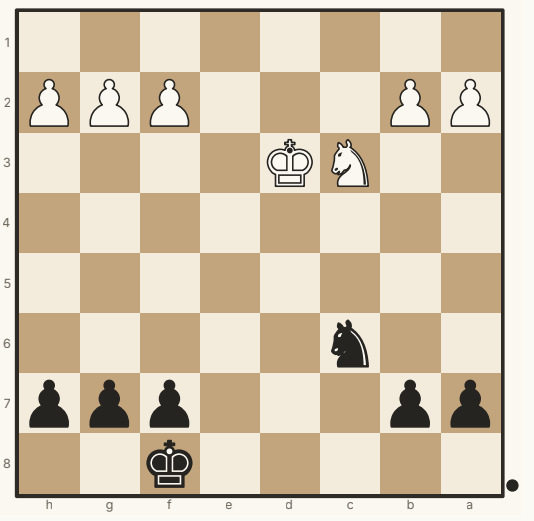
Black to move. Count the retreat squares of your knight on a5, then play the move that keeps it alive. Solution p. 695

19.5 STRATEGY



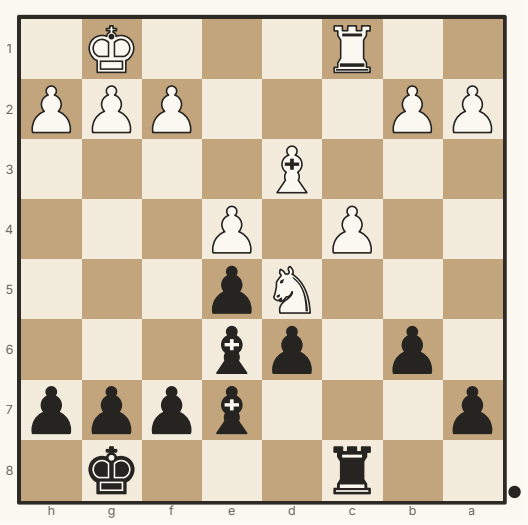
White to move. Find the best square for your knight. Solution p. 695

19.6 ENDGAME



Black to move. White's king is already on d3. What should Black do? Solution p. 695

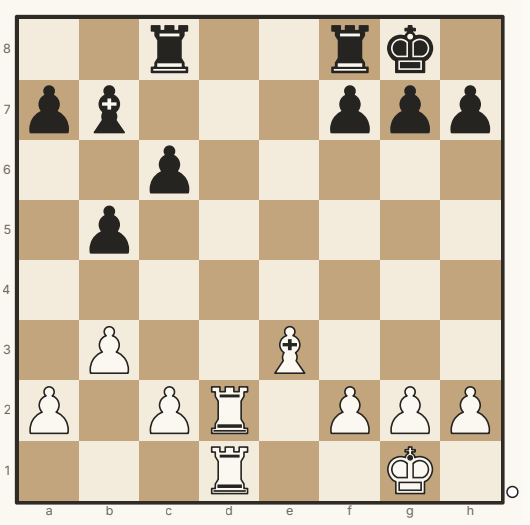
19.7 STRATEGY



Black to move. White's knight on d5 is the best piece on the board. Deal with it.

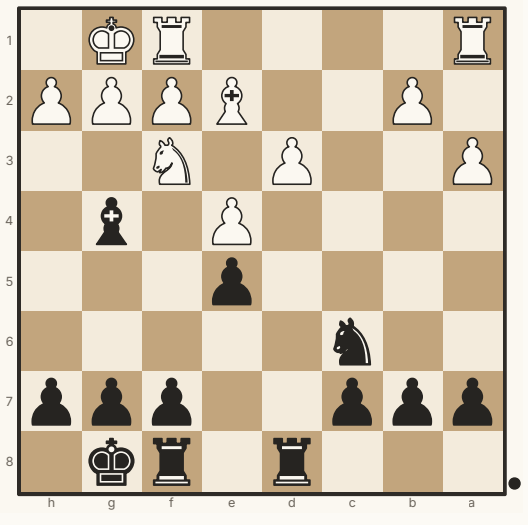
Solution p. 696

19.8 STRATEGY



White to move. Your rooks own the d-file. Make the most of it. Solution p. 696

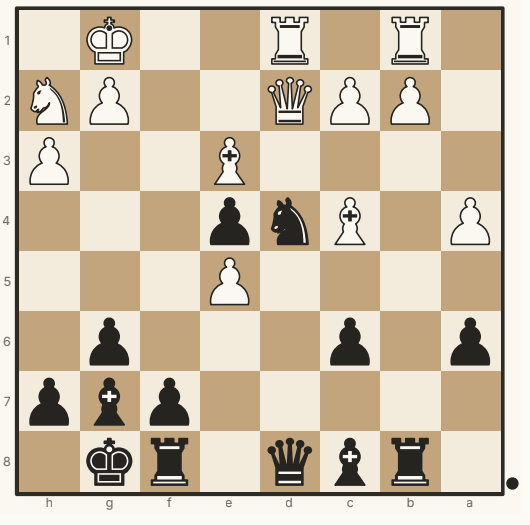
19.9 STRATEGY



Black to move. Your knight would love the d4 square. How do you get it there safely?

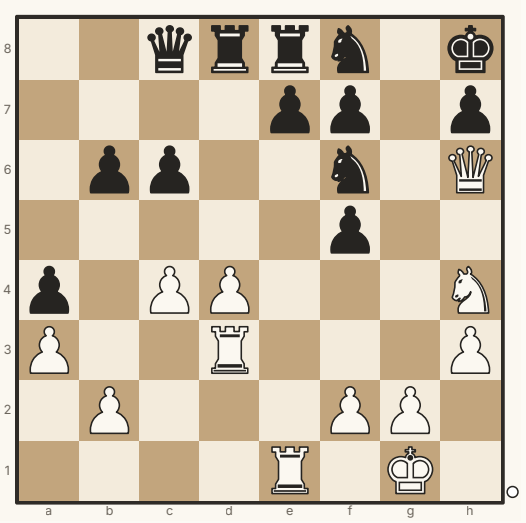
Solution p. 696

19.10 DEFENCE



Black to move. Your knight on d4 is attacked by the bishop, the queen and the rook. Keep it on its outpost. Solution p. 696

19.11 TACTIC



White to move. Your queen and knight are close to the king, but one piece is missing. Bring it in. Solution p. 697

HINTS

19.1 Which knight check also attacks a rook?

19.2 Which piece is still on its starting square doing nothing?

19.3 Where can your rook attack two things at once?

19.4 What happens to your knight after White's b2-b4?

19.5 Which square can no black pawn ever attack?

19.6 Count the moves each king needs to reach the queenside.

19.7 Which black piece can take the knight — and what does the recapture do to White's pawns?

19.8 Where is the seventh rank, and what does it attack?

19.9 Who guards d4 right now?

19.10 Which pawn can support d4?

19.11 Which square would let a rook attack g7?

CHAPTER 20 STAGE 1000–1200

Pawn Structure Essentials

Pawns can't move backwards — so read them before you push them

YOU WILL LEARN TO

- Count pawn islands and recognize isolated, doubled and backward pawns, and the weak squares they leave behind.
- Know when a weakness matters and when it is compensated by activity.
- Push passed pawns, blockade the opponent's with a knight, and use a pawn majority correctly.
- Attack a pawn chain at its base and choose the right lever (pawn break).
- Think before every pawn move, because pawns can never retreat.

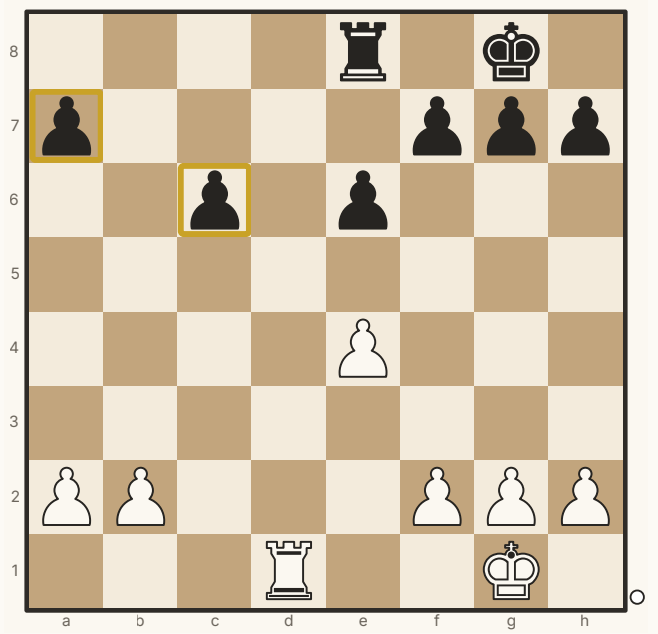
> **IN YOUR NEXT GAME**
>
> Before every pawn move, ask two questions — "which squares will this pawn never guard again?" and "does this pawn move create a weakness or fix one of my opponent's?"

Most of your moves in a game are piece moves, and you can take most of them back. A knight that went to the wrong square can return next move. A pawn cannot. Every pawn move changes the position for the rest of the game: it gains some squares and gives up others for ever.

This is why strong players read the pawns first. The pawn structure tells you where the weak squares are, which pieces are good, where the pawn breaks will come and how the endgame will go. Between 1000 and 1200 you start to win games not because your opponent hangs a piece, but because his pawns gave you a target and you knew what to do with it. This chapter gives you the vocabulary and the plans.

Pawn islands

A **pawn island** is a group of pawns on adjacent files, separated from the other pawns by at least one empty file. Fewer islands are usually better, because each island needs defending and the gaps between them are files where enemy rooks can enter.



Equal material, but White has two pawn islands (a2–b2 and e4–h2) and Black has three (a7, c6 and e6–h7). The isolated pawns on a7 and c6 need constant protection.

Count islands quickly: look along the board from the a-file to the h-file and count each time a group of pawns starts again after an empty file. In the diagram Black's a7 and c6 pawns are each alone, and White's rook will find them on the open files. In an endgame such a difference can decide the game. "Fewer islands are better" is one of the oldest guidelines in chess teaching, and it is a strong one — though, like every guideline, it can be outweighed by activity.

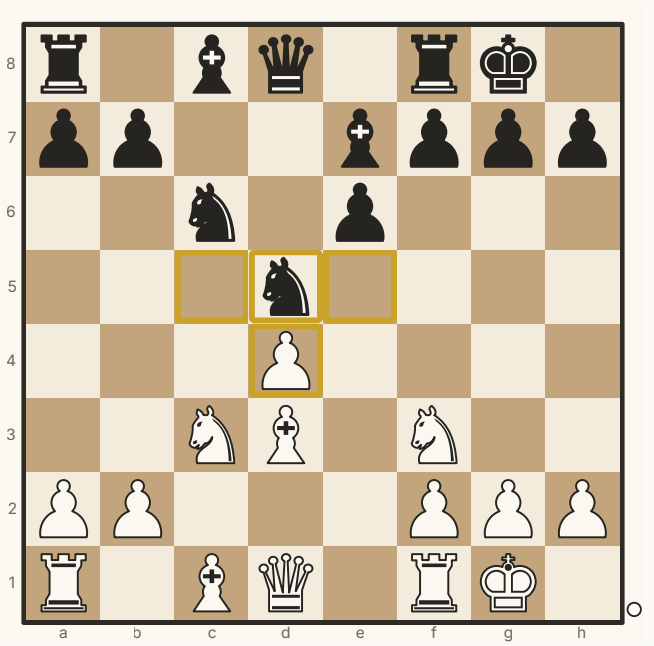
> **KEY IDEA** **Fewer islands, fewer worries**
>
> When two captures are possible, ask which one leaves you with fewer pawn islands. All else being equal, choose that one — and when your opponent has more islands than you, aim for the endgame.

The isolated pawn

An **isolated pawn** has no friendly pawns on the files next to it, so no pawn can ever defend it. The most important one is the **isolated queen's pawn** (often called the IQP), a pawn on d4 (or d5 for Black) with no c- or e-pawn beside it. It arises from many openings, including the Caro-Kann Panov, the Queen's Gambit and the Alapin Sicilian you play as White.

1. e4 c6 2. d4 d5 3. exd5 cxd5 4. c4 Nf6 5. Nc3 e6 6. Nf3 Be7 7. cxd5 Nxd5 8. Bd3 Nc6 9. O-O O-O

A typical IQP position from the Caro-Kann. White's d4 pawn is isolated; both sides have developed quickly.



The isolated pawn on d4 controls e5 and c5, ideal squares for White's knights. The square d5 in front of it can never be attacked by a white pawn, so it is a perfect home for a black knight.

The isolated pawn has two faces.

- **Its weakness** is static and shows up in the endgame. The pawn needs pieces to defend it, and the square in front of it is a hole where the opponent can plant a blockader. The more pieces are exchanged, the weaker the pawn gets.

- **Its strength** is dynamic and shows up in the middlegame. The pawn gives its owner space, open files for the rooks (the c- and e-files), outposts on c5 and e5, and attacking chances against the king. The thematic pawn break d4-d5 can open the position at the right moment.

So the plans write themselves. *With* the isolated pawn: keep pieces on, attack — often the king — and prepare d4-d5. *Against* it: blockade the square in front of it (ideally with a knight), exchange pieces, and win the pawn in the endgame.

> COACHING JUDGMENT **Which side to choose?**
>
> Coaching judgment: at this level, positions with an isolated pawn are fine to play from either side, as long as you know which plan you are following. The most common error is to sit passively with the pawn and drift into an endgame where it is just weak — or, from the other side, to forget the blockade and allow d4-d5 at a moment when it works.

Doubled pawns

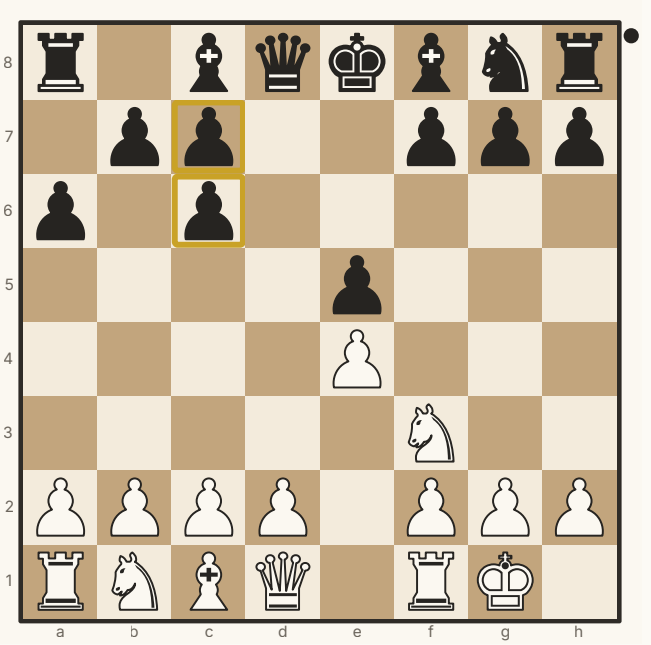
Doubled pawns are two pawns of the same colour on one file, created by a capture. They are often weak: the front pawn blocks the rear one, they cannot protect each other, and in a pawn ending they are worth less than two healthy pawns, because they cannot create a passed pawn against an equal number of enemy pawns.

But doubled pawns are not always bad. They come with compensation when:

- the capture that created them **opened a file** for a rook, or
- the pawns **control important central squares**, or
- the side with doubled pawns got **the bishop pair** or a lead in development in exchange.

1. **e4 e5** 2. **Nf3 Nc6** 3. **Bb5 a6** 4. **Bxc6 dxc6** 5. **O-O**

The Exchange Ruy Lopez. Black has doubled c-pawns, but in return gets two bishops and the open d-file. The doubled pawns are a long-term endgame concern, not an immediate problem.



If White later plays d4 and the d-pawns are exchanged, White's four kingside pawns against three can make a passed pawn, while Black's queenside majority, crippled by the doubled pawns, cannot. In return Black has the bishop pair and free development.

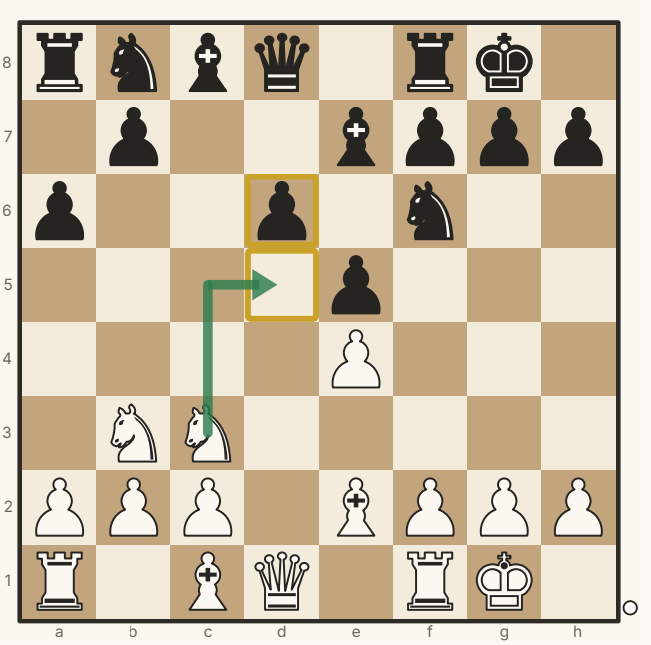
The practical rule: doubled pawns usually matter more the more pieces are traded. If your opponent has doubled pawns and nothing else to show for them, steer toward a pawn ending.

Backward pawns and holes

A **backward pawn** has fallen behind its neighbours and can no longer be protected by them; the square in front of it is usually controlled by the opponent, so it cannot safely advance. A **hole** (or weak square) is a square that can no longer be defended by any of your pawns. Backward pawns and holes come together: the square in front of a backward pawn is almost always a hole.

1. e4 c5 2. Nf3 d6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 a6 6. Be2 e5 7. Nb3 Be7 8. O-O O-O

A Najdorf Sicilian structure. Black's ...e5 gained space and kicked the knight, but left d6 backward and d5 a hole.



The d6 pawn is backward on a half-open file, and d5 is a hole that White will try to occupy with a knight. Black's counterplay is the freeing break ...d5 — if it can ever be prepared.

This is a good example of a weakness with compensation. Black gets space, a solid pawn on e5 and active piece play; plenty of strong players choose this structure on purpose. The battle is about one square. If White controls d5 for ever, Black suffers. If Black achieves ...d5 safely, the backward pawn disappears and Black is often better.

> **COMMON MISTAKE** **Breaking too early**
>
> The freeing break is the right idea, but only when you have counted the pieces on the file it opens. A typical error is ...d5 played one move too early: after exd5 Nxd5 Nxd5 the d-file opens and a black piece on d7 or d8 is suddenly loose. Prepare the break with a rook on the file first.

Using a hole

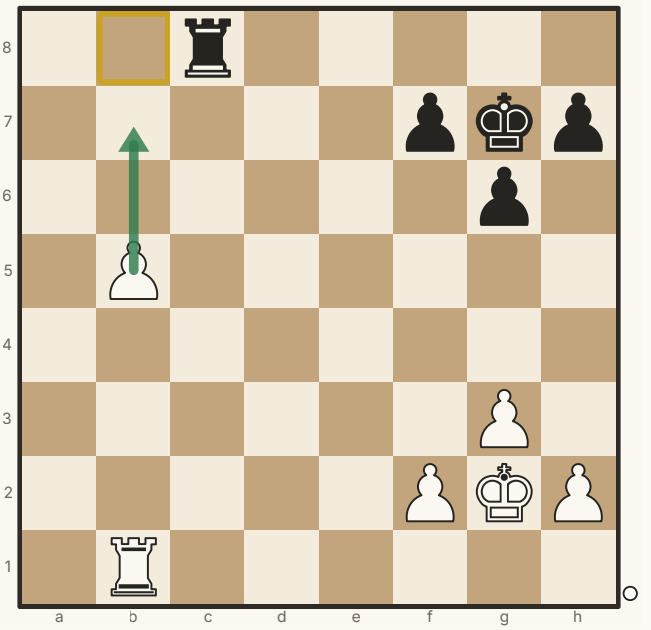
A hole is only a weakness if a piece can use it. The ideal occupant is a knight, for the reasons you met in Chapter 19: on a hole it cannot be chased by pawns, and from a central hole such as d5 or e6 (d4 or e3 for Black) it attacks squares deep in the enemy camp. Bishops and rooks can use holes too — a rook on a hole on the seventh rank, for example — but the knight is the classic tenant.

The plan against a hole has three steps. First, **control** the square with as many pieces as you can, so that the opponent cannot fill it with a pawn break such as ...d5. Second, **exchange** the defender that could take your piece there — often a knight on f6 or a bishop that covers the square. Third, **occupy** it, preferably with a knight supported by a pawn. If you are the side with the hole, you have the mirror plan: keep a minor piece that guards it, and prepare the pawn break that removes it.

Passed pawns

A **passed pawn** has no enemy pawn in front of it on its own file or the two adjacent files. Nothing but pieces can stop it, which is why passed pawns win endgames.

An old saying puts it simply: "passed pawns must be pushed." A passed pawn gets stronger with every step — it takes more and more of the opponent's force to stop it — and the opponent's best defence is to **blockade** it, placing a piece on the square directly in front. Aron Nimzowitsch built a whole theory of the blockade in his book *My System* (1925).



The rook belongs behind the passed pawn. As the b-pawn advances, White's rook gains squares, while Black's rook in front of it loses them.

40. **b6 Kf6** 41. **b7 Rb8** 42. **Kf3**

Black's rook is reduced to a blockader on b8. White's king now walks to the queenside or toward Black's pawns, and Black has no counterplay.

Three rules for passed pawns:

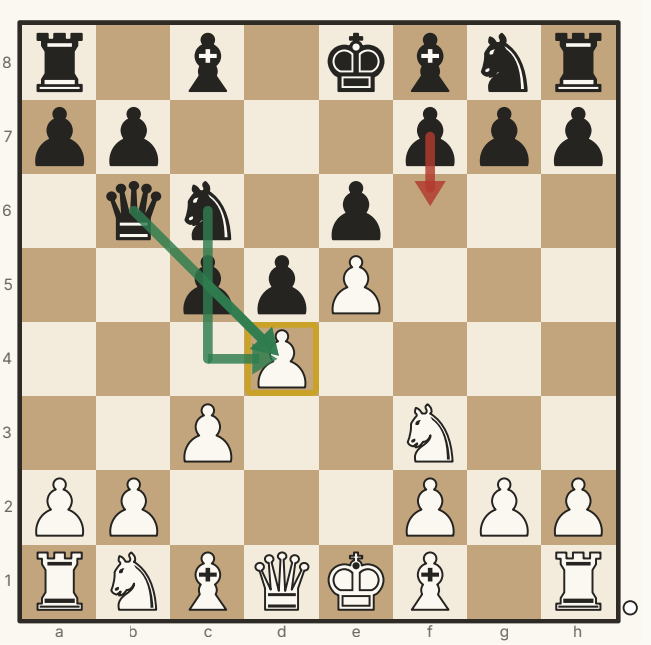
1. **Push it when it is safe** — every step forward ties down more enemy pieces. "Safe" matters: a push that drops a piece is not a push.
2. **Put rooks behind passed pawns** — yours or the opponent's. The rook behind gains activity as the pawn advances.
3. **Blockade enemy passed pawns with a knight.** A knight on the square in front of the pawn cannot be chased by a pawn and still attacks squares all around. A bishop is a decent blockader; a rook or queen is a poor one, because it is too valuable to be tied down.

Pawn chains and pawn breaks

A **pawn chain** is a diagonal line of pawns protecting each other, such as White's d4-e5 against Black's e6-d5 in the French Defence. The front of the chain is protected; the **base** — the pawn at the back, here d4 — is not protected by another pawn. Nimzowitsch's famous advice, also from *My System*, is to **attack the base**.

1. **e4 e6** 2. **d4 d5** 3. **e5 c5** 4. **c3 Nc6** 5. **Nf3 Qb6**

The French Advance. Black's pawn on c5, knight on c6 and queen on b6 all hit d4, the base of White's chain. White's c3 and Nf3 defend it.



The c5 pawn and the c6 knight attack d4 directly, and the queen on b6 stands behind c5 ready to join after ...cxd4. White defends with c3, Nf3 and the queen. The second break, ...f6, attacks the head of the chain on e5.

A pawn move that offers a pawn exchange like ...c5 here is called a **lever** or a **pawn break**. Breaks are how you open lines for your pieces and change a structure that does not suit you. Before you play one, answer three questions: which files and diagonals does it open, whose pieces are better placed for that, and what happens if the opponent does *not* take?

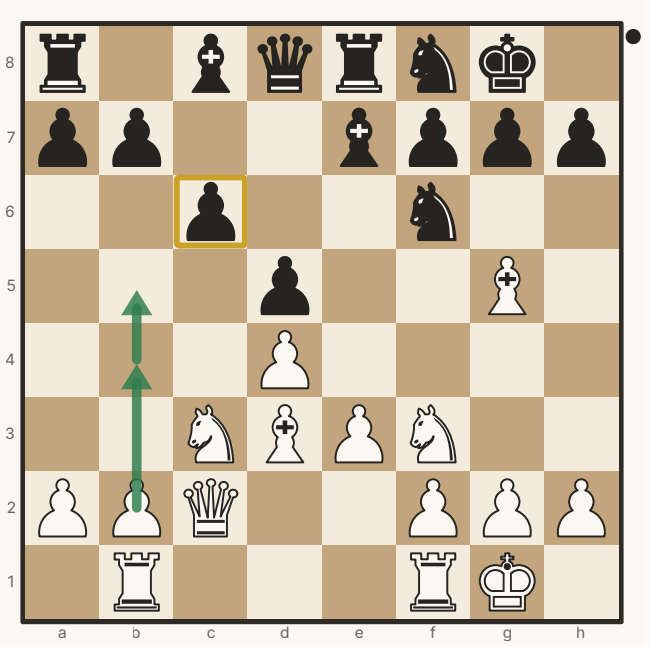
Majorities and the minority attack

When one side has more pawns on one wing than the opponent — for example three against two on the queenside — that side has a **pawn majority**. A healthy majority can create a passed pawn by itself. The method is to advance the **candidate**, the pawn without an opposing pawn in front of it, and let the others support it. Pushing the wrong pawn first lets the opponent hold two pawns with one.

Surprisingly, the side with *fewer* pawns on a wing can also attack there. The best-known example is the **minority attack** in the Queen's Gambit Exchange structure.

1.d4 d5 2.c4 e6 3.Nc3 Nf6 4.cxd5 exd5 5.Bg5 Be7 6.e3 c6 7.Bd3 O-O 8.Qc2 Nbd7 9.Nf3 Re8 10.O-O Nf8 11.Rab1

The Queen's Gambit Exchange. White's rook on b1 prepares b2-b4-b5, attacking Black's c6 pawn with two pawns against three.



The plan is b4 and b5. If Black takes on b5, the d5 pawn is isolated; if Black lets White take on c6, Black is left with a backward c-pawn on a half-open file. Either way White has created a target.

You will meet the minority attack again in Chapter 25. For now the lesson is general: a pawn advance can be used to *create* a weakness in the opponent's camp, not only to gain space.

Pawns can't move backwards

Every pawn move gives up control of the squares it used to guard. That is the whole reason pawn structure matters, and it leads to the most practical rule in this chapter: **think before every pawn move.**

Pawn moves near your own king are the most dangerous. The pawns in front of a castled king are its shield, and once they advance, the squares behind them are weak for ever — a theme for Chapter 21. In the middle of the board, a pawn move

can leave a hole for an enemy knight, like the d5 hole after ...e5 in the Sicilian. On the wings, a hasty pawn move can freeze your own majority.

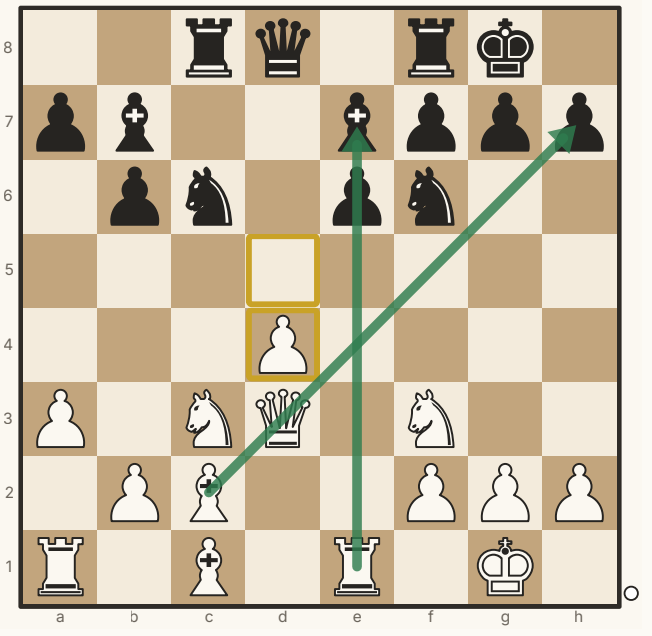
Before you push a pawn, ask:

- **What does this pawn stop guarding?** Name the squares.
- **Can an enemy piece use them?** A knight on a hole is the classic case.
- **Does the move fix a weakness — mine or my opponent's?**

> IN YOUR NEXT GAME **In your next game**
>
> Every time you are about to move a pawn, say (silently) the names of the squares it will never guard again. If one of them is next to your king or in the middle of the board, look for another move first.

Worked example: when the isolated pawn strikes

Here is the thinking process in an IQP position, from White's side.



White to move. A typical isolated-pawn middlegame: the queen and bishop aim at h7, the rook on e1 waits on the e-file behind Black's e6 pawn and the bishop on e7.

Evaluate. Safety scan first: nothing of White's is hanging, and Black's last moves (...b6, ...Bb7, ...Rc8) were natural development. The structure is an isolated pawn on d4. That tells me my plan before I look at any move: keep pieces on and use my activity before Black blockades d5 and trades pieces. My pieces are active: queen and bishop on the b1-h7 diagonal, rook on the e-file, knights covering e5 and d5. Black's knight on f6 is the only defender of h7.

Plan. The thematic break d4-d5 opens the e-file toward the bishop on e7 and the long diagonal for my queen. If the break works tactically, it is the best way to cash in my activity.

Candidate moves. 14.d5, 14.Bg5 (more pressure first) and 14.Bf4 (simple development).

Opponent's best reply to 14.d5. I check each capture. 14...Nxd5?? removes the only defender of h7: 15.Qxh7 is mate. 14...exd5 opens the e-file, and 15.Bg5 piles up on the bishop on e7 and the knight on f6.

14. **d5 exd5?!** 15. **Bg5 g6** 16. **Rxe7 Qxe7** 17. **Nxd5**

After 14...exd5 White wins material: the rook takes on e7, and after 16...Qxe7 17.Nxd5 the knight hits the queen and the f6-knight at once.

Black's best is to ignore the pawn and move the attacked knight, but even then White keeps the initiative.

14. **d5 Na5** 15. **Bg5**

White's pieces are all in play and Black's are still untangling; the isolated pawn has become an attacking weapon.

Move. 14.d5. The blunder check: the pawn on d5 is attacked three times, but each capture runs into a tactic, and no white piece hangs. If White had played slowly, Black would have played ...Nb4 or ...Nd5 next and the isolated pawn would have started to look like a weakness. The structure gave the plan; calculation confirmed the moment.

WORKED EXAMPLE **What the worked example teaches**

- Read the structure first: it tells you whether to play for activity (with the isolated pawn) or for simplification (against it).
- The thematic pawn break is correct when the files it opens are better for your pieces than for your opponent's.
- Check every capture of the advanced pawn before you push — tactics decide whether a break works.

Reading structures in your own games

When you review a game, pause after the opening and write down the pawn structure in one line: "isolated pawn on d4 for me", "his backward pawn on d6", "my doubled c-pawns". Then ask whether your next ten moves followed the plan that structure suggests. Players at this level often play good individual moves that belong to different plans. Choosing one plan from the pawns — and sticking to it — is a big step up.

Practice at the board

1. **Island counting.** FEN `4r1k1/p4ppp/2p1p3/8/4P3/8/PP3PPP/3R2K1 w - - 0 25`. Play White against a friend or an engine at a modest level and try to win one of the isolated black pawns. Then switch sides and try to defend with Black. Count how many moves your rook needs to reach each target.
2. **The rook behind.** FEN `2r5/5pkp/6p1/1P6/8/6P1/5PKP/1R6 w - - 0 40`. Push the b-pawn and keep your rook behind it. Check yourself: can Black's rook ever leave the b-file?
3. **The isolated pawn, both sides.** FEN `r1bq1rk1/pp2bppp/2n1p3/3n4/3P4/2NB1N2/PP3PPP/R1BQ1RK1 w - - 4 10`. Play ten moves as White aiming for d4-d5 or a kingside attack. Then reset and play Black aiming for a knight on d5 and piece trades. Ask the engine afterwards which plan went better.
4. **Majority drill.** Set up kings on e3 and e6, White pawns a2, b2, c2, g2, h2 and Black pawns a7, b7, f7, g7, h7 (`8/pp3ppp/4k3/8/8/4K3/PPP3PP/8 w - - 0 35`). Try to make a passed pawn on the queenside, first by starting with c4, then by starting with b4. Which works?

Chapter summary

- Pawns cannot move backwards: every pawn move gives up squares for ever. Think before you push.
- Fewer pawn islands are usually better; isolated pawns need pieces to defend them.
- The isolated queen's pawn is weak in the endgame and strong in the middlegame: with it, attack and prepare d4-d5; against it, blockade d5 and trade pieces.
- Doubled pawns are a weakness unless they bring open files, central control or the bishop pair.
- A backward pawn and the hole in front of it are targets; the freeing break must be prepared before it is played.
- Passed pawns must be pushed when it is safe; rooks go behind them; knights blockade them.
- Attack a pawn chain at its base; use levers to open lines for your better-placed pieces.
- Advance a majority with the candidate pawn first; a minority attack can create weaknesses on the opponent's wing.

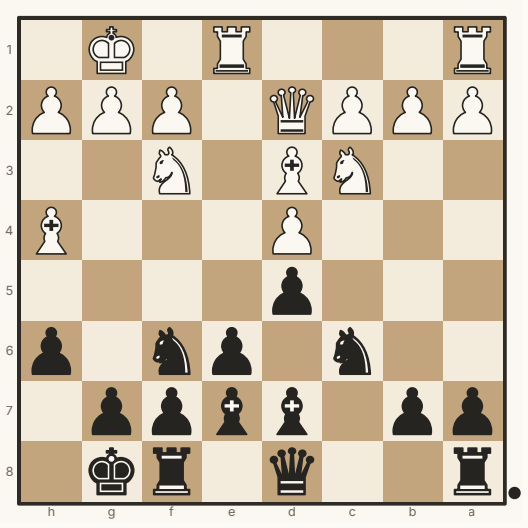
CHEAT CODE **Pawn structure — the decision procedure**

1. Describe the structure in one sentence: islands, isolated, doubled, backward, passed pawns.
2. List the weak squares (holes) for both sides, especially squares in front of weak pawns.
3. Choose the plan the structure suggests: attack the weakness, blockade it, or use the activity it gives you.
4. Before a pawn move, name the squares it will no longer guard and check whether an enemy piece can use them.
5. Before a pawn break, count the pieces on the lines it opens and check every capture.
6. Push passed pawns when safe, put rooks behind them, blockade enemy ones with a knight.

Exercises

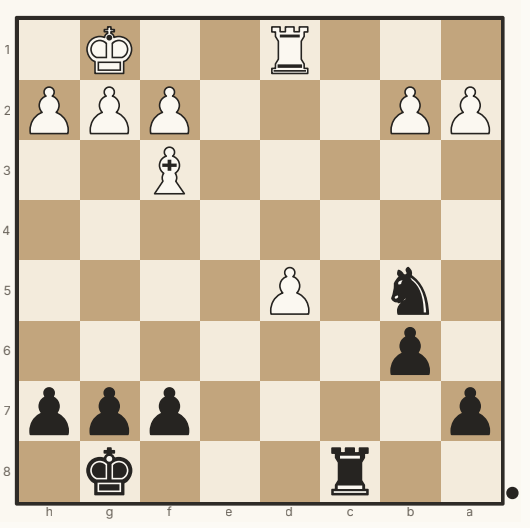
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

20.1 BLUNDER CHECK



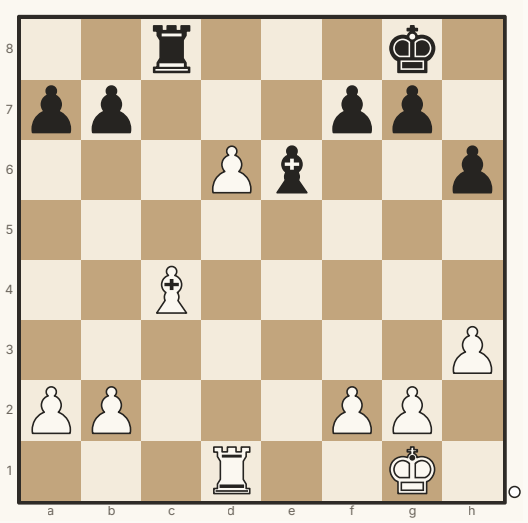
Black to move. The bishop on h4 pins your knight, and ...g5 would chase it away. Is that a good idea? Play the best move. Solution p. 697

20.2 DEFENCE



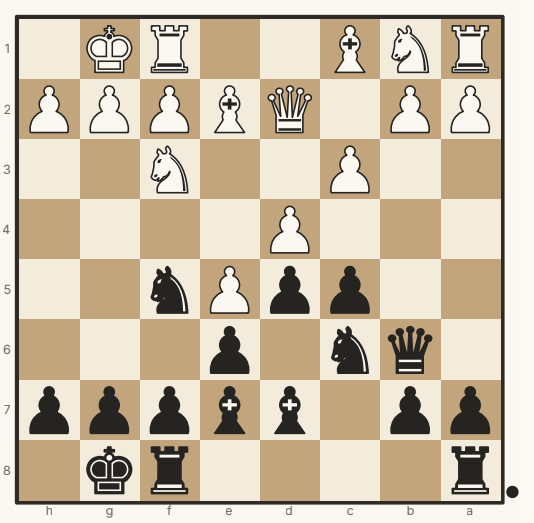
Black to move. White's d-pawn is passed and supported by the rook behind it. Stop it. Solution p. 697

20.3 CONVERSION



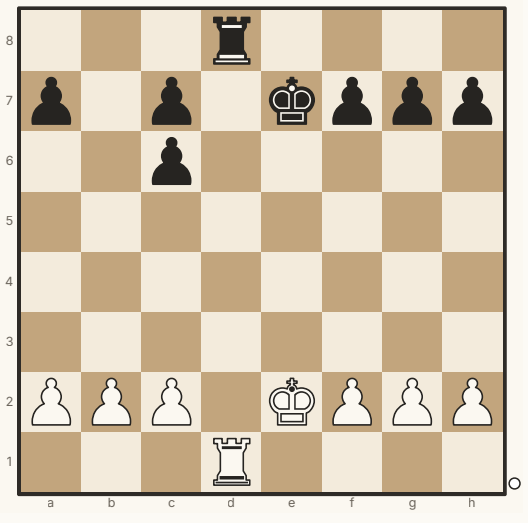
White to move. Your passed d-pawn is blockaded only by the bishop on e6. What do you do? Solution p. 698

20.4 STRATEGY



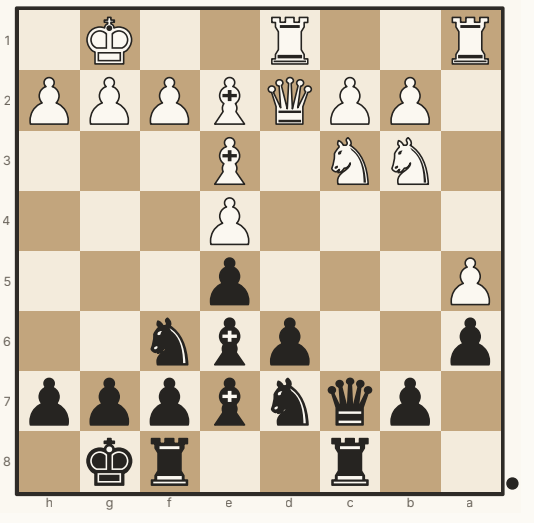
Black to move. White's pawn chain runs d4-e5. Where should you strike? Solution p. 698

20.5 ENDGAME



White to move. Black has doubled c-pawns. Should you trade rooks? Solution p. 698

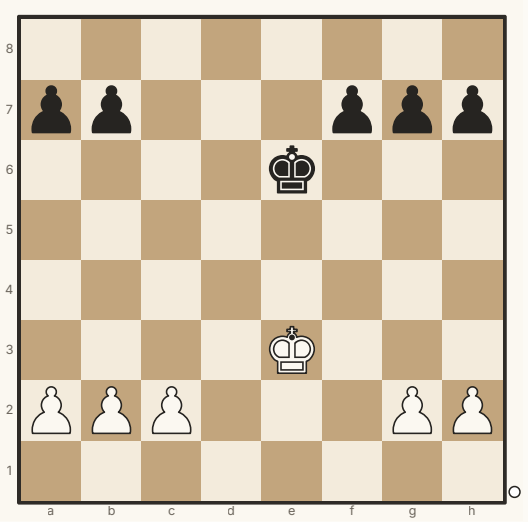
20.6 BLUNDER CHECK



Black to move. Your d6 pawn is backward and White's pieces eye d5. Is the freeing break ...d5 possible? Play the best move.

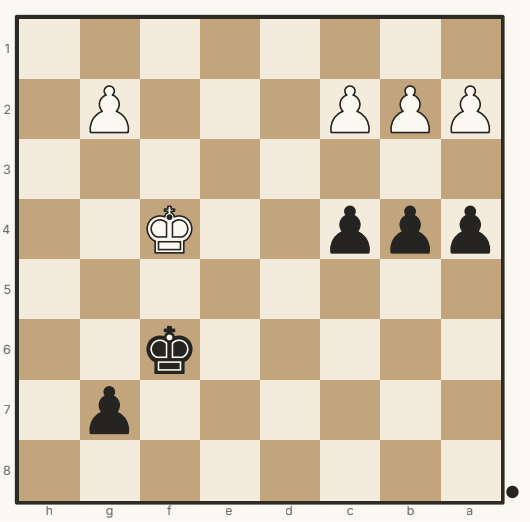
Solution p. 698

20.7 ENDGAME



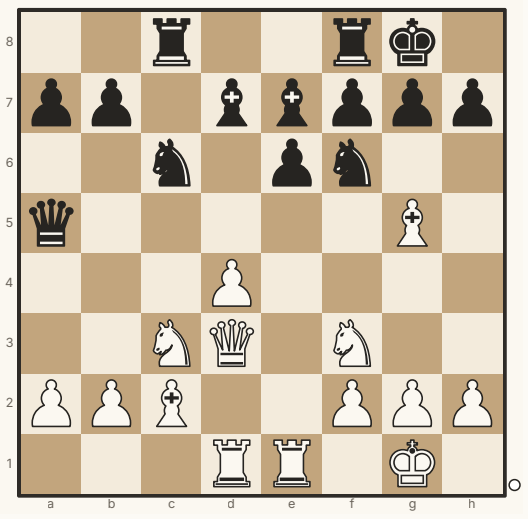
White to move. You have three queenside pawns against two. One natural pawn move throws the majority away. Avoid it and play a sound move. Solution p. 699

20.8 ENDGAME



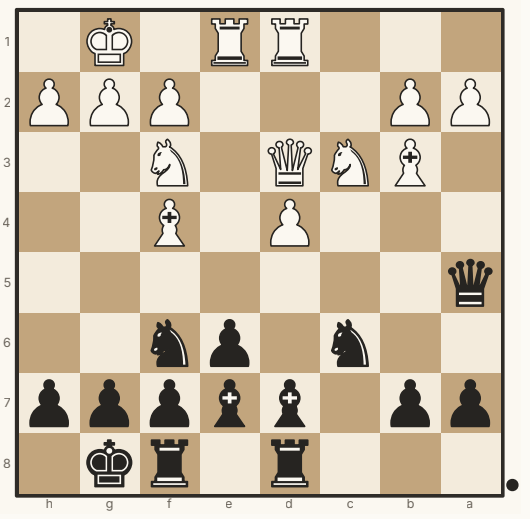
Black to move. White's king is close to your queenside pawns. Which move saves the game? Solution p. 699

20.9 TACTIC



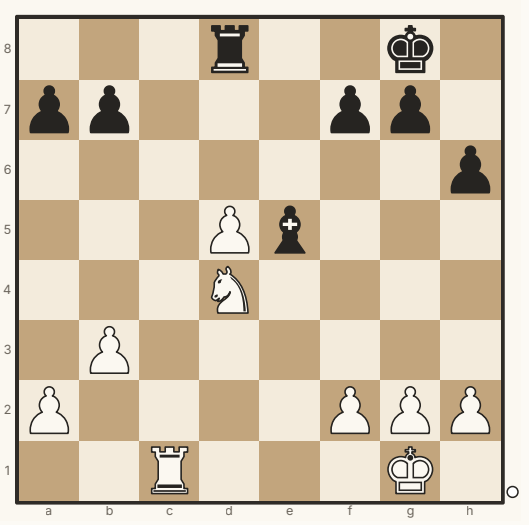
White to move. You have the isolated d-pawn — and the active pieces that come with it. Use them. Solution p. 699

20.10 STRATEGY



Black to move. You are playing against White's isolated d-pawn. The square d5 is the key. How do you get control of it? Solution p. 699

20.11 CALCULATION



White to move. Your passed d-pawn is attacked by the rook, and your knight is attacked by the bishop. Find the move that keeps both. Solution p. 700

HINTS

20.1 Before every pawn move near your king, ask what squares it leaves behind.

20.2 Which piece blockades best, and on which square?

20.3 What piece stops your pawn, and can you take it?

20.4 Count attackers and defenders of the base pawn.

20.5 Imagine the position with only kings and pawns. Whose pawns are healthier?

20.6 After the pawn break, which of your pieces would stand on the open d-file?

20.7 Which of your queenside pawns has no black pawn in front of it?

20.8 Which pawn move attacks two white pawns at once?

20.9 What does the f6-knight defend?

20.10 Which move gains time before you blockade?

20.11 Where can the knight go so that ...Rxd5 becomes a mistake?

CHAPTER 21 STAGE 1000–1200

King Safety and Attacking the King

Count the attackers, count the defenders, then strike

YOU WILL LEARN TO

- Judge king safety from the pawn shield, the open lines and the number of attackers and defenders nearby.
- Make luft with the right pawn and understand the weaknesses created by ...h6, ...g6 and ...f6.
- Attack with pawn storms when the kings have castled on opposite sides, and with pieces when they are on the same side.
- Recognize the typical sacrifices — Bxh7+, Nxf7, Rxf6 and Bxg7 — and the conditions that make them work.
- Punish a king stuck in the centre and defend your own king by trading attackers and counter-attacking.

> **IN YOUR NEXT GAME**
>
> Before every move, count the attacking pieces near each king — yours and your opponent's. If one side has more attackers than the other has defenders, that is where the game will be decided.

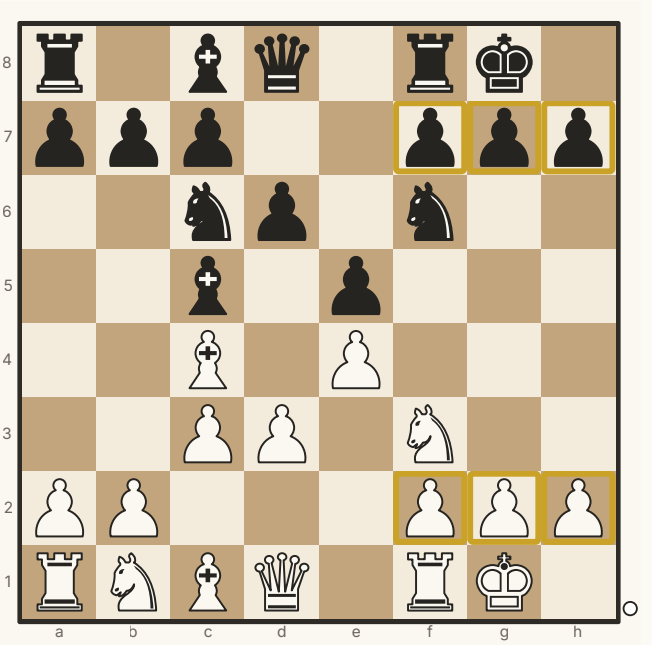
At every level, the quickest way to win a game is to checkmate the opponent's king, and the quickest way to lose one is to neglect your own. Between 1000 and 1200 you will start to win games with attacks that are not just one-move tactics: you bring pieces closer, open a file, sacrifice to break the pawn shield, and mate. You will also lose games because a single pawn move in front of your king gave your opponent a target.

This chapter is about both sides of that fight. You will learn to *read* king safety — pawns, open lines, attackers and defenders — and then the main methods of at-

tack and defence. The theme throughout is counting. An attack usually succeeds when the attacker has more pieces near the king than the defender can bring.

The pawn shield

After castling, the three pawns in front of the king — f2, g2 and h2 for White on the kingside — form its **pawn shield**. Unmoved, they cover all the squares the king needs and give enemy pieces no entry points.



Both kings are perfectly safe: their pawn shields are untouched, a knight stands on f3 or f6 to cover h2 or h7, and no files are open toward either king.

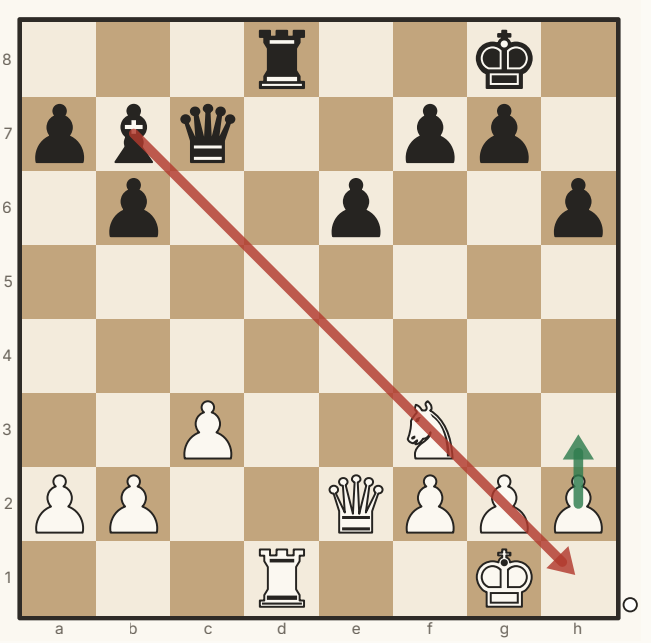
Three things make a castled king safe:

1. **Intact shield pawns.** Every pawn move in front of the king creates weak squares that can never be covered by pawns again.
2. **A defending piece nearby**, especially a knight on f3 (or f6), which guards h2 (h7) and stops many sacrifices.
3. **No open lines** toward the king — no open files for enemy rooks, no open diagonals for enemy bishops.

When you assess king safety, check those three in order, and then count: how many enemy pieces can reach the king's area in one or two moves, and how many of the defender's pieces can come to help?

Luft: when and which pawn

Sometimes you must move a shield pawn. The most common reason is the back rank (see Chapter 11): with rooks or queens still on the board, a king with no escape square can be mated by a single check on the first rank. A pawn move that gives the king an escape square is called **luft** (German for "air").



White to move. The back rank is a problem: the king has no square. h3 gives luft safely; g3? would weaken the light squares, and Black's bishop on b7 is waiting on the long diagonal.

24. **h3 Rxd1+** 25. **Qxd1**

With luft on h2, White's king is safe and the position is balanced. After 24.g3? Rxd1+ 25.Qxd1 Qc6 the queen and bishop on the long diagonal threaten f3 and h1, and White is in serious trouble.

The guideline: make luft with the **h-pawn** (h3 or ...h6) when the opponent's bishop controls the long light diagonal or could use the g3 square, and choose the pawn whose move creates squares the opponent *cannot* use. Make luft early, at a quiet moment, rather than when you are already in trouble.

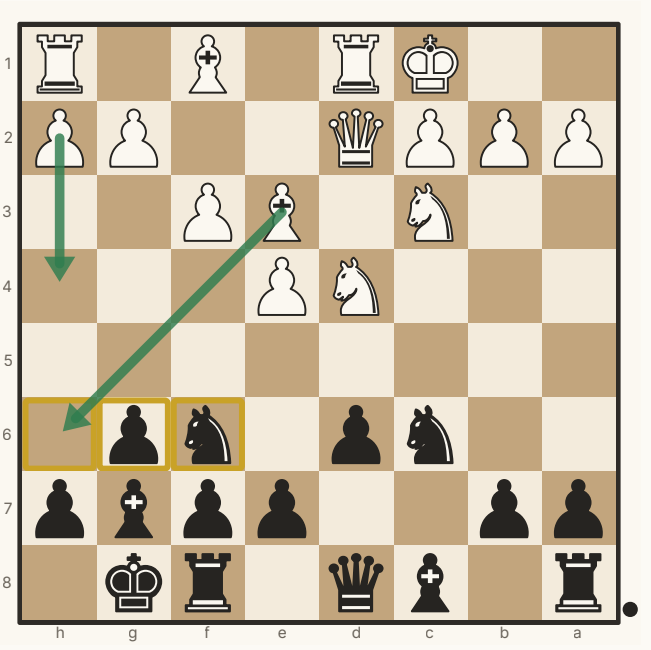
The weaknesses of ...h6, ...g6 and ...f6

Each shield pawn move gives up different squares. For Black (White is a mirror image):

- **...h6** weakens g6 and, more importantly, creates a **hook**: a pawn on h6 can be attacked by g4-g5, which opens the g-file. It also makes sacrifices on h6 possible (Bxh6 when the g7 pawn is pinned or overloaded).
- **...g6** weakens the dark squares f6 and h6. If Black's dark-squared bishop disappears, a white queen or knight on those squares can be deadly. It also creates a hook on g6 for h4-h5.
- **...f6** opens the a2-g8 diagonal to the king and weakens e6 and g6. A white bishop on c4 or b3 then points straight at the king.

1.e4 c5 2.Nf3 d6 3.d4 cxd4 4.Nxd4 Nf6 5.Nc3 g6 6.Be3 Bg7 7.f3 O-O 8.Qd2 Nc6 9.O-O-O

The Yugoslav Attack in the Sicilian Dragon, a famous opposite-castling battle.



Black's ...g6 put the bishop on g7, but it also made h6 and f6 sensitive. White's plan is Bh6 to trade the bishop that guards those squares, followed by h4-h5 against the hook on g6. Black's plan is a race on the other wing.

None of these moves is automatically bad. ...g6 with a bishop on g7 is a sound defensive setup, and ...h6 is often useful to stop Bg5 or Ng5. The rule is: make such a move for a reason, and then remember which squares it gave away.

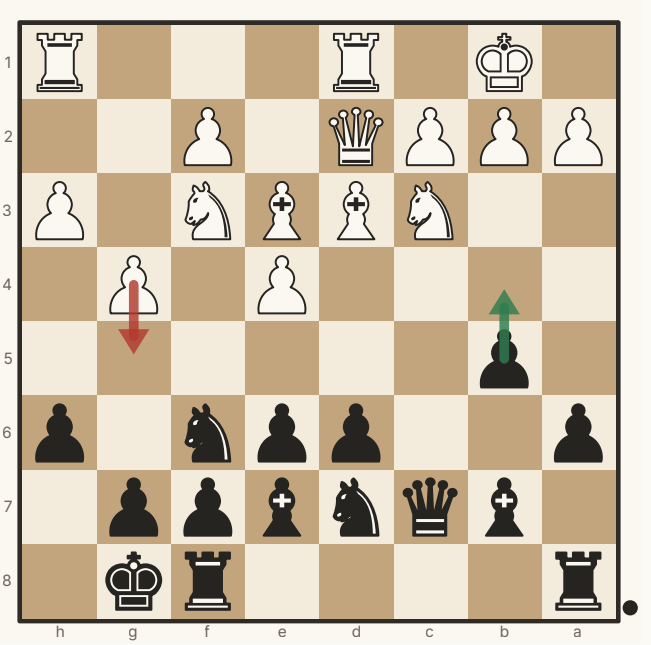
Opposite-side castling: the pawn storm

When the kings have castled on **opposite sides**, the game changes character. You can throw the pawns in front of the *opponent's* king forward without weakening your own king, because your own king is on the other wing. These advancing pawns form a **pawn storm**.

Opposite-side castling positions are races. Two practical rules:

- **Speed matters more than material.** A pawn sacrificed to open a file toward the enemy king is often well spent. A tempo spent on a "safety" move is often a tempo lost.
- **Use the hooks.** An advanced enemy pawn on h6 or g6 (h3 or g3 for White) gives your pawns something to attack, so files open faster.

When you are defending in such a race, do not only defend. Counter-attack on your own wing, and if the centre can be opened, do that — an open centre usually helps the side whose pieces are ready.



Black to move in a race. White's g4-g5 is coming against the hook on h6, but Black is first: ...b4 hits the knight that defends e4 and opens lines toward White's king.

15...**b4** 16.**Na4 Nxe4**

Black wins the e4 pawn and his attack is already rolling, while White's storm has not started.

Same-side castling: attacking with pieces

When both kings are on the same side, a pawn storm weakens your own king too. The attack must come mainly from **pieces**. The method is the counting from the start of this chapter:

1. **Bring more attackers than he has defenders.** Count the pieces that can reach the king's area in one or two moves.
2. **Remove or deflect the key defender** — usually the knight on f6, the bishop on g7, or the pawn on g7 or h7.
3. **Open a line** — a file for a rook, a diagonal for a bishop — or use a **rook lift** (Re3-g3, Rf3-h3) to bring a rook into the attack when the files are closed.

When is it time to attack?

Many attacks at this level fail because they start too early, with two pieces against four defenders. Before you commit — especially before you push pawns or sacrifice — run a short check:

- **Is my centre stable?** If the opponent can open the centre with a pawn break, your attacking pieces may be cut off or your own king exposed. Attacks on the wing work best when the centre is closed or firmly under your control.
- **Are my pieces pointing at the king?** Count the pieces that already aim at the king's area: bishops on the diagonals toward g7 or h7, a knight that can reach g5 or f5, a queen that can reach h5 or the g-file, a rook that can lift.
- **Is there a target?** A weakened shield pawn (h6, g6 or f6), a missing defender (no knight on f6), or an open file.

If two of the three answers are yes, start bringing pieces. If all three are yes, look for a forcing blow now — this is when sacrifices appear.

Typical sacrifices

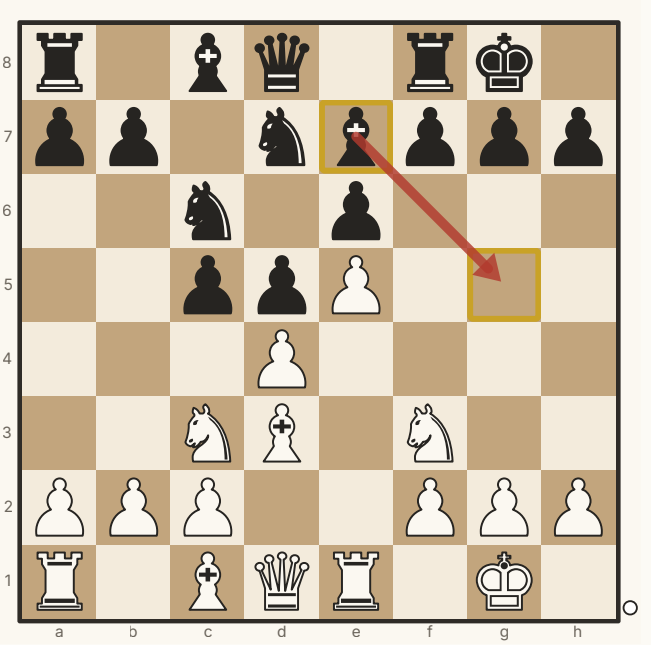
Most attacks against the castled king end with one of a handful of sacrifices. Learn the *conditions* for each, not just the move.

Bxh7+ (the Greek gift)

You met the Greek gift in Chapter 11. The pattern is Bxh7+ Kxh7 Ng5+ followed by Qh5 (or Qg4, Qd3+). It usually works when:

- the bishop is on d3 (or the b1-h7 diagonal) and nothing defends h7 except the king;
- a knight can reach g5 and **no black piece can take it there** (watch out for a bishop on e7 or a queen on d8 with the diagonal open);
- the queen can reach h5 quickly;
- Black has **no knight on f6** — often a white pawn on e5 has driven it away;
- the black king cannot run to g6 and then escape safely.

The second condition is the one players forget most often. Here is what happens when it fails:



Everything looks right — bishop on d3, knight on f3, pawn on e5 — but the bishop on e7 covers g5. The sacrifice fails.

9. **Bxh7+? Kxh7**

After 10.Ng5+ Bxg5 the knight is simply gone, and White has given a bishop for a pawn. Count the defenders of g5 before you sacrifice on h7.

Nxf7, Rxf6, Bxg7 and Rxh7

- **Nxf7** (or Bxf7+) works when f7 is defended only by the king — typically when the king is still on e8 or the rook has left f8. It is especially strong when the e6 pawn is pinned, so that the queen can follow on e6.
- **Rxf6** (an exchange sacrifice) removes the knight that defends h7. If Black recaptures with the g-pawn, the king's shield is wrecked; if he does not, he is a piece down. The idea works whenever h7 is attacked twice and defended only by the knight.
- **Bxg7** breaks the pawn shield when the queen can follow on g4, g5 or h6, or when a rook can swing to the g-file. It often comes after a first sacrifice on h7 — the famous "double bishop sacrifice", in which a bishop is given on h7, the second on g7, and the queen and a lifted rook finish the job. Check before playing it that the black king cannot run to f8 and e7, and that your queen really reaches the g-file with check.

- **Rxh7** or Qxh7+ with a rook on the h-file finishes many opposite-castling attacks once the h-file is open.

> KEY IDEA **Count before you sacrifice**
>
> List the attackers that will still be in play after the sacrifice, then list the defenders that can arrive. If the attackers do not outnumber the defenders near the king, the sacrifice is probably unsound — however beautiful.

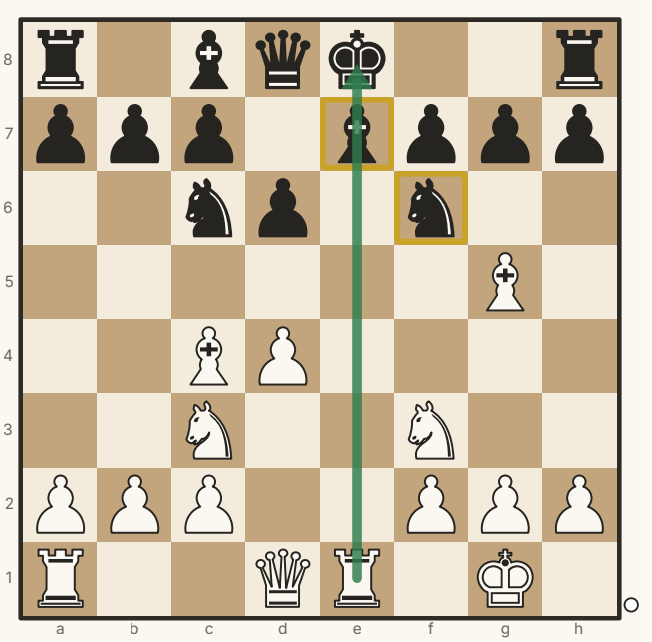
The king in the centre

A king that has not castled is vulnerable as soon as the centre opens. The e-file is the main highway: a rook or queen on e1 against a king on e8 pins anything that stands between them. The method for punishing an uncastled king is:

- **Open the centre** — with a pawn break (d4-d5, e4-e5) or a sacrifice — before the king escapes.
- **Use the pins on the e-file**, which stop the defender's pieces and pawns from capturing.
- **Prevent castling** by controlling f8 or f1, or by giving a check that forces the king to move.

If *your* king is still in the centre and the position is opening, castle — and do not grab a pawn that opens yet another line toward your own king.

Worked example: the uncastled king



White to move. Black has not castled, the e-file is open, and the bishop on e7 is pinned against the king by the rook on e1.

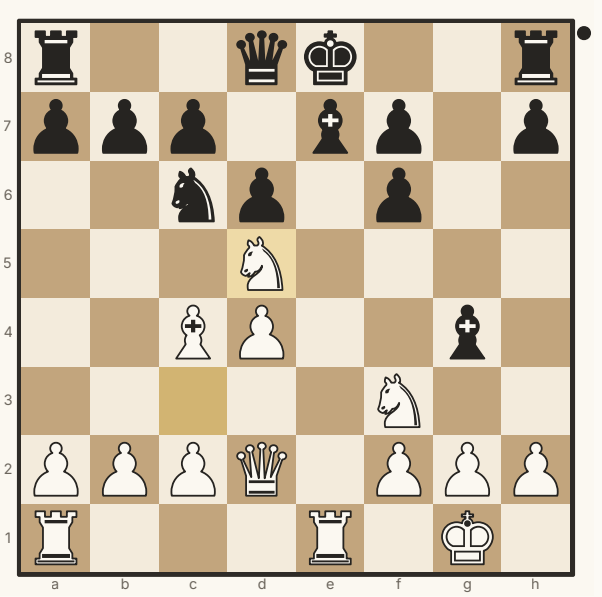
Evaluate. Safety scan first: nothing of White's is attacked, and Black's only threat is to castle next move, which would end most of White's chances. White has more pieces near the black king: the rook on e1, the bishops on c4 and g5, two knights. Black's king has no pawn shield problems yet, but it is still on e8, and the bishop on e7 is pinned — it cannot move without exposing the king to Re1.

Plan. Use the pin now, before ...O-O. The knight on f6 is the piece that holds Black's position together: it guards d5, e4 and h7, and it stands between the g5-bishop and the pinned bishop on e7.

Candidate moves. 10.Bxf6, 10.d5 (kicking the c6-knight and opening the centre) and 10.Qe2 (adding to the pressure on e7).

Opponent's best reply to each. After 10.Qe2 Black simply castles and is fine. After 10.d5 Ne5 11.Nxe5 dxe5 12.Rxe5 White wins a pawn — good, but there is more. After 10.Bxf6 the obvious recapture ...Bxf6 is *illegal*: the bishop on e7 is pinned. Black must take with the g-pawn.

10. **Bxf6 gxf6** 11. **Qd2 Bg4** 12. **Nd5**



Black's pawn shield is wrecked before the king has even castled, the e7-bishop is still pinned, and the knight on d5 joins the attack on e7 and f6. White is winning.

Move. 10.Bxf6. The blunder check: after 10...gxf6 no white piece is loose, and the pin on e7 remains. Notice the order of thought: the pin was the special feature of the position, so the first question was "what can I capture that he cannot recapture normally?" A king in the centre often hands you exactly that.

> **WORKED EXAMPLE** **What the worked example teaches**
>
> - An uncastled king turns e-file pins into weapons: pieces and pawns between the king and your rook cannot recapture.
> - Strike before the king castles; one tempo can decide.
> - Removing a key defender (here the f6-knight) is often stronger than winning a pawn.

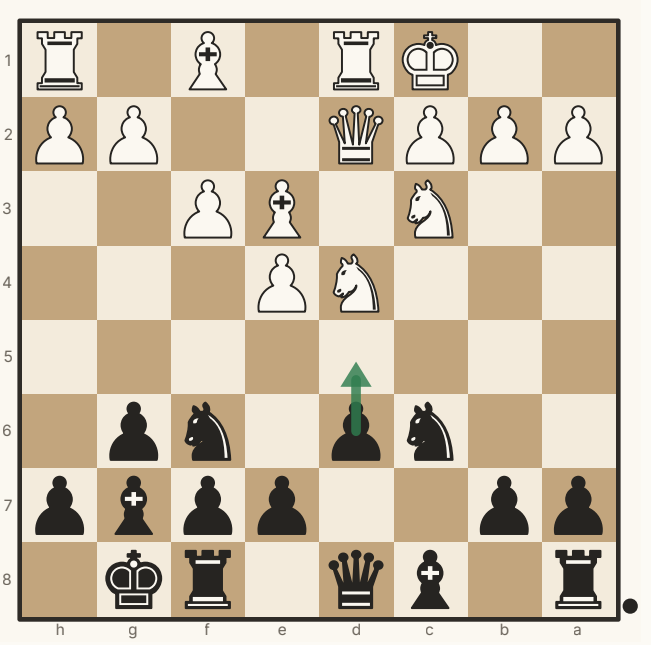
Defending your king

Defence is not passive waiting. The good defender has a plan too:

1. **Trade the attackers.** Every exchange of an attacking piece reduces the count near your king. A knight on g5 or a bishop on h6 is often worth taking even at a small cost.

2. **Bring defenders**, especially a knight to f6 (f3 for White) or a bishop back to g7 or f8.
3. **Avoid new pawn weaknesses.** If you must move a shield pawn, choose the one that blocks the attacking line or gains a tempo, and weakens the fewest squares.
4. **Counter-attack in the centre.** A flank attack is often best met by opening the centre, where the attacker's pieces may be stuck on the wrong side.
5. **Give back material** if it breaks the attack. A pawn returned to trade queens can save the game.

The fourth point deserves an example. Go back to the Dragon position from earlier in this chapter. White has castled long and wants Bh6 and h4-h5. Black's most effective answer is not a defensive move on the kingside at all, but a strike in the centre.



Black to move. Instead of waiting for Bh6 and h4-h5, Black strikes in the centre with ...d5 and fights for the initiative before White's storm arrives.

9...**d5** 10.**exd5 Nxd5** 11.**Nxc6 bxc6** 12.**Bd4**

The centre is open, Black's pieces are active and the bishop on g7 bites along the long diagonal toward White's queenside. White has had to give up the attacking plan to deal with the centre.

Notice what ...d5 did. White's planned pawn storm needed time, and opening the centre took that time away: White had to react to threats in the middle of the board before any pawn reached Black's king. A defender who only shuffles on the kingside gives the attacker all the time he needs.

> COMMON MISTAKE **The panic pawn move**
>
> When your king is threatened, the first move that comes to mind is often a pawn move in front of it — ...h6, ...g6 or ...f5 — because it "makes a wall". Each one creates new targets. Before playing it, look for a piece that can defend instead, and if a pawn move is needed, compare the squares each one gives away.

Attacking and defending in your own games

In your game reviews, mark every move where you moved a pawn in front of your castled king and write down why. Then mark every attack you started and count the attackers and defenders at the moment you began. You will usually find that your failed attacks started with too few pieces, and that most of your king's problems began with a pawn move you did not need to make.

> IN YOUR NEXT GAME **In your next game**
>
> Before every move, glance at both kings and count: attackers near the enemy king, defenders near yours. When your count is bigger, look for a sacrifice or a line-opening move. When his is bigger, bring a defender or trade an attacker before anything else.

Practice at the board

1. **The failed Greek gift.** FEN `r1bq1rk1/pp1nbppp/2n1p3/2ppP3/3P4/2NB1N2/PPP2PPP/R1BQR1K1 w - - 6 9`. Play 9.Bxh7+ and try to make the attack work against an engine; then take back and play a normal move such as 9.Bf4. Which defender of g5 ruined the sacrifice?
2. **Opposite-side race.** FEN `r1bq1rk1/pp2ppbp/2np1np1/8/3NP3/2N1BP2/PPPQ2PP/2KR1B1R b - - 4 9`. Play this Dragon position several times from both sides

against a friend or the engine at a modest level. As White, aim for Bh6 and h4-h5; as Black, for ...Rc8, ...Ne5 and ...b5. Who gets there first?

3. **Luft choice.** FEN `3r2k1/pbq2pp1/1p2p2p/8/8/2P2N2/PP2QPPP/3R2K1 w - - 0 24`. Play the position after 24.h3 and after 24.g3. Notice how often Black's bishop on b7 matters in the second case.
4. **Count drill.** Take three middlegame positions from your recent games. For each king, write down the number of enemy pieces that could reach its area in two moves and the number of defenders. Compare with what happened in the game.

Chapter summary

- A king is safe when its pawn shield is intact, a defending piece is nearby, and no lines are open toward it.
- Make luft early and with the pawn that weakens the fewest useful squares, usually the h-pawn.
- ...h6, ...g6 and ...f6 each give away specific squares and create hooks; play them for a reason.
- With opposite-side castling, storm with pawns and value speed over material; use the opponent's hooks.
- With same-side castling, attack with pieces: more attackers than defenders, remove the key defender, open a line.
- Learn the conditions of Bxh7+, Nxf7, Rxf6 and Bxg7, and count attackers and defenders before you sacrifice.
- Punish an uncastled king by opening the centre and using pins on the e-file.
- Defend by trading attackers, bringing defenders, avoiding panic pawn moves and counter-attacking in the centre.

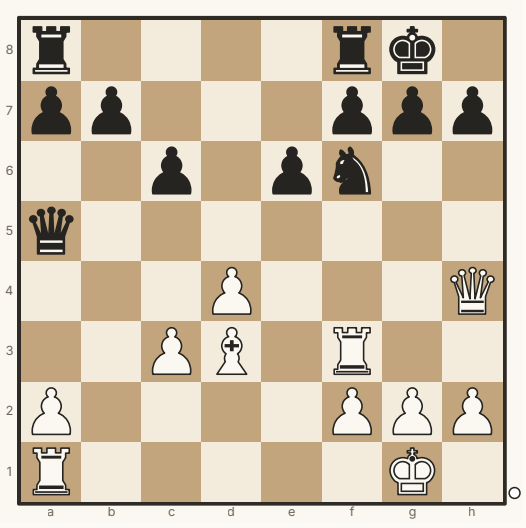
CHEAT CODE **King safety — the attack-and-defence procedure**

1. For both kings, check the shield pawns, the nearby defenders and the open lines.
2. Count attackers that can reach the king's area in one or two moves against the defenders that can arrive.
3. Opposite-side castling: storm with pawns at once; same-side: attack with pieces and a rook lift.
4. Before a sacrifice, list the conditions (defender of g5, knight on f6, escape squares) and count again.
5. Defending: trade the strongest attacker, bring a defender, and only then consider a shield pawn move — the one that blocks the line or gains a tempo.
6. If a king is still in the centre, open the centre (or castle, if it is yours).

Exercises

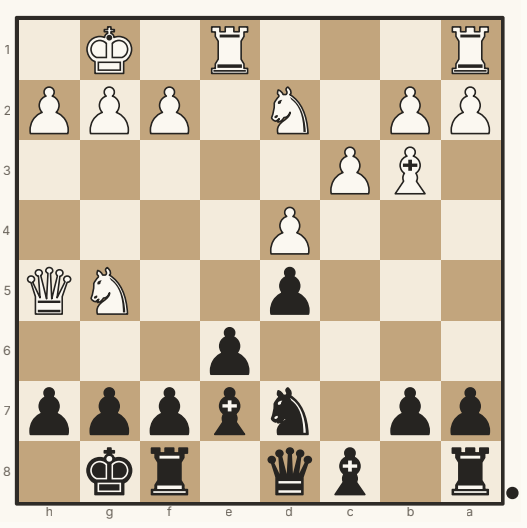
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

21.1 TACTIC



White to move. Your queen and bishop both aim at h7. Count its defenders and win material. Solution p. 700

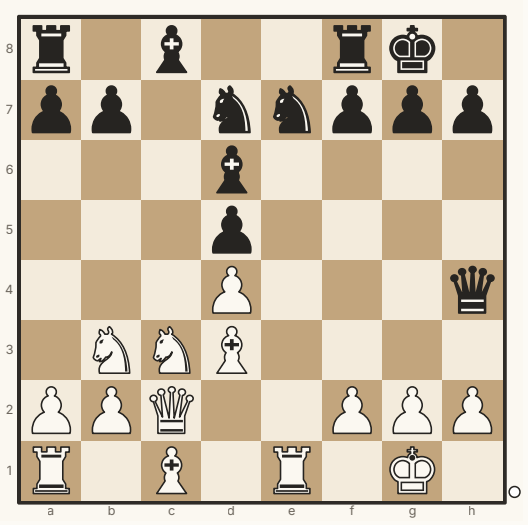
21.2 TACTIC



Black to move. White threatens Qxh7 mate. Find the defence that also wins material.

Solution p. 700

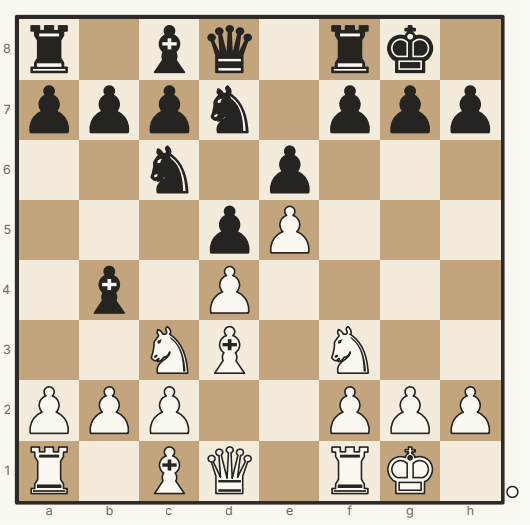
21.3 DEFENCE



White to move. Black threatens ...Qxh2+ and mate. Which pawn move defends best?

Solution p. 701

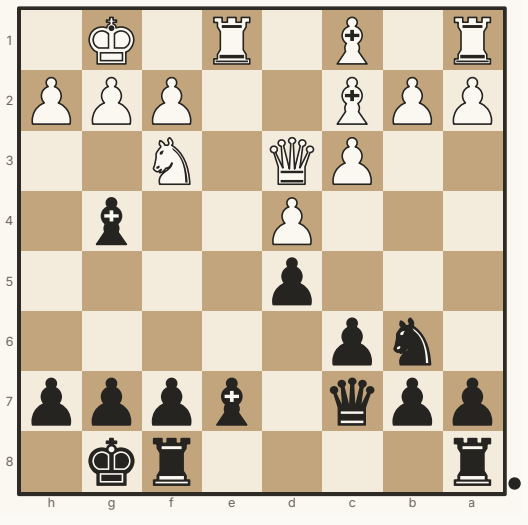
21.4 TACTIC



White to move. Check the conditions for the Greek gift sacrifice, then play it if it works.

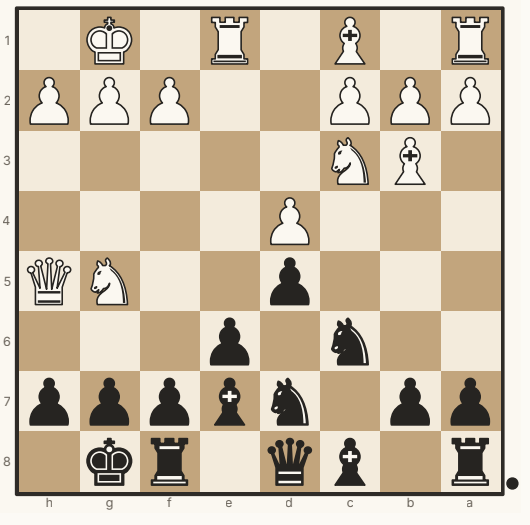
Solution p. 701

21.5 DEFENCE



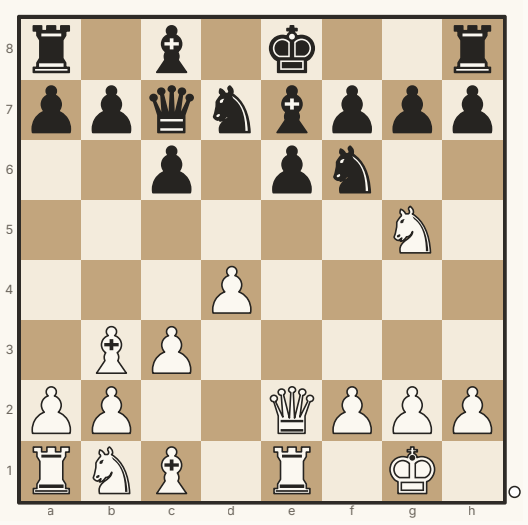
Black to move. White's queen and bishop threaten Qxh7 mate. Which pawn move weakens your king least? Solution p. 701

21.6 DEFENCE



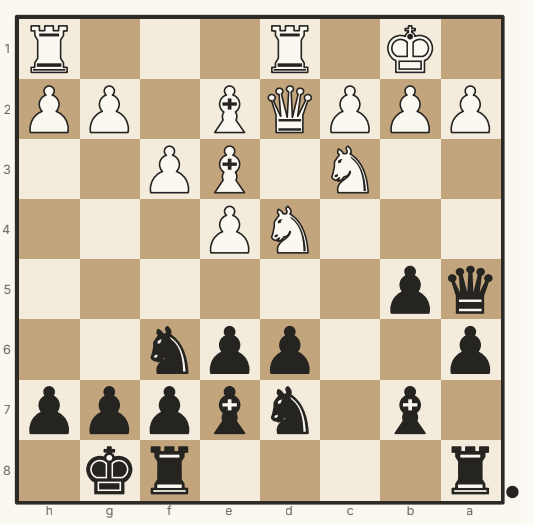
Black to move. White threatens Qxh7 mate. Find the best defence. Solution p. 701

21.7 TACTIC



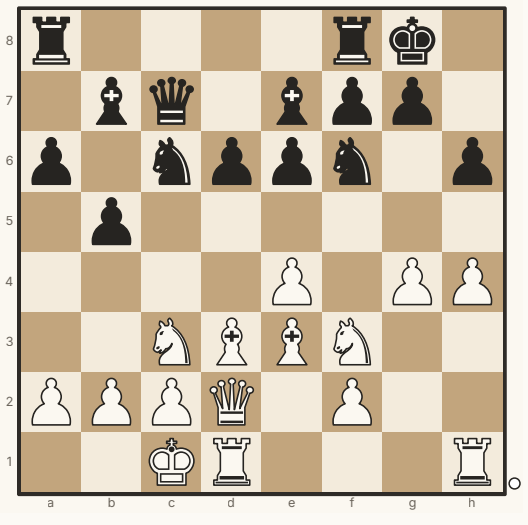
White to move. Black's king is still in the centre and the e6 pawn is pinned against it. Strike. Solution p. 702

21.8 STRATEGY



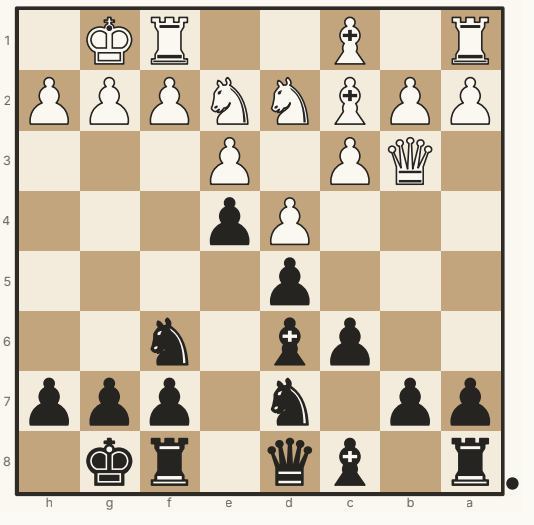
Black to move. The kings have castled on opposite sides. Choose your plan move.
Solution p. 702

21.9 STRATEGY



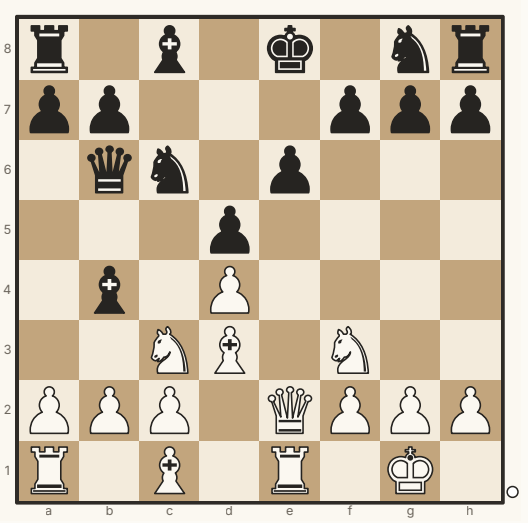
White to move. Kings on opposite wings, and Black's pawn on h6 is a hook for your pawns. Choose the best plan move.
Solution p. 702

21.10 TACTIC



Black to move. Your bishop, knight and queen are all pointed at White's king. Is there a sacrifice? Solution p. 702

21.11 TACTIC



White to move. Black's king is still on e8 and the e-file is half-open. Punish it.

Solution p. 703

HINTS

21.1 Which black piece is the only defender of h7?

21.2 Which attacking piece can you simply take?

21.3 Which pawn move also attacks something?

21.4 Bishop d3, knight g5, queen h5 — is anything missing?

21.5 Which squares lie between the bishop on c2 and h7?

21.6 Compare the two ways of getting rid of the knight on g5.

21.7 Which black pawn cannot move, and which square does only the king defend?

21.8 Which pawn can attack a defender of White's king or centre?

21.9 Which pawn move attacks a defender of Black's king?

21.10 Can any white piece take a knight on g4?

21.11 Which black pawn cannot move because of the queen on e2?

CHAPTER 22 STAGE 1000–1200

Making a Plan

What to do when there is nothing to calculate

YOU WILL LEARN TO

- Evaluate a position with a short checklist instead of a vague feeling.
- Find targets: weak pawns, weak squares, bad pieces and an exposed king.
- Choose a plan from the pawn structure, and improve your worst piece.
- Ask "What does my opponent want?" and stop it when it matters (prophylaxis).
- Make short, concrete plans of two or three moves, and change them when the position changes.

> **IN YOUR NEXT GAME**
>
> When there is no tactic for either side, run the evaluation checklist, name your worst piece and your opponent's main idea, and choose a move that improves the first or stops the second.

You have castled, developed your pieces, and checked for tactics. Nothing hangs. Nobody threatens anything. Now what? Many players at this level freeze here, or they play a "random sensible move," or they launch an attack that the position does not support. A few moves later they are worse and do not know why.

This chapter gives you a way to decide what to do in quiet positions. It is not magic and it is not a twenty-move master plan. It is a short routine: evaluate, find targets, read the pawn structure, improve your worst piece, and check what your opponent wants. Masters do this almost automatically. You can learn to do it deliberately, and with practice it becomes quick.

What a plan is (and is not)

A **plan** is a short sequence of moves with a purpose: "put my knight on f5," "push b4 and b5 to weaken the c6-pawn," "trade my bad bishop." A good plan comes *from the position*: it uses something that is already there, such as a weak pawn, an open file or a badly placed enemy piece.

A plan is not a dream. "I will attack the king, sacrifice on h7 and mate him" is not a plan if the pieces are not there and the position does not ask for it. At club level the most useful plans are **two or three moves long**. You make the plan, play it, then look again.

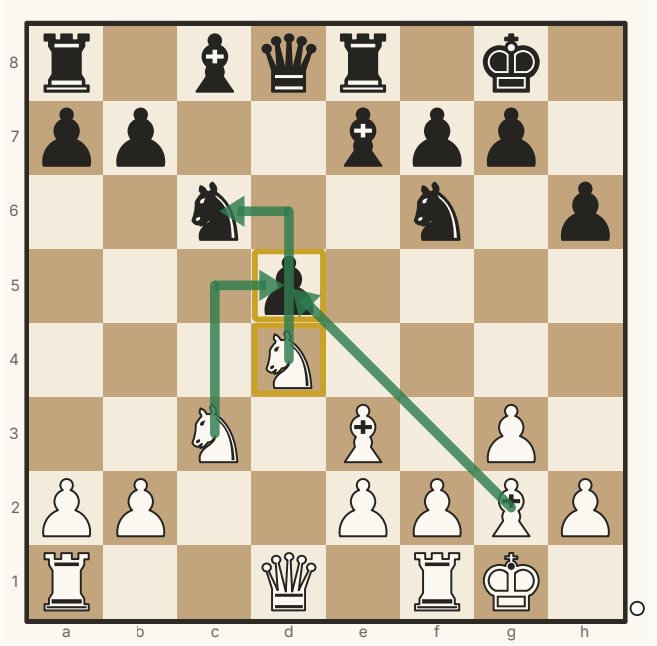
> KEY IDEA **Short plans beat long dreams**
>
> Coaching judgment: a 2–3 move plan that improves your position is worth far more than a 20-move scheme. Positions change every time your opponent moves. Plan a little, then re-evaluate.

Step 1: Evaluate with a checklist

To make a plan you first need to know what is going on. Instead of a vague "I think I'm fine," walk through a fixed list. This is your **evaluation checklist**:

1. **Material.** Count it (see Chapter 3). Is anyone ahead? Is there an imbalance, such as bishop against knight?
2. **King safety.** Whose king is more exposed? Are there holes or open files near either king? (See Chapter 21.)
3. **Piece activity.** Which pieces are active, and which are passive or blocked? Which is each side's worst piece? (See Chapter 19.)
4. **Pawn structure.** Isolated, doubled, backward or passed pawns? Pawn majorities? Which pawn breaks are possible? (See Chapter 20.)
5. **Space.** Whose pawns are further advanced, giving more room behind them?
6. **Files and key squares.** Who controls the open files, the important diagonals, and the squares that pawns can no longer guard?
7. **Threats.** What does each side threaten *right now*?

The last item is a safety check. If there is a threat or a tactic, deal with it first: tactics come before strategy. The checklist is for positions where your safety scan (Chapter 4) finds nothing urgent.



Run the checklist. Material is equal and both kings are safe, but Black has an isolated pawn on d5. White's pieces aim at it, and the d4-square in front of it is a perfect home for a white piece.

Here is the checklist in action:

- **Material:** equal.
- **King safety:** both kings are castled and safe.
- **Activity:** both sides are well developed. White's bishop on g2 already presses on d5.
- **Pawn structure:** Black has an **isolated pawn** on d5 (a pawn with no friendly pawns on the neighbouring files). It cannot be defended by another pawn, and the square in front of it, d4, can never be attacked by a black pawn.
- **Space:** roughly equal; Black's d-pawn gives a little central space.
- **Files and squares:** the c-file is open, and White can use it; d4 belongs to White.
- **Threats:** none for either side.

The verdict: the position is balanced, with a small, lasting edge for White because of the target on d5 and the square on d4. That verdict already suggests a

plan: keep a piece on d4, put a rook on the c-file, and line up more pieces against d5. The engine's favourite move here is Rc1, taking the open file, which fits that plan.

> COMMON MISTAKE **Skipping the checklist and "just playing"**
>
> Many games are lost not by blunders but by a series of moves without a purpose. If you cannot say in one sentence why you are playing a move, you probably skipped the evaluation. Run the seven items, even quickly. In your own games, look for stretches of three or four moves where you could not explain your idea afterwards.

Step 2: Find the targets

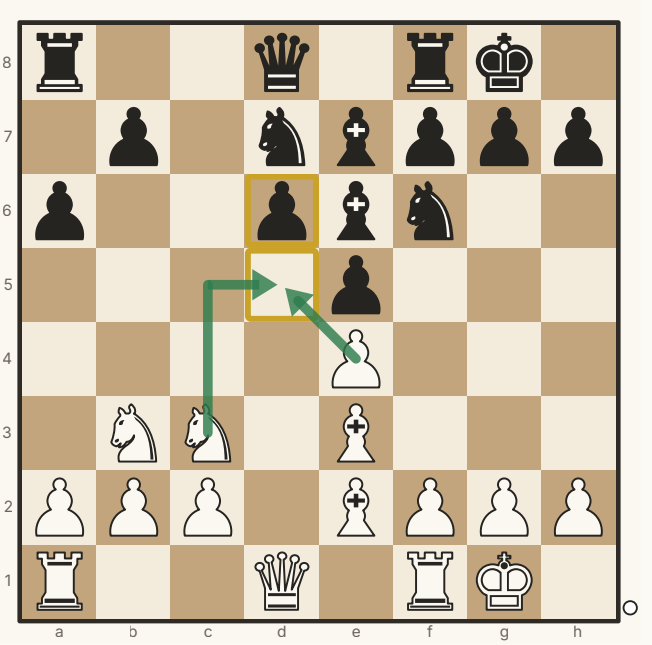
A plan usually means attacking something. Strategic targets are the long-term weaknesses in the enemy camp. There are four main kinds.

Weak pawns

A **weak pawn** is one that cannot easily be defended by another pawn and can be attacked by pieces: isolated pawns, backward pawns (a pawn that has fallen behind its neighbours and cannot safely advance), and doubled pawns. Weak pawns are good targets because the opponent must defend them with pieces, and pieces that defend are not doing anything else.

Weak squares

A **weak square** (or **hole**) is a square that the opponent's pawns can no longer attack. A knight that lands on such a square, especially in the centre or near the enemy king, is often worth more than a rook there.

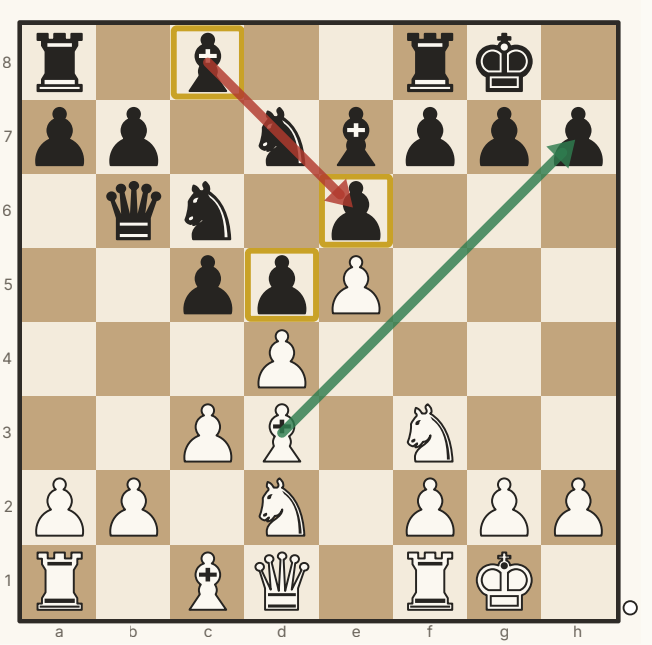


Black's pawns on d6 and e5 can no longer cover d5, so d5 is a hole, and the d6-pawn is backward. White's plan writes itself: control d5, put a piece there, and keep Black from playing ...d5.

In this structure White's pieces are drawn to d5. If a white knight lands there and cannot be removed, Black will be passive for a long time. Black, for his part, wants the freeing move ...d5, which would remove both weaknesses at once. That is the key to the position: one side wants d5, the other wants to play ...d5.

Bad pieces

A **bad piece** is one that has no good squares and little prospect of finding any. The classic example is the **bad bishop**: a bishop blocked by its own pawns, which stand on squares of the bishop's colour.

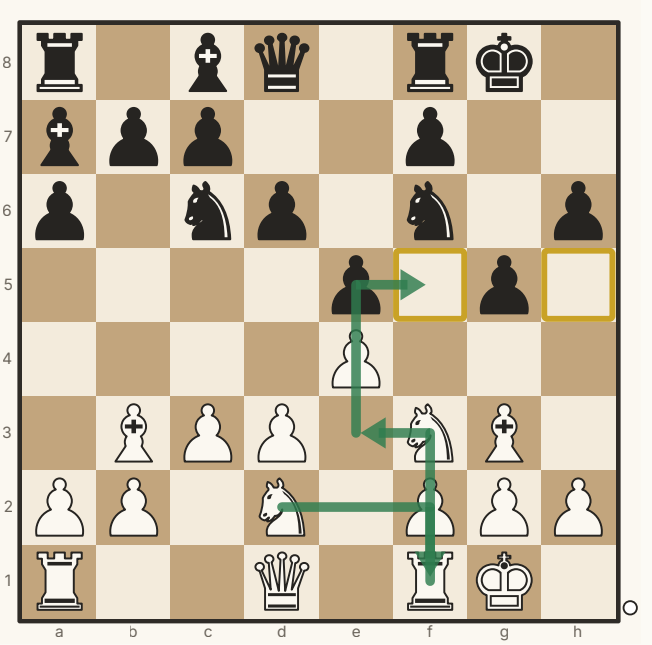


Black's bishop on c8 is hemmed in by its own pawns on e6 and d5, all on light squares. White's light-squared bishop on d3, by contrast, has an open diagonal towards Black's king.

When the opponent has a bad piece, try to keep it bad: do not trade it, and do not open the position for it. When *you* have a bad piece, either improve it (bring it outside the pawn chain, or break the chain with a pawn move) or trade it for one of the opponent's good pieces.

An exposed king

Holes and open lines near the king are long-term targets too. Pawn moves in front of your own castled king create squares that can never be covered by pawns again.



Black's ...h6 and ...g5 chased White's bishop but left holes on f5 and h5. They are long-term targets: a white knight on f5 would be a thorn next to Black's king.

No attack is possible right now, and the engine rates the position as only slightly better for White. But the holes will not go away. White's plan for the next few moves is clear: reroute a knight towards f5 and see whether Black can ever challenge it.

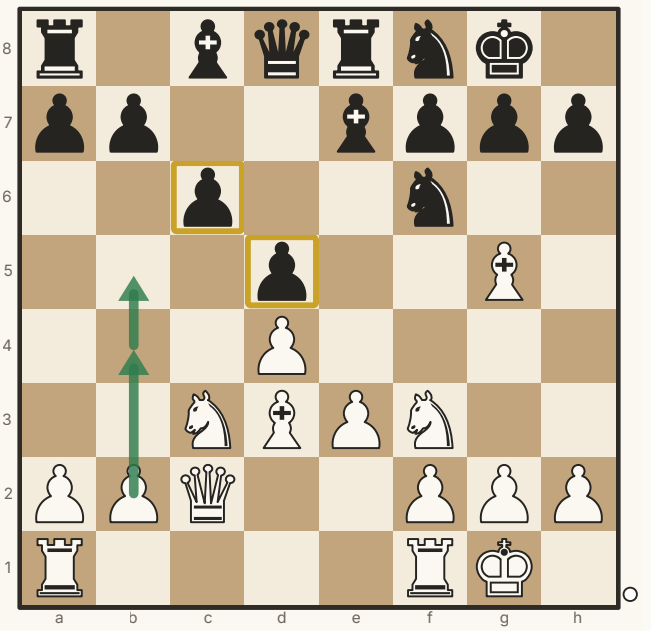
> **IN YOUR NEXT GAME** **Name the target**
>
> In your next game, at the first quiet moment, finish this sentence aloud in your head: "The biggest weakness in my opponent's position is ____." If you cannot name one, look for your own worst piece instead (Step 4).

Step 3: Let the pawn structure suggest the plan

Pawns are the slowest pieces and the hardest to move back, so the pawn structure changes slowly. That makes it a reliable guide to planning. Here are a few classic pairings of structure and plan. They are guidelines, not laws: tactics and piece placement can always override them.

Structure	Typical plan
Opponent has an isolated pawn	Blockade the square in front of it, trade pieces, attack the pawn
You have a pawn majority on one wing	Advance it to create a passed pawn
Opponent's pawns on one wing are fewer but healthy (Carlsbad, see below)	Minority attack: advance your smaller group to create a weakness
Locked centre	Play where your pawns point: attack on the side where you have more space
Hole in the opponent's camp	Put a knight there
Open file	Put a rook on it, double rooks, aim for the seventh rank

The Carlsbad structure (named after a well-known tournament town where it was often seen) is the clearest example of a structure that dictates a plan.



The Carlsbad structure: White's d4/e3 against Black's d5/c6. White's standard plan is the minority attack: b4 and b5, exchanging on c6 to leave Black with a weak pawn.

White has two queenside pawns (a and b) against Black's three (a, b and c). Why would the smaller group attack? Because when White plays b5 and exchanges on

c6, Black is left with either an isolated d5-pawn or a backward c6-pawn on a half-open file. A weak pawn has been *created* where none existed.

11. **Rab1 Ng6** 12. **b4 a6** 13. **a4 Ne4** 14. **Bxe7 Qxe7**

White prepares b5 with Rab1, b4 and a4. Black's ...a6 is prophylaxis against b5, and ...Ne4 looks for counterplay in the centre.

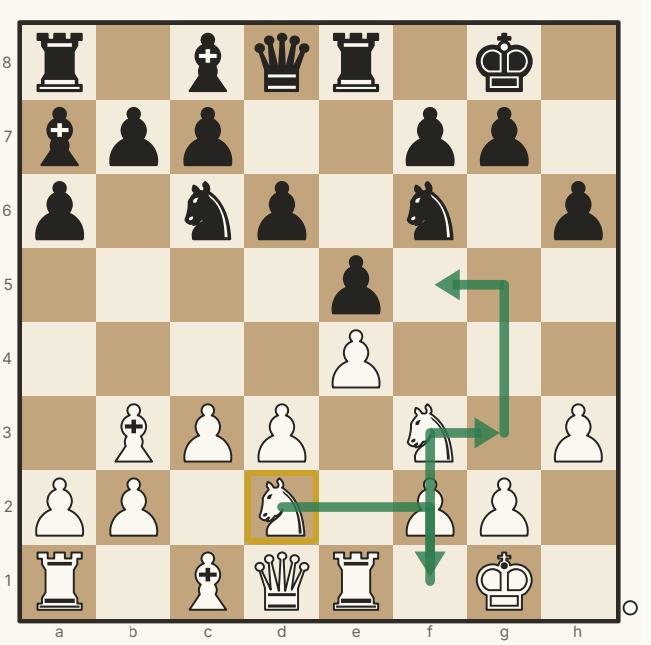
Exercise 22-02 asks you to find this plan, and Exercise 22-07 shows Black's prophylactic answer.

Step 4: Improve your worst piece

When you cannot find a target, or you are not sure what to do, use the most practical planning question in chess:

Which of my pieces is doing the least, and how can I improve it?

This idea is widely taught and works at every level. Your position is only as strong as your weakest piece. A rook still on its starting square, a knight on the rim, a bishop staring at its own pawns: each of these is a whole piece you are not using.



White's worst piece is the knight on d2, which blocks the c1-bishop. The standard Italian manoeuvre Nf1-g3 (or e3) brings it to the kingside, where it eyes f5.

How to execute it:

1. List your pieces and give each a quick grade: active, OK or poor.
2. Pick the poorest one and find its best realistic square: one where it attacks something, controls a key square or defends something important.
3. Find the route, square by square, and check that each step is tactically safe.

What commonly goes wrong: you move your *best* piece again because it is the one you are looking at. A knight that is already on a great square does not need another move. Moves with the passive pieces are usually the ones that improve your position most.

"Do nothing" moves that improve

Sometimes the best move is a small one that does not change the structure: a king step to safety, a rook to a better file, a pawn move that gives the king **luft** (an escape square, so that back-rank mates are no longer possible), or a quiet retreat that prepares a manoeuvre. These moves look like "doing nothing," but they make every later plan stronger.

> COACHING JUDGMENT **Patience is a skill**
>
> Coaching judgment: at this level, the player who is happy to make a quiet improving move, rather than forcing the action, usually ends up with the better position. If the position does not call for action, improve a piece and wait for your opponent to create a weakness.

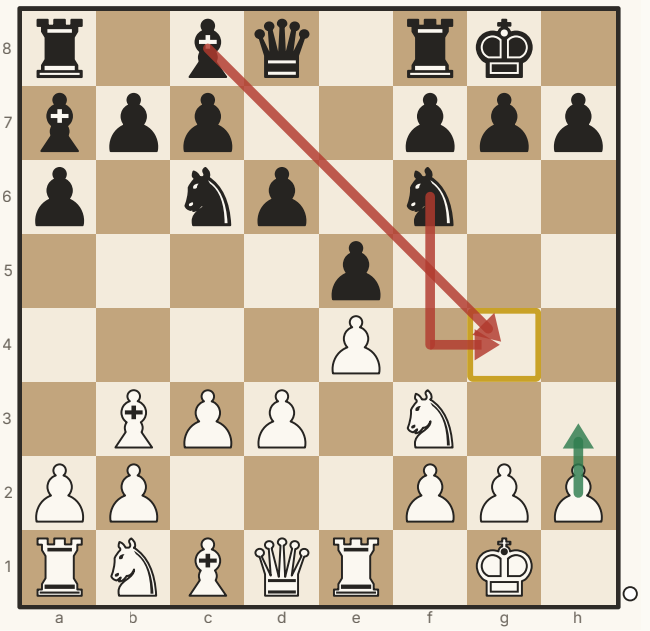
Step 5: Prophylaxis — what does my opponent want?

Prophylaxis means preventing your opponent's plan before it happens. The idea was developed and popularised by Aron Nimzowitsch in the early twentieth century, and it is one of the most important habits a strategic player can have.

The procedure is a single question, asked every move in quiet positions:

If it were my opponent's move right now, what would they play?

If their best move would hurt you, consider stopping it first. If it would not, carry on with your own plan.



If it were Black's move, ...Ng4 (hitting f2 together with the a7-bishop) or ...Bg4 (pinning the f3-knight) would be annoying. The quiet h3 rules out both and gives the king luft. The engine rates it among White's best moves.

Prophylaxis is not passivity. A good prophylactic move often improves your own position at the same time: h3 here also makes the later Nbd2-f1-g3 manoeuvre safe. The best prophylactic moves are the ones that **stop the opponent's idea and fit your own plan**.

What commonly goes wrong:

- Ignoring the question. You play your plan, your opponent plays his, and his was faster.
- Overdoing it. You stop every possible idea and never make progress. Stop the opponent's *main* idea, then return to your plan.

Exercises 22-06, 22-07 and 22-11 are all about asking this question. In two of them, the opponent's idea is a concrete threat that only one move stops.

Imbalances: what makes the two sides different

Another useful way to plan is to list the **imbalances**: the differences between White's and Black's positions. Material, minor pieces (bishop or knight), pawn structure, space, development, open files, king safety, the initiative. This way of thinking has been popularised especially by the American author and coach Jeremy Silman.

The point is that a plan should use *your* favourable imbalances. If you have the bishop and your opponent has the knight, open the position. If you have the knight and he has a bad bishop, keep the position closed and find an outpost. If you have more space, avoid exchanges, because the side with less space is cramped and wants to trade. If your opponent has a lead in development, keep the position closed until you catch up.

> KEY IDEA **Imbalances in one question**
>
> "What do I have that my opponent does not, and how do I make it matter?" The answer is usually your plan.

When the position changes, change the plan

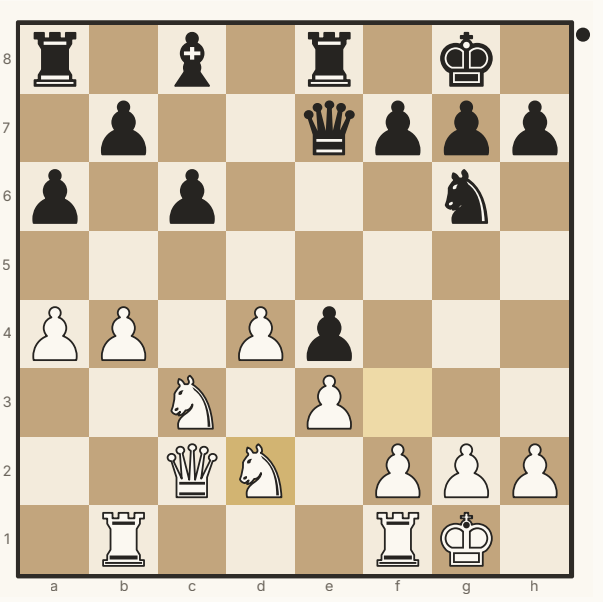
A plan is a tool, not a promise. Every time your opponent changes the pawn structure, trades a piece or creates a threat, run the checklist again. Good players change plans without regret.

Typical moments to re-evaluate:

- A pawn break opens or closes the centre.
- A piece trade removes the piece your plan depended on (for example, your opponent trades your good knight).
- Your opponent creates a weakness of his own somewhere else.
- A tactic appears.

The Carlsbad structure shows a typical switch. White's minority attack is a good plan while Black's pawns stand on c6 and d5. But in the line shown in Step 3, Black answered with ...Ne4, and once White exchanges on e4 the pawn structure changes, and so must the plan.

15. **Bxe4 dxe4** 16. **Nd2**



After the exchanges on e4 the d5-pawn has gone, so the old target has changed. The knight and the queen on c2 now both hit the e4-pawn, Black must defend it (the engine's choice is ...f5), and White's new plan is to press against e4 and use the open d-file and the c5-square.

Nothing was wrong with White's first plan. The position changed, and the checklist produced a new one.

Worked example: planning a quiet Italian

Let us put the whole routine together. This is the Giuoco Pianissimo from your White repertoire (Chapter 12), after 1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6 6.O-O O-O 7.Re1 a6 8.Bb3 Ba7. White is to move (see the diagram in Step 5).

First I check for tactics. Checks, captures, threats: none for White. If it were Black's move, ...Ng4 would attack f2 twice (knight and bishop), and ...Bg4 would pin my knight. Neither is deadly, but both are annoying.

Now the checklist. Material: equal. King safety: both castled, both fine. Activity: my b1-knight and c1-bishop are undeveloped; Black's c8-bishop is too. Pawn structure: symmetrical and closed; no weaknesses yet. Space: equal. Files: no open files. Threats: none.

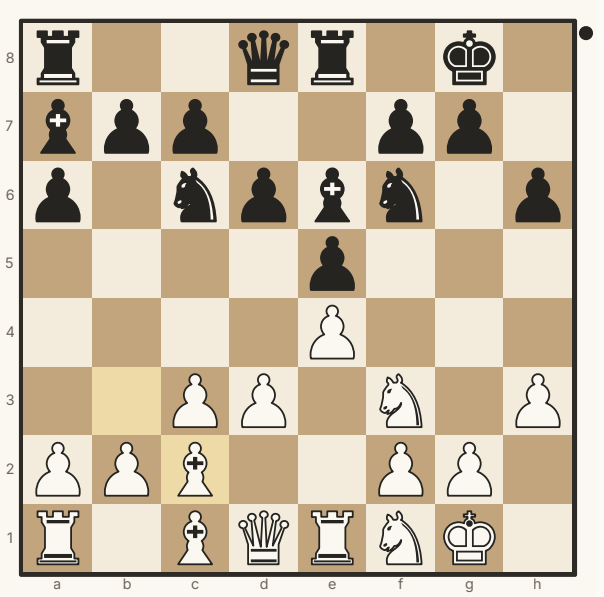
Targets. None in Black's camp yet. That means the plan will be about *improvement*, not attack.

Worst piece. The knight on b1. Its best realistic square is g3 (via d2 and f1), where it eyes f5 and h5 and defends my king. The route is safe.

Opponent's idea. ...Ng4 or ...Bg4. A quiet h3 stops both.

Candidate moves. h3 (prophylaxis plus luft), Nbd2 (start the manoeuvre at once), and perhaps d4 (a central break). The opponent's best reply to d4 is ...Bg4 with pressure on d4, and Black is comfortable. h3 first and Nbd2 next fits all my answers.

9. **h3 h6** 10. **Nbd2 Re8** 11. **Nf1 Be6** 12. **Bc2**



A two-step plan carried out: prophylaxis with h3, then the knight heads for g3. White keeps the bishop with Bc2. Now Black's ...d5 break is the critical question (Exercise 22-03).

Notice how short each plan was. h3 and the first two steps of Nbd2-f1-g3 took three moves. Meanwhile the position changed (Black's bishop came to e6, offering a trade on b3), so White re-evaluated and chose Bc2 to keep the bishop. The knight will finish its journey to g3 next. That is planning at club level: a few purposeful moves, then look again.

> **EVIDENCE** **Plans and pattern knowledge**
>
> Research on chess expertise suggests that strong players recognise familiar structures quickly and retrieve typical plans for them, rather than calculating everything from scratch. The classic studies are Adriaan de Groot's *Thought and Choice in Chess* (1965, English edition) and William Chase and Herbert Simon's "Perception in Chess" (*Cognitive Psychology*, 1973). Practical consequence: learning a handful of standard structures and their plans pays off in many games.

How to find planning mistakes in your own games

After each game, look for the quiet moments, not just the blunders:

- Find two or three moves where nothing tactical was happening. Could you say what your plan was?
- Ask: what was my worst piece at that moment? Did I improve it?
- Ask: what did my opponent want? Did I let him do it for free?
- Use an engine afterwards to see which plans it preferred, but write down the *idea*, not just the move. "Engine wanted b4 here: minority attack" is a lesson; "engine wanted b4" is not.

Record planning errors in your mistake journal under *positional*.

Practice at the board

1. **Checklist drill.** Set up `r1bqr1k1/pp2bpp1/2n2n1p/3p4/3N4/2N1B1P1/PP2PPBP/R2Q1RK1 w - - 2 12` (the isolated-pawn position). Write the seven checklist items on paper in under three minutes. Then compare with the evaluation in this chapter and with an engine's top three moves.
2. **Worst-piece drill.** Set up `r1bqr1k1/bpp2pp1/p1np1n1p/4p3/4P3/1BPP1N1P/PP1N1PP1/R1BQR1K1 w - - 2 11`. Grade every white piece as active, OK or poor, then find a route for the poorest piece. Check with the engine whether its first choice matches your piece.
3. **Prophylaxis drill.** In any position from your own games, before each move, play "your opponent's move" on the board first (give him a free move). What did he do? Was it dangerous? Then put the pieces back and decide whether to prevent it.
4. **Structure drill.** Set up the Carlsbad position `r1bqrnk1/pp2bppp/2p2n2/3p2B1/3P4/2NBPN2/PPQ2PPP/R4RK1 w - - 8 11` and play the minority attack against an engine at a moderate level. Your goal is simply to create a weak black pawn on the queenside and attack it.
5. **Play.** In your next three games, write your plan in two or three words on your scoresheet (or in your head) at move 15: "knight to f5," "minority attack," "rook to c-file." Check after the game whether you followed it, and whether it was right.

Chapter summary

- A plan is a short, purposeful sequence of moves that uses something already in the position; aim for 2–3 moves, then re-evaluate.
- Evaluate with the checklist: material, king safety, activity, pawn structure, space, files and key squares, threats.
- Look for targets: weak pawns, weak squares, bad pieces and an exposed king.
- Let the pawn structure suggest the plan; standard structures have standard plans.
- When in doubt, improve your worst piece; quiet improving moves are often the strongest.
- Prophylaxis: ask what your opponent wants and stop his main idea, ideally with a move that also helps your plan.
- List the imbalances and play to make your favourable ones matter.
- When the position changes, change the plan.

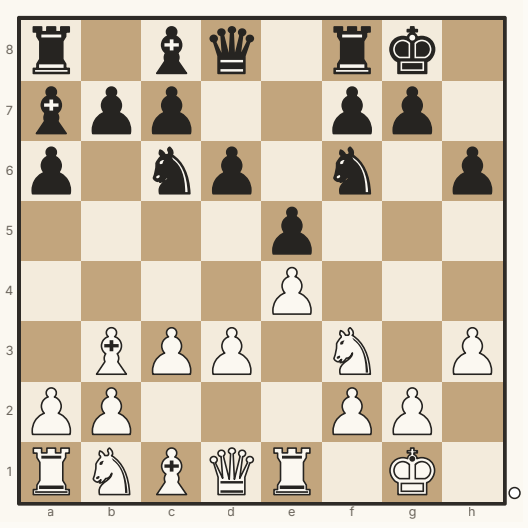
CHEAT CODE **Planning procedure for quiet positions**

1. Safety first: checks, captures and threats for both sides. Tactics before strategy.
2. Run the checklist: material, kings, activity, pawns, space, files/squares, threats.
3. Name one target in your opponent's camp (weak pawn, hole, bad piece, exposed king).
4. If there is no target, name your worst piece and find it a better square.
5. Ask "What does my opponent want?" and stop it if it hurts you.
6. Choose a 2–3 move plan that fits the pawn structure; re-evaluate after each change.

Exercises

Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

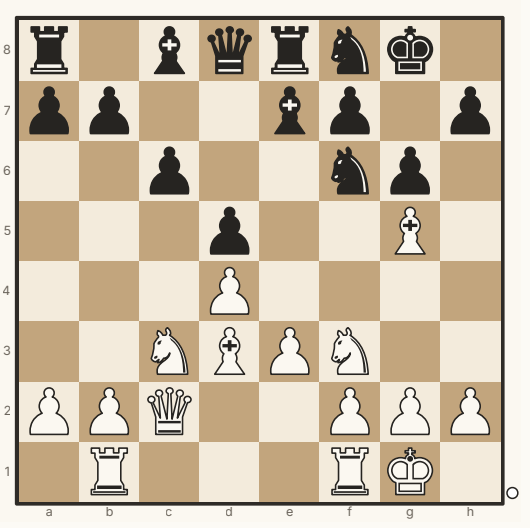
22.1 STRATEGY



White to move. Nothing is happening tactically. Find your worst piece and improve it.

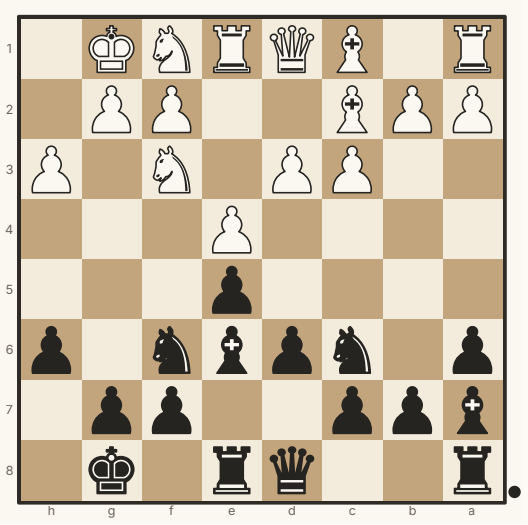
Solution p. 703

22.2 STRATEGY



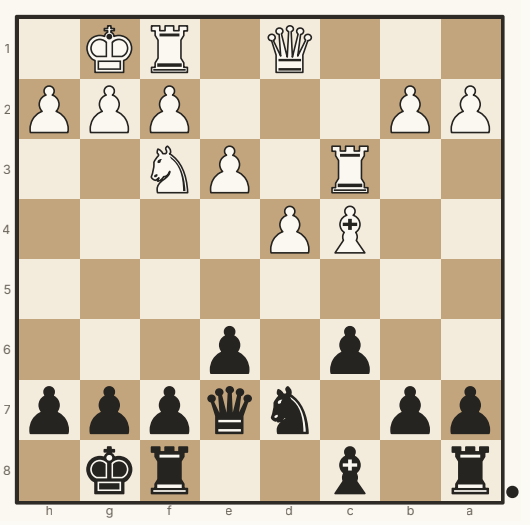
White to move. Read the pawn structure and choose a plan move. Solution p. 703

22.3 STRATEGY



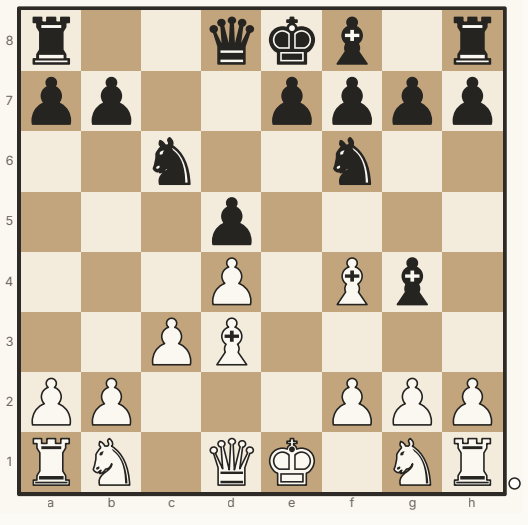
Black to move. Both sides have developed calmly. What is Black's most purposeful plan move? Solution p. 704

22.4 STRATEGY



Black to move. Black's position is solid but cramped. Which piece is Black's worst, and which move frees it? Solution p. 704

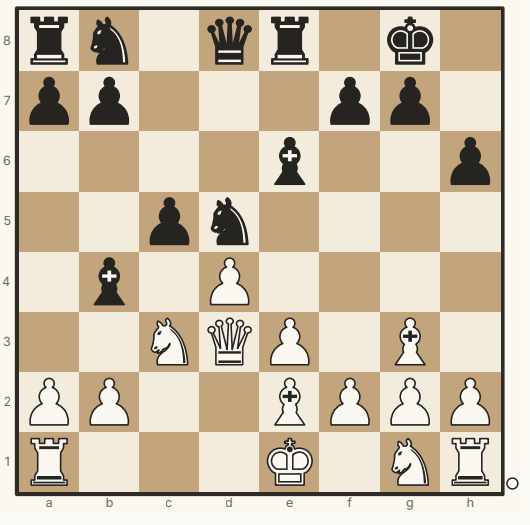
22.5 STRATEGY



White to move. Black's bishop has just left c8 to go to g4. Find a weakness in Black's camp and the move that targets it.

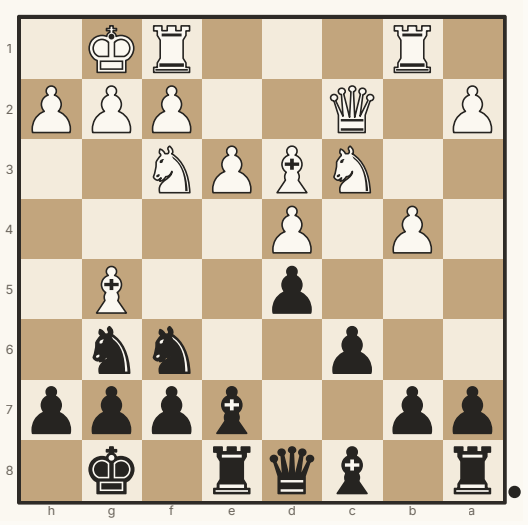
Solution p. 704

22.6 STRATEGY



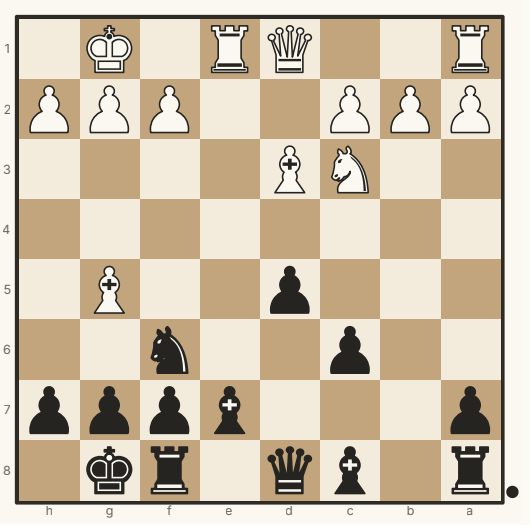
White to move. Ask first: what does Black want to do next? Then find the move that stops it. Solution p. 704

22.7 STRATEGY



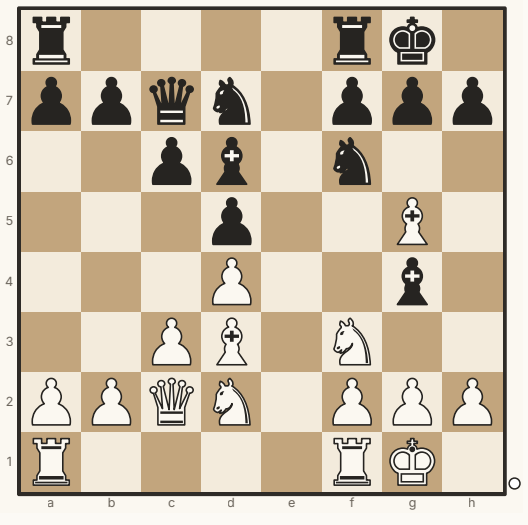
Black to move. White has started the minority attack with b4. Choose a sensible plan move. Solution p. 705

22.8 STRATEGY



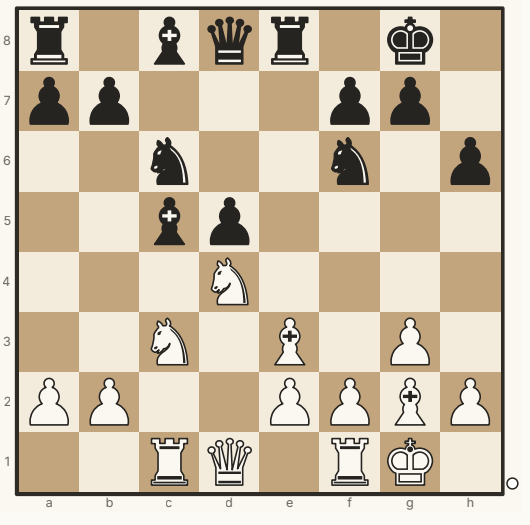
Black to move. Which open line belongs to your rook? Solution p. 705

22.9 QUIET MOVE



White to move. The structure is symmetrical and nothing is hanging. Choose a useful quiet move. Solution p. 705

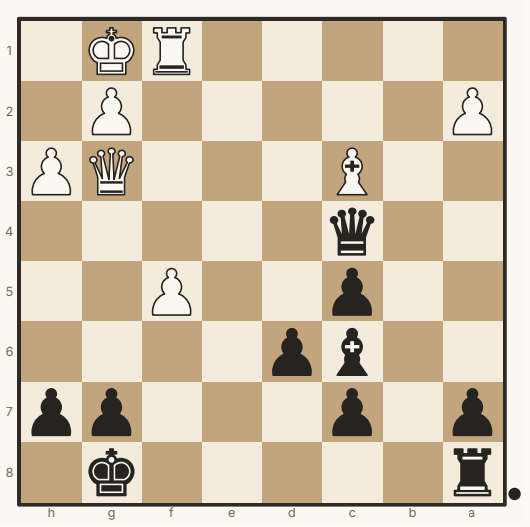
22.10 TACTIC



White to move. Black has just played ...Bc5, attacking your d4-knight. Before you think about plans, check for tactics. Win material.

Solution p. 705

22.11 DEFENCE



Black to move. You are two pawns up. Before you make a plan, ask: what does White want? Solution p. 706

HINTS

22.1 Which white piece has not moved yet?

22.2 The rook on b1 is a clue. Which pawn would it like to support?

22.3 Which pawn move fights for the centre and opens lines for your pieces?

22.4 Which pawn is blocking the c8-bishop?

22.5 When a piece leaves its post, what does it stop defending?

22.6 Which central pawn is Black attacking, and how can you defend it while developing?

22.7 What is White's next pawn move, and how can you prepare for it?

22.8 Look for a file with no black pawn on it.

22.9 Which black piece is slightly loose, and what small move questions it?

22.10 What would the e3-bishop attack if the knight on d4 moved away?

22.11 Which square do two white pieces attack?

CHAPTER 23 STAGE 1000–1200

Rook Endgames

The endings you will actually reach

YOU WILL LEARN TO

- Keep your rook active and put it behind passed pawns, yours and your opponent's.
- Cut off the defending king by a file or rank before you push the pawn.
- Win the Lucena position by building a bridge.
- Draw the Philidor position with third-rank defence and checks from behind.
- Defend from the long side and count tempi in rook-against-pawn races.

> **IN YOUR NEXT GAME**
>
> When a rook endgame appears, ask first "Which rook is more active?" and "Can I cut off the enemy king?" before you push a single pawn.

You have traded queens, then minor pieces, and now each side has a rook and a handful of pawns. Maybe you are a pawn up and expect to win. Maybe you are a pawn down and expect to lose. At club level both expectations are often wrong, because rook endgames reward a few specific techniques much more than they reward material.

Rook endgames are widely regarded as the most common type of endgame, because rooks are usually the last pieces to be traded. The good news is that a small amount of knowledge goes a long way: two famous positions, a handful of rules of thumb and the habit of keeping your rook busy will save you many half-points and win you many more. This chapter gives you exactly that toolkit.

Why rook endgames are different

In a king-and-pawn ending (Chapter 18) a single tempo decides everything. In a rook ending the rook is a long-range piece that can check from far away, attack

pawns from behind and cut the enemy king off from the action. That makes rook endings more forgiving for the defender and more technical for the attacker.

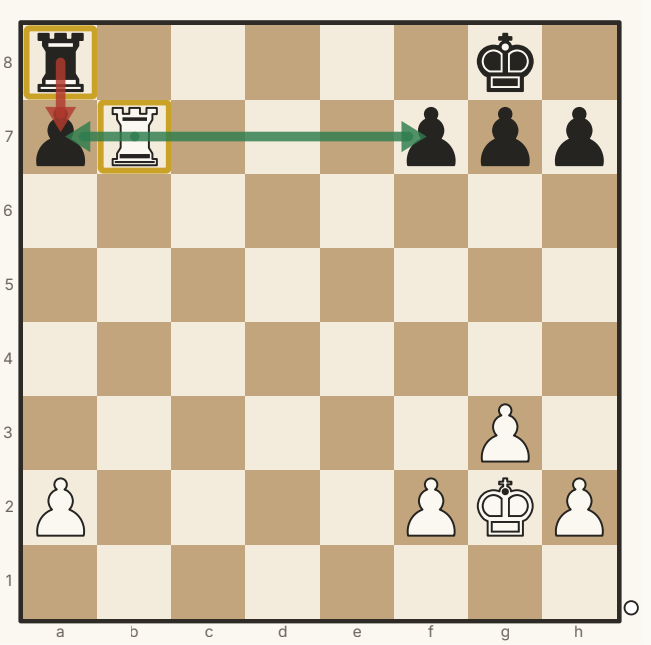
Three questions guide almost every decision:

1. **Which rook is more active?** An **active rook** attacks pawns, gives checks, or cuts off the enemy king. A **passive rook** is stuck defending a pawn or blocking a passed pawn from in front.
2. **Where are the kings?** A king that is cut off from the passed pawn cannot help to stop it.
3. **Where are the passed pawns, and where are the rooks relative to them?**

> KEY IDEA **The activity rule**
>
> In rook endgames, activity is usually worth more than a pawn. A coaching rule that serves club players well: if you must choose between keeping a pawn with a passive rook and giving up the pawn to activate the rook, look very hard at the active option first.

Active rook, passive rook

Look at the position below. Material is level, but the rooks are doing very different jobs.



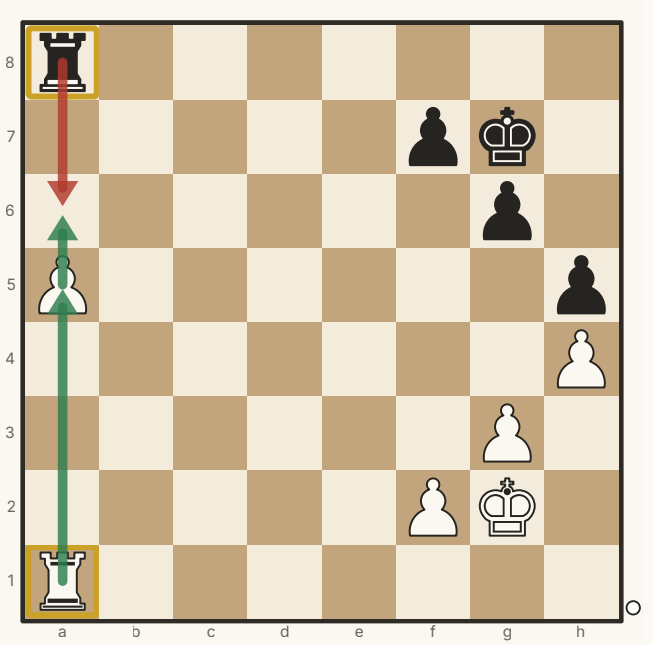
Material is equal, but White's rook on the seventh rank attacks a7 and cramps Black's king, while Black's rook on a8 is tied to the defence of the a-pawn.

White's rook sits on the **seventh rank** (the second rank from Black's point of view), where it attacks pawns that cannot easily be defended by other pawns and keeps the black king on the back rank. Black's rook can only stand guard on a8. The engine rates this position as only slightly better for White, so it is not a win, but it is clear who is playing for two results. If Black lets the passivity become permanent, the slight edge grows.

What commonly goes wrong: the defender clings to every pawn. You put your rook behind your weak pawn "to keep material," and your opponent's rook and king then do whatever they want. In your own games, look for moments where your rook spent many moves guarding one pawn. That is a sign you chose passivity.

Rooks belong behind passed pawns

The most famous rook-endgame guideline is usually credited to the German master and teacher Siegbert Tarrasch: **rooks belong behind passed pawns** — both your own and your opponent's.



White's rook on a1 is behind the a-pawn; Black's rook on a8 is in front of it. Every step the pawn takes gives White's rook more room and Black's rook less.

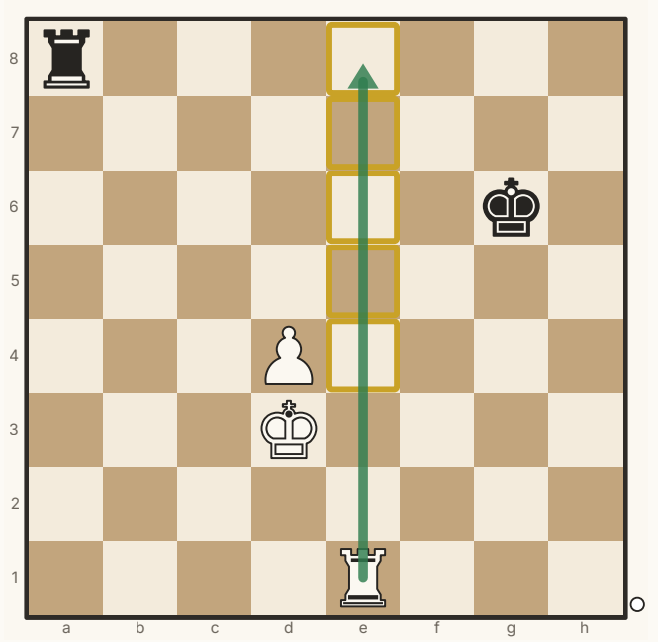
Why does it work? A rook behind its own passed pawn supports the pawn on every square of its journey and gains scope as the pawn advances. A rook in front of an enemy passed pawn is a blocker that loses scope as the pawn advances: with the pawn on a7, Black's rook is chained to the a-file in front of it, and every move it makes must watch for tricks. In the diagram the engine evaluates White as clearly winning: White's king walks to the kingside and eats pawns, because Black's rook and king are both busy with the a-pawn.

The same idea works in defence. If your opponent has a passed pawn, try to get your rook behind it, not in front of it. Exercise 23-07 shows a pawn-down position that is saved precisely by this.

> **COMMON MISTAKE** **Blocking a passed pawn with your rook from the front**
>
> It feels safe to stand in front of the pawn. It is usually the most passive square on the board. An exception: when the pawn is a rook pawn on the seventh rank and your king sits on g7 or h7, the rook in front is part of a known drawing set-up. Even then you must watch for the skewer trick in Exercise 23-01.

Cutting off the king

Before you push your passed pawn, ask whether you can **cut off** the enemy king: place your rook on a file (or rank) that the king cannot cross. A king that cannot reach the pawn cannot stop it, and the defending rook alone usually cannot hold.



White's rook on the e-file cuts the black king off by two files. The white king will escort the d-pawn, and the engine rates this as a clear win.

The wall on the e-file means the black king cannot reach d7 or d8. White's king will walk forward with the pawn, using the pawn and the rook as shields from checks. When the king is cut off by only one file, the defender has more chances: the king can attack the pawn from the side, and the defending rook may find enough checks. Then the exact position of every piece matters.

> COACHING JUDGMENT **How many files?**
>
> Coaching judgment, not a law: with the defender's king cut off by **two or more files**, the attacker usually wins with a centre or knight pawn. With a **one-file** cut-off, the result depends on how far the pawn has advanced and where the rooks are, so you need to calculate. Either way, cutting off first and pushing second is almost always the right order.

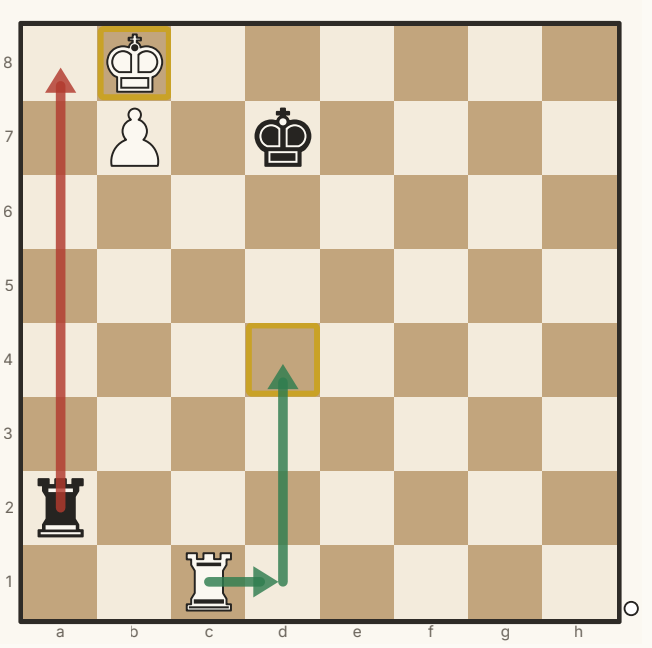
How to execute it: find the file between the enemy king and your pawn, and put your rook there, ideally from a distance so the king cannot attack it. Then bring your own king forward in front of the pawn. Push the pawn only when the king has prepared its path.

What commonly goes wrong: you push the pawn first because it feels like progress, and the enemy king walks across in front of it. Exercises 23-03 and 23-10 punish exactly that.

The Lucena position: building a bridge

Sooner or later a successful cut-off leads to the most important winning position in rook endings. It is traditionally named after the fifteenth-century Spanish author Luis Ramírez de Lucena, although chess historians doubt that he actually published it. The name has stuck regardless of who found it first.

In the **Lucena position** the attacker's pawn is on the seventh rank, his king stands on the queening square in front of it, the defending king is cut off by at least one file, and the defending rook waits to give checks.

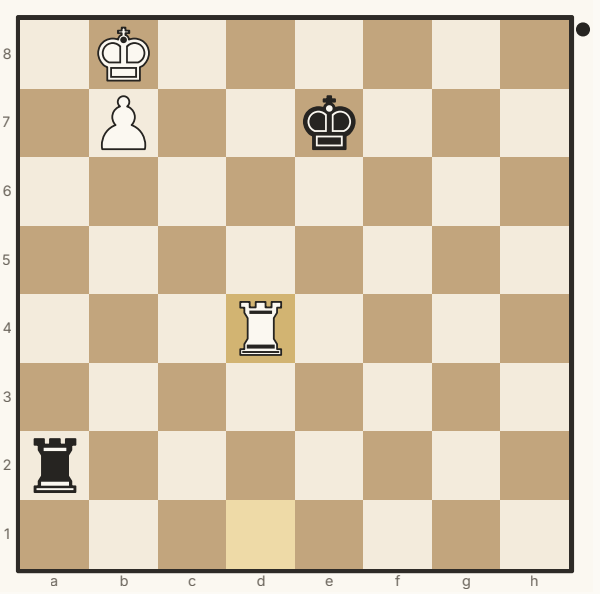


The Lucena position. White's king is stuck in front of the pawn: if it steps out, Black checks from the side. The solution is to build a bridge.

White's problem is simple: the king cannot get out. The black king covers c7 and c8, and the black rook controls the a-file. Even if the king could step out, the rook would check it from the side. White's solution has two steps:

1. **Drive the enemy king one more file away** with a check: Rd1+.
2. **Build a bridge**: put the rook on the fourth rank, so that it can later block the rook's checks.

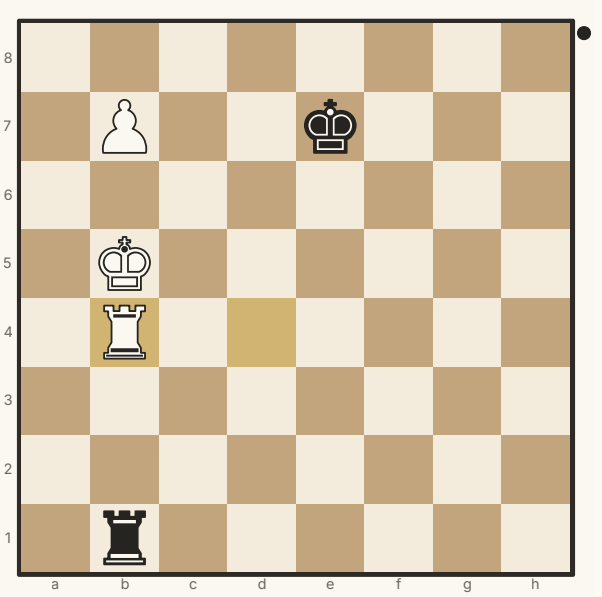
1. **Rd1+ Ke7** 2. **Rd4**



Step one and step two: the check pushes the black king to the e-file, and the rook goes to the fourth rank, ready to become the bridge.

Now the king walks out. The line below shows the natural defence, with Black checking for as long as possible. (The engine actually prefers to give up the rook for the pawn at some point, because the checks lead to mate faster; the bridge method is the same either way.)

2... **Ra1** 3. **Kc7 Rc1+** 4. **Kb6 Rb1+** 5. **Kc6 Rc1+** 6. **Kb5 Rb1+** 7. **Rb4**



The bridge is complete. The rook on b4 blocks the last check, and the pawn will promote.

Walk through the logic. After 2.Rd4 White threatens to bring the king out. Black waits with 2...Ra1, and the king comes out with 3.Kc7. Black checks from the side and from behind, but each check drives the king down towards the rook on the fourth rank. After 7.Rb4 there are no more useful checks, and the pawn queens. Why the fourth rank? Because the king needs that many squares to step down the board before the rook can interpose without being captured.

> **KEY IDEA** **Lucena in one sentence**
>
> Check the defending king away, put your rook on the fourth rank, walk your king out, and use the rook as a shield.

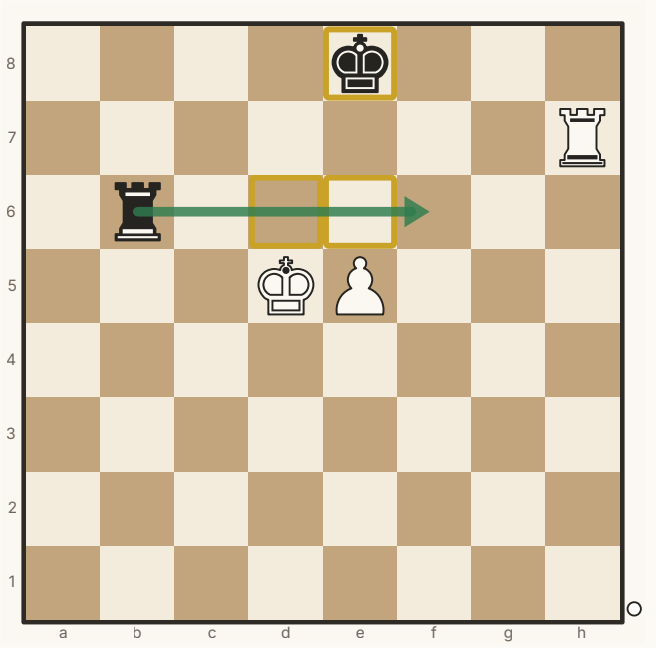
Practise this until you can do it without thinking. At the board you will reach the Lucena with little time on the clock, and knowing the method turns a nervous ending into a routine win.

The Philidor position: third-rank defence

If the Lucena is the attacker's dream, the **Philidor position** is the defender's. It is named after the eighteenth-century French player and composer François-André

Danican Philidor, who analysed it. The idea is simple enough to learn in five minutes and important enough to save you many games.

The defending king stands on the queening square, or next to it. The attacker's pawn has not yet reached the sixth rank. The defending rook stands on its **third rank** (the sixth rank for Black), stopping the attacking king from advancing.

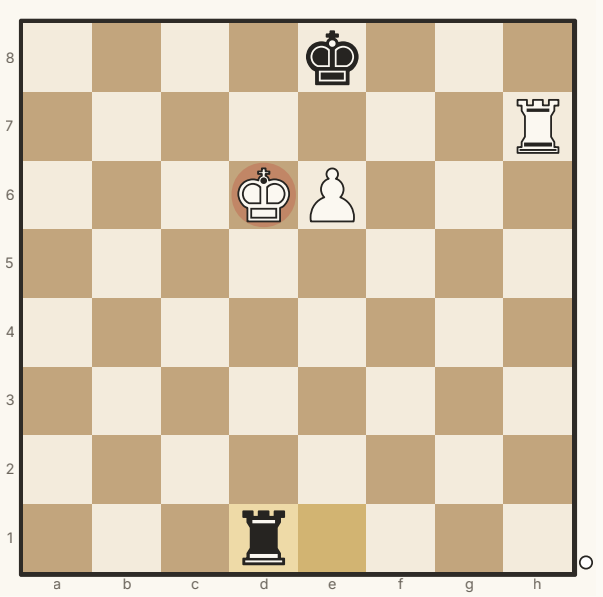


The Philidor position. Black's rook on b6 stops the white king from reaching d6 or e6 (the pawn on e5 cannot shield it yet). If the pawn advances, the rook drops back and checks from behind.

The defence has two stages:

- **Stage 1.** Keep the rook on the sixth rank (Black's third rank). The white king cannot advance because the rook covers the sixth. White's only way to make progress is to push the pawn to e6.
- **Stage 2.** The moment the pawn reaches e6, the king could hide in front of it, but the pawn has also taken away its own shelter. Now the rook goes to the far end of the board, to the first rank, and checks from behind. The white king has nowhere to hide.

1. **e6 Rb1** 2. **Kd6 Rd1+** 3. **Ke5 Re1+** 4. **Kd6 Rd1+**



The checks from behind never end. If the king runs to the rook, it leaves the pawn behind.

> **COMMON MISTAKE** **Waiting on the back rank**
>
> The most common club-level loss in this ending is the passive rook on the back rank. If the white king is allowed to reach the sixth rank, the defence hangs by a thread: with White's king on d6, pawn on e5, rook on h7, and Black's rook on b8 and king on e8 (Black to move), only ...Rb6+ saves the game; every other move loses. Do not live that close to the edge. Third rank first; checks from behind when the pawn moves.

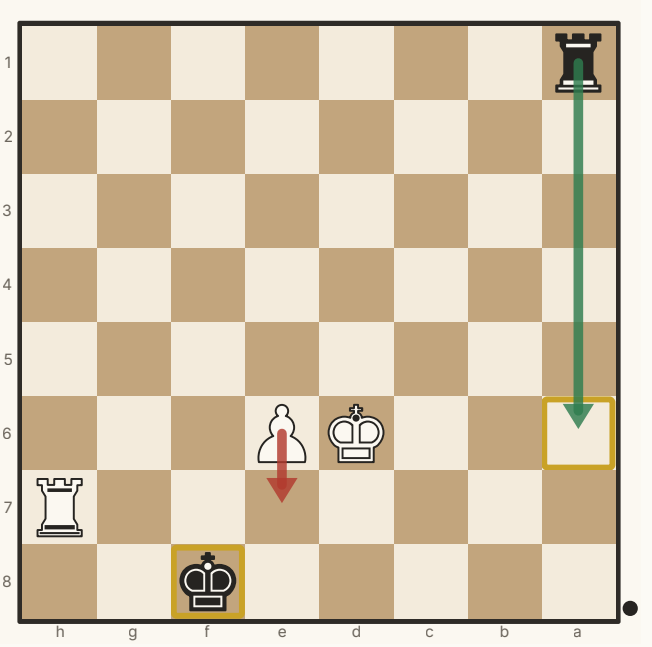
What if your king is not on the queening square at all? Then step one is to get it there. Exercise 23-04 shows that a king that is not yet on the queening square can still save the game, provided it heads there at once and your rook does not try to do the king's job.

Checking distance and the short side

Sometimes the defending king cannot stay in front of the pawn, and the defence relies on side checks. Side checks only work if the rook is far enough away. If the rook checks from too close, the attacking king walks up to it and the checks stop.

That leads to the idea of the **short side** and the **long side**. A pawn divides the board into two sides. For an e-pawn, the f-, g- and h-files are the short side (three files), and the a- to d-files are the long side (four files).

> **KEY IDEA** **The short-side rule**
>
> When you are defending and your king is pushed off the queening square, move the king to the **short side** and keep your rook on the **long side**. The rook then has room to check from at least three files away, which is usually enough to keep the attacking king from approaching.



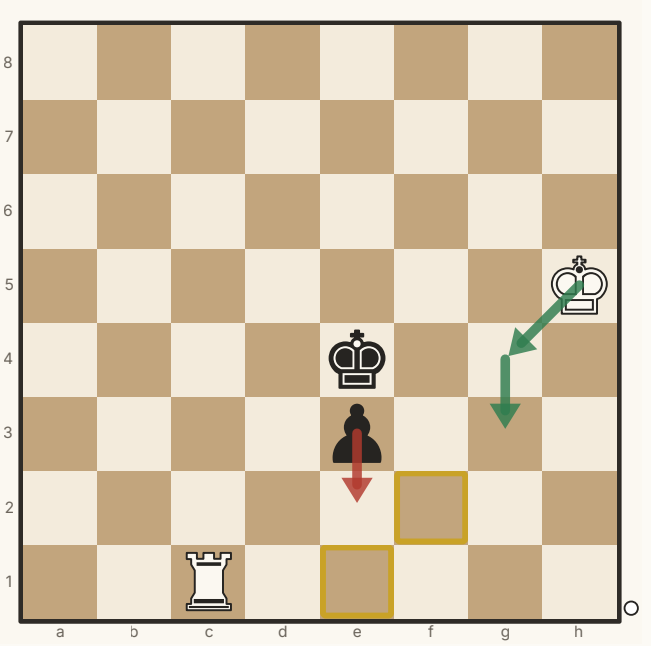
Black to move. The king is on the short side (f8), the rook on the long side (a-file). Ra6+ starts checking from three files away, and the engine confirms the draw.

Here White threatens e7+. Black's king is on the short side, so the rook can check from the long side: 1...Ra6+ and the king has no good way to approach. Checking from behind with 1...Rd1+ also holds here. Exercise 23-06 shows the same idea with the pawn on the other side of the board. If the roles were reversed, with the king on the long side and the rook squeezed onto the short side, the checks would often come from too close.

What commonly goes wrong: defenders send the king towards the “big” side because there seems to be more room there, and then the rook has nowhere to check from. Flip the instinct: king short, rook long.

Rook against pawn: count the race

When the defender gives up the rook for the last pawn, or the attacker has to give up the rook for a dangerous pawn, you reach **rook against pawn**. The rook is worth far more than a pawn, but a far-advanced pawn supported by its king can force a draw. The outcome depends on counting.



White to move. The rook alone cannot win the pawn; the white king must arrive in time. Only Kg4, heading for f2, wins.

Ask three questions:

1. **How many moves does the pawn need to promote?**
2. **How many moves does my king need to reach the pawn (or the square in front of it)?**
3. **Can my rook cut the enemy king off from the pawn**, along a rank or a file, to win time?

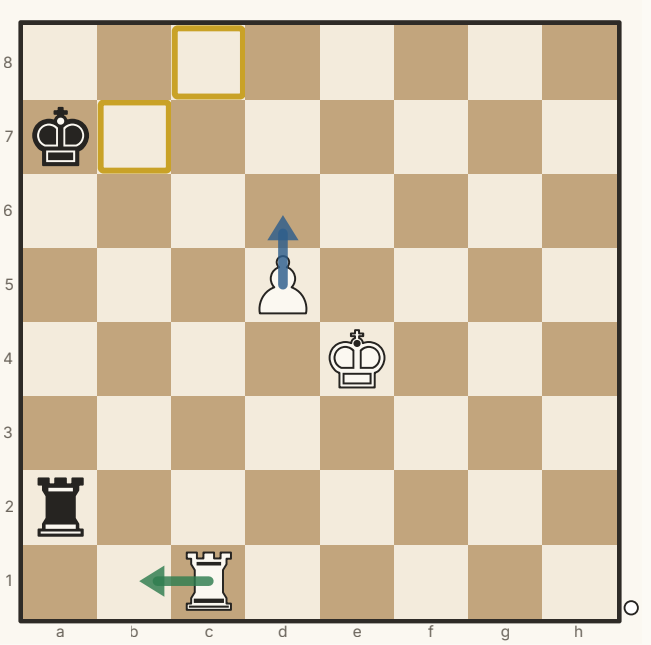
In the diagram the pawn needs two moves to queen, and the rook already controls e1 from c1. White's king needs to reach f2 or d2 in time to help the rook. Only 1.Kg4! does it. After 1...e2 2.Kg3 the king reaches f2 next. Any rook move wastes the tempo: after ...e2 and ...Kf3 the black king supports the pawn, and the rook must give itself up.

There are a few tricks worth knowing too. A rook can stop a pawn on the seventh by attacking it from in front while also covering the squares the defending king needs (Exercise 23-08). Sometimes the defending side can under-promote to a knight with check to avoid a skewer. And often the defender's king should stay *beside* or *in front of* its pawn, not behind it, so that the attacking king cannot cut it off.

> BOARD DRILL **Count before you move**
>
> Set up the diagram above. Before touching a piece, write down how many moves the pawn needs and how many your king needs. Then play Kg4 against an engine at full strength and check that you can win the pawn. Repeat with the white king on h6: now it is one step too slow, and the engine shows a draw. One tempo is the whole difference.

A worked example: thinking through a rook ending

Here is a full thinking process for a typical situation. Look at this position with White to move.



White to move. The pawn is on d5; the black king is far away on a7. What is the plan?

Step 1: Material and pawns. White has a rook and a passed d-pawn; Black has only a rook. A win is possible but not automatic: rook and pawn against rook is often a draw.

Step 2: The kings. White's king is right next to the pawn. Black's king is on a7, three files from the d-file. Can it get back? It wants b7, c7 or c8, and then d8.

Step 3: Candidate moves. The tempting move is 1.d6, because pushing passed pawns feels like progress. The other candidates are king moves such as 1.Ke5 and the rook move 1.Rb1, which puts a wall on the b-file.

Step 4: The opponent's best reply. After 1.d6 Kb7, Black's king heads for c8 and d8, or Black's rook starts checking, and the engine confirms that White has nothing. After 1.Rb1, the black king cannot cross the b-file at all. Black can check with the rook, but White's king will walk up the board with the pawn, and the black rook alone cannot stop both.

Step 5: Decide. 1.Rb1 wins; 1.d6 lets the win slip away.

1. **Rb1 Rh2** 2. **Ke5 Rh5+** 3. **Ke6 Rh6+** 4. **Kd7 Rh7+** 5. **Kc6**

After the cut-off the white king advances with the pawn. Once the king reaches c6 or c7, the checks run out and the pawn marches.

The same thinking applies in almost every rook ending: material, kings, candidate moves including the boring rook move that cuts off the king, the opponent's best reply, decision.

"All rook endings are drawn"

There is an old chess saying that all rook endings are drawn. It is half joke and half truth. The truth: a rook ending with an extra pawn is often a draw, especially when all the pawns are on one side of the board, or when the defender's rook is active. The joke: it is not literally true, and believing it makes you give up too easily as the attacker or relax too early as the defender.

> COACHING JUDGMENT **What the saying should mean for you**
>
> Coaching judgment: when you are a pawn down in a rook ending, do not resign in your head. Activate your rook, get it behind passed pawns, keep your king near the pawns that matter, and you will save many positions. When you are a pawn up, do not assume the win: you need the cut-off, the Lucena method and patience. The side that knows more technique wins most of the "drawn" rook endings at club level.

How to identify this in your own games: after each game that reached a rook ending, ask three questions. Was my rook active or passive? Did I cut off the king before pushing? Did I know whether the critical position was a Lucena or a Philidor? Record any "no" in your mistake journal under *endgame*.

Practice at the board

Rook endgames are learned with your hands. Use a physical board or any free chess app with an engine "play from position" feature.

1. **Lucena drill.** Set up `1K6/1P1k4/8/8/8/8/r7/2R5 w - - 0 1`. Play White against the engine at full strength. Win five times in a row, including after you vary the first move (for example, put the white rook on h1 and the black rook on h2). Your goal is to build the bridge without thinking.
2. **Philidor drill.** Set up `4k3/7R/1r6/3KP3/8/8/8/8 b - - 0 1` and play Black. Keep the rook on the sixth rank until the pawn moves, then go to the first rank and check from behind. Then set up `1r2k3/7R/3K4/4P3/8/8/8/8 b - - 0 1`,

where the white king has already reached d6, and find the only saving move (the mistake box above gives it away, so first try without looking).

3. **Short-side drill.** Set up `5k2/7R/3KP3/8/8/8/8/r7 b - - 0 1`. Hold the draw as Black against the engine. Then swap sides and try to beat the engine's defence: watch how it keeps its king on the short side.
4. **Cut-off drill.** From `r7/8/6k1/8/3P4/3K4/8/4R3 w - - 0 1`, win as White. Then move the black king to f5, only one file away: the engine still finds a win, but only with precise play. Find the first move yourself, then check.
5. **Check yourself.** After each attempt, ask the engine where the evaluation changed. If you lost the thread, find the move and write down the rule you broke.

Chapter summary

- Rook endgames are the most common endgames, and technique matters more than a single pawn.
- Activity first: an active rook usually outweighs a pawn; a rook stuck defending a pawn or blocking from the front is passive.
- Rooks belong behind passed pawns — yours and your opponent's.
- Cut the enemy king off by a file or rank before you push the pawn; two files usually wins, one file needs calculation.
- Lucena: check the king away, put your rook on the fourth rank, walk the king out, block the last check.
- Philidor: king on the queening square, rook on the third rank; once the pawn advances, check from behind.
- Defend with the king on the short side and the rook on the long side, checking from a distance.
- Rook against pawn is a race: count the pawn's moves, your king's moves and the cut-off possibilities.

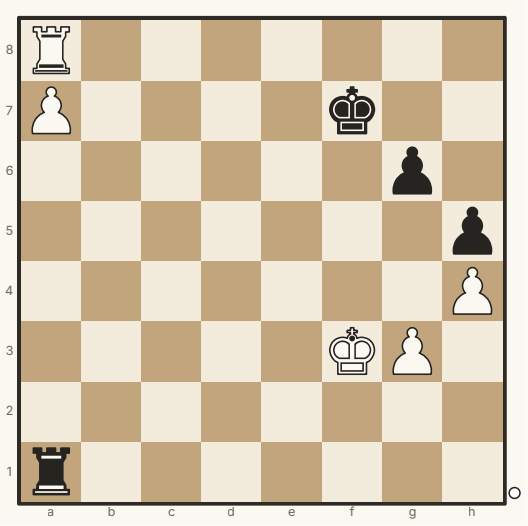
CHEAT CODE **Rook endgame procedure**

1. Compare the rooks: which is active, which is passive? Activate yours, even at the cost of a pawn.
2. Find the passed pawns and put your rook behind them (yours or theirs).
3. Attacking: cut off the enemy king by a file or rank before pushing; aim for the Lucena and build the bridge.
4. Defending: king in front of the pawn, rook on the third rank (Philidor); when the pawn advances, check from behind.
5. If your king is pushed aside, go to the short side and check from the long side.
6. In rook-against-pawn races, count tempi for the pawn and both kings before you move.

Exercises

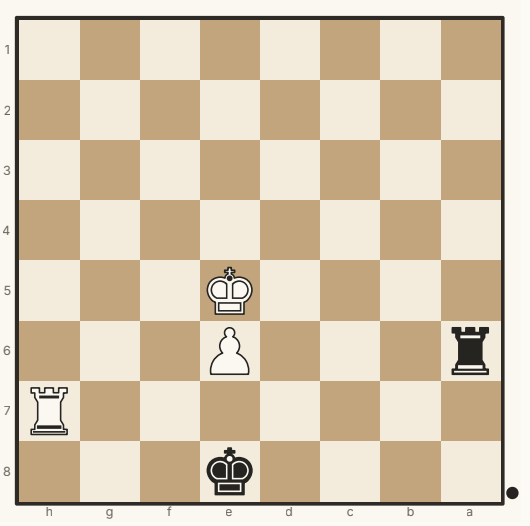
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

23.1 ENDGAME



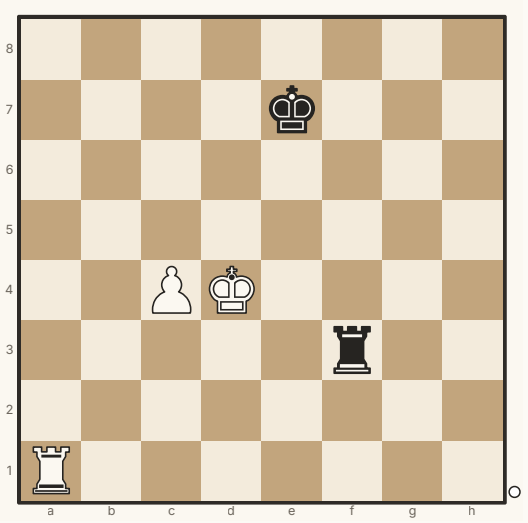
White to move. Black's king has wandered to f7, away from the safe squares g7 and h7. Win the game. Solution p. 706

23.2 ENDGAME



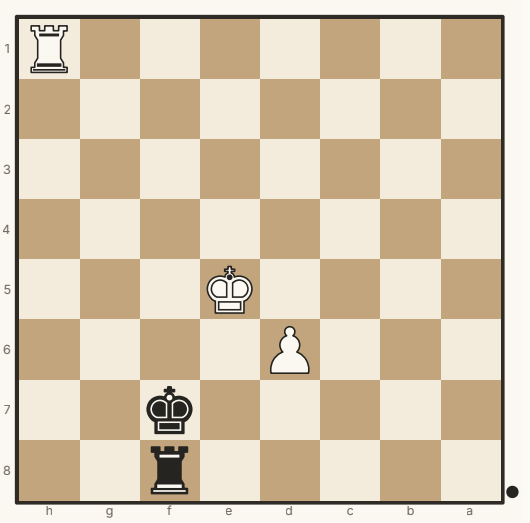
Black to move. You defended with the rook on the sixth rank, and White has just pushed the pawn to e6. Which move holds the draw? Solution p. 706

23.3 ENDGAME



White to move. The black king on e7 wants to come to d7 and stop the c-pawn. Choose the move that keeps it out. Solution p. 707

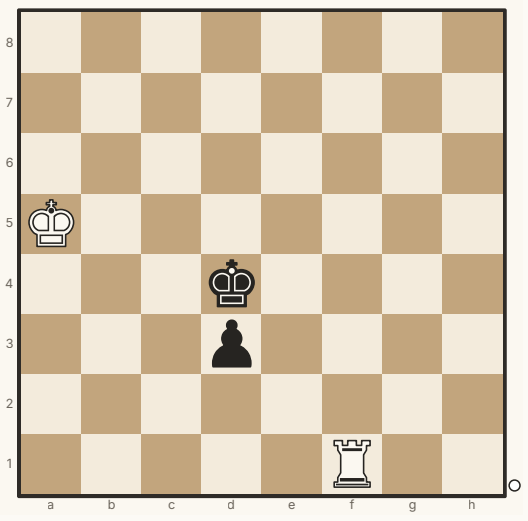
23.4 ENDGAME



Black to move. White's passed pawn is on d6. Find the only move that draws.

Solution p. 707

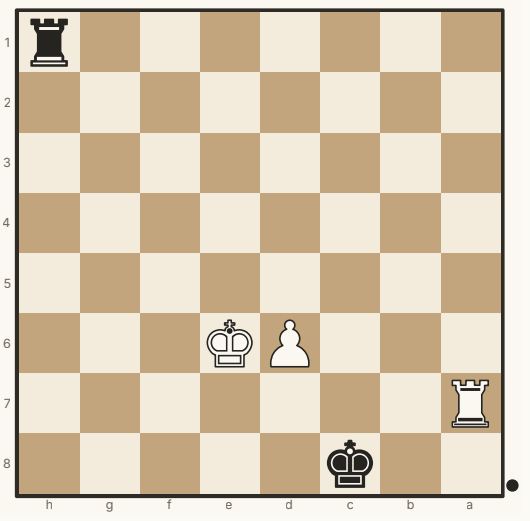
23.5 ENDGAME



White to move. Rook against pawn: count the race and find the only winning move.

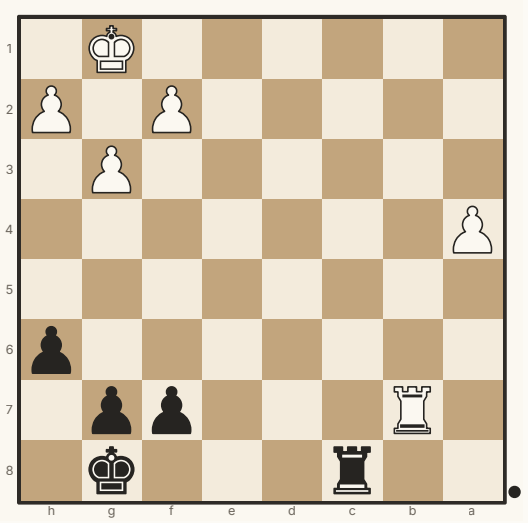
Solution p. 707

23.6 ENDGAME



Black to move. White threatens d7+. Which move saves the game? Solution p. 707

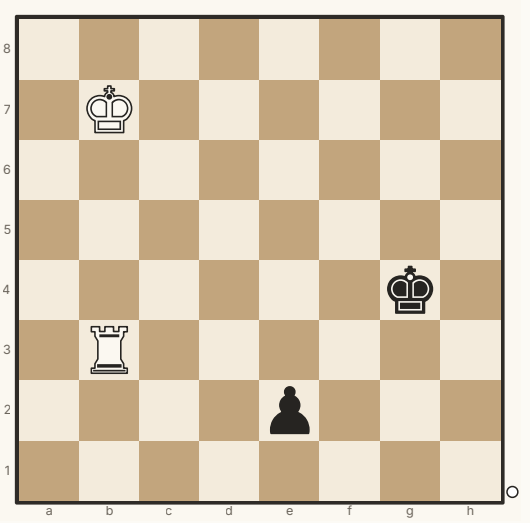
23.7 ENDGAME



Black to move. You are a pawn down and White has an outside passed a-pawn. Find the active plan that saves the game.

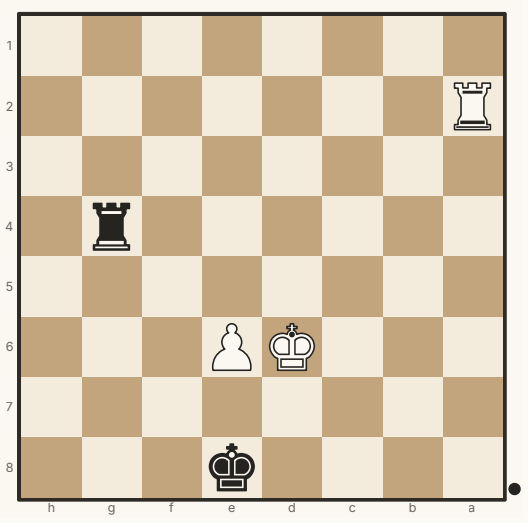
Solution p. 708

23.8 ENDGAME



White to move. Black's pawn is one step from queening. Is it a draw, or is there a win? Solution p. 708

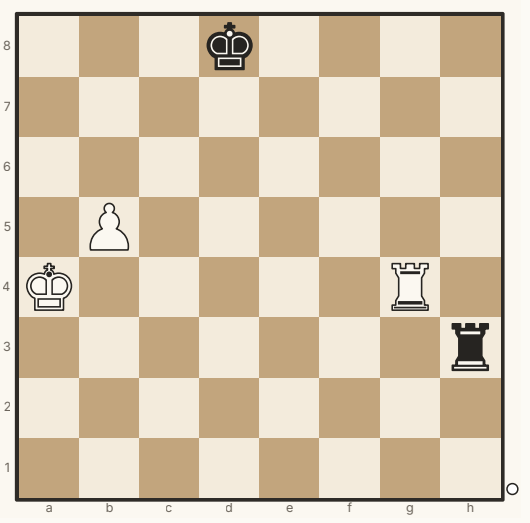
23.9 DEFENCE



Black to move. White threatens Ra8 mate. Find the only move that saves the game.

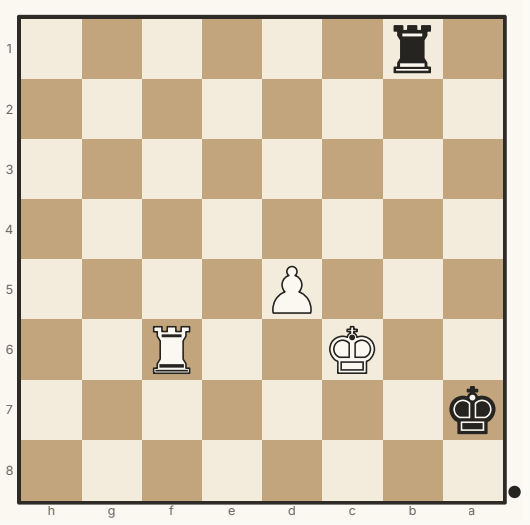
Solution p. 708

23.10 ENDGAME



White to move. Before you push the b-pawn, what must you do? Solution p. 708

23.11 ENDGAME



Black to move. The d-pawn is about to run.
Find the only drawing move. Solution p. 709

HINTS

23.1 Where must the black king stand so that a check on the seventh rank is impossible?

23.2 Once the pawn moves to the sixth rank, the rook's job changes. Where can it check from?

23.3 Which file must the black king cross to reach the pawn?

23.4 Which piece should stand on the queening square?

23.5 Which squares must the white king reach to help the rook?

23.6 Checks force the white king to answer instead of pushing the pawn.

23.7 Which square behind the a-pawn would your rook like to reach?

23.8 Attack the pawn from in front and check what happens if the king defends it.

23.9 What is White threatening? Then look for a check.

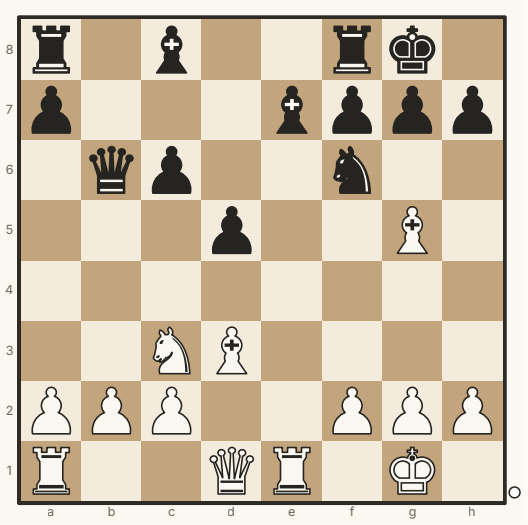
23.10 Build a wall between the black king and the pawn.

23.11 A check can gain the tempo you need, or force a trade that helps you.

Stage 4 Test

Fifteen mixed positions covering Part 4. Solve them without moving the pieces, write down your answers, then mark yourself. 12/15 or better: move on with confidence. 9–11: review the chapters behind your misses. Below 9: repeat this part's exercises before continuing.

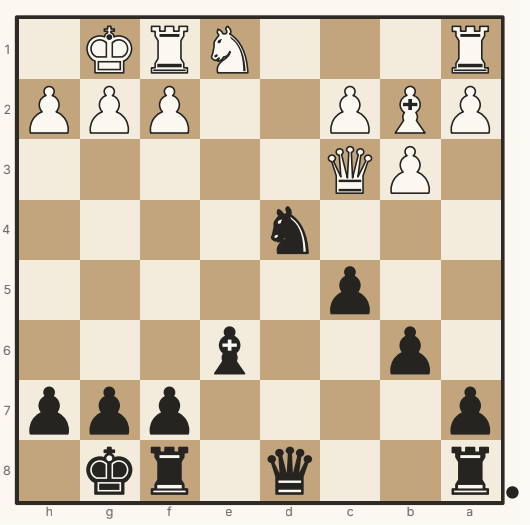
T4.1 TACTIC



White to move. Black has just played ...Qb6, attacking b2. Do you defend the pawn?

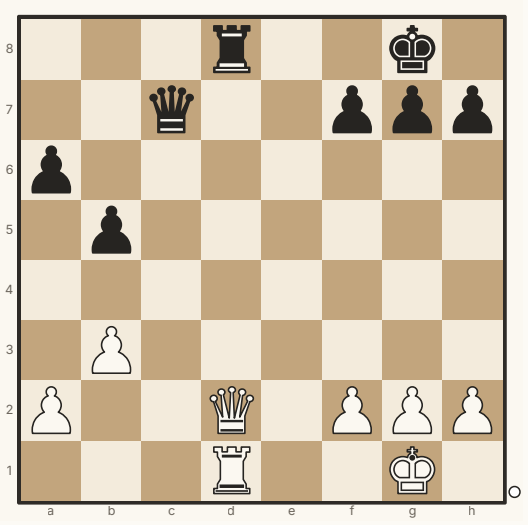
Solution p. 709

T4.2 TACTIC



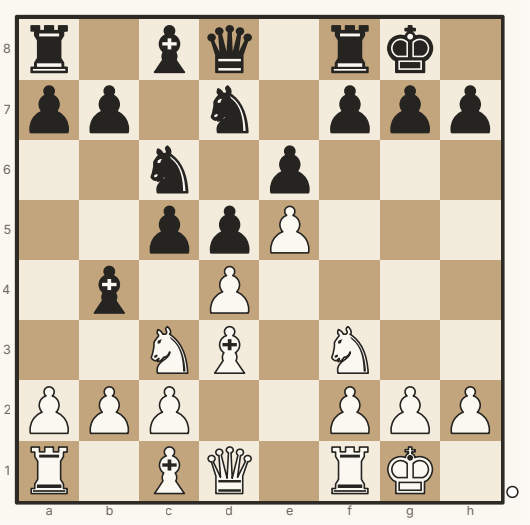
Black to move. White's queen and bishop are lined up against g7. Find the strongest move. Solution p. 709

T4.3 CHECKMATE



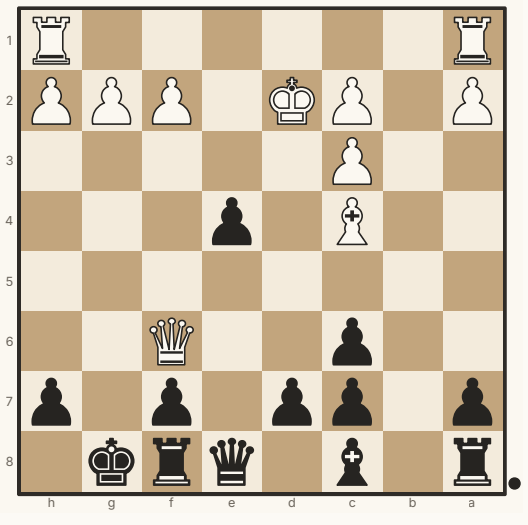
White to move. Black has no luft. Checkmate in two. Solution p. 710

T4.4 TACTIC



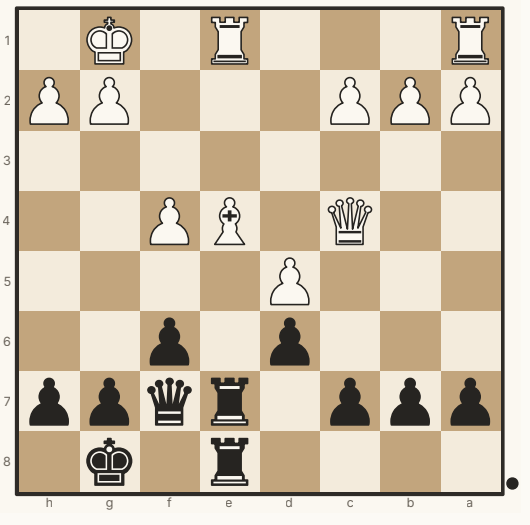
White to move. Black has castled, but no knight defends f6 or h7. Find the classic attacking sacrifice. Solution p. 710

T4.5 DEFENCE



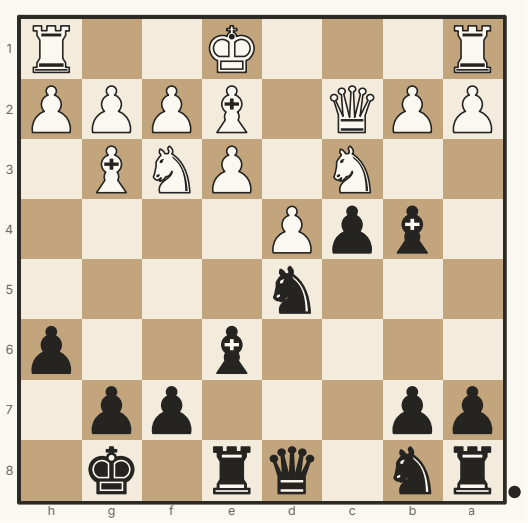
Black to move. White's queen sits on f6 next to your king, and White wants to bring a rook to the g-file. Find the only defence. Solution p. 710

T4.6 TACTIC



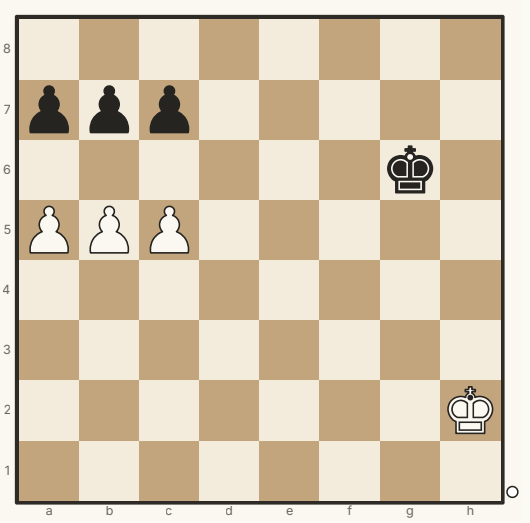
Black to move. You are a piece down, but White's e4-bishop is pinned to the rook on e1. Win the material back. Solution p. 710

T4.7 STRATEGY



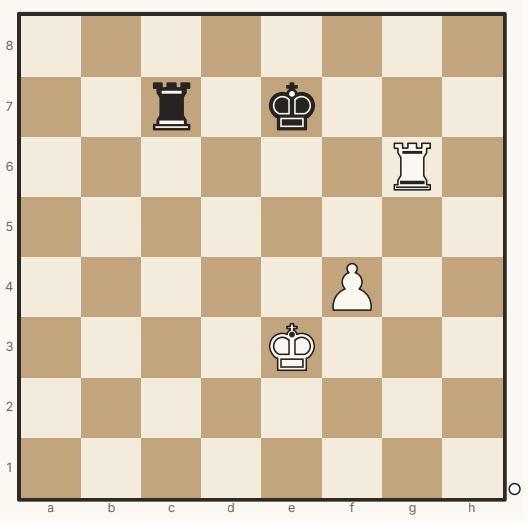
Black to move. White's king has not castled yet, and the c3-knight is pinned. What does White want to do next, and how do you stop it? Solution p. 711

T4.8 ENDGAME



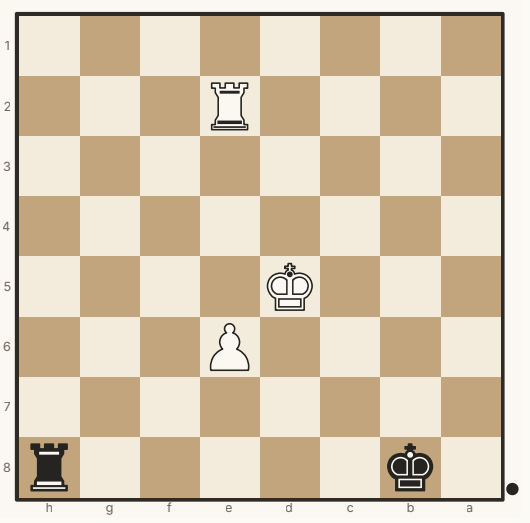
White to move. Both kings are far from the queenside pawns. Is there a way to make a queen? Solution p. 711

T4.9 ENDGAME



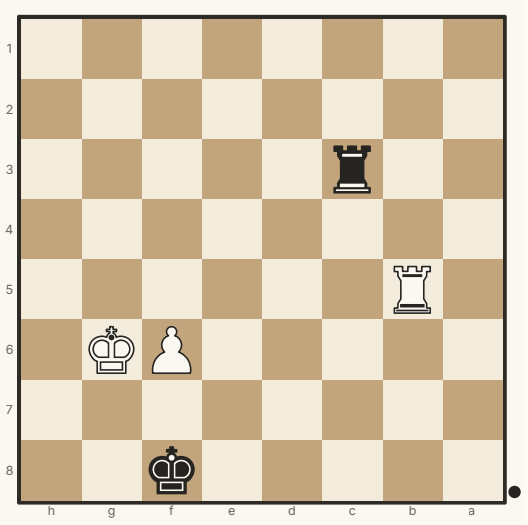
White to move. A rook ending with an extra pawn is often a draw. Is there a quick win here? Solution p. 711

T4.10 ENDGAME



Black to move. White's pawn is on e6. Your king must get in front of it. Which route draws? Solution p. 711

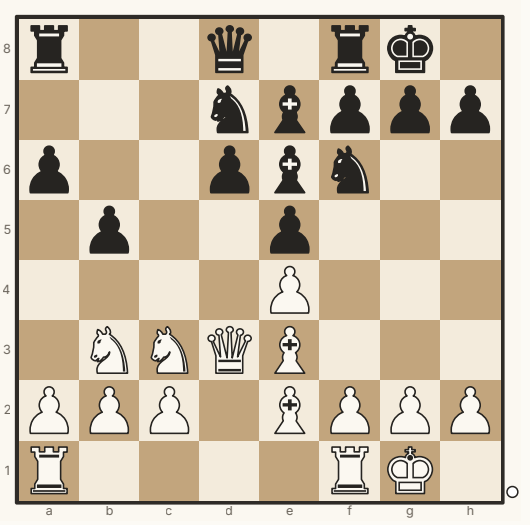
T4.11 ENDGAME



Black to move. White threatens Rb8 mate. Find the only move that saves the game.

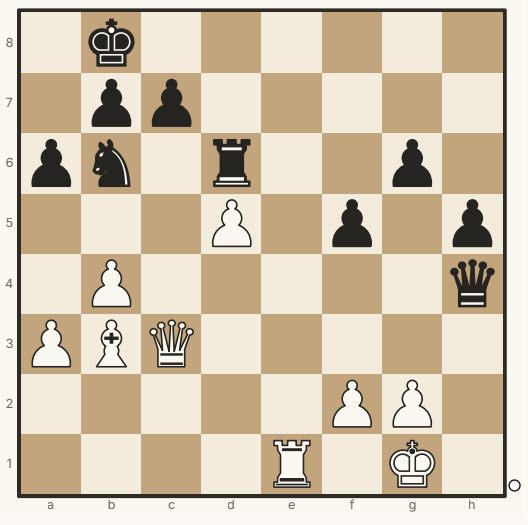
Solution p. 712

T4.12 STRATEGY



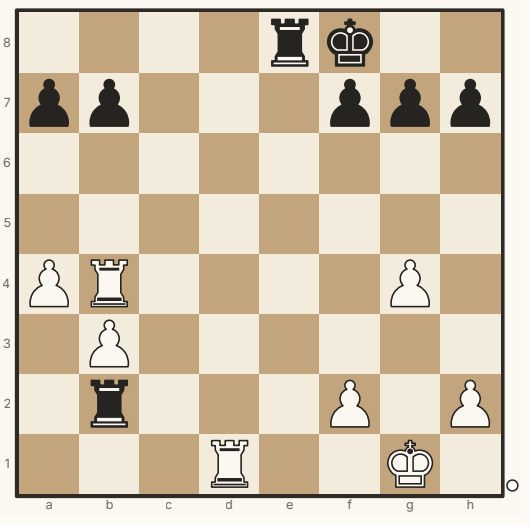
White to move. Black's d6-pawn is backward and d5 is a hole. Choose a plan move for your rook. Solution p. 712

T4.13 TACTIC



White to move. Black's queen and rook look active. Find the forcing sequence that overloads Black's queen. Solution p. 712

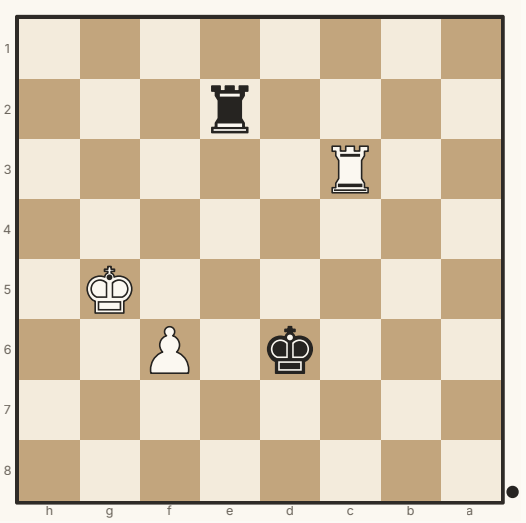
T4.14 ENDGAME



White to move. Both sides have active rooks. Which move uses your rooks best?

Solution p. 712

T4.15 ENDGAME



Black to move. White's f-pawn is close to promotion. Which move draws? Solution p. 713

HINTS

T4.1 What was the queen defending on d8?

T4.2 Which knight check attacks two pieces at once?

T4.3 Which piece defends Black's back rank, and can you remove it with check?

T4.4 Bishop to h7, knight to g5, queen to h5.

T4.5 Which pawn move attacks a piece and opens lines for your defenders?

T4.6 Attack the pinned piece with something cheap.

T4.7 Which of your pieces are loose, and how can one move protect them while adding pressure?

T4.8 Sacrifice the middle pawn first.

T4.9 King and rook on the same rank: is there a check along it?

T4.10 Which route keeps d6 and d7 under your control?

T4.11 When mate is threatened, look for checks.

T4.12 Which file points at Black's weak pawn?

T4.13 Check first, then ask which black piece is the only defender of the blocking rook.

T4.14 Look for a free pawn that also brings a rook to the seventh rank.

T4.15 Which piece should stand in front of the pawn?

PART FIVE

The Complete Player

Stage 5 · 1200–1500

The final part sharpens calculation for complex positions, deepens strategy and endgame technique, and gives you the practical, psychological and analytical tools to keep improving long after this book is finished.

Chapter 24 · Advanced Tactics and Calculation

Chapter 25 · Middlegame Strategy

Chapter 26 · Endgame Technique

Chapter 27 · Practical Play and Psychology

Chapter 28 · Analyzing Your Games and Training for Life

CHAPTER 24 STAGE 1200–1500

Advanced Tactics and Calculation

Chains of motifs, counter-shots and honest calculation

YOU WILL LEARN TO

- Find combinations that chain two motifs, such as a decoy that sets up a fork or a skewer.
- Use and defend against in-between moves, desperado captures, x-rays, interference and clearance.
- Spot quiet moves inside combinations and the opponent's best defensive resource.
- Calculate a small tree of candidate moves, stop when the position is quiet, and evaluate the end honestly.

> IN YOUR NEXT GAME
>
> Before you play any forcing sequence, list your opponent's checks and captures at every step of it, including the last move, and evaluate the final position as if you were your opponent.

At 1200 you already see forks, pins and back-rank mates. Your opponents see them too, which is why the simple one-move shot rarely lands any more. The games you now win and lose are decided by two-step and three-step ideas: a sacrifice that drags a piece onto a forkable square, a check slipped into the middle of an exchange, a capture that looks forced but is not.

This chapter is about those longer ideas and about the calculation that makes them safe. You will learn the advanced motifs, but the real skill is the discipline behind them: at every branch, look for your opponent's best reply, including the annoying in-between check, and judge the final position as it really is, not as you hoped it would be.

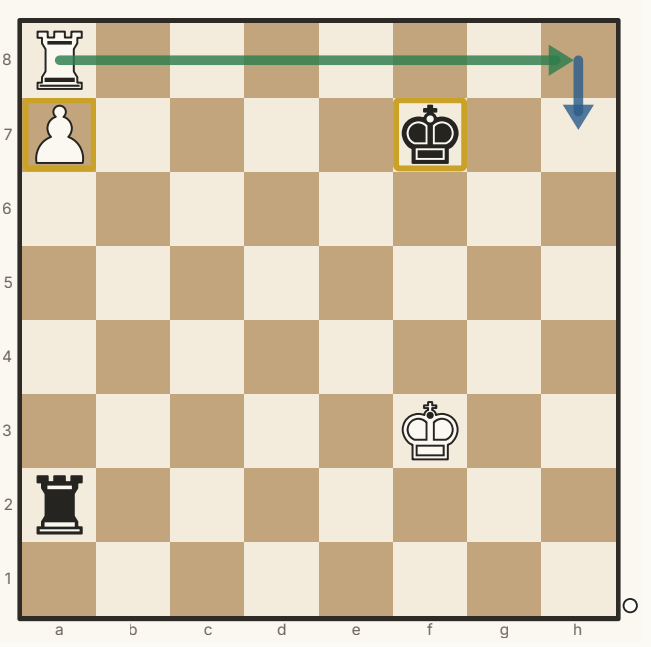
Combinations are chains of motifs

A **combination** is a forcing sequence, usually with a sacrifice, that uses two or more tactical motifs one after the other. The first motif rarely wins anything on its own. Its job is to prepare the second one: to put a piece on the right square, to remove a defender, or to open a line.

The two most common chains are:

- **Deflection or decoy, then fork.** A **deflection** drags a defender away from its job; a **decoy** lures a piece onto a particular square. Once the piece stands where you want it, a knight or queen forks it.
- **Decoy, then skewer.** You force the king onto a line, then attack it with a long-range piece so that, when it steps aside, you capture what stood behind it.

The way to find chains is to work backwards. First you notice a *target pattern*: "if their queen stood on d7, my knight could fork it from f6", or "if their king stood on the seventh rank, my rook would skewer it". Then you ask the second question: "Which forcing move puts it there?"



White to move. The rook on a8 is stuck in front of the pawn, but Rh8! threatens a8=Q. If Black takes the pawn, the rook on a7 and the king on the seventh rank are lined up.

This is a classic rook-ending trick and a perfect small chain. White plays 50.Rh8!, threatening 51.a8=Q. If Black grabs the pawn, the king and rook stand on the same rank and White skewers them:

50.**Rh8 Rxa7?** 51.**Rh7+**

The rook check on h7 drives the king off the seventh rank and the rook on a7 falls.

Notice the geometry: the trick works because Black's king is on f7. With the king on g7 or h7, Black answers Rh8 with ...Rxa7, because the check on h7 can simply be met by capturing the rook. Recognizing which square makes the chain work is half of the skill.

> KEY IDEA **Work backwards**
>
> Look for the pattern first (fork square, skewer line, mating square), then search for the check, capture or threat that forces an enemy piece into that pattern. Most combinations are found in that order, not by trying sacrifices at random.

The in-between move, in depth

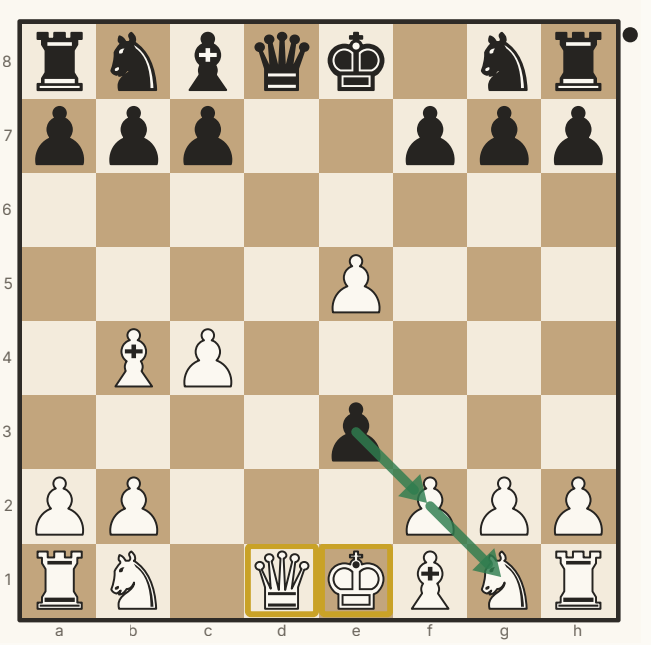
A **zwischenzug** (German for "in-between move") is a move inserted into a sequence that both players expected to go a certain way. Usually it is a check or a threat that must be answered, played *instead of* the "obvious" recapture. When the opponent answers it, the position has changed in your favour, and only then do you recapture.

You met the idea in earlier chapters. At this level you need to use it in two directions: to win material yourself, and to see it coming when you are the one who just captured.

The most instructive example is an old trap in the Albin Countergambit. White grabs material with a capture that looks completely forced, and Black answers with a zwischenzug that comes with check:

1.**d4 d5** 2.**c4 e5** 3.**dxe5 d4** 4.**e3?! Bb4+** 5.**Bd2 dxe3** 6.**Bxb4?? exf2+** 7.**Ke2 fxg1=N+!** 8.**Ke1 Qh4+**

Black never recaptured on b4. The check on f2 and the knight promotion with check leave White's king exposed and Black winning.



Black to move after 6.Bxb4??. White expects ...exf2+ to be met by a quiet king move. The real point is that the pawn checks, captures and then promotes, always with tempo.

Look at what White missed. After 6.Bxb4 White counted: "I took a bishop, Black's pawn on e3 is attacked; I am a piece up." White did not ask the most important question of this chapter: *what checks does my opponent have now?* The answer was ...exf2+, and after it the king can no longer defend the queen on d1 (7.Kxf2 Qxd1), or, after 7.Ke2, a new queen-or-knight appears on g1 with check.

When to look for a zwischenzug

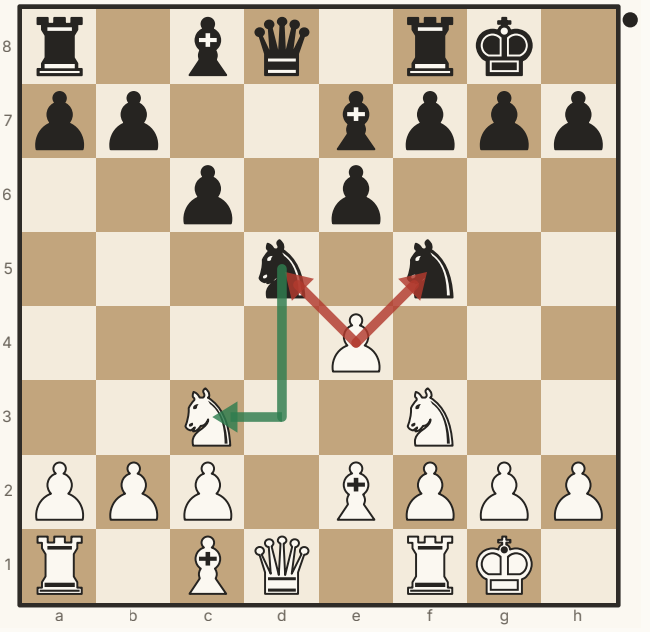
- **Every time you are about to recapture.** Before taking back, ask: do I have a check, a capture of something bigger, or a mate threat first?
- **Every time your calculation ends with your capture.** The opponent's move after your capture is part of the line. If it is a check, keep calculating.
- **Every time a line "forces" you.** "I must take back" is a feeling, not a rule. List the alternatives anyway.

COMMON MISTAKE **Stopping one move too early**

Most missed zwischenzugs happen when a player calculates “I take, he takes, I take, and I am a piece up” and stops. Your opponent gets a move after your last capture too. If that move is a check or a capture, your line is not finished.

Desperado: sell a lost piece dearly

A **desperado** is a piece that is going to be lost anyway, so it captures something on its way out. Desperados appear when two pieces are attacked at once, or when both sides have pieces hanging in an exchange. The rule is simple: if a piece is doomed, use its last move to take as much as possible, ideally with check or with a new attack.



Black to move. White has just played e3-e4, forking both black knights. Retreating one knight loses the other; the doomed knight on d5 must take something first.

Black’s knight on d5 cannot be saved, so it becomes a desperado: 10...Nxc3! takes a whole knight. After the recapture the second knight on f5 has time to step away:

10...**Nxc3** 11.**bxc3 Nd6**

Material is level again and Black's position is fine. Any knight retreat on move 10 would have lost a piece.

There is a second layer, and it shows how desperados interact with in-between moves. After 10...Nxc3 White might try the zwischenzug 11.Qxd8, trading queens before recapturing. But Black has a desperado of his own: the knight on c3 is still alive and takes with check.

10...**Nxc3** 11.**Qxd8? Nxe2+** 12.**Kh1 Rxd8**

The knight captures again with check before Black recaptures the queen. Black ends up a piece ahead.

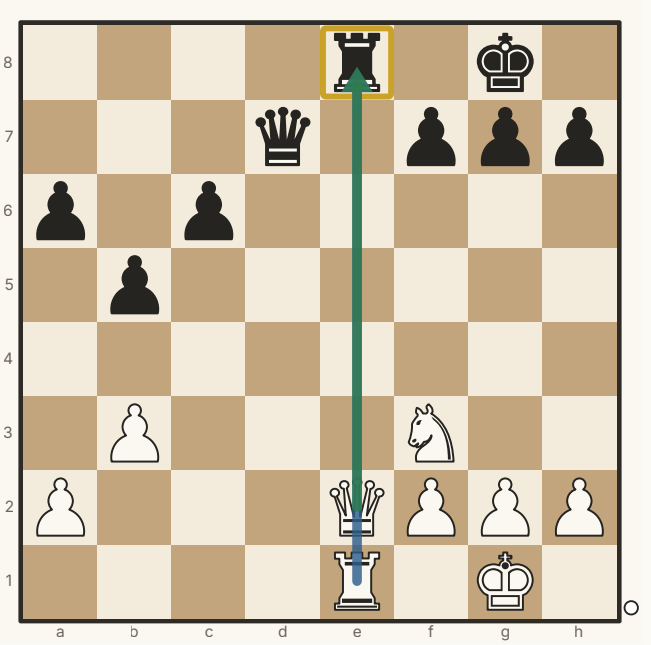
> COACHING JUDGMENT **Desperado thinking**
>
> In my experience, players at this level resolve hanging pieces in the order they notice them. Instead, list every piece that is attacked on both sides, then ask which capture gains the most, and whether it comes with check. The side that captures with tempo usually comes out ahead.

Lines through pieces: x-ray, interference and clearance

Long-range pieces act along lines. Three motifs are about what happens on those lines when other pieces stand in the way.

X-ray

An **x-ray** is the power of a piece *through* another piece. A rook behind your queen on a file defends the squares in front of the queen, even though the queen blocks it. The same is true of a rook behind an enemy piece: it attacks the square behind it as soon as the front piece leaves or is captured.



White to move. Black's rook on e8 is defended by the queen. White's rook on e1 stands behind the queen and x-rays e8 through it.

24. **Qxe8+ Qxe8** 25. **Rxe8#**

The queen trade is only a means: the rook behind it recaptures with mate on the back rank.

When you count attackers and defenders, count x-rays too. A queen with a rook behind it on the same file counts as two attackers on that file.

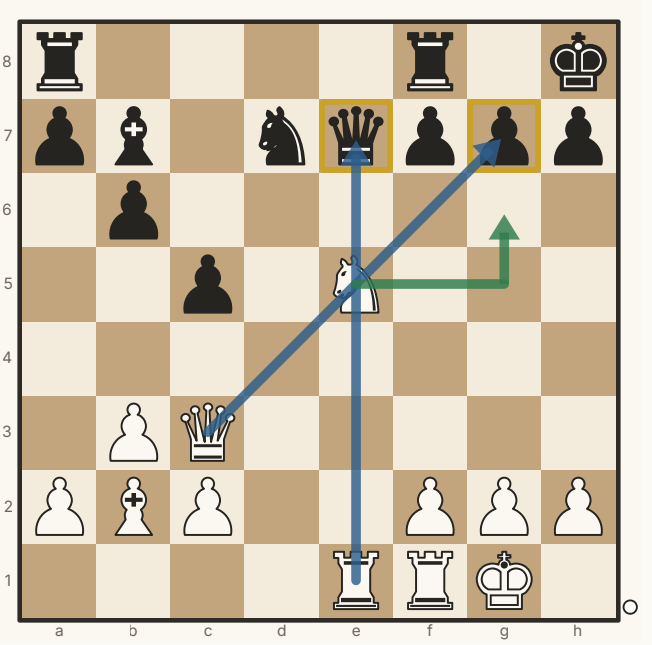
Interference

Interference means placing a piece on a line between an enemy piece and the square it guards. The defender is still on the board, but its line is cut. The classic case is a passed pawn on the seventh rank: the enemy rook guards the queening square from the side, and you drop a piece on the square between them. If the piece is captured, the pawn usually recaptures and promotes. You will find a clean example of this in the exercises.

To spot interference, look at the defender's line, not at the defender. Ask: "Which squares lie between this piece and the square it protects, and can I put something there with a threat?"

Clearance

Clearance is the opposite: you move your own piece out of the way, usually with a threat or a sacrifice, so that another of your pieces can use the square or the line. A knight standing on a battery's diagonal, a bishop in front of your own rook, a pawn blocking your queen: all of these are candidates for a clearance move.



White to move. The knight on e5 blocks two lines at once: the long diagonal from b2 and c3 to g7, and the e-file from e1 to Black's queen on e7.

We will calculate this position properly in the worked example below. For now, notice the idea: if the knight leaves e5 *with check*, Black has no time to deal with both opened lines.

Quiet moves inside combinations

A **quiet move** is a move that is not a check or a capture. Inside a combination, a quiet move is often the strongest move precisely because it is unexpected: your opponent is preparing to answer checks, and instead you make a threat that cannot be parried.

Typical quiet moves in combinations are:

- a queen move that creates two threats at once (a mate threat plus a discovered attack);
- a rook lift or king step that brings the last attacker in or takes away the opponent's only check;
- a retreat that keeps a pin or a bind instead of cashing in too early.

Quiet moves are easy to miss because our search naturally runs through checks, captures and threats in that order. The fix is simple: when the forcing moves do not quite work, ask "Is there a move that creates a threat my opponent cannot meet, even though it gives him a free move?" If the free move does not help him, you have found your quiet move.

> KEY IDEA **Checks, captures, threats, then quiet threats**
>
> Run the forcing-move scan as usual. When a promising attack seems to fall one move short, run it again looking for *quiet* moves that create two threats. A free move for your opponent is harmless if he has no useful way to spend it.

The opponent's counter-shot

Every combination has a weak point: your opponent also has moves. The defender's best resources are almost always forcing:

- an **in-between check** (the zwischenzug we met above);
- a **counter-attack** on something bigger than what you are attacking;
- a **desperado** capture by a piece you thought was trapped;
- a **perpetual check** when your king is exposed and you have spent material.

The habit to build is this: at every node of your calculation, before you look at your own next move, list your opponent's checks and captures. If one of them is dangerous, it is the move he will play, so it is the move you must calculate.

> **COMMON MISTAKE** **Calculating your own moves only**
>
> A very common pattern in club games: the attacker calculates five of his own moves and two of his opponent's, and the two he chose are the most convenient ones. A strong player calculates the opponent's *most annoying* reply first. If your combination survives that reply, it probably survives the others too.

When you are the defender, the same list is your rescue kit. Down material, with your opponent's king a little open? Look for checks before you look for defensive moves. Several exercises at the end of this chapter test exactly that.

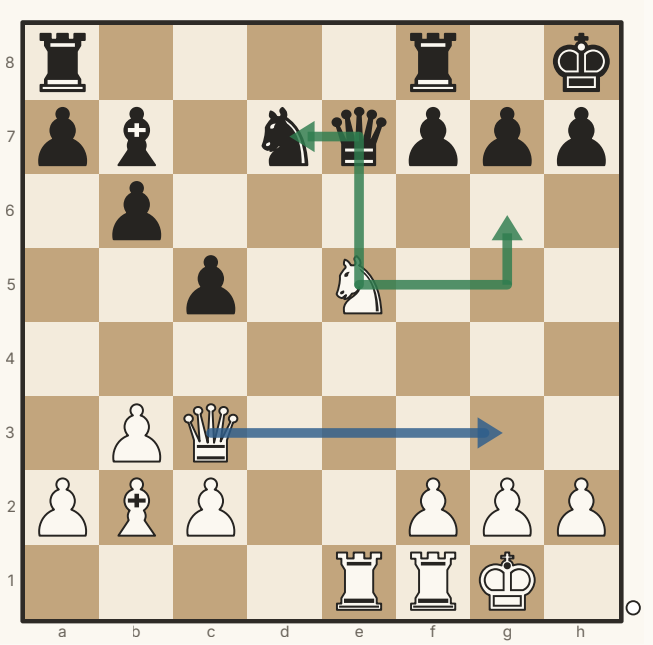
Calculating trees, not lines

So far we have spoken about lines. Real calculation is a **tree**: at each move there are a few sensible candidates, and each of them branches again. You cannot calculate every branch, so you need a method:

1. **At your move, choose two or three candidates** (forcing moves first).
2. **At your opponent's move, choose his most dangerous replies**, starting with checks and captures.
3. **Stop a branch when the position is quiet**: no checks, no captures that change the material balance, no immediate threats.
4. **Evaluate the quiet position** with material, king safety and activity.
5. **Compare the branches** and pick the move whose *worst* outcome is best for you.

Worked example: thinking through a tree

Return to the clearance position. White's knight on e5 blocks both the long diagonal and the e-file.



White to move. Candidate moves: Nxd7, Qg3 and Ng6+. Calculate each until the position is quiet.

First I check the position. Material is equal. Black's king on h8 is defended by the g7 pawn, which is attacked by my queen and bishop along the diagonal but shielded by my own knight. Black's queen on e7 is on the same file as my rook on e1, again behind my knight. Black's own threats: the knight on d7 attacks my knight on e5, and the queen also attacks it. So my knight is under fire; whatever I do, it should move with a purpose.

Candidate moves. The forcing moves are Nxd7 (capture), Ng6+ (check, also a sacrifice), Nxf7+ (check) and Qg3 (threat). Nxf7+ just loses the knight to ...Rxf7, so I keep three.

Branch 1: 18.Nxd7. The knight captures and uncovers my rook on the queen. Black must move the queen, for example 18...Qg5, and I am a piece up: Black cannot recapture on d7 and save the queen at the same time. Quiet position, plus a knight. Good, but let me look for more.

Branch 2: 18.Ng6+. Check, so Black's replies are few: ...hxg6, ...fxg6 and ...Kg8. I must look at each.

18. **Ng6+ hxg6??** 19. **Qxg7#**

The diagonal is open and g7 falls with mate.

18. **Ng6+ Kg8??** 19. **Qxg7#**

Same idea: the king cannot take on g7 because the bishop on b2 guards it.

18. **Ng6+ fxg6** 19. **Rxe7**

Black's best. With the f-pawn gone the queen guards g7 along the seventh rank, but the e-file has opened and the queen falls.

Stop and evaluate. After 18...fxg6 19.Rxe7 the position is quiet: no checks, no hanging pieces of note. I have given a knight for the queen, so I am about six points ahead. That is better than branch 1.

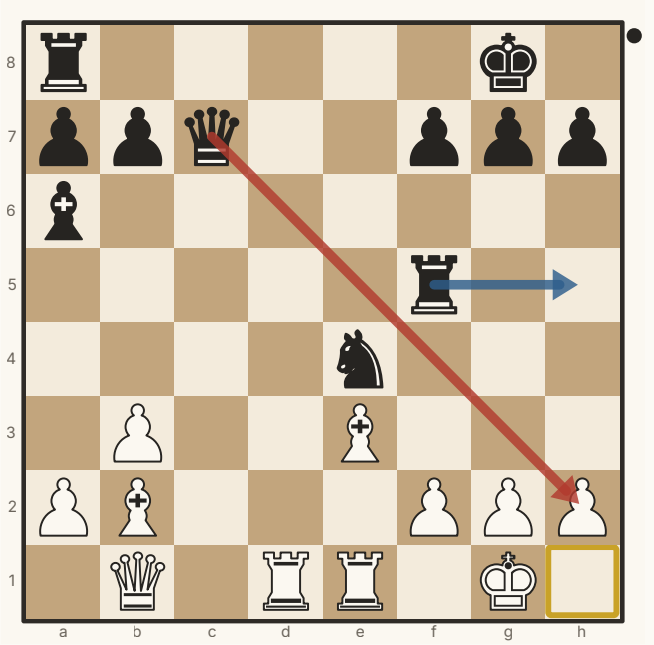
Branch 3: 18.Qg3 threatens Qxg7#, but Black answers 18...Nxe5 19.Bxe5 f5, and although I am still clearly better, I have won less than in branch 2.

Blunder-check before I move. In branch 2, after 19.Rxe7, does Black have a counter-shot? Black's checks: none. Captures: ...Bxg2 is answered by Kxg2, and my rook on e7 is defended by nothing but attacked by nothing either. My back rank: Black's rooks are on a8 and f8, not on open files towards my king. The line holds. I play 18.Ng6+.

Notice how small the tree was: three candidates, three replies in the critical branch, and every branch ended in a quiet position that I could count. That is what good calculation at this level looks like.

Evaluate the final position honestly

The last step of calculation is the one most often skipped. You reach the end of a line, you feel happy, and you play the first move. But the end of the line has to be judged as it is, from your opponent's side as well as your own.



Black to move. The sacrifice ...Qxh2+ Kxh2 Rh5+ Kg1 Rh1 looks like the famous rook mate. Check the last position honestly before you play it.

In this position Black's pieces are all aimed at White's king and the sacrifice looks thematic. Play it through to the end:

21...**Qxh2+??** 22.**Kxh2 Rh5+** 23.**Kg1 Rh1+?** 24.**Kxh1**

The rook on h1 is not protected, so the king simply takes it. Black has given away queen and rook for a pawn.

The pattern Black remembered needs a piece guarding h1, such as a bishop on the long diagonal or a knight on g3 or f2. Here nothing guards h1. The mistake was not in the calculation of moves; it was in the *evaluation* of the final position. Black saw "rook check on h1" and stopped thinking.

> KEY IDEA **Three questions at the end of every line**
>
> 1. Is it really over? (No checks or captures left for either side.)
> 2. What is the material count, honestly?
> 3. If I were my opponent, what would I play now?

Blunder-checking in complex positions

In quiet positions, a blunder-check is a quick look at checks, captures and threats. In sharp positions you need a slightly bigger routine, because many pieces are attacked at once:

- **What does my moving piece stop doing?** A queen that leaves the back rank may have been its only guard; a knight that jumps away may have blocked a file.
- **What do x-rays reveal?** If a piece leaves a line, a long-range piece behind it may now attack or defend through the gap.
- **What is the opponent's most forcing reply?** Calculate it first, to the end, before you calculate anything else.
- **Is every piece of mine defended after the move?** Loose pieces are the raw material of combinations.

> IN YOUR NEXT GAME **The two-question habit**
>
> In your next game, every time you are about to make a capture or a forcing move, ask two questions out loud in your head: "What does this piece stop guarding?" and "What is his most annoying reply?" Do it even when the move feels obvious.

Practice at the board

Set up each position on a board or in a free analysis app. Calculate for at most ten minutes without moving the pieces, write down your tree, then check with the engine.

1. **The skewer trick.** `R7/P4k2/8/8/8/5K2/r7/8 w - - 0 50`. Find White's winning move and calculate both of Black's main replies (taking on a7, or checking from the side). Then move Black's king to g7 and check why the trick no longer works.
2. **Desperado with two layers.** `r1bq1rk1/pp2bppp/2p1p3/3n1n2/4P3/2N2N2/PPP1BPPP/R1BQ1RK1 b - - 0 10`. Black to move. Calculate ...Nxc3 against both bxc3 and Qxd8. Then try one of the knight retreats and see how quickly a piece is lost.

3. **The clearance tree.** `r4r1k/pb1nqppp/1p6/2p1N3/8/1PQ5/PBP2PPP/4RRK1 w - - 0 18`. Write the tree for Nxd7, Qg3 and Ng6+, with Black's best reply in each branch. Compare your evaluations with the engine's.
4. **Honest evaluation.** `r5k1/ppq2ppp/b7/5r2/4n3/1P2B3/PB3PPP/1Q1RR1K1 b - - 0 21`. Play the sacrifice on h2 against the engine and see what goes wrong. Then decide which black piece would have to guard h1 for the sacrifice to work, and find a sound move for Black instead.

Check yourself: for each line you calculated, did you write down the opponent's checks and captures at every step? If not, repeat the exercise and do it.

Chapter summary

- Combinations are chains: the first motif (decoy, deflection, clearance) puts a piece or line in place for the second (fork, skewer, mate). Find the target pattern first, then the forcing move that creates it.
- A zwischenzug is a check or threat inserted before an expected recapture. Look for one every time you are about to recapture, and every time your calculation ends with your own capture.
- A doomed piece should become a desperado and capture something on its way out, preferably with check.
- X-rays count as attackers and defenders; interference cuts a defender's line; clearance vacates a square or line for another piece.
- Quiet moves inside combinations are strong when they create two threats and the opponent's free move does not help him.
- Calculate a tree: two or three candidates for you, the most dangerous replies for your opponent, and stop when the position is quiet.
- Evaluate the end of every line honestly, from your opponent's side, before you play the first move.

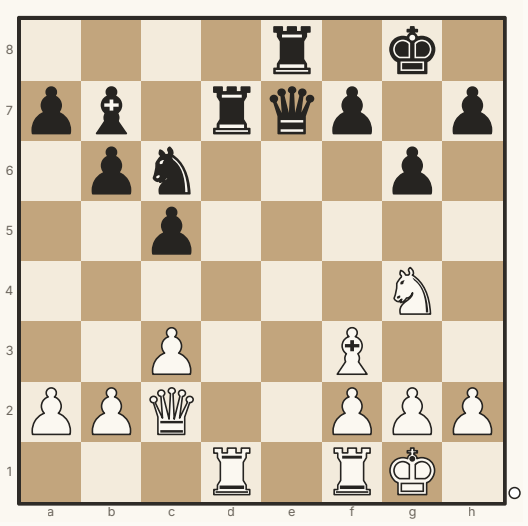
CHEAT CODE **The calculation routine**

1. Scan both sides' checks, captures and threats, and note every loose piece and every line (files, diagonals, x-rays).
2. Look for a target pattern (fork square, skewer line, mating square) and the forcing move that creates it.
3. Choose two or three candidate moves, forcing moves first, and include one quiet move with a double threat.
4. For each candidate, calculate your opponent's most annoying reply first, especially in-between checks and desperado captures.
5. Stop each branch when the position is quiet, count material and king safety, and ask what your opponent would play there.
6. Play the move whose worst branch is best for you, after a final check of what your moving piece stops defending.

Exercises

Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

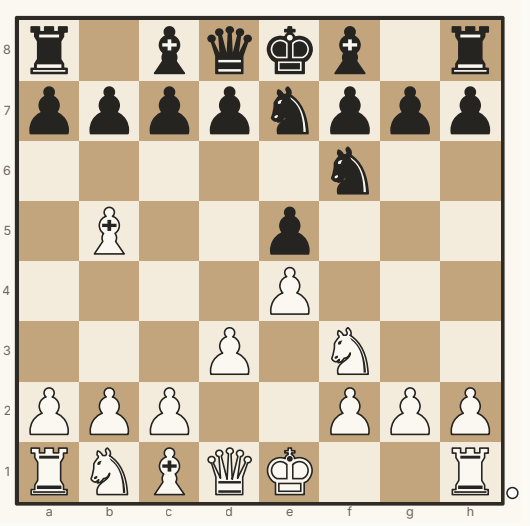
24.1 TACTIC



White to move. Two motifs are hiding here, one behind the other. Win material.

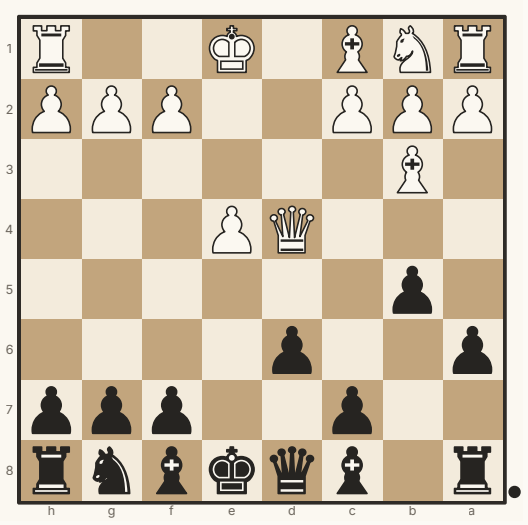
Solution p. 713

24.2 BLUNDER CHECK



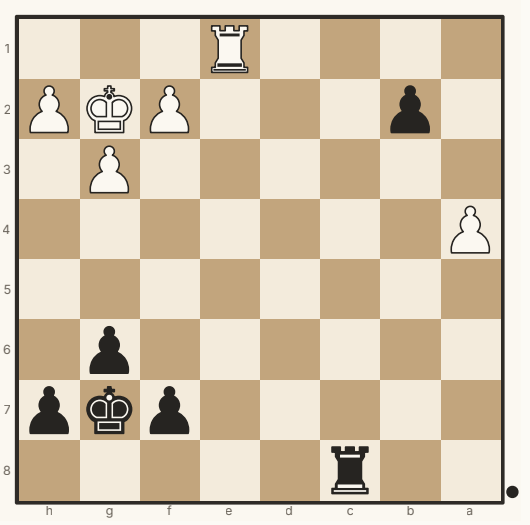
White to move. In the Ruy Lopez Black played 3...Nf6 4.d3 Ne7?!, and the e5-pawn is suddenly undefended. 5.Nxe5 seems to win a pawn. Play the right move. Solution p. 713

24.3 CALCULATION



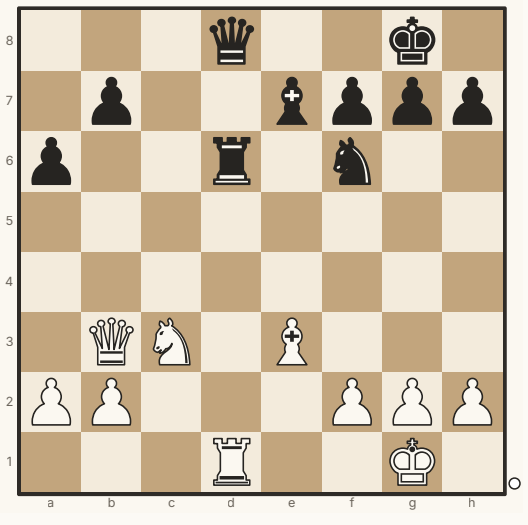
Black to move. White's queen has just taken on d4. Find the pawn moves that trap a piece. Solution p. 714

24.4 TACTIC



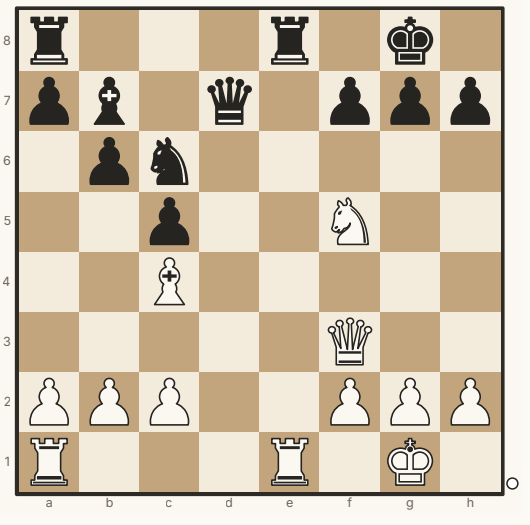
Black to move. The b-pawn wants to queen, but the rook on e1 guards b1. Cut the line.
Solution p. 714

24.5 BLUNDER CHECK



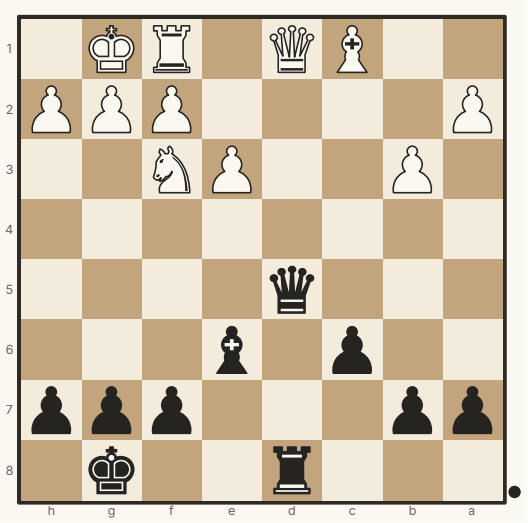
White to move. The b7 pawn is attacked twice and defended by nobody. Can you take it right now? Play the best move.
Solution p. 714

24.6 TACTIC



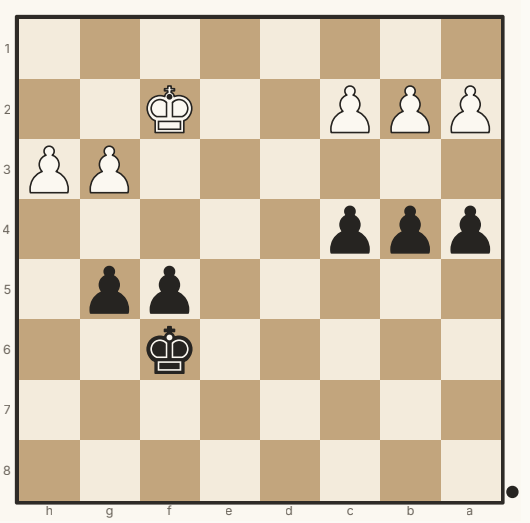
White to move. Find the quiet queen move that creates two threats at once. Solution p. 714

24.7 TACTIC



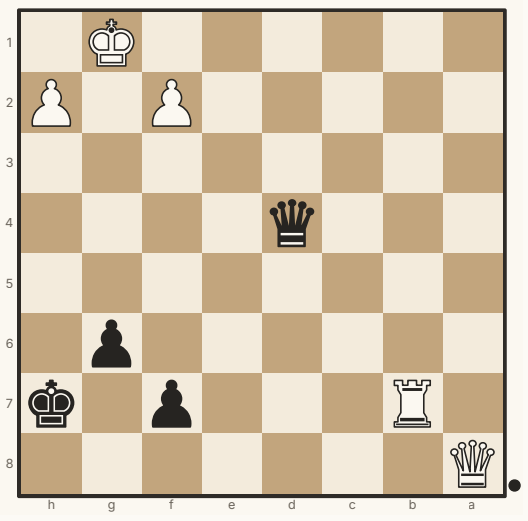
Black to move. You are a piece down, but both queens face each other on the d-file. Who x-rays whom? Solution p. 715

24.8 CALCULATION



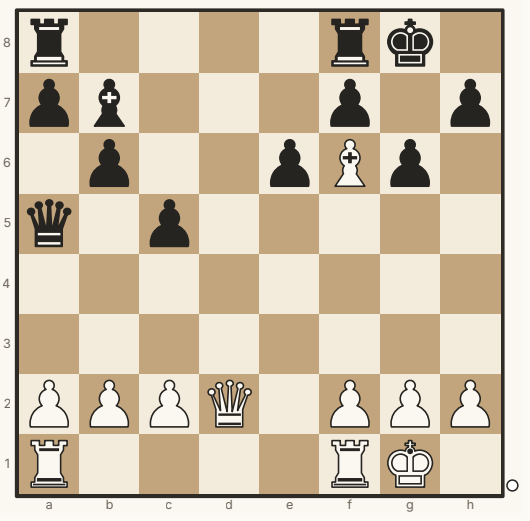
Black to move. Calculate the whole tree before you move: there is a way to make a queen by force. Solution p. 715

24.9 DEFENCE



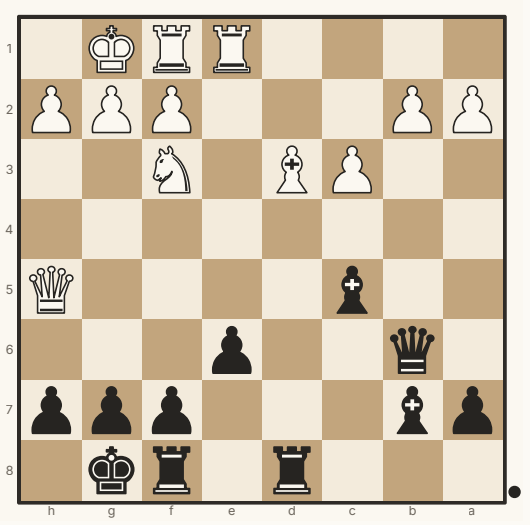
Black to move. You are a rook down and White is about to break through on f7. Which move saves the game? Solution p. 715

24.10 CHECKMATE



White to move. Your queen is attacked. Mate in 2. Solution p. 715

24.11 DEFENCE



Black to move. White threatens Qxh7#. You also have a flashy check on f2. What is the strongest move? Solution p. 716

HINTS

24.1 Which square would fork the king and the queen, and how can you bring the queen to it?

24.2 After Nxe5 c6 and a bishop retreat, which black check hits two things?

24.3 Where can the bishop on b3 go if you control c4 and a4?

24.4 Put a piece on the line between the defender and the queening square, on a square where the pawn can capture.

24.5 What does your queen defend right now?

24.6 Which square attacks g7 and also stands behind your knight on the d7 diagonal?

24.7 Count the attackers on d1, including pieces standing behind other pieces.

24.8 Sacrifice the middle pawn first, then answer each capture with the pawn that runs.

24.9 Can White's king ever escape your checks?

24.10 Which square next to g7 can your queen reach in one move?

24.11 Is the attacking piece defended?

CHAPTER 25 STAGE 1200–1500

Middlegame Strategy

Pieces, pawns and where to play

YOU WILL LEARN TO

- Judge bishop against knight, good and bad bishops, and the value of the bishop pair.
- Play the isolated queen's pawn from both sides and the Carlsbad minority attack.
- Handle space advantages and cramped positions, including freeing pawn breaks.
- Decide which pieces to exchange and on which side of the board to play.

> IN YOUR NEXT GAME
>
> When there is no tactic, name the pawn structure, decide which minor pieces are good or bad in it, and choose your plan and your exchanges from that.

Many games at your level are decided in the moments when nothing is happening. There is no tactic, no threat, and you play a "useful" move: a pawn push on the wrong wing, a trade that frees your cramped opponent, a knight on a beautiful square that is chased away two moves later. Twenty moves later you are losing and cannot say why.

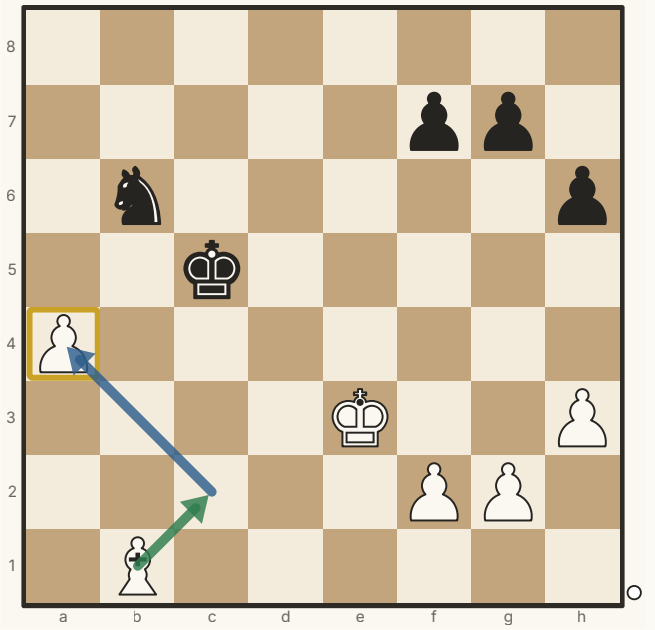
Strategy is the answer to "what do I do when there is nothing to calculate?". It rests on a few stable facts: which minor pieces suit the pawn structure, which pawns are weak, where you have more space, and which exchanges help whom. This chapter gives you those facts and turns each into questions you can ask at the board. Strategy never replaces calculation: every plan move still has to pass the blunder-check of Chapter 24.

Bishop against knight

The bishop and the knight are both worth about three points, but they are good at different things. A **bishop** is a long-range piece: it reaches both sides of the board quickly but can only ever use squares of one colour. A **knight** is a short-range piece: it is slow to cross the board, but it can reach every square and jump over pawns.

From that follow the main guidelines:

- **Open positions favour bishops.** With few central pawns, long diagonals are open and the bishop's range counts.
- **Closed positions favour knights**, especially when there is a protected square in the centre for them.
- **Pawns on both wings favour the bishop.** It can defend on one side and attack on the other; the knight needs several moves to switch.
- **A knight needs outposts.** An **outpost** is a square that no enemy pawn can attack, ideally protected by your own pawn. A knight there can be stronger than any bishop.



White to move. Pawns on both wings: the bishop can guard the a-pawn from far away (Bc2) while the king goes to the kingside. The knight cannot do two jobs at once.

Here 45.Bc2! defends the a4 pawn along the diagonal, and White keeps a clear advantage; slow moves that allow ...Nxa4 throw it away. The bishop works at long distance; the knight on b6 must stay close to whatever it attacks.

> COACHING JUDGMENT **How to judge a minor piece**
>
> In my experience, the quickest test is: "How many safe squares does this piece have, and do they reach the important part of the board?" A bishop staring at its own pawns, or a knight with no square closer than the fourth rank, is a weak piece, whatever its point value.

Outposts in practice

A knight on an outpost is strong only if it can stay there. Before you put a knight on a beautiful square, check three things: can an enemy pawn ever attack it, can the opponent simply trade it off for a bad piece, and if it is exchanged, does your recapture keep the square? A knight on e5 supported by a pawn on d4 can be traded, but after dxe5 your new pawn on e5 may be weak, or it may cramp the opponent; that is the real question.

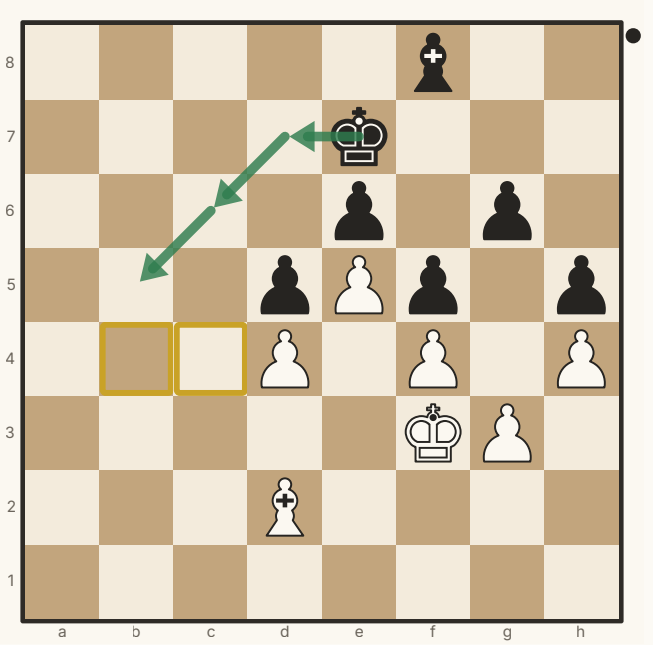
The defender's methods against an outpost are the mirror image: trade the knight for a minor piece of yours that is doing less, undermine the supporting pawn, or prepare a pawn move that controls the square in advance.

Good and bad bishops

A **bad bishop** is a bishop whose own pawns are fixed on squares of its colour. It is blocked by them, and it cannot attack the enemy pawns, which usually stand on the other colour. A **good bishop** is one whose pawns stand on the opposite colour, leaving its diagonals open.

Two important refinements:

- A bad bishop can still be a good *defender*: it protects the pawns that block it. It becomes truly bad when the position simplifies and it has nothing to do.
- "Bad bishop" is about **fixed** pawns. If the pawns can still move, the problem may be solved by a pawn break.



Black to move. All White's pawns stand on dark squares, so White's dark-squared bishop is bad. Black's king heads for the dark squares on the queenside: ...Kd7, ...Kc6 and ...Kb5.

In this ending Black wins with 45...Kd7! and the king march to b5 and c4. The white bishop cannot attack a single black pawn, and the white king cannot guard both d4 and the queenside entry squares. Waiting moves draw: the plan is the whole point.

What to do with a bad bishop:

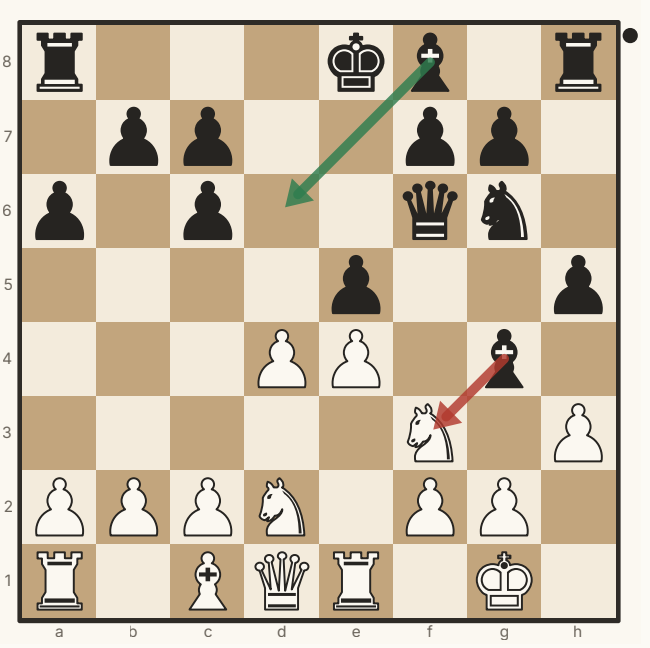
1. **Trade it**, ideally for the opponent's good bishop or a strong knight.
2. **Put it outside the pawn chain** (for example ...Bd7-e8-h5 in some French structures) so it attacks something.
3. **Change the pawn structure** with a break that opens its diagonals.

What to do against a bad bishop: keep the pawns fixed, avoid trading it, and, as the position simplifies, invade on the squares it cannot cover.

The bishop pair

Two bishops together cover both colours, and they are often worth more than bishop plus knight, especially in open positions and endgames. The side with the **bishop pair** usually wants to open the position with pawn breaks and to avoid

trading one of the bishops. The other side wants to keep the position closed, set up outposts for the knight, and trade one of the bishops when it can.



Black to move, Ruy Lopez Exchange. Black has the bishop pair in return for the doubled c-pawns. Developing moves such as ...Bd6 or ...Nf4 keep it; ...Bxf3 gives it up.

In positions like this, giving up a bishop "because it is attacked" or "to double the pawns" is a typical strategic slip. The engine is not dramatic about it here, but it rates the trade lower than the developing moves, and the bishop pair is exactly what Black got in return for the structure.

> **COMMON MISTAKE** **Trading the wrong minor piece**
>
> Before every exchange of minor pieces, ask: "Which of my pieces will be left, and which of his?" Trading your good bishop for his knight can leave you with a bad bishop against a dominant knight. Trading your bishop pair away without a reason gives back the main trump of your position.

Opposite-coloured bishops in the middlegame

When each side has one bishop and they run on different colours, endgames tend to be drawish (see Chapter 26). In the middlegame the opposite is true: the attacker's bishop cannot be opposed by the defender's bishop, so on the attack-

er's colour he effectively has an extra piece. If you have the initiative against the king, keep the queens and rooks on and aim your bishop at squares the enemy bishop can never guard. If you are the one defending, trading queens is usually your best route to safety.

The isolated queen's pawn

An **isolated queen's pawn** (IQP) is a pawn on d4 (or d5 for Black) with no friendly pawns on the c- and e-files. It arises from many openings you play, including the Alapin Sicilian (1.e4 c5 2.c3) and lines of the Queen's Gambit and the Exchange French.

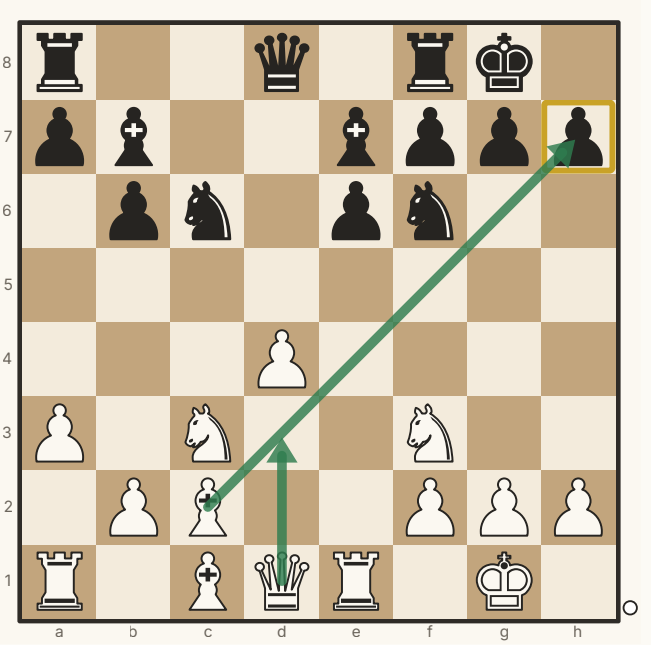
The IQP is a two-sided thing:

- **Strengths:** it gives space, controls e5 and c5 as outposts for your knights, and gives your pieces open files and diagonals. The side with the IQP usually has more active pieces in the middlegame.
- **Weaknesses:** it cannot be protected by a pawn, and the square in front of it (d5 for White, d4 for Black) is a perfect blockade square for the opponent. In the endgame the IQP is often simply weak.

Playing with the IQP

Use your activity before it fades. The two main plans are:

1. **A kingside attack with pieces.** The typical set-up is a queen and bishop battery against h7 (Qd3 with Bc2 or Bb1), a rook on e1 or e3, and a knight on e5.
2. **The d4-d5 break.** When your pieces are ready, d5 opens files and diagonals at the exact moment the opponent's pieces are least coordinated.



White to move, Alapin Sicilian. With Bc2 already in place, Qd3! completes the battery against h7 and forces a weakening.

13. **Qd3 g6** 14. **Bh6 Re8**

Black must weaken the dark squares around the king, and the white pieces gain targets.

Playing against the IQP

The defender has three goals: **stop the attack, blockade, and trade pieces.**

- **Blockade** the square in front of the pawn, usually with a knight (Nd5 against White's IQP). A blockading knight cannot be chased by pawns.
- **Trade pieces**, especially the IQP side's attacking pieces: its light-squared bishop and its knights. Every trade brings you closer to an endgame in which the pawn is weak.
- **Neutralize the attack first.** A blockade is no use if you are mated. Sometimes the right move is prophylactic, such as covering the square the enemy queen wants.

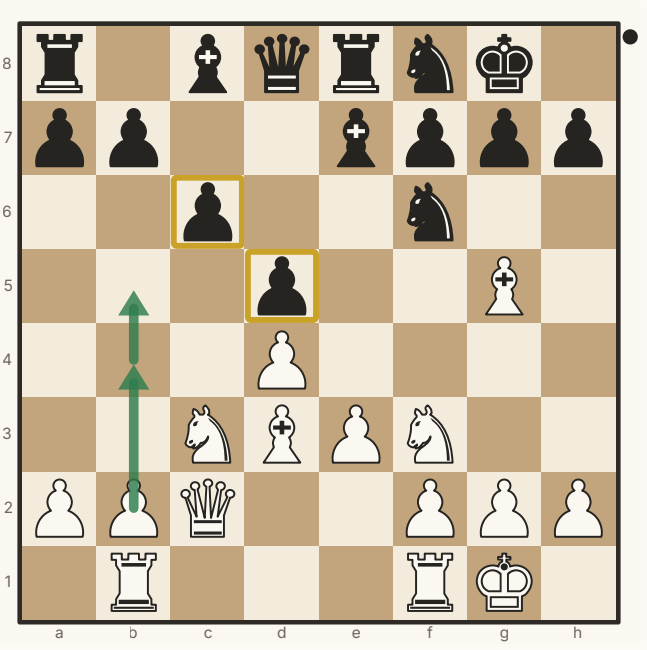
> KEY IDEA **IQP in one line**
>
> With the IQP: keep pieces, attack, and look for d4-d5. Against it: blockade, trade pieces, and head for the endgame.

The Carlsbad structure and the minority attack

The **Carlsbad structure** arises from the Queen's Gambit Declined Exchange Variation: White has pawns on a2, b2, d4 and e3; Black has a7, b7, c6 and d5. Black has three pawns on the queenside against White's two.

1.**d4 d5** 2.**c4 e6** 3.**Nc3 Nf6** 4.**cxd5 exd5** 5.**Bg5 c6** 6.**e3 Be7** 7.**Bd3 O-O** 8.**Qc2 Nbd7** 9.**Nf3 Re8** 10.**O-O Nf8** 11.**Rab1**

A typical Carlsbad position. The rook on b1 shows White's plan.



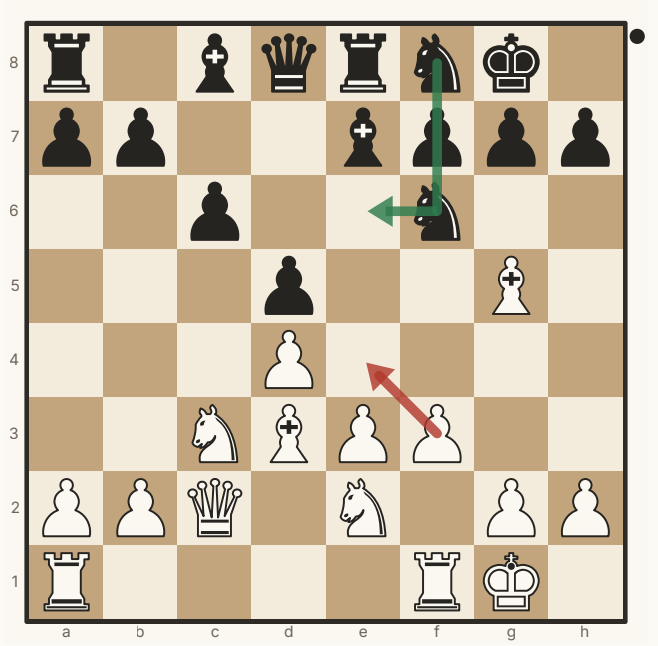
Black to move. White plans b2-b4-b5. After b5xc6 Black is left with a weak pawn on c6, or, after ...cxb5, with a weak isolated pawn on d5.

This is the **minority attack**: the side with fewer pawns on a wing advances them to create a weakness in the opponent's majority. White does not try to win a pawn immediately; it aims to leave Black with a permanent target, which White then attacks with rooks on the c-file and a knight heading for c5 or a4.

11...**Ng6** 12.**b4 a6** 13.**a4**

White prepares b4-b5. Black's ...a6 makes White spend a move on a4 before the pawn can arrive.

White's other plan in the Carlsbad is central: Nge2, f3 and e4, using the extra central pawn. It is more ambitious and weakens White's own king a little, so Black must react actively.

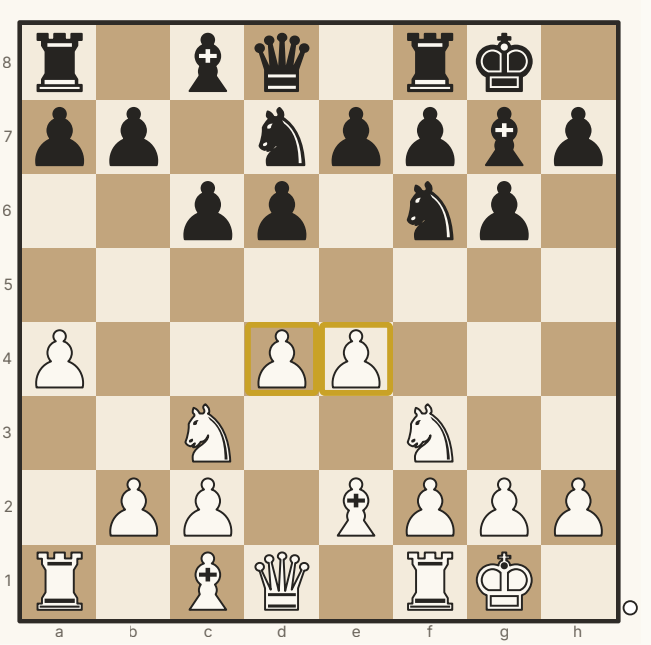


Black to move. White has chosen the central plan with Nge2 and f3, preparing e4. Black keeps the pieces flexible, for example with ...g6 and ...Ne6 or ...Ng6; the routine trade ...Ne4 is impossible here because of fxe4.

Black's usual counterplay is in the centre and on the kingside: ...Ne4 to trade pieces, ...Ng6 and ...Bd6 towards White's king, and sometimes ...f5 to support a knight on e4. White also has a central plan, f3 and e4, but it must be prepared; played too early it often runs into tactics.

Space and cramped positions

Your **space** is the area behind your pawn chain where your pieces can move freely. More space means more good squares for your pieces and easier regrouping. The cramped side's pieces get in each other's way.



White to move, a Pirc set-up met by our repertoire (Nc3, Nf3, Be2, O-O). White's central pawns on d4 and e4 give a comfortable space advantage; Black must eventually play ...e5 or ...c5 to free the game.

With more space:

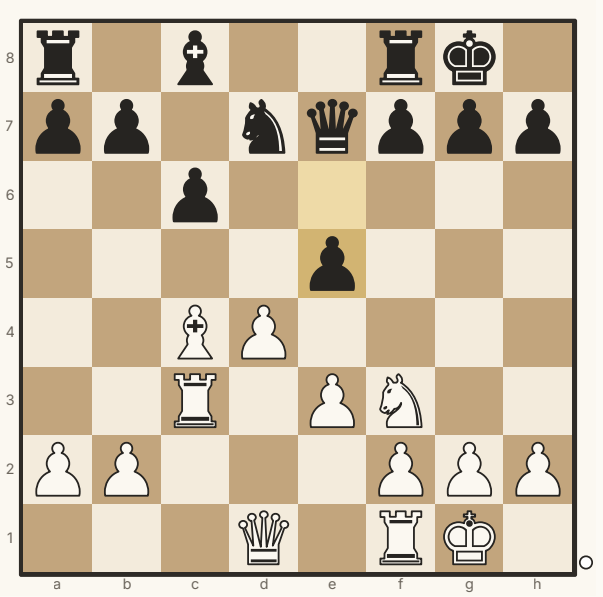
- **Keep pieces on the board.** Each exchange gives the cramped side more room.
- **Prevent the freeing breaks.** Know which pawn moves free the opponent (here ...e5 and ...c5) and make sure they cost something.
- **Use the space to regroup** towards the side where you want to play.

When cramped:

- **Trade pieces**, especially the ones that get in each other's way.
- **Prepare one freeing break** and make sure it works tactically. A break that loses a pawn frees nothing.

A model of the freeing idea is Capablanca's method in the Queen's Gambit Declined, part of your Black repertoire. Black trades two pairs of minor pieces with ...dxc4 and ...Nd5, and then frees the position with ...e5 or ...c5:

1. **d4 d5** 2. **c4 e6** 3. **Nc3 Nf6** 4. **Bg5 Be7** 5. **e3 O-O** 6. **Nf3 Nbd7** 7. **Rc1 c6** 8. **Bd3 dxc4** 9. **Bxc4 Nd5** 10. **Bxe7 Qxe7** 11. **O-O Nxc3** 12. **Rxc3 e5**



Two pairs of minor pieces are gone and ...e5 opens the diagonal for the c8 bishop. Black's cramp is over.

> **COMMON MISTAKE** **The break at the wrong moment**
>
> A freeing break must be prepared. The same pawn move that frees your game in one position can simply lose a pawn in another, if one more enemy piece attacks the square. Before any pawn break, count the attackers and defenders of the pawn that advances and of the squares it leaves behind.

When to exchange

Exchanges are strategic decisions. A useful set of guidelines:

- **Trade the defenders of weaknesses.** If a weak pawn or square is guarded by one piece, exchanging that piece makes the weakness fall. In the Carlsbad structure, for example, White sometimes plays Bxf6 to remove the knight that guards d5 and e4.
- **Trade your bad pieces for his good ones.** A bad bishop for a good bishop or a dominant knight is usually a good deal.
- **Trade attackers when you are under attack**, and avoid trades when you are the attacker.
- **The side with more space keeps pieces; the cramped side trades.**

- **When ahead in material, trade pieces, not pawns.** When behind, trade pawns and keep pieces (see Chapter 16 and Chapter 17).
- **Check what the recapture changes.** Every exchange involves a recapture, and the recapturing piece or pawn lands somewhere. A trade that brings an enemy pawn to a strong square, opens a file for his rook, or kicks your knight is not neutral.

A small example of the recapture question: if you take a knight on c6 with your bishop, the recapture decides what you have really done. After ...bxc6 your opponent gets the bishop pair and a half-open b-file, but also doubled pawns; after ...dxc6 the d-file opens for his rooks and his light-squared bishop gets an open diagonal. The same capture can be a good trade in one case and a gift in the other, so picture the position after the recapture before you exchange.

Exchanges are also the main tool of the defender in a worse position. If you are cramped or under pressure, every trade reduces the attacking potential of your opponent, but choose the trades that remove his *active* pieces. Trading your passive rook for his active one is good; trading your active knight for his passive bishop usually helps him.

Open and closed centres: where to play

The centre decides where the battle is fought.

- **Open centre** (few central pawns, open files): piece play dominates. Flank attacks with pawns are dangerous because the opponent can counter in the middle. Develop, centralize, and fight for open files.
- **Closed centre** (locked pawn chains): the centre cannot be opened quickly, so both sides play on the wings. Flank pawn storms become possible, even against a castled king.
- **Fluid centre** (pawns can still be exchanged or advanced): the timing of central breaks is everything. Do not lock it or open it without a reason.

In closed positions two rules help you choose the wing:

1. **Play on the side your pawns point to.** A pawn chain "points" in the direction it advances diagonally. White's chain d4-e5 points to the kingside; Black's d6-e5 points to the kingside too, so in King's Indian structures Black attacks with ...f5 and White with c4-c5 on the queenside.

2. **Play where you are stronger**: where you have more space, more pieces, or a pawn majority. Attacking where your opponent is stronger usually backfires.

The **base of the pawn chain** is its last pawn, the one that is not protected by another pawn. That is where the chain is most vulnerable: in the French, Black attacks d4 with ...c5, ...Nc6 and ...Qb6; in the King's Indian, White attacks the base d6 with c4-c5 and Black attacks the base e4 with ...f5. Attacking the head of a chain only makes it more solid.

> **IN YOUR NEXT GAME** **Name the structure**
>
> In your next three games, at move 15, write down the pawn structure in two or three words ("IQP", "Carlsbad", "closed centre, my pawns point kingside"), which minor pieces are good and bad in it, and which wing you will play on. Then compare with the engine after the game.

A worked example: choosing a plan

Here is how the procedure runs in a real decision. Take the Carlsbad position above with Black to move after 11.Rab1.

First I name the structure. Carlsbad: White has a minority on the queenside, Black has the majority there and more space on the kingside. The centre is closed by d4 and d5.

Then I judge the minor pieces. Black's light-squared bishop on c8 is the problem piece: my pawns on c6 and d5 stand on light squares and hem it in, so it needs a diagonal outside the chain (f5, g4 or e6). White's dark-squared bishop on g5 presses on my knight on f6. My knight on f8 is heading for g6 or e6.

Then I look at plans. White's plan is clear: b4-b5. My plans: (a) slow it down with ...a6, (b) trade pieces with ...Ne4 to lessen the pressure, (c) play on the kingside with ...Ng6, ...Bd6 and perhaps ...Ne4.

Then I check the candidate moves tactically. 11...Ne4 looks thematic, but after the trade on e7 and Bxe4 dxe4, my e4 pawn may become a target; I would need to calculate it. 11...Ng6 develops and prepares ...Bd6 without committing. 11...a6 is useful prophylaxis.

Finally I blunder-check. After 11...Ng6 my pieces are all defended and White has no direct tactic. I play it and keep ...a6 for the moment White plays b4.

The engine agrees: ...Ng6, ...a5 and ...Ne4 are the top choices, all close together. In strategic positions there is often no single best move, but there is always a *reason* for the move you choose.

Practice at the board

1. **IQP attack and defence.** `r2q1rk1/pb2bppp/1pn1pn2/8/3P4/P1N2N2/1PB2PPP/R1BQR1K1 w - - 2 13`. Play White against an engine at a modest level for ten moves, aiming for the battery on h7 and the d4-d5 break. Then switch sides and play Black, aiming to blockade on d5 and trade pieces.
2. **Minority attack.** `r1bqrnk1/pp2bppp/2p2n2/3p2B1/3P4/2NBPN2/PPQ2PPP/1R3RK1 b - - 10 11`. Play White's plan (b4, a4, b5) against the engine and try to leave Black with a weak c6 or d5 pawn. Write down when you would trade bishops on f6 or e7.
3. **Good against bad bishop.** `5b2/4k3/4p1p1/3pPp1p/3P1P1P/5KP1/3B4/8 b - - 0 45`. Win as Black against the engine. Then give White the move and see if White can hold.
4. **Bishop against knight.** `8/5pp1/1n5p/2k5/P7/4K2P/5PP1/1B6 w - - 0 45`. Find White's best move, then play the ending out. Count how many moves the knight needs to get from one wing to the other.

Check yourself by asking the engine for its evaluation after each phase. If your evaluation and the engine's differ by more than about a pawn, find the moment where your plan went wrong.

Chapter summary

- Bishops prefer open positions and pawns on both wings; knights prefer closed positions and need protected outposts.
- A bishop is bad when its own pawns are fixed on its colour. Trade it, activate it outside the chain, or change the structure; against it, fix the pawns and invade on the other colour.
- The bishop pair is a long-term trump: open the position and avoid trading a bishop without a reason.
- With an IQP, attack with pieces and look for d4-d5; against it, stop the attack, blockade the square in front, and trade pieces.

- In the Carlsbad structure, White's minority attack (b4-b5) creates a weak pawn on c6 or d5; Black's play is in the centre and on the kingside.
- More space: keep pieces and prevent the freeing breaks. Cramped: trade pieces and prepare one break that works tactically.
- In closed centres play on the side your pawns point to, where you are stronger, and attack the base of the enemy chain.

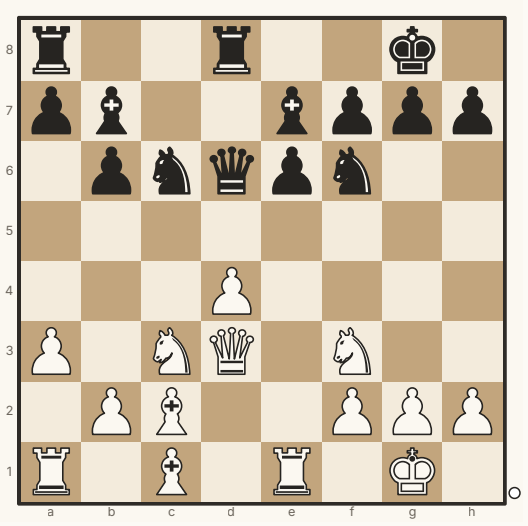
CHEAT CODE **The strategic routine**

1. Name the pawn structure and its weak pawns and squares, for both sides.
2. Judge each minor piece: good or bad, and how many useful squares it has.
3. Decide where to play: in the centre if it is open, otherwise on the wing your pawns point to and where you are stronger.
4. Pick a plan of two or three moves and the exchanges that help it (trade defenders, bad pieces and attackers; keep pieces when you have space).
5. Check every plan move tactically, including what the recapture of any exchange gives the opponent.

Exercises

Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

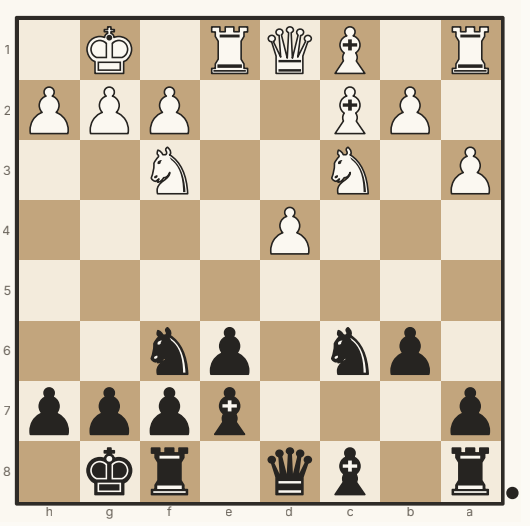
25.1 STRATEGY



White to move. You have the isolated queen's pawn and your pieces are set up behind it. Choose the move that cashes in.

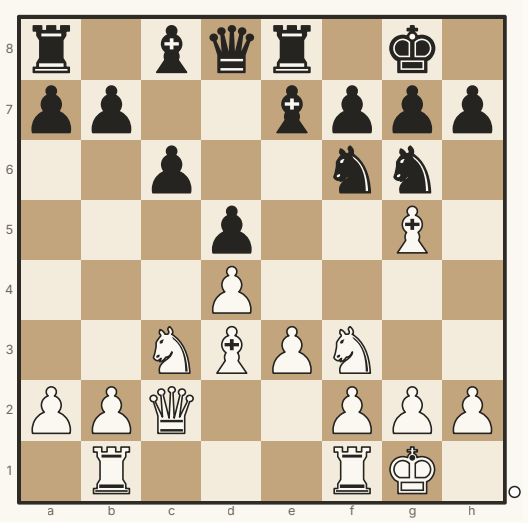
Solution p. 716

25.2 STRATEGY



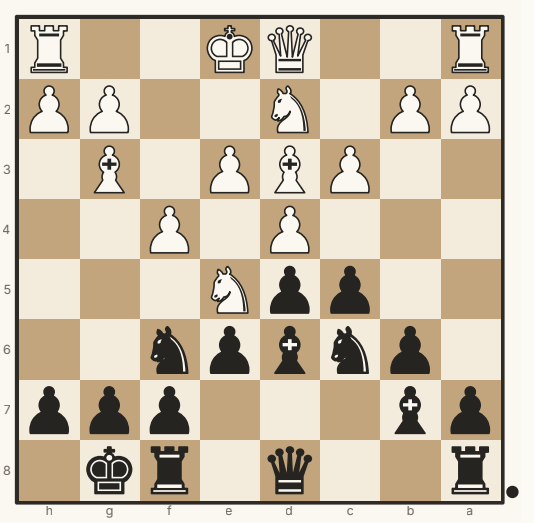
Black to move. White's bishop on c2 and a queen coming to d3 would aim at h7. Find the move that prevents White's attacking set-up. Solution p. 716

25.3 STRATEGY



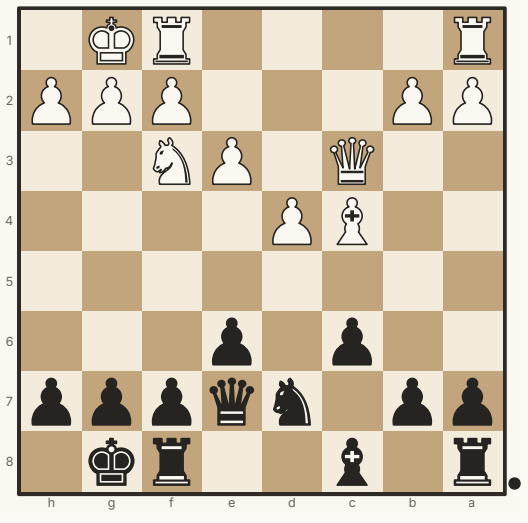
White to move. This is the Carlsbad structure from the Queen's Gambit Exchange. Choose the plan move. Solution p. 717

25.4 STRATEGY



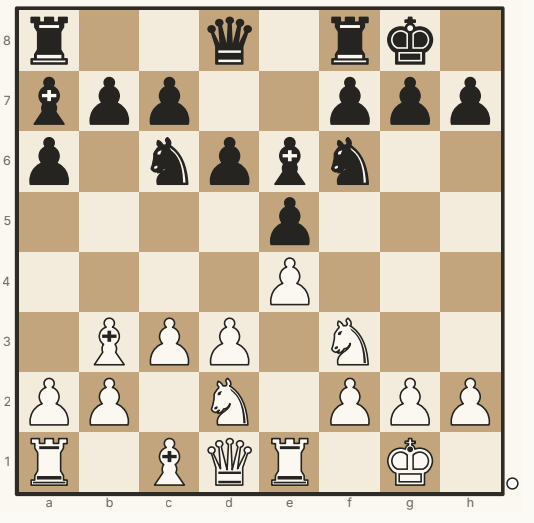
Black to move. White's knight on e5 is annoying. Should you take it with the bishop? Choose the best move. Solution p. 717

25.5 STRATEGY



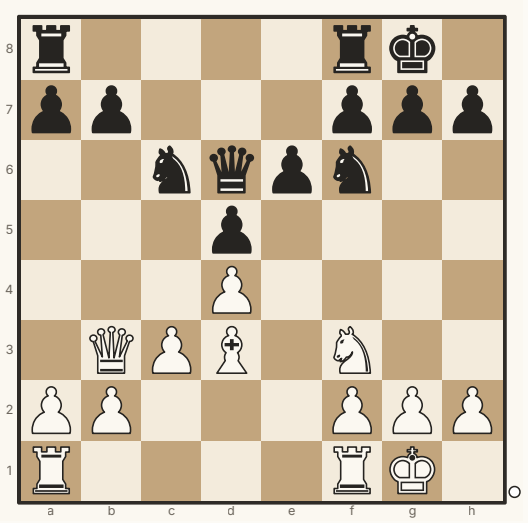
Black to move. You are a little cramped and your c8 bishop is shut in. Prepare the right freeing break. Solution p. 717

25.6 STRATEGY



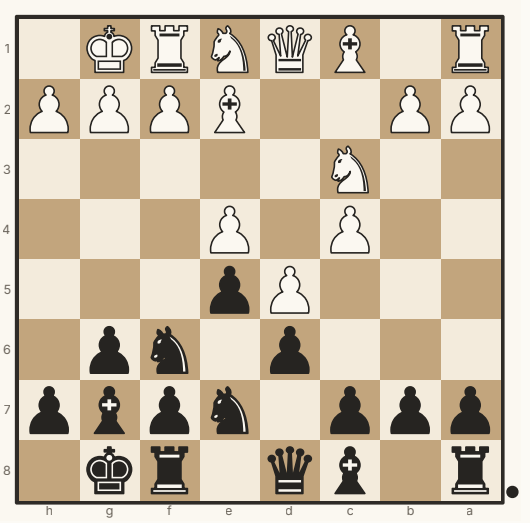
White to move. Italian Game, slow centre. Choose a good plan move. Solution p. 717

25.7 STRATEGY



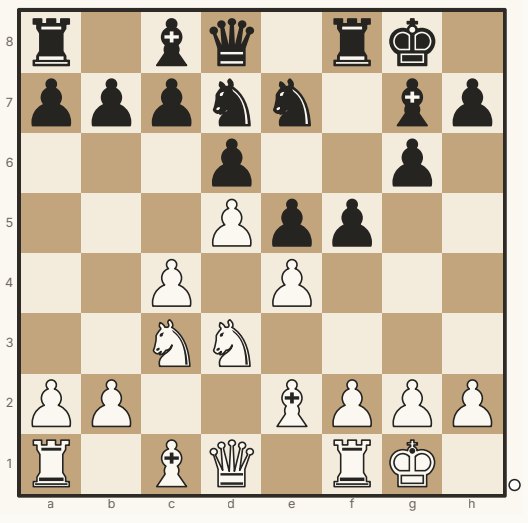
White to move. Exchange Caro-Kann structure. Ne5 looks like a dream outpost. Is it? Play a sound move. Solution p. 718

25.8 STRATEGY



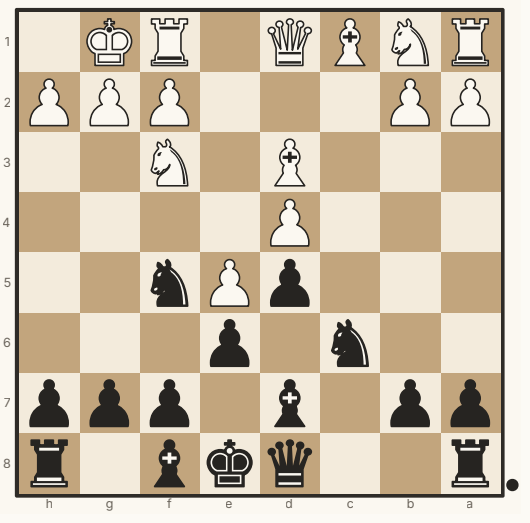
Black to move. The centre is closed. Your pawns on d6 and e5 point to the kingside; White's point to the queenside. Choose the plan move. Solution p. 718

25.9 STRATEGY



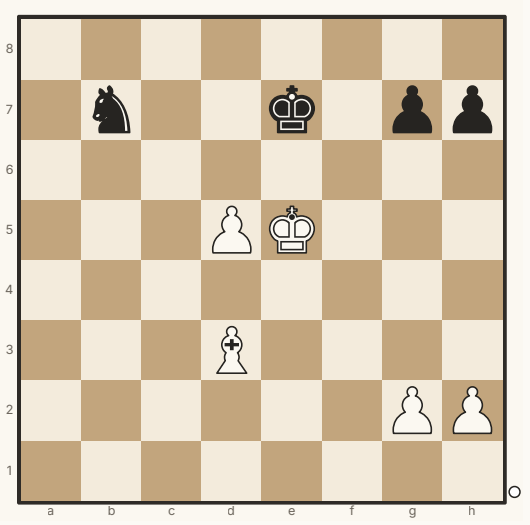
White to move. Black has started ...f5 on the kingside. Choose the move that keeps your position healthy for queenside play.
Solution p. 718

25.10 STRATEGY



Black to move. French structure: White's chain d4-e5. Attack the base of the chain.
Solution p. 718

25.11 STRATEGY



White to move. Bishop against knight with pawns on both wings. Use the bishop's long range. Solution p. 719

HINTS

25.1 With the IQP, which pawn move opens the position for all your pieces?

25.2 Which square does White's queen need?

25.3 Your rook stands on b1. Which pawn does it support?

25.4 After a trade on e5, which White pawn recaptures, and what does it attack?

25.5 Which break opens the long diagonal for your bishop?

25.6 Which of your pieces is doing the least?

25.7 After Nxe5 and dxe5, who defends the pawn on e5?

25.8 Which pawn break attacks the head of White's chain, and what is in its way?

25.9 Which pawn move makes your centre immune to ...fxe4?

25.10 Count attackers and defenders of d4.

25.11 Which black pawn is out of the knight's reach?

CHAPTER 26 STAGE 1200–1500

Endgame Technique

Kings, passed pawns and the minor-piece endings

YOU WILL LEARN TO

- Apply the general principles of endgame technique, including king activity, pushing passed pawns and the principle of two weaknesses.
- Play the main minor-piece endings, the wrong-coloured bishop and opposite-coloured bishops correctly.
- Know when queen against a pawn on the seventh wins and when the stalemate defence draws.
- Handle queen endings practically and calculate endgames by counting tempi.

> IN YOUR NEXT GAME
>
> When the queens come off, bring your king towards the centre at once, find your passed pawn or your opponent's weakest pawn, and count tempi before every pawn race.

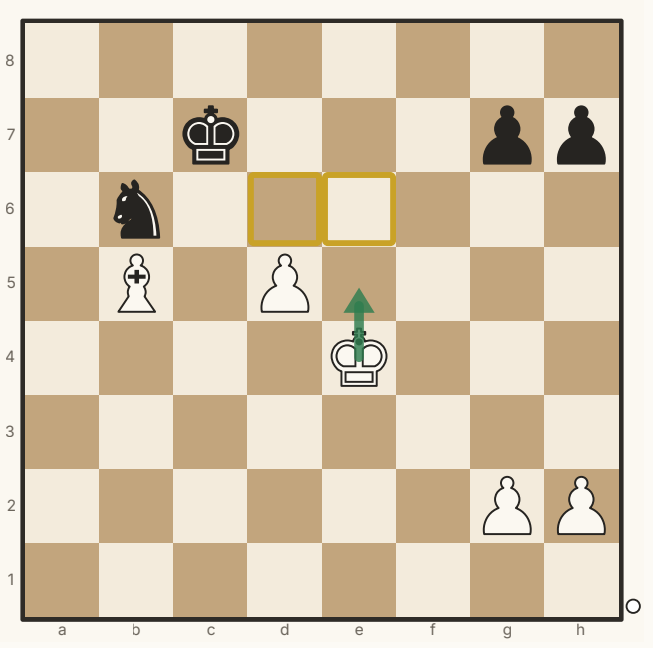
You reach an endgame a pawn up, or with a better minor piece, and you feel relieved. Twenty moves later it is a draw, or worse. At this level more half-points are lost in "winning" endings than in any other phase, and the reason is rarely a lack of knowledge. It is technique: the king stays at home, a passed pawn waits too long, a pawn is exchanged that should not have been, a check is overlooked.

This chapter gives you the working principles of endgame technique, then the minor-piece endings, the queen against a far-advanced pawn, and practical queen endings. You already know king and pawn endings from Chapter 18 and rook endings from Chapter 23; here we build on both.

The principles of technique

Activate the king

In the middlegame the king hides. In the endgame, with fewer pieces left to attack it, the king becomes a strong fighting piece, worth roughly a minor piece in many positions. Its jobs are to support your passed pawns, to stop your opponent's, and to attack pawns.



White to move. Bishop against knight with a passed d-pawn. The king belongs on e5, supporting the pawn from in front and eyeing the black kingside.

45.Ke5! is the move. From e5 the king supports d5-d6, controls e6 and d6, and prepares to march to the kingside pawns while the knight has to watch the passed pawn.

45. **Ke5** **Nc8** 46. **Bd3** **g6** 47. **Ke6**

The king supports the pawn and heads for Black's kingside while the knight is tied to d6 and b6. Bishop moves instead of Ke5 would let Black's king reach d6 and blockade.

> **KEY IDEA** **The king first**
>
> When the queens come off, ask: "Where would my king be most useful in ten moves?" and start walking. A king that arrives two moves earlier than the opponent's often decides the game on its own.

Passed pawns must be pushed

A passed pawn is your main winning asset, and it gets stronger the further it advances: it ties down enemy pieces and makes them passive. Two practical rules:

- **Push when the pawn is safe and the push costs the opponent something**, for example when it drives a piece to a worse square or gains a tempo with check.
- **Do not let it be blockaded cheaply.** A pawn that waits on the fourth rank gives the opponent time to put a king or knight in front of it.

The opposite advice also holds: a passed pawn is pushed, not *thrown*. If it advances far beyond the support of your king and pieces, it can simply be rounded up.

Do not rush, but do not waste chances

"Do not rush" is a famous endgame motto. When your opponent has no counterplay, improve your pieces fully before committing, because committal moves (pawn exchanges, sacrifices) are irreversible. A good practical method is to make the "free" improving moves first: king to the centre, rooks or minor pieces to their best squares, pawns away from the colour of the enemy bishop.

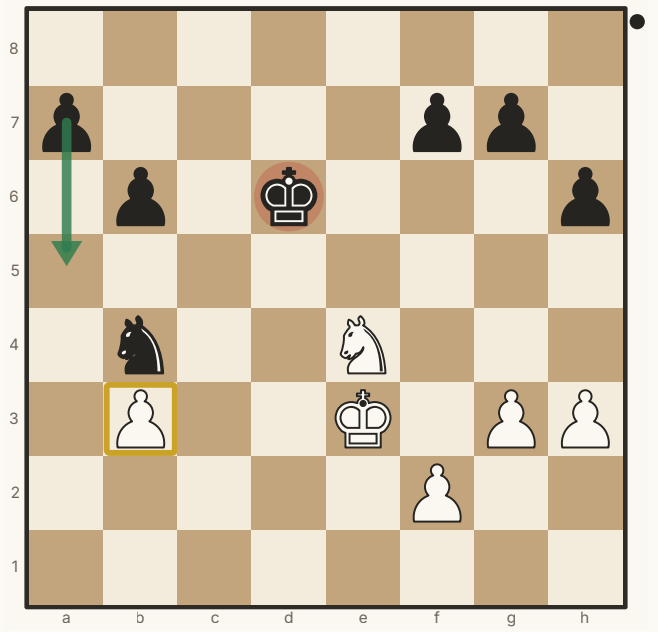
The second half of the advice is just as important: *do not throw away the chances the opponent might give you*. While you improve, choose moves that set problems. A move that keeps your position just as good but gives the opponent a chance to go wrong is better than one that gives him nothing to think about.

> **COACHING JUDGMENT** **Technique is calm**
>
> My coaching advice: in a winning endgame, write down in one sentence what your plan is and which counterplay you must prevent. If you cannot say it, you are not ready for committal moves. Make an improving move instead.

The principle of two weaknesses

A single weakness can usually be defended: the defender puts his pieces around it, and your attack stalls. The **principle of two weaknesses** says that you win by creating a *second* weakness far away from the first. The defender's pieces cannot be on both sides at once, and something gives.

The most common second weakness is an **outside passed pawn**: a passed pawn far from the other pawns, which drags the enemy king or knight away.



Black to move. Black's two pawns against one on the queenside will produce an outside passed pawn. While White's king and knight stop it, Black's king goes to the kingside: two weaknesses, one on each wing.

Black is clearly better here. The plan is simple to state: create a passed a-pawn, push it to draw White's pieces to the queenside, then use the king against the weakened kingside pawns.

Minor-piece endings

Knight against pawns

The knight is the worst piece against a far-advanced passed pawn, especially a rook pawn, because it has so few squares near the edge. Two rules help:

- **A knight in front of a pawn**, on its path, usually holds it easily.
- **A knight chasing a pawn** must count moves exactly. Count the knight moves to a square on the pawn's path, and remember that the knight needs both time and a safe square: a knight move to a square next to the enemy king may simply be captured.

Typical knight resources are worth knowing by heart. A knight that cannot reach the pawn's path can sometimes reach a square from which it *checks* the enemy king and then jumps in front of the pawn, gaining the decisive tempo. And a knight defending against a rook pawn on the seventh is in real danger, because from the corner it can be cut off from almost every useful square. When you are the side with the pawn, aim for the rook's file and keep your king between the knight and the queening square.

As the side with the knight against several pawns, a good defence is often to attack the pawns from behind or from the side, where the knight cannot be shouldered away, while your king blockades from in front.

Bishop against knight

The middlegame guidelines from Chapter 25 still apply, and they become stronger in the endgame. With pawns on both wings and an open board, the bishop is usually better: it guards one wing from a distance while the king attacks on the other. With pawns on one wing only, especially if they are fixed, the knight is often as good or better, because it can attack pawns of either colour.

Good bishop against bad bishop

With bishops of the same colour, the side whose pawns stand on the *other* colour has the good bishop. The winning method in such endings is almost always the same: the king invades on the colour that the bad bishop cannot control, and the bad bishop's pawns become targets. The defender must keep his king active and look for a pawn break that changes the structure; passive waiting usually loses.

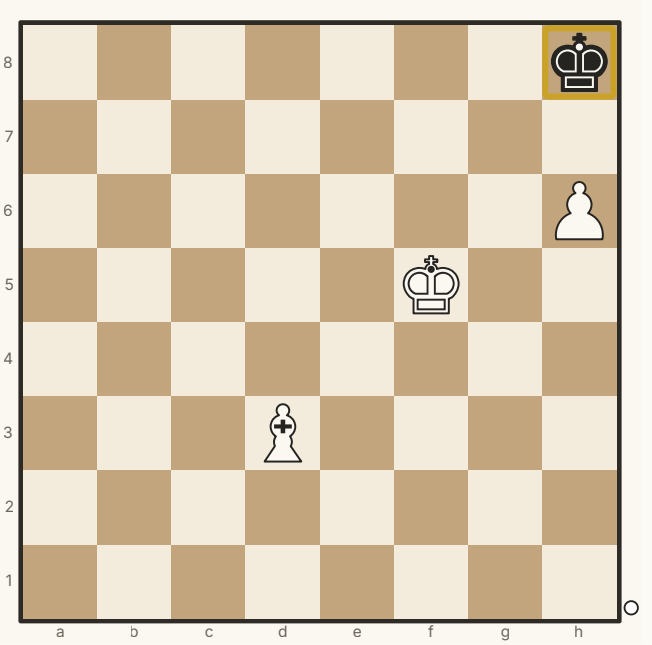
A practical checklist for the defender with the bad bishop: keep your king as active as the pawns allow, place your pawns (if any can still move) on the colour of the enemy bishop so that they are protected by the structure rather than by your bishop, and look for the one pawn break that can open a second front. A bad bishop that defends one weak pawn may hold; a bad bishop that must defend two will not.

Converting extra material

Most technical endings are about converting an extra pawn or a better piece. The methods from Chapter 16 apply directly: trade pieces rather than pawns, keep pawns on both wings when you are ahead (they make the opponent's defence harder), and avoid leaving your pawns on the colour of your own bishop where they can be blockaded. Before each exchange ask: "Is the ending after this trade easier to win than the one I have now?" If the answer is not a clear yes, keep improving instead.

The wrong-coloured bishop and the rook pawn

King, bishop and rook pawn against a lone king is a famous exception: if the bishop does not control the queening square (a light-squared bishop with an h-pawn, since h8 is dark), and the defending king reaches the corner, it is a draw. The king cannot be driven out of h8 without stalemate.



A dead draw. The bishop cannot control h8, the king cannot be forced out of the corner, and pushing h7 only stalemates. The engine shows a small plus, but no progress is possible.

The consequence works in both directions. As the attacker, you must keep the defending king *away from the corner*, usually by cutting it off with the bishop on the diagonal that leads to the corner. As the defender, run for the corner, even at the cost of other pawns. And remember: the same applies to the a-pawn with the "wrong" dark-squared bishop, since a8 is light.

Opposite-coloured bishops

When the bishops run on different colours, the endgame becomes famously drawish. Each bishop controls squares the other can never contest, so the defender can set up an unbreakable **blockade**: the king stands in front of the passed pawns, and the bishop controls the squares of its colour on which the pawns would have to advance. Even two extra pawns are often not enough.

Practical rules for these endings:

- **For the defender:** put the king in front of the pawns, keep the bishop on a long diagonal that crosses the pawns' path, and do not grab pawns if it lets the passed pawns run.

- **For the attacker:** you usually need two passed pawns far apart, two or more files, so that the defending king and bishop cannot both stop them. Connected pawns that are already blockaded rarely win.
- **In the middlegame the opposite is true:** as we saw in Chapter 25, opposite-coloured bishops favour the attacker, because the defender's bishop cannot guard the squares the attack needs.

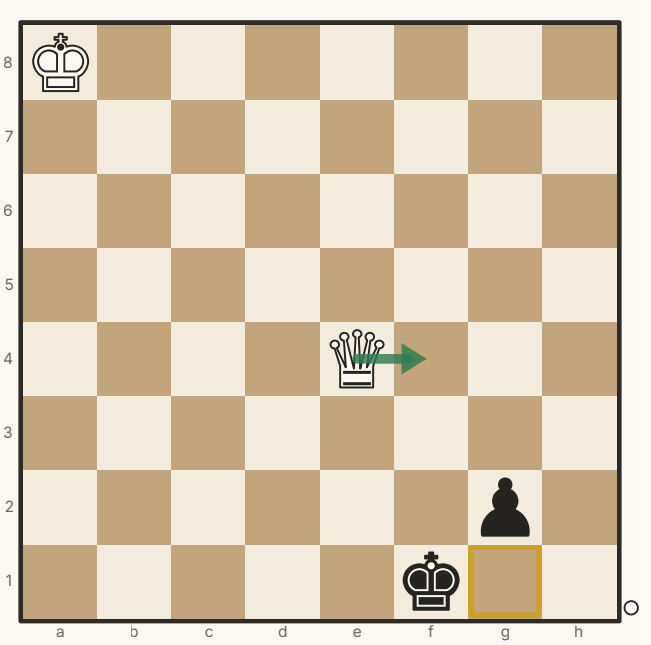
> COMMON MISTAKE **Trading into a drawn ending**
>
> A frequent way to throw away a winning middlegame is to trade pieces "to simplify" and end up in an opposite-coloured bishop ending a pawn or two ahead. Before trading the last minor pieces, check which bishops will remain.

Queen against a pawn on the seventh

A queen normally beats a pawn easily, but a pawn on the seventh rank supported by its king is a special case.

Centre pawns and knight pawns: the queen wins

The method against a centre pawn or a knight pawn is a staircase of checks and attacks that forces the defending king to stand **in front of its own pawn.** Each time that happens, the pawn cannot move, and the attacking king gains a tempo to step closer. Repeat until the king is close enough to help win the pawn or mate.



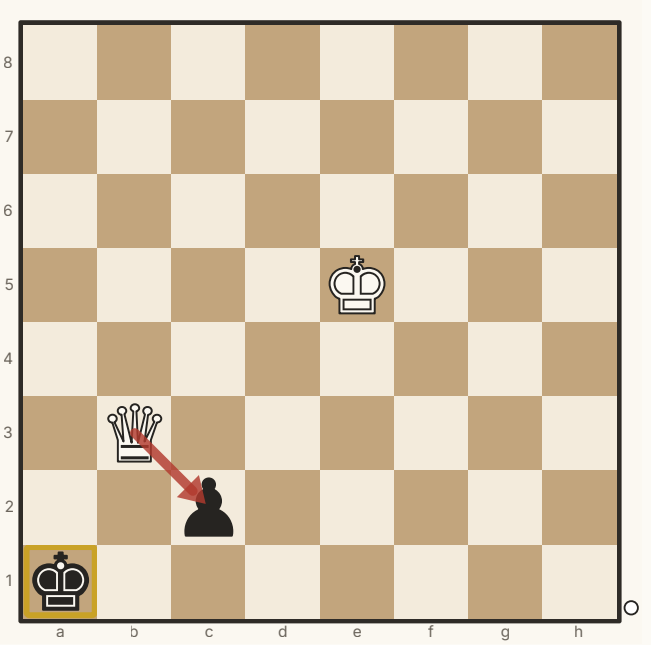
White to move. The g-pawn is on the seventh, but with checks the queen either wins it or drives the king in front of it, then the white king approaches.

60. **Qf4+ Ke2** 61. **Qg3 Kf1** 62. **Qf3+ Kg1** 63. **Kb7**

The checks drive the king to g1, in front of its own pawn. That gives White the free move Kb7, and the process repeats until the white king arrives.

Bishop pawns and rook pawns: the stalemate defence

Against a **bishop pawn** or a **rook pawn** on the seventh, the defence has a stalemate resource, and with the attacking king far away the game is a draw.



White to move. Against the c-pawn Black's king has gone to a1 instead of protecting the pawn. If Qxc2, it is stalemate, so White cannot make progress.

60. **Qa3+ Kb1** 61. **Qb3+ Ka1**

Every time White checks, Black returns to the corner. White never gets the free move needed to bring the king closer.

With a rook pawn the defence is even simpler: the defending king hides in front of the pawn in the corner, and the attacker has to be careful not to stalemate.

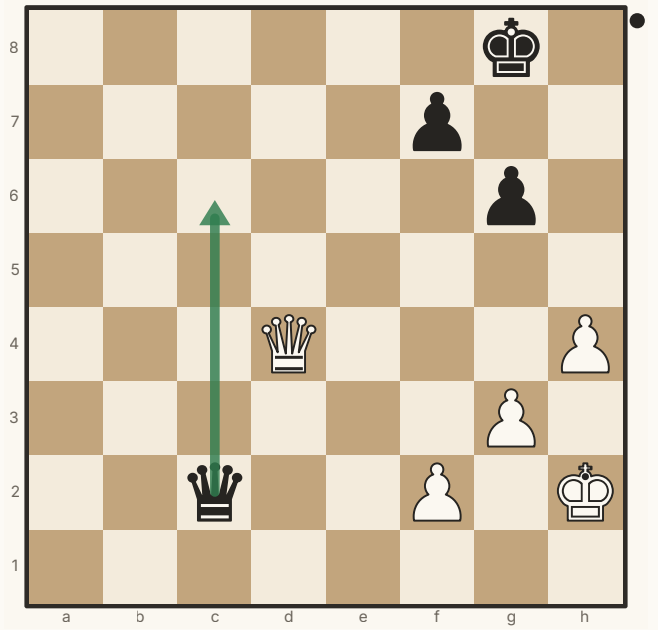
The exception to the exception: if the attacking king is **already close**, the queen and king together win even against these pawns, usually by a mating net. The rule of thumb is that the queen wins against a bishop or rook pawn only when the king is near enough to take part within a few moves; count it.

Queen endings in practice

Queen endings are the most tactical endings. Practical advice:

- **Perpetual check is everywhere.** The defending side's best hope is usually a perpetual, and the attacking side's main job is to avoid it.

- **King safety still matters.** The attacker's king needs shelter, often in front of or beside the enemy pawns, so that checks run out.
- **Centralize the queen.** A queen in the centre both attacks and defends against checks.
- **A passed pawn far advanced** is worth more than material elsewhere: it ties the enemy queen down.



Black to move. White is a pawn up, but White's king is open and Black's queen is active. The engine sees this as close to equal: the checks are the defence.

IN YOUR NEXT GAME **Count the checks**

In your next queen ending, before every move, count your opponent's checks. If he has more than two in a row, ask where your king could hide and whether your move takes away one of them.

Calculating in the endgame

Endgame calculation is different from middlegame calculation: lines are longer, but there are fewer pieces and fewer branches. Three tools do most of the work:

1. **Count tempi.** In any race, count how many moves each side needs to promote, and include checks that gain or lose a tempo. Many "obvious" races are decided by one tempo.
2. **Use geometry.** The rule of the square, the knight's route, the king's diagonal shortcuts: shapes on the board let you skip move-by-move calculation.
3. **Look for the only move.** In sharp endings, often only one move holds or wins. When the position is critical, list *all* candidate moves, including king moves and quiet bishop moves, before you choose.

A short example of tempo counting: your passed a-pawn stands on a4, the enemy king is on e5, and it is your move. The pawn needs four moves to queen (a5, a6, a7, a8). The king needs to reach the a8 corner or the pawn's path: from e5 to b7 is three king moves (d6, c7, b7). Because you move first, the pawn reaches a5, a6, a7 while the king reaches d6, c7, b7, and the king arrives in time. Move the king one file further away and the result changes. Counting like this takes ten seconds and replaces a lot of guessing; the rule of the square from Chapter 18 is a shortcut for exactly this count.

Common technical mistakes

Most lost half-points in good endings come from a handful of habits. Recognize them in your own games:

- **The passive king.** The king stays on g1 "for safety" while the opponent's king marches into the centre. Unless there are still queens or heavy mating threats, the safety argument no longer applies.
- **Trading the wrong pawns.** When you are ahead, pawn trades bring the opponent closer to a draw (a lone extra minor piece cannot win). Trade pieces; keep pawns.
- **Hurrying the passed pawn.** The pawn runs ahead of its support and is collected by the enemy king or knight. Advance it when the king or a piece can follow it.

- **Allowing the "bad" ending.** A trade that leads to opposite-coloured bishops, to the wrong-coloured bishop with a rook pawn, or to a lost pawn ending can turn a win into a draw or a loss in one move.
- **Ignoring the opponent's checks.** In queen and rook endings the defender's main weapon is the check. Ask before every move how many checks your opponent has.
- **Playing on autopilot.** Technique is not the same as playing fast. A winning ending still needs a blunder-check on every move.

> COMMON MISTAKE **"Any move wins"**
>
> When the position is winning, players stop calculating. Then a single careless move, such as a pawn push that gives up a key square or a king step into a fork, throws away the work of the whole game. In a won ending, the blunder-check matters more, not less, because there is no more counterplay to warn you.

A worked example: choosing the winning move

Return to the bishop-against-knight position from the start of the chapter.

First I check the material and the pawns. Material is level in pawns, with bishop against knight. My trump is the passed d-pawn; Black has two kingside pawns against my two.

Then I look at the opponent's resources. Black wants to blockade the pawn with the king on d6. Black's knight on b6 attacks d5 and is close to the queening path.

Candidate moves. 45.Ke5 (activate the king), 45.d6+ (push the pawn), 45.Bc6 or 45.Be8 (improve the bishop).

45.d6+? looks forcing, but after 45...Kxd6 the pawn is simply gone: my king on e4 is too far away to protect it. A passed pawn is pushed with support, not thrown.

45.Bc6 keeps the bishop active, but after 45...Kd6 Black's king blockades and my king cannot get past. The engine evaluates this as only slightly better for White.

45.Ke5! prevents ...Kd6, supports d6 next move and keeps the king close to Black's kingside pawns. After 45...Nc8 46.Bd3 g6 47.Ke6, the pawn is safe, my king is close to Black's pawns and the knight is reduced to guarding d6.

Blunder-check. Does Black have a knight fork or check? From b6 the knight reaches d7 (check on e5) and c4 (check on e5). After 45.Ke5 Nd7+ my king steps to e6 or d4 and the knight has abandoned the queenside; 45...Nxd5 is met by 46.Kxd5. No danger. I play 45.Ke5.

Practice at the board

1. **King activation.** `8/2k3pp/1n6/1B1P4/4K3/8/6PP/8 w - - 0 45`. Play White against an engine and win. Then repeat with 45.Bc6 instead of 45.Ke5 and see how much harder it becomes.
2. **Wrong-coloured bishop.** `7k/8/7P/5K2/8/3B4/8/8 w - - 0 60`. Try to win against the engine for ten moves and convince yourself it is impossible. Then set the position up again with Black's king on e8 instead of h8 and work out which squares your bishop must control to keep the king out of the corner.
3. **Queen against a pawn.** `K7/8/8/8/4Q3/8/6p1/5k2 w - - 0 60`. Win against the engine. Then move the pawn to c2 and the black king to b2, keep White's king on a8, and see why the same method only draws.
4. **Two weaknesses.** `8/p4pp1/1p1k3p/8/1n2N3/1P2K1PP/5P2/8 b - - 0 40`. Play Black against the engine and win by creating a passed a-pawn and then switching to the kingside.

Check yourself: did you activate your king within the first three moves of each ending? If not, replay the ending and do it.

Chapter summary

- In the endgame the king is a fighting piece: bring it to the centre and towards the pawns at once.
- Passed pawns must be pushed when it is safe and costs the opponent something, but not thrown forward without support.
- Do not rush: make all useful improving moves before committal ones, and choose moves that give the opponent chances to go wrong.
- Win by creating a second weakness far from the first; an outside passed pawn is the most common one.
- Bishops prefer pawns on both wings; knights are poor against far-advanced rook pawns; a good bishop wins by king invasion on the other colour.

- King, wrong-coloured bishop and rook pawn draw if the defending king reaches the corner; opposite-coloured bishop endings are drawish even with extra pawns.
- Queen against a centre or knight pawn on the seventh wins; against a bishop or rook pawn the stalemate defence draws unless the attacking king is close.
- In queen endings, perpetual check is the main defensive resource; count checks before every move.

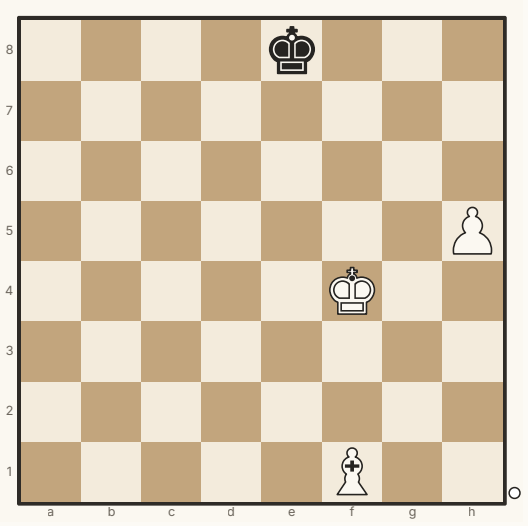
CHEAT CODE **The endgame routine**

1. Activate the king: decide where it is most useful in ten moves and start walking.
2. Find the trumps: your passed pawns and the opponent's weakest pawns; plan a second weakness.
3. Improve every piece before committal moves, and keep setting problems.
4. Before any trade, check which ending remains (opposite bishops, wrong-coloured bishop, lost pawn ending).
5. In races and sharp positions, count tempi and list all candidates, including quiet king and bishop moves; in queen endings, count the opponent's checks.

Exercises

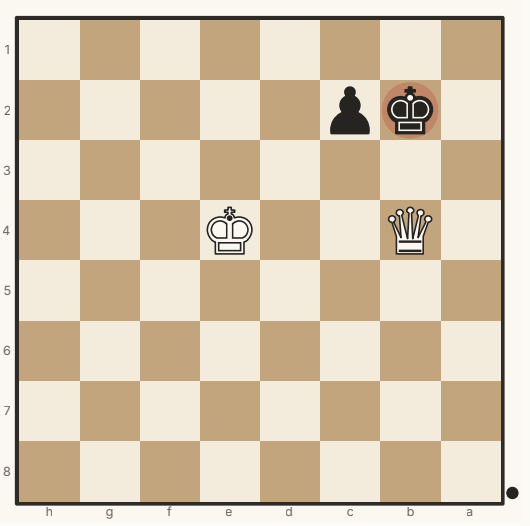
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

26.1 ENDGAME



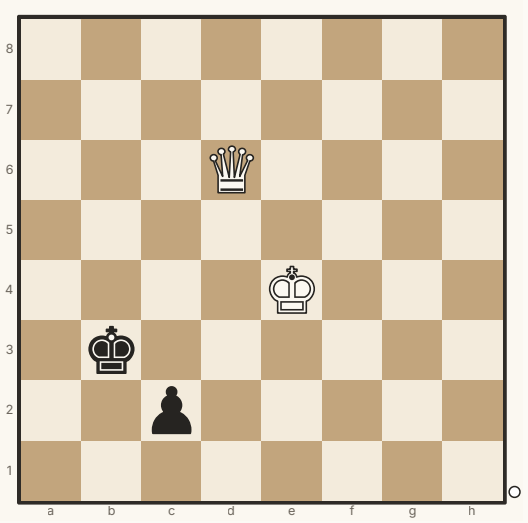
White to move. Your bishop does not control h8, the queening square. Keep the black king out of the corner and win. Solution p. 719

26.2 ENDGAME



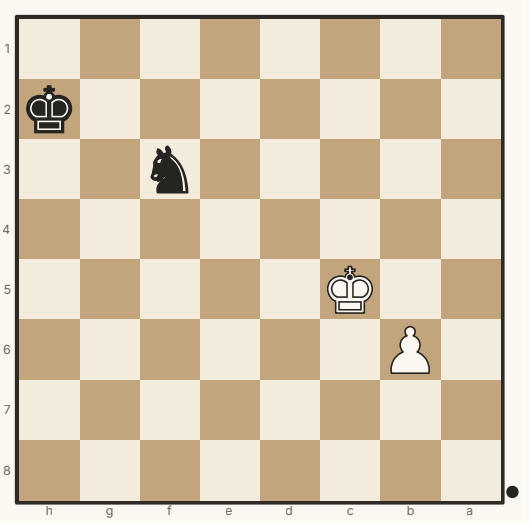
Black to move. Queen against a c-pawn on the seventh. Which king move saves the game? Solution p. 719

26.3 ENDGAME



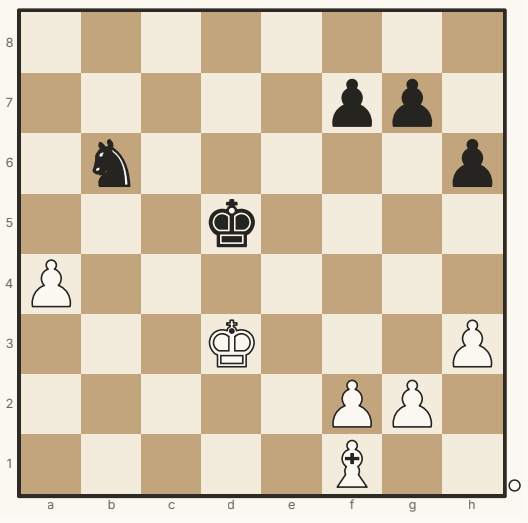
White to move. Your king is close enough to win against the c-pawn, but only one move does it. Find it. Solution p. 720

26.4 ENDGAME



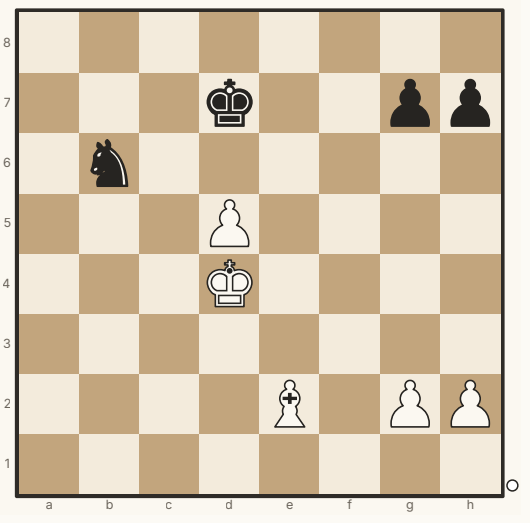
Black to move. White's b-pawn is running and your king is far away. Find the only knight move that stops it. Solution p. 720

26.5 ENDGAME



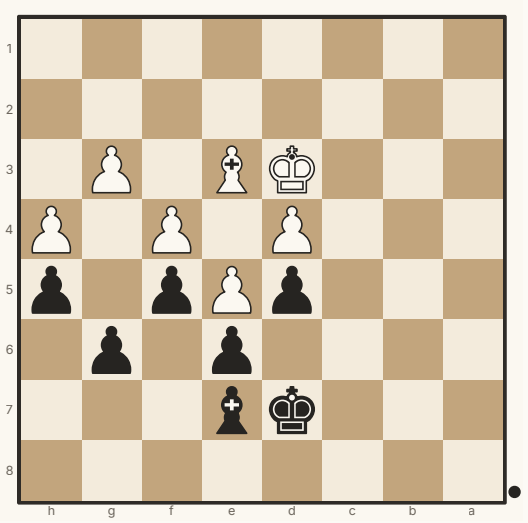
White to move. Bishop against knight, and you have an outside passed pawn. Use it. Solution p. 720

26.6 ENDGAME



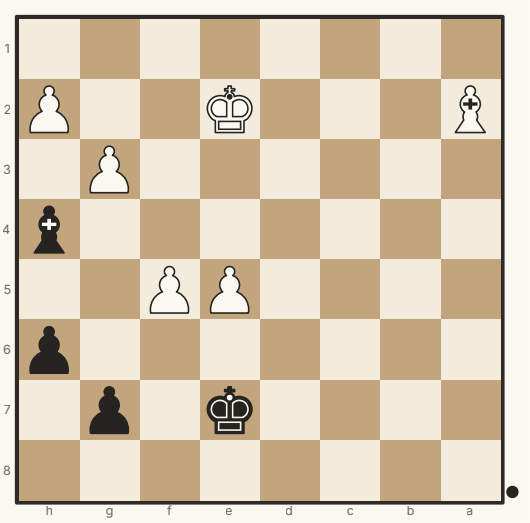
White to move. Bishop against knight, and you have a passed d-pawn. Find the king move that wins. Solution p. 720

26.7 ENDGAME



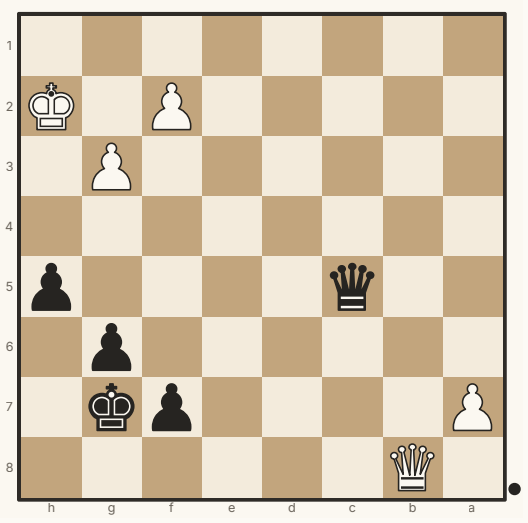
Black to move. White's bishop is bad: all White's pawns stand on dark squares. Find the winning plan move. Solution p. 721

26.8 ENDGAME



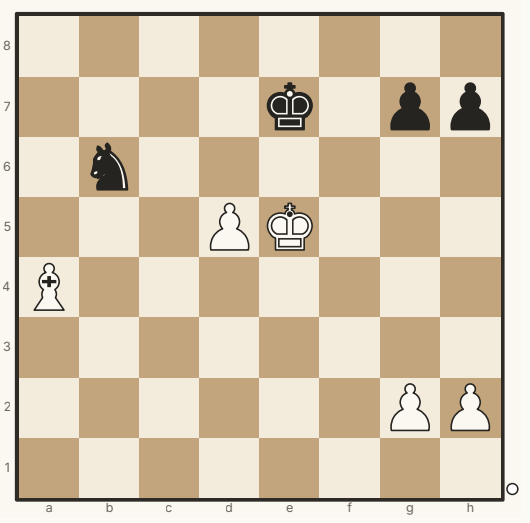
Black to move. Opposite-coloured bishops and White has two connected passed pawns. Your bishop is attacked. Where does it go? Solution p. 721

26.9 ENDGAME



Black to move. White's a-pawn is about to queen. Save the game. Solution p. 721

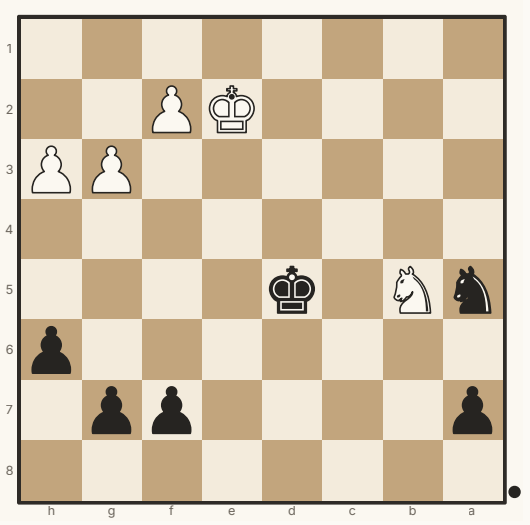
26.10 ENDGAME



White to move. Bishop against knight. Push the passed pawn at the right moment.

Solution p. 721

26.11 ENDGAME



Black to move. Knight ending: your a-pawn is your trump, and White's knight attacks it. Keep it. Solution p. 722

HINTS

26.1 Which squares does the black king need on its way to h8?

26.2 If White takes the pawn while your king is on a1, what happens?

26.3 Find a quiet queen move that both attacks the pawn and controls its queening square.

26.4 Count the knight moves to b8 from each square it can reach.

26.5 Your a-pawn is attacked. Defend it or push it?

26.6 In the endgame the king is a fighting piece. Where does it belong?

26.7 Which route lets your king reach the white pawns?

26.8 Which dark squares must stay under control so that the pawns cannot advance?

26.9 Which white pawn shelters the king?

26.10 Which forcing pawn move gains a tempo?

26.11 Which square defends a7 and stays near the pawn?

CHAPTER 27 STAGE 1200–1500

Practical Play and Psychology

Playing the opponent, the clock and yourself, as well as the board

YOU WILL LEARN TO

- Spend thinking time where it matters, and handle a clock and time trouble without hope moves (if you play with a clock).
- Recognize tilt and use a stop rule and a reset routine so one bad game does not become five.
- Make sound decisions about draw offers, resignation and playing on, with good etiquette.
- When winning, kill counterplay before collecting material; when losing, set problems the engine would also approve of.
- Run the "sit on your hands" check before every move that feels obvious, especially after a gift or a big gain.

> IN YOUR NEXT GAME
>
> Before every move that feels obvious, especially after you win material or your opponent leaves something hanging, sit on your hands and name your opponent's best reply to it.

You have spent twenty-six chapters learning chess. This one is about something slightly different: playing *games*. Games are played by tired, hopeful, nervous people with clocks, ratings and egos. At 1200–1500 the evidence of this is everywhere. Players lose won positions because they relaxed after winning a piece. They take a "free" bishop and get mated on the back rank. They resign positions that had a stalemate trick, and they lose four games in a row because they kept clicking "new game" while angry.

None of that is a lack of chess knowledge. It is a lack of *practical* skill: deciding where to spend effort, keeping your emotions out of your moves, and handling the moments when the game changes. These skills can be trained just like tactics.

Thinking time: where to spend it

This book never requires you to play with a clock. Untimed games, or very slow games, are a perfectly good way to improve, and many of the habits in this chapter work without one. But most club and online chess is timed, so here is how to use a clock well when you have one.

The budget idea

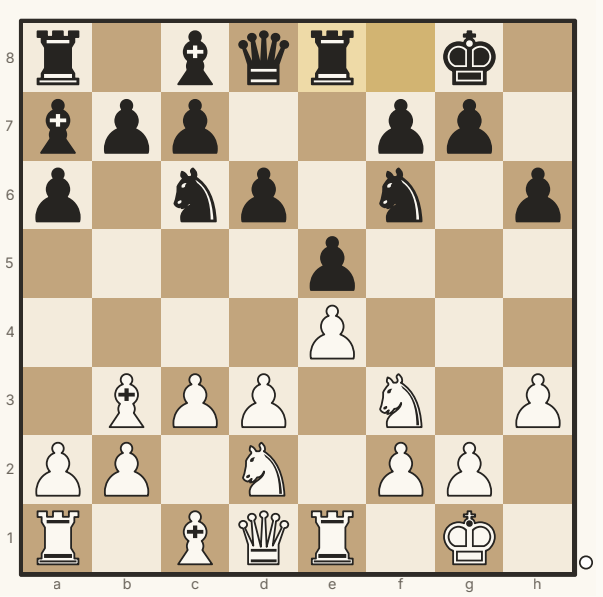
A **time control** is the amount of thinking time each player gets, such as "15+10": fifteen minutes each, plus ten seconds added after every move. That added time is the **increment**. Without an increment, a game can be lost on time however good your position is; with one, a scramble is survivable if your moves stay simple.

A useful way to think about the clock is as a budget. You will not spend the same amount on every move, so decide roughly where the money goes:

- **Known opening moves:** quickly. If you have played the position many times, a few seconds per move is enough.
- **Critical moments:** this is where most of your time belongs (see below).
- **Simple, forced moves** (recaptures, the only legal reply to a check): quickly, after a short safety check.
- **A reserve:** keep some time for the ending. The ending is where many won games are thrown away in a scramble.

> COACHING JUDGMENT **A starting budget (coaching judgment)**
>
> For a 15+10 game, a reasonable target is to spend no more than about two minutes on the opening, to have at least a third of your time left when the middlegame becomes simple, and never to spend more than three or four minutes on one move unless the game is decided by it. These numbers are a starting point, not a law. Adjust them after you look at your own clock usage in Chapter 28.

1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3 d6 6.O-O O-O 7.Re1 a6 8.Bb3 Ba7 9.h3 h6 10.Nbd2 Re8



A familiar Giuoco Pianissimo structure. Nothing is attacked and both sides have normal plans. This is a position to play at a steady pace, not to burn ten minutes on.

Recognizing a critical moment

A **critical moment** is a position where the choice you make now changes the character of the game, and a mistake will be hard to repair. You cannot afford to think deeply on every move, so learn the signs that say "this one matters":

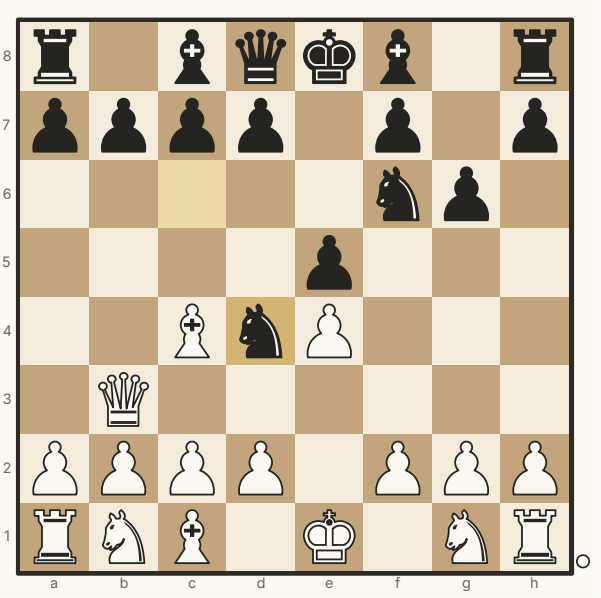
1. **Forcing moves exist** for either side: checks, captures, or threats against big pieces.
2. **A transition is possible:** a queen trade, a trade into a king and pawn ending, or opening the position with a pawn break.
3. **Your opponent's last move surprised you.** A surprise means your plan was based on a picture that has just changed.
4. **The evaluation could swing.** One side is about to win material, or a king is suddenly exposed.
5. **You feel the urge to play fast** because the move is "obvious". This is the most dangerous sign of all, and it gets its own section later in this chapter.

When you see two or more of these signs, slow down and run your full move-selection routine from Chapter 14: threats, candidate moves, calculate the forcing ones, blunder-check.

Time trouble without hope

Time trouble means having so little time that you cannot think normally. The typical bad habit is the **hope move**: a move whose only merit is a cheap threat that you *hope* your opponent will miss. Hope moves feel active, but they usually make your position worse if the opponent sees the threat, and a stronger opponent usually does.

1. **e4 e5** 2. **Bc4 Nc6** 3. **Qh5 g6** 4. **Qf3 Nf6** 5. **Qb3?! Nd4**



Three queen moves, three one-move threats against f7, and three calm answers. Now 5...Nd4 hits the queen and c2: the hope moves have only cost White time.

In time trouble, follow a simplified routine instead:

- **Checks, captures and threats for the opponent first.** Five seconds is enough to see a mate in one or a hanging queen.
- **Prefer moves you understand.** Recapture, trade when ahead, defend when attacked, and keep the king covered.
- **Make the safe move rather than the brilliant one.** A move that keeps your position intact is worth more than a combination you cannot check.
- **Use the increment.** If you have an increment, play the simple moves quickly to build time back.

COMMON MISTAKE **Spending your time in the wrong place**

A very common pattern at club level is to spend eight minutes choosing between two equally good developing moves in the opening, then lose a won ending with thirty seconds on the clock. If your games keep ending in scrambles, the fix is almost always earlier in the game: play known positions faster.

Tilt and emotional control

Tilt is a poker word for playing badly because you are upset. In chess it looks like this: you lose a game you should have won, immediately start another, play too fast and too aggressively, lose again, and repeat. Your chess knowledge has not changed between the first game and the fifth. Your judgment has.

Common signs of tilt:

- You start a new game within seconds of losing the last one.
- You want to "win your rating back" rather than play a good game.
- You play your moves faster than usual, or play openings you do not normally play.
- You feel angry at your opponent, the site, or yourself while you are moving.

COACHING JUDGMENT **The two-losses stop rule (coaching judgment)**

After two losses in a row, stop playing rated games for the session. Do something else: go for a walk, solve a few easy puzzles, or analyze the two games. This rule is a coaching habit, not a research finding, but it is cheap insurance. Almost nobody plays their best chess immediately after two painful defeats, and the rating points you save are real.

A simple **reset routine** between games helps even when you are not tilting. Stand up, drink some water, and before the next game ask one question: "What is my one focus for this game?" (for example, "blunder-check every capture"). This turns your next game back into practice instead of revenge.

Stronger and weaker opponents

Playing someone stronger

Against a stronger player, two opposite errors are common. Some players freeze and play passively, assuming every move of the opponent is deep. Others try wild complications from move one. Both throw away your best weapon, which is simply playing good chess.

- **Trust your calculation, not their rating.** If you check a pawn carefully and it is free, take it. Stronger players also make mistakes; they just make fewer of them. Exercise 27-07 shows a case where the "suspicious" pawn really is free.
- **Keep the game normal.** Play your repertoire, develop, castle. You will learn more from a sound game you lose than from a gamble.
- **Do not offer early draws out of fear.** You are there to learn and to win if they slip.

Playing someone weaker

Against a weaker player, the danger is **overpressing**: trying to win too quickly, or forcing play in an equal position until it becomes a lost one.

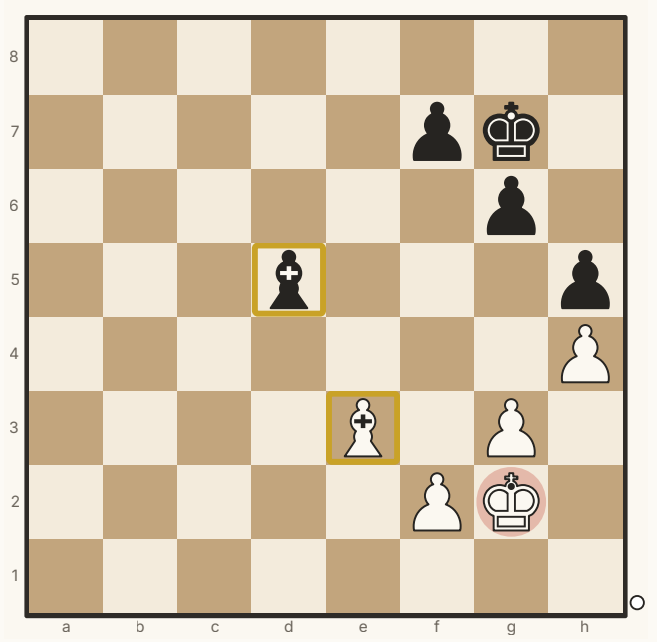
- **Be patient.** A weaker opponent is more likely to make mistakes in a long game, so give them the chance.
- **Do not play "for tricks".** Tricks are what weaker players fall for, but playing for them means making slightly bad moves on purpose. Sound pressure produces the mistakes for you.
- **Accept that a draw is sometimes the correct result.** If the position is dead equal, pressing on with a bad move to avoid the draw is how upsets happen.

Draw offers and when to keep playing

A **draw offer** is a proposal that the game end as a draw. In over-the-board play, the correct procedure under the FIDE Laws of Chess is to make your move, offer the draw, and then press your clock. Your opponent may accept, decline by saying so, or decline by simply making a move. Offering repeatedly after one refusal is poor etiquette and can be treated as annoying the opponent.

When you are *offered* a draw, treat it as information. Your opponent thinks the position is no better than equal for them, or they are tired, or short of time. Before you answer:

1. **Run your safety scan as if it were your move.** A draw offer often comes at a moment when the opponent has missed something (Exercise 27-03).
2. **Ask who is better, and who has the easier position to play.** A small edge in a simple position is worth playing on for.
3. **Check the clocks.** If your opponent is short of time and you are not, playing on is usually right.
4. **Then decide.** If the position is genuinely equal and dry, a draw is a perfectly respectable result.



Opposite-coloured bishops with equal, symmetrical pawns. Neither side can create a passed pawn or attack anything. Accepting a draw here is sensible chess, not cowardice.

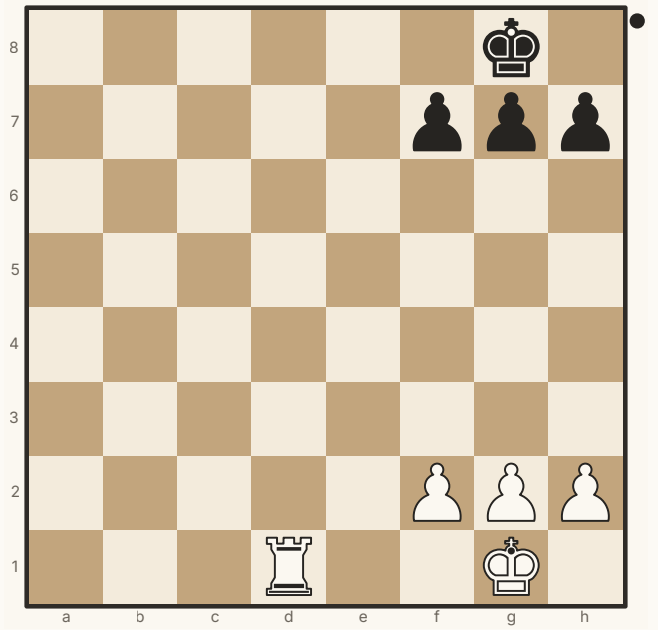
KEY IDEA **Draw decisions in one line**

Play on when you have a safe edge, the easier position, or more time. Accept when the position is equal and dry and you have nothing to play for. Never accept a draw without first looking at the board.

Resignation etiquette

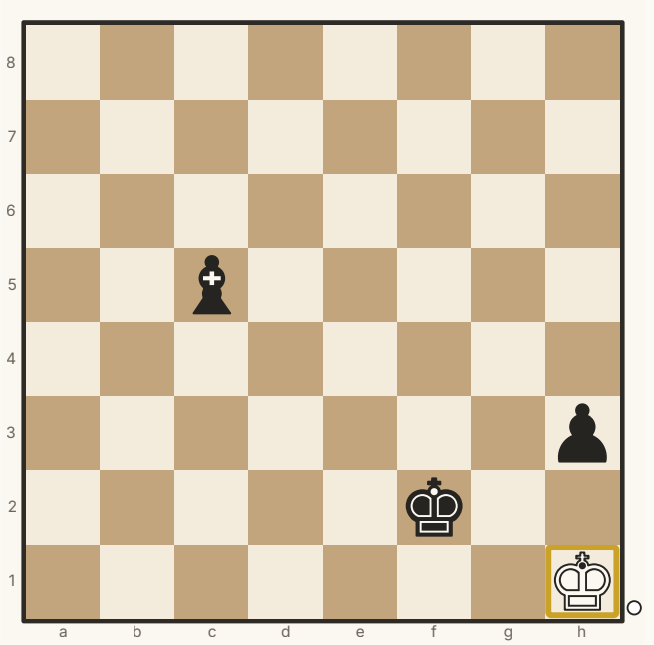
Resigning means conceding the game. At the board you can say "I resign", stop the clock, or offer your hand. Online you click the resign button. Letting your clock run out in a hopeless position, or simply leaving, is considered rude in both settings.

When should you resign? A practical guideline at 1200–1500 is: resign when you are completely lost *and* you have checked for your last resources: stalemate, perpetual check, a fortress, or an opponent who is very short of time. Before then, playing on is not rude. Your opponent has to prove the win, and at this level many wins are not proved.



Black is a whole rook down with no passed pawns and no counterplay. Against an opponent of your level or above, resigning here is reasonable; playing on for twenty moves hoping for a blunder mostly wastes both players' time.

Compare that with a position that looks hopeless but is not:



White is a bishop and a pawn down, but the h-pawn queens on a light square and Black's bishop is dark-squared. If White's king shuttles between h1 and h2, Black cannot force it out. This is a known draw, and resigning would throw half a point away.

When you are worse: set problems

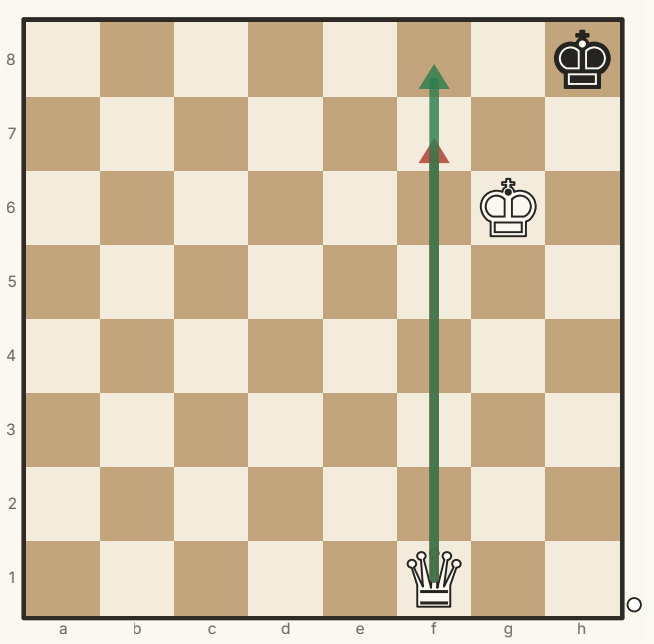
When you are worse, your goal changes. You are no longer looking for the objectively best plan to win; you are looking for the move that keeps the game alive and gives your opponent the most chances to go wrong. This is called **setting problems**. It is not the same as hope chess, because the moves you choose must still be sound. A swindle that loses immediately when the opponent sees it is just a quicker loss.

Good practical defence usually means:

- **Keep the pieces that give counterplay.** Avoid trades that lead to simple, lost endings, such as a king and pawn ending a pawn down against an outside passed pawn (Exercise 27-08).
- **Create a concrete threat every move if you can.** Your opponent must spend time and energy answering it, and each answer is another chance to err.

- **Look for drawing mechanisms.** Perpetual check, stalemate, a fortress, opposite-coloured bishops, or a rook ending with all pawns on one side. Exercise 27-04 is a stalemate rescue a rook down.
- **Keep the position complicated, but not lost.** A position with many pieces and some tactics is harder for your opponent than a clean technical ending.
- **Use the clock if you have one.** If your opponent is short of time, simple solid moves that keep threats on the board can be more effective than a desperate sacrifice.

Stalemate is the most common last-chance resource. Here is the pattern that you, as the stronger side, must also know:



White to move. 70.Qf7?? takes every square and leaves Black with no legal move: stalemate. 70.Qf8# ends the game properly. When your opponent has only a king, give check or leave him a move.

COMMON MISTAKE **Resigning too early**

Many players at this level resign positions with a stalemate trick or a perpetual check still on the board. Before resigning, ask two questions: "Can I give checks forever?" and "If my last mobile piece disappeared, would I have any legal moves?"

When you are winning: do not relax

The most expensive emotion in chess is relief. You win a piece, the hard part seems over, and your attention drops just as your opponent starts looking for tricks. A player who is losing has nothing to lose by trying everything, so this is exactly the moment to *raise* your attention.

Conversion was covered in detail in Chapter 16. The practical rules are:

1. **Name your opponent's last chance.** Ask, "If I were them, how would I try to save this game?" Usually there are only one or two ideas: a back-rank trick, a perpetual check, a passed pawn, or an attack on your king.
2. **Kill that chance first.** A flight square for your king, a queen trade, a blockade. Only then continue with your own plan.
3. **Trade pieces, not pawns.** Every trade of pieces reduces the opponent's chances of counterplay; pawns are what you will eventually win with.
4. **Prefer the simple win to the brilliant one.** You get the same point for both.

WORKED EXAMPLE **Worked example: a piece up, five minutes left**

XSTASHX6X

First I check what my opponent wants. Black's pieces point at f3 and h2. If ...Bxf3 and I recapture with the g-pawn, my king is open and Black's queen gets a target. That is Black's only real idea.

Next, my candidate moves. I can see Ng5 (moving the knight with an attack on f7), h3 (asking the bishop), Bxf7+ (grabbing a pawn with check), and Qf4 (offering a queen trade).

Then I compare them practically. Bxf7+ looks forcing, but after 16...Qxf7 I have given a bishop for a pawn and must find more tactics to justify it. Ng5 is strong but leaves a complicated position with queens on. Qf4 offers a trade: if Black takes nothing, my queen defends f3 and h2; if Black plays ...Bxf3, I recapture with the queen and the queens come off.

Finally I blunder-check Qf4. Is the queen safe on f4? Nothing attacks it. Does it allow a check or a capture? ...Bxf3 is met by Qxf3, keeping my pawn structure intact.

XSTASHX7X

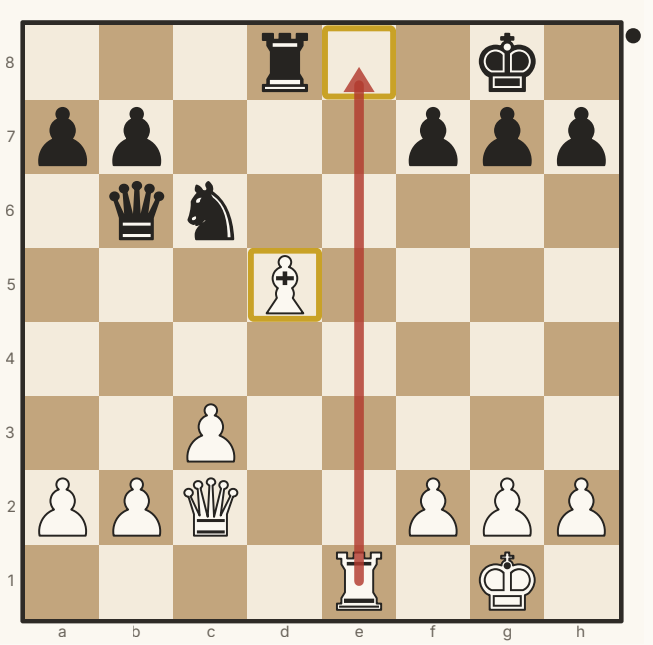
Stockfish slightly prefers 16.Ng5, but 16.Qf4 keeps an overwhelming advantage and is far easier to play with little time. In a practical game, choosing the move you can fully check is often the stronger decision.

The "sit on your hands" check

The phrase comes from an old piece of chess advice: when you have found a move you like, physically sit on your hands so that you cannot play it immediately. Then look at the board once more, this time from your opponent's side.

This habit matters most in exactly the situations where you want to move fast:

- **Your opponent has just left something hanging.** Before you take it, ask what they gain if you do (Exercises 27-01 and 27-06).
- **You have just won material.** Ask what their best reply would be if they were desperate (Exercise 27-02).
- **The move is a natural recapture or a "free" pawn grab.** Check whether a stronger in-between move exists for them.



White's bishop on d5 seems to be a gift. But after 22...Rxd5?? the rook has left the back rank and 23.Re8# is mate, because Black's king has no flight square.

KEY IDEA **The sit-on-your-hands routine**

1. You have found a move. Do not play it yet.
2. Imagine it on the board. List your opponent's checks, captures and threats in that new position.
3. Ask: "What is the most annoying reply?" Name one concrete move.
4. If you have a good answer to it, play your move. If not, look for a different move.

The check takes about fifteen seconds. Over a season of games it will save more points than any opening you could study in the same time.

Game day and online routine

What happens before and around a game affects the moves you make. None of the following is mysterious, and much of it is coaching judgment rather than research, but it is consistently useful.

Before you play

- **Sleep.** A long game after a short night is a test of concentration you have already half failed. If you are exhausted, an unrated game or a training session is a better choice than a rated game.
- **Eat and drink sensibly.** Hunger and thirst late in a long game make you impatient. Have water and a light snack available in long over-the-board games.
- **Warm up.** Solve five easy puzzles before you play. The aim is not to train; it is to switch on your pattern recognition.
- **Decide on one focus.** "Blunder-check every capture" or "spend my time at critical moments" is enough.

During a session

- **Play one game at a time,** with no other tabs, messages or videos. Divided attention is a reliable source of blunders.
- **Take breaks.** Stand up between games. In a long game, a minute away from the board during your opponent's turn can reset your focus.
- **Apply the stop rule.** Two losses in a row ends the rated session.
- **Prefer slower time controls for improvement.** Fast games are fun, but games with time to think are where you practise the habits in this book.

After you play

- **Write down your thoughts before you open an engine.** This is the start of the analysis loop in Chapter 28.
- **Record the result without judging yourself.** A loss with three good decisions and one blunder is useful information, not a verdict on your talent.

> IN YOUR NEXT GAME **Your practical game plan**
>
> In your next game, pick one practical focus: either "sit on my hands before every capture" or "stop after two losses in a row". Write it at the top of your scoresheet or on a sticky note next to your screen, and check afterwards whether you kept to it.

Practice at the board

Set these positions up on a board or in any free analysis app (Lichess's analysis board and Chess.com both let you paste a FEN). Use them to practise practical decisions, not just tactics.

1. **The free bishop.** FEN: `3r2k1/pp3ppp/1qn5/3B4/8/2P5/PPQ2PPP/4R1K1 b - - 0 22`. Black to move. Before you think about the bishop, name White's best reply to ...Rxd5. Then find a move that keeps everything safe (a flight square for your king is a good start). Check with an engine.
2. **Stalemate or mate.** FEN: `7k/8/6K1/8/8/8/8/5Q2 w - - 0 70`. Find every queen move that stalemates and every queen move that mates in one. Say the rule aloud: "give check or leave him a move."
3. **The draw that looks lost.** FEN: `8/8/8/2b5/8/7p/5k2/7K w - - 0 70`. Play White against an engine or a friend for ten moves. Keep your king on h1 and h2 and see whether Black can make progress.
4. **The convert-and-trade drill.** FEN: `r3r1k1/pp3ppp/2p5/7q/4P1b1/1BNP1N2/PP1Q1PPP/R4RK1 w - - 0 16`. Play White against an engine at a moderate level with a five-minute clock. Your goal is not to find the best move but to trade queens and win without allowing a single check against your king.
5. **The timed routine.** Play three games at a slow time control. After each game, mark on your scoresheet the moves where you spent the most time. Were they the critical moments, or just the moments when you were unsure about an opening move?

Chapter summary

- Spend thinking time at critical moments: when forcing moves exist, when a transition is possible, or when your opponent's move surprised you.
- In time trouble, play simple, safe moves and do not rely on hope moves that only work if your opponent misses a one-move threat.
- Tilt is real. A stop rule (coaching judgment: two losses in a row) and a short reset routine protect your rating and your learning.
- Against stronger players, trust your calculation; against weaker players, be patient and avoid overpressing.
- Before answering a draw offer, look at the board as if it were your move. Accept equal, dry positions; play on when you have a safe edge or more time.

- Resign only when you are lost and have checked for stalemate, perpetual check and fortresses. Never let the clock run out on purpose.
- When winning, name and kill the opponent's last chance before collecting more material; when losing, keep sound counterplay and look for drawing mechanisms.
- Sit on your hands before every obvious move, especially after a gift or a big gain.

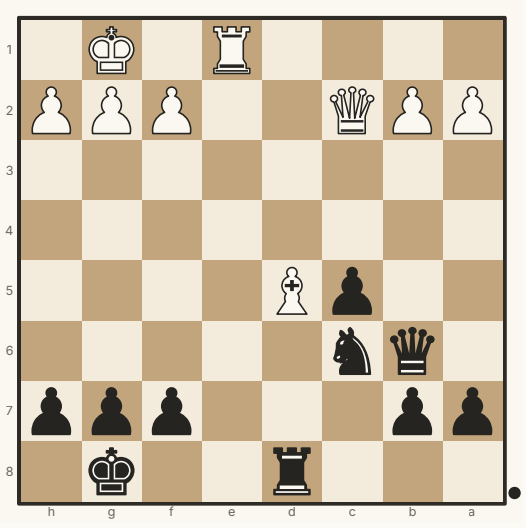
CHEAT CODE **Practical decision procedure**

1. Is this a critical moment (forcing moves, a transition, a surprise)? If yes, slow down and use your full routine.
2. What is my opponent trying to do? Name their best idea or last chance.
3. If I am winning: stop their idea first, then trade pieces and simplify.
4. If I am losing: keep sound counterplay, avoid trades into lost endings, look for perpetual check and stalemate.
5. Before playing the move, sit on your hands: what is the most annoying reply?
6. Between games: after two losses in a row, stop the rated session.

Exercises

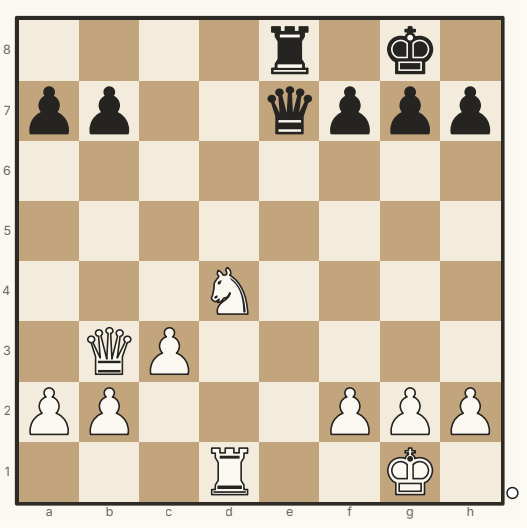
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

27.1 BLUNDER CHECK



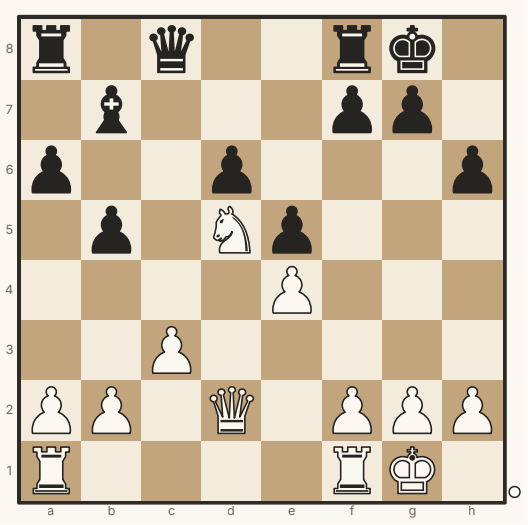
Black to move. You are a pawn up and White has just put a bishop on d5 where only your rook attacks it. Is it a free piece? Find the best move. Solution p. 722

27.2 BLUNDER CHECK



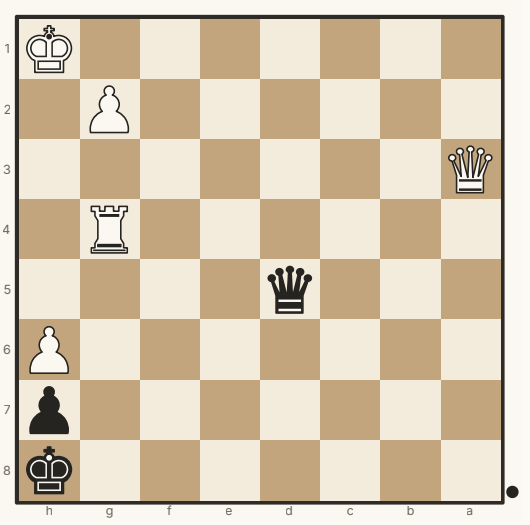
White to move. You won a knight a few moves ago and want to keep pressing. What does Black threaten, and what is your best move? Solution p. 722

27.3 TACTIC



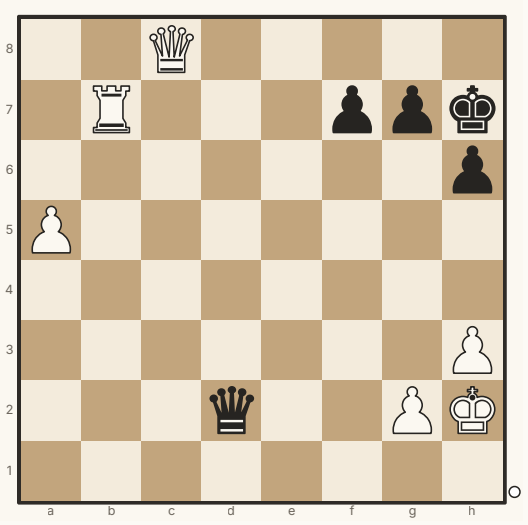
White to move. Your higher-rated opponent has just offered a draw. Before you answer, look at the board: is there a reason to keep playing? Solution p. 723

27.4 DEFENCE



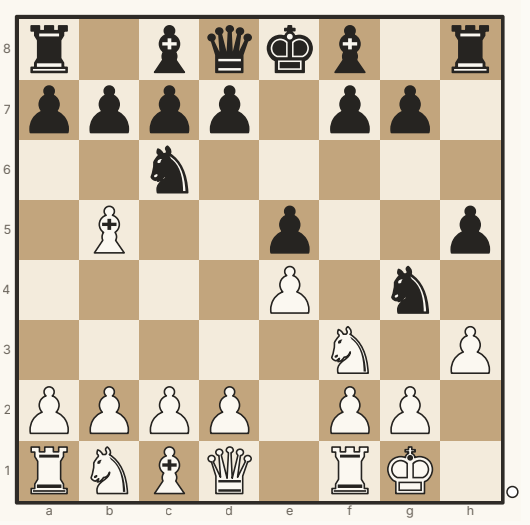
Black to move. You are a rook down and White threatens mate. Your king has no moves and your pawn is blocked. Find the move that saves the game. Solution p. 723

27.5 CHECKMATE



White to move. You are a rook up and your a-pawn is ready to run. Black's queen is hovering near your king. Finish the game cleanly: mate in 3. Solution p. 723

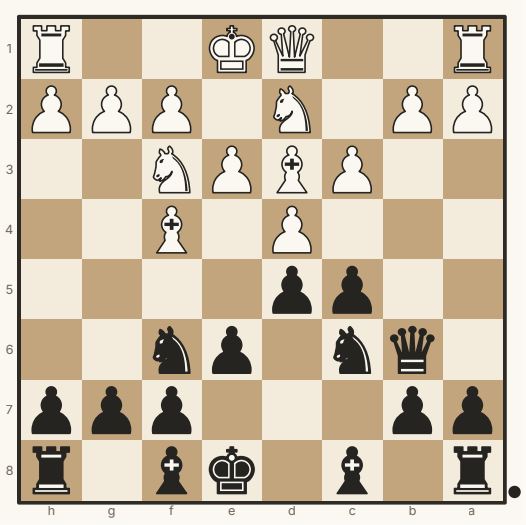
27.6 BLUNDER CHECK



White to move. In a Ruy Lopez, Black has played the odd-looking ...Ng4 and ...h5, and the knight on g4 is attacked by your h-pawn. Is it a free piece? Choose your move.

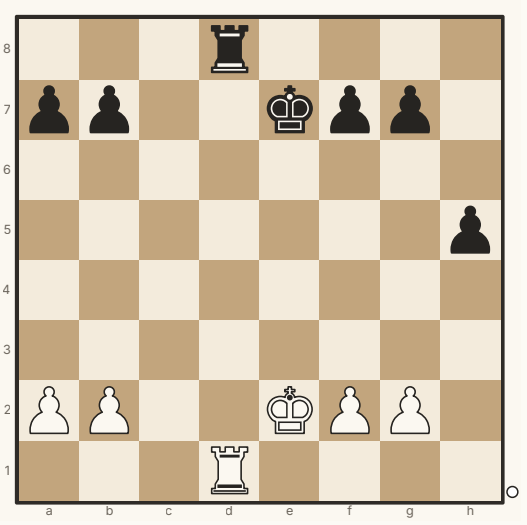
Solution p. 723

27.7 CALCULATION



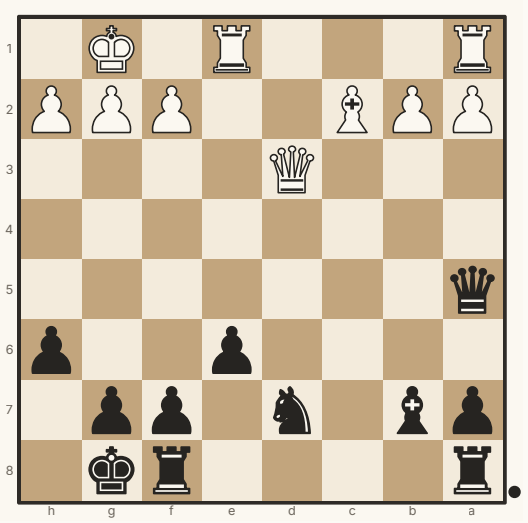
Black to move. Your opponent is rated 300 points higher and has just played 7.Bd3, leaving the b2-pawn hanging. Many players assume a stronger opponent never blunders. Check it yourself: what is your best move? Solution p. 724

27.8 DEFENCE



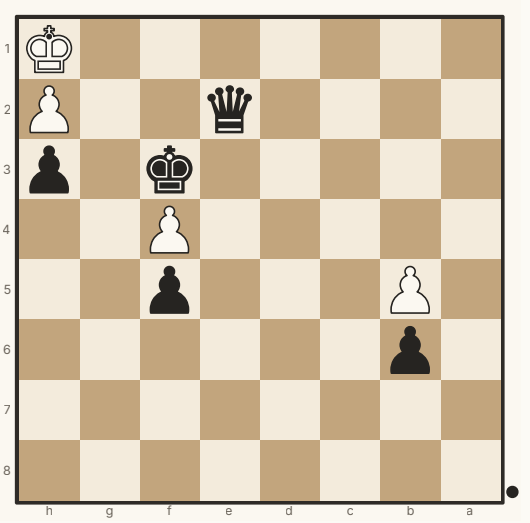
White to move. You are a pawn down and Black's h-pawn is passed. Black has just offered a rook trade on the d-file. Which move gives you the best practical chances? Solution p. 724

27.9 CONVERSION



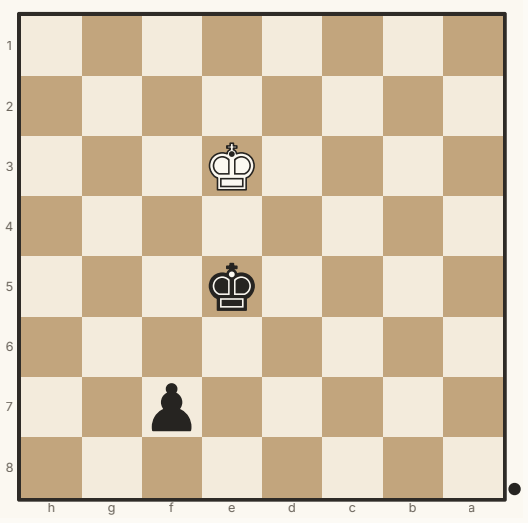
Black to move. You are a piece up. Both a2 and g2 can be taken. What is the safest way to keep your advantage? Solution p. 724

27.10 CHECKMATE



Black to move. You have just queened and your clock is running low. White's king is boxed in and every white pawn is blocked. Mate in 1, and do not stalemate! Solution p. 724

27.11 ENDGAME



Black to move. Your opponent offers a draw. You have an extra pawn but the kings are face to face. Should you play on? Find the winning move. Solution p. 725

HINTS

27.1 If your rook leaves d8, who guards e8?

27.2 Look at the e-file and at your king's flight squares.

27.3 Which square attacks both g8 and c8?

27.4 Count your legal moves if your queen were gone.

27.5 Checks first: which check also attacks f7?

27.6 If you take on g4, which black pieces suddenly see your king?

27.7 List White's best replies to ...Qxb2. Do any of them trap the queen?

27.8 Which ending would you rather defend: rooks on, or bare kings and pawns?

27.9 What is the only way White can still win this game?

27.10 Only consider checks.

27.11 Who wants to move here, and do you have a waiting move?

CHAPTER 28 STAGE 1200–1500

Analyzing Your Games and Training for Life

Turning every game into the right training

YOU WILL LEARN TO

- Analyze your own games in the right order, recording your thoughts before you use an engine.
- Find the turning points of a game and classify each mistake with a fixed set of categories.
- Read engine evaluations sensibly, find the first big swing, and ask why a move was bad.
- Keep a mistake journal and a prioritized weakness list, and turn mistakes into spaced, mixed review exercises.
- Understand what ratings, accuracy scores and performance figures can and cannot tell you, and plan your training beyond 1500.

> **IN YOUR NEXT GAME**
>
> After your next game, write down your thoughts at the three moments that felt most important before you open any engine, then log every mistake with one category from the list in this chapter.

You have reached the last chapter, but not the end of the book's method. Every chapter has ended with the same loop: LEARN, SOLVE, PLAY, ANALYZE, FIX, REPEAT. This chapter is about the second half of that loop, the part most players skip.

Most improving players play far more than they analyze. They glance at the engine's bars, sigh at the blunder on move 23, and click "new game". Two weeks later the same blunder happens again, because nothing has changed in how they think. Players who keep improving treat their own games as their main text-

book: they know their typical mistakes, build exercises from their own positions, and review them until the patterns stick. This chapter shows you how.

The analysis loop at a glance

Here is the whole process. Each step gets its own section below.

1. **Record your thoughts** right after the game, before any engine.
2. **Find the turning points:** the moments where the result was really decided.
3. **Categorize each mistake** using one fixed list.
4. **Check with the engine:** find the first big swing and ask why.
5. **Log it** in your mistake journal.
6. **Prioritize** with a weakness list (how often × how costly).
7. **Fix it** by turning mistakes into exercises and reviewing them on a schedule.

> COACHING JUDGMENT **How much time to spend (coaching judgment)**
>
> A useful rule of thumb is to spend at least as long analyzing a serious game as you spent playing it, and to analyze every slow game you play. Fast games are fine for fun, but you do not need to analyze them all; pick the ones where something important happened.

Step 1: Your thoughts first, the engine second

The most valuable information about a game is not in the moves. It is in your head: what you were afraid of, which candidates you considered, what you missed and why. That information fades within hours, and an engine destroys it in seconds. Once you have seen the "right" move, you cannot honestly remember whether you would have found it.

So before you open any engine, write short notes at the moments that mattered. You do not need beautiful prose. For each moment, answer:

- **What did I think was going on?** ("I thought my attack was faster.")
- **Which candidate moves did I consider?** ("Nxg5 and Bg3.")
- **What was I afraid of?** ("...Qd7 and ...Kg7 defending.")

- **How much time did I use, and how did I feel?** ("Six minutes, nervous, short of time later.")

Use your scoresheet, or an analysis board with the engine switched off. Analyzing with your opponent or a friend helps too: explaining your thinking aloud reveals its gaps.

> COMMON MISTAKE **Letting the engine think for you**
>
> If the engine is on while you go through the game, you will mostly learn *what* the better move was. You will not learn *why* you missed it, which is the only part that helps in your next game. Turn the engine off for the first pass, every time.

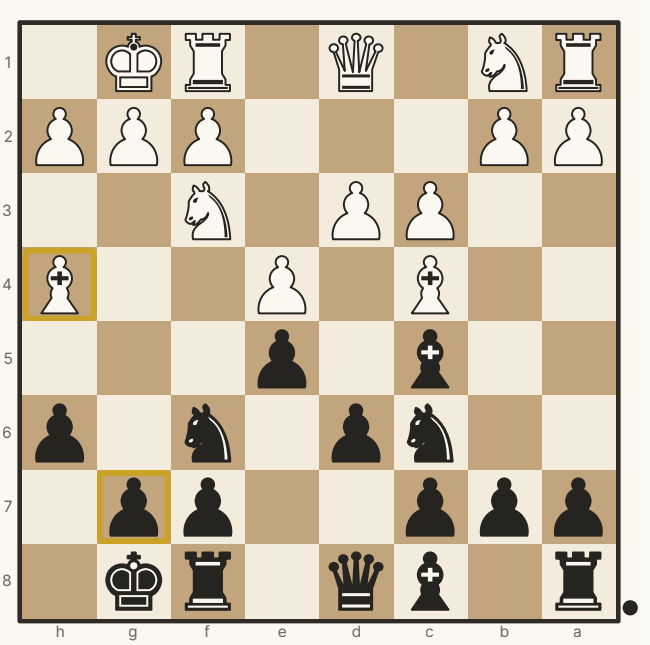
Step 2: Find the turning points

A **turning point** is a moment where the game changed character: the evaluation swung, a plan began or failed, or one side started to lose control. Most games have only one to three real turning points. Everything else is detail.

Before you use an engine, you can usually find them by asking:

- Where did I first feel uncomfortable, or first feel I was winning?
- Where did I spend the most time?
- Where did material change hands, or the pawn structure change?
- Where did the type of position change (queens off, a pawn break, an ending)?

Mark those moves. They are where your analysis effort belongs.



A typical turning-point candidate. Black is deciding whether to break the pin with ...g5. Pawn moves in front of a castled king cannot be taken back, so this is a moment to slow down and, afterwards, a moment to analyze.

Step 3: Categorize every mistake

A mistake you do not name is hard to fix. This book uses one fixed list of **mistake categories**. Use exactly these labels, and choose the *main* reason when more than one applies:

Category	Use it when...	Example
hanging-piece	You left a piece or pawn undefended, or missed that the opponent's was	The student's 6...Ng4? in Exercise 28-04
missed-tactic	A tactic existed (fork, pin, discovered attack...) for you or against you	Exercises 28-05, 28-06, 28-07, 28-10
missed-mate	A forced mate was available, for you or against you	Exercises 28-01, 28-11
king-safety	You weakened your king, or ignored danger to it	Moving pawns in front of a castled king
calculation	You saw the right idea but calculated it wrongly	Wrong move order, missed defensive reply

Category	Use it when...	Example
opening	The problem came from the opening: a trap, a bad plan, a lost tempo	Exercise 28-04
endgame	You misplayed a known endgame or misjudged a transition to one	Exercise 28-03
missed-win	You had a clearly winning line and played something weaker	Exercise 28-08
conversion	You were winning but let the advantage slip by playing loosely	Exercise 28-09
positional	A plan, structure or piece-placement error with no tactical cause	Bad trade, weak square, passive piece
time-psychology	Time trouble, tilt, fear, over-confidence or resigning too early	Exercise 28-02

Record the deeper cause. “I blundered because I had ten seconds” is a time-psychology mistake, even though the move itself hung a piece; mention the surface cause in your notes.

Step 4: Use the engine well

Only now open the engine. An engine is an extraordinary tool, but it answers a different question from the one you need. It tells you what the strongest move was. You need to know what you should change in your thinking.

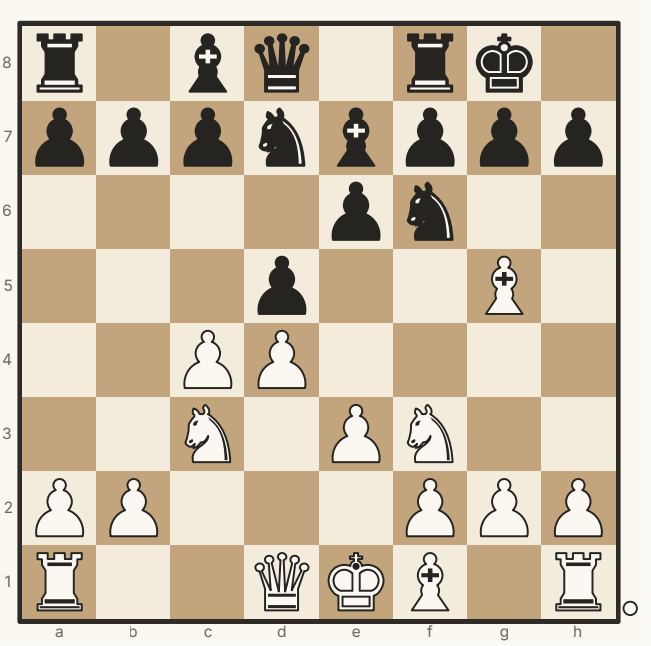
How to read evaluations

An engine **evaluation** is a number from White’s point of view, measured roughly in pawns. +1.00 means White is about a pawn better; −2.50 means Black is clearly better; a score such as “M4” or “#4” means a forced mate in four moves. The number is the engine’s judgment at a certain search depth; it is not a law of nature, and it can change as the engine searches deeper.

For practical purposes at your level, the following bands are a good guide (coaching judgment):

- **Within about ±0.5:** roughly equal; the game is about who plays better from here.
- **About 0.5 to 1.5:** a clear but not decisive advantage.

- **About 1.5 to 3:** a large advantage that usually wins with good technique.
- **Beyond 3, or a mate score:** normally decisive, unless the other side has a tactical rescue.



A normal Queen's Gambit Declined position. Stockfish rates Qc2, Qb3, Rc1 and Bh4 all within about a tenth of a pawn of each other. Differences this small are not mistakes; do not spend your analysis time on them.

Do not obsess over small differences

If the engine prefers +0.6 to your +0.3, that is not a mistake worth logging; at club level such differences are decided by plans, time and nerves. Look for swings of roughly a pawn or more, or swings that change the band the position is in.

Look for the first big swing

Run the engine through the whole game and find the **first big swing**: the first move where the evaluation changed by roughly a pawn or more against you. Later mistakes often happen *because* of that first one (you were worse, so you became nervous or desperate), so fixing the first swing often fixes several later moves too.

Ask “why”, not just “what”

When the engine shows a better move, play it out and ask:

- **Why was my move bad?** What does the opponent do against it, concretely?
- **Why didn’t I consider the better move?** Did I not see it, see it and reject it, or not look at that kind of move at all?
- **What rule would have found it?** (“Check every check.” “Look at the opponent’s best reply before playing an attacking move.”)

The rule is what goes into your journal. The engine’s move alone is not enough.

WORKED EXAMPLE **Worked example: analyzing a sacrifice that failed**

The student was White in a Giuoco Pianissimo and wrote these notes before opening any engine: "After 8...g5 I remembered that Nxg5 often works against this pawn push. I calculated 9.Nxg5 hxg5 10.Bxg5 and thought my attack would be very strong. Spent four minutes."

XSTASHX2X

First I check the student's notes against the board. The idea was a known pattern: when a castled king's pawns advance with ...h6 and ...g5, a knight sacrifice on g5 can sometimes break the position open. The student did calculate a few moves. So the problem is not laziness; something in the evaluation went wrong.

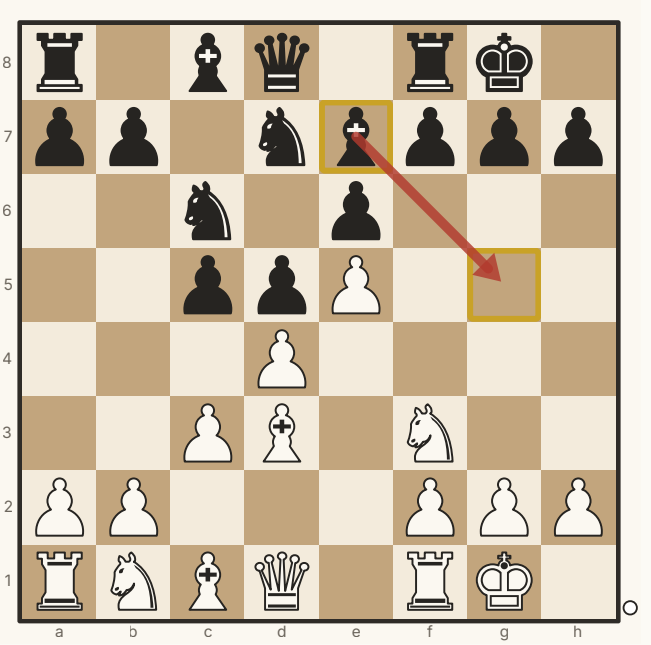
Next I let the engine speak. Before 9.Nxg5 the position is roughly equal (about +0.2 for White). After it, the evaluation drops to about −1, a swing of more than a pawn. This is the first big swing of the game.

XSTASHX3X

Then I ask why. Playing the line out shows the reason: White has only the queen and one bishop near the black king. The knight on b1 is far away, the bishop on c4 is not aimed at the kingside, and Black's king simply walks to g7 while the rook takes over the h-file. The pattern needs more attackers than White has here.

Finally I write the journal entry. Category: *calculation* (right pattern, wrong assessment of the final position). Rule for next time: "Before a sacrifice, count the attackers that can join in the next two moves; if it is only the queen and one piece, look for the calm move." The calm 9.Bg3 keeps the balance.

The same check, counting the pieces that can join the attack, also explains why the Greek gift sacrifice works in one position and fails in a very similar one:



This looks like a textbook Greek gift, but 9.Bxh7+? Kxh7 10.Ng5+ fails because the bishop on e7 covers g5: after 10... Bxg5 11.Bxg5 Qxg5 White has simply lost two pieces for a pawn. Compare Exercise 28-08, where Black has no dark-squared bishop and the sacrifice wins.

Step 5: The mistake journal

A **mistake journal** is a simple record of every significant mistake, with the position and the lesson. Keep it in a notebook, a spreadsheet or a text file. What matters is that you use the same columns every time, so that patterns become visible.

Here is a template you can copy:

Date	Game (opponent, colour, time control)	Move	FEN	I played	Better	Category	Why it happened	Rule for next time	Review dates
									1 / 3 / 7 / 21 days
									1 / 3 / 7 / 21 days

And here is the entry from the worked example, filled in:

Date	Game	Move	FEN	I played	Better	Category	Why it happened	Rule for next time	Review dates
(date)	Club game, White, 30+0	9	`r1bq1rk1/ppp2p2/2np1n1p/2b1p1p1/2B1P2B/2PP1N2/PP3PPP/RN1Q1RK1 w - - 0 9`	Nxg5	Bg3	calculation	Remembered a pattern; did not count attackers after ...Kg7	Count attackers that can join within two moves before sacrificing	✓ / ✓ / – / –

Three practical points:

- **Save the FEN.** Every free analysis board can show the FEN of the current position. With the FEN saved, the position becomes an exercise you can set up again at any time.
- **Write the rule in your own words.** "Count attackers before sacrificing" is a rule you will remember; "Bg3 was +0.2" is not.
- **One line per mistake.** A long essay is not more useful. If an entry takes more than two or three minutes, you are writing too much.

Step 6: The weakness list (frequency × cost)

After ten or twenty analyzed games, your journal will show patterns. Some categories appear again and again; some appear rarely but cost you whole games. A **weakness list** turns that information into a training priority.

For each category, count how often it appeared and estimate how much it cost each time. A simple cost scale works well (coaching judgment):

- **1:** a noticeable but small loss (a good position became equal).
- **2:** a big swing (an equal position became clearly worse, or a win became a draw).
- **3:** it decided the game.

Multiply frequency by average cost to get a priority score. Train the top one or two categories first.

Category	Times (last 15 games)	Average cost (1–3)	Score (times × cost)	Priority	Training fix	Re-check on
hanging-piece						
missed-tactic						
missed-mate						
king-safety						
calculation						
opening						
endgame						
missed-win						
conversion						
positional						
time-psychology						

For example, a student might find 6 hanging-piece mistakes with an average cost of 2.5 (score 15), 4 conversion mistakes with an average cost of 2 (score 8), and 3 opening problems with an average cost of 1 (score 3). That student should spend the next month on the blunder-check routine and conversion technique, not on learning more opening theory, however tempting that is.

> KEY IDEA **Train what costs you points**
>
> The weakness list protects you from training what you enjoy instead of what you need. Opening study is fun, but if your games are decided by hanging pieces and lost conversions, the list will show it in black and white.

Step 7: Turn mistakes into exercises

The journal becomes powerful when you use it as a training set. Every entry with a FEN is a ready-made exercise taken from your own games, which makes it more relevant to you than any puzzle collection.

Building your personal exercise set

1. **Set up the position from the FEN** in an analysis board or on a physical board.
2. **Write the question** you should have asked during the game: "White to move. Is the sacrifice on g5 sound?" or "Black to move. Find the move that saves the game."
3. **Hide the answer** (on the back of a card, in a separate column, or in a study chapter you do not open until you have answered).
4. **Add two or three similar positions** from this book's exercises with the same theme. A tag such as "fork" or "back-rank" makes these easy to find.

A free online study (for example a Lichess study with one chapter per position) is a convenient home for the set.

The review schedule

Review each new exercise roughly **1, 3, 7 and 21 days** after you create it. If you solve it confidently, move it to the next interval; if you fail it, start again from 1 day. After the 21-day review, put it in a "maintenance" pile that you sample from now and then.

> COACHING JUDGMENT **Why 1, 3, 7 and 21? (coaching judgment)**
>
> These exact intervals are a practical choice, not a research finding. Research on spacing (see the evidence box below) shows that spreading reviews out helps long-term retention, and that the best gap depends on how long you want to remember the material. Expanding gaps of a day, a few days, a week and a few weeks are simple to manage and work well in practice. If a different rhythm suits your week better, use it.

Mix your practice

When you review, **mix the themes**. Do not solve ten forks in a row, then ten pins. In a real game, nobody tells you which theme is on the board; recognizing *which*

idea applies is part of the skill. A mixed set (one of your own positions, a fork from Chapter 7, a rook ending from Chapter 23, a mate pattern from Chapter 11) trains exactly that.

Why this works: the evidence, and its limits

This chapter's method is built on a small number of well-studied ideas from the science of learning. Here is what they say, and what they do not.

> **EVIDENCE** **Deliberate practice**
>
> Ericsson, Krampe and Tesch-Römer (1993), "The role of deliberate practice in the acquisition of expert performance" (*Psychological Review*), described **deliberate practice**: effortful activity designed to improve specific weaknesses, usually with feedback, rather than simply doing the activity for enjoyment. In chess specifically, Charness, Tuffiash, Krampe, Reingold and Vasyukova (2005), "The role of deliberate practice in chess expertise" (*Applied Cognitive Psychology*), found that the amount of serious study alone (studying games, positions and books by yourself) was the strongest predictor of chess skill among the activities they measured.

> **EVIDENCE** **Retrieval practice**
>
> Roediger and Karpicke (2006), "Test-enhanced learning: Taking memory tests improves long-term retention" (*Psychological Science*), showed that testing yourself on material produces better long-term retention than spending the same time re-studying it. For chess, this supports solving positions (retrieving the idea) over re-reading annotations.

> **EVIDENCE** **Spacing**
>
> Cepeda, Pashler, Vul, Wixted and Rohrer (2006), "Distributed practice in verbal recall tasks: A review and quantitative synthesis" (*Psychological Bulletin*), reviewed a large body of studies and found that spreading study sessions out over time generally produces better retention than massing them together, with the best gap depending on how long the material must be remembered. The research concerns mostly verbal material; applying it to chess positions is a reasonable extension, not a tested result.

EVIDENCE **Interleaving**

Kornell and Bjork (2008), "Learning concepts and categories: Is spacing the 'enemy of induction'?" (*Psychological Science*), found that interleaving examples of different categories (in their study, paintings by different artists) improved people's ability to classify new examples. Rohrer and Taylor (2007), "The shuffling of mathematics problems improves learning" (*Instructional Science*), found that mixing different types of mathematics problems in practice led to better test performance than practising one type at a time, even though practice felt harder. Mixed chess exercises aim at the same benefit: learning to recognize which idea fits.

EVIDENCE **The limits of the evidence**

Practice is not the whole story. Macnamara, Hambrick and Oswald (2014), "Deliberate practice and performance in music, games, sports, education, and professions: A meta-analysis" (*Psychological Science*), found that deliberate practice explained an important part, but far from all, of the differences between people's performance, and the authors argued that other factors must also matter. There is also very little research on adults improving from 400 to 1500 in particular. So the honest summary is: the methods in this chapter are well supported as ways to learn and remember, but no method guarantees a particular rating, and people improve at different speeds.

What your numbers mean (and what they do not)

Online chess gives you a flood of numbers. Here is how to read the common ones.

- **Online rating.** Your rating is an estimate of your playing strength *within one rating pool*: one site, one time control. Different sites use different systems and starting points, and their players are different, so a rating on one site does not translate exactly into a rating on another site or into an official over-the-board rating. Ratings also move up and down by chance from game to game; judge trends over months, not days.
- **Puzzle rating.** Your puzzle rating measures how you solve puzzles *when you know there is a puzzle*. It is useful for tracking tactical training, but it is usually

not the same number as your playing rating, and it should not be compared with it directly.

- **Engine accuracy scores.** "Accuracy" percentages summarize how close your moves were to the engine's, but they depend heavily on the position. A quiet, simple game can produce a high accuracy score even with poor play, and a wild tactical game can produce a low one even when you played well. Use them as a rough hint, never as a grade.
- **Single-game "performance rating" or "estimated Elo".** Some tools estimate a rating from one game's moves. One game is far too small a sample for that to mean much. It is not your Elo, and a high or low number after one game tells you almost nothing.
- **Real mastery.** The best evidence of real progress is in your games and your journal: fewer hanging pieces, more converted wins, better-chosen plans, categories disappearing from your weakness list.

> KEY IDEA **Measure what you control**
>
> You cannot control your rating from one week to the next. You can control how many games you analyze, how many journal entries you review, and whether you ran your blunder-check before every capture. Track those process numbers, and let the rating follow in its own time.

Beyond 1500: a plan for the long term

This book was designed to take you from your first moves to around 1500, and to leave you with a method that keeps working afterwards. Here is what the next stage of training typically looks like (coaching judgment).

Play slower games

Classical games, with an hour or more each, are where you practise everything in this book under realistic conditions. If you can, play at least some slow games every month, online or over the board.

Deepen your openings by understanding, not memorizing

Your repertoire from Chapter 12 and Chapter 13 gives you sound positions. The next step is to understand them: the typical plans, pawn breaks and endgames

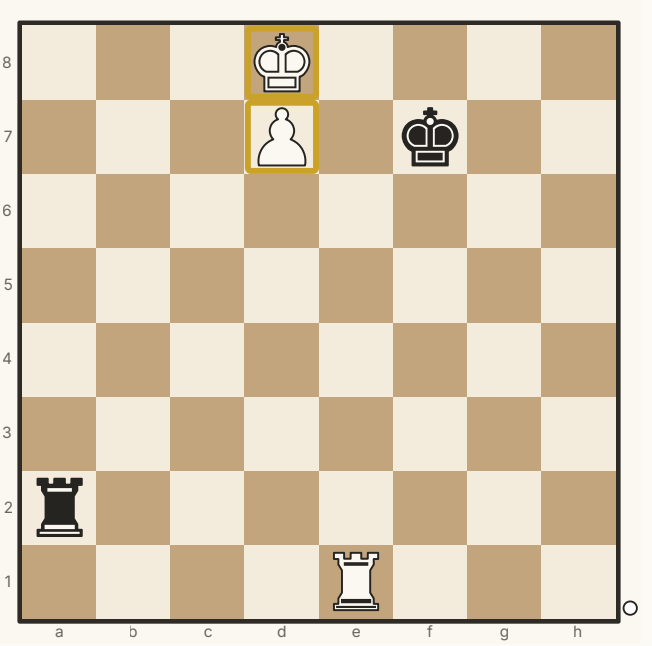
that come from your openings. Collect model games in your lines and play through them.

Study annotated master games

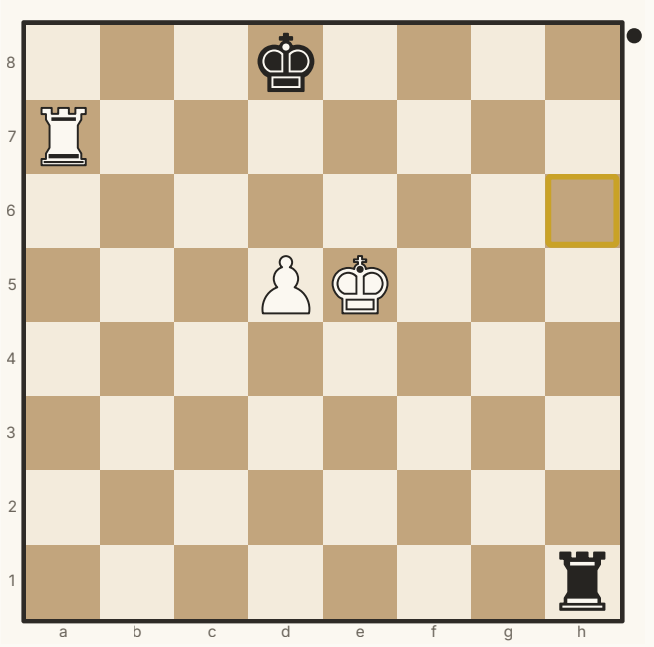
Play through annotated master games, guessing each move before you look. Books with explained games (see Further Reading) tell you *why*, not just *what*.

Make endgame study a habit

A small number of endgames come up again and again. Know them so well that you can play them quickly:

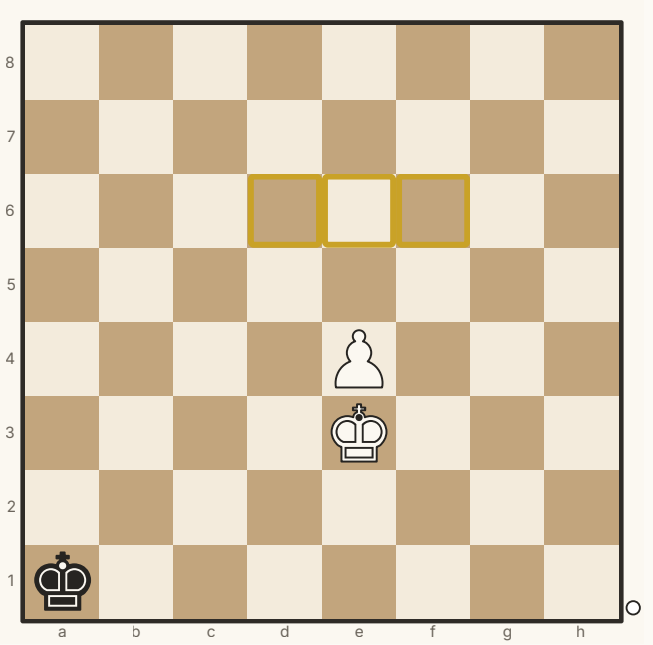


The Lucena position: White wins. The standard method is to bring the rook to the fourth rank to "build a bridge" that shields the king from checks once it leaves d8.



A Philidor-type defence: Black draws by keeping the rook on the sixth rank (...Rh6) until the pawn advances, then checking from behind. Both this and the Lucena position belong in every player's long-term memory.

Key squares from Chapter 18 are part of the same toolkit:



With a pawn on e4, the key squares are d6, e6 and f6. If White's king reaches any of them, the pawn promotes whatever Black does.

Get feedback from people

A coach, a stronger club player or a study partner sees patterns in your games that you cannot. Even occasional lessons add the feedback that deliberate practice needs.

Join a club or play over the board

Over-the-board chess adds a scoresheet, a real opponent, post-game analysis together and longer time controls. Clubs also give you training partners.

> **IN YOUR NEXT GAME** **Your analysis assignment**
>
> Play one slow game this week. Within an hour of finishing, write notes at the three most important moments with the engine off. Then use the engine to find the first big swing, and make one journal entry with a FEN, a category and a rule. Review that entry tomorrow.

Practice at the board

1. **Analyze the worked example yourself.** FEN: `r1bq1rk1/ppp2p2/2np1n1p/2b1p1p1/2B1P2B/2PP1N2/PP3PPP/RN1Q1RK1 w - - 0 9`. Before you look at an engine, play 9.Nxg5 hxg5 10.Bxg5 on a board and try to find a way for White to bring a third attacker. Then check with an engine and see how Black defends.
2. **Spot the difference.** Set up the Greek gift diagram above (`r1bq1rk1/pp1nbppp/2n1p3/2ppP3/3P4/2PB1N2/PP3PPP/RNBQ1RK1 w - - 0 9`) and Exercise 28-08's position side by side. Write one sentence explaining why the sacrifice works in one and fails in the other.
3. **Your first ten journal entries.** Take your last five slow games, analyze them using Steps 1 to 4, and log at least ten mistakes with FENs and categories. Fill in the weakness list template.
4. **Build a mixed set.** From your journal and this book's exercises, build a set of twelve positions with at least four different categories. Shuffle them, solve them, and schedule the reviews at 1, 3, 7 and 21 days.
5. **Learn two endgames for life.** Set up the Lucena and Philidor positions above. Play each side against an engine until you can win the Lucena and hold the Philidor three times in a row without hesitation.

Chapter summary

- Analyze in the right order: your own thoughts first, then turning points, then the engine.
- Use one fixed list of mistake categories (hanging-piece, missed-tactic, missed-mate, king-safety, calculation, opening, endgame, missed-win, conversion, positional, time-psychology), and record the deeper cause.
- Read evaluations in bands; ignore differences of a few tenths; look for the first big swing; ask why your move was bad and why you missed the better one.
- Keep a mistake journal with a FEN, a category and a rule in your own words for every significant mistake.
- Prioritize training with a weakness list: frequency × cost.
- Turn mistakes into exercises, review them on an expanding schedule such as 1, 3, 7 and 21 days, and mix themes when you review.

- Deliberate practice, retrieval practice, spacing and interleaving are well-supported learning principles, but none of them guarantees a rating, and people improve at different speeds.
- Ratings, puzzle ratings, accuracy scores and single-game estimates are different measurements; judge your progress by trends and by your games.

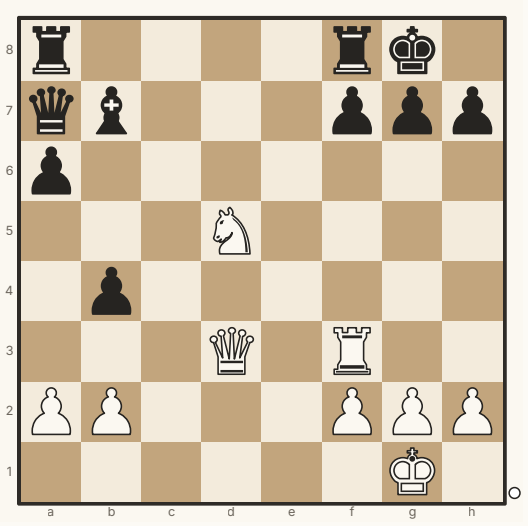
CHEAT CODE **The post-game procedure**

1. Engine off: write your thoughts at the three most important moments.
2. Mark the turning points: discomfort, long thinks, material or structure changes.
3. Engine on: find the first big swing (about a pawn or more); ignore small differences.
4. For each real mistake, ask why, choose one category, and write a rule in your own words.
5. Log it in the journal with a FEN, and schedule reviews at 1, 3, 7 and 21 days.
6. Every ten to twenty games, update the weakness list and train the top category.

Exercises

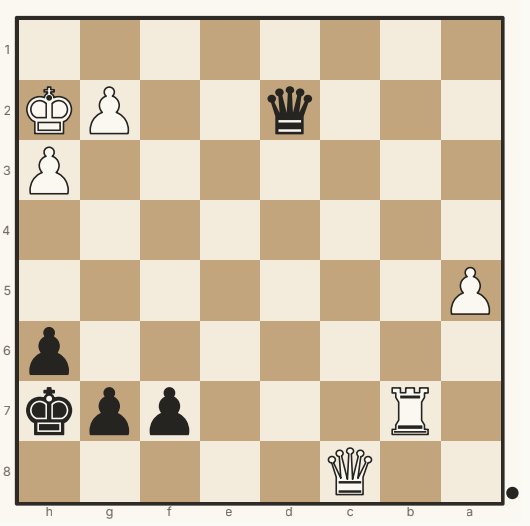
Set up each position on a board or solve it from the diagram. Decide on your move *and* the opponent's best reply before you check. Record your result: first-try correct, correct after the hint, or missed.

28.1 CHECKMATE



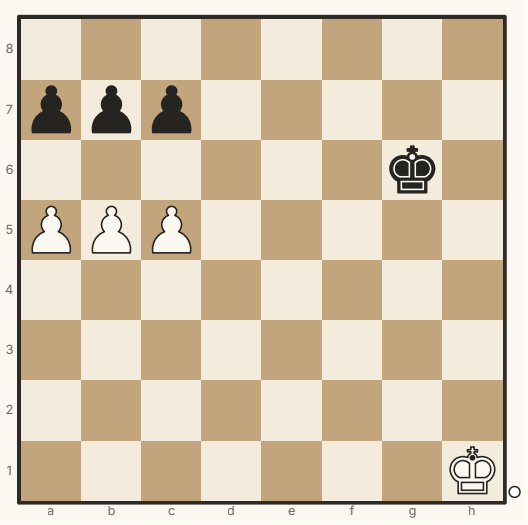
White to move. From a student's game (illustrative): White played 22.Qxh7+? here and the attack fizzled out. Find the mate in 3 that was missed. Solution p. 725

28.2 DEFENCE



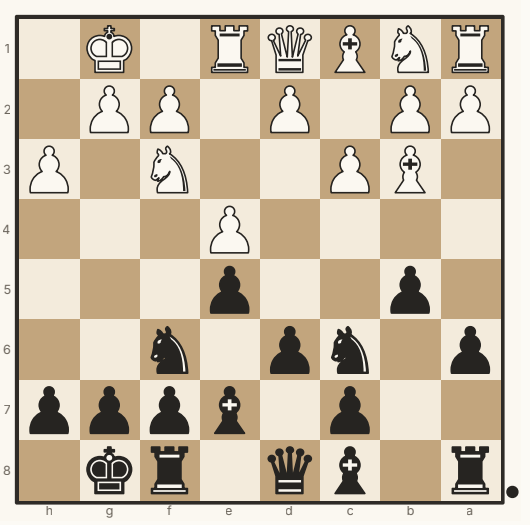
Black to move. From a student's game (illustrative): Black, a rook down, resigned here. Was resignation correct? Find the move that saves the game. Solution p. 725

28.3 ENDGAME



White to move. From a student's game (illustrative): White played 40.Kg2 here, trying to bring the king over, and lost. Both kings are far from the queenside. Find the win. Solution p. 726

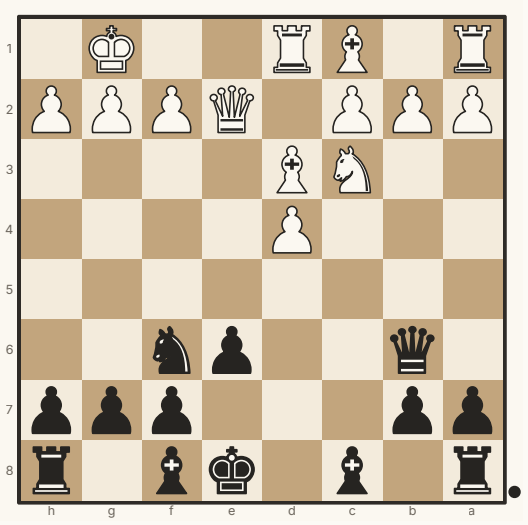
28.4 OPENING



Black to move. From a student's game in the Closed Ruy Lopez, your repertoire as Black (illustrative): after 9.h3 the student played 9...Bg4?, a move that works in many other openings. What should Black play instead?

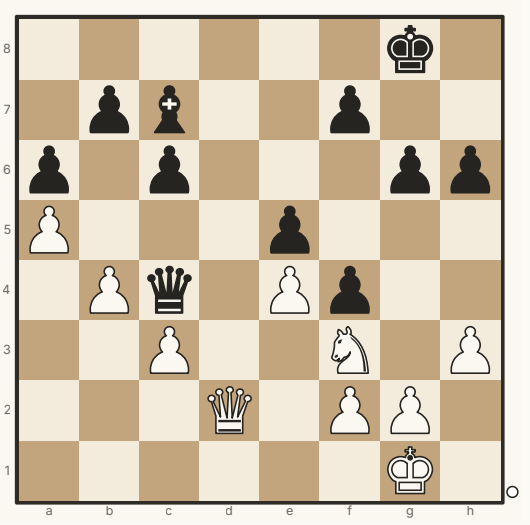
Solution p. 726

28.5 BLUNDER CHECK



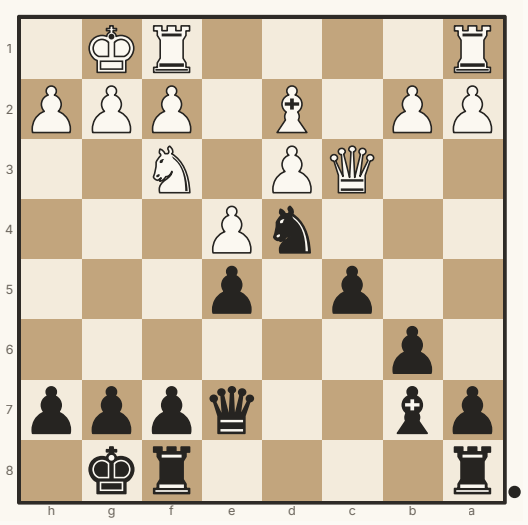
Black to move. From a student's game (illustrative): Black played 10...Qxd4 here, winning back a pawn, and resigned two moves later. What is the problem, and what is a sound move? Solution p. 726

28.6 TACTIC



White to move. From a student's game (illustrative): White played 29.Kh2 and drifted into a worse ending. Black's bishop on c7 is undefended. Find the move that wins material. Solution p. 726

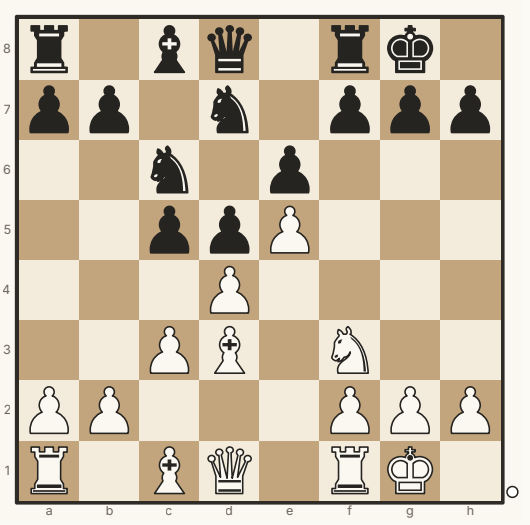
28.7 TACTIC



Black to move. From a student's game (illustrative): Black played 15...Nxf3+ here and was pleased to wreck White's pawns. A much bigger win was available. Find it.

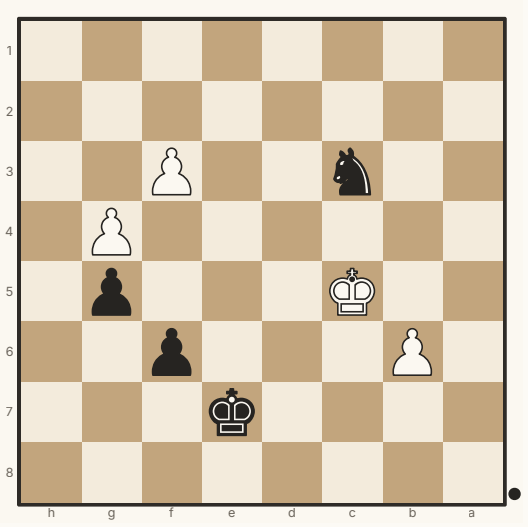
Solution p. 727

28.8 TACTIC



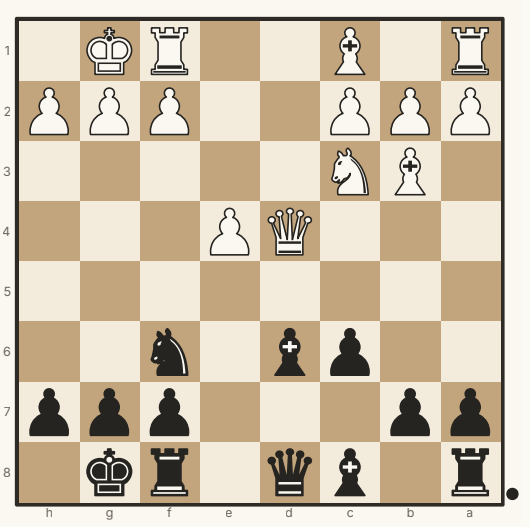
White to move. From a student's game (illustrative): White played 10.Re1 and the game stayed level. Black's kingside has no knight on f6 and no dark-squared bishop. Find the attacking blow. Solution p. 727

28.9 CONVERSION



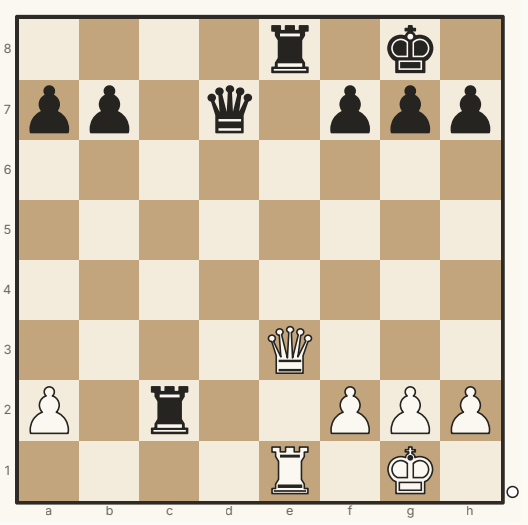
Black to move. From a student's game (illustrative): Black, a knight up, played 50...Kd7 here, and White's passed b-pawn held Black's king and knight tied up. Find the cleanest way to convert. Solution p. 727

28.10 TACTIC



Black to move. From a student's game (illustrative): Black played 11...Ng4, attacking h2, and the game stayed balanced. White's queen on d4 has no defender. Find the winning blow. Solution p. 727

28.11 CHECKMATE



White to move. From a student's game (illustrative): White, a rook down, played 28.Qxa7?? and was mated by 28...Rxe1#. What should White have played? Mate in 2.

Solution p. 728

HINTS

28.1 Which check takes away g6 and g8 before the queen sacrifice?

28.2 Count the white king's squares. Can you check on all of them?

28.3 Sacrifice two pawns to make the third a queen.

28.4 What did White's last move, h3, take away from you?

28.5 What stands between White's rook on d1 and your queen if it lands on d4?

28.6 Which black piece has no defender?

28.7 List every check you have.

28.8 Bishop on d3, knight on f3, queen on d1, and h7 defended only by the king.

28.9 Which knight move comes with check?

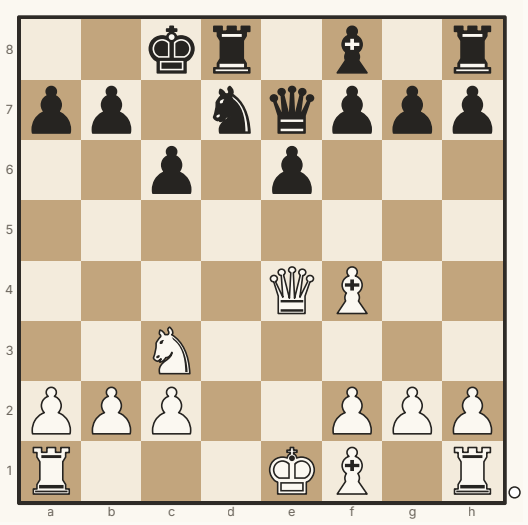
28.10 What is behind your bishop on the d-file?

28.11 Count the defenders of e8.

Stage 5 Test

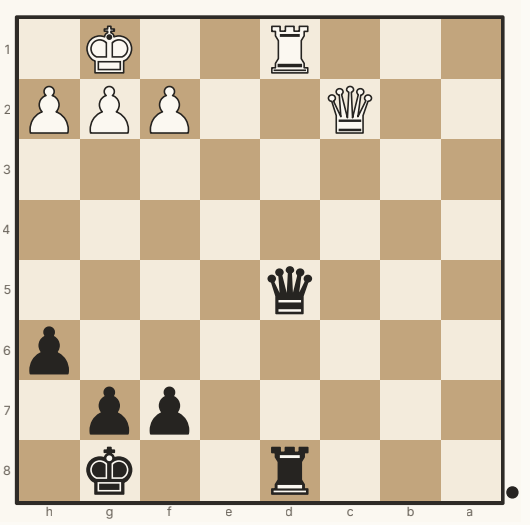
Fifteen mixed positions covering Part 5. Solve them without moving the pieces, write down your answers, then mark yourself. 12/15 or better: move on with confidence. 9–11: review the chapters behind your misses. Below 9: repeat this part's exercises before continuing.

T5.1 CHECKMATE



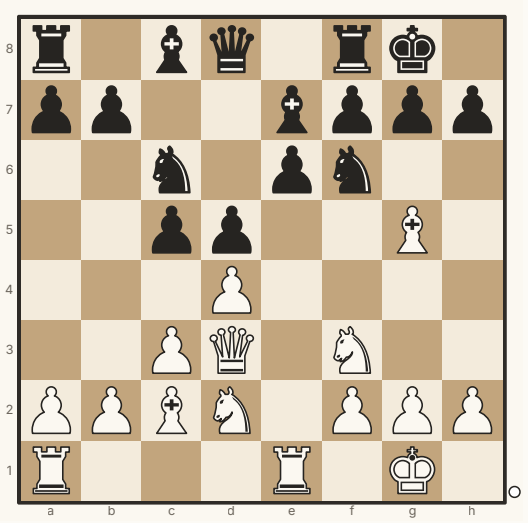
White to move. Black has castled queenside behind the c6-pawn, and both your bishops are still able to reach the long diagonals. Mate in 2. Solution p. 728

T5.2 CHECKMATE



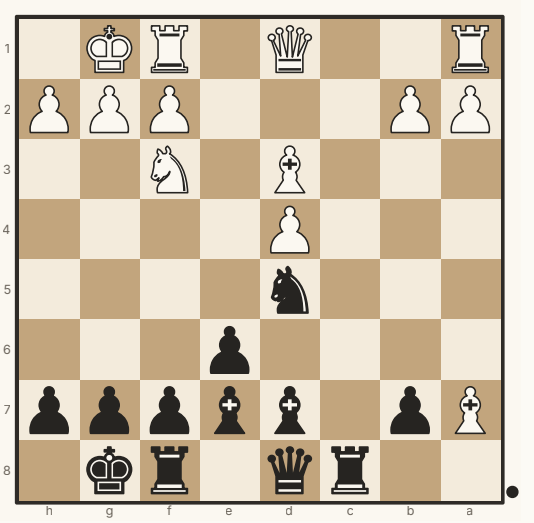
Black to move. White's rook on d1 is defended by the queen. Count attackers and defenders on d1, then mate in 2. Solution p. 728

T5.3 TACTIC



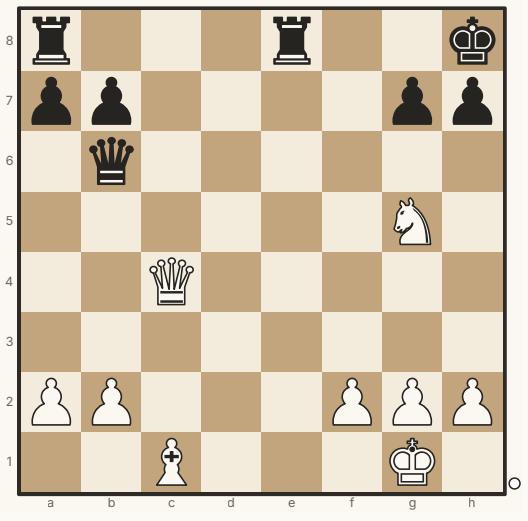
White to move. Your queen and bishop are lined up on h7, but the knight on f6 defends it. Win material. Solution p. 729

T5.4 TACTIC



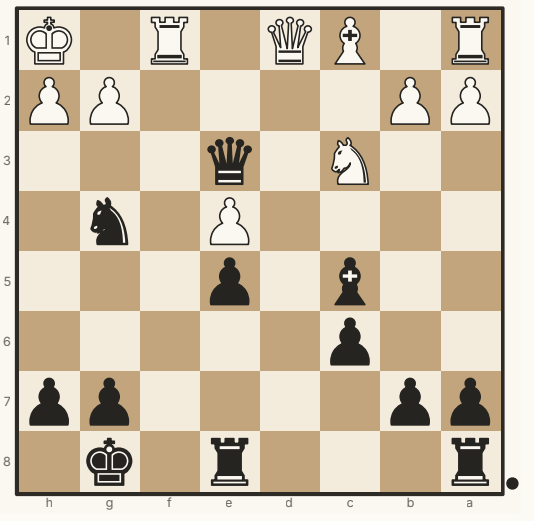
Black to move. White's bishop has just grabbed a pawn on a7. Is it in danger? Find the best move. Solution p. 729

T5.5 CHECKMATE



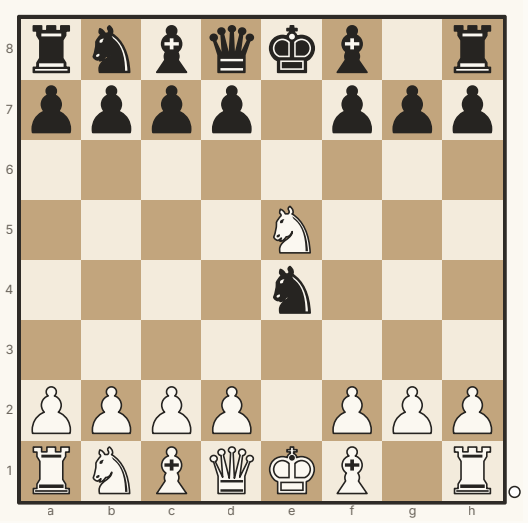
White to move. You are behind in material, but Black's king is boxed in by its own pawns and your queen and knight are close. Mate in 4. Solution p. 729

T5.6 CHECKMATE



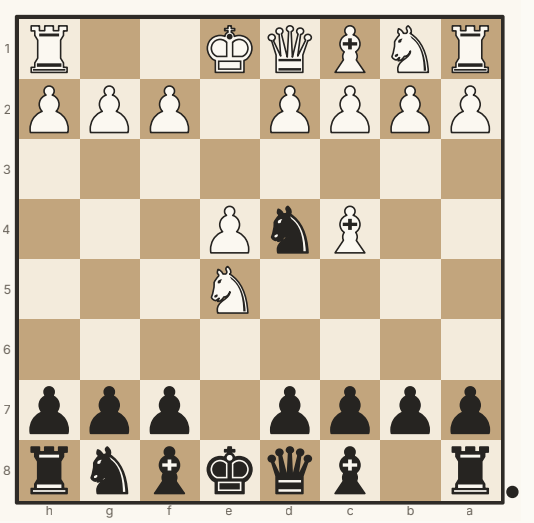
Black to move. White's king on h1 is surrounded by its own pieces and pawns. Mate in 2. Solution p. 729

T5.7 OPENING



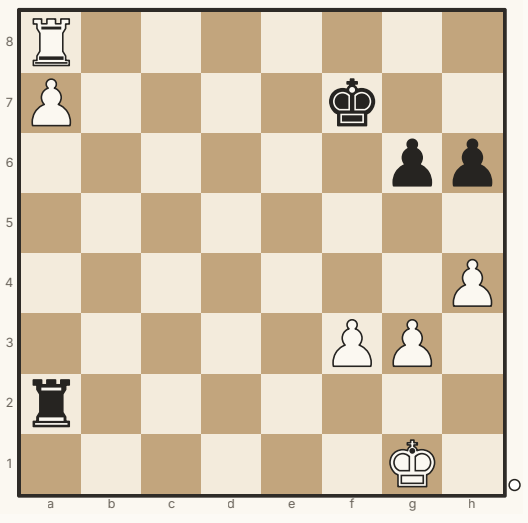
White to move. In your Petrov line, Black has answered 3.Nxe5 by copying you with 3...Nxe4 instead of 3...d6. What is the most testing reply? Solution p. 730

T5.8 OPENING



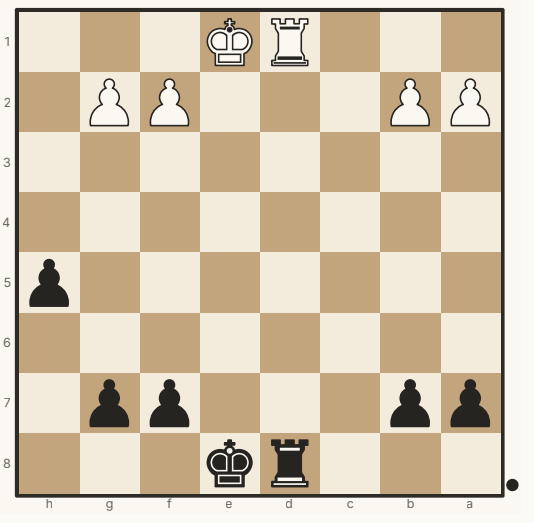
Black to move. You tried the dubious 3... Nd4, and your opponent grabbed the e5-pawn with 4.Nxe5. Find the move that punishes the greed. Solution p. 730

T5.9 ENDGAME



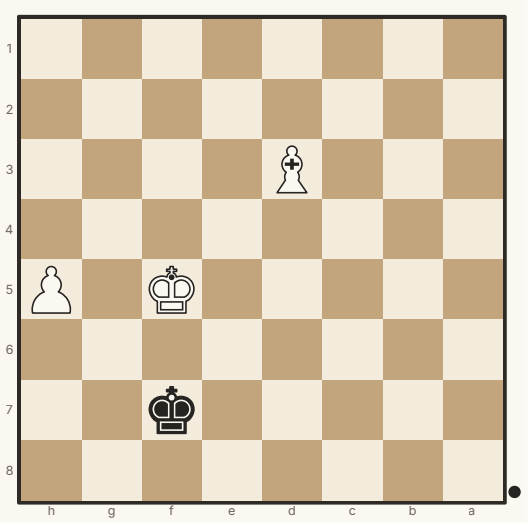
White to move. Your rook is stuck in front of the a-pawn and Black's rook sits behind it. Black's king is on f7. Find the trick that wins. Solution p. 730

T5.10 CONVERSION



Black to move. You are a pawn up and your h-pawn is passed. Choose the clearest way to win. Solution p. 730

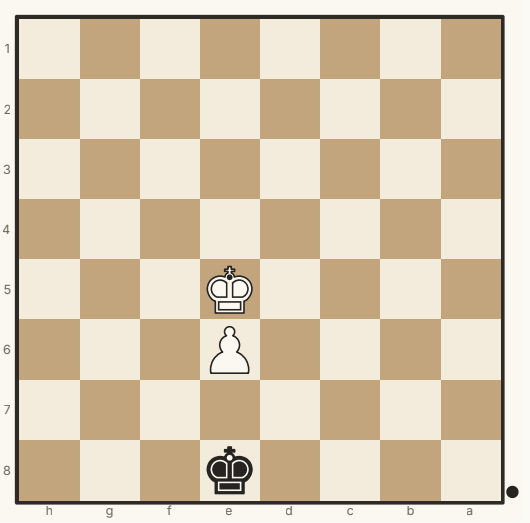
T5.11 DEFENCE



Black to move. White has a bishop and an h-pawn. Find the move that holds the draw.

Solution p. 731

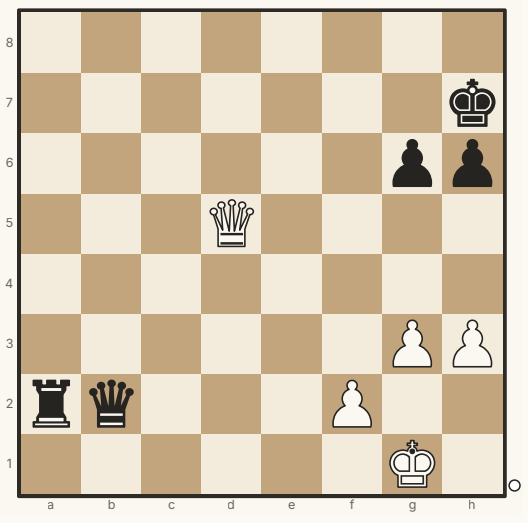
T5.12 DEFENCE



Black to move. White's king and pawn are advanced. Only one move draws. Find it.

Solution p. 731

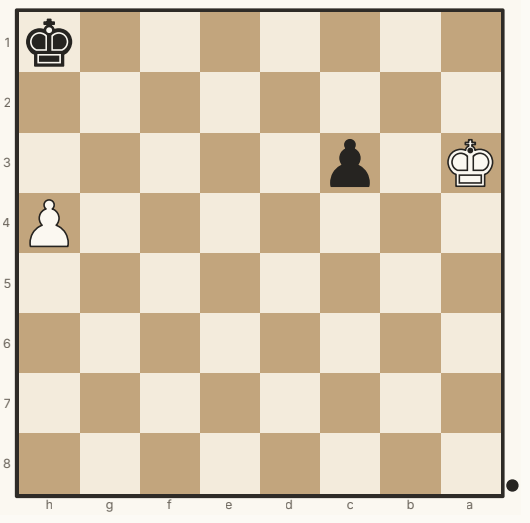
T5.13 DEFENCE



White to move. You are a rook down and Black threatens ...Qxf2+ with mate to follow. Find the move that saves the game.

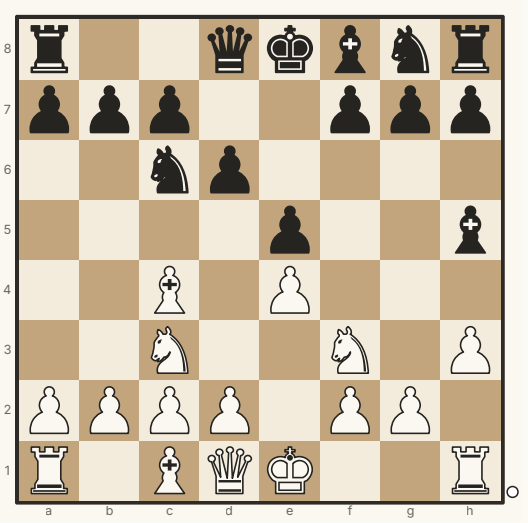
Solution p. 731

T5.14 DEFENCE



Black to move. White's h-pawn looks unstoppable and White's king is ready to take your c-pawn. Find the only move that draws. Solution p. 731

T5.15 OPENING



White to move. Black has just retreated 5... Bh5, keeping the pin on your knight. Find the strongest move. Solution p. 732

HINTS

T5.1 The c6-pawn is the only thing shielding the a6-c8 diagonal.

T5.2 Which white piece has two jobs?

T5.3 If the f6-knight disappeared, what would happen on h7?

T5.4 Count the bishop's escape squares.

T5.5 Start with a check that drives the king back to g8, then use a double check.

T5.6 Which square next to the king can you fill with a piece that the king cannot capture?

T5.7 Both kings are on the e-file. Who gets there first?

T5.8 Find a queen move that attacks two things.

T5.9 Move the rook off the a-file with an eye on h7.

T5.10 Which ending is easiest to win with an outside passed pawn?

T5.11 What colour is h8, and what colour is White's bishop?

T5.12 Kings face to face, with one square between them.

T5.13 Which squares does the black king have, and can one queen check on both?

T5.14 A diagonal king move gets closer to two places at once.

T5.15 If Black takes your queen, which two minor pieces deliver mate?

Solutions

Each solution gives the main line with best defence, the reasoning, engine-approved alternatives, and the tempting wrong answers with the reason they fail.

Chapter 1 · The Rules, Completely

1.1 20. **Rd8#** Exercise 1.1 (p. 33)

Checkmate ends the game at once, so the attack on your queen does not matter. After Rd8+ the black king cannot escape because its own pawns block f7, g7 and h7, nothing can block on e8 or f8, and nothing can capture the rook: the bishop on c8 blocks the rook on a8.

TEMPTING BUT WRONG

- **Qxb2?** Wins the queen, but Rd8 was checkmate; always look for checks before you take material.

SKILL: Recognize that checkmate beats any material gain, and check every check first.

1.2 21... **Qxg2#** Exercise 1.2 (p. 33)

The queen goes down the open g-file and captures on g2 with check. The white king cannot take the queen because the knight on f4 defends g2, and every escape square (f1, h1, h2) is covered by the queen. A king can never capture a protected piece.

TEMPTING BUT WRONG

- **Nxh3+?** A check, but after gxh3 or Kf1 the game goes on; Qxg2 ended it.

SKILL: Deliver a queen mate supported by another piece.

1.3 11. **exd6 Qd8** 12. **dxc7** Exercise 1.3 (p. 34)

Black's pawn passed over d6, so your e5-pawn may capture it en passant, but only on this move. The pawn lands on d6 and attacks both the queen on e7 and the knight on c7. If the queen takes on d6, Qxd6 wins it, because your queen on d1 guards d6 down the open file. So the queen must move away and you collect the knight.

TEMPTING BUT WRONG

- **h3?** Chases the bishop but lets the en passant chance disappear forever.
- **Bg5?** Attacks the queen, but the queen simply moves and the chance to capture en passant is gone.

SKILL: Spot an en passant capture and use it immediately.

1.4 32... **dxe3+** 33. **Ke1 exd2+** Exercise 1.4 (p. 34)

Your d4-pawn is on Black's fifth rank (the fourth rank on the board), and White's pawn has just passed e3, so you may capture en passant. The pawn lands on e3 with check and also attacks the rook on d2. The king cannot take on e3 because your rook on e8 now guards it along the open e-file. White loses the rook.

TEMPTING BUT WRONG

- **Rxe4?** Wins a pawn, but taking en passant wins a whole rook.
- **Kf6?** A quiet move that lets the en passant chance slip away.

SKILL: Use en passant as a forcing move that gives check and forks.

1.5 11. O-O Exercise 1.5 (p. 34)

Castling is legal: the king has not moved, f1 and g1 are empty and not attacked, and the king is not in check. An attacked rook does not stop castling. O-O moves the rook out of the bishop's line and tucks your king away in one move, and g2 can be covered later by Bf3 or f3 if needed. Moving the rook to g1 also saves it, but leaves your king in the centre.

Also correct: Rf1, Rg1

TEMPTING BUT WRONG

- **O-O-O?** Castles the wrong way: the rook on h1 is still attacked and Black wins the exchange with ...Bxh1.
- **Bd3?** Develops, but ignores the threat, and ... Bxh1 wins a whole rook.

SKILL: Know that an attacked rook may castle, and use castling to solve two problems at once.

1.6 4...Bd7 Exercise 1.6 (p. 34)

You must deal with the check, and you cannot castle out of it. Look for a block that also saves the attacked bishop: Bd7 does both, because the bishop retreats along the g4-d7 diagonal into the line of the check. Your castling rights are untouched because the king did not move. Blocks with the knights or the c-pawn leave the bishop on g4 hanging to Qxg4.

TEMPTING BUT WRONG

- **Nc6?** Blocks the check, but Qxg4 wins the bishop for nothing.
- **c6?** Blocks the check, but the bishop on g4 is still undefended and falls to Qxg4.
- **Qd7?** Saves the bishop, but after Bb5 the queen is pinned and Black loses material.

SKILL: Escape check in the way that also solves the second threat, and know that blocking keeps your castling rights.

1.7 40. c8=N+ Kd7 41. Nxe7 Exercise 1.7 (p. 35)

A new queen on c8 would give no check, and Black would mate you with ...Qe1. A knight on c8 gives check to the king on d6 and attacks the queen on e7 at the same time. Black must answer the check, and then you take the queen, leaving you a bishop up with an easily winning endgame.

TEMPTING BUT WRONG

- **c8=Q?** No check, so Black plays ...Qe1, which is checkmate.
- **h3?** Makes room for the king, but Black simply takes your pawn with ...Kxc7 or ...Qxc7 and is a queen up.

SKILL: Underpromote to a knight when the knight's check or fork is worth more than a queen.

1.8 50...g1=R Exercise 1.8 (p. 35)

Count the white king's squares after each promotion. A queen on g1 would cover g2, g3 and h2, while your king covers g4 and h4: White would have no legal move and no check, which is stalemate. A rook covers the g-file and the first rank but not h2, so White still has a move, and King and Rook against King is a forced win.

TEMPTING BUT WRONG

- **g1=Q?** Stalemate: White has no legal move, and the game is drawn on the spot.
- **Kg5?** Avoids stalemate but loses the pawn to Kxg2, and the blocked a-pawns give a draw.

SKILL: Underpromote to a rook to avoid stalemate.

1.9 60. **Qg7#** Exercise 1.9 (p. 35)

Black's pawns cannot move, so Black's only possible moves are king moves. Qg7 gives check, and the queen is defended by your king, so it is mate. The natural-looking Qg6 takes away g8, g7 and h7 without giving check, and Black has no legal move at all: stalemate.

TEMPTING BUT WRONG

- **Qg6?** Stalemate: Black is not in check and has no legal move, so the game is a draw.

SKILL: Avoid stalemate by checking whether the opponent has a legal move.

1.10 40... **Qf4+** 41. **Kg1 Qc1+** 42. **Kh2 Qf4+** Exercise 1.10 (p. 35)

Look for checks: the white king has no shelter on the dark squares f4 and c1. After Qf4+ the king can only go to g1 or h1, and Qc1+ (or Qf1+) drives it back. If White blocks with g3, Qxf2+ keeps checking. The same position will repeat three times, which is a draw by repetition.

TEMPTING BUT WRONG

- **Qc5?** Defends nothing important and gives White time to attack with Qb8 or Rxf7; Black is lost.
- **f5?** A quiet move; White's queen and rook pick off pawns and mate soon follows.

SKILL: Save a lost position with perpetual check and threefold repetition.

1.11 40. **hxg6#** Exercise 1.11 (p. 36)

Your h5-pawn may capture the g5-pawn en passant, landing on g6. That move also clears the h-file, so the rook on h1 gives check. The king cannot take on g6 (the f5-pawn guards it), g7 and g5 are covered by the knight, and h7 and h5 are on the rook's file. Capturing with the other pawn, fxg6, gives no check.

TEMPTING BUT WRONG

- **fxg6?** Also captures en passant, but the h-file stays closed, so there is no check.

SKILL: See en passant as a move that can open lines and deliver a discovered check.

Chapter 2 · Reading the Board: Notation and Board Vision

2.1 14. Bxe5 Exercise 2.1 (p. 53)

Follow the long diagonal from your bishop on b2: c3 and d4 are empty, so the bishop reaches e5. Nothing defends the knight there: the queen on b6, the bishop on e7 and the knight on f6 all point elsewhere, and there is no black pawn on d6 or f6. Bxe5 wins a clean piece.

TEMPTING BUT WRONG

- **d4?** Attacks the knight with a pawn, but it simply retreats; capturing it was free.
- **e4?** Gains space in the centre, but leaves a free piece on the board.

SKILL: Use the full length of a long diagonal.

2.2 10... Qxh4 Exercise 2.2 (p. 53)

Pieces on the edge of the board are easy to overlook, and so are long backward-looking diagonals. Your queen on d8 sees down the diagonal e7, f6, g5 to h4, and the knight there has no defender: the bishop on e2 points at h5, not h4, and there is no white pawn on g3. Qxh4 wins a whole piece.

TEMPTING BUT WRONG

- **Bxh2+?** A flashy check, but after Kxh2 Qxh4+ Kg1 you have only won a pawn instead of a piece.
- **a5?** A slow move: White takes your bishop on g6 and the chance is gone.

SKILL: Spot captures of undefended pieces on the edge of the board.

2.3 7. Qxh5 Exercise 2.3 (p. 54)

Before you rescue the bishop, look at the whole board. Your queen on d1 looks down the diagonal e2, f3, g4, h5, and all three squares are empty because your king's knight is still on g1. The knight on h5 is not defended, so Qxh5 wins a piece, and the bishop on f4 is no longer attacked.

TEMPTING BUT WRONG

- **Bg3?** Saves the bishop, but misses that the knight on h5 was free.
- **Be5?** Also rescues the bishop, but Black is allowed to keep the knight.

SKILL: Scan for long diagonal captures before making a natural retreat.

2.4 15... Bxc7 Exercise 2.4 (p. 54)

Pieces moving backwards are the easiest to miss. Your bishop on f4 can travel back along the diagonal e5, d6 to c7, and both squares are empty. Bxc7 removes the forking knight, so you have won a piece instead of losing material.

TEMPTING BUT WRONG

- **Qd6?** Saves the queen, but Nxa8 wins the rook.
- **Qg4?** Moves the queen to an active square, but Nxa8 still wins the exchange.

SKILL: See backward captures by long-range pieces.

2.5 15. **Ne7+ Kh8** 16. **Nxc8** Exercise 2.5 (p. 54)

Picture the eight squares your knight on d5 can reach. One of them, e7, is a dark square that touches both g8 and c8 by a knight's jump. Ne7+ checks the king and attacks the queen; nothing guards e7, so after the king moves you take the queen.

TEMPTING BUT WRONG

- **Nb6?** Attacks the rook and queen, but Nxb6 removes the knight.
- **Rac1?** A sensible developing move, but it misses the knight fork.

SKILL: Visualize all of a knight's target squares, including checks.

2.6 12... **Qxg2#** Exercise 2.6 (p. 54)

Your bishop on b7 looks all the way down the long diagonal to g2, because c6, d5, e4 and f3 are all empty now that White's knight has left f3. So your queen can take on g2 down the g-file, and the king cannot recapture a defended queen. The squares f1 and h1 are covered too.

TEMPTING BUT WRONG

- **Bxe5?** Wins nothing much and misses the mate; White recaptures with dxe5.

SKILL: Use a long diagonal as hidden support for a mating move.

2.7 18. **Rxh4** Exercise 2.7 (p. 55)

Rooks see whole ranks, not just files. Your rook on d4 looks along the fourth rank through e4, f4 and g4 to h4, and the knight on the edge has no defender. Rxh4 wins a piece. Players often only look at a rook's file and miss the sideways capture. (Bf4, hitting the queen first, also keeps a large advantage, but the direct capture is simplest and strongest.)

TEMPTING BUT WRONG

- **Qg4?** Attacks the knight and g7, but the knight is free right now, and after ...Nf3+ White loses material.

SKILL: See horizontal rook moves across the whole board.

2.8 20... **g6** Exercise 2.8 (p. 55)

Follow the long diagonal from the bishop on b2: c3, d4, e5 and f6 are all empty, so the bishop guards g7, and Qxg7 would be mate. Moving the g-pawn to g6 closes the g-file so the queen can no longer reach g7. Blocking the diagonal with ...e5 also works. Blocking with ...f6 fails because it opens e6 to Qxe6+.

Also correct: e5, g5

TEMPTING BUT WRONG

- **Qc2?** Attacks the bishop, but Qxg7 is checkmate.
- **Rfd8?** Fights for the d-file, but Qxg7 is checkmate.
- **f6?** Blocks the diagonal, but Qxe6+ wins material and keeps the attack.

SKILL: Spot an opponent's long diagonal mating threat and stop it.

2.9 16. **Bxc2** Exercise 2.9 (p. 55)

You do not have to choose which rook to save. Your bishop on f5 looks backwards along the diagonal e4, d3 to c2, and both squares are empty. Bxc2 removes the forking knight and leaves you a piece up. Backward diagonal moves are among the hardest to see, so scan in every direction from every piece.

TEMPTING BUT WRONG

- **Re2?** Saves one rook, but ...Nxa1 wins the other; you lose the exchange instead of winning a piece.
- **Rb1?** Saves the a1 rook, but ...Nxe1 wins the other.

SKILL: See backward diagonal captures.

2.10 4... **e6** Exercise 2.10 (p. 55)

Before grabbing, scan what White's pieces can reach. Your queen has left d8, so nothing defends the d5-pawn, and the knight on c3 attacks it. After ...Qxb2 or ...Nc6, White plays Nxd5, and the knight then threatens Nc7+, a square the bishop on f4 already watches along its diagonal, forking your king and rook. A quiet move such as ...e6 defends d5 and opens a diagonal for your bishop; ...Nf6 also guards d5.

Also correct: Nf6, cxd4

TEMPTING BUT WRONG

- **Qxb2?** Nxd5 attacks c7 with the knight and bishop together, and Nc7+ is coming; Black loses more than the pawn.
- **Nc6?** Develops, but leaves d5 undefended: Nxd5 wins a pawn and threatens Nc7+.

SKILL: Scan the opponent's reachable squares, including knight routes and long diagonals, before grabbing material.

2.11 55. **Qe8#** Exercise 2.11 (p. 56)

The queen travels the whole e-file to e8, giving check along the eighth rank; your king covers g7 and h7, so it is mate. Review from Chapter 1: Black's pawns cannot move, so the tempting Qe6, which takes g8 without checking, would be stalemate.

TEMPTING BUT WRONG

- **Qe6?** Black is not in check and has no legal move: stalemate.

SKILL: See long queen moves and avoid stalemate.

Chapter 3 · Piece Values and Counting

3.1 7. Nxe4 Exercise 3.1 (p. 72)

Count: the knight on e4 is attacked once, by your knight on c3, and defended by nothing, because the d6-pawn covers e5 and c5, not e4. Nxe4 wins a knight for a pawn. If Black then plays ...d5 to attack your knight, it simply retreats, since your bishop is on b3 and not on c4 where ...d5 would hit it as well.

TEMPTING BUT WRONG

- **Nd5?** A pleasant-looking central move, but it leaves Black's extra pawn and lets the knight escape.
- **Re1?** Pins nothing and lets Black trade the knight with ...Nxc3; you never win the piece.

SKILL: Count attackers and defenders to see that a piece is undefended.

3.2 6... d6 Exercise 3.2 (p. 72)

The d4-pawn is attacked once and defended once, but the defender is the c3-pawn, worth much less than your knight. After ... Nxd4 cxd4 you would have given a knight for a pawn. Leave the pawn alone and develop: ...d6 controls e5 and opens a diagonal for your c8-bishop.

Also correct: Be7, Ne7, b6, d5, g6

TEMPTING BUT WRONG

- **Nxd4?** cxd4 wins the knight for a pawn: the defender is a cheap pawn.

SKILL: Never capture with a piece when a pawn defends the target.

3.3 9. gxf3 Exercise 3.3 (p. 73)

Count the f3-square before recapturing: the bishop there is attacked by your queen and your g-pawn, and defended by the knight on e5. If you take with the queen, ...Nxf3+ captures it with check. Take with the cheapest attacker, the pawn: your kingside pawns are a little loose afterwards, but you have won back the piece.

TEMPTING BUT WRONG

- **Qxf3?** The knight on e5 defends f3, so ...Nxf3+ wins your queen.
- **Bxc5?** Grabs a bishop, but ...Bxd1 takes your queen first.

SKILL: Recapture with the lowest-value piece when the square is still defended.

3.4 9... Nxe4 Exercise 3.4 (p. 73)

Count: e4 is attacked twice (knight on f6, rook on e8) and defended once (knight on c3); the bishop on e2 and the queen do not reach it. Capture with the cheaper piece first: after ...Nxe4 Nxe4 Rxe4 you have won a pawn. Starting with the rook, ...Rxe4, lets White take it with the knight and you lose the exchange.

TEMPTING BUT WRONG

- **Rxe4?** Nxe4 Nxe4 leaves you with a knight and pawn for a rook: the wrong capture order.

SKILL: Capture with the lowest-value attacker first.

3.5 14. Nxd6 Exercise 3.5 (p. 73)

Two attackers, one defender, so the bishop can be won, but only in the right order. Nxd6 first: if ...Rxd6, Qxd6 wins the rook too, so Black cannot recapture and you are a piece up. Starting with the queen, Qxd6?, allows ...Rxd6 Nxd6: you have traded your queen for a bishop and a rook.

TEMPTING BUT WRONG

- **Qxd6?** Rxd6 Nxd6 gives Black a rook and bishop for your queen; capture with the knight first.

SKILL: Use the lowest-value attacker first so your opponent cannot profit from recapturing.

3.6 5...Bxd4 Exercise 3.6 (p. 73)

The knight on d4 is attacked twice, by your knight on c6 and your bishop on c5, and defended only once, by the queen on d1. After ...Bxd4 the queen cannot recapture because your knight guards d4 as well. White's last move developed a piece but forgot the usual Be3 or Nxc6, which keep d4 covered.

Also correct: Nxd4

TEMPTING BUT WRONG

- **Nf6?** Develops, but White plays Nxc6 or Be3 and the extra piece is gone.
- **Qf6?** Adds a third attacker but gives White time to defend with Be3.

SKILL: Win a piece by counting attackers and defenders.

3.7 4. O-O Exercise 3.7 (p. 74)

Counting says e5 is free, but step 4 of the safety procedure asks what your opponent can do afterwards. After Nxe5, Black plays ...Qg5, attacking the knight on e5 and the pawn on g2 at once, and your position collapses. Castle instead, or simply trade on d4. A sound developing move keeps your advantage without walking into a trap.

Also correct: Nc3, Nxd4

TEMPTING BUT WRONG

- **Nxe5?** After ...Qg5, Black hits both e5 and g2, and White loses material.

SKILL: Check the opponent's forcing replies before grabbing a 'free' pawn.

3.8 4...Nfd7 Exercise 3.8 (p. 74)

The attacker is a pawn, worth less than your knight, so adding defenders is useless: White would simply play exf6 and you would recapture a pawn for your knight. The knight must move. Nfd7 is the standard move in this French Defence position, and ...Ne4 is also playable.

Also correct: Ne4

TEMPTING BUT WRONG

- **Be7?** Defends the knight, but exf6 wins it for a pawn anyway.
- **Qe7?** Another defender does not help against a pawn attack; exf6 wins the knight.

SKILL: Move a piece attacked by a cheaper attacker instead of defending it.

3.9 18. **Nxd5** Exercise 3.9 (p. 74)

The knight on d5 is defended by the bishop on c6 and the rook on d8. It is attacked by your knight, your rook on d2 and, behind it, the rook on d1, an x-ray attacker. Three against two: after Nxd5 Bxd5 Rxd5 Rxd5 Rxd5 you finish a piece up, so Black should not recapture at all. Starting with the knight, the cheapest attacker, keeps the most material.

Also correct: Rxd5

TEMPTING BUT WRONG

- **Ne4?** Avoids the trade but misses that d5 can be won outright.
- **Qe4?** Adds a fourth attacker but gives Black time to defend with ...Nf6 or ...Nb4.

SKILL: Include x-ray attackers in your count.

3.10 10... **g6** Exercise 3.10 (p. 74)

Material means nothing if you are mated: White threatens Qxh7#, with the bishop on d3 supporting the queen. ...g6 blocks the bishop's diagonal and attacks the queen at the same time. After the queen moves, ... Bxc3 still wins the knight, because nothing has started defending it.

TEMPTING BUT WRONG

- **Bxc3?** Wins a knight, but Qxh7 is checkmate.
- **Re8?** Gives the king f8, but Qxh7+ still wins a pawn and keeps the attack.

SKILL: Recognize that a mate threat outweighs any material gain, and defend with tempo.

3.11 9. **Qxa4** Exercise 3.11 (p. 75)

Review from Chapter 2: look along every diagonal from your queen, including towards the edge. The diagonal d1, c2, b3, a4 is open, and no black piece defends the knight. Qxa4 wins a piece.

TEMPTING BUT WRONG

- **b3?** Attacks the knight with a pawn, but it retreats; it was free to take.
- **Nbd2?** Develops, but leaves a free knight on the board.

SKILL: Spot undefended pieces on the edge of the board.

Chapter 4 · The Safety Scan: Checks, Captures, Threats

4.1 9.axb4 Exercise 4.1 (p. 93)

Ask the first scan question: what is attacking the square the piece just landed on? Your a3-pawn attacks b4, and nothing Black has defends the knight except the bishop on e7, which can only recapture a pawn. After 9.axb4 Bxb4 you have won a knight for a pawn. Retreating the bishop instead keeps material level and wastes the gift.

TEMPTING BUT WRONG

- **Bb1?** Saves the bishop, but you only had to take the knight that attacked it.
- **Be2?** Same mistake: you react to the threat and miss that the attacker itself is hanging.

SKILL: When the opponent's piece attacks yours, first check whether the attacker can simply be taken.

4.2 7...hxg5 Exercise 4.2 (p. 93)

White played the move from habit and forgot your pawn on h6, which already guards g5. 7...hxg5 wins a bishop. After 8.Nxg5 White has a pawn and threatens f7, but Black can meet that with ...O-O or ...Rf8 and is a piece up. Always look at what attacks the square your opponent's piece just moved to.

TEMPTING BUT WRONG

- **Bxf2+?** A flashy check that gives back material; the free bishop on g5 was the point.
- **O-O?** Ignores the gift; White retreats Bh4 and nothing has been won.

SKILL: Spot an opponent's piece that moved to an attacked, undefended square.

4.3 4...Qa5+ 5.Nc3 Qxg5 Exercise 4.3 (p. 94)

Undefended means target. 4...Qa5+ checks the king along a5-e1 and, from the same square, attacks the bishop along the fifth rank. White must answer the check, so there is no time to save the bishop: 5.Nc3 Qxg5 (or 5.Nd2 Qxg5). Checks that also attack a loose piece are among the most common ways material is won.

TEMPTING BUT WRONG

- **Ne4?** Attacks the bishop but gives White time to retreat it; you win nothing.
- **h6?** Asks the bishop to move, and it does (Bh4). A check first would have won it.

SKILL: Use a check to win an undefended piece (loose pieces drop off).

4.4 6.e3 Exercise 4.4 (p. 94)

The pawn is poisoned. After 6.Nxd5? Nxd5! the 'pinned' knight moves anyway, because 7.Bxd8 Bb4+ 8.Qd2 Bxd2+ 9.Kxd2 Kxd8 leaves Black a piece up. The pin is only relative (the queen is behind the knight, not the king), so ask what your opponent's checks are after you capture. Simple development with 6.e3 keeps a pleasant game.

Also correct: Bh4, Nf3, Qc2, Rc1, a3

TEMPTING BUT WRONG

- **Nxd5?** 6...Nxd5! 7.Bxd8 Bb4+ and Black wins a piece (the Elephant Trap).

SKILL: Run the blunder check on captures: look for the opponent's checks after the exchange.

4.5 8...Qd8 Exercise 4.5 (p. 94)

Two questions answer this. What line did it open? The knight left c3 and uncovered the bishop on d2, which now attacks your queen. What does the knight attack? It hits f6 and threatens Nxf6+. The queen must move, and 8...Qd8 is the only square that also guards f6 and d5. Taking the knight with 8...cxd5 or 8...exd5 loses the queen to Bxa5.

TEMPTING BUT WRONG

- **cxd5?** Captures the attacker but forgets the bishop on d2: 9.Bxa5 wins the queen.
- **Qxd2+?** Gives the queen for a bishop.

SKILL: Ask what line the opponent's last move opened (a discovered attack).

4.6 15.h3 Exercise 4.6 (p. 94)

Black threatens 15...Qe1+! 16.Rxe1 Rxe1#, a back-rank mate, because your king has no square and your rook on d1 is the only guard of e1. 15.h3 gives the king a flight square and ends the threat for good; blocking the e-file with a bishop move also works. 15.Qxb7?? is exactly the greedy move that walks into mate.

Also correct: Bd2, Be3, Bf4, Bg5, Qc4

TEMPTING BUT WRONG

- **Qxb7?** 15...Qe1+ 16.Rxe1 Rxe1# — the pawn costs the game.
- **Qxf7+?** 15...Qxf7 and you have given away your queen for a pawn.

SKILL: Check the opponent's forcing moves (especially back-rank checks) before grabbing material.

4.7 7...d5 Exercise 4.7 (p. 95)

Count attackers and defenders before you capture. e4 is attacked once (Nf6) and defended by the bishop on d3, which is not pinned. After 7...Nxe4? 8.Bxe4 Black has lost a knight for a pawn. 7...d5! challenges the centre and develops instead, the main line of this Scotch Four Knights.

Also correct: O-O

TEMPTING BUT WRONG

- **Nxe4?** 8.Bxe4 and you are a knight for a pawn down; the pin on c3 did not matter because d3 also guards e4.

SKILL: Count all defenders before capturing a 'free' pawn.

4.8 8.Nxe5 Exercise 4.8 (p. 95)

8.axb4?? loses at once to 8...Nd3#: the knight checks, your own knight on d2 and pawn on e2 box the king in, and the e2-pawn cannot take on d3 because the queen on e7 pins it once the e5-knight has moved. This is the classic trap of the Budapest Gambit. Exchanging the dangerous knight with 8.Nxe5 is calm and good; the bishop on b4 is not going anywhere.

Also correct: Bxe5, Qb3, Qc2, Rc1, e3, g3

TEMPTING BUT WRONG

- **axb4?** 8...Nd3# — the 'free' bishop was bait.
- **Ra2?** Unpins nothing and still allows ...Nd3+ ideas; it also wastes a tempo.

SKILL: Check your opponent's checks after your capture (the blunder check).

4.9 5...Bb4 Exercise 4.9 (p. 95)

Run the question 'after my move, what is my opponent's strongest reply?' If 5...Bd6?, then 6.e5! attacks the knight and the bishop at once and wins a piece. 5...Bb4 develops with a pin and puts pressure on e4; 5...Be7 is also safe. Pawn forks are easy to walk into when two pieces stand side by side on the same rank in front of an enemy pawn.

Also correct: Be7, d5, d6

TEMPTING BUT WRONG

- **Bd6?** 6.e5 forks the knight on f6 and the bishop on d6.
- **Bc5?** 6.dxc5 — the bishop lands on a square your opponent's pawn attacks.

SKILL: Blunder-check your developing move against pawn forks and pawn attacks.

4.10 8.c3 Exercise 4.10 (p. 95)

After 8.Qxd4? c5! the queen is attacked and must move, and then ...c4 traps the bishop on b3 behind Black's pawns (the 'Noah's Ark' trap). 8.c3 offers a pawn to open lines; 8.a4, attacking the b5 pawn, and 8.Bd5 are fine too. The habit to build: before an automatic recapture, look for the opponent's checks, captures and threats against the recapturing piece and against your other pieces.

Also correct: Bd5, O-O, a4

TEMPTING BUT WRONG

- **Qxd4?** 8...c5 and ...c4 trap the bishop on b3; you lose a piece.

SKILL: Look at the opponent's strongest reply before an automatic recapture.

4.11 7...Bh5 Exercise 4.11 (p. 96)

The first scan question is 'what does the last move attack?' The h3-pawn now hits your bishop on g4, and the bishop has no defender except the knight on f6, which only recaptures a pawn. Retreat it (7...Bh5 keeps the pin) or exchange it. Castling first, 7...O-O??, just loses the bishop to 8.hxg4.

Also correct: Bc8, Bd7, Be6, Bxf3

TEMPTING BUT WRONG

- **O-O?** 8.hxg4 and you have lost a bishop for a pawn.
- **h5?** Offering the bishop does not work here: 8.hxg4 hxg4 9.Ng5 and Black's attack is too slow.

SKILL: Answer the question 'what does my opponent's last move attack?' every move.

Chapter 5 · The Basic Checkmates

5.1 1.Qb7# Exercise 5.1 (p. 114)

The queen goes right next to the enemy king, protected by your own king on c6. From b7 it covers a8, a7, c8 and c7, and the king cannot take it. This 'kiss' mate, queen in front of the king with the king guarding it, is the most common finish of K+Q against K.

TEMPTING BUT WRONG

- **Qh8+?** Check along the back rank, but the king escapes to a7.
- **Qc7+?** Protected check, but a8 is free: ...Ka8.

SKILL: Recognize the protected-queen mate in front of the king.

5.2 1...Ra1# Exercise 5.2 (p. 114)

The White king is on the edge and your king on g3 takes away g2 and h2. A rook check along the first rank covers g1 as well. This is the basic rook mate: the kings face each other (or nearly), and the rook checks along the edge.

TEMPTING BUT WRONG

- **Rh4+?** Check, but along the h-file; the king steps to g1.
- **Ra2?** Takes away the second rank, but that was already covered by your king. It is not check.

SKILL: Deliver the basic rook mate on the edge.

5.3 1.Rb7+ Kd8 2.Ra8# Exercise 5.3 (p. 115)

This is the ladder (or 'lawnmower'). The rook on a6 already guards the sixth rank, so 1.Rb7+ drives the king to the back rank, and 2.Ra8# finishes while the b7-rook keeps the seventh rank closed. The rooks work far away from the enemy king, so it cannot attack them.

TEMPTING BUT WRONG

- **Ra7+?** This rook check leaves the sixth rank open: the king goes back to d6 or e6.
- **Rbb6?** Doubling on the sixth rank wastes a move; there is no mate next move.

SKILL: Execute the two-rook ladder mate.

5.4 1...Qc1# Exercise 5.4 (p. 115)

1...Qc1# (or 1...Qf1#) checks along the first rank; a2 and b2 are covered by your king and b1 by the queen. The trap is 1...Qc2??, which takes away every square but does not give check: White has no legal move and the game is a draw by stalemate.

Also correct: Qf1#

TEMPTING BUT WRONG

- **Qc2?** Stalemate: White's king has no legal move and is not in check.
- **Qc3+?** Check, but the king escapes to b1.

SKILL: Avoid stalemate when finishing with the queen.

5.5 1...Kf2 2.Kh2 Qh7# Exercise 5.5 (p. 115)

The queen cannot mate alone; the king must help. 1...Kf2 takes g1, g2 and g3 away and leaves the White king only h2. After 2.Kh2 the queen mates along the h-file with 2...Qh7#. The tempting queen move 1...Qg3?? is stalemate.

TEMPTING BUT WRONG

- **Qg3?** Stalemate: the White king has no move and is not in check.
- **Qd1+?** A check that only drives the king to h2; it is not mate next move.

SKILL: Use the king to support the queen mate; avoid stalemate.

5.6 1.**Re1 Kc8** 2.**Re8#** Exercise 5.6 (p. 115)

1.Re1! cuts the king off from the e-file, so its only move is 1...Kc8, straight into opposition with your king. Now 2.Re8# is mate: c7, b7 and d7 are covered by the king and the whole back rank by the rook. Forcing the king to step in front of yours is the key skill of the rook mate.

TEMPTING BUT WRONG

- **Rh8+?** Checks too early: the king walks out to e7 and the box gets bigger.
- **Kd6?** Kings in opposition but it is Black's move: ...Ke8 or ...Kc8 and no mate follows at once.

SKILL: Force the king into opposition before the rook mates.

5.7 1.**Kc6 Ka7** 2.**Qb7#** Exercise 5.7 (p. 116)

Your queen already guards the b-file, so the black king has only a7 and a8 to shuffle between. 1.Kc6! brings the king close enough to protect b7, and after 1...Ka7 2.Qb7# is mate. 1.Ka6?? looks just as close but takes away a7 too: Black has no legal move and it is stalemate.

TEMPTING BUT WRONG

- **Ka6?** Stalemate: the king on a8 has no legal move and is not in check.
- **Qh7?** Keeps the box but frees b8 for the king; there is no mate next move.

SKILL: Bring the king while leaving the enemy king a legal move.

5.8 1...**Re8** 2.**Kc1 Re1#** Exercise 5.8 (p. 116)

The White king on d1 has three squares: c1, e1 and e2 (your king already covers c2 and d2). 1...Re8! takes away the e-file, so 2.Kc1 is forced, into opposition with your king, and 2...Re1# finishes on the edge. Checking first with 1...Rh1+ lets the king escape to e2.

TEMPTING BUT WRONG

- **Rh1+?** Check too early: the king escapes to e2 and the box grows.
- **Kd3?** A natural king move, but now White has e1 and c1 and there is no mate next move.

SKILL: Cut off the king with the rook so it must step into opposition.

5.9 1.**Rh6+ Kd7** 2.**Rg7+ Ke8** 3.**Rh8#** Exercise 5.9 (p. 116)

The rook on g5 guards the fifth rank. 1.Rh6+ forces the king up to the seventh, 2.Rg7+ forces it to the eighth, and 3.Rh8# ends it. The rooks stay on the far side of the board, so the king can never attack them; that is why the ladder works without any help from your own king.

TEMPTING BUT WRONG

- **Rg6+?** Check on the wrong side: after ...Kd7 your rooks are in each other's way and the ladder takes longer.
- **Rxa5?** Grabbing a pawn when you have a forced mate.

SKILL: Run the two-rook ladder in a real endgame.

5.10 1.Kc6 Kd8 2.Re1 Kc8 3.Re8# Exercise 5.10 (p. 116)

The rook on b1 already keeps the king off the b-file, but it cannot mate without the king's help. 1.Kc6! brings your king to the sixth rank and leaves Black only one move, 1...Kd8. Now the cut 2.Re1 takes away the e-file, so 2...Kc8 is forced, straight into opposition, and 3.Re8# is mate. Approach with the king, cut off the side the king wants to run to, and let it step in front of your king.

TEMPTING BUT WRONG

- **Rd1?** Gives up the b-file: the king can go back to b7 or b8 and the box gets bigger.
- **Ka6?** The king approaches from the wrong side and the black king slips toward the centre with ...Kc7 or ...Kd7.

SKILL: Combine the king approach, the cut-off and the waiting move.

5.11 1...Rh2+ Exercise 5.11 (p. 117)

Your king is stuck: g8 is covered by the queen, g7 by the h6-pawn, and the h7-pawn is blocked. If your rook disappears, you will have no legal move. 1...Rh2+! offers it: 2.Kxh2 is stalemate, and if the king runs, the rook keeps checking (2.Kg1 Rg2+ 3.Kf1 Rf2+) and must be taken sooner or later. This 'crazy rook' trick is the reason the side with the queen must always check for stalemate.

Also correct: Rb1+

TEMPTING BUT WRONG

- **Rc2?** Attacks the queen but is not check: 2.Qd4+ Kg8 3.Qg7# follows.
- **Rg2?** Not check: 2.Qc8+ Rg8 3.Qc3+ and mate on g7.

SKILL: Use stalemate as a defensive resource (review of Chapter 1).

Chapter 6 · Opening Principles and Early Traps

6.1 3.Qh5# Exercise 6.1 (p. 136)

Black's two pawn moves opened the diagonal h5-e8 and took away every way to block it. Qh5 is check, f7 and g6 cannot be filled by a Black piece, and the king has no flight square. This is Fool's Mate with colours reversed, and it is the clearest lesson in why the f- and g-pawns should usually stay at home early.

TEMPTING BUT WRONG

- **Bxg5?** Wins a pawn, but you had checkmate in one move.
- **Nc3?** A good developing move in general, but forcing moves come first and Qh5 ends the game.

SKILL: Recognize the Fool's Mate pattern when the f- and g-pawns have moved.

6.2 3...g6 Exercise 6.2 (p. 136)

White threatens Qxf7#, and the queen also attacks e5. 3...g6 blocks the queen's attack on f7 and gains time by hitting the queen; after 4.Qf3 Nf6 Black develops with a good game. 3...Qe7 also defends f7 and e5. The move to avoid is the natural-looking 3...Nf6??, which develops but does nothing about f7.

Also correct: Qe7, Qf6

TEMPTING BUT WRONG

- **Nf6?** Attacks the queen but ignores the threat: 4.Qxf7# is checkmate.
- **d6?** Defends e5 but not f7: 4.Qxf7# is checkmate.

SKILL: Defend f7 correctly against an early queen and bishop battery.

6.3 4.Qh5+ Ke7 5.Qxe5+ Exercise 6.3 (p. 137)

With the f-pawn gone, h5-e8 is wide open. 4.Qh5+ forces 4...Ke7 (4...g6 5.Qxe5+ forks king and rook on h8), and 5.Qxe5+ wins back the piece with a pawn and a king hunt still to come. Black's king is stuck in the centre with no pieces developed.

TEMPTING BUT WRONG

- **d4?** Developing quietly lets Black keep the extra knight; the open diagonal needed to be used now.
- **Bc4?** Aims at the king but gives Black time to play ...Nc6 and ...Nf6; you stay a piece down.

SKILL: Punish an early f-pawn move with a queen check on the opened diagonal.

6.4 5...Nd4 Exercise 6.4 (p. 137)

5...Nd4! develops with gain of time: it attacks the queen on b3 and threatens ...Nxc2+, forking king and rook. White has no good answer: 6.Qg3 allows 6...Nxc2+, 6.Qc3 runs into 6...Nxe4 hitting the queen again, and 6.Bxf7+ Ke7 leaves the bishop stranded. Punishing an early queen means developing with threats against it.

TEMPTING BUT WRONG

- **Qe7?** Defends f7 and is playable, but it is passive and lets White's queen escape without further loss of time.
- **Bc5?** Natural development, but it misses the chance to hit the queen with tempo; White gets time to regroup with Ne2 and d3.

SKILL: Punish an early queen sortie by developing with gain of tempo.

6.5 5...dxe5 Exercise 6.5 (p. 137)

No. 5...Bxd1?? runs into Légal's Mate: 6.Bxf7+ Ke7 7.Nd5#. Black has to accept the loss of a pawn with 5...dxe5 6.Qxg4 and play on a pawn down; 5...Be6 is about as good. When an opponent offers you the queen, the first question is what their pieces can do to your king after you take.

Also correct: Be6

TEMPTING BUT WRONG

- **Bxd1?** 6.Bxf7+ Ke7 7.Nd5# — Légal's Mate.
- **Nh6?** Defends f7 but leaves the bishop on g4 and the pawn on f7 under fire; White wins a piece with Nxg4.

SKILL: Refuse a poisoned queen when your king is exposed (Légal's pattern).

6.6 6.Bxf7+ Ke7 7.Nd5# Exercise 6.6 (p. 137)

This is Légal's Mate. 6.Bxf7+ forces 6...Ke7 (the only legal move), and 7.Nd5# finishes: the d5-knight gives check and covers f6, the e5-knight covers d7 and guards the bishop, and the bishop takes away e6 and e8. Three minor pieces against a king that has not moved its defenders out of the way are enough.

TEMPTING BUT WRONG

- **Kxd1?** Recapturing leaves you a queen for a bishop down; you had mate in two.
- **Nxf7?** Forks queen and rook but there is no queen left there to fork; Bxf7+ first leads to mate.

SKILL: Execute Légal's Mate.

6.7 4...d5 Exercise 6.7 (p. 138)

White threatens Nxf7, forking queen and rook. 4...d5! blocks the bishop's line to f7 and hits e4, which is why it is the main move. After 5.exd5 Black should play 5...Na5 and not 5...Nxd5?, which allows the Fried Liver attack 6.Nxf7. This is one reason the book's repertoire chooses 3...Bc5 instead of 3...Nf6 (see Chapter 13).

TEMPTING BUT WRONG

- **Nxe4?** Counter-attacks, but 5.Bxf7+ Ke7 leaves Black's king stuck in the centre.
- **Qe7?** Defends f7 but blocks the bishop on f8 and allows Bxf7+ anyway; Black's position becomes passive and loose.

SKILL: Meet a double attack on f7 by blocking the bishop's diagonal.

6.8 3...Nf6 Exercise 6.8 (p. 138)

Here the queen is on f3, so 3...Nf6 is exactly right: the knight blocks the f-file and stands on its best square. Against Qh5 the same move would lose to Qxf7#, so always check where the queen's line runs before choosing the defence. 3...Qf6 also holds but takes the knight's best square.

Also correct: Qf6, f5

TEMPTING BUT WRONG

- **Bc5?** Develops toward f2 but ignores f7: 4.Qxf7# is mate.
- **Nd4?** Attacks the queen but does not stop it: 4.Qxf7# is mate.

SKILL: Choose the right defence of f7 depending on where the queen stands.

6.9 5.Nc6+ Qe7 6.Nxe7 Exercise 6.9 (p. 138)

The queen on e2 and the king on e8 share the open e-file, with only White's own knight in between. Moving the knight gives a discovered check, and from c6 it also attacks the queen on d8. Black must deal with the check, and White takes the queen next move. Copying moves in the opening does not work when the position is no longer symmetrical.

TEMPTING BUT WRONG

- **Ng6+?** Also a discovered check and wins the rook, but Nc6+ wins the queen.
- **Qe3?** Defends the knight but misses the discovered check that wins the queen.

SKILL: Use a discovered check on the open e-file against an uncastled king.

6.10 4.Nf3 Exercise 6.10 (p. 138)

4.Nf3 returns the knight to its natural square. Black will win the e4-pawn back with ...Qxe4+, but after Be2 White is well ahead in development and Black's king is airy. 4.Nd3? blocks the d-pawn and the c1-bishop, and 4.d4? simply drops the knight.

Also correct: Nc4, Ng4

TEMPTING BUT WRONG

- **d4?** Defends nothing: 4...fxe5 wins the knight for a pawn.
- **Nd3?** Saves the knight but blocks your d-pawn and bishop; Black regains the pawn with ...Qxe4+ and is fine.

SKILL: Retreat to a square that keeps development flowing.

6.11 6.O-O Exercise 6.11 (p. 139)

6.O-O tucks the king away and brings the rook toward the centre, the most useful thing White can do. Nothing is attacked, so there is no reason for violence. The 'attacking' tries all lose material: 6.Bxf7+? Kxf7 gives a bishop for a pawn, and 6.Nxe5? dxe5 simply loses a knight because e5 is defended.

Also correct: Bb3, Nbd2, a4, b4, h3

TEMPTING BUT WRONG

- **Bxf7+?** 6...Kxf7 and White has given a bishop for a pawn; the king on f7 is perfectly safe.
- **Nxe5?** e5 is defended twice: 6...dxe5 (or ...Nxe5) and you are a knight for a pawn down.
- **g4?** A pointless pawn move that weakens your king and drops the pawn to ...Bxg4.

SKILL: Castle and develop when there is no forcing move.

Stage 1 Test

T1.1 9.exd6 Be6 10.dxe7 Exercise T1.1 (p. 140)

Because the d-pawn passed d6 on its two-square move, your e5-pawn may capture it en passant, but only right now. 9.exd6 does two things: the pawn on d6 attacks the bishop on e7, and the e-file opens so your rook pins that bishop to the king. The bishop cannot take on d6 or escape, and 10.dxe7 wins it.

TEMPTING BUT WRONG

- **c4?** Attacks the d5-pawn but lets the en passant chance disappear forever.
- **Nbd2?** Normal development, but the en passant capture is only possible this move.

SKILL: Know the en passant rule and spot when it wins material (Chapter 1).

T1.2 10.Bxa8 Exercise T1.2 (p. 140)

Follow the long diagonal from g2: f3, e4, d5, c6, b7 are all empty now, so your bishop sees the rook on a8. 10.Bxa8 Qxa8 wins the exchange (a rook for a bishop). Long-range pieces are the ones beginners overlook, so sweep every diagonal and file from your bishops, rooks and queen each move.

TEMPTING BUT WRONG

- **b4?** A plan move on the queenside, but the free rook on a8 was the point.
- **Bf4?** Develops, but misses that the c6-knight was blocking the diagonal to a8.

SKILL: See long-range captures along diagonals (Chapter 2).

T1.3 12...Nxe5 Exercise T1.3 (p. 141)

Count: Nd7 and Bd6 attack e5; only the d4-pawn defends it. So 12...Nxe5 13.dxe5 Bxe5 wins a pawn, and the bishop on e5 now also eyes the h2-pawn. Taking with the knight first keeps your bishop for the last capture, where it lands on a strong square.

TEMPTING BUT WRONG

- **c5?** Ignores the count; White keeps the strong knight and you win nothing.
- **Rad8?** A natural developing move that misses the free pawn.

SKILL: Count attackers and defenders, and choose the capturing order (Chapter 3).

T1.4 4.Qe2 Exercise T1.4 (p. 141)

Black threatens 4...Qxf2#: the bishop on c5 protects the queen, and your king has no escape. 4.Qe2 defends f2 and develops; 4.g3 (attacking the queen) and 4.Qf3 also work. The natural 4.Nf3?? attacks the queen but does nothing about f2.

Also correct: Qf3, g3

TEMPTING BUT WRONG

- **Nf3?** Attacks the queen but allows 4...Qxf2#.
- **d3?** Opens a square for the king on d2, but the queen covers it: 4...Qxf2# anyway.

SKILL: Spot the opponent's mating threat before making your own plan (Chapter 4).

T1.5 1...Qh3# Exercise T1.5 (p. 141)

Your king on f2 covers g1, g2 and g3. 1...Qh3# checks down the h-file, which also covers h2, so the king has no square. The tempting 1...Qg3?? covers every square but is not check, and White, whose b-pawn is blocked, has no legal move: stalemate.

TEMPTING BUT WRONG

- **Qg3?** Stalemate: White has no legal move and is not in check.
- **Qd1+?** Check, but the king slips out to h2.

SKILL: Finish the queen mate without stalemating (Chapter 5).

T1.6 1.Rc1 Ka8 2.Rc8# Exercise T1.6 (p. 141)

Your king on a6 covers a7 and b7. 1.Rc1! takes away the c-file, so the only legal move is 1...Ka8, and 2.Rc8# mates on the back rank. A rook check right away (1.Rb1+) only chases the king to c7 or c8.

TEMPTING BUT WRONG

- **Rb1+?** The king escapes to c7 or c8 and the box grows.
- **Kb6?** Opposition, but it is Black's move: ...Kc8 and there is no mate next move.

SKILL: Use a rook cut-off to force the king into a mating net (Chapter 5).

T1.7 5...O-O Exercise T1.7 (p. 142)

5...O-O! puts the king in safety and brings the rook to f8, where it defends f7. Now Nxf7 just loses the knight to ...Rxf7. Castling is often the most natural defence of f7, because f7 is weak mostly while the king stands on e8.

TEMPTING BUT WRONG

- **d6?** Develops, but 6.Nxf7 forks your queen and rook on h8.
- **Qe7?** Defends f7 once but 6.Bxf7+ still wins a pawn and keeps your king in the centre.

SKILL: Castle early, and use it to defend f7 (Chapter 6).

T1.8 1.e8=N+ Kg8 2.Nxc7 Exercise T1.8 (p. 142)

A pawn may promote to a queen, rook, bishop or knight. Here 1.e8=Q?? is not check, and Black mates with 1...Qxh2#. 1.e8=N+! is check, so Black has no time for the mate, and the new knight also attacks the queen on c7. After 2.Nxc7 White is a rook up.

TEMPTING BUT WRONG

- **e8=Q?** Not check: 1...Qxh2# ends the game.
- **Re1?** Defends nothing: 1...Qxh2+ and Black's queen and knight hunt your king.

SKILL: Use underpromotion to a knight when the queen is not the best choice (Chapter 1).

T1.9 6...Bb4 Exercise T1.9 (p. 142)

Instead of running with the queen, 6...Bb4! pins the bishop on c3 to the White king, so it cannot take your queen. Now ...Qxa1 is threatened, and the natural 7.Qd2 runs into 7...Bxc3 8.Qxc3 Qc1#. This is the classic trap of the Englund Gambit; it works because White chased the queen instead of developing.

TEMPTING BUT WRONG

- **Qb6?** Saves the queen, but Black is a pawn down after the gambit and the chance is gone.
- **Qxc3+?** Gives the queen for a bishop.

SKILL: Answer an attack on your queen with a counter-threat (pin) when it exists (Chapter 6).

T1.10 1.Qh4# Exercise T1.10 (p. 142)

Your king on f7 covers g8 and g7, so a check along the h-file finishes: 1.Qh4# (1.Qh1# also mates). The quiet 1.Qg6?? takes h7 away but gives no check; Black's king has no move and the a5-pawn is blocked, so it is stalemate. Blocked pawns do not save the opponent from stalemate.

Also correct: Qh1#

TEMPTING BUT WRONG

- **Qg6?** Stalemate: Black has no legal move and is not in check.
- **Qe8+?** Check, but the king escapes to h7.

SKILL: Remember that blocked pawns do not prevent stalemate (Chapters 1 and 5).

T1.11 9...f5 Exercise T1.11 (p. 143)

White threatens Qxh7#: the bishop on d3 protects the queen and h7 is defended only by your king. 9...f5 blocks the bishop's diagonal and gives your rook the f-file; 9...g6 also blocks, though it weakens the dark squares. 9...Nf6? looks natural but runs into 10.exf6, and 9...b6?? loses to Qxh7#.

Also correct: g6

TEMPTING BUT WRONG

- **Nf6?** The e5-pawn simply takes the knight: 10.exf6.
- **f6?** Opens the wrong diagonal: 10.Qxh7+ Kf7 11.Bg6#.
- **b6?** Ignores the threat: 10.Qxh7#.

SKILL: Run the safety scan and stop a mating threat on h7 (Chapter 4).

T1.12 8.dxe4 Exercise T1.12 (p. 143)

No: your d3-pawn guards e4, so 8.dxe4 wins a knight for a pawn. Black may have hoped for ...Bxf2+ afterwards, but after 8...Bxf2+ 9.Kxf2 Black has only given away more material. The first question after every opponent move: is the piece that just moved defended, and can I take it?

TEMPTING BUT WRONG

- **Rxe4?** 8...d5! forks the rook on e4 and the bishop on c4, and Black gets the material back.
- **d4?** Attacks the bishop but lets the knight live; Black keeps the extra pawn.

SKILL: Take a piece that lands on a square you defend (Chapter 4).

T1.13 14...Nb6 Exercise T1.13 (p. 143)

Run the safety scan on the position after the capture: what checks would White have? After 14...Qxd4?? 15.Bxh7+! Kxh7 16.Rxd4 the bishop's check uncovers the rook on d1, and your queen is lost. The pawn was defended by the rook standing behind the bishop. Develop instead with 14...Nb6, freeing the bishop on c8 and watching d5; the game is level.

Also correct: Qd6, Qf4, h6

TEMPTING BUT WRONG

- **Qxd4?** 15.Bxh7+! Kxh7 16.Rxd4: the check uncovers the rook and your queen is lost.

SKILL: Blunder-check a pawn grab with the queen by looking at the opponent's checks after it (Chapters 4 and 6).

T1.14 5...Qh4+ 6.g3 Nxg3 Exercise T1.14 (p. 143)

5.f3 opened the e1-h4 diagonal next to White's king, and your queen can use it at once. After 5...Qh4+ 6.g3 Nxg3! the knight cannot be taken safely, because 7.hxg3 Qxh1 wins a whole rook, and Black is winning in any case. 6.Ke2 is no better: 6...Qf2+ 7.Kd3 dxc4+ and the white king is hunted in the middle of the board. The attacked knight did not need to retreat; a check came first.

TEMPTING BUT WRONG

- **Nxc3?** A normal reply, but it misses the queen check that wins material.
- **Nc5?** Retreating gives White a big centre for free; you missed 5...Qh4+.

SKILL: Scan checks first: a pawn move next to the king can open a diagonal for a queen check (Chapters 4 and 6).

T1.15 6.Qd5 Nh6 7.Bxh6 Exercise T1.15 (p. 144)

The forcing-move scan finds 6.Qd5!, which threatens Qxf7#. Black's only sensible way to cover f7 is 6...Nh6, and 7.Bxh6 wins that knight: 7...gxh6?? would allow 8.Qxf7#, because the knight was the only extra defender of f7. A queen and bishop aimed at f7 create threats that normal developing moves cannot meet.

TEMPTING BUT WRONG

- **O-O?** A good principle move, but here a forcing move wins material.
- **Qe2?** Develops quietly and lets Black finish development with ...Ngf6.

SKILL: Look for checks, captures and threats against f7 in the opening (Chapters 4 and 6).

Chapter 7 · Forks

7.1 10. Nc7+ Kd8 11. Nxa8 Exercise 7.1 (p. 161)

The king on e8 and the rook on a8 are a knight's fork apart: both are attacked from c7, and nothing guards c7. Nc7+ checks, the king must move, and Nxa8 wins a full rook. Even if the knight is later trapped in the corner, White has won at least the exchange.

TEMPTING BUT WRONG

- **Nd6+?** A check, but d6 is guarded by the bishop on f8: Bxd6 wins the knight.
- **Bf4?** Developing toward c7 is too slow; Black simply plays ...Rc8 or ...Kd8 and the fork is gone.

SKILL: Recognize a knight fork of king and rook on an unguarded square.

7.2 4...e4 Exercise 7.2 (p. 161)

The bishop on d3 and the knight on f3 stand side by side, exactly where one black pawn can hit both. ...e4 is defended by the pawn on d5 and the knight on f6, so White cannot take it safely and must lose a piece. Pawn forks are the cheapest forks of all: a pawn costs 1 and wins 3.

TEMPTING BUT WRONG

- **Bd6?** A normal developing move, but it lets White play e4 or Be2 and the fork disappears.
- **Nc6?** Natural, but it misses the immediate win of a piece with ...e4.

SKILL: Spot a pawn fork against two minor pieces side by side.

7.3 5...Qa5+ Exercise 7.3 (p. 162)

The bishop on g5 has no defender: no knight on f3, no pawn on h4, and the queen on d1 cannot reach it. ...Qa5+ checks the king along the a5-e1 diagonal and attacks g5 along the fifth rank. White must deal with the check (Nd2, c3, Nbc3 or Qd2), and then ...Qxg5 wins a clean piece. A queen fork almost always means check plus a loose piece.

TEMPTING BUT WRONG

- **h6?** Asks the bishop a question, but White just takes on f6 or retreats; nothing is won.
- **Be7?** Breaks the pin calmly but misses that the pinning bishop itself is loose.

SKILL: Find a queen fork that combines a check with an attack on an unprotected piece.

7.4 15. Ne7+ Kh8 16. Nxc6 Exercise 7.4 (p. 162)

Start from the targets: the king is on g8 and the queen on c6. A knight on e7 attacks both, and no black piece guards e7. Ne7+ forces the king to h8, and Nxc6 wins the queen for a knight. This is a royal fork (king and queen).

TEMPTING BUT WRONG

- **Nc7?** Attacks the rook on a8, but it is not check, so Black simply moves the rook or the queen takes on c7.
- **Nb4?** Attacks the queen, which just steps away; it wins nothing.

SKILL: Work backwards from two targets to find the royal fork square.

7.5 7...**Qxd1+** 8.**Kxd1 Nxf2+** 9.**Ke1 Nxh1** Exercise 7.5 (p. 162)

The knight on g4 already eyes f2, and from f2 it would attack d1 and h1. The trade ...Qxd1+ forces the white king onto d1, because nothing else can recapture. Now ...Nxf2+ forks the king on d1 and the rook on h1, and Black wins the exchange and a pawn. The capture with check drives the king onto the forkable square.

TEMPTING BUT WRONG

- **Nxf2?** Wrong move order: White answers Qxd8+ and Kxf2, and Black has lost a knight for a pawn.
- **O-O?** Safe, but after Qxd8 the chance is gone for good.

SKILL: Use a forcing capture to drive the king onto a fork square.

7.6 8...**Bb4+** Exercise 7.6 (p. 162)

The fork square is c7, and right now nothing guards it. ...Bb4+ gains time with a check, and after c3 the bishop drops back to a5, where it guards c7. Stepping the king to d8 or d7 also covers c7 and is fine. What you must not do is make a move that ignores c7, such as ...Rb8 or ...Bc5.

Also correct: Kd7, Kd8

TEMPTING BUT WRONG

- **Bc5?** Develops, but ignores the threat: Nc7+ Ke7 Nxa8 wins the rook.
- **Rb8?** Saves the rook from the fork but leaves it stuck; Bf4 attacks it again and Black loses the exchange.
- **Nb4?** A counter-threat that comes too late: Nc7+ Kd8 Nxa8 and White is a rook up.

SKILL: Identify the opponent's fork square and cover it before it is used.

7.7 5.**e4** Exercise 7.7 (p. 163)

Before you make any routine move, ask what your opponent's pawn moves would attack. ...e4 would hit d3 and f3 together. Playing e4 yourself takes the square away, and moving the bishop off d3 (Be2, or Bb5+ first) also takes the sting out of the push. Castling here is the natural blunder.

Also correct: Bb5+, Be2

TEMPTING BUT WRONG

- **O-O?** Castling ignores the threat: ...e4 forks the bishop and knight and wins a piece.
- **Qe2?** It does nothing about the threat: after ...e4 the bishop and the knight are both still attacked.
- **b3?** A slow move: ...e4 wins a piece.

SKILL: Defend against a pawn fork by moving a target or controlling the fork square.

7.8 11...**Nxe3** Exercise 7.8 (p. 163)

The pawn on e3 has no defender: the f-pawn has moved to f4, the d-pawn is on d4 and the dark-squared bishop is on b2. From e3 a knight attacks both d1 and f1, so ...Nxe3 takes a pawn and forks the queen and rook. White saves the queen with Qe2, and ...Nxf1 wins the exchange as well.

TEMPTING BUT WRONG

- **f5?** A normal Stonewall move, but it lets White play Qe2 or Qe1 and the fork never happens.
- **Bxf4?** After exf4 Black has simply given a bishop for a pawn, and the fork square e3 is no longer a target.

SKILL: Spot a knight fork that also captures material.

7.9 20. **Rh8+ Kxh8** 21. **Nxf7+ Kg8** 22. **Nxd6** Exercise 7.9 (p. 163)

A knight on f7 would attack d6 and h8, so the question is how to get the black king onto h8. Rh8+ is a sacrifice with check: the king must take, because h7 is covered by the rook. Then Nxf7+ forks king and queen, and Nxd6 leaves White a piece up. Look for checks that drive the king onto the fork square.

TEMPTING BUT WRONG

- **Nxf7?** Without the check first, Black simply plays ...Kxf7 and the knight is gone.
- **Qe3?** A reasonable attacking move, but it gives Black time to defend; the fork is there only now.

SKILL: Sacrifice with check to drive the king onto a knight-fork square.

7.10 12... **Qxc7** Exercise 7.10 (p. 163)

A fork only works if the forking piece is safe. Before you panic about your rooks, run the safety scan from Chapter 4: what attacks the knight on c7? Your queen on d8 does, and nothing defends it. ...Qxc7 simply wins a piece.

TEMPTING BUT WRONG

- **Rf8?** Saves one rook but lets the knight take the other: Nxa8 and White is ahead.
- **Rb8?** Moves the wrong rook; Nxe8 wins the exchange when the knight could simply have been captured.

SKILL: Check whether the forking piece can simply be captured.

7.11 6. **c4** Exercise 7.11 (p. 164)

Nc7+ looks like a fork of king and rook, but c7 is guarded by the queen on d8, so it just loses the knight. Nxd6+ fails for the same reason: the bishop on f8 guards d6. There is no tactic, so play a sound move: c4 (or simply develop the other knight).

Also correct: Bd3, N1a3, N1c3, Nd2

TEMPTING BUT WRONG

- **Nc7+?** The queen guards c7: ...Qxc7 wins the knight.
- **Nxd6+?** The bishop guards d6: ...Bxd6 wins the knight for a pawn.

SKILL: Reject a fork when the fork square is defended.

Chapter 8 · Pins and Skewers

8.1 7. Nxd4 Exercise 8.1 (p. 182)

The knight on c6 is pinned by your bishop on b5 to the king on e8, so it cannot recapture on d4: that would expose the king. A pinned piece is a defender in name only. Nxd4 simply wins a bishop. Bxc6+ first, removing the pinned knight and then taking on d4, wins the same piece.

Also correct: Bxc6+

TEMPTING BUT WRONG

- **Nd5?** Centralizes the knight but leaves the bishop on d4 alone; Black castles and the pin is gone.
- **Bg5?** Pins the other knight, but Black castles and the knight on c6 becomes a real defender of d4.

SKILL: Recognize that an absolutely pinned piece does not defend anything.

8.2 30... Rb3+ 31. Kc4 Rxh3 Exercise 8.2 (p. 182)

The king on d3 and the rook on h3 are lined up on the third rank with nothing between them. ...Rb3+ checks the king; when it steps off the rank, the rook on h3 is exposed and ...Rxh3 wins it. That is a skewer: the more valuable piece in front is attacked, moves, and the piece behind is lost.

TEMPTING BUT WRONG

- **Rb4?** Attacks the e4 pawn, but misses the skewer; White keeps everything.
- **Rd8+?** A check on the wrong line: the king steps away and nothing stands behind it.

SKILL: Spot a rook skewer of king and rook along a rank.

8.3 10. d5 Exercise 8.3 (p. 183)

The bishop on e6 is pinned by the rook on e1 to the king on e8, so it cannot take on d5 or run away. Attack a pinned piece with a pawn: d5 hits the bishop, and Black cannot save it. Castling now does not help either, because the bishop is still attacked and White takes it next move.

TEMPTING BUT WRONG

- **Ng5?** Attacks the pinned bishop a second time, but Black castles and the e-file pin is gone; White wins much less.
- **h3?** A useful-looking waiting move, but it gives Black time to castle out of the pin.

SKILL: Exploit an absolute pin by attacking the pinned piece with a pawn.

8.4 10... d4 11. Nf1 dxe3 Exercise 8.4 (p. 183)

Your rook on e8 pins the bishop on e3 to the white king on e1. A pinned piece cannot run, so attack it with a pawn: ...d4 hits the bishop, and nothing White does saves it. Even if White castles now, the bishop is still attacked and ...dxe3 wins a piece.

TEMPTING BUT WRONG

- **Qd6?** Develops, but lets White castle and unpin; the chance is gone.
- **Ng4?** Too slow: White castles, the bishop is no longer pinned, and Black wins far less.

SKILL: Use a pawn to win an absolutely pinned piece.

8.5 5...Qe7 Exercise 8.5 (p. 183)

Your e-pawn has gone, so the e-file is open between White's queen and your king: the knight on e4 is pinned and may not move. White threatens 6.d3, attacking a piece that cannot run away. 5...Qe7! puts your queen on the file, so the knight is now only pinned to a defended queen: after 6.d3 Nf6 the queens face each other and the game is level. 5...d5? defends the knight but leaves it pinned, so 6.d3 wins it for a pawn, and 5...Be7? blocks the file but leaves the knight undefended against 6.Qxe4.

TEMPTING BUT WRONG

- **d5?** It defends the knight but does not break the pin: 6.d3 wins it for a pawn.
- **f5?** It supports the knight, but it stays pinned and 6.d3 still wins it.
- **Be7?** It breaks the pin, but now nothing defends the knight: 6.Qxe4.

SKILL: Break an absolute pin before the opponent can attack the pinned piece with a pawn.

8.6 6...Nxd5 Exercise 8.6 (p. 183)

This is the Elephant Trap in the Queen's Gambit Declined. The knight on f6 is only pinned to the queen, so it may move. After ...Nxd5, if White takes the queen with Bxd8, ...Bb4+ forces Qd2, ...Bxd2+ Kxd2 and ...Kxd8 leaves Black a piece up. If White does not take, Black is simply a knight up.

TEMPTING BUT WRONG

- **h6?** Too slow: White keeps the extra pawn after Bxf6.
- **c6?** Kicks the knight but misses that it can simply be taken.

SKILL: Use a relatively pinned piece when moving it wins more than it loses.

8.7 45.Rh8 e3 46.a8=Q Rxa8 47.Rxa8 Exercise 8.7 (p. 184)

Rh8! clears the a8 square for the pawn and threatens a8=Q. If Black takes the pawn with ...Rxa7, Rh7+ skewers the king on f7 and the rook on a7. So Black cannot take, and the pawn promotes; the best Black can do is give up the rook for the new queen. This skewer trick works whenever the defending king stands on the seventh rank with no piece in between.

TEMPTING BUT WRONG

- **Kf1?** Slow: Black's king goes to g7 or h7 and the skewer trick is no longer possible; the ending is close to a draw.
- **g4?** Changes nothing: Black keeps the king near g7 and holds.

SKILL: Use the rook skewer trick to promote a pawn on the seventh rank.

8.8 15.Bc5+ Ke8 16.Bxf8 Exercise 8.8 (p. 184)

The king on e7 and the rook on f8 sit on the same diagonal, a3-f8. Bc5+ checks the king, nothing can block on d6, and when the king steps away the rook behind it falls. That is a skewer with a bishop: White wins the exchange.

TEMPTING BUT WRONG

- **Rfe1+?** A check, but the king steps to d7 or d8 and there is nothing behind it to win.
- **Nxd5+?** Nxd5+ Nxd5 just gives a knight for a pawn.

SKILL: Find a bishop skewer of king and rook.

8.9 6.d5 a6 7.Ba4 b5 8.Nxb5 Exercise 8.9 (p. 184)

Attack the pinned knight with a pawn: d5 hits c6. Black's best try is ...a6 and ...b5, chasing the bishop, but Nxb5! keeps the pin alive: after ...axb5 Bxb5 the knight on c6 is pinned again and still attacked, so White wins it back with interest. Taking on c6 at once (Bxc6+) throws the pin away.

TEMPTING BUT WRONG

- **Bxc6+?** Exchanges the pinning bishop; after ... bxc6 Black has no problems.
- **dxe5?** Opens the position but releases the tension; White is better but wins nothing.

SKILL: Keep a pin alive when the opponent tries to break it.

8.10 11...Ne2+ 12.Kh1 Nxc3 Exercise 8.10 (p. 184)

Work backwards from the targets: the king on g1 and the queen on c3. A knight on e2 attacks both, and no white piece guards e2. ...Ne2+ is a royal fork; after the king moves, ...Nxc3 wins the queen.

TEMPTING BUT WRONG

- **Nxf3+?** A check, but Bxf3 recaptures and nothing is won.
- **c5?** Supports the knight but misses the fork.

SKILL: Review: find a royal fork by working backwards from the targets.

8.11 4.Nc3 Exercise 8.11 (p. 185)

The pin is only relative, so the knight is allowed to move, but count what stands behind it: after 4.Nxe5?? Bxd1 Black has won your queen for a pawn. Unlike Légal's trap, there is no mate to make up for it, because you have no bishop on c4 and no knight on c3. Leave the pawn and develop with 4.Nc3, or ask the bishop a question with 4.h3, or break the pin with 4.Be2: you stay a healthy pawn up.

Also correct: Bb5+, Be2, h3

TEMPTING BUT WRONG

- **Nxe5?** 4...Bxd1 wins your queen for a pawn.

SKILL: Before moving a pinned piece, check what is standing behind it.

Chapter 9 · Discovered Attacks and Double Check

9.1 16.Bxh7+ Nxh7 17.Rxd6 Exercise 9.1 (p. 200)

The bishop on d3 is the front piece of a battery: the rook on d1 is aimed at the queen on d6 behind it. Move the bishop with a threat and the rook's attack is uncovered. Bxh7+ takes a pawn with check, so Black has no time to save the queen, and Rxd6 wins it.

TEMPTING BUT WRONG

- **Bb2?** Develops, but lets Black move the queen off the d-file; the battery fires at nothing.
- **Ba6?** A discovered attack too, but without check: Black simply moves the queen off the d-file and the trade on b7 wins nothing.

SKILL: Fire a battery: move the front piece with check so the back piece wins material.

9.2 8...Nc3+ Exercise 9.2 (p. 200)

Any knight move uncovers a check from the rook on e8, so the knight may go anywhere, even to a square where it can be captured. ...Nc3+ is attacked by the b2 pawn, but White must answer the check first, and then ...Nxd1 wins the queen. Discovered check gives the front piece a free move.

TEMPTING BUT WRONG

- **Nxf2+?** Also a discovered check, but White answers with Kxf2: the king captures the knight and steps off the e-file at the same time.
- **Ng3+?** After Be3 Nxh1 the knight is trapped in the corner, White's development is far ahead, and Black ends up worse.

SKILL: Use a discovered check to let the front piece attack the queen from an unprotected square.

9.3 8.exf6 Qxe2+ 9.Bxe2 Exercise 9.3 (p. 201)

The two queens face each other on the e-file with only your e5-pawn between them. 8.exf6! captures the knight and at the same time uncovers your queen's attack on Black's queen: a discovered attack. Black must deal with the queen first, 8...Qxe2+ 9.Bxe2, and only then can take back on f6 with 9...gxf6, so White has won a knight for a pawn. The point to remember: when a pawn or piece masks a line, look at what it can capture as it moves away.

TEMPTING BUT WRONG

- **exd6?** It opens lines but wins nothing: after 8...cxd6 the game is level.
- **Bf4?** It defends the e5-pawn but misses that the pawn can capture with a discovered attack.

SKILL: Use a pawn capture to unmask a discovered attack on the queen.

9.4 17...Bxe5 Exercise 9.4 (p. 201)

The bishop on d6 is the front piece of a battery aimed at the white queen. When it moves, the rook attacks d2. Make the front piece's move count: ...Bxe5 captures the knight, and White cannot both save the queen and recapture. After the queen moves, ...Bxb2 leaves Black a piece up.

TEMPTING BUT WRONG

- **Bxg2?** Grabs a pawn, but the battery has not fired; White calmly answers and Black has won far less.
- **Ng4?** Attacks the knight, but White has time to move the queen off the d-file.

SKILL: Use a discovered attack where the front piece also captures material.

9.5 11. **Nc6+ Be6** 12. **Nxd8** Exercise 9.5 (p. 201)

The knight blocks your rook's line to the king, so every knight move is a discovered check. Pick the square with the biggest second threat: Nc6+ attacks the queen on d8. The pawn on b7 attacks c6, but it cannot capture, because that does not answer the check. After ...Be6 to block, Nxd8 wins the queen.

TEMPTING BUT WRONG

- **Ng6+?** Also a discovered check, but it only attacks the rook on h8; Black blocks with ...Be6 and keeps the queen.
- **Nd7+?** The king simply takes: ...Kxd7 steps off the e-file and the knight is lost.

SKILL: Choose the best destination for the front piece of a discovered check.

9.6 16... **Qb8** Exercise 9.6 (p. 201)

White's battery threatens Bxh7+, uncovering the rook on your queen with check. The cure is to step off the line: ...Qb8 (or ...Qc7, ...Qc5 and similar) takes the queen off the d-file. Taking the f3 knight is also fine. Moves that ignore the line lose the queen.

Also correct: Bxf3, Qb4, Qc5, Qc6, Qc7

TEMPTING BUT WRONG

- **h6?** Removes the check idea on h7 but not the battery: Bh7+ is still possible, and after ...Nxh7 Rxd6 the queen is gone.
- **Rfd8?** Defends the queen, but Bxh7+ Kxh7 Rxd6 Rxd6 still wins queen for rook and bishop.

SKILL: Spot an enemy battery and move the target off the line.

9.7 9. **Nf6+ Kd8** 10. **Re8#** Exercise 9.7 (p. 202)

Nf6+ is a double check: the knight checks from f6 and the rook on e1 checks along the e-file. The knight is attacked by the g7 pawn and the g8 knight, but against a double check only a king move helps. The king must go to d8, and Re8# finishes.

TEMPTING BUT WRONG

- **Nd6+?** Also a double check, but the king escapes to d7 or d8 and there is no mate.
- **Ng3+?** A discovered check that wins the queen after ...Be6 Nxh5, but it is not mate, and mate was available.

SKILL: Use double check to force the king onto a mating square.

9.8 18... **Nxf2+** 19. **Kg1 Nh3#** Exercise 9.8 (p. 202)

Any knight move uncovers the bishop's check on h1. ...Nxf2+ is the right one: the king cannot take on f2 because your rook covers the f-file, and g2 is covered by the bishop. After Kg1, ...Nh3# checks from h3; f1 is blocked, f2 and g2 are covered, and nothing can capture the knight.

TEMPTING BUT WRONG

- **Nxd2+?** Wins material, but after f3 it is not mate.
- **Qxd1?** Takes the queen, but after Rfxd1 the battery has been wasted and there is no mate.

SKILL: Combine a discovered check with a knight check to mate.

9.9 15. **Nh6+ Kh8** 16. **Qg8+ Rxg8** 17. **Nf7#** Exercise 9.9 (p. 202)

Nh6+ is a double check from the knight and the queen on c4, so the g7 pawn cannot capture the knight and the king must go to h8. Then the queen sacrifice Qg8+ forces ... Rxg8 (the king cannot take, because the knight guards g8), and Nf7# is a smothered mate. Taking the queen on d8 wins material but misses the mate.

TEMPTING BUT WRONG

- **Nxd8+?** Wins the queen with check, but after ... d5 Black is still fighting; the mate was faster and certain.
- **Nxd6+?** A discovered check that wins only a pawn; ...Kh8 and the attack has run out.

SKILL: Use a double check to start a forced mating sequence.

9.10 6... **Nxc2+** Exercise 9.10 (p. 202)

With the queen gone from d1, nothing guards c2. ...Nxc2+ takes a pawn and forks the king on e1 and the rook on a1. After the king moves, ...Nxa1 wins the exchange and a pawn; even if the knight is later trapped in the corner, Black stays ahead.

TEMPTING BUT WRONG

- **c5?** A good central move, but it misses the free fork on c2.
- **Be7?** Develops, but lets White play Qd1 and the fork is gone.

SKILL: Review: find a knight fork of king and rook after the queen has left its post.

9.11 12. **cxd4** Exercise 9.11 (p. 203)

Run the safety scan from Chapter 4 first: the knight on d4 is attacked by your c3 pawn and is defended only by the e5 pawn. Capturing the front piece of a battery is the cleanest way to defuse it, and here cxd4 wins a piece outright. Grabbing the e5 pawn instead lets the battery fire.

TEMPTING BUT WRONG

- **Nxe5?** Moves the knight away from f3 and walks into trouble: after ...Qxe5 cxd4 Rxd4 Black's rook hits your queen and Black has regained the material.
- **Nxd4?** Takes the front piece, but with the wrong unit: after ...exd4 cxd4 White has won only a pawn instead of a piece.

SKILL: Defuse an enemy battery by capturing the front piece.

Chapter 10 · Removing the Defender, Overloading, Deflection and Decoys

10.1 14.Bxf6 Exercise 10.1 (p. 219)

Your queen and bishop on the b1-h7 diagonal aim at h7, and the knight on f6 is its only defender. After Bxf6 Black cannot recapture: 14...Bxf6 or 14...gxf6 allows 15.Qxh7#. Black has to block the diagonal with 14...g6, and White is simply a knight up.

TEMPTING BUT WRONG

- **Rxe6?** Grabs a bishop for a rook; after ...fxe6 the knight on f6 still guards h7 and you have given away the exchange.
- **Ne5?** A nice square, but it does nothing to the defender of h7 and Black has time to play ... Ne4 and trade pieces.

SKILL: Capture a defender whose recapture would allow mate.

10.2 20...Bxd4 Exercise 10.2 (p. 219)

The rook on d1 defends the knight on d4 and also guards White's back rank against ...Re1#. It cannot do both. After 20...Bxd4 the recapture 21.Rxd4?? runs into 21...Re1#, so White must leave Black a clean piece ahead.

TEMPTING BUT WRONG

- **Nd5?** Centralises the knight, but gives White time to play h3 or c3 and the overload disappears.
- **Re4?** Attacks the knight a second time but lets White defend calmly with Be3 or c3; you had the win of a piece immediately.

SKILL: Spot an overloaded rook that guards both a piece and the back rank.

10.3 8.Bf7+ Kxf7 9.Qxd8 Exercise 10.3 (p. 220)

The queens face each other on the open d-file and the king on e8 is the black queen's only guard. 8.Bf7+ is a check that must be answered: the king cannot go to d7 (the white queen covers it) and e7 and f8 are blocked, so 8...Kxf7 is forced. Now the king has left e8 and 9.Qxd8 wins the queen for a bishop.

TEMPTING BUT WRONG

- **Qxd8+?** A plain queen trade; after ...Kxd8 the material is level and the chance is gone.
- **Qe2?** A developing move that lets Black trade queens or defend the d8 queen next move.

SKILL: Use a check to deflect the only defender of a valuable piece.

10.4 24...Rxf1+ 25.Kxf1 Nd2+ 26.Kf2 Nxb3 Exercise 10.4 (p. 220)

On g1 the white king cannot be forked from d2, but on f1 it can. 24...Rxf1+ forces 25.Kxf1, and then 25...Nd2+ attacks the king on f1 and the queen on b3 at the same time. After the king steps aside, 26...Nxb3 leaves Black a queen against a knight.

TEMPTING BUT WRONG

- **Nd2?** Attacks the queen without check, so White answers Rxf8+ or simply moves the queen with tempo.
- **Rc8?** A natural developing move that misses the decoy; the chance disappears once White's queen or rook moves.

SKILL: Use a sacrifice to decoy the king onto a forking square.

10.5 14.g4 Exercise 10.5 (p. 220)

Count the knight's escape squares before you attack it: f4 is covered by the bishop on e3, g3 by the f2-pawn, f6 by the e5-pawn, and g7 is blocked by Black's own bishop. 14.g4, supported by the h3-pawn, attacks the knight and it has nowhere to go. Black can only win a pawn or two for it, for example 14...f5 15.Bg5 Qe8 16.gxh5.

TEMPTING BUT WRONG

- **Rae1?** Useful, but it gives Black time for ...f5 or ...f6, after which the knight escapes through f6.
- **Ng5?** Aims at the kingside but lets Black free the knight with ...f6 and ...Ng7.

SKILL: Trap a piece by checking that every escape square is covered before attacking it.

10.6 14...g6 Exercise 10.6 (p. 220)

The knight on f6 is the only defender of h7, and Bxf6 would remove it. 14...g6 shuts the b1-h7 diagonal, so Bxf6 no longer threatens mate and Black can recapture. 14...Ne4, blocking the diagonal on e4, is the other good defence. Moves that ignore the diagonal lose at least a piece.

Also correct: Ne4

TEMPTING BUT WRONG

- **h6?** It attacks the bishop, but 15.Bxf6 Bxf6 16.Qh7# still works: the pawn has left h7 and the square is empty for the queen.
- **Rfe8?** It gives the king f8, but 15.Bxf6 Bxf6 16.Qxh7+ wins a pawn and leaves Black's king exposed.

SKILL: Defend against a removing-the-defender threat by cutting the attacker's line.

10.7 15.Rxd8+ Qxd8 16.Qxb4 Exercise 10.7 (p. 221)

List the queen's duties: it guards the rook on d8 and the bishop on b4. 15.Rxd8+ forces 15...Qxd8, and the queen has been dragged away from b4. 16.Qxb4 collects the bishop and White is a piece up.

TEMPTING BUT WRONG

- **Bxf6?** Trades off a minor piece but gives Black time; after ...Rxd1+ and ...gxf6 the bishop on b4 is safe.
- **Qa4?** Attacks the bishop a second time, but Black simply trades rooks on d1 and retreats with ...Bc5.

SKILL: Exploit an overloaded queen by forcing it to choose between two duties.

10.8 18...Ne2+ 19.Kh1 Nxc3 Exercise 10.8 (p. 221)

From e2 a knight attacks both g1 and c3, and nothing White has covers e2. 18...Ne2+ is check, so White has no time to save the queen, and 19...Nxc3 wins it. Before you play any knight move, look at every square it could reach, especially the ones that give check.

TEMPTING BUT WRONG

- **Nxb3?** Takes a bishop for a knight; that's an even trade when you could have won the queen.
- **Nxf3+?** A check, but after Qxf3 you have only traded knights.

SKILL: Recognize a knight fork of king and queen.

10.9 22.**Nf3** Exercise 10.9 (p. 221)

Black threatens 22...Qe1+ 23.Rxe1 Rxe1# and also 22...Bxd4, since Rxd4 would allow ...Qe1+. 22.Nf3 takes the knight away from the attacked square and covers e1, so the rook on d1 is down to one job again. Making luft with g3 is also fine.

Also correct: Ne2, g3, g4

TEMPTING BUT WRONG

- **Bxf7+?** It wins a pawn but drags the bishop away from defence; after ...Kxf7 Black's threats on the e-file and against d4 win material.
- **Kf1?** The king steps off the back rank, but after ...Bxd4 the f2-pawn falls and White's position collapses.

SKILL: Defend by relieving an overloaded piece before the opponent exploits it.

10.10 21...**Rh1+** 22.**Kxh1 Nxf2+** 23.**Kh2 Nxd3** Exercise 10.10 (p. 221)

Nxf2 would already attack the queen and rook, but with the king on g1 White could defend calmly. 21...Rh1+! forces 22.Kxh1 because g2 and f2 block every escape. Now 22...Nxf2+ is a fork of the king on h1, the queen on d3 and the rook on d1, and 23...Nxd3 leaves Black a queen against a rook.

TEMPTING BUT WRONG

- **Nxf2?** Without check White replies Qe2 or Qf3 and keeps everything; the king had to be lured to h1 first.
- **Bc4?** Attacks the queen, but after the queen moves Black has only a slight edge.

SKILL: Combine a decoy sacrifice with a knight fork.

10.11 12...**Bxd4** 13.**Qf3 Nc2+** 14.**Kd1 Nxa1** Exercise 10.11 (p. 222)

The square c2 is where your knight on b4 would fork the king and the rook on a1, and only the knight on d4 guards it. 12...Bxd4 removes that guard and hits the queen. If 13.Qxd4, then 13...Nc2+ forks king and queen. After the better 13.Qf3, 13...Nc2+ forks king and rook and wins the exchange on top of the extra piece.

TEMPTING BUT WRONG

- **Nc2+?** Too early: the knight on d4 simply takes it with Nxc2.
- **Nc5?** A good square, but it misses the combination and White consolidates with castling.

SKILL: Remove a defender to make a fork possible.

Chapter 11 · Checkmate Patterns

11.1 25.**Re8#** Exercise 11.1 (p. 241)

Black's king is boxed in by its own pawns on f7, g7 and h7, and no black piece guards the back rank: the knight on c6 covers d8 but not e8. Re8# is the classic back-rank mate. Look for checks before you take free material.

TEMPTING BUT WRONG

- **Qxc2?** It wins a rook, but you had mate in one.

SKILL: Recognize a back-rank mate when the king has no luft.

11.2 22...**Qe1+** 23.**Rxe1 Rxe1#** Exercise 11.2 (p. 241)

White's back rank is guarded only by the rook on d1. 22...Qe1+! sacrifices the queen to deflect that rook: after 23.Rxe1 Rxe1# the king on g1 is trapped by its own f-, g- and h-pawns. The queen sacrifice works because the black rook stands behind it on the open e-file.

TEMPTING BUT WRONG

- **Qxc3?** It wins a knight, but White replies Qxa7 or h3 and the mate is gone.
- **Qxh2+?** After Kxh2 you have lost your queen for a pawn.

SKILL: Use a queen sacrifice to deflect the last back-rank defender.

11.3 27...**Qg1+** 28.**Rxg1 Nf2#** Exercise 11.3 (p. 242)

This is the smothered mate. The knight on h3 covers g1, so the king cannot take the queen and 27...Qg1+ forces 28.Rxg1. Now g1, g2 and h2 are all filled with White's own pieces, and 28...Nf2# gives a knight check that cannot be blocked.

TEMPTING BUT WRONG

- **Nf2+?** A check, but after Rxf2 the knight is gone and so is the mate.
- **Qxc3?** Grabs a knight, but White plays gxh3 and is no longer in danger.

SKILL: Execute the smothered-mate pattern with a queen sacrifice on g1.

11.4 23.**Nh6+ Kh8** 24.**Qg8+ Rxg8** 25.**Nf7#** Exercise 11.4 (p. 242)

23.Nh6+ is a double check from the knight and the queen on c4, so the king must move. 23...Kf8 allows 24.Qf7#, so 23...Kh8 is forced. Then 24.Qg8+! is protected by the knight, 24...Rxg8 is forced, and 25.Nf7# is a smothered mate.

TEMPTING BUT WRONG

- **Ng5+?** A discovered check, but Black blocks with ...Be6 and the attack fizzles.
- **Nd8+?** Also a discovered check, but ...Be6 blocks it and the knight on d8 is out of play.

SKILL: Execute the full smothered-mate sequence: double check, queen sacrifice, knight mate.

11.5 20.Qxh7+ Kxh7 21.Rh3# Exercise 11.5 (p. 242)

This is Anastasia's mate: a knight on e7 covers g8 and g6, the g7-pawn blocks the king, and a rook mates on the h-file. 20.Qxh7+ forces 20...Kxh7, and 21.Rh3# finishes because nothing can block on h4, h5 or h6.

TEMPTING BUT WRONG

- **Bxf7?** Wins a pawn and threatens things, but Black survives with ...h6 and you have missed a forced mate.
- **Ng6+?** After ...Kg8 you can take the rook on f8, but that wins only material when mate was available.

SKILL: Recognize Anastasia's mate: knight on e7 plus a rook or queen on the h-file.

11.6 26...Rxh2# Exercise 11.6 (p. 242)

This is the Arabian mate: the rook gives check on h2 next to the cornered king, and the knight on f3 protects the rook and covers g1. White's own g2-pawn takes the last square. Rook and knight together are a very strong mating team against a king in the corner.

TEMPTING BUT WRONG

- **Qxd1?** It wins material, but after Qxf3 or Rxd1 White is fighting, and you had mate on the spot.
- **Nxh2?** Takes a pawn but gives up the knight's control of g1 and h2; there is no mate any more.

SKILL: Recognize the Arabian mate with rook and knight.

11.7 14.Qxc6+ bxc6 15.Ba6# Exercise 11.7 (p. 243)

This is Boden's mate: two bishops on criss-crossing diagonals. The bishop on f4 already covers b8 and c7, and d7 and d8 are blocked by Black's own pieces. 14.Qxc6+ forces 14...bxc6, which opens the a6-c8 diagonal, and 15.Ba6# is mate.

TEMPTING BUT WRONG

- **Bb5?** Attacks the knight, but Black has time for ...Nde5 or ...Kb8 and the mating net is gone.
- **Nb5?** A threat-looking move that allows ...e5 or ...Kb8 and does not force anything.

SKILL: Recognize Boden's mate against a king castled queenside.

11.8 28...Qh3 29.Qe2 Qh1# Exercise 11.8 (p. 243)

The pawn on f3 covers g2, so a queen on h3 threatens both Qg2# and Qh1#. 28...Qh3! is not a check, but White has no way to cover both squares and has no checks of his own. If 29.Qe2, then 29...Qh1#, and other moves allow Qg2#.

TEMPTING BUT WRONG

- **Rh1+?** This also mates, but it takes three moves (Kxh1 Qh3+ Kg1 Qg2#); the quiet queen move is faster.
- **Qe4?** Threatens Qg2#, but White defends with Qd3 or Qe3 and trades queens.

SKILL: Use an advanced pawn to support a queen mate next to the king.

11.9 16. **Rh8+ Kxh8** 17. **Qh5+ Kg8** 18. **Qh7#** Exercise 11.9 (p. 243)

16.Rh8+! is a decoy: it drags the king into the corner where the queen can check it along the h-file. After 16...Kxh8 17.Qh5+ Kg8 18.Qh7#, the bishop on d3 protects the queen and the rook on f8 takes the king's last square.

TEMPTING BUT WRONG

- **Qh5?** It threatens mate, but Black defends with ...f5 and the attack slows down; the rook sacrifice was forcing.
- **Bh7+?** After ...Kh8 you have only repositioned the bishop; the direct rook sacrifice mates faster.

SKILL: Use a rook sacrifice on the h-file to set up a queen-and-bishop mate on h7.

11.10 13... **Nf6** Exercise 11.10 (p. 243)

The queen on h5 and the bishop on d3 both aim at h7, which only the king defends. 13...Nf6 adds a defender of h7 and attacks the queen with gain of time. Pawn moves in front of the king either fail or leave serious weaknesses.

TEMPTING BUT WRONG

- **h6?** It does not cover h7 for long: 14.Bxh6! smashes the pawn shield and White's pieces pour in.
- **f5?** It blocks the bishop, but 14.Rxe6 rips open the light squares and Black's king is soon in trouble.

SKILL: Meet a mate threat on h7 by adding a defender with tempo.

11.11 11. **Nf3** Exercise 11.11 (p. 244)

Black threatens 11...Qxh2+ and mate next, because the queen on h4 and the bishop on d6 both hit h2. 11.Nf3 develops, guards h2 and attacks the queen, so Black loses time. g3 (blocking the bishop's diagonal) and h3 (giving the king room) also hold, but moves that ignore h2 lose.

Also correct: g3, h3

TEMPTING BUT WRONG

- **Bd3?** A natural developing move, but it ignores the threat: 11...Qxh2+ 12.Kf1 Qh1+ and White's king is torn apart.
- **f4?** It blocks nothing on h2 and weakens e3 and g3; after ...Bg4 White is in serious trouble.

SKILL: Spot a queen-and-bishop battery against h2 and defend it in time.

Chapter 12 · Your Repertoire as White

12.1 3.Nc3 Exercise 12.1 (p. 269)

Black's queen came out early, so you develop a piece with gain of time. 3.Nc3 attacks the queen, which must move again, and next you play d4 and Nf3. This is the opening principle 'develop with tempo' in its purest form.

TEMPTING BUT WRONG

- **Bc4?** The bishop steps onto a square attacked by the queen and is simply lost: 3...Qxc4.

SKILL: Develop with tempo against an early queen.

12.2 4.c3 Exercise 12.2 (p. 269)

The repertoire move is 4.c3, preparing d3 or d4 and giving the bishop a retreat square. Castling, d3 and Nc3 are also sound developing moves. The trap to avoid is 4.Ng5??, which attacks f7 but leaves the knight undefended: 4...Qxg5 wins a piece because the d2 pawn blocks the c1 bishop.

Also correct: Nc3, O-O, a4, d3, h3

TEMPTING BUT WRONG

- **Ng5?** The knight on g5 is unprotected and 4...Qxg5 wins it.

SKILL: Follow the Italian set-up and reject an early knight raid.

12.3 7.Qh5 Exercise 12.3 (p. 270)

After the trades on e5 the black pawn on e5 has no defender, and f7 is guarded only by the king. 7.Qh5! attacks both at once: it threatens 8.Qxf7+ (your bishop on c4 supports it) and 8.Qxe5. Black cannot cover everything: 7...g6 8.Qxe5 wins the pawn and hits the rook on h8, while 7...Nf6 or 7...Qd7 allows 8.Qxf7+ and the black king is dragged into the open. Trading queens with 7.Qxd8+? throws the chance away.

TEMPTING BUT WRONG

- **Qxd8+?** Trading queens removes your attacker; Black has an easy game after 7...Bxd8.
- **Qd5?** It also eyes f7, but 7...Qxd5 simply trades queens.

SKILL: Look for a queen move that hits two loose targets (f7 and an unprotected pawn) at once.

12.4 8.Nd5 Exercise 12.4 (p. 270)

Your bishop on d2 is aimed at the queen on a5, and only your own knight on c3 is in the way. 8.Nd5! moves it with threats of its own: now the bishop attacks the queen, and the knight hits c7 (Nxc7+ would fork king and rook), b6 and f6. Every queen move either lands on a square the knight or bishop covers or allows Nxc7+, so Black's best is to give the queen for two pieces, for example 8...Qxd2+ 9.Qxd2 exd5. That is why 7...c6, taking d5 and b5 away from the knight, is the standard move.

TEMPTING BUT WRONG

- **Qe2?** Natural development, but it gives Black time for ...c6 and the chance is gone.
- **Nb5?** It also uncovers the bishop, but after 8...Qb6 the queen is safe and still guards c7.

SKILL: Spot a discovered attack on the queen by moving the piece that blocks your bishop.

12.5 8.a5 Exercise 12.5 (p. 271)

Count the bishop's squares: c5 is covered by the b4 pawn, a7 is blocked by Black's own pawn, and a5 is the square you now attack it from. After 8.a5 the bishop is trapped, and the best Black can get is two pawns for it. The correct reply to 7.a4 was 7...a6 or 7...a5.

TEMPTING BUT WRONG

- **O-O?** Castling is useful but lets Black play ...a6 or ...a5 and free the bishop.
- **b5?** Chasing the knight looks active, but the bishop gets the a5 and c5 squares back.

SKILL: Trap a bishop that has run out of retreat squares.

12.6 4.Nf3 Exercise 12.6 (p. 271)

The knight must leave e5, and Black will then win back the pawn with ...Nxe4. 4.Nf3 is the repertoire move: after 4...Nxe4 5.d4 d5 6.Bd3 White develops easily and keeps a small lead. Do not try to hold the knight with 4.d4?, because after 4...dxe5 5.dxe5 Qxd1+ 6.Kxd1 Nxe4 you are a piece down for a pawn. The sacrifice 4.Nxf7?! (the Cochrane Gambit) gives a knight for two pawns, and Black's king usually survives.

Also correct: Nc4, Nd3

TEMPTING BUT WRONG

- **d4?** 4...dxe5 5.dxe5 Qxd1+ 6.Kxd1 Nxe4 and you have lost a knight for a pawn.
- **Nxf7?** The Cochrane Gambit: a knight for two pawns, and Black keeps the extra piece.

SKILL: Retreat an attacked knight to its best square instead of propping it up with a pawn.

12.7 8.Re2 Exercise 12.7 (p. 272)

After Re1, the f2 pawn is protected only by the king, and now the knight on g4 and the bishop on c5 both hit it. 8.Re2 adds a defender (8.Rf1, returning the rook, does the same), and the knight has no follow-up: it will retreat after h3. Chasing the knight at once with 8.h3 fails to 8...Nxf2, which forks your queen and the h3 pawn while the bishop on c5 protects the knight.

Also correct: Rf1

TEMPTING BUT WRONG

- **h3?** 8...Nxf2 wins a pawn and forks the queen and the h3 pawn; the knight cannot be taken safely.
- **d4?** Opening the centre while f2 is under fire allows 8...exd4 and ...Nxf2 tactics.

SKILL: Defend the f2 square against ...Ng4 and ...Bc5.

12.8 4.Qe2 Exercise 12.8 (p. 272)

4.Qe2 pins the e4 pawn to the black king, so the pawn cannot capture or advance with tempo and the knight on f3 is safe. Next you play d3 to open the centre and keep the extra pawn or a lead in development. Retreating the knight instead is passive and lets Black regain the pawn comfortably.

TEMPTING BUT WRONG

- **Nd4?** Playable, but after 4...Qxd5 Black regains the pawn with an active queen.
- **Ng1?** Undeveloping the knight hands Black the initiative.

SKILL: Use a pin to meet a pawn attack with gain of time.

12.9 4...Qe7 Exercise 12.9 (p. 273)

Your queen attacks the knight on e4, and if it moves away, Nc6+ is a discovered check that wins the queen on d8. 4...Qe7 defends the knight and blocks the e-file. After 5.Qxe4 d6 6.d4 dxe5 7.dxe5 White still has the more pleasant position. Knowing the opponent's best defence tells you what to expect when you set this trap.

TEMPTING BUT WRONG

- **Nf6?** 5.Nc6+ is a discovered check that also attacks the queen on d8.
- **Bc5?** Developing ignores the knight: 5.Qxe4 wins it, and Black's king is still on the open e-file.

SKILL: Understand the opponent's best defence to your trap.

12.10 7.Qb3 Exercise 12.10 (p. 273)

Black has given up the light-squared bishop, so f7 and b7 are both weak. 7.Qb3 attacks them at the same time; Black must defend with 7...Qe7, and after 8.Nc3 White is far ahead in development. This is the famous pattern of Morphy's Opera Game.

TEMPTING BUT WRONG

- **O-O?** A natural move, but it gives Black time for ...Nc6 and ...Qe7; the double attack is gone.

SKILL: Exploit the weaknesses left behind when a bishop is traded for a knight.

12.11 7...a5 Exercise 12.11 (p. 274)

White threatens 8.a5, after which the bishop on b6 has no square: c5 is covered by the b4 pawn and a7 is blocked. 7...a5 (or 7...a6) gives the bishop room and keeps the game level. If your opponents do not see this, your plan with b4 and a4 wins a piece, so always check whether they have answered the threat.

Also correct: a6

TEMPTING BUT WRONG

- **O-O?** 8.a5 traps the bishop and wins a piece for two pawns at most.

SKILL: Recognize a threatened trap of a piece and give it an escape square.

Chapter 13 · Your Repertoire as Black

13.1 4...Qxg5 Exercise 13.1 (p. 297)

Before defending against an attack, check whether the attacker is defended. The knight on g5 is not: White's c1 bishop is blocked by the d2 pawn, and the queen on d1 does not reach g5. 4...Qxg5 simply wins a piece, and the queen also eyes g2.

TEMPTING BUT WRONG

- **Nh6?** It defends f7 but misses that the knight on g5 is free to take.
- **d5?** Active, but it gives back the chance to win a whole piece at once.

SKILL: Check whether an attacking piece is defended before you defend.

13.2 6...O-O Exercise 13.2 (p. 297)

f7 is attacked twice and defended only by your king, so 7.Nxf7 is threatened, forking your queen and the rook on h8. 6...O-O! solves everything at once: the rook on f8 now guards f7, your king is safe, and White's knight on g5 has wasted a tempo (it usually goes back to f3). Slow moves such as 6...a6? or 6...Bd7? lose material to 7.Nxf7.

TEMPTING BUT WRONG

- **a6?** It does nothing for f7: 7.Nxf7 forks your queen and the rook on h8.
- **Bd7?** Development, but f7 is still weak: 7.Nxf7 wins material.
- **Na5?** It attacks the bishop, but 7.Qa4+ c6 8.Bxf7+ wins a pawn and drives your king out.

SKILL: Meet an early attack on f7 by castling, so that the rook defends it.

13.3 3...Nc6 Exercise 13.3 (p. 298)

White's queen came out early, so develop with tempo. 3...Nc6 attacks the queen, which has to move again, and you continue with ...Nf6, ...Bb4 and ...O-O. 3...Nf6 is also fine, but the knight on c6 gains a tempo immediately.

Also correct: Nf6

TEMPTING BUT WRONG

- **d6?** A passive pawn move that gives White time to develop; the queen stays in the centre unchallenged.

SKILL: Develop with tempo against an early queen.

13.4 5...Ng4 Exercise 13.4 (p. 298)

5...Ng4! moves the attacked knight to a square where it hits White's queen on e3 and the pawn on e5. White has to move the queen for the third time, and Black wins the e5-pawn back with a lead in development: 6.Qe4 Ngxe5, or 6.Qe2 d6 opening lines. 5...Nd5? also hits the queen, but after 6.Qe4 the knight is attacked again and must retreat. 5...Nxe5?? loses the knight to 6.Qxe5+.

TEMPTING BUT WRONG

- **Nd5?** After 6.Qe4 the knight is attacked again and has to retreat; White keeps the e5-pawn.
- **Nxe5?** The pawn was defended by the queen: 6.Qxe5+ wins the knight.
- **Ng8?** It saves the knight but undevelops it; White gains space for free.

SKILL: Punish an early queen sortie by developing with gain of tempo.

13.5 5...Qd4 Exercise 13.5 (p. 299)

The pawn on e5 was not really hanging. 5...Qd4 forks the knight and the e4 pawn; after 6.Nf3 Qxe4+ 7.Qe2 Qxe2+ 8.Kxe2 material is level and Black has the bishop pair. 5...Qe7 and 5...Qg5 regain the pawn by similar double attacks.

Also correct: Qe7, Qg5

TEMPTING BUT WRONG

- **Be6?** Calm development, but White simply keeps the extra pawn.

SKILL: Use a queen fork to regain a gambit pawn.

13.6 8...Nxg5 Exercise 13.6 (p. 299)

The knight on c3 is pinned to the king by your queen, and your queen also looks along the fifth rank towards g5; only your own d5-pawn is in the way. 8...Nxg5! wins a piece. If 9.Nxg5, then 9...dxc4! opens the rank: the queen attacks the knight on g5 and the pawn attacks the bishop on d3, and White cannot save both. After 9.Nd2 the knight simply returns with 9...Ne4 and Black is a piece up.

TEMPTING BUT WRONG

- **Nxc3?** It only trades knights: after 9.bxc3 the bishop on g5 survives and the game is level.
- **Bb4?** It adds to the pin but gives White time: 9.cxd5 and the chance to win the bishop is gone.

SKILL: Combine a pin with a hidden attack along the queen's rank.

13.7 6...Qe7 Exercise 13.7 (p. 300)

6...Qe7 pins the e5 pawn to the white king: the pawn cannot take on f6 at all, because that would expose the king on the e-file, and the queen attacks it. White has to defend with 7.Qe2, and after 7...Nd5 your knight is centralized and the e5 pawn remains a target. 6...Nd5 at once is also playable. The check 6...Bb4+? fails, because after 7.c3 the bishop is attacked and the knight on f6 still hangs.

Also correct: Nd5

TEMPTING BUT WRONG

- **Bb4+?** After 7.c3 both the bishop and the knight on f6 are attacked, and Black loses a piece.
- **Ng8?** Retreating to the back rank gives White a free lead in development.

SKILL: Meet a pawn attack on your knight with counter-pressure.

13.8 8...d5 Exercise 13.8 (p. 300)

8...d5! attacks e4 before White can push e5 or d5. After 9.exd5 Nxd5 Black's knight sits on the blockading square in front of White's isolated d-pawn, and Black has equalized. Taking at once with 8...Nxe4 9.Nxe4 d5 is a similar idea. Slow moves like 8...O-O or 8...d6 let White build up with e5 or d5 and gain space.

Also correct: Nxe4

TEMPTING BUT WRONG

- **Na5?** Chasing the bishop costs time; White keeps the big centre and Black's knight is offside.

SKILL: Break up a big pawn centre with a timely pawn strike.

13.9 3...Qh4+ Exercise 13.9 (p. 300)

With the f-pawn gone, the e1-h4 diagonal is open. 3...Qh4+ forces 4.g3, and then 4...Qxe4+ forks the king and the rook on h1. After 5.Qe2 Qxh1 Black is a whole rook ahead.

TEMPTING BUT WRONG

- **dxe4?** A normal recapture, but it misses the chance to win a rook with a queen check and fork.
- **Nc6?** Development is good, but there is a forcing win available right now.

SKILL: Exploit the weakened e1-h4 diagonal with a check and fork.

13.10 7...Nxe4 Exercise 13.10 (p. 300)

With the knight on c3 pinned by your bishop, the e4 pawn has lost a defender. 7...Nxe4 takes it, and after 8.O-O Bxc3 9.d5 Bf6 10.Re1 Ne7 11.Rxe4 d6 Black has a solid, equal position. The automatic strike 7...d5 is weaker here: after 8.exd5 Nxd5 9.O-O White's pieces are very active and the pins work against Black.

TEMPTING BUT WRONG

- **d5?** After 8.exd5 Nxd5 9.O-O White's development and the pressure on d5 and the e-file are dangerous.
- **d6?** Too slow: White plays O-O, e5 or d5 and the centre rolls forward with tempo.

SKILL: Notice when a pin removes a defender and take the pawn it was guarding.

13.11 5...Bb4+ Exercise 13.11 (p. 301)

Answer a fork with a check or a capture. 5...Bb4+ saves the bishop with tempo, and after 6.c3 the knight escapes by capturing: 6...Nxc4. Black keeps the extra piece. 5...Nxc4 first (6.dxc5) is also good. Simply retreating the bishop loses the knight on e5, and retreating the knight loses the bishop.

Also correct: Nxc4

TEMPTING BUT WRONG

- **Bd6?** After 6.dxe5 Bxe5 7.Qd5 White's queen attacks f7 and e5 at once and wins material back.
- **Bb6?** The bishop is safe, but 6.dxe5 wins the knight and White has regained the piece.

SKILL: Escape a pawn fork with a check that gains a tempo.

Stage 2 Test

T2.1 20.Ne7+ Kh8 21.Nxc6 Exercise T2.1 (p. 302)

Black's queen and bishop line up on g2, so you need a forcing move. 20.Ne7+ is a knight fork: it checks the king on g8 and attacks the queen on c6. After 20...Kh8 21.Nxc6 the mate threat has gone with the queen.

TEMPTING BUT WRONG

- **f3?** It stops the mate, but Black takes the knight on d5 and you are a piece down.
- **Bd3?** It defends nothing important; Black wins the knight with ...Qxd5.

SKILL: Answer a mate threat with a forcing fork.

T2.2 10...Bb4 11.O-O Bxd2 Exercise T2.2 (p. 302)

The queen on d2 and the king on e1 stand on the same diagonal, and nothing can be put on c3 to block it. 10...Bb4 pins the queen to the king, and the bishop is protected by the knight on c6, so Qxb4 only gives the queen for a bishop. Castling gets the king out, but the queen is still attacked and 11...Bxd2 wins it.

TEMPTING BUT WRONG

- **Ne4?** Attacks the queen, but after Qd1 or Qc2 White is fine.
- **Bg4?** Develops with a mild pin on f3, but misses the much stronger pin on the queen.

SKILL: Pin a queen to its king with a protected piece.

T2.3 10.Bb5+ Ke7 11.Bxe8 Exercise T2.3 (p. 303)

The king on d7 and the queen on e8 sit on the a4-e8 diagonal. 10.Bb5+ is a skewer: the king must step off the line and the queen behind it falls. Nothing can block on c6 because Black's c-pawn has already gone to c5.

TEMPTING BUT WRONG

- **Ne5+?** A check, but after ...Kc7 Black's king walks to safety and the queen stays.
- **Bf4?** Develops, but gives Black time to play ...Kc7 or ...Qe7.

SKILL: Recognize a bishop skewer of king and queen.

T2.4 14...Bxh2+ 15.Kf1 Rxd4 Exercise T2.4 (p. 303)

Any bishop move uncovers the rook on d8 against the queen on d4. But your rook is unprotected, so a quiet bishop move would allow Qxd8+. 14...Bxh2+ discovers the attack with check, so White has no time to deal with the queen and 15...Rxd4 wins it.

TEMPTING BUT WRONG

- **Be5?** A discovered attack without check: White answers Qxd8+ and wins your rook instead.
- **Qc7?** Defends the rook but wastes the discovered attack; White moves the queen away.

SKILL: Use a discovered attack with check to win the queen.

T2.5 6.Nd6# Exercise T2.5 (p. 303)

Nd6 is a double check: the knight attacks e8 and the queen on e2 checks down the opened e-file. Against a double check only a king move helps, and every square next to Black's king is blocked by its own pieces or covered. This is a well-known opening trap.

TEMPTING BUT WRONG

- **Nxf6+?** A single check that Black answers with ...Nxf6; the mating idea is gone.

SKILL: Spot a double-check mate in the opening.

T2.6 10...Bxf3 Exercise T2.6 (p. 303)

The knight on f3 attacks your queen, but it is also the only defender of h2, which your queen and bishop both hit. 10...Bxf3 removes it: if 11.Bxf3 or 11.gxf3, then 11...Qxh2#. White has to play something like h3 or g3, and you stay a piece up.

TEMPTING BUT WRONG

- **Qh5?** The queen escapes, but you have missed winning a piece.
- **Bxh2+?** After Nxh2 the h2 square is defended again and you have lost a bishop.

SKILL: Capture a defender that guards a mating square.

T2.7 16.Rxd5 Exercise T2.7 (p. 304)

The knight on f6 guards both the bishop on d5 and the h7-square, where your queen and bishop battery points. 16.Rxd5! wins a piece, because 16...Nxd5 allows 17.Qxh7#. Black must block the diagonal with ...g6 and stay a piece down.

TEMPTING BUT WRONG

- **Nd4?** Centralizes the knight but gives Black time to play ...g6 or ...Be4 and the overload disappears.
- **Qxh7+?** Premature: after ...Nxh7 you have lost your queen.

SKILL: Exploit a knight overloaded with two defensive duties.

T2.8 24...Qxc2 Exercise T2.8 (p. 304)

The rook on c1 is overloaded: it protects the queen on c2 and also the e1-square. 24...Qxc2! wins the queen, because 25.Rxc2 allows 25...Re1#. Deflecting the rook first with 24...Re1+ 25.Rxe1 Qxc2 wins the queen too.

Also correct: Re1+

TEMPTING BUT WRONG

- **Qe6?** Avoids the queen trade but wastes the chance; White plays g3 or h3 and the back rank is safe.
- **Qd7?** A passive retreat that lets White trade or make luft.

SKILL: Exploit a rook that guards both a piece and the back rank.

T2.9 19.Rh8+ Kxh8 20.Nxf7+ Kg8 21.Nxd8 Exercise T2.9 (p. 304)

On g8 the king is safe from the knight, but on h8 it is not: a knight on f7 attacks h8 and d8 together. 19.Rh8+! forces 19...Kxh8, and 20.Nxf7+ forks the king and queen. After 20...Kg8 21.Nxd8 White has won the queen for a rook.

TEMPTING BUT WRONG

- **Nxf7?** Without the decoy Black recaptures with ...Kxf7 and you are just a knight down.
- **d5?** A central break, but it allows ...exd5 or ...Rxc1 and Black is fine.

SKILL: Combine a decoy sacrifice with a knight fork.

T2.10 24...Rd1+ 25.Rxd1 Rxd1# Exercise T2.10 (p. 304)

White's only back-rank guard is the rook on c1, and the king has no luft. 24...Rd1+ forces 25.Rxd1, and the second rook recaptures with mate: 25...Rxd1#. Doubled rooks on an open file are a back-rank weapon.

TEMPTING BUT WRONG

- **Rxb7?** Wins the queen, but you had mate in two.
- **Qe6?** Offers a queen trade and lets White off the hook.

SKILL: Execute a back-rank mate with doubled rooks.

T2.11 22.Qg8+ Rxg8 23.Nf7# Exercise T2.11 (p. 305)

22.Qg8+! cannot be taken by the king because the bishop on c4 guards g8, so 22...Rxg8 is forced. Black's own rook now blocks the king's only flight square, and 23.Nf7# is a smothered mate.

TEMPTING BUT WRONG

- **Qxa5?** Wins the queen, but Black answers ...Rxe1+ and you have missed mate.
- **Nf7+?** After ...Kg8 the mate takes longer; the queen sacrifice mates at once.

SKILL: Recognize the smothered-mate finish with a queen sacrifice.

T2.12 12...Qxc3+ 13.bxc3 Ba3# Exercise T2.12 (p. 305)

Your bishop on f5 already covers b1 and c2, and White's own pieces block d1 and d2. 12...Qxc3+! forces 12...bxc3, which opens b2, and 13...Ba3# is Boden-style mate from two bishops. The queen sacrifice clears the path for the second bishop.

TEMPTING BUT WRONG

- **Nc5?** Brings another piece, but White defends with e4 or Qc4 and the chance passes.
- **Ba3?** Without the sacrifice White takes the bishop with bxa3 and nothing happens.

SKILL: Use a queen sacrifice to open a diagonal for a two-bishop mate.

T2.13 3...Qe7 Exercise T2.13 (p. 305)

White has two threats: 4.Qxf7# and 4.Qxe5+, which would also hit your bishop on c5 and the rook on h8. 3...Qe7 defends both f7 and e5 (3...Qf6 does the same). 3...g6? blocks the attack on f7 but drops e5: 4.Qxe5+ forks king and rook. Natural developing moves such as 3...Nf6?? or 3...Nc6?? ignore f7 and allow 4.Qxf7#.

Also correct: Qf6

TEMPTING BUT WRONG

- **Nf6?** It attacks the queen but ignores f7: 4.Qxf7#.
- **g6?** It saves f7 but drops e5: 4.Qxe5+ forks your king and the rook on h8.
- **Nc6?** It guards e5 but not f7: 4.Qxf7#.

SKILL: Find the one move that parries two threats at once.

T2.14 5.Nxe5 Exercise T2.14 (p. 305)

The pin on f3 is only a pin to the queen, not the king, so the knight may move. 5.Nxe5! is the Légal pattern: if 5...Bxd1?? 6.Bxf7+ Ke7 7.Nd5#. If 5...dxe5, 6.Qxg4 and White has won a pawn with a fine position.

TEMPTING BUT WRONG

- **h3?** Reasonable, but after ...Bxf3 Qxf3 Black is fine and the tactical chance is gone.
- **d3?** Solid but passive; it lets Black off the hook.

SKILL: Recognize when a relative pin can be broken with a tactic (Légal's pattern).

T2.15 5...c4 Exercise T2.15 (p. 306)

Before you attack a piece, count its escape squares. After 5...c4 the bishop on d3 cannot go back to e2 or c2 (its own pieces are there), e4 is covered by the d5-pawn, and on f5, g6 or h7 it would be captured. White gets only one pawn for the bishop, for example 6.Bxc4 dxc4.

TEMPTING BUT WRONG

- **h6?** Questions the other bishop, but it misses that the d3-bishop can be trapped.
- **Nc6?** A good developing move, but White can play c3 and the bishop gets c2 back.

SKILL: Trap a piece with a pawn move after counting its escape squares.

Chapter 14 · How to Think: Candidate Moves and the Move-Selection Routine

14.1 5...Na5 Exercise 14.1 (p. 322)

Before recapturing, ask what White's forcing replies would be. After 5...Nxd5? White has 6.Nxf7! Kxf7 7.Qf3+, the Fried Liver Attack: your king is dragged into the open and the pinned knight on d5 is attacked again. Engines say Black can survive this with perfect defence, but over the board the attack is very hard to meet, and you would be playing White's game. 5...Na5! instead hits the bishop on c4, and after 6.Bb5+ c6 7.dxc6 bxc6 Black gets quick development for the pawn. 5...b5 and 5...Nd4 are sharp alternatives that also avoid the trap.

Also correct: Nd4, b5

TEMPTING BUT WRONG

- **Bg4?** It attacks the queen, but 6.Nxf7 forks your queen and rook first.

SKILL: Check the opponent's forcing replies before making a 'natural' recapture.

14.2 25...g6 Exercise 14.2 (p. 322)

Step one of the routine is 'what does my opponent want?' Your back rank has no defender: the queen on b6 and the rook on c2 are far away, and your king has no flight square, so White threatens Re8#. That is why 25...Qxb2?? and 25...Rxb2?? both lose at once to 26.Re8#. 25...g6 (or 25...h6) gives the king an escape square; after that White has no threat, and the pressure on b2 and f2 is still yours.

Also correct: h6

TEMPTING BUT WRONG

- **Qxb2?** 26.Re8# — the free pawn does not count if you are mated.
- **Rxb2?** 26.Re8# — nothing can block or capture on the back rank.

SKILL: Start every move by asking what the opponent threatens, even when you have tempting captures.

14.3 8.O-O Exercise 14.3 (p. 323)

f2 is attacked twice (Bc5 and Ng4) and defended only by the king. Castling adds the f1-rook as a defender and takes the king off the e1-a5 diagonal and the e-file, so ...Bxf2+ is no longer a threat; White can follow with h3 to kick the knight. 8.Qe2? also guards f2, but 8...Bxf2+ 9.Kf1 still wins a pawn and leaves the king stuck in the middle.

TEMPTING BUT WRONG

- **Qe2?** 8...Bxf2+ 9.Kf1 — the pawn falls anyway and your king can no longer castle.
- **d4?** Opening the centre while f2 is under fire only helps Black's pieces; after 8...exd4 the weaknesses multiply.

SKILL: Count attackers and defenders of a target square before choosing a defence.

14.4 9.b3 Exercise 14.4 (p. 323)

The 'free' knight is a trap. The bishop on a6 is aimed at your queen on e2, and only the c4-pawn stands between them, so the pawn is pinned. If it captures, 9.cxd5? Bxe2 10.Bxe2 Qxe5, Black has won the queen for a knight. The main line is 9.b3, defending c4 so that Black's knight has to move and the pin causes no harm; 9.Nd2 and 9.Qe4 are also fine. Your first idea is only a candidate until you have checked the opponent's best reply.

Also correct: Nd2, Qe4, g3

TEMPTING BUT WRONG

- **cxd5?** The c4-pawn was pinned to your queen: 9...Bxe2 wins the queen for a knight.

SKILL: Compare candidates by looking at the opponent's best reply, not your hoped-for one.

14.5 20...h6 Exercise 14.5 (p. 324)

White's doubled rooks threaten Rd8+! Rxd8 Rxd8+ Ne8 Rxe8#. That threat exists already, so ...Qxa2 is not a candidate at all. Give the king an escape square with 20...h6 (or ...g6/...h5); then Black's extra piece will tell.

Also correct: Kh8, Re8, g6, h5

TEMPTING BUT WRONG

- **Qxa2?** 21.Rd8+! Rxd8 22.Rxd8+ Ne8 23.Rxe8# — your queen is on a2 and cannot help.

SKILL: Spot an opponent's threat that already exists before grabbing material.

14.6 15.Qxh7# Exercise 14.6 (p. 324)

The fork is good; checkmate is better. The queen on h5 and the bishop on d3 both hit h7, the f6-square is empty so no knight defends, and the black king's escape squares f8 and h8 are blocked or covered. Always finish the candidate list — checks first — before you play the first strong move you see.

TEMPTING BUT WRONG

- **Ne7+?** It wins the queen, but Qxh7# ends the game at once.

SKILL: Keep looking after finding a good move; checks that mate beat checks that win material.

14.7 4...Na6 Exercise 14.7 (p. 325)

Knight on b5 plus bishop on f4 both hit c7: White threatens Nxc7+ forking king and rook. 4...Na6 guards c7 with a developing move. 4...Bd6 (challenging the bishop) and 4...Bb4+ also work. The routine 4...Be7?, played on autopilot, loses material after 5.Nxc7+.

Also correct: Bb4+, Bd6

TEMPTING BUT WRONG

- **Be7?** 5.Nxc7+ forks king and rook — a normal developing move that ignores the threat.
- **e5?** Hitting the bishop does not cover c7, and 5.dxe5 leaves Black a pawn down with problems.

SKILL: Name the opponent's threat before choosing a developing move.

14.8 13.Rxe4 Exercise 14.8 (p. 325)

Ask what the d5-pawn is doing. It guards the knight on e4, but it also blocks the d-file between your queen and Black's unprotected queen on d7. 13.Rxe4! exploits this double duty: if 13...dxe4, the d-file opens and 14.Qxd7 wins the queen; if Black does not take back, White has simply won a knight. The automatic 13.f3? allows 13...Bc5+ with active play for Black. This is the Tarrasch Trap: before playing the obvious move, check the captures.

TEMPTING BUT WRONG

- **f3?** 13...Bc5+ and Black's pieces come to life; the chance to win material is gone.
- **Be3?** Natural development, but 13...Nxe5 wins a pawn for Black.

SKILL: Treat the obvious move as one candidate among several: first check captures that overload a defender.

14.9 8...Ba6 Exercise 14.9 (p. 326)

A knight retreat is not the only candidate: 8...Ba6 pins the c4-pawn to the queen on e2, so the knight is safe for now and Black develops with tempo. 8...Nb6 is the solid retreat. 8...Nb4? and 8...Nf4? look active, but the knight lands on a square where it can be harassed and has no future.

Also correct: Nb6, Qb4+

TEMPTING BUT WRONG

- **Nb4?** After 9.a3 the knight must go back to a6, a passive square, and Black has lost time.
- **Nf4?** 9.Qe4 (or 9.Qe3) attacks the knight, which has no safe retreat, and Black loses material.

SKILL: Include counter-attacking and pinning moves in the candidate list, not just retreats.

14.10 6...d5 Exercise 14.10 (p. 326)

The natural candidates are knight moves (...Ne4, ...Ng4), but a counter-strike belongs on the list too. 6...d5! attacks the bishop on c4 and opens the c8-bishop; after 7.Bb5 Ne4 Black's knight sits on a strong central square. 6...Ne4? runs into 7.Bd5! with the knight in trouble; 6...Ng4 is playable but after 7.cxd4 White gets a comfortable centre, so ...d5 is the more precise choice.

TEMPTING BUT WRONG

- **Ne4?** 7.Bd5! attacks the knight a second time and Black's position falls apart.

SKILL: Add counter-attacking pawn breaks to the candidate list when a piece is attacked.

14.11 10.Re2 Exercise 14.11 (p. 326)

f2 is attacked by Ba7 and Ng4 and defended only by the king. 10.Re2 defends it calmly (10.Rf1, admitting the rook was misplaced, also works). The attacking-looking 10.h3? fails to 10...Nxf2, and 10.Qe2? to 10...Bxf2+ winning the exchange. Under pressure, compare the quiet defensive candidates; the flashy one often loses.

Also correct: Rf1

TEMPTING BUT WRONG

- **h3?** 10...Nxf2! — the knight takes on f2 instead of retreating and Black wins material.
- **Qe2?** 10...Bxf2+ 11.Kh1 Bxe1 wins the exchange.

SKILL: Prefer a quiet defence that keeps the count in your favour over a 'tempo' move that does not.

Chapter 15 · Calculation Fundamentals

15.1 5...d5 Exercise 15.1 (p. 344)

This is the 'fork trick'. 5...d5 attacks the bishop on c4 and the knight on e4 at the same time; White can save only one. After 6.Bd3 dxe4 7.Bxe4 material is level and Black has a free, central position. Every other move simply leaves Black a piece down.

TEMPTING BUT WRONG

- **Be7?** A quiet developing move while a piece down — White keeps the extra knight.
- **d6?** One square short: it attacks nothing, and White stays a piece up.

SKILL: Look for forcing moves (attacks on two pieces) first when material is missing.

15.2 6...Bb4+ 7.Nd2 Qxd4 Exercise 15.2 (p. 344)

Your bishop on d6 stands between your queen and the knight on d4. Move it with check: 6...Bb4+! The check must be answered first, and every answer puts a piece on d2, cutting the line between White's queen and the knight. After 7.Nd2 Qxd4 (or 7.Bd2 Nxd2 8.Nxd2 Qxd4) Black has won a knight. To find it you must look one move past the check and ask what the reply blocks.

TEMPTING BUT WRONG

- **Qf6?** It eyes f2, but after 7.Nf3 White simply keeps the extra pawn.
- **Qh4?** It threatens f2, but 7.g3 defends and nothing has been won.

SKILL: Use an in-between check to cut a defender's line before capturing.

15.3 16. **Qb8+ Nxb8** 17. **Rd8#** Exercise 15.3 (p. 345)

This is the finish of Morphy's famous Opera game (Paris, 1858). 16.Qb8+!! deflects the knight from d7; after 16...Nxb8 17.Rd8# the rook mates, guarded by the bishop on g5, because the black queen and bishop block their own king. Forcing moves first: the queen sacrifice is a check, so it must be examined even though it 'loses' the queen.

TEMPTING BUT WRONG

- **Qb7?** Strong, but not mate; Black can defend with ...f6 and play on.
- **Rxd7?** The rook trade gives up the mating pattern and allows Black to untangle.

SKILL: Examine every check, including sacrifices, before quieter moves.

15.4 30. **Nf7+ Kg8** 31. **Nh6+ Kh8** 32. **Qg8+ Rxg8** 33. **Nf7#** Exercise 15.4 (p. 345)

'Philidor's legacy': 30.Nf7+ Kg8 31.Nh6+ (double check from knight and queen, so the king must move) Kh8 (31...Kf8 32.Qf7#) 32.Qg8+! Rxg8 33.Nf7#, a smothered mate. Each move is a check, which keeps Black's choices to one or two — that is what makes a four-move line calculable.

TEMPTING BUT WRONG

- **Qg8+?** 30...Kxg8! — the king, not the rook, takes, and there is no follow-up.
- **Qf7?** Threatens mate but gives Black a tempo; the forced mate is gone.

SKILL: Calculate a sequence of checks to the end, picturing where every piece stands.

15.5 20. **Rxd8 Raxd8** 21. **Qh6** Exercise 15.5 (p. 345)

20.Qh6? at once fails to 20...Qxd1+, because the black queen both escapes and checks. So first remove it: 20.Rxd8 Raxd8 and only now the quiet 21.Qh6!, after which nothing stops Qg7# — Black's 21...Rd1+ 22.Bxd1 only delays it. A quiet move in the middle of a forcing line is easy to miss; look for it when the threat it creates cannot be parried.

TEMPTING BUT WRONG

- **Qh6?** 20...Qxd1+ — the queen you wanted to deflect takes your rook with check.
- **Qf4?** Too slow; Black trades on d1 and the mating idea disappears.

SKILL: Find a quiet move inside a forcing sequence after removing the only defender.

15.6 15. **Nxd5** Exercise 15.6 (p. 345)

d5 is attacked three times (Nc3, Bf3, Rd1) and defended twice (Nf6, Be6), so White wins it — provided the cheap pieces capture first. 15.Nxd5 Nxd5 16.Bxd5 Bxd5 17.Rxd5 leaves White a clean pawn up. Starting with the rook, 15.Rxd5?, gives up a rook for a knight because every recapture is by a cheaper piece.

Also correct: Bxd5

TEMPTING BUT WRONG

- **Rxd5?** 15...Nxd5 16.Nxd5 Bxd5 17.Bxd5 — you have traded a rook for a pawn and a knight.

SKILL: Count a capture sequence and order the captures from cheapest to most valuable.

15.7 45.Kf1 Exercise 15.7 (p. 346)

Count: the h3-pawn needs two moves to queen, the a4-pawn four. In a race Black queens first (45.a5? h2 46.a6 h1=Q). But the white king is inside the h-pawn's 'square': 45.Kf1 h2 46.Kg2 catches it, and then the a-pawn walks home because the black king is far outside its square.

Also correct: Kf2

TEMPTING BUT WRONG

- **a5?** 45...h2 and Black queens two moves before you do.
- **Ke2?** One step too slow: 45...h2 and the pawn cannot be caught.

SKILL: Count tempi in a pawn race before pushing.

15.8 18...Rxe1+ 19.Kh2 Rxd7 Exercise 15.8 (p. 346)

An intermediate move: 18...Rxe1+! takes a whole rook with check (the bishop on c1 blocks the a1-rook), and only after 19.Kh2 does Black recapture the queen with 19...Rxd7. Black ends up a rook ahead. The automatic 18...Rxd7 19.Rxe8+ Nxe8 just trades everything off.

TEMPTING BUT WRONG

- **Rxd7?** Recaptures, but 19.Rxe8+ follows and the position is level.
- **Nxd7?** Also recaptures, but leaves your e8-rook hanging to 19.Rxe8+.

SKILL: Look for an in-between check before making the obvious recapture.

15.9 52...Kf7 Exercise 15.9 (p. 346)

52...b1=Q? 53.e8=Q+ gives White a queen too, and with checks available the game is drawn. 52...Kf7! takes e8 away: if 53.e8=Q+ Kxe8, and otherwise Black collects the e-pawn; either way the b-pawn then queens and Black is a queen up. Calculation includes counting what the opponent does with his tempo.

TEMPTING BUT WRONG

- **b1=Q?** 53.e8=Q+ — White queens with check and it is queen against queen.

SKILL: In a race, include the opponent's best reply before queening.

15.10 50...a3 Exercise 15.10 (p. 346)

Count: the a-pawn needs three moves, the white c-pawn four, and the white king on d5 is outside the a-pawn's square. 50...a3! 51.c5 a2 52.c6 a1=Q wins easily. 50...Kxh4? spends the decisive tempo: 51.c5 a3 52.c6 a2 53.c7 a1=Q 54.c8=Q and both sides have a queen.

TEMPTING BUT WRONG

- **Kxh4?** Grabbing costs a tempo; White queens right after you and the game is drawn.

SKILL: Count tempi in a pawn race instead of grabbing material.

15.11 5...Qa5+ 6.Bd2 Qxe5 Exercise 15.11 (p. 347)

Forcing moves first, and queen checks are the most forcing of all. 5...Qa5+! checks the king along the a5-e1 diagonal (White's d-pawn has gone) and attacks the knight on e5 along the fifth rank, with the bishop on g5 behind it. White must answer the check, for example 6.Bd2 or 6.Nd2, and then 6...Qxe5 wins a whole piece. A developing move such as 5...Nc6 lets White off the hook.

TEMPTING BUT WRONG

- **Nc6?** Natural development, but you miss the queen check that wins a piece.
- **Qc7?** It attacks the knight, but there is no check: 6.Nf3 and White is fine.

SKILL: Scan queen checks for double attacks that win material.

Chapter 16 · Converting a Winning Position

16.1 35.Rxd8 Exercise 16.1 (p. 363)

Trade into the king and pawn ending: 35.Rxd8 Kxd8 36.Ke4 and the outside a-pawn decides. Black's king must run to stop it, and the white king then eats the kingside pawns. With rooks on, Black's rook can attack your pawns and the win takes far longer. Before trading, check that the pawn ending is won — here it clearly is.

TEMPTING BUT WRONG

- **Ra1?** Keeps rooks and lets Black's rook become active; still better for White, but far from the clean win.
- **Rd2?** A passive retreat that hands the d-file to Black's rook.

SKILL: Simplify into a won king and pawn ending when it is clearly winning.

16.2 40...Rxc1 Exercise 16.2 (p. 363)

40...Rxc1 41.Kxc1 h3 and the h-pawn runs: it needs three moves to queen, while the white king on c1 is far outside its square. Without rooks White has no way to stop it. Keeping the rooks with 40...Rh8 is still good for Black but gives White's rook time to get behind the pawn.

TEMPTING BUT WRONG

- **Rh8?** Supports the pawn, but 41.Ke3 and Rh1 slow it down; the clean win has gone.

SKILL: Use the rule of the square to decide whether a trade into a pawn ending wins.

16.3 60.Qf8# Exercise 16.3 (p. 364)

Count Black's legal moves before you move: the king has g8 and the c-pawn can advance; nothing else moves. 60.Qf8# covers g8 and g7 with check. The 'tidy' 60.Qxc4?? takes away Black's only pawn move and covers g8 without giving check — stalemate.

TEMPTING BUT WRONG

- **Qxc4?** Stalemate: the black king has no squares and the last free pawn is gone.

SKILL: Count the opponent's legal moves before every move in a won ending.

16.4 50...Qxd6 Exercise 16.4 (p. 364)

The white king has no moves: g1 is covered by your queen and g2 by the h3-pawn, while White's pawns are blocked. White's last hope is a 'desperado' rook. 50...Kxd6?? leaves White with no legal move — stalemate. 50...Qxd6! takes with the queen, which unblocks g1, and mate follows soon.

Also correct: Ke5, Ke7, Kf5

TEMPTING BUT WRONG

- **Kxd6?** Stalemate — the rook was White's only piece that could move.

SKILL: Spot desperado stalemate tricks and capture with the piece that frees the king.

16.5 40...b3 41.cxb3 a3 Exercise 16.5 (p. 364)

The classic breakthrough: 40...b3! 41.cxb3 a3! (or 41.axb3 c3! 42.bxc3 a3) and one black pawn runs through, because the white king is miles away. If White ignores 40...b3, Black takes on a2 or c2 and queens. Slow king moves give White time to bring his king across.

TEMPTING BUT WRONG

- **Kf6?** Too slow: the breakthrough only works while the white king is far away.
- **a3?** 41.bxa3 bxa3 and White's c-pawn and king hold everything.

SKILL: Create a passed pawn with a pawn breakthrough when the enemy king is far away.

16.6 40.Rb1 Exercise 16.6 (p. 364)

Four pawns against three on one side is usually a draw with rooks on the board, because your rook keeps checking and attacking. The pawn ending after 40.Rxd8? Kxd8 is a different story: Black's extra pawn becomes a passed pawn and his king walks in. When you are behind, avoid simplifications that help your opponent — move the rook off the file and keep it active.

Also correct: Ra1, Rc1, Re1, Rf1, Rh1

TEMPTING BUT WRONG

- **Rxd8?** 40...Kxd8 gives Black a winning pawn ending.

SKILL: When behind in material, avoid trades that simplify into a lost ending.

16.7 16.g3 Exercise 16.7 (p. 365)

First the threat, then the extras. 16.g3 blocks the bishop's diagonal and you can collect b7 later. The 'developing' 16.Nf3? runs into 16...Rxf3!, and Black's follow-up sacrifices on g3 force at least a perpetual check. 16.Qxb7?? ignores the threat: 16...Qxh2#. When you are winning, the move that kills counterplay is usually better than the move that grabs more.

Also correct: f4, h3

TEMPTING BUT WRONG

- **Nf3?** 16...Rxf3! and Black's sacrifices against your king force a draw by perpetual check.
- **Qxb7?** 16...Qxh2# — greed with a mate threat on the board.

SKILL: Consolidate against threats before taking more material.

16.8 20.gxf4 Exercise 16.8 (p. 365)

Remove the attacker: 20.gxf4 kills the mate threat for good. Black can check (20...Qg4+) but cannot do more, and your extra rooks decide. 20.Qd5?, covering g2, loses the queen to 20...Nxd5, and 20.Qxf7+? Kxf7 21.gxf4 Qg4+ only gives a perpetual check. Against a desperate attack, take the attacking piece if you safely can.

Also correct: Re8+

TEMPTING BUT WRONG

- **Qd5?** Covers g2, but the knight on f4 simply takes the queen.
- **Qxf7+?** A check that gives back the queen: after 20...Kxf7 Black has a perpetual check.

SKILL: Meet a desperate counter-attack by removing the key attacking piece.

16.9 45...Qd4 Exercise 16.9 (p. 365)

45...Qd4 blocks the check with a centralised queen protected by the c5-pawn and offers a queen trade; after 46.Qxd4+ cxd4 the d-pawn runs and the pawn ending is easily won, so White must retreat and Black keeps the initiative. 45...Qe5?? blocks with an unprotected queen: 46.Qxe5+. A king move such as 45...Kg8 lets White's queen roam and pick up the a7-pawn.

TEMPTING BUT WRONG

- **Qe5?** The blocking queen is not protected: 46.Qxe5+ wins it.
- **Kg8?** Gives White a free hand: 46.Qa5 and the queen attacks your pawns.

SKILL: Offer queen trades when ahead, but only on protected squares.

16.10 38...Kf5 Exercise 16.10 (p. 365)

In pawn endings the king is a fighting piece: 38...Kf5 (or 38...Ke5) heads for e5, d4 and the white queenside, and White's king cannot cover both wings. Pawn moves such as 38...b5? or 38...a5? only fix the structure while wasting the tempo the king needs.

Also correct: Ke5

TEMPTING BUT WRONG

- **b5?** A pawn move that does not help; White's king reaches e3 and the advantage is gone.
- **a5?** The same problem: the king, not the pawns, must move first.

SKILL: Activate the king before pushing pawns in the endgame.

16.11 16...Nxd3 Exercise 16.11 (p. 366)

When ahead, trade pieces — especially attacking ones. 16...Nxd3 removes the bishop that supports Qxh7#, so the threat disappears and White's attack is over. 16...Qxb2?? walks into 17.Qxh7#. 16...h6? stops the mate but weakens the king: 17.Bxh6! gives White a dangerous attack.

TEMPTING BUT WRONG

- **Qxb2?** 17.Qxh7# — the threat comes before the pawn.
- **h6?** 17.Bxh6! — the pawn is pinned against nothing, and White's attack is back.

SKILL: Trade the opponent's attacking pieces when you are ahead.

Chapter 17 · Defense and Resourcefulness

17.1 3...Nf6 Exercise 17.1 (p. 383)

White threatens Qxf7#. Nf6 blocks the queen's line to f7 and develops a piece at the same time, so it is both defence and progress. Moves that ignore the threat lose at once, and that includes counter-attacks: a threat against the queen is useless when the opponent can give mate.

TEMPTING BUT WRONG

- **Nd4?** Attacks the queen, but 4.Qxf7# comes first. A counter-threat must be bigger than the threat.
- **d6?** Natural developing move that ignores the threat: 4.Qxf7#.

SKILL: Spot a mate threat on f7 and meet it with a defending move that also develops.

17.2 1.g3 Exercise 17.2 (p. 383)

Before grabbing, check your opponent's threats: Black threatens Rd1#, because White's king has no escape square behind its pawns. A quiet move such as g3 (or h3, f3, Kf1) removes the back-rank problem, and the a7-pawn will still be there next move. 1.Rxa7?? Rd1# is the classic back-rank disaster.

Also correct: Kf1, f3, f4, g4, h3, h4

TEMPTING BUT WRONG

- **Rxa7?** 1...Rd1# - the king is trapped on the back rank.
- **Rb8?** Offering a trade on a square that is not defended: 1...Rxb8 wins a whole rook.

SKILL: Check the opponent's threat before taking material, and make luft against back-rank mate.

17.3 1.axb4 Exercise 17.3 (p. 384)

Run the defensive menu in order and start with captures: can you take the attacker? Yes, the a3-pawn simply captures the bishop. After 1.axb4 Nxb4 White has won a bishop for a pawn. Moving the queen away is safe but gives Black's mistake back.

TEMPTING BUT WRONG

- **Qxb4?** Captures the attacker with the wrong piece: 1...Nxb4 and the queen is gone.
- **Nc3?** Blocks the attack, but leaves the bishop alive and lets Black play ...Ne4 with pressure; you missed a free piece.

SKILL: When a piece is attacked, check first whether you can capture the attacker safely.

17.4 3...d5 Exercise 17.4 (p. 384)

Instead of defending e5 passively, strike back in the centre. 3...d5! attacks the e4-pawn: after 4.fxe5 Nxe4 your knight sits on a strong central square and the f-pawn's advance has only loosened White's king. Guarding e5 with 3...Nc6? fails to 4.fxe5 Nxe5 5.d4, and White's pawns chase your knights back. Developing the bishop with 3...Bc5? ignores the threat: 4.fxe5 drives the knight away and wins the e5-pawn.

TEMPTING BUT WRONG

- **Nc6?** After 4.fxe5 Nxe5 5.d4 White's centre pawns drive your knights back.
- **Bc5?** It ignores the threat: 4.fxe5 hits the knight and Black loses the e5-pawn with a bad position.

SKILL: Answer an attack in the centre with a counter-strike when one is available.

17.5 1.Qg5+ Kh8 2.Qf6+ Exercise 17.5 (p. 384)

The black king has no g-pawn and its own rooks block e7 and e8. That makes perpetual check: 1.Qg5+ Kh8 2.Qf6+ Kg8 3.Qg5+, and if 1...Kf8 2.Qh6+ Kg8 3.Qg5+. Black cannot interpose and cannot escape, so it is a draw by repetition. Anything slower loses to Black's attack on f2 and the back rank.

TEMPTING BUT WRONG

- **h3?** Making luft is too slow when you are two rooks down: 1...Qxd2 and Black wins easily.
- **Qd8?** 1...Rxd8 - the queen is lost and there is no check to follow.

SKILL: Use perpetual check to save a lost position when the enemy king is exposed.

17.6 1...Rc1+ Exercise 17.6 (p. 384)

Look at the black king: it has no legal move (f7-queen covers g8 and g7) and the h7-pawn is blocked. If Black can get rid of the rook, it is stalemate. 1...Rc1+! forces 2.Kxc1, because a1 and c2 are also covered by the rook, and Black has no legal move: stalemate. Every normal defence loses to mate on g7 or f8.

TEMPTING BUT WRONG

- **Rg8?** 2.Qf6+ Rg7 3.Qxg7# - the rook only delays the mate.
- **Rc7?** 2.Qf8# - the back rank is unguarded.

SKILL: Spot a stalemate resource when your king has no legal moves.

17.7 1...Rb8 Exercise 17.7 (p. 385)

Rook endings with all pawns on one side are often drawn even a pawn down, because the defending rook stays active. The pawn ending is different: after 1...Rxd1? 2.Kxd1 the white king and the extra pawn win by force. So keep the rooks: move your rook off the d-file to an open file (b8, c8 or a8) and stay active.

Also correct: Ra8, Rc8, Re8, Rf8, Rg8

TEMPTING BUT WRONG

- **Rxd1?** 2.Kxd1 gives White a winning pawn ending; trading pieces helps the side with more pawns.

SKILL: Avoid trading into a lost pawn ending when you are behind; keep the pieces that give you counterplay.

17.8 1.g3 Exercise 17.8 (p. 385)

g3 blocks the bishop's diagonal to h2 and attacks the queen at the same time. Black cannot take on g3: 1...Bxg3? 2.hxg3 (or fxg3) and Black has simply lost a bishop. After 1...Qg5 White continues 2.Ne4, hitting the queen and the bishop again, so the defence gains time. h3 and Nf3 also stop the mate, but they defend without gaining anything.

TEMPTING BUT WRONG

- **h3?** Stops the mate but gains nothing: 1...Re8 and Black keeps the initiative.
- **Qh5?** 1...Bxh2+! uncovers the black queen with check, and after 2.Kh1 Qxh5 Black has won the queen.

SKILL: Prefer a defence that gains a tempo over a purely passive one.

17.9 1...Ke6 Exercise 17.9 (p. 385)

White's bishop is light-squared, but the h-pawn queens on h8, a dark square. If the black king reaches h8 (or g8/g7), the bishop can never drive it out: a fortress. 1...Ke6! heads for f7 and g8 in time, for example 2.h5 Kf7 3.h6 Kg8. Any move that does not run to the corner at once is too slow, because the pawn and bishop cut the king off.

TEMPTING BUT WRONG

- **Ke5?** Looks active, but after 2.h5 Kf6 3.h6 Kf7 4.Bh7! the bishop keeps the king out of g8 and the pawn queens.
- **Kc5?** Goes the wrong way; the h-pawn simply runs.

SKILL: Know the wrong-coloured bishop fortress and run your king to the corner in time.

17.10 1...Qf2+ Exercise 17.10 (p. 385)

Black is down material and White's rooks are rampaging, so look for checks. 1...Qf2+ 2.Kh1 Qf1+ 3.Kh2 Qf2+ repeats: the white king cannot escape and no white piece can block on g1 or g2. Qc2+ with checks along the second and first ranks also draws. Making luft with ...h6 or ...h5 is natural but too slow.

Also correct: Qc2+

TEMPTING BUT WRONG

- **h6?** Luft is not enough: 2.Re8+ Kh7 and White's rooks and bishop win.
- **Kf8?** 2.Rxf7+ Ke8 3.Rde7+ and White wins the queen.

SKILL: Look for perpetual check with your queen when you are materially lost.

17.11 1.Nxf6+ Exercise 17.11 (p. 386)

When two pieces are attacked, use one of them as a desperado. 1.Nxf6+ captures with check, so Black must recapture, and then the bishop is free to deal with d5: 1...Qxf6 2.Bxd5 and White has won a pawn. Saving one piece the slow way (Bb3 or Ng3) loses the other one.

TEMPTING BUT WRONG

- **Bxd5?** 1...Qxd5 and the e4-knight is still attacked; at best White trades it off and ends up a bishop for a pawn down.
- **Ng3?** Saves the knight but 1...dxc4 wins the bishop.

SKILL: Use a desperado capture with check when two of your pieces are attacked.

Chapter 18 · King and Pawn Endgames

18.1 1.Ke4 Exercise 18.1 (p. 406)

Draw the square of the h4-pawn: h4-h1-e1-e4. Only Ke4 steps inside it, and from there the king catches the pawn: 1.Ke4 h3 2.Kf3 h2 3.Kg2. Every other move leaves the king one step outside the square and the pawn promotes.

TEMPTING BUT WRONG

- **Kd4?** Looks central, but d4 is outside the square: 1...h3 2.Ke3 h2 3.Kf2 h1=Q.
- **Ke5?** The fifth rank is outside the square; the king is one tempo short after 1...h3.

SKILL: Use the rule of the square to decide instantly whether your king catches a passed pawn.

18.2 1...Kd6 Exercise 18.2 (p. 406)

Kd6 takes the opposition: the kings face each other on the d-file with one square between them and White to move. White's king cannot advance to d5, c5 or e5 without giving up the key squares, so Black keeps blocking. Any other move lets the white king step forward, for example 1...Kf6? 2.Kd5 and White reaches the key squares in front of the pawn.

TEMPTING BUT WRONG

- **Kf6?** Gives up the opposition: 2.Kd5 and the white king walks to d6 or e6.
- **Ke7?** Retreating hands White the key squares after 2.Ke5 or 2.Kd5.

SKILL: Take the direct opposition when defending king and pawn against king.

18.3 1.Ke5 Exercise 18.3 (p. 407)

Ke5 takes the opposition in front of the pawn. Black must now give way, and the white king reaches a key square (d6, e6 or f6): 1.Ke5 Kd7 2.Kf6 and the pawn marches. Pushing the pawn first or wandering to the side lets Black take the opposition and draw.

TEMPTING BUT WRONG

- **e5?** Pushing too early: 1...Ke6 2.Ke4 Ke7 and Black blocks the pawn with the opposition; it is a draw.
- **Kg5?** 1...Ke6 takes the opposition and White makes no progress.

SKILL: Win king and pawn against king by leading with the king and taking the opposition.

18.4 1...Kc7 Exercise 18.4 (p. 407)

Against an a-pawn (or h-pawn) the defender draws by reaching the corner in front of it: once the black king sits on a8/b8, White can only stalemate or let the pawn be blocked. Kc7 heads for b8 and a8 and gets there in time. Moves away from the corner allow the white king to b7 or b6 and cut Black off.

Also correct: Kd7

TEMPTING BUT WRONG

- **Ke7?** Walks away from the corner: 2.Kc6 and the white king shields the pawn all the way to a8.
- **Kd5?** Chasing behind the pawn is too slow: 2.a5 and White's king guards the a-pawn's path.

SKILL: Know that king against king and rook pawn is drawn once the defender reaches the corner.

18.5 1...Kd7 Exercise 18.5 (p. 407)

The black king must stay directly in front of the pawn. After Kd7 the white king cannot get to c7 or e7, and if White plays Kd5 Black answers Kd8 and keeps the blockade until stalemate or the pawn falls. Stepping to the side lets the white king take the square in front of the pawn: 1...Ke8? 2.Ke6 and 3.d7, or 1...Kc8? 2.Ke6 and 3.Ke7.

TEMPTING BUT WRONG

- **Ke8?** 2.Ke6 and the pawn reaches d7 with White's king guarding the promotion square.
- **Kc8?** 2.Ke6 followed by Ke7 controls d8; the pawn queens.

SKILL: Defend king and pawn against king by keeping your king in front of the pawn.

18.6 1...g3 2.hxg3 f3 3.gxf3 h3 Exercise 18.6 (p. 407)

This is the classic three-against-three breakthrough. 1...g3! sacrifices the middle pawn to open lines: after 2.hxg3 f3! 3.gxf3 h3 the h-pawn runs to h1, and after 2.fxg3 h3! 3.gxh3 f3 the f-pawn does the same. The white king on c5 is far outside the square of either runner. Calm king moves let White's king come back and the chance is gone.

TEMPTING BUT WRONG

- **Kc2?** Natural but slow: 2.Kd4 brings the white king to the kingside and the breakthrough no longer works.
- **h3?** The wrong pawn first: 2.gxh3 and the black pawns can no longer break through; White's extra pawn decides.

SKILL: Recognize and execute the three-against-three pawn breakthrough.

18.7 1...h4 Exercise 18.7 (p. 408)

Count: the h-pawn needs four moves to queen, the a-pawn needs four too, but Black moves first. After 1...h4 2.a5 h3 3.a6 h2 4.a7 h1=Q+ Black queens with check, and the new queen on h1 also covers a8 along the long diagonal. The black king cannot catch the a-pawn (h8 is outside its square), so chasing it loses.

TEMPTING BUT WRONG

- **Kg7?** The king is outside the a-pawn's square: 2.a5 and White queens first.
- **Kg8?** Same problem: the king is too slow to catch the a-pawn, and Black has wasted the race.

SKILL: Count tempi in a pawn race, including checks and control of the other queening square.

18.8 1.Kd3 Exercise 18.8 (p. 408)

Kd3 takes the opposition in front of the pawn. Because the pawn is still far back, White has spare pawn moves to win any opposition battle later. After 1.Kd3 Kc5 2.Ke4 or 1...Ke5 2.Kc4 the white king advances and reaches the key squares. Pushing the pawn first lets the black king blockade it.

TEMPTING BUT WRONG

- **d4?** Pushing first throws away the spare tempo: 1...Kd6! and Black keeps the opposition in front of the pawn; it is a draw.
- **Ke3?** 1...Ke5 takes the opposition; with the king beside the pawn instead of in front, White cannot make progress.

SKILL: Lead with the king and use the opposition when the pawn is still behind.

18.9 1.Kb5 Exercise 18.9 (p. 408)

White wants the king on d6 or f6, but the direct approach meets the opposition. Kb5! outflanks: if 1...Kc7 2.Kc5 Black must give way, and if 1...Ke7 2.Kc6 and the king walks to d6. The black king cannot cover both sides because the e5-pawn takes d6 and f6 away from it. Central moves like Kd4 or Kc5 let Black take the opposition and draw.

TEMPTING BUT WRONG

- **Kd4?** 1...Kc6! takes the opposition and White cannot break through.
- **Kc5?** 1...Kc7 takes the opposition; White has no spare pawn move and it is a draw.

SKILL: Use outflanking to win a blocked king and pawn ending.

18.10 1...Kf6 Exercise 18.10 (p. 408)

Kf6 takes the direct opposition (f4 against f6). Now White's king cannot go to g5 or e4-e5 without Black mirroring it; for example 2.Ke4 Kg5 or 2.Kg4 Ke5 and the d6-pawn stays defended. Kf8 (distant opposition, three squares apart) also holds. Moving toward the queenside allows 2.Kg5 or 2.Kf5 and the king outflanks to e6.

Also correct: Kf8

TEMPTING BUT WRONG

- **Ke8?** 2.Kg5 Kf7 3.Kf5 and White's king reaches e6, winning the d6-pawn.
- **Kd8?** 2.Kf5 and the white king goes to e6 next; the d6-pawn falls.

SKILL: Use the opposition defensively to stop an outflanking manoeuvre.

18.11 1.Rxd8 Exercise 18.11 (p. 409)

Count the pawn ending before you go in: after 1.Rxd8 Kxd8 White has an outside passed a-pawn. The white king heads for d5, the a-pawn drags the black king to the queenside, and then White's king eats the kingside pawns. With rooks on, Black's rook stops the a-pawn from behind or in front and the game is close to equal. The trade turns a roughly equal rook ending into a winning pawn ending.

TEMPTING BUT WRONG

- **Ra1?** Keeps rooks: 1...Ra8 and the black rook blockades the a-pawn; White has only a small edge.
- **Re1+?** A check with no follow-up: 1...Kf8 2.Ra1 Ra8 and Black holds.

SKILL: Trade into a pawn ending only after calculating that the outside passed pawn wins it.

Stage 3 Test

T3.1 7.Ndb5 Exercise T3.1 (p. 410)

Look at every knight jump: 7.Ndb5! attacks the queen on c7 and the bishop on d6 at the same time, and nothing can take the knight on b5. If the queen moves, for example 7...Qb8, then 8.Nxd6+ wins the bishop with check. Black's best try, 7...Bxh2, gives up the queen after 8.Nxc7+ Bxc7 for a knight and a pawn. 7.Ncb5 makes the same fork and also wins material, but it is less precise: 7...Qa5+ escapes with check.

Also correct: Ncb5

TEMPTING BUT WRONG

- **Be3?** A normal developing move; Black can play ...a6 and the fork is gone.
- **O-O?** Safe, but it lets Black untangle with ...a6 or ...Be5; the fork was available now.

SKILL: Review: spot a knight fork of queen and bishop by checking every knight jump.

T3.2 1...Kf4 Exercise T3.2 (p. 410)

Draw the square of the b4-pawn: b4-b8-f8-f4. Only Kf4 steps into it, and from there the king catches the pawn: 1...Kf4 2.b5 Ke5 3.b6 Kd6 4.b7 Kc7. The white king on h8 is far too slow to help. Kf3, Kg4 and every other move leave the black king one tempo short.

TEMPTING BUT WRONG

- **Kf3?** The third rank is outside the square: 2.b5 and the pawn queens.
- **Kg4?** The g-file is outside the square; 2.b5 Kf5 3.b6 Ke6 4.b7 and Black is one move late.

SKILL: Review the rule of the square in a race against a passed pawn.

T3.3 1.Qxd5 Exercise T3.3 (p. 411)

When you are a piece up, trading queens removes your opponent's only real counterplay. 1.Qxd5 exd5 leaves a simple bishop-up ending that White wins by bringing the king to the centre. The queen trade also solves the back-rank problem: as long as queens are on, Black's queen is eyeing d1.

Also correct: Qe5+, Qf4+, f4

TEMPTING BUT WRONG

- **Qa4?** Moving the queen off the d-file allows 1...Qd1#; White's back rank is weak.

SKILL: Convert a material advantage by trading queens and removing counterplay.

T3.4 1...Nf6 Exercise T3.4 (p. 411)

White threatens Qxh7#. Nf6 defends h7 and attacks the queen at the same time, so White loses a tempo. ...g6 (blocking and hitting the queen) is equally good. What matters is that you name the threat before playing your own plan: the natural ...cxd4 or ...c4 is met by Qxh7#.

Also correct: f5, g6, h6

TEMPTING BUT WRONG

- **cxd4?** Continues the plan and ignores the threat: 2.Qxh7#.
- **c4?** Attacks the bishop, but 2.Qxh7# comes first.

SKILL: Spot a mate threat before continuing your own plan, and defend with tempo.

T3.5 1.Qxd8+ Qxd8 2.Rxd8# Exercise T3.5 (p. 411)

The black queen is overloaded: it must both protect the rook on d8 and keep the back rank closed. 1.Qxd8+! forces 1...Qxd8, and 2.Rxd8# mates because the king on g8 is boxed in by its own pawns. Any quiet move lets Black trade rooks on d1 and the chance is gone.

TEMPTING BUT WRONG

- **Qa4?** 1...Rxd1+ 2.Qxd1 and the back-rank pressure has disappeared.
- **Qg4?** Threatens nothing serious; 1...g6 or ...Rxd1+ and Black is fine.

SKILL: Calculate a forcing two-move sequence against an overloaded defender.

T3.6 1...Ke6 Exercise T3.6 (p. 411)

Lead with the king and take the opposition: after 1...Ke6 the kings face each other with White to move, and the black king will reach a key square in front of the pawn (1...Ke6 2.Kf4 Kd5 or 2.Kd4 Kf5). The pawn on e7 still has spare tempo moves for later opposition battles. Pushing the pawn first, or putting the king beside it, lets White blockade.

TEMPTING BUT WRONG

- **Kd6?** 2.Kd4 takes the opposition; with the king beside the pawn Black cannot make progress.
- **Kc6?** Walks away from the pawn: 2.Ke5 and the white king blockades in front of it.

SKILL: Review: win king and pawn against king with the opposition and spare pawn tempi.

T3.7 8.Nd5 Exercise T3.7 (p. 412)

8.Nd5! attacks the pinned knight a second time and also hits the bishop on b4. Calculate Black's replies. If the pinned knight takes, 8...Nxd5 9.Bxd8 Nxd8 10.exd5, White has won the queen for a bishop. If the bishop on b4 moves away, for example 8...Bc5, then 9.Bxf6 Qxf6 10.Nxf6+ wins a piece with check. A pin is worth most when you can attack the pinned piece again before the opponent can break it.

TEMPTING BUT WRONG

- **h3?** Useful in general, but it gives Black time for ...Bxc3 and ...Qe7, and you win much less.
- **Bxc6?** It damages Black's pawns but does nothing to the pinned knight; Black regroups.

SKILL: Calculate the opponent's best replies to find out whether a pin really holds.

T3.8 9...Nd4 10.hxg4 Nxe2+ Exercise T3.8 (p. 412)

Before retreating, look at forcing moves. 9...Nd4! attacks the queen on e2 and sets up ...Qh2#, because your queen on c7 and the knight on g4 both aim at h2, and only the knight on f3 defends it. If 10.Nxd4?? then 10...Qh2# is mate. White's best is 10.hxg4 Nxe2+ 11.Bxe2, and Black has won the queen for a knight. This is the Siberian Trap.

TEMPTING BUT WRONG

- **Nf6?** A safe retreat, but you miss the chance to win the queen.
- **Nge5?** It trades a pair of knights and you stay a pawn down; Nd4 wins the queen.

SKILL: Calculate forcing moves before retreating an attacked piece: a counter-threat can be bigger than the threat.

T3.9 1.Kf5 Exercise T3.9 (p. 412)

White's king wants d6 or b6, but the direct approach meets the opposition. Kf5! outflanks: after 1...Ke7 2.Ke5 Black must give way, and after 1...Kd7 2.Kf6 the king heads for e7-d6. The c5-pawn controls d6 and b6, so the black king can only defend from d7 or c7 and runs out of squares.

TEMPTING BUT WRONG

- **Ke5?** 1...Ke7 takes the opposition and White cannot get past.
- **Kd4?** 1...Kd7 and the black king holds the c6-pawn; draw.

SKILL: Review outflanking in a blocked king and pawn ending.

T3.10 1...Rxe1 Exercise T3.10 (p. 412)

Calculate the pawn ending: after 1...Rxe1 2.Kxe1 Black has an outside passed h-pawn. It drags the white king to the kingside while the black king marches to the queenside pawns. With rooks on, White's rook stops the h-pawn and the game is roughly equal.

TEMPTING BUT WRONG

- **Rh8?** Keeps rooks; 2.Rh1 and the white rook holds the h-pawn back.
- **Rd8+?** A check that leads nowhere: 2.Kc1 and White's rook still guards the kingside.

SKILL: Simplify into a pawn ending only after calculating that the outside passed pawn decides.

T3.11 1.Rf8+ Exercise T3.11 (p. 413)

Look at your own king first: it has no legal move, and the a2-pawn is blocked. If your rook disappears, it is stalemate. 1.Rf8+! Kxf8 is forced (h8 and f7 are covered by the rook) and White has no legal move. Any normal move loses to ...Qb2#.

TEMPTING BUT WRONG

- **Rb1?** Covers b2 for a moment, but 1...Qc3+ 2.Rb2 Qxb2# mates anyway.
- **Rc1?** 1...Qxc1+ and Black wins easily.

SKILL: Review the stalemate resource: sacrifice your last piece when your king has no moves.

T3.12 1...Kd5 Exercise T3.12 (p. 413)

The a-pawn queens on a8, a light square, and White's bishop is dark-squared, so it can never drive a king out of a8. Black must get to the corner at once: 1...Kd5 2.Kd3 Kc6 3.Kd4 Kb7 and the king reaches a8. Moves to the kingside are one step too slow.

TEMPTING BUT WRONG

- **Kf4?** Runs the wrong way: 2.a5 and the pawn cannot be caught.
- **Kf5?** 2.a5 Ke6 3.a6 and the pawn queens.

SKILL: Review the wrong-coloured bishop fortress and reach the corner in time.

T3.13 1.Rd8+ Rxd8 2.cxd8=Q# Exercise T3.13 (p. 413)

Black's rook has two jobs: stop the c-pawn and guard the back rank. 1.Rd8+! deflects it: after 1...Rxd8 2.cxd8=Q# the new queen mates on the back rank. Slower moves still win, but checks first would have found the immediate mate.

TEMPTING BUT WRONG

- **Rd7?** Strong, but not mate: Black defends with ...Rf8 and the game goes on.
- **Kf3?** Calm, but it lets Black organise ...Kf8 and ...Ke7.

SKILL: Use a deflecting check to promote with mate.

T3.14 1...Re8 Exercise T3.14 (p. 413)

White's queen and king stand on the same open file. 1...Rhe8! pins the queen to the king; it cannot leave the e-file, and 2.Qxe8+ Kxe8 gives Black a queen for a rook. Every other move leaves Black simply behind.

TEMPTING BUT WRONG

- **Qf6?** Offers a trade but misses the pin; after 2.O-O White stays a piece up.
- **Rf8?** Defends f7 passively; 2.Qf4+ and White is still a knight ahead.

SKILL: Review: spot a pin of the queen against the king.

T3.15 1.Ke5 Exercise T3.15 (p. 414)

The black king is on b7, defending c6 from the queenside. Ke5! heads for d6 by the side door: if 1...Kc7 2.Ke6 and 3.Kd6 wins the pawn, and if 1...Kc8 2.Kd6. The black king cannot cover both d7 and d6 because the c5-pawn takes d6 and b6 away from it. Kc4 or Kd3 waste the moment and Black regroups with ...Kc7.

TEMPTING BUT WRONG

- **Kc4?** 1...Kc7 and the black king is back in front of the pawn; draw.
- **Ke4?** Too slow: 1...Ka6! and the black king attacks c5 in time; it is a draw.

SKILL: Calculate the king route in a blocked pawn ending.

Chapter 19 · Piece Activity

19.1 26. **Ne7+ Kh7** 27. **Nxc8** Exercise 19.1 (p. 431)

An active knight on d5 attacks e7, c7, b6, f6 and more — and e7 is a forking square. Ne7+ checks the king and hits the rook on c8 at the same time; after the king moves, Nxc8 wins a whole rook for the knight. Taking on b6 instead runs into ...Rxc1+, because your rook was guarded only by your back rank.

TEMPTING BUT WRONG

- **Nxb6?** Black answers ...Rxc1+ and wins your rook; the knight capture left c1 undefended.
- **Rxc8+?** Trading rooks is safe but throws away the fork; you win nothing.

SKILL: Look for forks from an active, centralized knight.

19.2 30. **Ke2** Exercise 19.2 (p. 431)

In a knight ending the king is the most important piece. Ke2 heads for d3 and c4, keeping Black's king out of c5 and b4; the position stays level. A queenside pawn move such as a4 or b4 creates a target and loses the king race: Black's king goes d6-c5-b4.

Also correct: Ke1, f4, g3, h4

TEMPTING BUT WRONG

- **a4?** After ...Kd6 and ...Kc5, Black's king reaches b4 first and the a4 pawn becomes a target.
- **b4?** The pawn is simply lost to ...Nxb4.

SKILL: Centralize the king as soon as the queens are off.

19.3 24... **Rd2** Exercise 19.3 (p. 432)

The second rank is Black's 'seventh rank'. ...Rd2 attacks the bishop on e2 and the pawn on c2; after Bd3 Black takes on f2 and the rook keeps eating from behind. White's rook on b1 is a spectator.

TEMPTING BUT WRONG

- **Kf8?** Too slow: White plays Rd1 and trades off your only active piece.
- **Nd4?** Centralizes the knight but lets White's rook into the d-file; Black's advantage is gone.

SKILL: Invade the seventh (for Black, the second) rank with a rook on an open file.

19.4 12... **c5** Exercise 19.4 (p. 432)

The knight on the rim has no safe square: b3 and c4 are covered, and b7 and c6 are occupied by Black's own pieces. White threatens b2-b4, trapping it. ...c5 frees c6 for the knight and meets b4 with ...cxb4 cxb4 Nc6. Clearing b7 with ...Bc8 also works.

Also correct: Bc8

TEMPTING BUT WRONG

- **Re8?** A natural developing move, but b4 wins the knight.
- **Qc7?** Does nothing for the knight; b4 traps it.
- **Nd7?** Brings the other knight toward c5, but b4 comes first and the a5-knight is lost.

SKILL: Check a rim knight's retreats and make room before it is trapped.

19.5 25. Nd5 Exercise 19.5 (p. 432)

d5 is a classic outpost: Black has no c-pawn and the e-pawn has passed e6, so no pawn can ever chase the knight, and the e4 pawn supports it. From d5 the knight eyes b6, c7, e7 and f6, while Black's bishop is hemmed in by its own pawns on d6 and e5. b3, protecting c4 first, is also good. Nb5 hits a7, but after ...a6 the knight has to retreat to the rim; Kf1 simply drops c4.

Also correct: b3

TEMPTING BUT WRONG

- **Nb5?** Hits a7, but after ...a6 the knight must go back to a3 on the rim, and White's advantage is gone.
- **Kf1?** Centralizing the king is a good idea later, but now ...Rxc4 wins a pawn.

SKILL: Recognize an outpost and occupy it with a knight.

19.6 30... Ke7 Exercise 19.6 (p. 432)

White's king wants c4 and c5, where it would attack Black's queenside pawns. Black must answer by centralizing his own king at once: ...Ke7 and ...Kd6 keeps the white king out and the game stays balanced. A pawn move like ...a5 loses a tempo and creates a target: Kc4 and Kc5 (or Kb5) follow.

Also correct: Ke8, f6, g6

TEMPTING BUT WRONG

- **a5?** Loses the king race: Kc4, Kb5 and the a5 pawn and b7 pawn become targets.

SKILL: Answer the opponent's king activity with your own king.

19.7 22... Bxd5 Exercise 19.7 (p. 433)

Trade your opponent's best piece. ...Bxd5 removes the monster knight; after exd5 (or cxd5) White's pawns stand on light squares and White's remaining bishop on d3 becomes the bad bishop, while Black's position is solid. Black is slightly better. Attacking the knight with ...Rc5 instead allows Nxe7+, forking king and rook on the way out.

TEMPTING BUT WRONG

- **Rc5?** Nxe7+ wins a piece: the knight escapes with check and the bishop is gone.

SKILL: Exchange the opponent's strongest piece, especially an outpost knight.

19.8 22. Rd7 Exercise 19.8 (p. 433)

Doubled rooks on an open file are made for invading. Rd7 attacks the bishop on b7 and the pawns on a7 and f7; after ...Ba6 the second rook follows and a7 falls. Bc5, first driving the f8-rook away, is equally strong. Grabbing a7 with the bishop is less precise: it wins a pawn but lets Black play ...Ra8 with the bishop on the edge, and a quiet move like f3 lets Black untangle.

Also correct: Bc5

TEMPTING BUT WRONG

- **f3?** Luft is not needed yet; Black uses the time to untangle with ...Ra8 and ...Bc8.

SKILL: Invade the seventh rank with doubled rooks on an open file.

19.9 15...**Bxf3** 16.**Bxf3** **Rxd3** Exercise 19.9 (p. 433)

d4 is an outpost for Black: White has no c-pawn and the e-pawn has passed e3. But the knight on f3 guards d4, so ...Nd4 at once loses a piece: after Nxd4 Rxd4 the bishop on g4 is left hanging to Be2xg4. First remove the guard: ...Bxf3 Bxf3 Rxd3 wins the d3 pawn as well, because the bishop on e2 was its only defender. The square d4 still waits for the knight.

Also correct: Rfe8, a6

TEMPTING BUT WRONG

- **Nd4?** Nxd4 Rxd4 Bxg4 wins a piece: once the f3-knight is gone, your g4-bishop hangs.

SKILL: Remove the defender of an outpost square before occupying it.

19.10 19...**c5** Exercise 19.10 (p. 433)

Count: the knight is attacked three times and defended only by the queen. ...c5 adds a pawn defender, which is what makes a square a true outpost — now Bxd4 cxd4 gives Black a protected passed pawn on d4. Retreating with ...Ne6 lets White's knight jump to g4 and f6 with a strong attack, and ...Nf5 loses material after Qxd8.

TEMPTING BUT WRONG

- **Ne6?** Gives up the outpost; after Ng4 White's pieces flood the dark squares near your king.
- **Nf5?** Abandons the defence of your queen: Qxd8 starts a sequence that leaves White a piece up.
- **Rb7?** Ignores the knight's problem; White wins material on d4.

SKILL: Support an outpost piece with a pawn when it is attacked more often than defended.

19.11 26.**Rg3** Exercise 19.11 (p. 434)

A rook lift: the d3-rook swings to g3 and threatens Qg7#. Black has no good defence — ...Ng4 hxg4, ...Ng6 Rxg6 and ...Nh5 Qxh5 all give up material, and White's attack continues. Rf3 lifts the rook to the wrong file, and grabbing a pawn with Nxf5 lets the queen recapture with tempo.

TEMPTING BUT WRONG

- **Rf3?** The rook does not hit g7 from f3; Black regroups with ...Ng8 and is a clear exchange up.
- **Nxf5?** ...Qxf5 and your attack is over; you are simply down material.

SKILL: Use a rook lift to add an attacker on a closed position.

Chapter 20 · Pawn Structure Essentials

20.1 11...a6 Exercise 20.1 (p. 449)

Pawns cannot move backwards. ...g5 would break the pin, but it tears open the pawn shield in front of your own king: Bxg5! hxg5 Nxg5 gives White two pawns, a raging attack and Qd2 is ready to join on h6. A quiet move such as ...a6 (or ...Nb4, hitting the d3-bishop) keeps the kingside intact and the position is level.

Also correct: Nb4, Qb6, Rc8

TEMPTING BUT WRONG

- **g5?** Bxg5! hxg5 Nxg5 and White's queen comes to h6; the weakened king cannot be defended.

SKILL: Think before every pawn move: a pawn advance near your king can never be taken back.

20.2 28...Nd6 Exercise 20.2 (p. 449)

The best blockader of a passed pawn is a knight on the square directly in front of it. On d6 the knight cannot be attacked by a pawn, it stops the pawn dead, and it still controls c4, e4, b5 and f5. Black is a pawn down but holds comfortably. Any slower move lets White play d6, and the pawn gets much stronger with every step.

TEMPTING BUT WRONG

- **Kf8?** d6! and the pawn runs: ...Rd8 d7 and Black's pieces are tied up for good.
- **Nc7?** The knight looks at d5 but does not block it; after Rc1 and d6 White is close to winning.
- **Rd8?** A rook is a poor blockader: it is tied to d8 and White improves at leisure.

SKILL: Blockade a passed pawn with a knight on the square in front of it.

20.3 28.Bxe6 Exercise 20.3 (p. 450)

Passed pawns must be pushed, and the first step is to remove what stands in the way. Bxe6 fxe6 eliminates the blockader; now the d-pawn has a clear road to d7 and Black's rook is tied to the back rank. d7 at once, threatening Bxe6, is just as strong. Retreating the bishop lets ...Bxb3 or ...Bd5 trade it off with the blockade intact.

Also correct: d7

TEMPTING BUT WRONG

- **Bb3?** Offers a trade on Black's terms: after ... Bxb3 axb3 the d-pawn is stuck and the advantage is mostly gone.
- **Bd5?** ...Bxd5 Rxd5 trades the wrong pieces; Black's rook and king can now hold the pawn.

SKILL: Remove the blockader of a passed pawn, then push.

20.4 11...cxd4 Exercise 20.4 (p. 450)

Attack the base of the chain. After ...cxd4 the d4 pawn is attacked by the knights on c6 and f5 and the queen on b6, and defended only by the knight on f3 and the queen; after cxd4 Nfxd4 Black wins it. White's best is to leave d4 and play Bd3, but Black is still a pawn up after ...dxc3. Taking on d4 with a knight first, before the pawn exchange, loses material, and ...c4 releases the pressure.

TEMPTING BUT WRONG

- **c4?** Closes the position and stops attacking d4; White's chain is safe again.
- **Nfxd4?** Premature: cxd4 cxd4 and White's pawns hold; you have given a knight for two pawns.

SKILL: Attack a pawn chain at its base and count before capturing.

20.5 30. Rxd8 Kxd8 31. Kd3 Exercise 20.5 (p. 450)

In a pawn ending, doubled pawns are a long-term weakness: Black's three queenside pawns cannot create a passed pawn against White's three, while White's healthy majority can. After Rxd8 Kxd8 Kd3 the white king heads for c4 and c5, and Black's crippled pawns fall or let White through. With rooks on, the weakness matters much less.

TEMPTING BUT WRONG

- **Rd3?** Keeps rooks on; Black's rook stays active and the doubled pawns are only a small minus.
- **Ke3?** Leaves your rook undefended: ...Rxd1 wins it.

SKILL: Recognize when doubled pawns make a pawn ending winning.

20.6 13... Rfd8 Exercise 20.6 (p. 450)

...d5 is the dream move against a backward pawn, but check it first. After ...d5 exd5 Nxd5 Nxd5 Bxd5 the d-file opens and your knight on d7 is hanging to the queen on d2 with the rook behind it. Prepare the break instead: ...Rfd8 puts a rook on the d-file and defends d7, and only then does ...d5 become a real option.

Also correct: Ng4, Rfe8, h6

TEMPTING BUT WRONG

- **d5?** exd5 and Black cannot recapture safely: after the exchanges on d5 the knight on d7 falls.

SKILL: Prepare a freeing pawn break before playing it, and check the opened file.

20.7 35. c4 Exercise 20.7 (p. 451)

A pawn majority creates a passed pawn when the pawn without an opponent — the candidate, here the c-pawn — leads the advance. c4, followed by b4 and c5, keeps the majority flexible. b4? is the classic mistake: after ...Kd5 and ...b5 one black pawn holds two white ones, White's majority is frozen, and Black's kingside majority decides.

Also correct: Kd4, Ke4, Kf4, a4, h4

TEMPTING BUT WRONG

- **b4?** ...b5 fixes a2, b4 and c2 with just two black pawns; White's majority is useless and Black wins on the kingside.

SKILL: Advance a pawn majority with the candidate pawn first.

20.8 40... b3 Exercise 20.8 (p. 451)

A lever — a pawn move that offers an exchange — can break open a blocked structure. ...b3! forces matters: after cxb3 axb3 axb3 cxb3 Black gets a passed pawn, which is enough to hold the draw. Every slow move lets White's king walk to the queenside and eat the pawns, and ...c3 or ...a3 are the wrong levers: after the exchange White's remaining pawn stops the rest.

TEMPTING BUT WRONG

- **c3?** bxc3 bxc3 and the c3 pawn falls to the white king; Black's a-pawn is not enough.
- **a3?** bxa3 bxa3 and White's c-pawn and king hold everything.
- **Kg6?** Too slow: White's king takes the c4 pawn and the queenside collapses.

SKILL: Use the right lever to create a passed pawn in a blocked pawn structure.

20.9 15. **Bxf6** Exercise 20.9 (p. 451)

The isolated pawn's dynamic side: White's queen and bishop form a battery against h7, and the knight on f6 is its only defender. Bxf6 removes it; Black cannot recapture (... Bxf6 or ...gxf6 allows Qxh7#), so ...g6 is forced and Bxe7 wins a piece. The thematic d5 break is premature here: ...Nb4 hits the queen and the c2-bishop and Black equalizes.

TEMPTING BUT WRONG

- **d5?** ...Nb4! attacks the queen and the bishop on c2; the battery is broken and the position is level.
- **Ne5?** Natural centralization, but ...Nb4 again neutralizes the battery.

SKILL: Use the piece activity an isolated pawn gives you to attack the king.

20.10 15... **Nb4** Exercise 20.10 (p. 451)

The square in front of an isolated pawn is the natural home for a blockader, but you must get there with time to spare. ...Nb4 hits the queen first; after the queen moves, Black follows with ...Bc6 and ...Nbd5 or ...Nfd5, and the isolated pawn is frozen. The immediate ...Nd5 loosens the defence of h7, and White replies Ng5! with Bxd5 and Qg3 ideas.

TEMPTING BUT WRONG

- **Nd5?** Ng5! hits h7 and e6; after Bxd5 exd5 Qg3 White's pieces are much more active.

SKILL: Blockade the square in front of an isolated pawn, gaining tempo where possible.

20.11 26. **Nf5** Exercise 20.11 (p. 452)

Pushing a passed pawn is the rule, but only when it is safe. Here d6? loses the knight to ...Bxd4. Nf5 saves the knight and protects d5 tactically: ...Rxd5?? runs into Ne7+, forking king and rook. White keeps the extra pawn and a lasting edge.

TEMPTING BUT WRONG

- **d6?** ...Bxd4 wins the knight; the pawn push ignored the attacked piece.
- **Nc6?** ...bxc6 dxc6 and you have given a knight for a pawn.

SKILL: Defend a passed pawn tactically before pushing it.

Chapter 21 · King Safety and Attacking the King

21.1 20. **Rxf6** Exercise 21.1 (p. 468)

h7 is attacked by the queen and the bishop and defended only by the knight on f6 (and the king). Rxf6 removes that defender. If Black recaptures with ...gxf6, Qxh7 is mate; so Black must play ...g6 and is simply a piece down. Moving the rook to g3 or pushing g4 first gives Black time to bring more defenders. (The slower g4 also keeps a large advantage, but Rxf6 settles matters at once.)

TEMPTING BUT WRONG

- **Rg3?** Black answers ...Kh8 or ...Qh5 and the knight on f6 holds everything.

SKILL: Remove the last defender of a square next to the king.

21.2 12...**Bxg5** Exercise 21.2 (p. 468)

Defend by trading the attacker. The knight on g5 is the piece that makes Qxh7 mate possible, and your bishop on e7 can take it. After ...Bxg5 the threat is gone, and if Qxg5 then ...Qxg5 wins White's queen because nothing recaptures on g5. White is simply a piece down. Kicking the knight with ...h6 also defends but wins nothing, and ...g6?? allows Qxh7 mate anyway.

TEMPTING BUT WRONG

- **h6?** Defends, but after Ngf3 White is fine; you missed a free piece.
- **Nf6?** Covers h7 but leaves the attacking knight on g5; White keeps the initiative.
- **g6?** Qxh7 is still mate: g6 does not stop the queen and knight.

SKILL: Defend by exchanging the key attacking piece.

21.3 12. **g3** Exercise 21.3 (p. 469)

Of the three pawn moves that meet the threat, g3 is the only good one: it blocks the bishop's diagonal *and* hits the queen with tempo, so Black has to spend a move retreating. White ends up better. h3 also stops the mate but gains nothing and lets Black regroup with ...Nf6, and f4 is a disaster: the f2 pawn was blocking the queen's diagonal to e1, and ...Qxe1+ wins the rook.

TEMPTING BUT WRONG

- **h3?** Stops the mate but gains no time; Black keeps a comfortable attacking position.
- **f4?** ...Qxe1+: the f2 pawn was shielding the rook on e1 from the queen.
- **Bxh7+?** ...Qxh7 Qxh7+ Kxh7 and you have given a bishop for a pawn.

SKILL: Choose the defensive pawn move that weakens the king least and gains time.

21.4 9. **Bxh7+ Kxh7** 10. **Ng5+ Kg8** 11. **Qh5** Exercise 21.4 (p. 469)

All the conditions are there: bishop on d3, knight ready for g5, queen ready for h5, pawn on e5 keeping Black's knight off f6, and no black bishop on e7 to take on g5. After Bxh7+ Kxh7 Ng5+ Kg8 Qh5 White threatens Qh7 mate, and Black's only way out is to give back the queen with ...Qxg5. If 10...Kg6 instead, Qd3+ or Qg4 continues the attack.

TEMPTING BUT WRONG

- **Ng5?** Without the sacrifice, ...h6 kicks the knight and White's chance for a decisive attack is gone.
- **a3?** Gains the bishop pair but gives Black time for ...Be7 or ...f6; the chance is gone.

SKILL: Recognize and execute the Greek gift sacrifice Bxh7+.

21.5 13...g6 Exercise 21.5 (p. 469)

Only a move that blocks the b1-h7 diagonal stops the mate. ...g6 does it and keeps the pawn shield almost intact; the dark squares around the king are covered by the bishop on e7. ...f5 also blocks, but it leaves e6 and e5 weak for ever and White's knight jumps to e5. ...h6?? does not block anything: Qh7 is mate.

TEMPTING BUT WRONG

- **h6?** Qh7 is mate — moving the h-pawn does not cut the diagonal.
- **f5?** Blocks the diagonal but weakens e6 and e5; after Ne5 Black's position is very hard.
- **Bf5?** Qxf5: the bishop is simply lost.

SKILL: Pick the defensive pawn move that creates the fewest new weaknesses.

21.6 12...h6 Exercise 21.6 (p. 469)

Here the trade on g5 is not the answer: after ...Bxg5 Bxg5 White's bishop arrives with tempo on the queen. Instead ...h6 asks the knight where it is going; after Nf3 Black plays ...Nf6, covering h7, and the attack has run out of pieces. ...Nf6, hitting the queen, is equally good. ...g6?? allows Qxh7 mate.

Also correct: Nf6

TEMPTING BUT WRONG

- **Bxg5?** Bxg5 recaptures with tempo on the queen, and White's dark-squared bishop joins the attack.
- **g6?** Qxh7 is mate: the g-pawn move does not stop queen and knight.

SKILL: Defend by challenging the attacking pieces, choosing the version that does not help the opponent develop.

21.7 12.Nxf7 Exercise 21.7 (p. 470)

With the king on e8, the pawn on e6 is pinned by the queen on e2, and f7 is guarded only by the king. Nxf7! Kxf7 runs into Qxe6+ Kf8 Qf7 mate, so Black must castle and leave the knight on f7, after which Qxe6 wins more material and keeps the attack. Bxe6, the same idea in a different order, is just as strong. Nxe6 is the wrong piece: after ...fxe6 Qxe6 the knight on d7 defends and White has only some pawns for the piece.

Also correct: Bxe6

TEMPTING BUT WRONG

- **Nxe6?** ...fxe6 Qxe6 and the f7 square is no longer a target; White's attack is far weaker.

SKILL: Punish a king left in the centre with a sacrifice on f7 or e6.

21.8 13...b4 Exercise 21.8 (p. 470)

With opposite-side castling, the side whose pawn storm arrives first usually wins. ...b4 hits the knight on c3, the defender of e4, and opens lines toward White's king; after Nb3 Qc7 Black's attack is well ahead of White's. Developing moves like ...Rfc8 or ...Nc5 are too slow — White plays Nd5 and starts his own play.

TEMPTING BUT WRONG

- **Rfc8?** Too slow: Nd5! hits the queen and the e7-bishop and White takes over.
- **Nc5?** Nd5 again: Black has to trade pieces and his attack is gone.

SKILL: In opposite-side castling positions, launch the pawn storm without delay.

21.9 14. **g5** Exercise 21.9 (p. 470)

The pawn on h6 gives White a target to bite on: g5 attacks the knight on f6 and opens the g- and h-files. After ...hxg5 hxg5 the rook on h1 is already on an open file pointed at Black's king, and the knight must leave f6. Safety moves such as Kb1 lose a tempo in a race, a3 weakens White's own king, and Bxh6 is a premature sacrifice.

TEMPTING BUT WRONG

- **Kb1?** A useful move in quiet positions, but in a pawn race Black plays ...b4 and gets there first.
- **a3?** Gives Black's pawns a hook: ...b4 opens lines against your own king.
- **Bxh6?** ...gxh6 and the attack is not ready; White has just given away a piece.

SKILL: Use the opponent's advanced pawn as a hook for your pawn storm.

21.10 11... **Bxh2+** 12. **Kxh2 Ng4+** 13. **Kg3 Qg5** Exercise 21.10 (p. 470)

The Greek gift works for Black too. The pawn on e4 keeps White's knight off f3, and no white piece can take a knight on g4. After ...Bxh2+ Kxh2 Ng4+ the king cannot go back to g1 (...Qh4 wins on h2 and f2), so Kg3 is the best try, and ...Qg5 keeps the attack going with ...Nxe3+ in the air. Slow moves such as ...Qc7 or ...Re8 give White time for h3 or f3, and the chance is gone.

Also correct: a5

TEMPTING BUT WRONG

- **Qc7?** Lines up on h2 but gives White time for h3 or f3 to defend.
- **Re8?** Too slow; White breaks with f3 and the e4 pawn is no longer a strong point.

SKILL: Recognize the Greek gift sacrifice from the black side.

21.11 10. **Nxd5** Exercise 21.11 (p. 471)

The king in the centre makes Black's pawn on e6 pinned: it cannot recapture on d5 because the queen on e2 is staring at e8. So Nxd5 wins a pawn and attacks the queen on b6 and the bishop on b4 at the same time. After ...Qd8 Nxb4 Nxb4 Bb5+ White keeps a big lead in development against the uncastled king. Grabbing on h7 instead loses a bishop and the attack.

TEMPTING BUT WRONG

- **Bxh7?** ...Bxc3 bxc3 Rxh7 and you have lost a piece.
- **Qe3?** A quiet move that lets Black castle; the pin on e6 disappears and with it your chance.

SKILL: Exploit pins and open lines against a king that has not castled.

Chapter 22 · Making a Plan

22.1 10. Nbd2 Exercise 22.1 (p. 489)

Run the worst-piece test: the knight on b1 has never moved and blocks the rook on a1 from joining the game. Nbd2 starts the classic Italian manoeuvre Nd2-f1-g3 (or e3), where the knight eyes f5 and d5 and helps defend the king. Flashy alternatives fail: the sacrifice Bxh6 gives a piece for two pawns and no attack, and Ng5 runs into ... hxg5.

Also correct: Be3, a4, d4

TEMPTING BUT WRONG

- **Bxh6?** After ...gxh6 White has only two pawns for the piece and nothing to follow up with.
- **Ng5?** The knight jump attacks nothing that matters and ...hxg5 simply wins a piece.

SKILL: Identify the worst piece and give it a better square.

22.2 12. b4 Exercise 22.2 (p. 489)

This is the Carlsbad structure: White's pawns on d4 and e3 face Black's on d5 and c6. The standard plan is the minority attack: push the b-pawn to b5, exchange it for the c6-pawn, and leave Black with a weak pawn on c6 or d5 to target. The rook already waits on b1, so b4 (or a3 first) is exactly in tune with the structure. Opening the centre with e4 loses material here, and g4 weakens White's own king.

Also correct: Bh4, Rfc1, a3, h3

TEMPTING BUT WRONG

- **e4?** After ...dxe4 Bxe4 Nxe4 Nxe4 Bxg5 the exchanges leave White's pieces loose, and Black wins material.
- **g4?** A kingside pawn storm with no support; ... Nxg4 simply wins a pawn and exposes White's king.

SKILL: Choose the plan that the pawn structure suggests (minority attack).

22.3 12... d5 Exercise 22.3 (p. 490)

In this quiet Italian structure, the pawn break ...d5 is Black's natural plan: it challenges White's centre, frees Black's pieces and uses the fact that White's knight on f1 has not yet reached g3. After exd5 Bxd5 Black's pieces are active and the a7-bishop bears down on f2. The bishop sacrifice on h3 is unsound: White simply takes and Black has no follow-up.

Also correct: Rb8

TEMPTING BUT WRONG

- **Bxh3?** After gxh3 Black has one pawn for the bishop and no pieces ready to attack the king.
- **Nh5?** The knight goes to the rim hoping for ... Nf4, but d4 takes over the centre and Black is worse.

SKILL: Find the central pawn break that fits the structure.

22.4 12... e5 Exercise 22.4 (p. 490)

Black's worst piece is the bishop on c8, locked in by the pawn on e6. The freeing break ...e5 opens the c8-h3 diagonal and challenges White's centre at the same time; the other freeing break, ...c5, is also reasonable. Random pawn moves on the kingside, such as ...f5 or ...g5, create permanent weaknesses (the e5-square and e6-pawn, or the king) and leave the bishop just as bad.

Also correct: b5, b6, c5, h6

TEMPTING BUT WRONG

- **f5?** It weakens e5 and e6 for good and still leaves the c8-bishop locked in.
- **g5?** An attack with no pieces behind it; it loosens Black's own king and achieves nothing.

SKILL: Improve the worst piece with a freeing pawn break.

22.5 7. Qb3 Exercise 22.5 (p. 490)

Two things changed when the bishop left c8: your queen on d1 is now attacked, and the b7-pawn has lost its defender. Qb3 solves the first problem and creates a problem for Black, attacking b7. This is the standard idea in the Exchange Caro-Kann. Kicking the bishop with h3 is a typical autopilot move that forgets the queen is already attacked.

Also correct: Nf3, Qc2

TEMPTING BUT WRONG

- **h3?** Your queen is attacked: ...Bxd1 wins it.

SKILL: Spot the weakness created by the opponent's last move.

22.6 12. Nf3 Exercise 22.6 (p. 490)

Black's idea is ...cxd4: after exd4, the pinned c3-knight can be exchanged on c3 and White's queenside falls apart, and the e-file opens against White's uncastled king. Nf3 is prophylaxis and development in one move: the knight defends d4 so White can recapture with a piece, and it clears the way to castle and break the pin. The greedy dxc5 opens the d-file for Black's queen and runs into ...Nxc3 with a discovered attack.

TEMPTING BUT WRONG

- **dxc5?** After ...Nxc3 the queen on d3 is attacked by the queen on d8 and White loses material.
- **O-O-O?** Castling into the open c-file: ...Bxc3 and ...cxd4 rip the queenside open.
- **a3?** Kicking the bishop allows ...Qa5, increasing the pressure on the pinned knight.

SKILL: Use prophylaxis: find the opponent's plan and stop it while developing.

22.7 12...a6 Exercise 22.7 (p. 491)

Ask what White wants: b4-b5 to break up your c6-d5 chain. ...a6 is prophylaxis: if White plays b5, Black can recapture with the a-pawn and keep a healthier structure. ...Bd6, activating the bishop towards White's kingside, is an equally good plan. The pawn grab ...Bxb4 simply loses a piece to the rook on b1, and ...h6 allows Bxh6! with a strong attack.

Also correct: Bd6, Ne4, Qd6

TEMPTING BUT WRONG

- **Bxb4?** The b4-pawn is defended by the rook on b1: Rxb4 wins the bishop.
- **h6?** After Bxh6! gxh6 Bxg6 fxg6 Qxg6+ White's attack is too strong.

SKILL: Anticipate the opponent's plan and prepare against it.

22.8 11...Rb8 Exercise 22.8 (p. 491)

Black's pawn structure has given you a half-open b-file, and at the end of it stands White's b2-pawn. ...Rb8 puts the rook on that file at once, where it presses on b2 and ties White's pieces down. Developing with ...Be6, kicking the bishop with ...h6 or contesting the e-file with ...Re8 are also good. The tempting ...Ng4 is a blunder: the f6-knight was blocking the bishop on g5, and after Bxe7 Black loses a piece.

Also correct: Bd6, Be6, Re8, a5, h6

TEMPTING BUT WRONG

- **Ng4?** It uncovers the g5-bishop's attack on e7; Bxe7 wins a piece because the e-file is also controlled by White's rook.

SKILL: Put rooks on open and half-open files.

22.9 11. **h3** Exercise 22.9 (p. 491)

In a quiet, symmetrical position, small improvements decide. h3 asks the g4-bishop a question: it must take on f3 (giving White the bishop pair), or retreat, and the white king gets luft for later. Putting a rook on the half-open e-file with Rfe1 or Rae1 is an equally sensible plan. The Greek-gift sacrifice Bxh7+ fails here, because the knight on f6 simply recaptures on h7.

Also correct: Bh4, Rae1, Rfe1, a3

TEMPTING BUT WRONG

- **Bxh7+?** The f6-knight guards h7: after ...Nxh7 White has given a bishop for a pawn and has no attack.

SKILL: Find a quiet improving move when nothing forcing is available.

22.10 13. **Nxc6 bxc6**
14. **Bxc5** Exercise 22.10 (p. 491)

Tactics come before strategy. The knight on d4 stands between your e3-bishop and the black bishop on c5. Nxc6! moves it with a threat (the queen on d8) and uncovers the attack on c5 at the same time. After ...bxc6 Bxc5 White has won a whole piece. Retreating the knight, or protecting it, wastes the chance.

TEMPTING BUT WRONG

- **Nb3?** Retreating gives up the discovered attack; Black's bishop simply goes back or trades, and material stays equal.
- **Nxd5?** After ...Nxd5 Bxd5 Bxd4 Black is fine; the wrong piece was captured.

SKILL: Check for tactics before choosing a plan; spot a discovered attack.

22.11 28... **Qf7** Exercise 22.11 (p. 492)

White threatens Qxg7 mate: the queen on g3 and the bishop on c3 both hit g7. ...Qf7 defends g7 and also blocks the f-pawn, the only move that keeps your advantage. The natural ...g6 loses to fxg6, opening the f-file for White's rook, and grabbing the bishop with ...Qxc3 drops the queen after Qxc3.

TEMPTING BUT WRONG

- **g6?** After fxg6 the f-file opens and White's rook and queen crash through.
- **Qxc3?** Qxc3 simply wins your queen for a bishop.

SKILL: Ask what the opponent wants before choosing a plan.

Chapter 23 · Rook Endgames

23.1 1. Rh8 Exercise 23.1 (p. 510)

When your rook stands in front of an a-pawn on the seventh rank, watch for the skewer trick. Rh8! clears the a-file: if Black takes with ...Rxa7, then Rh7+ checks along the seventh rank and wins the rook on a7 next move. Against other replies the a-pawn simply promotes or costs Black the rook. Every other move lets the black king return to g7, after which the position is a well-known draw.

TEMPTING BUT WRONG

- **Kg2?** Bringing the king is too slow; Black plays ...Kg7 and the skewer chance is gone for good.
- **Rb8?** Black simply takes ...Rxa7 and the check on b7 would be met by ...Rxb7, because the black rook guards the seventh rank.

SKILL: Use the Rh8 skewer trick when the defending king leaves g7/h7.

23.2 1...Ra1 Exercise 23.2 (p. 510)

This is the second stage of the Philidor defence. Once the pawn reaches e6, the white king can no longer hide in front of it, so the rook drops to the far end of the board and checks from behind. After ...Ra1 the king has no shelter: every time it approaches the pawn, another check arrives. Staying on the sixth rank is now useless because the pawn blocks it.

Also correct: Ra2, Ra3, Ra4

TEMPTING BUT WRONG

- **Rb6?** The sixth rank no longer stops anything; Kf6 follows with mating threats on the back rank.
- **Kf8?** Passive king moves lose: Kf6 and Rh8+ drive the king away while the pawn advances.

SKILL: Switch from third-rank defence to checks from behind in the Philidor position.

23.3 1. Rd1 Exercise 23.3 (p. 511)

Cut the king off first, push the pawn second. Rd1! builds a wall on the d-file: the black king can never cross to the c-pawn's side of the board. White's king then walks forward with the pawn, and the black rook alone can only give checks until the king hides. Pushing c5 at once, or stepping forward with Kc5, lets ...Kd7 bring the black king in front of the pawn, and the position is a draw.

TEMPTING BUT WRONG

- **c5?** Pushing first lets the king come across with ...Kd7 and ...Kc7; the pawn is blockaded and the game is drawn.
- **Kc5?** Natural, but after ...Kd7 the black king reaches the pawn first.

SKILL: Cut off the defending king along a file before advancing the pawn.

23.4 1...Ke8 Exercise 23.4 (p. 511)

The defender's king belongs in front of the pawn. ...Ke8 heads for d8, the queening square, and from there the rook can keep checking or cut the white king off. If the king stays on f7, White plays Rh7+ or Ke6 and the pawn marches with the king's help, because the rook alone cannot block and defend at the same time.

TEMPTING BUT WRONG

- **Rd8?** Blocking with the rook leaves the king outside; Rh7+ drives it away and White wins the rook or queens.
- **Re8+?** This check drives the king forward to d5 and c6, and the black king is still cut off.

SKILL: Put the defending king in front of the passed pawn.

23.5 1.Kb4 Exercise 23.5 (p. 511)

In rook against pawn, the stronger side's king usually decides. Kb4! is the only move that brings the king close enough in time: after ...d2, Kb3 and Kc2 arrive just as the pawn reaches the second rank, and the rook can then win it with the king's help. Rook moves waste the one tempo White has, and after ...d2 and ...Kc3 the pawn costs White the rook.

TEMPTING BUT WRONG

- **Rd1?** The rook already controls the queening square from f1; this move wastes time and the king arrives too late.
- **Kb5?** Moving the king away from c3/d2 loses a tempo in the race, and Black draws.

SKILL: Count tempi in rook-versus-pawn endings and bring the king towards the pawn.

23.6 1...Rh6+ Exercise 23.6 (p. 511)

Your king is on the short side of the pawn (the c-file side), so your rook has room on the long side to check from a distance. ...Rh6+ starts checking along the sixth rank from three files away; the white king cannot approach the rook without leaving the pawn. Checking from behind with ...Re1+ also holds. Anything slow allows d7+ and the pawn promotes.

Also correct: Re1+

TEMPTING BUT WRONG

- **Rd1?** Getting behind the pawn is too slow; Ra8+ and Re8 support the pawn and White wins.
- **Rh3?** This waits instead of checking; d7+ follows and the pawn queens.

SKILL: Defend from the long side with checks from a distance.

23.7 1...Rc1+ 2.Kg2 Ra1 Exercise 23.7 (p. 512)

Rooks belong behind passed pawns, including your opponent's. ...Rc1+ followed by ...Ra1 puts the rook behind the a-pawn: the further the pawn advances, the more squares the rook gains, and White's rook must stay in front of its own pawn. Passive defence with the rook on a8 or f8 lets White push the pawn with tempo and bring the king.

TEMPTING BUT WRONG

- **Rf8?** Guarding f7 passively lets the a-pawn run: a5-a6 and the rook is tied down.

SKILL: Place the rook behind the opponent's passed pawn and stay active.

23.8 1.Re3 Exercise 23.8 (p. 512)

Re3! attacks the pawn from in front, and the black king cannot defend it: the rook also covers f3, the only square from which the king could protect e2. If ...e1=Q, the rook captures on e1; if ...Kf4 attacks the rook, Rxe2 removes the pawn anyway. Ordinary moves like Rb1 allow ...Kf3 and ...Kf2, and Black draws.

TEMPTING BUT WRONG

- **Rb1?** Black answers ...Kf3 and ...Kf2, the pawn is protected, and White must give the rook for it.
- **Kc6?** The king is far too slow; ...e1=Q gives Black a queen for nothing.

SKILL: Use the rook's long-range tricks against a far-advanced pawn.

23.9 1...Rd4+ Exercise 23.9 (p. 512)

Run your safety scan first: White threatens Ra8#, because the e6 pawn and the white king cover every flight square. ...Rd4+ checks the king and gains the time needed; after the king moves, the rook can return to the first rank or keep checking. Guarding the back rank with ...Rg8 fails because the black king on e8 blocks the rook's path to a8.

TEMPTING BUT WRONG

- **Rg6?** Pinning the pawn ignores the threat; Ra8 is mate.
- **Rg8?** It looks like back-rank defence, but the king on e8 blocks the rook, so Ra8 is still mate.

SKILL: Spot back-rank mate threats in rook endgames and answer with a check.

23.10 1.Rc4 Exercise 23.10 (p. 512)

The black king on d8 is close enough to reach the pawn via c7 and b7. Rc4! cuts it off along the c-file, and then White's king and pawn advance together while the black rook can only give checks. Pushing b6 at once lets the king come to c8 and b7, and the game is drawn.

TEMPTING BUT WRONG

- **b6?** The pawn advances but ...Kc8 and ...Kb7 put the king in front of it: a draw.
- **Kb4?** Natural, but ...Kc7 brings the black king to the pawn first.

SKILL: Cut off the defending king along a file before advancing.

23.11 1...Rb6+ Exercise 23.11 (p. 513)

...Rb6+! uses a check to change the position. If Kc7, Black gives the rook for White's rook with ...Rxf6 and, after d6, the black king and remaining pieces deal with the pawn: the line ...Rxf6 d6 Rf7+ d7 Rxd7+ Kxd7 leaves bare kings. If Rxb6, ...Kxb6 brings the king next to the pawn. Slow moves let d6 and d7 come with the white king protecting them.

TEMPTING BUT WRONG

- **Rb2?** Too slow: d6 and the pawn reaches the seventh rank with the king's support.
- **Kb8?** The king heads the wrong way; d6 and d7 win for White.

SKILL: Use checks and exchanges to reach a drawn ending.

Stage 4 Test

T4.1 12.**Rxe7** Exercise T4.1 (p. 514)

Before defending, ask what the opponent's move gave up. The queen on d8 was the only defender of the e7-bishop, which is attacked by both the g5-bishop and the e1-rook. Rxe7 wins a piece at once; if Black takes on b2, White is still a piece up and the rook on e7 is very active. Defending b2 with Rb1 or Qc1 lets Black cover e7 again.

Also correct: Bxf6

TEMPTING BUT WRONG

- **Rb1?** Passive defence: Black has time for ...Bd6 or ...Re8 and the chance is gone.
- **Na4?** Kicking the queen lets it return with ...Qd8, defending e7.

SKILL: Check what an opponent's move stops defending.

T4.2 16...**Ne2+** 17.**Kh1 Nxc3** Exercise T4.2 (p. 514)

Checks first. ...Ne2+ is a knight fork of the king on g1 and the queen on c3, and the e2-square is not covered by any white piece. After Kh1 Nxc3 Bxc3 Black has won the queen for a knight. Grabbing a pawn with ...Bxb3 is far weaker.

TEMPTING BUT WRONG

- **Bxb3?** It wins a pawn, but misses a queen-winning fork.
- **Re8?** Developing is fine in general, but a forcing win is available.

SKILL: Look for forcing knight forks involving the king.

T4.3 25.**Qxd8+ Qxd8** 26.**Rxd8#** Exercise T4.3 (p. 515)

Black's back rank is guarded only by the rook on d8, and that rook is defended only by the queen. Qxd8+! forces ...Qxd8, and Rxd8 is mate because the pawns on f7, g7 and h7 trap their own king. Notice that your own queen stands in front of your rook, so the queen must capture first.

TEMPTING BUT WRONG

- **Qe3?** It eyes e8, but leaves the d1-rook unprotected: ...Rxd1+ wins it with check.
- **Qe1?** Too slow; Black trades rooks on d1 and the back-rank chance disappears.

SKILL: Exploit a back-rank weakness by removing the last defender.

T4.4 8.**Bxh7+ Kxh7** 9.**Ng5+ Kg8** 10.**Qh5** Exercise T4.4 (p. 515)

All the conditions for the Greek gift are present: a bishop aimed at h7, a knight ready for g5, a queen that can reach h5, and the e5-pawn keeping Black's knights away from f6. After Bxh7+ Kxh7 Ng5+ Kg8 Qh5 White threatens Qh7 mate, and Black must give back material with ...Qxg5 to survive. Note that Black's dark-squared bishop is on b4, not e7, so nothing guards g5.

TEMPTING BUT WRONG

- **Nb5?** A one-move threat that Black meets calmly with ...a6; the attacking chance is gone.
- **a3?** Asking the bishop a question gives Black time to defend with ...f5 or ...Nf6.

SKILL: Recognize the conditions for the Greek-gift sacrifice.

T4.5 15...d5 Exercise T4.5 (p. 515)

Ask what White wants: Rhe1-e3-g3 (or Rag1) with mate on g7. ...d5 gains time by attacking the c4-bishop, shuts the bishop off from f7, and opens the c8-bishop and the e8-queen for defence (...Be6 or ...Qe6). White has nothing better than repeating checks. Slow moves such as ...h5 or ...Bb7 allow the rook lift and the attack crashes through.

TEMPTING BUT WRONG

- **h5?** Creating luft weakens g6: Qg6+ and Qxh5 follow with a winning attack.
- **Bb7?** The bishop develops to the wrong diagonal; White's rook joins the attack via b1 or e1.

SKILL: Defend an exposed king by bringing pieces with tempo.

T4.6 24...f5 Exercise T4.6 (p. 515)

The bishop on e4 cannot move without exposing the e1-rook to your doubled rooks. Attack the pinned piece with a pawn: ...f5! If the bishop takes, ...Qxf5 and the pin is still there; after Rxe7 Rxe7 material is level. Other moves let White unpin with Kf2 or Re2 and stay a piece up.

TEMPTING BUT WRONG

- **b5?** Chasing the queen with ...b5 achieves nothing; after Qb4 White keeps the extra piece.
- **Qh5?** The queen lunge leaves the pin unused and White consolidates.

SKILL: Exploit a pin by attacking the pinned piece with a pawn.

T4.7 13...Qa5 Exercise T4.7 (p. 516)

White wants to castle, ending the pin, and then play e4 to kick your d5-knight, which also defends the loose bishop on b4. ...Qa5 defends b4 a second time and piles up on the pinned knight, so White's plan no longer wins material. After O-O Bxc3 e4 Bb4 exd5 Qxd5 the material stays level. Slow moves such as ...Nc6 allow O-O and e4 with a clear advantage for White.

TEMPTING BUT WRONG

- **Nc6?** Natural development, but after O-O and e4 the loose pieces on d5 and b4 drop.
- **Bxc3+?** Giving up the pinning bishop releases the pressure; after bxc3 White has the bishop pair and a strong centre.

SKILL: Use prophylaxis: anticipate the opponent's plan and defend loose pieces in advance.

T4.8 1.b6 Exercise T4.8 (p. 516)

This is the classic pawn breakthrough. b6! forces a passed pawn: after ...axb6 c6! bxc6 a6 the a-pawn runs, and after ...cxb6 a6! bxa6 c6 the c-pawn does. The black king is too far away to catch it. Any slow move lets the black king approach, and then White's pawns fall.

TEMPTING BUT WRONG

- **Kg3?** The king is too slow to help, and the black king reaches the queenside in time to stop the breakthrough.
- **a6?** After ...bxa6 there is no breakthrough left; Black's king will win the pawns.

SKILL: Use a pawn majority to force a passed pawn.

T4.9 1.Rg7+ Exercise T4.9 (p. 516)

The black king and rook stand on the same rank. Rg7+ is a skewer: the king must step off the seventh rank, and Rxc7 wins the rook. With rook and pawn against a bare king the rest is easy. Other moves leave a rook ending that is close to a draw.

TEMPTING BUT WRONG

- **Kf3?** Bringing the king is natural, but Black has time to get the king and rook off the same line.
- **Ra6?** Attacking from the side gains nothing; the skewer was available now.

SKILL: Spot skewers in rook endgames.

T4.10 1...Kc7 Exercise T4.10 (p. 516)

The defending king belongs on the queening square. ...Kc7 heads for d8 and also controls d6 and d7, so the white king cannot shoulder it away. After Rc2+ Kd8 Black reaches the Philidor-type defence. The natural-looking ...Kc8 loses, because Kd6 takes the key squares and the black king is cut off.

TEMPTING BUT WRONG

- **Kc8?** After Kd6 the white king takes d7 and the black king never reaches e8 or d8 in time.
- **Kb7?** The king walks away from the pawn; e7 follows and the pawn promotes.

SKILL: Bring the defending king to the queening square by the right route.

T4.11 1...Rg3+ Exercise T4.11 (p. 517)

Rb8 would be mate, so you need a forcing move. ...Rg3+ checks the king from behind; once it moves away from g6, your rook can return to the first rank or keep checking, and the white king cannot both escape the checks and support the f-pawn. Defending the back rank with ...Rc8 fails to Rh5 and Rh8+, which trades rooks into a lost pawn ending.

TEMPTING BUT WRONG

- **Rc8?** Passive: Rh5 and Rh8+ force a rook trade, and the f-pawn promotes.
- **Re3?** Too slow: Rb8+ Re8 Rxe8+ Kxe8 Kg7 and the pawn queens.

SKILL: Defend rook endings with checks from behind.

T4.12 12.Rfd1 Exercise T4.12 (p. 517)

The d-file is where Black's weaknesses live: the backward pawn on d6 and the hole on d5. Rfd1 puts the rook behind your queen on that file and prepares Nd5. Grabbing the b5-pawn with the knight is a blunder: ... axb5 Qxb5 leaves White with two pawns for a piece.

Also correct: Nd5, Rfb1, Rfc1, Rfe1, a4

TEMPTING BUT WRONG

- **Nxb5?** The b5-pawn is defended by the a6-pawn: after ...axb5 Qxb5 White has given a knight for two pawns.

SKILL: Put rooks on files that point at weaknesses.

T4.13 32. **Re8+ Rd8** 33. **g3** Exercise T4.13 (p. 517)

Re8+ forces ...Rd8 (...Ka7 drops the c7-pawn and more to Qxc7, and ...Qd8 simply loses the queen to Rxd8+). Now the rook on d8 is defended only by the queen on h4, along the h4-d8 diagonal. g3! attacks the queen, which must stay on that diagonal to keep guarding d8; after ...Qg5 f4 it runs out of squares and Black loses material. One defender, two jobs: that is overloading.

TEMPTING BUT WRONG

- **Qg7?** It looks active, but Black regroups with ...Nc8 and the back rank is safe.
- **b5?** A slow pawn move; Black's queen and knight get time to attack d5 and White's king.

SKILL: Recognize overloading: a single defender with two jobs.

T4.14 32. **Rxb7** Exercise T4.14 (p. 517)

The b7-pawn is undefended because White's own pawn on b3 blocks Black's rook. Rxb7 wins a pawn and puts a rook on the seventh rank; Rdd7 next doubles rooks there, attacking f7 and a7 and tying Black's king to the back rank. Quiet moves such as Rd7 or a5 give Black time for ...Ree2, and then Black's rooks are the active ones.

TEMPTING BUT WRONG

- **Rd7?** It reaches the seventh rank, but Black answers ...Re1+ and ...Ree2 with full counterplay.
- **Rc1?** Defensive: Black gets time for ...b6 and ...Ree2, and White's chance is gone.

SKILL: Activate rooks on the seventh rank in rook endings.

T4.15 1... **Ke6** Exercise T4.15 (p. 518)

Your king must fight the pawn directly. ...Ke6 attacks f6 and heads for f7 or f8, the squares in front of the pawn. After Rc6+ Kf7 the king blocks the pawn and the position is a draw. Checks such as ...Rg2+ only drive the white king where it wants to go, and the black king stays cut off.

TEMPTING BUT WRONG

- **Rg2+?** The check drives the white king forward to f5 and g6, supporting the pawn while your king is still far away.
- **Kd7?** The king goes the wrong way; Kg6 and f7 follow, and the pawn queens.

SKILL: Bring the defending king to the passed pawn before checking.

Chapter 24 · Advanced Tactics and Calculation

24.1 22.Rxd7 Qxd7 23.Nf6+ Kg7 24.Nxd7 Exercise 24.1 (p. 536)

The knight on g4 can reach f6, a square that forks g8, e8 and d7, and the g6 pawn no longer guards f6. Rxd7! lures the queen onto d7: after Qxd7 Nf6+ the king must move and Nxd7 wins the queen. If Black declines with a queen move, White is simply a rook up.

TEMPTING BUT WRONG

- **Nf6+?** The fork comes one move too early: Qxf6 and the knight is gone.
- **Bxc6?** Trading on c6 wins nothing and lets Black exchange rooks on d1.

SKILL: Use a decoy capture to bring an enemy piece onto a forkable square.

24.2 5.O-O Exercise 24.2 (p. 536)

Blunder-check the capture by looking at Black's forcing replies after it. 5.Nxe5? c6! attacks the bishop on b5, and when it retreats (6.Ba4 or 6.Bc4), 6...Qa5+ is a check that also hits the knight on e5, so Black wins a piece. 6.Nxf7 Kxf7 7.Ba4 only gives two pawns for the knight. This is the Mortimer Trap. Calm development such as 5.O-O, 5.c3 or 5.Nc3 keeps a pleasant game.

Also correct: Ba4, Bc4, Nc3, c3, d4

TEMPTING BUT WRONG

- **Nxe5?** 5...c6! and after the bishop retreats, 6...Qa5+ forks your king and the knight on e5.

SKILL: Blunder-check a capture by looking for the opponent's in-between check.

24.3 8...c5 9.Qc3 c4 Exercise 24.3 (p. 537)

...c5 hits the queen with tempo, and the follow-up ...c4 traps the bishop on b3: c4 and a4 are covered by Black's pawns and a2 is blocked by White's own pawn. White can collect a pawn or two for it, but the bishop is lost. This trap in the Ruy Lopez has a nickname, Noah's Ark, and the method transfers: gain a tempo on the queen, then close the cage.

TEMPTING BUT WRONG

- **Nf6?** Natural development, but it lets the bishop escape and White is fine.
- **Bb7?** Develops, but misses the pawn chase that traps the bishop.

SKILL: Chain a tempo gain with a trapping pawn move.

24.4 38...Rc1 39.Rxc1 bxc1=Q Exercise 24.4 (p. 537)

...Rc1! puts a rook between e1 and b1. If Rxc1, the pawn captures bxc1=Q; if White leaves the rook alone, ...b1=Q follows. The interfering piece must stand on c1, the one square where the pawn can recapture; quiet moves give White time for Rb1 and a draw.

TEMPTING BUT WRONG

- **Rb8?** Defends the pawn but cannot help it queen: Rb1 rounds it up and the game is a draw.
- **Ra8?** Grabs the a-pawn while White's rook blockades on b1; a draw.

SKILL: Recognize interference: blocking a defender's line of control.

24.5 22. Rxd6 Bxd6 23. Qxb7 Exercise 24.5 (p. 537)

Your rook on d1 is guarded by the queen and the knight, but Black attacks it twice with the rook and the x-raying queen. 22.Qxb7?? Rxd1+ 23.Nxd1 Qxd1# is mate on the back rank. First 22.Rxd6 removes the attacker, and only then 23.Qxb7 wins the pawn.

TEMPTING BUT WRONG

- **Qxb7?** Rxd1+ Nxd1 Qxd1# - the queen was guarding the back rank.

SKILL: Before a capture, check what the capturing piece stops defending.

24.6 18. Qg4 Exercise 24.6 (p. 537)

Qg4! threatens Qxg7# and also lines the queen up behind the knight against Black's queen on d7. If ...g6, the knight jumps with check (Nh6+) and uncovers an attack on d7; if ...Qxf5, Qxf5 wins the queen for a knight. A check or capture first, such as Rxe8+, gives Black time to organize the defence.

TEMPTING BUT WRONG

- **Rxe8+?** Forcing, but after Qxe8 Black's queen can reach e5 and guard g7; much of the advantage is gone.
- **Red1?** Attacks the queen, but ...Ne5 comes with tempo and White keeps only a modest edge.

SKILL: Find a quiet move that creates two threats (mate and a discovered attack).

24.7 20... Qxd1 Exercise 24.7 (p. 538)

Black's rook on d8 stands behind the queen, so it defends d1 through the queen. After ... Qxd1 White cannot recapture: Rxd1 Rxd1+ Ne1 Rxe1# is mate on the back rank, so Black simply wins the queen. Any other move lets White trade queens on d5 and stay a piece up.

TEMPTING BUT WRONG

- **Qa5?** Saves the queen but leaves you a piece down; the d-file shot works only now.
- **f6?** Making luft is too slow: Qxd5 trades queens and White stays a piece up.

SKILL: Recognize an x-ray: a piece that attacks or defends through another piece.

24.8 40... b3 41. cxb3 a3 42. bxa3 c3 Exercise 24.8 (p. 538)

...b3! is the breakthrough. If cxb3, ...a3! and after bxa3 the c-pawn runs (...c3, ...c2, ... c1=Q); if axb3, ...c3! and after bxc3 the a-pawn runs. White's king on f2 is one move too slow in every branch. Stop calculating only when no capture or check is left and a pawn is unstoppable.

TEMPTING BUT WRONG

- **a3?** Premature: bxa3 bxa3 and White's king collects the loose queenside pawns.
- **Ke5?** Centralizes, but it gives White time to defuse the breakthrough and the game is level.

SKILL: Calculate a branching tree (two recaptures) to the end.

24.9 38...**Qg4+** 39.**Kf1 Qd1+** 40.**Kg2 Qg4+** Exercise 24.9 (p. 538)

White's king has no shelter: Qg4+ and Qd1+ chase it between g2 and f1, and White's queen and rook are too far away to block. Any slow defensive move loses to the mating attack, so the only resource is to keep checking. When you are losing, look for your opponent's open king before you look for defensive moves.

Also correct: Qd1+

TEMPTING BUT WRONG

- **Qd5?** Centralizing is too slow: Qc8 and White's extra rook decides.
- **Qf4?** Too slow: Rb1 tidies up and White's king is safe while yours is not.

SKILL: Find perpetual check as a defensive resource when material is lost.

24.10 18.**Qh6 Qd2** 19.**Qg7#** Exercise 24.10 (p. 538)

Ignore the threat to your queen: Qh6! is a quiet move that threatens Qg7#, and nothing in Black's camp can cover g7 or create a flight square. The bishop on f6 is the key piece; trading queens on a5 would throw the attack away.

TEMPTING BUT WRONG

- **Qxa5?** Trading queens rescues Black: bxa5 and the mating idea has disappeared.
- **Qg5?** Also aims at h6, but gives Black a tempo; there is no mate next move.

SKILL: See a quiet move that creates an unstoppable mate threat.

24.11 19...**Rxd3** Exercise 24.11 (p. 539)

The bishop on d3 is both the attacker of h7 and an undefended piece: ...Rxd3 removes the threat and wins a piece outright. The check ...Bxf2+ looks forcing, but after Rxf2 Rxd3 Black has only won a pawn net; evaluate the final position honestly rather than falling for the check. Passive moves like ...h6 hold but waste the chance.

TEMPTING BUT WRONG

- **Bxf2+?** Forcing but not best: Rxf2 Rxd3 nets only a pawn compared with winning a whole piece.
- **h6?** Defends h7 but leaves the d3 bishop alive; the advantage is gone.

SKILL: Defend by capturing the attacker, and compare the end positions of forcing lines honestly.

Chapter 25 · Middlegame Strategy

25.1 14.d5 Exercise 25.1 (p. 555)

The IQP side's main break is d4-d5, played when the pieces are ready: here the queen and bishop aim at h7, the rook stands on e1 and both knights are active. 14.d5! opens lines: after 14...exd5 15.Bg5 the pin on f6 and the threat Nb5 are hard to meet, and 14...Nxd5?? removes the f6 defender, so 15.Qxh7+ Kf8 16.Qh8# follows. Wait one more move and Black blockades on d5.

TEMPTING BUT WRONG

- **Rd1?** Passive: Black completes development, blockades on d5 and the pawn becomes a target.
- **h3?** A tempo too slow: ...Nxd4 and the isolated pawn simply falls.

SKILL: Recognize when the d4-d5 break works for the side with the isolated pawn.

25.2 12...Ba6 Exercise 25.2 (p. 555)

Against the IQP you want to blockade and trade, but first you must stop the attack that the IQP side gets from its active pieces. 12...Ba6! covers d3, so Qd3 and the queen-and-bishop battery against h7 are no longer possible. The standard blockade 12...Nd5? is premature here: Nxd5 exd5 Qd3 g6 and White has removed your best kingside defender.

TEMPTING BUT WRONG

- **Nd5?** Blockading at the wrong moment: Nxd5 exd5 Qd3 and the f6 defender is gone.
- **Bd7?** Too passive: Qd3 forces ...g6 and Bh6 follows with a strong attack.

SKILL: Against the IQP, neutralize the attacking set-up before blockading.

25.3 12.b4 Exercise 25.3 (p. 556)

With your rook already on b1, 12.b4 starts the minority attack: b4-b5 will hit c6 and leave Black with a weak pawn on c6 or d5. The central break 12.e4? fails tactically here: dxe4 Bxe4 Nxe4 Nxe4 Bf5 pins the knight to your queen. A pawn storm with g4 is on the wrong side and just drops a pawn.

Also correct: Bxf6, Nh4, Rfc1, Rfe1, h3

TEMPTING BUT WRONG

- **e4?** dxe4 Bxe4 Nxe4 Nxe4 Bf5 pins the knight against your queen and wins material.
- **g4?** Bxg4: the pawn is simply lost and your own king is exposed.

SKILL: Execute the minority attack in the Carlsbad structure.

25.4 10...Ne7 Exercise 25.4 (p. 556)

10...Bxe5? gives up your good bishop for the knight, and after fxe5 White gains space, kicks your knight from f6 and opens the f-file for the attack. 10...Ne7! reroutes the knight towards f5 or g6, where it challenges White's pieces without giving away the bishop. Exchange to remove a strong piece only when the recapture does not improve your opponent's position.

Also correct: Be7, Qc7

TEMPTING BUT WRONG

- **Bxe5?** fxe5 gains space with tempo on f6 and opens the f-file for White's attack.
- **Ne4?** Nxe4 dxe4 Bxe4 and Black has simply lost a pawn.

SKILL: Judge an exchange by what the recapture gives the opponent.

25.5 12...b6 Exercise 25.5 (p. 556)

Cramped positions need a freeing pawn break, but the break must work. 12...b6 prepares ...Bb7 and ...c5, freeing the bishop safely; 12...c5 at once is also fine. The other break 12...e5? fails right now: Nxe5 Nxe5 dxe5 and your queen cannot regain the pawn because the white queen guards e5.

Also correct: a5, c5, h6

TEMPTING BUT WRONG

- **e5?** Nxe5 Nxe5 dxe5: the pawn on e5 is guarded by White's queen and Black is a pawn down.
- **f5?** Creates permanent holes on e5 and e6 and frees nothing; the c8 bishop stays bad.

SKILL: Choose a freeing break that actually works in a cramped position.

25.6 10.Nf1 Exercise 25.6 (p. 556)

In the closed Italian centre nothing is urgent, so improve your worst piece: Nf1 heads for g3 or e3 and frees the c1 bishop. The pawn break 10.d4? is premature: exd4 cxd4 Bxd4 and the a7 bishop wins a pawn. 10.c4? gives up control of d4 for ever and blocks your own bishop.

Also correct: Ba4, Bxe6, Qc2, a3, a4, h3

TEMPTING BUT WRONG

- **d4?** Premature: exd4 cxd4 Bxd4 and the pawn is lost to the bishop on a7.
- **c4?** Leaves a permanent hole on d4 for Black's knight and shuts in your bishop.

SKILL: In a closed centre, improve pieces before breaking.

25.7 13.Rae1 Exercise 25.7 (p. 557)

An outpost is only good if the piece can stay there. Here 13.Ne5? Nxe5 14.dxe5 Qxe5 wins a pawn, because after the recapture the e5 pawn has no support. Bring the rooks to the centre first; Ne5 can become strong later, once a rook on e1 or a pawn on f4 backs it up.

Also correct: Rfe1, a3, a4, g3, h3, h4

TEMPTING BUT WRONG

- **Ne5?** Nxe5 dxe5 Qxe5: the outpost was not supported and a pawn is lost.
- **c4?** Opens the centre for no reason: ...Na5 and ...dxc4 leave you with a weak isolated pawn.

SKILL: Check that an outpost piece can be maintained before occupying it.

25.8 9...Ne8 Exercise 25.8 (p. 557)

In a closed centre play on the side your pawns point to: Black prepares ...f5 by moving the f6 knight out of the way (Ne8 or Nd7). 9...b5? starts play on the queenside, where White is stronger, and after cxb5 it simply loses a pawn.

Also correct: Kh8, Nd7, a5, b6, h6

TEMPTING BUT WRONG

- **b5?** Wrong wing: cxb5 and you have lost a pawn where White is strongest.

SKILL: With a closed centre, play on the wing your pawns point to.

25.9 11.f3 Exercise 25.9 (p. 557)

11.f3 supports e4 so that Black's ...fxe4 never opens lines against your king, and it frees you to play c4-c5 on the queenside, where your pawns point. Lashing out with 11.h4? weakens your own king on the side where Black is stronger.

Also correct: Bd2, Bg5, Kh1, Rb1, b3

TEMPTING BUT WRONG

- **h4?** Weakens your king on Black's strong side; after ...Nf6 and ...fxe4 Black is better.
- **Nb5?** The knight achieves nothing on b5 while ...fxe4 opens Black's play.

SKILL: Stabilize the centre and play on your own strong wing.

25.10 9...Nfxd4 Exercise 25.10 (p. 557)

Against a pawn chain, strike at its base. The d4 pawn is attacked by both knights and defended only by the f3 knight, because your bishop on d3 blocks the queen's protection. So 9...Nfxd4 wins it: after 10.Nxd4 Nxd4 Black is a clean pawn up. Slow moves give White time for Bxf5 and Nc3, after which d4 is safe again.

Also correct: Ncxd4

TEMPTING BUT WRONG

- **Qb6?** Adds pressure but gives White time: Bxf5 exf5 and d4 holds.
- **Be7?** Too slow: White trades on f5, develops the knight and the pawn is safe.

SKILL: Attack the base of a pawn chain when it is under-defended.

25.11 45.Bxh7 Exercise 25.11 (p. 558)

The knight on b7 is stuck on the queenside and needs several moves to reach the kingside, while the bishop strikes at once: 45.Bxh7 wins a pawn, and the knight is too far away to trap the bishop. With pawns on both wings the long-range bishop is usually stronger than the knight; use its range before the knight can come back.

TEMPTING BUT WRONG

- **Kf4?** Too slow: the knight returns to defend and the chance is gone.
- **d6+?** The pawn is blockaded and lost after ... Nxd6; the advantage evaporates.

SKILL: Exploit the bishop's range against a knight when pawns are on both wings.

Chapter 26 · Endgame Technique

26.1 60.Bc4 Exercise 26.1 (p. 574)

With the wrong-coloured bishop the rook pawn wins only if the defending king never reaches h8. 60.Bc4! takes f7 and g8 away, so the king cannot cross towards the corner, and the h-pawn simply runs: 60...Ke7 61.Ke5 and h6-h7-h8. Any slow move lets ... Kf7-g7 and the game is a dead draw.

TEMPTING BUT WRONG

- **h6?** Kf7 and the king reaches g8 and h8: a draw, because the bishop can never drive it out.
- **Kg5?** Kf7 and the king gets to the corner in time; draw.

SKILL: Use the wrong-coloured bishop to cut the defending king off from the corner.

26.2 60...Ka1 Exercise 26.2 (p. 574)

Against a bishop's pawn the defender has a stalemate trick: with the king in the corner, capturing the pawn (or blocking it from c1) leaves Black with no moves. So ...Ka1! ignores the pawn's protection: after Qa3+ Kb1 Qb3+ Ka1 White cannot gain the tempo needed to bring the king closer. 60...Kc1? steps in front of the pawn and gives White time to approach.

Also correct: Ka2

TEMPTING BUT WRONG

- **Kc1?** Blocks your own pawn: White checks, gains time and brings the king to b3 or d2 for mate.

SKILL: Know the stalemate defence against queen versus bishop's pawn.

26.3 60.Qd2 Exercise 26.3 (p. 575)

60.Qd2! is a quiet move: it controls c1 and attacks the pawn on c2 without giving a check that drives the king towards the corner. If 60...Kc4, then 61.Qxc2+ wins the pawn; if 60...Kb2, then 61.Kd3 and king and queen together win. Checks such as Qb4+ or Qd5+ let the black king reach a1 or b1, where the stalemate defence holds.

TEMPTING BUT WRONG

- **Qb4+?** Drives the king towards a1 where the stalemate trick holds the draw.
- **Kd3?** The king steps up too early: ...c1=Q and the game is a draw.

SKILL: Win queen versus bishop's pawn when the king is close, without allowing the stalemate defence.

26.4 60...Ne5 Exercise 26.4 (p. 575)

The knight must reach a square from which it can control b8 in time. 60...Ne5! eyes d7 and c6, both of which guard b8: after 61.b7 Nd7+ 62.Kd6 Nb8 the pawn is stopped and the king comes back. Knight routes are short in squares but long in moves, so count them one by one.

TEMPTING BUT WRONG

- **Nd4?** Looks central, but the knight is simply captured: Kxd4.
- **Kg3?** The king is far too slow; the pawn promotes.

SKILL: Calculate knight routes to stop a passed pawn.

26.5 45.a5 Exercise 26.5 (p. 575)

Passed pawns must be pushed. 45.a5! drives the knight from b6 to c8 or a8, a passive square, and the pawn on a5 now ties the knight down for good. White's king and bishop then attack the kingside. Slow moves let Black play ...Nxa4 or ...Nd7 with a drawish position.

TEMPTING BUT WRONG

- **Ke3?** Too slow: ...Nxa4 and the outside passed pawn is gone.
- **g3?** Black simply captures on a4; the trump disappears.

SKILL: Push the outside passed pawn to tie down the opponent's knight.

26.6 45.Ke5 Exercise 26.6 (p. 575)

Activate the king: 45.Ke5! supports the d-pawn from in front and heads for e6 or f7, where it attacks Black's kingside pawns while the knight is busy. The knight cannot both stop the pawn and defend the kingside. Bishop checks or pawn moves waste the moment and Black's king blockades on d6.

TEMPTING BUT WRONG

- **Bg4+?** A check for nothing: ...Kd6 and Black's king blockades the pawn; draw.
- **d6?** The pawn runs too early and is rounded up by king and knight.

SKILL: Activate the king to support a passed pawn and attack weaknesses.

26.7 45...Kc6 Exercise 26.7 (p. 576)

Your bishop is good and White's is bad, so the dark squares in White's camp are yours. 45...Kc6! starts the king march to b5 and c4 or b4; White's king cannot cover both the queenside entry squares and d4, and the white bishop cannot attack anything. Bishop moves or waiting moves let White's king hold the entry squares and the game is drawn.

TEMPTING BUT WRONG

- **Bb4?** Activity without a target: White's king stays on d3 and covers the entry squares.
- **g5?** Opens the kingside for White's pieces and throws away the bind; Black is worse.

SKILL: With good bishop against bad bishop, use the king to invade on the colour the bad bishop cannot cover.

26.8 50...Bg5 Exercise 26.8 (p. 576)

In opposite-coloured bishop endings the defender sets up a blockade: the king stands in front of the pawns and the bishop controls the dark squares they must cross. 50...Bg5 keeps f6 and f4 under control, so neither pawn can advance safely while the king guards e6. 50...Bxg3? grabs a pawn but hxg3 leaves the connected pawns free to roll.

TEMPTING BUT WRONG

- **Bxg3?** Grabbing loses: hxg3 and the connected pawns advance with the king's help.
- **Bf6?** exf6+ and the bishop is simply lost.

SKILL: Blockade connected passed pawns in an opposite-coloured bishop ending.

26.9 45...Qxf2+ Exercise 26.9 (p. 576)

Queen endings are full of perpetual checks, and the defender should look for them first. 45...Qxf2+ takes the pawn that shelters White's king, and after Kh3 Qf1+ or Kh1 Qf1+ White cannot escape the checks. Any slow move loses to a8=Q.

TEMPTING BUT WRONG

- **Qf8?** Stops nothing for long: White's queen supports the pawn and it promotes.
- **Kh7?** A waiting move while the pawn queens; lost.

SKILL: Look for perpetual check in queen endings when you are worse.

26.10 45.d6+ Exercise 26.10 (p. 576)

45.d6+! gains a tempo with check: the pawn on d6 is supported by the king and cramps Black, and after 45...Kd8 46.Bb5 the knight is reduced to passive squares. White's king then turns to the kingside pawns. Quiet bishop moves give Black time for ...Kd7 and a solid blockade in front of the pawn.

TEMPTING BUT WRONG

- **Bc2?** Too slow: ...Kd7 and the blockade on d6 holds.
- **Kf5?** Leaves the pawn: ...Nxd5 and Black has won the trump.

SKILL: Push a passed pawn with tempo when the king supports it.

26.11 40...Nc6 Exercise 26.11 (p. 577)

An outside passed pawn wins knight endings because it drags the enemy king or knight away from the other wing. 40...Nc6! defends a7 and keeps the knight close to the pawn; next ...a5-a4 pushes. If the a-pawn falls, the remaining pawns are level and the game is drawn.

TEMPTING BUT WRONG

- **Nc4?** Active-looking, but Nxa7 removes your only winning chance.
- **a6?** Nc7+ forks king and pawn: the a-pawn falls next.

SKILL: Protect the outside passed pawn; it is the winning trump.

Chapter 27 · Practical Play and Psychology

27.1 22...g6 Exercise 27.1 (p. 594)

Before taking a gift, ask what the giver gains if you accept. After 22...Rxd5?? the rook leaves the back rank and 23.Re8# follows, because your king has no flight square. 22...g6 creates a flight square and keeps your extra pawn; only after that does the threat of ...Rxd5 become real. Nb4, chasing the bishop, is also fine.

Also correct: Nb4

TEMPTING BUT WRONG

- **Rxd5?** Takes the 'free' bishop and is mated at once by Re8#.
- **h6?** Also makes a flight square, but after Qc4 Black cannot keep f7 guarded: Bxc6 and Qxf7 win the pawn back with an active queen.

SKILL: Treat an unexpected 'free' piece as a question: check what your opponent gains if you take it.

27.2 24.Nf3 Exercise 27.2 (p. 594)

Black threatens 24...Qe1+! 25.Rxe1 Rxe1#, because your king has no flight square. Nf3 guards e1 and blocks nothing important; a flight square with h3, g3 or h4 also solves the problem. Once the back rank is safe, your extra piece decides the game at leisure.

Also correct: Qb5, Qc4, Rf1, g3, h3, h4

TEMPTING BUT WRONG

- **Nf5?** Attacks the queen, but 24...Qe1+ 25.Rxe1 Rxe1 is mate.
- **Qxb7?** Grabs a pawn and walks into the same mate: 24...Qe1+ 25.Rxe1 Rxe1#.

SKILL: After winning material, scan for the opponent's checks and back-rank tricks before pressing on.

27.3 18.Ne7+ Kh7 19.Nxc8 Exercise 27.3 (p. 595)

A draw offer is information, not an instruction: always look at the position before replying. Here 18.Ne7+ forks king and queen. After 18...Kh7 (or 18...Kh8) 19.Nxc8 you have won the queen for a knight, and the game is no longer close to a draw.

TEMPTING BUT WRONG

- **Nb6?** Forks queen and rook, but Black's queen simply moves and the knight on a8 is lost later; it wins far less than the check.
- **Rfd1?** A reasonable-looking move that misses the fork; the position stays roughly balanced.

SKILL: When offered a draw, run your normal checks-captures-threats scan before answering.

27.4 45...Qxg2+ Exercise 27.4 (p. 595)

When you are lost, look for stalemate: your king has no legal move and your pawn cannot move, so if your queen disappears you have no moves at all. 45...Qxg2+! leaves White only Kxg2 or Rxg2, and both are stalemate because the rook on g4 (or g2) still covers g7 and g8. 45...Qd1+ and 45...Qh5+ also draw by the same idea: either White takes the queen and it is stalemate, or the checks never stop.

Also correct: Qd1+, Qh5+

TEMPTING BUT WRONG

- **Qf5?** Natural centralizing, but 46.Qa1+ forces ...Qf6 and 47.Qxf6 is mate.
- **Qf3?** Also allows 46.Qa1+ Qf6 47.Qxf6#; a quiet move gives White the time he needs.

SKILL: In a lost position, look for stalemate and perpetual-check resources before resigning.

27.5 38.Qf5+ g6 39.Qxf7+ Kh8 40.Qg7# Exercise 27.5 (p. 596)

When you are winning, the most dangerous thing on the board is usually your opponent's counterplay, here perpetual check with ...Qd6+ and ...Qf4+. The safest way to stop checks is to give mate first. 38.Qf5+ g6 39.Qxf7+ Kh8 40.Qg7# works because the rook on b7 guards g7 once f7 has gone.

TEMPTING BUT WRONG

- **a6?** Relaxes: 38...Qd6+ starts a perpetual check that your king cannot escape.
- **Qc2+?** Offers a queen trade that is not a trade: the black queen on d2 simply takes on c2.

SKILL: When winning, eliminate the opponent's counterplay (here perpetual check) before pushing your own plan.

27.6 6.Ba4 Exercise 27.6 (p. 596)

This is the 'fishing pole' trap. After 6.hxg4? hxg4 the h-file is open for Black's rook and queen, and your knight on f3 is attacked; when it moves, ...Qh4 and mate on h2 or h1 follow, and Black also regains the piece. Simply ignoring the knight with 6.Ba4, 6.d4, 6.d3 or 6.c3 keeps a clear advantage, because Black's adventure has cost time.

Also correct: c3, d3, d4

TEMPTING BUT WRONG

- **hxg4?** Takes the bait: 6...hxg4 opens the h-file, and after the knight on f3 is lost or moves, ...Qh4 brings mating threats.

SKILL: Before taking a piece, check which enemy pieces gain lines towards your king.

27.7 7...Qxb2 Exercise 27.7 (p. 597)

Respect your opponent, but trust your calculation. White has no trap: after 7...Qxb2 the natural 8.Rb1 is met by 8...Qxc3, and Black is two pawns up with a safe queen. 7...c4, gaining space with tempo before taking b2, is just as good.

Also correct: c4

TEMPTING BUT WRONG

- **Bd7?** A timid developing move that lets White play Rb1 or Qb3; you leave a clean pawn on the table.
- **cxd4?** Releases the tension and gives White time to cover b2; the extra pawn is gone.

SKILL: Against stronger opponents, decide by calculation, not by reputation.

27.8 30.Rc1 Exercise 27.8 (p. 597)

When you are worse, avoid trades that lead to simple, lost endings, and keep pieces that can make trouble. After 30.Rxd8? Kxd8 the king and pawn ending is close to hopeless, because Black's outside passed h-pawn will drag your king away. 30.Rc1 keeps rooks on and heads for the c-file and the seventh rank; Rh1 (attacking the h-pawn) is a similar active idea.

Also correct: Rh1

TEMPTING BUT WRONG

- **Rxd8?** Trades into a king and pawn ending where Black's outside passed pawn should decide.

SKILL: When worse, keep the pieces that give counterplay and avoid simplifying into a lost ending.

27.9 20...Nf6 Exercise 27.9 (p. 598)

White has one idea left: Qh7#, with the bishop on c2 supporting the queen. 20...Nf6 covers h7, improves your worst piece and leaves White with nothing. Greedy captures lose instantly: winning extra pawns is worthless if your king is mated.

Also correct: Rfd8

TEMPTING BUT WRONG

- **Qxa2?** Takes a pawn and allows 21.Qh7#.
- **Bxg2?** Takes a pawn and allows 21.Qh7#.

SKILL: When ahead, find and kill the opponent's last threat before collecting more material.

27.10 60...Qf1# Exercise 27.10 (p. 598)

Every white pawn is blocked, so if the white king has no move and is not in check, it is stalemate. 60...Qf1# gives check along the first rank while your king and the h3-pawn cover g2. The moves that 'take away squares', ...Qf2 or ...Kf2, leave White with no legal move and no check: a draw.

Also correct: Qd1#, Qe1#, Qg2#

TEMPTING BUT WRONG

- **Qf2?** Takes g1 and g2 without giving check: stalemate, a draw.
- **Kf2?** Same mistake: White has no legal move and is not in check, so it is stalemate.

SKILL: When converting with a big material lead, give check or leave the opponent a move: avoid stalemate.

27.11 60...f6 61.Kd3 Kf4 Exercise 27.11 (p. 598)

Before accepting a draw, check whether you have a clear way to win. Here you do: the kings stand in opposition with White to move after a pawn move, so 60...f6! (a spare tempo) forces the white king to step aside, and after 61.Kd3 Kf4 your king walks in front of the pawn. 60...Kf5, heading straight for the squares in front of the pawn, also wins.

Also correct: Kf5

TEMPTING BUT WRONG

- **f5?** Pushes too far: after 61.Kf3 White holds the opposition in front of the pawn and it is a draw.
- **Kd5?** Lets 61.Kf4 put White's king in front of your pawn; a draw.

SKILL: Use a spare pawn tempo to win the opposition, and do not accept draws in won endings.

Chapter 28 · Analyzing Your Games and Training for Life

28.1 22.Ne7+ Kh8 23.Qxh7+ Kxh7 24.Rh3# Exercise 28.1 (p. 620)

Order matters. 22.Ne7+ first drives the king to h8 and puts the knight on e7, where it covers g8 and g6. Only then does 23.Qxh7+ Kxh7 24.Rh3# work: Anastasia's mate, with the knight taking the king's escape squares. In the game, 22.Qxh7+? Kxh7 23.Rh3+ Kg6 let the king walk out. Journal category: missed-mate.

TEMPTING BUT WRONG

- **Qxh7+?** The right idea in the wrong order: after 22...Kxh7 23.Rh3+ Kg6 the king escapes because the knight is not yet on e7.
- **Rh3?** Too slow: Black answers 22...f5 and the mating net never closes.

SKILL: Find the move order that makes a mating pattern work, and log missed mates in your journal.

28.2 38...Qf4+ Exercise 28.2 (p. 620)

White's king has only g1, h1 and h2, and White's queen and rook are far away. 38...Qf4+ (or 38...Qd6+) starts a perpetual check: for example 39.Kh1 Qf1+ 40.Kh2 Qf4+, and White cannot escape or block without losing material. The student's notes said 'down a rook, no chances', which is exactly why the habit of checking for perpetual before resigning matters. Journal category: time-psychology.

Also correct: Qd6+

TEMPTING BUT WRONG

- **Qxa5?** Grabs a pawn, but 39.Rxf7 creates mating threats and Black has wasted the moment for checks.
- **Qd5?** Centralizes without checking; White has time for Qc7 and the a-pawn, and Black is lost.

SKILL: Before resigning, look for perpetual check, and note 'resigned too early' honestly in your journal.

28.3 40.b6 axb6 41.c6 bxc6 42.a6 Exercise 28.3 (p. 621)

This is the classic three-against-three breakthrough. 40.b6! forces a capture, and after 40...axb6 41.c6! bxc6 42.a6 the a-pawn cannot be stopped (after 40...cxb6, 41.a6! bxa6 42.c6 wins the same way). It works only because Black's king is too far away; the student's 40.Kg2? gave Black time to bring the king closer, and then Black's own majority decided. Journal category: missed-win (endgame).

TEMPTING BUT WRONG

- **Kg2?** Natural but too slow: Black's king approaches and the breakthrough no longer works.
- **a6?** The wrong pawn first: after 40...bxa6 Black has enough pawns left to stop everything.

SKILL: Recognize pawn-breakthrough patterns and check whether the defending king is in the square.

28.4 9...Na5 Exercise 28.4 (p. 621)

9...Na5 is the classical plan: the knight hits the bishop on b3, and after 10.Bc2 c5 Black gains space on the queenside (9...Bb7, 9...Re8 and 9...Nb8 are also main lines). The student's 9...Bg4? simply drops a piece to 10.hxg4 Nxg4, because White's last move h3 was played to stop exactly this pin. Journal category: opening (with a hanging-piece cause: the opponent's last move was not checked).

Also correct: Bb7, Be6, Nb8, Nd7, Re8, h6

TEMPTING BUT WRONG

- **Bg4?** 10.hxg4 and Black has only a pawn for the piece.
- **Nxe4?** 10.Rxe4 and you are simply a piece down for a pawn.

SKILL: Know the main plan in your repertoire, and always ask what the opponent's last move prevents.

28.5 10...**Be7** Exercise 28.5 (p. 622)

The d4-pawn is a bait on the d-file. After 10...Qxd4? 11.Bb5+! the bishop moves with check and uncovers the rook on d1 against the queen: a discovered attack. Black must deal with the check and loses the queen. 10...Be7 (or 10...Bd7, covering b5) develops and gets the king ready to castle. Journal category: missed-tactic (the opponent's).

Also correct: Bd7, a6

TEMPTING BUT WRONG

- **Qxd4?** 11.Bb5+ uncovers the rook on d1 and wins the queen.

SKILL: Before capturing on a line, check what pieces stand behind the target and what can move off that line with check.

28.6 29.**Qd7 Qxc3** 30.**Qc8+ Kg7** 31.**Qxc7** Exercise 28.6 (p. 622)

Loose pieces drop off. 29.Qd7! attacks the undefended bishop on c7 and also threatens Qd8+ and Qe8. If 29...Qxc3, then 30.Qc8+ Kg7 31.Qxc7 wins the bishop, and the e5-pawn falls too. The engine's verdict swings from slightly worse to clearly winning on this one move, which is exactly the kind of turning point to look for. Journal category: missed-tactic.

TEMPTING BUT WRONG

- **Kh2?** A passive safety move; Black continues ...c5 and White is slightly worse.
- **Nxe5?** Grabs a pawn but drops the knight: 29...Bxe5, and Black is winning.

SKILL: Scan for loose pieces (undefended pieces) and use queen double attacks against them.

28.7 15...**Ne2+** 16.**Kh1 Nxc3** Exercise 28.7 (p. 623)

The knight on d4 can reach e2 with check, and from e2 it also attacks c3. 15...Ne2+ 16.Kh1 Nxc3 wins the queen for a knight. The student's move was not bad, but it was a 'what' move (it looked nice) rather than the result of checking every check. Journal category: missed-tactic.

TEMPTING BUT WRONG

- **Nxf3+?** Damages White's pawns and is fine, but it wins nothing; the fork on e2 wins the queen.

SKILL: Always list all checks: a knight check is often a fork.

28.8 10.**Bxh7+ Kxh7** 11.**Ng5+ Kg8** 12.**Qh5** Exercise 28.8 (p. 623)

This is the Greek gift sacrifice. The checklist is all there: bishop aimed at h7, a knight that can reach g5 with check, the queen able to reach h5, and no black knight on f6 to defend. After 10.Bxh7+ Kxh7 11.Ng5+ Kg8 12.Qh5, mate on h7 is threatened and Black must give up material (12...Qxg5 13.Bxg5). If 11...Kg6, the attack continues with h4-h5 or Qg4. Journal category: missed-win.

TEMPTING BUT WRONG

- **Ng5?** Without the sacrifice first, Black plays ...h6 or ...f5 and the attack is gone.
- **Re1?** The game move: solid, but it lets Black play ...Nf6 or ...f6 and the chance disappears.

SKILL: Recognize the Greek gift checklist and log missed attacking chances as 'missed-win'.

28.9 50...**Na4+** 51.**Kd4** **Nxb6** Exercise 28.9 (p. 624)

When converting, first remove the opponent's only counterplay. 50...Na4+ forks king and pawn, and after 51.Kd4 Nxb6 White has nothing left; Black's king then goes after the kingside pawns. The student's 50...Kd7 is still good for Black, but it allowed play with the passed pawn and made the win much harder. Journal category: conversion.

TEMPTING BUT WRONG

- **Kd7?** Keeps a winning advantage, but the b-pawn stays alive and Black's king and knight become tied to it; the conversion is much harder.
- **Nd5?** Covers b6 but simply hangs the knight to 51.Kxd5.

SKILL: When ahead in the endgame, eliminate the opponent's passed pawn with tempo.

28.10 11...**Bxh2+** 12.**Kxh2** **Qxd4** Exercise 28.10 (p. 624)

The bishop on d6 is standing between your queen and White's undefended queen. Moving it with check gives White no time to react: 11...Bxh2+ 12.Kxh2 Qxd4 wins the queen for a bishop. Whenever your own piece blocks a line to an enemy target, ask whether it can move with check. Journal category: missed-tactic.

TEMPTING BUT WRONG

- **Ng4?** The game move; it threatens h2 but gives White time to cover everything.
- **Bc5?** Attacks the queen but simply hangs the bishop: 12.Qxc5.

SKILL: Spot discovered attacks created by your own pieces, especially with a checking first move.

28.11 28.**Qxe8+** **Qxe8** 29.**Rxe8#** Exercise 28.11 (p. 625)

Both kings have weak back ranks, so the first player to use the e-file wins. Black's rook on e8 is defended only by the queen. 28.Qxe8+! Qxe8 29.Rxe8# exploits that single defender. The student's notes said 'I was lost anyway, so I grabbed a pawn': a time-psychology cause behind a missed-mate result. Journal category: missed-mate.

TEMPTING BUT WRONG

- **Qxa7?** Grabs a pawn and walks into 28...Rxe1#.
- **h3?** Makes a flight square, but 28...Rxe3 simply wins the queen.

SKILL: Look for checks first, especially on an open file where both back ranks are weak.

Stage 5 Test

T5.1 11.Qxc6+ bxc6 12.Ba6# Exercise T5.1 (p. 626)

This is Boden's mate: two bishops on crossing diagonals. 11.Qxc6+! bxc6 is forced (b8 and c7 are covered by the bishop on f4, and d7 is blocked by Black's own knight). Then 12.Ba6# finishes, because the bishop on f4 still covers b8 and c7.

TEMPTING BUT WRONG

- **Ba6?** The right bishop, but without the queen sacrifice Black simply plays ...Nb8 or ...bxa6 and keeps the king covered.
- **O-O-O?** A natural developing move that misses the forced mate.

SKILL: Recognize Boden's mate pattern against a king castled on the queenside.

T5.2 30...Qxd1+ 31.Qxd1 Rxd1# Exercise T5.2 (p. 626)

d1 is attacked twice (queen and rook, doubled on the d-file) and defended twice (rook and queen), but White's queen is overloaded: it must recapture and also guard the back rank. 30...Qxd1+! 31.Qxd1 Rxd1# mates because White's king has no flight square.

TEMPTING BUT WRONG

- **Qd4?** Offers a trade but drops the queen to 31.Rxd4.
- **Qg5?** Avoids the trade but misses the forced mate; the position is just equal.

SKILL: Use overloading and the back rank: when a defender must recapture, check what it stops guarding.

T5.3 11.Bxf6 g6 12.Bxe7 Exercise T5.3 (p. 627)

Remove the defender. 11.Bxf6! threatens Qxh7# immediately, and Black cannot recapture: after 11...Bxf6 or 11...gxf6, 12.Qxh7# follows. Black must block with 11...g6, and 12.Bxe7 leaves White a clean piece up.

TEMPTING BUT WRONG

- **dxc5?** Wins a pawn for a moment, but gives Black time for ...g6 or ...h6 and the tactic disappears.
- **Ne5?** A good square for the knight, but it misses that the f6-knight can simply be removed.

SKILL: Remove the single defender of a mating square.

T5.4 13...b6 Exercise T5.4 (p. 627)

13...b6! traps the bishop: its only exit square, b8, is covered by your rook and queen, and the b6-pawn is protected by the knight and queen. Black will follow with ...Ra8 or ...Qc7 and win the bishop for at most a pawn or two. White's best practical try is to look for desperate counterplay with Bxh7+ ideas, which do not change the result.

TEMPTING BUT WRONG

- **Bc6?** A reasonable developing move, but it lets the bishop escape with Bc5 or Be3.
- **Qa5?** Attacks the bishop, but Bc5 gets it out and trades it off.

SKILL: Trap a piece that has wandered into the corner by taking away its last exit.

T5.5 25.Nf7+ Kg8 26.Nh6+ Kh8 27.Qg8+ Rxg8 28.Nf7# Exercise T5.5 (p. 627)

This is the smothered mate (also called Philidor's legacy). 25.Nf7+ Kg8 26.Nh6+ is a double check, so the king must move: 26...Kh8. Then 27.Qg8+! Rxg8 forces Black's rook onto g8, and 28.Nf7# mates a king that is smothered by its own pieces.

TEMPTING BUT WRONG

- **Qg8+?** Too early: Black simply plays 25...Kxg8, because the knight is not yet on h6 to protect g8.
- **Be3?** Attacks the queen but gives Black time; with a rook less, White is simply lost.

SKILL: Execute the smothered-mate sequence in the right order.

T5.6 20...Qg1+ 21.Rxg1 Nf2# Exercise T5.6 (p. 627)

20...Qg1+! cannot be taken by the king because the bishop on c5 covers g1, so 21.Rxg1 is forced. Now the king is completely boxed in and 21...Nf2# is a smothered mate. The direct 20...Nf2+ only wins material after 21.Rxf2.

TEMPTING BUT WRONG

- **Nf2+?** A fork, but after 21.Rxf2 Qxf2 White is only losing material, not mated.
- **Qf2?** Threatens mate on g2 and g1, but 21.Qb3+ and Rxf2 defuse everything.

SKILL: Recognize the queen-sacrifice smothered mate.

T5.7 4.Qe2 Exercise T5.7 (p. 628)

4.Qe2! attacks the knight on e4 and, after it moves or is defended by 4...Qe7, threatens to open the e-file against Black's king. After 4...Qe7 5.Qxe4 d6 6.d4 dxe5 7.dxe5 White has won a pawn. Copying moves in open positions is dangerous because the side that moves first gets to the open file first.

Also correct: d4

TEMPTING BUT WRONG

- **Bc4?** Develops, but 4...d5 kicks the bishop and Black is at least equal.
- **Nxf7?** Grabs a pawn, but 4...Qe7 hits the e-file and Black's threats (...Nc3+ ideas) win material.

SKILL: Punish a symmetrical opening mistake by using the open e-file.

T5.8 4...Qg5 Exercise T5.8 (p. 628)

4...Qg5! attacks both the knight on e5 and the g2-pawn. If White ignores it with 5.Nxf7?, 5...Qxg2 6.Rf1 Qxe4+ 7.Be2 Nf3# is a famous mate. White's best is 5.Bxf7+ Kd8 6.O-O Qxe5, when Black has the better game. Remember that 3...Nd4 is a trap, not a sound opening; the book's repertoire move is 3...Bc5.

TEMPTING BUT WRONG

- **Qe7?** Attacks e5 and e4, but after 5.Nxf7 the knight hits the rook on h8 and White keeps the extra pawn and the better game.
- **Nf6?** Develops, but 5.Bxf7+ wins a pawn and exposes Black's king.

SKILL: Punish a pawn grab with a double attack, and know which opening tricks are unsound.

T5.9 50.Rh8 Exercise T5.9 (p. 628)

50.Rh8! clears the a8-square and threatens a8=Q. If Black takes the pawn with 50...Rxa7, then 51.Rh7+ skewers king and rook along the seventh rank. Black's other tries fail too: after 51.a8=Q Rxa8 Rxa8 White is a rook up. The trick works because Black's king is on the seventh rank but not on g7 or h7, where it would guard h7.

TEMPTING BUT WRONG

- **g4?** Creates kingside play slowly; Black's king returns to g7 and the skewer trick disappears. White is still better, but the immediate win is gone.
- **Kf1?** Bringing the king over is too slow: Black's king heads for g7, where the trick no longer works.

SKILL: Know the rook-endgame skewer trick with a rook in front of an a-pawn.

T5.10 30...Rxd1+ 31.Kxd1 Kd7 Exercise T5.10 (p. 628)

Trade into the king and pawn ending. After 30...Rxd1+ 31.Kxd1 Kd7 your outside passed h-pawn will pull White's king to the kingside, and your king will then win the queenside pawns. With rooks on, White's rook stays active and the win is much slower.

TEMPTING BUT WRONG

- **f6?** Safe, but keeps rooks on and gives White's rook time to become active; the win is far from clear.
- **Rc8?** Avoids the trade that wins most simply; White's rook takes the open d-file.

SKILL: Convert by trading into a won king and pawn ending when you have an outside passed pawn.

T5.11 60...Kg7 Exercise T5.11 (p. 629)

The pawn queens on h8, a dark square, and White's bishop is light-squared, so it can never drive your king out of the corner. Head for h8: 60...Kg7 (or 60...Kf8/60...Kg8) reaches the corner in time and the game is a draw. Moving away lets the white king take the corner squares.

Also correct: Kf8, Kg8

TEMPTING BUT WRONG

- **Ke7?** Moves away from the corner: 61.Kg6 and the h-pawn promotes.
- **Ke8?** Also too far from h8: 61.h6 and the pawn runs with the king's support.

SKILL: Know the wrong-coloured bishop and rook pawn draw and head for the corner.

T5.12 60...Ke7 Exercise T5.12 (p. 629)

Take the opposition directly in front of the pawn. After 60...Ke7 White's king cannot advance to d6 or f6 without allowing your king to stay in front of the pawn. 60...Kd8? or 60...Kf8? lets 61.Kd6 or 61.Kf6 seize the key squares and the pawn queens.

TEMPTING BUT WRONG

- **Kd8?** 61.Kd6 takes the opposition and the key square; e7 and e8 follow.
- **Kf8?** 61.Kd6 again: White's king reaches the seventh rank beside the pawn.

SKILL: Use the opposition to defend king and pawn endings.

T5.13 40.Qf7+ Kh8 41.Qf8+ Exercise T5.13 (p. 629)

Black's king has only h7 and h8, and both g7 and g8 are covered from f7 and f8. 40.Qf7+ Kh8 41.Qf8+ Kh7 42.Qf7+ is a perpetual check. Any quiet move loses: Black's queen and rook mate on f2 or g2.

TEMPTING BUT WRONG

- **Qd3?** Defends nothing: 40...Qxf2+ 41.Kh1 Qg2 is mate.
- **Qd7+?** A check, but after 40...Qg7 the checks run out and Black's extra rook wins.

SKILL: When losing, look for perpetual check before anything else.

T5.14 60...Kg2 61.h5 Kf3 Exercise T5.14 (p. 629)

This is the Réti idea: a king moving diagonally can chase two goals at once. 60...Kg2! heads towards the h-pawn and towards your own c-pawn. After 61.h5 Kf3 Black's king is inside the h-pawn's square, and if White spends time on c3, Black's king catches the pawn; if White pushes the pawn, ...Ke4 supports the c-pawn. Moving straight along the edge with ...Kh2 or pushing ...c2 at once loses.

TEMPTING BUT WRONG

- **Kh2?** Chases only the pawn, and from the wrong direction: 61.h5 and the pawn outruns the king.
- **c2?** 61.Kb2 stops the pawn and the h-pawn runs.

SKILL: Use diagonal king moves to pursue two goals at the same time (the Réti manoeuvre).

T5.15 6.Nxe5 Exercise T5.15 (p. 630)

The pin is not real, because the queen on d1 is a bait. 6.Nxe5! and if 6...Bxd1?? 7.Bxf7+ Ke7 8.Nd5# is Legal's mate. Black's best is 6...Nxe5 7.Qxh5, and White has won a pawn with the better position.

TEMPTING BUT WRONG

- **O-O?** Safe, but lets Black keep the pin and the e5-pawn; White's chance is gone.
- **d3?** Natural, but after ...Na5 Black trades off your bishop and equalizes.

SKILL: Recognize Legal's mate pattern from the attacking side and calculate the best defence.

Quick Reference: The Cheat Codes

The master decision checklist

Run this before every move in a slow game. With practice it takes seconds in quiet positions and you spend your real thinking time where it matters.

CHEAT CODE **Before every move**

1. **What is my opponent threatening?** Ask of their last move: what does it attack, what did it stop defending, what line did it open?
2. **Are any pieces undefended?** Mine *and* theirs. Loose pieces are where tactics happen.
3. **Do I have forcing moves?** Every check, every capture, every threat — for both sides.
4. **What changes after my candidate move?** What does the moved piece stop guarding? Which squares and lines open?
5. **What is my opponent's strongest reply?** Look for their checks, captures and threats against the position *after* your move. Only then play it.

CHEAT CODE **When ahead in material**

1. Remove counterplay first: check the opponent's threats every move.
2. Trade pieces, not pawns (usually). Each trade makes your extra material count for more.
3. Keep your king safe; do not grab more material if it opens your king.
4. Activate your king in the endgame and create a passed pawn.
5. Before every move near the end: **check for stalemate.**

CHEAT CODE **When behind in material**

1. Seek activity: put pieces where they create threats.
2. Keep pieces on the board; avoid trades that simplify into a lost ending.
3. Look for tactical resources: checks, perpetual check, stalemate tricks, forks.
4. Set practical problems — make the opponent find good moves.
5. Do not resign out of frustration; resign only when you would be embarrassed to play on against a strong player.

CHEAT CODE **When you do not know what to do**

1. Which is my worst-placed piece? Improve it.
2. Which enemy piece or pawn is weakest? Attack it.
3. What does my opponent want? Prevent it, if it is cheap to do so.
4. Is there a pawn break that opens lines for my better-placed pieces?
5. Can I safely improve my king (endgame) or make it safer (middlegame)?

COACHING JUDGMENT **How to use heuristics**

Every procedure in this book is a guideline that is right most of the time. The concrete position always wins the argument: if a check, capture or threat changes the picture, calculate it.

Cheat codes by chapter

CHAPTER 1 **Special-move checklist**

1. **Before castling:** have the king or this rook moved? Are the squares between them empty? Am I in check? Does the king cross or land on an attacked square?
2. **After an enemy pawn moves two squares next to your pawn on your fifth rank:** consider capturing en passant now, because next move it is too late.
3. **Before promoting:** would a queen be stalemate? Would a knight give a winning check or fork? Otherwise choose a queen.
4. **When far ahead:** after my move, does my opponent have a legal move or am I giving check? If neither, it is stalemate.
5. **When far behind:** look for perpetual check and stalemate tricks before resigning.

CHAPTER 2 **The board scan**

1. Trace every ray from the piece my opponent just moved, and from the square it left.
2. Trace every enemy queen, rook and bishop to the edge, forwards, sideways and backwards.
3. Picture every enemy knight's star of squares.
4. Look along both edges and both back ranks for loose pieces.
5. Repeat steps 2 to 4 for my own pieces to find my chances.

CHAPTER 3 **Is it safe to take?**

1. Count attackers and defenders of the target, including x-rays.
2. Is any defender cheaper than my capturing piece? If so, think twice.
3. Play out the exchange, lowest-value attacker first, adding up values.
4. Check the final position for loose pieces, checks, forks and mate threats.
5. Take only if the count favours me and step 4 is clean; otherwise play a solid developing move.

CHAPTER 4 The safety scan

1. Opponent's move: What does it attack? What does it no longer defend? What line did it open?
2. Undefended pieces: list mine and theirs.
3. Forcing moves: my checks, captures, threats; then theirs.
4. Choose a candidate. What changes after it?
5. Blunder check: after my move, what can my opponent capture or check? If something bad, go back to step 3.

CHAPTER 5 Mating a lone king

1. Box: cut the king off with your queen or rook.
2. Shrink: queen at a knight's distance; rook with checks only when the kings face each other.
3. Bring your king. Wait with the rook (far away) when needed.
4. Before every move near the edge: is it check? Does the opponent have a legal move?
5. Mate with the protected queen, the edge check, or the ladder.

CHAPTER 6 Opening move check

1. What is my opponent threatening? Is f7 (or f2) attacked twice?
2. Can I develop a piece toward the centre, preferably a knight?
3. Can I castle, or prepare to castle?
4. Am I moving a piece twice, moving the queen early, or making a pawn move that does not help? If so, can I name the concrete reason?
5. Blunder check: after my move, what can my opponent capture or check?

CHAPTER 7 The fork check

1. List the targets: the enemy king, the queen, rooks, and every loose piece.
2. For each pair of targets, find the squares from which one of your pieces would attack both.
3. Ask: can a piece of mine reach that square now, or after a check or forced capture?
4. Check that the square is safe and that the opponent has no forcing answer (a check, a capture, or one move that saves both).
5. Play the forcing moves first and the fork second.
6. Then run the same scan for your opponent before you commit to your move.

CHAPTER 8 The line check

1. List every line (rank, file, diagonal) that runs through an enemy king or queen.
2. On each line, find what stands between, and what stands behind.
3. A piece between your line piece and the king or queen is pinned: attack it again, preferably with a pawn.
4. A king or queen in front of another piece can be skewered: find a safe check or attack along the line.
5. Before relying on a pin, ask what the pinned piece would do if it moved anyway.
6. Run the same scan for your own king and queen, and break dangerous pins early.

CHAPTER 9 **The battery check**

1. Follow each of your bishops, rooks and queen through your own pieces: is an enemy king, queen, rook or loose piece behind?
2. If so, list the front piece's moves: checks first, then captures, then attacks.
3. Choose the move that leaves the target no escape and no counter-capture.
4. If the king is the target, check whether the front piece can also give check: a double check forces a king move.
5. Then run the same scan on your opponent's line pieces and defuse any battery aimed at your king or queen.

CHAPTER 10 **Before you attack a piece, count its exits**

1. Name every square the piece can move to.
2. Cross out the squares your pieces or pawns cover.
3. Cross out the squares its own pieces block.
4. If none are left, find the cheapest attacker — usually a pawn — and attack it.
5. Check your attacker is protected and that the opponent has no strong counter-attack in the meantime.

CHAPTER 10 **The duty-list procedure**

1. After checks, captures and threats, pick the enemy pieces that defend something important.
2. For each, list its duties: which pieces and which squares it guards.
3. Mark any piece that is the *only* defender of something valuable, and any piece with *two* duties.
4. Find a forcing move (capture, check, threat) that removes, chases, blocks, deflects or overloads that piece.
5. Calculate the opponent's best reply, then blunder-check before you play.

CHAPTER 11 Finding (and stopping) a mate

1. Count the enemy king's flight squares and note which are blocked by its own pieces.
2. Name the pattern that is "nearly there" and the ingredient that is missing.
3. Look for a check, sacrifice or quiet threat that supplies the missing ingredient — usually a deflection or decoy.
4. Calculate the opponent's best defence, including counter-checks and declining a sacrifice.
5. Run the same scan on your own king before you commit to the attack.

CHAPTER 12 Opening checklist as White

1. After every opponent move, run the safety scan: checks, captures, threats.
2. If the move broke a principle (early queen, weakening pawn move, undefended piece), look for development with tempo or a safe capture.
3. Otherwise, place your pieces on this chapter's squares: centre pawns first, knights, bishops, castle, then Re1.
4. Play the prophylactic moves at the right time: h3 against ...Bg4/...Ng4; c3 so the bishop has a retreat.
5. Choose the plan that fits the centre: closed → manoeuvre (Nbd2-f1-g3, a4); open → rooks to open files, d3-d4 or attack.
6. Before any early attack (Ng5, Nxf7, g4), check the opponent's most natural defence.

CHAPTER 13 **Opening checklist as Black**

1. What did White's last move threaten? Look at f7 (after 1.e4) and pins on your f6 knight (after 1.d4).
2. Can you develop with tempo against an exposed white piece? Do it.
3. Otherwise, put your pieces on their set-up squares: knights, the bishop that is free, castle.
4. Prepare your second pawn break: ...d5 against 1.e4 structures; ...c5 or ...e5 in the QGD and against the London.
5. Before every pawn break, check that it does not hang a piece or a pawn, and that your king is safe.

CHAPTER 14 **The move-selection routine**

1. What does my opponent's last move attack, threaten or prepare?
2. Safety scan: loose pieces, counted attacks, my king.
3. Forcing moves — checks, captures, threats — for me *and* for him.
4. Choose 2–4 candidate moves (forcing moves, answers to his idea, plan moves).
5. For each candidate, find his best reply; pick the move whose worst case is best.
6. Blunder check the new position, then play — and think on his time.

CHAPTER 15 **Calculating a line**

1. List forcing moves — checks, captures, threats — for the side to move.
2. Follow the most forcing line; at each step, find the opponent's best reply, starting with his checks.
3. Look for intermediate moves and quiet moves that add an attacker.
4. Stop when the position is quiet; picture where every piece stands.
5. Evaluate in one sentence (material, king safety, activity) and compare with your other candidates.

CHAPTER 16 Converting an advantage

1. Ask: what is his counterplay (threats, checks, passed pawns, active pieces, stalemate)?
2. Remove it first: prophylaxis, luft, blockade, or trade his most active piece.
3. Trade pieces, keep pawns — but check the ending you are trading into.
4. Before trading into a pawn ending, count passers, kings and races.
5. In the endgame, activate the king and create a passed pawn; switch to a second target if needed.
6. Before every move in a won ending, count his legal moves and his checks.

CHAPTER 17 Defensive routine

1. Name the threat: checks, captures, mate or tactics, and what the last move stopped defending.
2. List the menu: capture the attacker, move, defend, block, counter-attack, interpose with tempo.
3. Compare: is my counter-threat bigger and faster? Which defence also improves my position?
4. Before any trade, ask who the resulting position helps; calculate pawn endings first.
5. If you are losing, look for checks (perpetual), stalemate ideas and fortresses before playing on passively.

CHAPTER 18 **Pawn ending procedure**

1. Count: material, passed pawns, and the square of every passed pawn.
2. Kings first: bring your king towards the centre and in front of your pawns.
3. Name the key squares and ask who has the opposition with the move.
4. Look for a decoy (outside passer), a breakthrough, or a protected passer.
5. In a race, count tempi including checks and control of the queening squares.
6. Before trading into a pawn ending, run steps 1-5 on the position after the trade.

CHAPTER 19 **Piece activity — the quiet-move procedure**

1. Safety scan first: checks, captures and threats for both sides.
2. Count squares and targets for each of your pieces; name your worst piece.
3. Find a two- or three-move route that improves it (outpost, open file, seventh rank, centre).
4. If an enemy piece guards the target square, plan to remove or exchange it first.
5. Before any exchange, ask which piece leaves and which stays — trade bad for good.
6. Blunder-check the first move of the route, then play it.

CHAPTER 20 Pawn structure — the decision procedure

1. Describe the structure in one sentence: islands, isolated, doubled, backward, passed pawns.
2. List the weak squares (holes) for both sides, especially squares in front of weak pawns.
3. Choose the plan the structure suggests: attack the weakness, blockade it, or use the activity it gives you.
4. Before a pawn move, name the squares it will no longer guard and check whether an enemy piece can use them.
5. Before a pawn break, count the pieces on the lines it opens and check every capture.
6. Push passed pawns when safe, put rooks behind them, blockade enemy ones with a knight.

CHAPTER 21 King safety — the attack-and-defence procedure

1. For both kings, check the shield pawns, the nearby defenders and the open lines.
2. Count attackers that can reach the king's area in one or two moves against the defenders that can arrive.
3. Opposite-side castling: storm with pawns at once; same-side: attack with pieces and a rook lift.
4. Before a sacrifice, list the conditions (defender of g5, knight on f6, escape squares) and count again.
5. Defending: trade the strongest attacker, bring a defender, and only then consider a shield pawn move — the one that blocks the line or gains a tempo.
6. If a king is still in the centre, open the centre (or castle, if it is yours).

CHAPTER 22 Planning procedure for quiet positions

1. Safety first: checks, captures and threats for both sides. Tactics before strategy.
2. Run the checklist: material, kings, activity, pawns, space, files/squares, threats.
3. Name one target in your opponent's camp (weak pawn, hole, bad piece, exposed king).
4. If there is no target, name your worst piece and find it a better square.
5. Ask "What does my opponent want?" and stop it if it hurts you.
6. Choose a 2–3 move plan that fits the pawn structure; re-evaluate after each change.

CHAPTER 23 Rook endgame procedure

1. Compare the rooks: which is active, which is passive? Activate yours, even at the cost of a pawn.
2. Find the passed pawns and put your rook behind them (yours or theirs).
3. Attacking: cut off the enemy king by a file or rank before pushing; aim for the Lucena and build the bridge.
4. Defending: king in front of the pawn, rook on the third rank (Philidor); when the pawn advances, check from behind.
5. If your king is pushed aside, go to the short side and check from the long side.
6. In rook-against-pawn races, count tempi for the pawn and both kings before you move.

CHAPTER 24 The calculation routine

1. Scan both sides' checks, captures and threats, and note every loose piece and every line (files, diagonals, x-rays).
2. Look for a target pattern (fork square, skewer line, mating square) and the forcing move that creates it.
3. Choose two or three candidate moves, forcing moves first, and include one quiet move with a double threat.
4. For each candidate, calculate your opponent's most annoying reply first, especially in-between checks and desperado captures.
5. Stop each branch when the position is quiet, count material and king safety, and ask what your opponent would play there.
6. Play the move whose worst branch is best for you, after a final check of what your moving piece stops defending.

CHAPTER 25 The strategic routine

1. Name the pawn structure and its weak pawns and squares, for both sides.
2. Judge each minor piece: good or bad, and how many useful squares it has.
3. Decide where to play: in the centre if it is open, otherwise on the wing your pawns point to and where you are stronger.
4. Pick a plan of two or three moves and the exchanges that help it (trade defenders, bad pieces and attackers; keep pieces when you have space).
5. Check every plan move tactically, including what the recapture of any exchange gives the opponent.

CHAPTER 26 The endgame routine

1. Activate the king: decide where it is most useful in ten moves and start walking.
2. Find the trumps: your passed pawns and the opponent's weakest pawns; plan a second weakness.
3. Improve every piece before committal moves, and keep setting problems.
4. Before any trade, check which ending remains (opposite bishops, wrong-coloured bishop, lost pawn ending).
5. In races and sharp positions, count tempi and list all candidates, including quiet king and bishop moves; in queen endings, count the opponent's checks.

CHAPTER 27 Practical decision procedure

1. Is this a critical moment (forcing moves, a transition, a surprise)? If yes, slow down and use your full routine.
2. What is my opponent trying to do? Name their best idea or last chance.
3. If I am winning: stop their idea first, then trade pieces and simplify.
4. If I am losing: keep sound counterplay, avoid trades into lost endings, look for perpetual check and stalemate.
5. Before playing the move, sit on your hands: what is the most annoying reply?
6. Between games: after two losses in a row, stop the rated session.

CHAPTER 28 The post-game procedure

1. Engine off: write your thoughts at the three most important moments.
2. Mark the turning points: discomfort, long thinks, material or structure changes.
3. Engine on: find the first big swing (about a pawn or more); ignore small differences.
4. For each real mistake, ask why, choose one category, and write a rule in your own words.
5. Log it in the journal with a FEN, and schedule reviews at 1, 3, 7 and 21 days.
6. Every ten to twenty games, update the weakness list and train the top category.

Glossary

Short, plain-English definitions of the chess terms used in this book. Where a term has a fuller explanation in a chapter, the definition here is only a reminder.

A

Active piece — A piece that controls many squares, attacks targets or supports a plan. The opposite of a passive piece.

Advantage — A lasting plus for one side: extra material, a safer king, better pieces or a better pawn structure.

Alapin Variation — The Sicilian Defence line 1.e4 c5 2.c3, which prepares d2-d4. It is the book's choice for White against the Sicilian.

Algebraic notation — The standard way of writing chess moves, naming each square by a file letter (a–h) and a rank number (1–8), such as `Nf3` or `exd5`.

Analysis board — A digital board, found on most chess sites and apps, where you can set up positions, try moves and (optionally) switch on an engine.

Annotation symbols — Marks added to moves to judge them: `!` good, `!!` brilliant, `?` mistake, `??` blunder, `!?` interesting, `?!` dubious.

Attack — A threat to capture a piece, or a coordinated effort with several pieces against the enemy king.

B

Back-rank mate — Checkmate by a rook or queen on the first or last rank, against a king trapped behind its own pawns.

Backward pawn — A pawn that has fallen behind its neighbours and cannot safely advance, often standing on a half-open file in front of an enemy piece.

Bad bishop — A bishop hemmed in by its own pawns fixed on squares of the same colour.

Battery — Two pieces lined up on the same line, such as a queen behind a bishop on a diagonal or two rooks doubled on a file.

Bishop pair — Having both bishops when the opponent does not. In open positions it is usually a small but real advantage.

Blitz — A fast time control, typically a few minutes per player for the whole game.

Blockade — Placing a piece, often a knight, directly in front of an enemy passed pawn to stop it.

Blunder — A serious mistake that loses material or the game, usually marked `??`.

Blunder-check — The habit of looking at your intended move from the opponent's side (checks, captures, threats) before you play it.

Boden's mate — A checkmate by two bishops on crossing diagonals, typically against a king castled on the queenside.

Breakthrough — A pawn sacrifice (or several) that clears the way for one pawn to promote.

Building a bridge — The winning method in the Lucena position: the rook goes to the fourth rank so that it can block the defending rook's checks.

C

Candidate moves — The short list of moves you seriously consider before choosing one.

Capture — Removing an enemy piece from the board by moving onto its square.

Castling — A special move of king and rook that brings the king to safety and the rook towards the centre. It has strict conditions (see Chapter 1).

Centre — The squares d4, d5, e4 and e5, and more loosely the squares around them.

Check — A direct attack on the king. The player in check must get out of it at once.

Checkmate — A check that cannot be escaped. It ends the game.

Checks, captures, threats — The order of the safety scan: look first at all checks, then all captures, then all threats, for both sides.

Chunk — A familiar group of pieces that a player recognizes as one pattern rather than as separate pieces.

Clearance — A sacrifice or move that empties a square or line so that another piece can use it.

Closed position — A position with locked pawn chains and few open lines, where manoeuvring matters more than quick tactics.

Connected passed pawns — Two or more passed pawns on adjacent files that can protect each other.

Conversion — Turning an advantage into a win, usually by stopping counterplay and simplifying.

Counterplay — Active chances for the side that is worse, which force the stronger side to react.

Counting — Comparing the number of attackers and defenders of a square to see whether a capture there wins material.

Critical moment — A position where the choice you make now changes the course of the game, so that it deserves extra thinking time.

Cut off (the king) — Using a rook on a file or rank to keep the enemy king away from the important part of the board.

D

Decoy — A sacrifice that lures an enemy piece, often the king, onto a bad square.

Deflection — Forcing an enemy piece away from a square or line it was defending.

Deliberate practice — Focused training aimed at specific weaknesses, with feedback, rather than simply playing for enjoyment.

Desperado — A piece that is lost anyway and captures as much as it can before it is taken.

Development — Bringing pieces from their starting squares into play.

Discovered attack — Moving one piece so that another piece behind it attacks something.

Discovered check — A discovered attack in which the uncovered piece gives check.

Double attack — One move that creates two threats at once. A fork is the most common type.

Double check — A check by two pieces at once. The king must move; nothing can block or capture both.

Doubled pawns — Two pawns of the same colour on the same file.

Draw — A game that ends with neither side winning, for example by agreement, stalemate, repetition, the fifty-move rule or insufficient material.

Draw offer — A proposal to end the game as a draw. Over the board it is made after making your move and before pressing your clock.

E

Elo rating — A rating system, named after Arpad Elo, that estimates a player's strength from results against rated opponents.

En passant — A special pawn capture: a pawn that has just advanced two squares can be captured as if it had moved only one, but only on the very next move.

Endgame — The final phase of the game, with few pieces left, where king activity and passed pawns become decisive.

Engine — A computer program that plays and analyzes chess, such as Stockfish.

Evaluation — A judgment of who stands better. Engines express it as a number in pawns from White's point of view, such as +1.20.

Exchange — A trade of pieces. "Winning the exchange" means winning a rook for a bishop or knight.

F

FEN — Forsyth–Edwards Notation: a one-line text description of a position, including whose move it is, which you can paste into any analysis board.

Fianchetto — Developing a bishop on b2, g2, b7 or g7 after moving the knight's pawn one square.

Fifty-move rule — A draw can be claimed when fifty moves by each side have been played with no capture and no pawn move.

File — A vertical column of the board, named a to h.

First big swing — In engine analysis, the first move where the evaluation changed by roughly a pawn or more.

Flight square — A square to which a king can escape, for example after a pawn move in front of a castled king. Also called luft.

Fork — One piece attacking two or more enemy pieces at the same time.

Fortress — A defensive set-up that the stronger side cannot break through, even with extra material.

G

Gambit — An opening in which one side offers material, usually a pawn, for development or attack.

Giuoco Piano — The Italian Game with 1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5. The slow version with c3 and d3 is called the Giuoco Pianissimo.

Good bishop — A bishop whose own pawns are mostly on squares of the opposite colour, so its diagonals are open.

Greek gift — The bishop sacrifice Bxh7+ (or ...Bxh2+) against a castled king, usually followed by a knight check on g5 and the queen joining the attack.

H

Half-open file — A file with pawns of only one colour on it. It is open for the side without a pawn there.

Hanging piece — A piece that is attacked and not adequately defended.

Hole — A square that can no longer be controlled by your own pawns, which an enemy piece may occupy.

Hope move — A move played only because the opponent might miss a cheap threat, rather than because it improves the position.

I

Increment — Time added to a player's clock after each move, for example the "+10" in a 15+10 time control.

Initiative — The ability to make threats that the opponent must answer, so that you control the course of the game.

Insufficient material — A position in which neither side can possibly checkmate, such as king against king, which is a draw.

Interference — Placing a piece on a line between an enemy piece and the square or piece it defends.

Interleaving — Mixing different types of problems in one practice session instead of practising one type at a time.

Isolated pawn — A pawn with no friendly pawns on the adjacent files.

K

Key squares — In king and pawn endings, squares that guarantee promotion if the attacking king reaches them.

King safety — How well protected a king is from attack: pawn shelter, nearby defenders and the absence of open lines towards it.

Kingside — The half of the board on the king's side (files e to h).

L

Legal's mate — A mating pattern in which a player sacrifices the queen to a pin and then mates with minor pieces, for example Nxe5, Bxf7+ and Nd5#.

Loose piece — An undefended piece, which is a natural target for double attacks.

Lucena position — The basic winning position in rook and pawn against rook, won by building a bridge.

Luft — German for "air": a flight square for the king, usually made by a pawn move such as h3 or ...h6.

M

Major piece — A queen or rook.

Mate pattern — A familiar arrangement of pieces that gives checkmate, such as the back-rank mate or the smothered mate.

Middlegame — The phase after the opening, when most pieces are developed and plans and tactics dominate.

Minor piece — A bishop or knight.

Minority attack — Advancing a smaller group of pawns against a larger one in order to create weaknesses in the opponent's structure.

Mistake categories — The eleven labels this book uses to classify errors: hanging-piece, missed-tactic, missed-mate, king-safety, calculation, opening, endgame, missed-win, conversion, positional and time-psychology.

Mistake journal — A record of your significant mistakes with the position (FEN), the category, the cause and a rule for next time.

Mobility — The number of useful moves your pieces have.

O

Open file — A file with no pawns on it, ideal for rooks.

Open position — A position with few pawns in the centre and many open lines, where piece activity and tactics matter most.

Opening — The first phase of the game, in which both sides develop their pieces and fight for the centre.

Opening repertoire — The set of openings you choose to play with each colour.

Opposite-coloured bishops — A position in which each side has one bishop and they move on different colours. In endings this often leads to a draw.

Opposition — In king endings, the situation in which the kings stand facing each other with one square between them and the side *not* to move holds the opposition.

Outpost — A square, usually in enemy territory, that cannot be attacked by enemy pawns and can be occupied by your piece, often a knight.

Outside passed pawn — A passed pawn far from the other pawns, which can drag the enemy king away.

Overloading — Giving an enemy piece more defensive duties than it can carry out at once.

Overpressing — Trying too hard to win an equal position, taking risks that make your own position worse.

P

Passed pawn — A pawn with no enemy pawns in front of it on its own file or the adjacent files.

Pawn break — A pawn move that challenges the opponent's pawn chain and opens lines.

Pawn chain — A diagonal line of pawns protecting each other.

Pawn majority — More pawns than the opponent on one side of the board, which can create a passed pawn.

Pawn structure — The arrangement of the pawns, which shapes the plans for both sides.

Performance rating — A rating estimated from results or moves over a set of games. From a single game it means very little.

Perpetual check — A series of checks that the opponent cannot escape, leading to a draw.

Philidor position — A key drawing method in rook and pawn against rook: the defending rook stays on its third rank until the pawn advances, then checks from behind.

Pin — An attack on a piece that cannot move without exposing a more valuable piece behind it. If the piece behind is the king, the pin is absolute.

Plan — A goal for the next several moves, based on the features of the position.

Promotion — A pawn reaching the last rank and becoming a queen, rook, bishop or knight.

Prophylaxis — A move that prevents the opponent's plan before it becomes a threat.

Protected passed pawn — A passed pawn defended by another pawn.

Puzzle rating — A rating for solving puzzles on a website. It measures something different from your playing rating.

Q

Queen's Gambit Declined — The opening 1.d4 d5 2.c4 e6, in which Black keeps the pawn on d5. It is the book's main defence for Black against 1.d4.

Queenside — The half of the board on the queen's side (files a to d).

Quiet move — A move that is not a check, capture or direct threat, often the hardest kind of move to find in a combination.

R

Rank — A horizontal row of the board, numbered 1 to 8 from White's side.

Rating — A number estimating playing strength within one pool of players, such as one website or one federation.

Removing the defender — Capturing or driving away a piece that guards an important square or piece.

Repetition — A draw that can be claimed when the same position occurs three times with the same player to move and the same possible moves.

Resignation — Conceding the game before checkmate.

Réti manoeuvre — A diagonal king march that pursues two goals at once, named after the endgame study composed by Richard Réti.

Retrieval practice — Learning by recalling information from memory, for example by solving positions, rather than by re-reading.

Rook lift — Bringing a rook forward along a file, for example to the third rank, so that it can swing across to attack.

Rule of the square — A quick way to see whether a king can catch a passed pawn: draw the square from the pawn to its promotion rank and check whether the king can step into it.

S

Sacrifice — Deliberately giving up material for an advantage such as an attack, a passed pawn or a draw.

Safety scan — The book's routine of checking all checks, captures and threats for both sides before moving.

SAN — Standard Algebraic Notation, the move format used in this book, such as `Nf3`, `exd5` or `O-O`.

Scholar's mate — The early mate on f7 with queen and bishop (for example Qh5, Bc4 and Qxf7#).

Seventh rank — The opponent's second rank. A rook there often attacks pawns and confines the enemy king.

Simplification — Trading pieces, usually to make a winning advantage easier to convert.

Skewer — An attack on a valuable piece that, when it moves, exposes a less valuable piece behind it.

Smothered mate — A knight checkmate against a king surrounded by its own pieces.

Space — Control of squares, especially in the opponent's half of the board.

Spaced repetition — Reviewing material at increasing intervals, such as 1, 3, 7 and 21 days.

Stalemate — A position in which the player to move is not in check but has no legal move. It is a draw.

Swindle — Saving or winning a bad position by setting practical problems that the opponent fails to solve.

T

Tempo — A unit of time in chess: one move. Gaining a tempo means making a useful move while forcing the opponent to respond.

Threat — A move that will win something, or give mate, if it is not answered.

Tilt — Playing worse because you are upset, usually after a painful loss.

Time control — The amount of thinking time each player has, such as 15+10 (fifteen minutes plus ten seconds per move).

Time trouble — Having so little time left that you cannot think normally.

Touch-move — The over-the-board rule that a piece you deliberately touch must be moved (or, if it is an opponent's piece, captured) if that is legal.

Trade — Exchanging pieces of roughly equal value.

Trapped piece — A piece that cannot escape attack because all its squares are covered or blocked.

Triangulation — In king endings, losing a move by walking the king around a triangle of squares to put the opponent in zugzwang.

Turning point — A moment in a game where the evaluation or the character of the position changed decisively.

U

Underpromotion — Promoting a pawn to a rook, bishop or knight instead of a queen, usually to avoid stalemate or to give check.

W

Weak square — A square that cannot be defended by pawns and can be used by enemy pieces.

Weakness list — A ranking of your mistake categories by how often they occur and how much they cost you.

Wrong-coloured bishop — A bishop that does not control the promotion square of its own rook pawn, so that the defending king in the corner holds the draw.

X

X-ray — An attack or defence by a piece through another piece on the same line.

Z

Zugzwang — A position in which any move worsens your position, so that you would prefer to pass.

Zwischenzug — German for "in-between move": an unexpected move, often a check or threat, played before the expected recapture.

Further Reading and References

This section lists the research the book relies on, a short list of chess books for each stage, and the free tools that work best with the book's exercises. It ends with a note on how claims in the book were labelled and checked.

Learning-science and chess-expertise references

These are the studies cited in the book's evidence boxes. Each is a published work; the book uses them for their general findings, not for precise numbers.

- Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. *Psychological Bulletin*, 132. (Spacing: spreading practice out improves long-term retention.)
- Charness, N., Tuffiash, M., Krampe, R., Reingold, E., & Vasyukova, E. (2005). The role of deliberate practice in chess expertise. *Applied Cognitive Psychology*, 19. (Serious solitary study as a strong predictor of chess skill.)
- Chase, W. G., & Simon, H. A. (1973). Perception in chess. *Cognitive Psychology*, 4. (Strong players perceive familiar patterns, or chunks.)
- de Groot, A. D. (1965). *Thought and Choice in Chess*. The Hague: Mouton. (English edition of the classic think-aloud study of chess players.)
- Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100. (The original description of deliberate practice.)
- Kornell, N., & Bjork, R. A. (2008). Learning concepts and categories: Is spacing the "enemy of induction"? *Psychological Science*, 19. (Interleaving examples improves classification of new cases.)
- Macnamara, B. N., Hambrick, D. Z., & Oswald, F. L. (2014). Deliberate practice and performance in music, games, sports, education, and professions: A meta-analysis. *Psychological Science*, 25. (Practice matters, but explains only part of the differences between people.)
- Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17. (Retrieval practice: testing yourself beats re-reading.)

- Rohrer, D., & Taylor, K. (2007). The shuffling of mathematics problems improves learning. *Instructional Science*, 35. (Mixed practice outperforms blocked practice on later tests.)

Recommended chess books by stage

All of these are well-known books that have stayed in print or in wide use for years. Pick one or two per stage; you do not need them all. The one-line reasons are coaching judgment.

Stage 1 (400–600): rules, safety and first mates

- **Yasser Seirawan, *Play Winning Chess*** (first book of Seirawan's *Winning Chess* series). A friendly, clear introduction to the ideas behind the moves.
- **László Polgár, *Chess: 5334 Problems, Combinations and Games*.** A huge collection; its many mate-in-one and mate-in-two problems are ideal daily practice at this stage and the next.

Stage 2 (600–800): basic tactics and a first repertoire

- **Yasser Seirawan, *Winning Chess Tactics*.** Explains each tactical motif with clear examples before testing you.
- **Dan Heisman, *Everyone's 2nd Chess Book*.** Practical advice on safety, thinking habits and the mistakes that decide games below 1200.

Stage 3 (800–1000): thinking, calculation and conversion

- **Irving Chernev, *Logical Chess: Move by Move*.** Complete games with every move explained in plain language; excellent for learning why moves are played.
- **Yasser Seirawan, *Winning Chess Strategies*.** The strategic companion to the tactics volume, at the right level for this stage.

Stage 4 (1000–1200): activity, structure, plans and rook endings

- **José Raúl Capablanca, *Chess Fundamentals*.** A short classic on endings, middlegame principles and planning by a world champion.
- **Jeremy Silman, *Silman's Complete Endgame Course*.** Endgames organized by rating level, so you learn exactly what you need now and what can wait.

Stage 5 (1200–1500) and beyond

- **Jesús de la Villa, *100 Endgames You Must Know*.** The essential theoretical endings, including Lucena and Philidor, explained carefully.
- **Axel Smith and Hans Tikkanen, *The Woodpecker Method*.** A large tactics collection designed for repeated solving cycles; best started near the end of this stage.
- **Alexander Kotov, *Think Like a Grandmaster*.** A classic on candidate moves and calculation; read it after Chapters 14, 15 and 24.
- **Aron Nimzowitsch, *My System*.** The famous classic on positional concepts such as blockade, prophylaxis and the open file. Demanding but rewarding once you are comfortable with Chapters 19–25.

Free tools

Lichess (lichess.org)

- **Analysis board** (lichess.org/analysis): set up any position, play moves, and switch the engine on or off with one click. To load a position from this book, paste its FEN into the FEN field below the board.
- **Studies** (lichess.org/study): create chapters with positions, comments and questions. A study with one chapter per journal entry is an easy way to store and review your own mistakes (Chapter 28).
- **Puzzle themes** (lichess.org/training/themes): puzzles filtered by theme, such as fork, pin, back-rank mate or endgame, which makes it easy to practise one weakness or to build a mixed set.
- **Practice against the computer and game analysis:** after any game, the analysis view shows engine evaluations; use it only after you have written your own notes.

Chess.com (chess.com)

- **Analysis board:** paste a FEN or PGN into the load option of the analysis tool to set up a position.
- **Puzzles and lessons:** large puzzle collections and themed lessons; the free tier is enough to support this book.

How to load a FEN anywhere

A FEN is one line of text such as `r1bqkbnr/pppp1ppp/2n5/4p3/4P3/5N2/PPPP1PPP/RNBQKB1R w KQkq - 2 3`. In almost every chess app, look for an option called "Load FEN", "Set up position", "Paste FEN" or "Board editor". Copy the whole line, including the part after the first space: it says whose move it is, which castling rights remain and whether en passant is possible. If an app refuses the FEN, check that you copied every character.

Evidence vs. coaching judgment in this book

The book distinguishes two kinds of advice.

- **Established evidence** is marked in `evidence` boxes and always names a real, published source (listed above). These sources mostly study learning and memory in general, or chess expertise in general. Applying them to an adult improving from 400 to 1500 is a reasonable extension, and the book says so where it matters; none of them promises a particular rating.
- **Coaching judgment** is marked in `coach` boxes or labelled in the text. It covers practical rules such as the 1-3-7-21 day review schedule, time budgets for a clock, the "stop after two losses in a row" rule and the frequency-times-cost weakness list. These are the author's recommendations based on common teaching practice, not research findings. Treat them as starting points and adjust them to your own experience.

Ratings mentioned in the book are milestones, not promises, and statistics have not been invented: where no reliable figure exists, the book gives none.

Every position and move sequence in the book was checked by computer. Legality, notation and the correct side to move were verified with the python-chess library, and every exercise solution, every "wrong answer" and every engine-checked line was evaluated with the Stockfish 18 engine. Illustrative positions described as "from a student's game" are constructed teaching positions, not records of real games; real games are only quoted where the players, year and moves are certain.